Indian Statistical Institute Series

Ravindra B. Bapat ·
Manjunatha Prasad Karantha ·
Stephen J. Kirkland ·
Samir Kumar Neogy · Sukanta Pati ·
Simo Puntanen *Editors*



Applied Linear Algebra, Probability and Statistics

A Volume in Honour of C. R. Rao and Arbind K. Lal

Springer

Indian Statistical Institute Series

Editors-in-Chief

Abhay G. Bhatt, Indian Statistical Institute, New Delhi, India

Ayanendranath Basu, Indian Statistical Institute, Kolkata, India

B. V. Rajarama Bhat, Indian Statistical Institute, Bengaluru, India

Joydeb Chattopadhyay, Indian Statistical Institute, Kolkata, India

S. Ponnusamy, Indian Institute of Technology Madras, Chennai, India

Associate Editors

Arijit Chaudhuri, Indian Statistical Institute, Kolkata, India

Ashish Ghosh, Indian Statistical Institute, Kolkata, India

Atanu Biswas, Indian Statistical Institute, Kolkata, India

B. S. Daya Sagar, Indian Statistical Institute, Bengaluru, India

B. Sury, Indian Statistical Institute, Bengaluru, India

C. R. E. Raja, Indian Statistical Institute, Bengaluru, India

Mohan Delampady, Indian Statistical Institute, Bengaluru, India

Rituparna Sen, Indian Statistical Institute, Bengaluru, Karnataka, India

S. K. Neogy, Indian Statistical Institute, New Delhi, India

T. S. S. R. K. Rao, Indian Statistical Institute, Bengaluru, India

The **Indian Statistical Institute Series,** a Scopus-indexed series, publishes high-quality content in the domain of mathematical sciences, bio-mathematics, financial mathematics, pure and applied mathematics, operations research, applied statistics and computer science and applications with primary focus on mathematics and statistics. Editorial board comprises of active researchers from major centres of the Indian Statistical Institute.

Launched at the 125th birth Anniversary of P.C. Mahalanobis, the series will publish high-quality content in the form of textbooks, monographs, lecture notes, and contributed volumes. Literature in this series are peer-reviewed by global experts in their respective fields, and will appeal to a wide audience of students, researchers, educators, and professionals across mathematics, statistics and computer science disciplines.

Ravindra B. Bapat · Manjunatha Prasad Karantha ·
Stephen J. Kirkland · Samir Kumar Neogy ·
Sukanta Pati · Simo Puntanen
Editors

Applied Linear Algebra, Probability and Statistics

A Volume in Honour of C. R. Rao and Arbind K. Lal



Springer

Editors
Ravindra B. Bapat
Indian Statistical Institute Delhi Center
New Delhi, India

Stephen J. Kirkland
Department of Mathematics
University of Manitoba
Winnipeg, MB, Canada

Sukanta Pati
Department of Mathematics
Indian Institute of Technology Guwahati
North Guwahati, Assam, India

Manjunatha Prasad Karantha
Department of Data Science, CARAMS,
Manipal Academy of Higher Education
Manipal, Karnataka, India

Samir Kumar Neogy
Indian Statistical Institute Delhi Center
New Delhi, India

Simo Puntanen
Faculty of Information Technology
and Communication Sciences
Tampere University
Tampere, Finland

ISSN 2523-3114 ISSN 2523-3122 (electronic)
Indian Statistical Institute Series
ISBN 978-981-99-2309-0 ISBN 978-981-99-2310-6 (eBook)
https://doi.org/10.1007/978-981-99-2310-6

Mathematics Subject Classification: M11094, M27004, S11001, M29020, M13011

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors, and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Singapore Pte Ltd.
The registered company address is: 152 Beach Road, #21-01/04 Gateway East, Singapore 189721, Singapore

Dedicated to
A great statistician, linear algebraist and living legend who is in his 103rd year and still inspiring the young generation

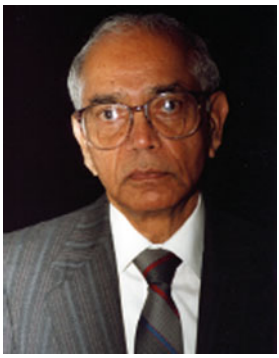
Padma Vibhushan
Calyampudi Radhakrishna Rao
(September 10, 1920–)

and our beloved friend who left us behind due to his untimely demise

Lal da
Arbind K. Lal
(January 01, 1966–March 07, 2021)

About Rao and Lal

Prof. Calyampudi Radhakrishna Rao



Calyampudi Radhakrishna Rao (1920–). Courtesy: Simo Puntanen

Rao[1] was born on September 10, 1920, in a small town in south India called Hoovina Hadagali, then in the integrated Madras Province of British India, but now in the Bellary district of state Karnataka. He was the eighth child (out of ten: six boys and four girls) of his parents, C. D. Naidu and Laxmikanthamma.

CRR joined the course 'training in statistics' offered by the ISI in January 1941. Six months later, Rao joined Calcutta University, which had just started a Master's degree program in statistics, the first of its kind in India. He completed the course with first-class honors, first rank and a gold medal in 1943. In August 1946, CRR visited England on the invitation of J. C. Trever to work in the Anthropological

[1] Thanks to Simo Puntanen for the contents of this section of Rao.

Museum and registered for a Ph.D. degree under Ronald Aylmer Fisher (1890–1962). Based on the work he did at the Museum, CRR received the Ph.D. degree from Cambridge University in 1948 with the thesis entitled 'Statistical Problems of Biological Classifications'.

Rao returned to India in August 1948, marrying Bhargavi (April 28, 1925–July 24, 2017) on September 09, 1948. Professor Mahalanobis then offered Rao a Professorship at ISI as well as the headship of the Research and Training School in 1949. He served in this position till 1963. Being an inspiring teacher, over the years about fifty students have completed their Ph.D. thesis under his guidance. CRR was the Director of ISI from 1963 to 1972, and, later, he was the Director and Secretary of the Indian Statistical Institute till 1976. He then continued in the ISI as a Jawaharlal Nehru Professor till 1984. During this period, he took leave from ISI in 1979 and went to the United States to accept a University Professorship at the University of Pittsburgh. He moved to The Pennsylvania State University in 1988, where he was offered several chair positions. He retired from Penn State at the age of 81, but continued doing research as the Director of the Center for Multivariate Analysis at Penn State until 2008. After that, he has the position as a Research Professor at the University at Buffalo.

In 1954, C. R. Rao collected some data from Japan in order to study the long-term effects of radiation caused by the Hiroshima and Nagasaki atom bomb explosions. In the statistical analysis, he encountered the situation of finding a matrix to replace the inverse of $X'X$, where X is the design matrix appearing in the linear model; here the matrix X is not a full column rank matrix and $X'X$ is singular and so the inverse was not defined. This led to a pseudo-inverse, which Rao introduced in his 1955 paper 'Analysis of dispersion for multiply classified data with an unequal number of cells' in Sankhya. This was the same year that R. A. Penrose published his paper on the generalized inverse. [Proc. Cambridge Philos. Soc. 51: 406–413 (1955)]. An interesting note to be made here is that the notion of pseudo-inverse, initially defined for the operators, was introduced by Moore in 1920, the year in which Rao was born. C. R. Rao then discovered the key condition for a generalized inverse G of a matrix A satisfying the equation $AGA = A$ and introduced the notation $G = A^{-}$. The calculus of g-inverses and the unified theory of linear estimation were then presented in 1962. The subject of g-inverses was further developed with the help of his colleagues at the ISI, leading to the full-length monograph with Sujith Kumar Mitra, Generalized Inverse of Matrices and its Applications, published in 1971. Using the concept of the g-inverse, Rao developed a unified theory for linear estimation, noting that g-inverses were particularly helpful with explicit expressions for projectors. He introduced the Inverse Partitioned Matrix (IPM) method [Unified theory of linear estimation, Sankhya Ser. A 33: 371–394 (1972)] for computing summary statistics for the general linear model. Linear models, which have wide applications in statistics, have also provided outlets for some basic research in Linear Algebra: the special issues on linear algebra and statistics of Linear Algebra and its Applications [Vols. 67 (1985),

70 (1985), 82 (1986), 127 (1990), 176 (1992), 210 (1994)] bear witness to this. Some of Rao's contributions are a general matrix inequality, separations for singular values and eigenvalues, matrix approximations and reduction of dimensionality, and Kantorovich type inequalities.

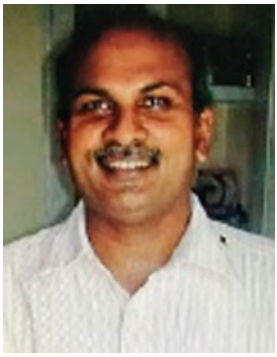
C. R. Rao has developed statistical estimation theory in small samples, extending the scope of statistical methods in practice. Many results in this area bear his name, such as the Cramer-Rao lower bound, Rao-Blackwell theorem, Rao-Blackwellization, Fisher-Rao theorem, Rao's second order efficiency and the Geary-Rao theorem on Pitman closeness criterion. Rao's orthogonal arrays were a major contribution to research in coding theory and in experimental design—especially in industrial experimentation as developed by Genichi Taguchi. Rao has made pioneering contributions to the development of multivariate statistical analysis, especially with the concept of Rao's quadratic entropy, Rao's U-test, Rao's F-approximation to the likelihood-ratio criterion and canonical coordinates. He has also introduced a broad class of asymptotic tests of hypothesis, including Rao's score statistic and the Neyman-Rao statistic; he has used differential-geometric techniques in discussing problems of statistical inference, based on Rao's distance function.

2023 International Prize in Statistics.

On 1 April 2023, it was announced on the website https://statprize.org/2023-International-Prize-in-Statistics-Awarded-to-C-R-Rao.cfm that C. R. Rao, a professor whose work more than seventy five years ago continues to exert a profound influence on science, has been awarded the 2023 International Prize in Statistics. The Award citation further reads: "In his remarkable 1945 paper published in the Bulletin of the Calcutta Mathematical Society, C. R. Rao demonstrated three fundamental results that paved the way for the modern field of statistics and provided statistical tools heavily used in science today". Here is the complete bibliographical information of the paper: Information and accuracy attainable in the estimation of statistical parameters, Bulletin of Calcutta Mathematical Society, Vol 37, No 3, 81-91, 1945.

The Prize was established in 2016 and is awarded once every two years to an individual or team "for major achievements using statistics to advance science, technology and human welfare". The Prize is managed by a foundation comprising representatives of the five major statistical organizations working cooperatively to develop this prestigious award which are American Statistical Association; Institute of Mathematical Statistics; International Biometric Society; International Statistical Institute; and Royal Statistical Society. The Prize is being considered as the statistics' equivalent of the Nobel Prize.

Arbind Kumar Lal



Arbind Kumar Lal (1966–2021). Courtesy: Sukanta Pati

Lal[2] known as Lalda for the younger generation, was in long association with CARAMS, MAHE and its activities. He is known for his nature to help everyone (students, teaching and nonteaching staff) in trouble, and for his ability to resolve issues within the department. Professor Lal was popularly tagged as the 'permanent HOD' by his colleagues. Born in Uttara (Madhubani district, Bihar) to Smt. Sushila Devi and Shri. Lekh Narayan Lal, his upbringing had made him a thorough gentleman and a person with a huge heart.

He completed his schooling at Sainik School, Tilaiyaa, in 1983 and obtained his BA with Mathematics, Honors in 1986 from Hansraj College, Delhi. He went to do his MStat (1986–1988) at ISI Delhi, and this is where his growth as a mathematician started. Because of his frank nature, being a sports-loving person and being a thoughtful person, he created many friends for life here. He completed his Ph.D. (1988–1993) from the same place, ISI Delhi. He was a visiting fellow at TIFR Mumbai for a year during which he worked on Combinatorics. After that, he became a visiting fellow at HRI Allahabad for two years during which he worked on coding theory. After that, he joined IIT Kanpur as a faculty in the Department of Mathematics in 1996 and served the department in various capacities till his untimely demise in 2021.

He guided seven Ph.D. students and his contribution 'Aq-analogue of the Distance Matrix of a Tree', Linear Algebra and its Applications, 416, 2–3, 2006, 799–814, has motivated many researchers to work on various aspects of the distance matrix. He was an editor of the journal 'Proceedings Mathematical Sciences' published by Springer and Indian Academy of Sciences.

CARAMS, MAHE salutes him and documents his tribute, besides praying to the Almighty to bless Rao with good health, prosperity and undiminished energy to inspire young generations of the future.

[2] Thanks to Sukanta Pati for the contents of this section on Lal.

Foreword

It gives me immense pleasure that a special volume dedicated to the conferences organized by the 'Center for Advanced Research in Applied Mathematics & Statistics, MAHE' is being published by Springer Nature's Indian Statistical Institute series in the form of a book-volume in honor of Profs. Calyampudi Radhakrishna Rao and Arbind K. Lal. CARAMS, MAHE initiated the activities in honor of Padma Vibhushan C. R. Rao, a great statistician and a living legend, in the year 2020. Due to the prevailing Covid situation, the events could not be organized in the normal physical format, but Rao's hundred years of life was celebrated by organizing the 'International Conference on Applied Linear Algebra, Probability and Statistics' in the online format. However, the conferences IWMS 2021 and ICLAA 2021, which were earlier scheduled in December 2020, have been rescheduled and organized in the hybrid format in December 2021. I congratulate the entire team led by the scientific advisory committee consisting of Ravindra B. Bapat, Manjunatha Prasad Karantha, Steve Kirkland, and Simo Puntanen for the great success of the events IWMS 2021 and ICLAA 2021 which brought several collaborators, disciples, and the followers of Rao together and connected with the young minds of India and abroad, remembering the achievements of Rao and further developments. Arbind K. Lal, an eminent young mathematician serving at IIT Kanpur, who had been continuously associated with the ICLAA series of conferences of MAHE, left us behind due to his untimely demise, and CARAMS, MAHE dedicated a session of ICLAA 2021 to remember him.

The book-volume 'Applied Linear Algebra, Probability, and Statistics—A Volume in Honor of C. R. Rao and Arbind K. Lal' is a part of several outcomes of the series of activities held by CARAMS, MAHE in honor of Rao. I understand that several associates of Rao and Lal and young scholars have contributed their work in the form

of book-chapters. I thank the entire editorial team and the authors for their efforts and contribution in the preparation of the book-volume ALAPS being published in the Indian Statistical Institute series of Springer Nature.

April 2023

Narayana Sabhahit
Pro Vice Chancellor—Technology and Science
Manipal Academy of Higher Education
Manipal, India

Preface

Linear Algebra, Probability Theory and Statistics are subjects having applications in almost every branch of science including Biology, Economics, Engineering and Physics, which are consistent with the focus research activities of CARAMS, MAHE. ICLAA 2021 is the fourth in its sequel, which started with CMTGIM 2012, and the fifth is ICLAA 2023 scheduled in December 2023. The initial intention was to dedicate the common day between IWMS 2020 and ICLAA 2020 to honor C. R. Rao, a living legend among Indian Statisticians, who was celebrating the birth centenary in the year 2020. Unfortunately, the events could not be organized as per schedule in the normal format due to the prevailing situation. With the hope of having a normal life in 2021, the events in the physical formats have been rescheduled to December 2021, and CARAMS organized ALAPS 2020 (International Conference on Applied Linear Algebra, Probability, and Statistics) in the online format (164 participants—25 invited, 40 contributory), in which many collaborators, disciples and followers of Rao participated. However, IWMS 2021 (192 participants—20 invited, 14 contributory) and ICLAA 2021 (242 participants—25 invited, 28 contributory) were organized successfully in the hybrid format and decided to publish a special issue in 'AKCE International Journal of Graphs and Combinatorics', and a book-volume 'Applied Linear Algebra, Probability, and Statistics—A volume in honor of Rao and Lal' (ALAPS) in the ISI series of Springer Nature.

The present volume is dedicated to Prof. Rao, who completed 100 years of legendary life and continues to inspire all of us, and Prof. Arbind Lal, who sadly departed us too early. The book focuses on, but is not limited to, linear algebra, statistics, matrices, graphs and applications—the areas to which Rao and Lal contributed. The volume is rich with contributions from collaborators, students, colleagues and admirers of Profs. Rao and Lal.

Many chapters feature new findings due to applications of matrix and graph methods, while others present rediscoveries of the subject using new methods. Chapters from Simo Puntanen, Steve Haslet and their collaborators reflect the continuation of Rao's legacy in the use of matrix methods in statistics, whereas two from TES Raghavan describe the use of matrix-graph methods in the computation of the nucleolus and Shapley value, which appear in game theory. The contributions from

Raghavan are based on his lectures delivered at the workshop and conferences held in honor of Rao.

Some chapters are expository, summarizing the development of subjects and presenting new open problems. BLS Prakasa Rao's 'On some matrix versions of covariance, harmonic mean and other inequalities: an overview' is one such article. Jeffery Hunter's contribution 'The impact of Prof. Rao's research used in solving problems in applied probability' describes how Rao's work influenced the author's research.

Neogy et al. (On some special matrices and its applications in linear complementarity problem) and Eagambaram (Characterization of Q-matrices using bordered matrix algorithm) discuss special classes of matrices having applications in the linear complementarity problem. The chapters by Saumyadipta Pyne and Sudeep Bapat present key applications of statistical data analysis to real-life problems. The book also includes one of Lal's last works 'Some observations on algebraic connectivity of graphs'.

With many more chapters on Generalized Inverses, Matrix Analysis, Matrices and Graphs, and the History of Ancient Mathematics, this volume offers a diverse array of mathematical results, techniques, applications and perspectives. Linear Algebra, Probability and Statistics are important branches of mathematics having applications in every dimension of science. The book promises to be especially rewarding for readers with an interest in the focus areas of ALAPS.

Manipal, Karnataka, India
April 2023

Ravindra B. Bapat
Manjunatha Prasad Karantha
Stephen J. Kirkland
Samir Kumar Neogy
Sukanta Pati
Simo Puntanen

Acknowledgements

We, the CARAMS, MAHE and the editorial team, thankfully acknowledge the support of the leadership of Manipal Academy of Higher Education, Manipal, particularly, Narayana Sabhahit, Pro Vice Chancellor—Technology & Science, MAHE (current position), for their continuous and incredible support in organizing ALAPS 2020, IWMS 2021 and ICLAA 2021. Also, we appreciate the efforts put in by the 'International Organizing Committee' of IWMS and the other committees which supported IWMS and ICLAA. Professor Narayanaswamy Balakrishnan, McMaster University, Canada, who led the organizing team from the front in organizing ALAPS 2020, coordinated with Rao's family and helped Prof. Rao join the gathering virtually at ALAPS 2020, and we are thankful for his efforts. We acknowledge that the 'International Linear Algebra Society' endorsed ICLAA 2021, and offered fund support for the Hans-Schneider ILAS lecture. We also thank the speaker, Stephen J Kirkland, for his virtual presence and for delivering the Hans-Schneider ILAS lecture.

The editors acknowledge the incredible contribution of all the authors (see the list of contributors), which made the book-volume so rich. Also, the editors acknowledge the association and the support of

1. R. Balaji, IIT Madras
2. N. Balakrishnan, McMaster University, Canada
3. Arup Bose, ISI Kolkata
4. Debajit Kalita, Tezpur University, India
5. Rajesh Kannan, IITKGP, India
6. Yimin Wei, Fudan University, PRC,

at different stages of editorial work. The selfless efforts of the reviewers helped the editorial team and the authors to a great extent in enhancing the quality of work. The editors acknowledge the following reviewers, other than the editors, for spending their quality time in reading the articles and offering their comments.

1. P. Anisha, Department of Data Science, MAHE, Manipal
2. R. K. Amit, Indian Institute of Technology Madras, India
3. Ramamurthy Balaji, Indian Institute of Technology Madras, India

4. Arathi Bhat, Manipal Institute of Technology, MAHE, Manipal, India
5. Raghavendra Bhat, Poornaprajna College, Udupi, India
6. S. Parameshwara Bhatta, Adjunct Professor, Mangalore University, Konaje, India
7. Shreemayee Bora, Indian Institute of Technology, Guwahati, India
8. Ramesha C., Manipal Institute of Technology, MAHE, Manipal, India
9. S. S. Chala, Indian Institute of Technology Hyderabad, India
10. Shih Yu Chang, San José State University, USA
11. Maolin Che, Southwestern University of Finance and Economics, Chengdu, China
12. N. Eagambaram, Former DDG, MOSPI, India
13. Katarzyna Filipiak, Poznan University of Technology, Poland
14. Paolo Gibilisco, Universit'a di Chieti-Pescara 'G. D'Annunzio', Italy
15. Frank Hansen, University of Copenhagen, Denmark
16. Surender Kumar Jain, Ohio University, Ohio, USA
17. S. R. Jammalamadaka, University of California, Santa Barbara, USA
18. P. Sam Johnson, National Institute of Technology Karnataka, India
19. Mohana K. S., Mangalore University, Konaje, India
20. Michael Karow, Technische Universität Berlin, Germany
21. Vinay Madhusudhanan, Manipal Institute of Technology, MAHE, Manipal
22. Debasis Mishra, Indian Statistical Institute Delhi, India
23. Naren Naik, Indian Institute of Technology Kanpur, India
24. Rajendra Pawale, University of Mumbai, India
25. Kuncham Syam Prasad, Manipal Institute of Technology, India
26. T. S. S. R. K. Rao, Adjunct Professor, MAHE, Manipal, India
27. Gopinath Sahoo, Indian Institute of Technology Bhubaneswar, India
28. Sharad Sane, Chennai Mathematical Institute, Chennai, India
29. Kuldeep Sarma, Tezpur University, Assam, India
30. Bhavanari Satyanarayana, Acharya Nagarjuna University, India
31. Arunava Sen, Indian Statistical Institute Delhi, India
32. Ranveer Singh, Indian Institute of Technology, Indore
33. Martin Singull, Linköping University, Linköping, Sweden
34. JuliaVolaufova, Louisiana State University Health Sciences Center, New Orleans, (USA)
35. Xiaoxiang Zhang, Nanjing University, PR China
36. Xuze Zhang, University of Maryland, USA

The following team of research scholars

1. Umashankara Kelathaya
2. Raksha Poojary
3. Savitha Varkady

has offered continuous support to the editorial team without which the completion of the book-volume project wouldn't have been so effective.

We are thankful to Springer Nature India Pvt. Ltd., particularly Shamim Ahmad—Executive Editor, and the production team for their support in publishing this book-volume in its Indian Statistical Institute series.

Manipal, Karnataka, India
April 2023

Ravindra B. Bapat
Manjunatha Prasad Karantha
Stephen J. Kirkland
Simo Puntanen

Contents

1 On Some Matrix Versions of Covariance, Harmonic Mean and Other Inequalities: An Overview 1
B. L. S. Prakasa Rao

2 The Impact of Prof. C. R. Rao's Research Used in Solving Problems in Applied Probability 11
Jeffrey J. Hunter

3 Upper Bounds for the Euclidean Distances Between the BLUEs Under the Partitioned Linear Fixed Model and the Corresponding Mixed Model 27
Stephen J. Haslett, Jarkko Isotalo, Augustyn Markiewicz, and Simo Puntanen

4 Nucleolus Computation for Some Structured TU Games via Graph Theory and Linear Algebra 45
T. E. S. Raghavan

5 From Linear System of Equations to Artificial Intelligence—The Evolution Journey of Computer Tomographic Image Reconstruction Algorithms 95
Rekha, Santosh Singh, and Samir Kumar Neogy

6 Shapley Value and Other Axiomatic Extensions to Shapley Value 117
T. E. S. Raghavan

7 An Accelerated Block Randomized Kaczmarz Method 133
Anjie Qiao, Xuezhong Wang, and Yimin Wei

8 Nullity of Graphs—A Survey and Some New Results 155
S. Arumugam, K. Arathi Bhat, Ivan Gutman, Manjunatha Prasad Karantha, and Raksha Poojary

9 Some Observations on Algebraic Connectivity of Graphs 177
Ravindra B. Bapat, A. K. Lal, and S. Pati

10 Orthogonality for Biadjoints of Operators 191
T. S. S. R. K. Rao

11 Permissible Covariance Structures for Simultaneous Retention of BLUEs in Small and Big Linear Models 197
Stephen J. Haslett, Jarkko Isotalo, Augustyn Markiewicz, and Simo Puntanen

12 On Some Special Matrices and Their Applications in Linear Complementarity Problem 215
Samir Kumar Neogy and Gambheer Singh

13 On Nearest Matrix with Partially Specified Eigen-Structure 245
Rafikul Alam

14 Equality of BLUEs for Full, Small, and Intermediate Linear Models Under Covariance Change, with Links to Data Confidentiality and Encryption 273
Stephen J. Haslett and Simo Puntanen

15 Statistical Inference for Middle Censored Data with Applications 293
Sudeep R. Bapat

16 Matrix Partial Orders Based on the Secondary-Transpose 317
Umashankara Kelathaya, Manjunatha Prasad Karantha, and Ravindra B. Bapat

17 On Products of Graph Matrices 337
G. Sudhakara, Vinay Madhusudanan, and K. Arathi Bhat

18 On Rao's Weighted Distributions for Modeling the Dynamics of Wildfires and Air Pollution 379
Saurav Guha, Alexander Petersen, Sumanta Ray, and Saumyadipta Pyne

19 Characterization of Q-Matrices Using Bordered Matrix Algorithm 395
N. Eagambaram

20 Descending Endomorphisms of Some Families of Groups 409
Vinay Madhusudanan, Arjit Seth, and G. Sudhakara

21 On Circulant Partial Hadamard Matrices 425
Pankaj Kumar Manjhi and Mahendra Kumar Rana

22 On Weak Hypervector Spaces Over a Hyperfield 435
Pallavi Panjarike, Syam Prasad Kuncham, Madeleine Al-Tahan, Vadiraja Bhatta, and Harikrishnan Panackal

23 Generalized Lie Triple Derivations of Trivial Extension Algebras 461
Mohammad Ashraf, Mohammad Afajal Ansari, and Md Shamim Akhter

24 On Ideals of Compatible ΘΓ N-Group 473
Hamsa Nayak, Syam Prasad Kuncham, and Babushri Srinivas Kedukodi

25 The Range Column Sufficiency and the Pseudo-SSM Property of Linear Transformation on Euclidean Jordan Algebra 483
Punit Kumar Yadav and K. Palpandi

26 On the Proofs of Formulae by Mahāvīra and Brahmagupta 501
Narayana Acharya

Editors and Contributors

About the Editors

Ravindra B. Bapat is Visiting Professor at the Indian Statistical Institute, Delhi Center, New Delhi, India, where he earlier served for more than 35 years. He is also Adjunct Professor at the Manipal Academy of Higher Education (MAHE), Manipal, Karnataka, India. On the editorial board of the *Indian Statistical Institute Series* (a Springer book series), he is on the editorial board of several journals including *Linear and Multilinear Algebra*, *AKCE International Journal of Graphs and Combinatorics*, *The Electronic Journal of Linear Algebra* and *Indian Journal of Pure and Applied Mathematics*. A contributor of more than 150 research articles, he has authored books, *Linear Algebra and Linear Models* (Springer, 2012) and *Graphs and Matrices* (Springer, 2010).

Manjunatha Prasad Karantha is Professor and Coordinator of the Center for Advanced Research in Applied Mathematics and Statistics at the Manipal Academy of Higher Education, Manipal, Karnataka, India. His research interests are in matrices over rings, generalized inverses and matrix methods in graphs. He is on the editorial board of several reputed journals, published several research articles and edited several books and special issues dedicated to the International Conference on Linear Algebra and its Applications series of conferences in reputed journals.

Stephen J. Kirkland is Professor at the Department of Mathematics and Associate Dean of the Faculty of Graduate Studies at the University of Manitoba, Canada. His research interests include non-negative matrix theory, spectral graph theory and combinatorial matrix theory. For more than six years, he served as President of the International Linear Algebra Society. Besides being Editor-in-Chief of the *Linear and Multilinear Algebra* journal, he has been serving on editorial boards of journals, *Linear Algebra and its Applications* and the *Electronic Journal of Linear Algebra.*

Samir Kumar Neogy is Professor at the Indian Statistical Institute and Head of its Delhi Center. He obtained his Ph.D. from the same institute, and his primary areas of research are mathematical programming, special matrices and game theory. He is Co-editor of several books and special issues of several journals.

Sukanta Pati is Professor at the Indian Institute of Technology Guwahati, Assam, India. His research interests are in matrices in graphs. He is also on the editorial board of journals, *Linear Multilinear Algebra*, *The Mathematics Student* and *Mathematics Newsletter*.

Simo Puntanen is—after retirement—Visiting Researcher at Tampere University, Finland. He is Founding Member, with George P. H. Styan, of the International Workshop on Matrices and Statistics, held regularly from 1990 onwards. He is Co-author (with George P. H. Styan and Jarkko Isotalo) of the book, *Matrix Tricks for Linear Statistical Models: Our Personal Top Twenty* (Springer, 2011).

Contributors

Narayana Acharya Retired Professor, Bhandarkars' Arts & Science College Kundapura, Udupi, KA, India

Md Shamim Akhter Department of Mathematics, Aligarh Muslim University, Aligarh, India

Madeleine Al-Tahan Department of Mathematics, Abu Dhabi University, Abu Dhabi, UAE

Rafikul Alam Department of Mathematics, IIT Guwahati, Guwahati, India

Mohammad Afajal Ansari Department of Mathematics, Aligarh Muslim University, Aligarh, India

K. Arathi Bhat Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

S. Arumugam Department of Computer Science and Engineering, Ramco Institute of Technology, Rajapalayam, Tamil Nadu, India

Mohammad Ashraf Department of Mathematics, Aligarh Muslim University, Aligarh, India

Ravindra B. Bapat Indian Statistical Institute, New Delhi, India

Vadiraja Bhatta Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

N. Eagambaram Former Deputy Director General, Indian Statistical Service; Guest faculty, Indian Institute of Technology, Tirupati, India

Saurav Guha ICAR-Indian Agricultural Statistics Research Institute, New Delhi, India;
Health Analytics Network, Pittsburgh, PA, USA

Ivan Gutman Faculty of Science, University of Kragujevac, Kragujevac, Serbia

Stephen J. Haslett School of Mathematical and Computational Sciences & Research; Centre for Hauora and Health, Massey University, Palmerston North, New Zealand

Jeffrey J. Hunter Auckland University of Technology, Auckland, New Zealand

Jarkko Isotalo Faculty of Information Technology and Communication Sciences, Tampere University, Tampere, Finland

Manjunatha Prasad Karantha Department of Data Science, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal, Karnataka, India; Centre for Advanced Research in Applied Mathematics and Statistics, Manipal Academy of Higher Education, Manipal, Karnataka, India

Babushri Srinivas Kedukodi Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

Umashankara Kelathaya Department of Data Science, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal, Karnataka, India

Syam Prasad Kuncham Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

A. K. Lal IIT Kanpur, Kanpur, India

Vinay Madhusudanan Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India

Pankaj Kumar Manjhi University Department of Mathematics, Vinoba Bhave University, Hazaribag, India

Augustyn Markiewicz Department of Mathematical and Statistical Methods, Poznań University of Life Sciences, Poznań, Poland

Hamsa Nayak Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

Samir Kumar Neogy Statistical Quality Control and Operation Research Unit, Indian Statistical Institute, New Delhi, India

K. Palpandi Malaviya National Instiute of Technology, Jaipur, India

Harikrishnan Panackal Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

Pallavi Panjarike Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India

S. Pati I I T Guwahati, Guwahati, India

Alexander Petersen Department of Statistics, Brigham Young University, Provo, UT, USA

Raksha Poojary Department of Data Science, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal, Karnataka, India

B. L. S. Prakasa Rao CR Rao Advanced Institute of Mathematics, Statistics, and Computer Science, Hyderabad, India

Simo Puntanen Faculty of Information Technology and Communication Sciences, Tampere University, Tampere, Finland

Saumyadipta Pyne Health Analytics Network, Pittsburgh, PA, USA;
Department of Statistics and Applied Probability, University of California Santa Barbara, Santa Barbara, CA, USA

Anjie Qiao School of Mathematical Sciences, Fudan University, Shanghai, People's Republic of China

Sudeep R. Bapat Indian Institute of Management, Indore, India

T. E. S. Raghavan University of Illinois at Chicago, Chicago, IL, USA

Mahendra Kumar Rana University Department of Mathematics, Vinoba Bhave University, Hazaribag, India

T. S. S. R. K. Rao Department of Mathematics, Shiv Nadar Institution of Eminence, Goutam Buddha Nagar, UP, India;
Manipal Academy of Higher Education, Manipal, India

Sumanta Ray Health Analytics Network, Pittsburgh, PA, USA;
Department of Computer Science and Engineering, Aliah University, Kolkata, WB, India

Rekha Department of Mathematics, Shiv Nadar University, Gautam Budha Nagar, India

Arjit Seth Department of Mechanical and Aerospace Engineering, Hong Kong University of Science and Technology, Hong Kong, China

Gambheer Singh Department of Mathematics, University of Delhi, Delhi, India

Santosh Singh Department of Mathematics, Shiv Nadar University, Gautam Budha Nagar, India

G. Sudhakara Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India

Xuezhong Wang School of Mathematics and Statistics, Hexi University, Zhangye, People's Republic of China

Yimin Wei School of Mathematical Sciences and Key Laboratory of Mathematics for Nonlinear Sciences, Fudan University, Shanghai, People's Republic of China

Punit Kumar Yadav Malaviya National Instiute of Technology, Jaipur, India

Chapter 1
On Some Matrix Versions of Covariance, Harmonic Mean and Other Inequalities: An Overview



B. L. S. Prakasa Rao

Mathematics Subject Classification (2010) 62B20

1.1 Introduction

C. R. Rao has extensively used matrix algebra in his study of unified theory of linear models. He has shown that some of the important results in matrix algebra can be derived from statistical results in Rao [19]. Dey et al. [6] gave proofs of some results in matrix algebra using ideas derived from statistics. Mitra [13] gave statistical proofs of some results for non-negative definite matrices. Rao [18] derived seven inequalities in statistical estimation theory one of which deals with the harmonic mean of positive random variables. A generalization of these results for random matrices was derived in Prakasa Rao [17]. Rao [19] gave an alternate proof of this result using ideas from statistics. Kimeldorf and Sampson [11] derived a class of covariance inequalities. Matrix versions of these inequalities were obtained in Prakasa Rao [16]. We will now give a short overview of some of these results dealing with the harmonic mean for positive random variables and some covariance inequalities and the recent work of Gibilisco and Hansen [9] generalizing the results of Rao [18] and Prakasa Rao [17] and present some open problems.

B. L. S. Prakasa Rao (✉)
CR Rao Advanced Institute of Mathematics, Statistics, and Computer Science, Hyderabad, India
e-mail: blsprao@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_1

1.2 Covariance Inequalities

Let $\mathcal{A}$ be an open convex subset of R^2 and (X, Y) be a bivariate random vector with support $\mathcal{A}$. Further suppose that $E(X^2)$ and $E(Y^2)$ are finite. Suppose $g(x, y)$ is a function satisfying the condition

$$(x_2 - x_1)(y_2 - y_1) \leq [g(x_2, y_2) - g(x_1, y_1)]^2, (x_i, y_i) \in \mathcal{A}, i = 1, 2.$$

Kimeldorf and Sampson [11] proved that

$$Cov(X, Y) \leq Var[g(X, Y)].$$

A matrix version of this result was proved in Prakasa Rao [16]. We will now review this result and give some applications.

Hereafter, we consider only random matrices with real entries.

Suppose ξ and η are two random matrices of order $r \times s$ and further suppose that all the entries of these matrices have finite second moments. Let $E(\xi)$ be the matrix of expectations of the entries of the matrix ξ. Let ξ' denote the transpose of the matrix ξ. For any two random matrices ξ and η, define the covariance and variance by the following relations:

$$Cov(\xi, \eta) = \frac{1}{2}\{E[(\xi - E(\xi))(\eta - E(\eta))')] + E[(\eta - E(\eta))(\xi - E(\xi))']\} \quad (1.1)$$

and

$$Var[\xi] = E[(\xi - E(\xi))(\xi - E(\xi))']. \quad (1.2)$$

It is clear from the definitions given above that the matrix $Var(\xi)$ is a non-negative definite matrix and that the covariance matrix $Cov(\xi, \eta)$ is a symmetric matrix. If the matrices ξ and η are random commuting symmetric matrices (and hence $r = s$), then

$$\begin{aligned} Cov(\xi, \eta) &= E[(\xi - E(\xi))(\eta - E(\eta))'] \\ &= E(\xi\eta) - E(\xi)E(\eta) \end{aligned} \quad (1.3)$$

which is analogous to the standard definition of the covariance for two random variables. For any two square matrices A and B of the same order, we say that $A \leq B$ if the matrix $B - A$ is non-negative definite. We say that $A < B$ if $B - A$ is positive definite.

The following result is proved in Prakasa Rao [16].

Theorem 1.1 *Let ξ and η be random matrices of order $r \times s$ such that $P[(\xi, \eta) \in \mathcal{A}] = 1$ where $\mathcal{A}$ is a subset of $R^{r\times s} \times R^{r\times s}$. Let $g(A, B)$ be a function defined over the space $R^{r\times s} \times R^{r\times s}$ taking values in $R^{r\times \ell}$. Then*

$$Cov(\xi, \eta) \leq Var[g(\xi, \eta)] \tag{1.4}$$

if and only if, for all $(A_i, B_i) \in \mathcal{A}, i = 1, 2,$

$$\begin{aligned} &\frac{1}{2}[(A_2 - A_1)(B_2 - B_1)' + (B_2 - B_1)(A_2 - A_1)'] \\ &\leq [g(A_2, B_2) - g(A_1, B_1)][g(A_2, B_2) - g(A_1, B_1)]'. \end{aligned} \tag{1.5}$$

For proof, see Prakasa Rao [16]. We now consider some interesting special cases of this theorem.

Corollary 1.1 *Suppose* ξ *and* η *are random matrices of order* $r \times r$. *Let* $g(A, B) = \frac{A+B}{2}$. *It is easy to check that the condition (1.5) holds using the fact that*

$$[\frac{A_2 - A_1}{2} - \frac{B_2 - B_1}{2}][\frac{A_2 - A_1}{2} - \frac{B_2 - B_1}{2}]' \geq 0.$$

Hence, an application of Theorem 1.1 shows that

$$Cov(\xi, \eta) \leq Var[\frac{\xi + \eta}{2}]. \tag{1.6}$$

Corollary 1.2 *Suppose* ξ *and* η *are positive definite matrices of order* $r \times r$. *Then* $A^{\frac{1}{2}}$ *and* $B^{\frac{1}{2}}$ *are symmetric matrices. Suppose they commute. Define* $g(A, B) = A^{\frac{1}{2}}B^{\frac{1}{2}} = B^{\frac{1}{2}}A^{\frac{1}{2}}$ *by commutativity. It can be checked that the condition (1.5) holds and hence, by Theorem 1.1,*

$$Cov(\xi, \eta) \leq Var[\xi^{\frac{1}{2}}\eta^{\frac{1}{2}}]. \tag{1.7}$$

This inequality can also be obtained as a consequence of the following matrix version of the Cauchy-Schwartz inequality (cf. Ibragimov and Khasminskii [10, p. 73]).

Theorem 1.2 *Let* ξ *and* η *be two random matrices of order* $r \times s$ *with the entries of these matrices having finite variances. Further suppose that* $E(\xi) = E(\eta) = 0$ *and* $E(\eta\eta') > 0$. *Then*

$$E(\xi\xi') \geq E(\xi\eta')[E(\eta\eta')]^{-1}E(\eta\xi'). \tag{1.8}$$

Example 1

It is easy to see that, for any positive definite matrix D,

$$D + D^{-1} \geq 2I \tag{1.9}$$

since $(D^{\frac{1}{2}} - D^{-\frac{1}{2}})^2 \geq 0$. In particular, for any two positive definite matrices A_1 and A_2,

$$(A_1 - A_2)(A_1^{-1} - A_2^{-1}) = 2I - A_2A_1^{-1} - A_1A_2^{-1} = 2I - D - D^{-1} \leq 0 \quad (1.10)$$

where $D = A_2A_1^{-1}$. Let $g(A, B) = AB$. It can be checked that the condition (1.5) holds for the function $g(A, B)$ and hence, by Theorem 1.1,

$$Cov[\xi, \xi^{-1}] \leq Var[g(\xi, \xi^{-1})] = 0. \quad (1.11)$$

This in turn implies that

$$E[\xi]E[\xi^{-1}] \geq I \quad (1.12)$$

for any positive definite random matrix ξ. This was proved by alternate arguments in Babrovsky et al. [4].

Remark 1.1 For any two positive definite matrices A and B, define

$$A \triangle B = \frac{A + B}{2}, \quad (1.13)$$

$$A \# B = A^{\frac{1}{2}}(A^{-\frac{1}{2}}BA^{-\frac{1}{2}})^{\frac{1}{2}}A^{\frac{1}{2}}, \quad (1.14)$$

and

$$A ! B = [\frac{1}{2}(A^{-1} + B^{-1})]^{-1}. \quad (1.15)$$

Ando [2] proved that

$$A ! B \leq A \# B \leq A \triangle B \quad (1.16)$$

generalizing the arithmetic mean-geometric mean-harmonic mean inequality for positive real numbers. It was also proved by Ando [1] that

$$A \triangle B \geq [\frac{1}{2}(A^q + B^q)]^{1/q} \quad (1.17)$$

for $\frac{1}{2} \leq q \leq 1$. Under the commutativity assumption on $A^{\frac{1}{2}}$ and $B^{\frac{1}{2}}$, it can be checked that $A \# B = (AB)^{\frac{1}{2}} = A^{\frac{1}{2}}B^{\frac{1}{2}}$. It was shown earlier that Theorem 1.1 holds for the function $g_1(A, B) = A \triangle B$ and for the function $g_2(A, B) = A^{\frac{1}{2}}B^{\frac{1}{2}}$ under the commutativity condition. Theorem 1.1 does not hold the function $g_3(A, B) = A ! B$ even for real-valued random variables from Kimeldorf and Sampson [11]. They have proved that Theorem 1.1 holds for the function

$$g(x, y) = [\frac{1}{2}(x^\beta + y^\beta)]^{1/\beta}, 0 < \beta \leq 1$$

for real-valued random variables.

It is an open problem to find whether Theorem 1.1 holds for the function

$$g_4(A, B) = [\frac{1}{2}(A^q + B^q)]^{1/q}, \frac{1}{2} \leq q \leq 1,$$

and for the function

$$g_5(A, B) = A\#B$$

in view of Ando's inequality (1.17).

For other remarks on such inequalities for random matrices, see Prakasa Rao [16].

1.3 Harmonic Mean Inequalities

Rao [18] proved the following result.

Theorem 1.3 *If X and Y are positive-valued random variables with finite expectations, then*

$$E[(\frac{1}{X} + \frac{1}{Y})^{-1}] \leq (\frac{1}{E(X)} + \frac{1}{E(Y)})^{-1}. \tag{1.18}$$

Rao's proof involves the use of Holder's inequality and the dominated convergence arguments. We have extended this result to positive definite random matrices in Prakasa Rao [17].

Theorem 1.4 *If A and B are almost surely positive definite random matrices with $E(A)$, $E(A^{-1})$, $E(B)$ and $E(B^{-1})$ finite and either $E(A^{-1}) - (E[A])^{-1}$ or $E(B^{-1}) - (E[B])^{-1}$ is non-singular, then*

$$E((A^{-1} + B^{-1})^{-1}) < ((E[A])^{-1} + (E[B])^{-1})^{-1}. \tag{1.19}$$

Theorem 1.4 can be proved in several ways. The first method uses the Jensen inequality for random matrices. For the notion of convexity for functions defined on the space of matrices, see Davis [5].

Proof For any fixed positive definite matrix $\mathbf{M}$, the matrix-valued function $g(\mathbf{X}) = (\mathbf{M} + \mathbf{X}^{-1})^{-1}$ is concave in $\mathbf{X}$ over the class of positive definite matrices $\mathbf{X}$, and hence

$$E[(\mathbf{M} + \mathbf{X}^{-1})^{-1}] < [\mathbf{M} + E(\mathbf{X})^{-1}]^{-1}.$$

Taking expectations with respect to the probability measure of the random matrix $\mathbf{M}$ on both sides, the result follows.

Other proofs of Theorem 1.4 are based on the following lemmas which are of independent interest.

Lemma 1.1 *For any two positive definite matrices* $\mathbf{G}$ *and* $\mathbf{H}$,

$$\mathbf{G}^{-1} - \mathbf{H}^{-1} \geq \mathbf{H}^{-1}(\mathbf{H} - \mathbf{G})\mathbf{H}^{-1}. \tag{1.20}$$

Lemma 1.2 *Let* $\mathbf{G}$ *any positive definite random matrix such that* $E[\mathbf{G}]$ *is finite and non-singular. Then*

$$E[\mathbf{G}^{-1}] \geq (E[\mathbf{G}])^{-\mathbf{1}}. \tag{1.21}$$

Lemma 1.3 *Let* $\mathbf{G}$ *and* $\mathbf{H}$ *be positive definite random matrices with finite expectations. Then*

$$E(\mathbf{G}\mathbf{H}^{-\mathbf{1}}\mathbf{G}) \geq \mathbf{E}[\mathbf{G}](\mathbf{E}[\mathbf{H}])^{-\mathbf{1}}\mathbf{E}[\mathbf{G}]. \tag{1.22}$$

Lemma 1.1 is a consequence of the fact that

$$(\mathbf{H}^{-1} - \mathbf{G}^{-1})\mathbf{G}(\mathbf{G}^{-1} - \mathbf{H}^{-1}) > 0.$$

Lemma 1.2 follows from the inequality proved in (1.12) given earlier (see Bobrovsky et al. [4] and Prakasa Rao [16]).

Lemma 1.3 is a consequence of the fact that the function $g(A, B) = AB^{-1}A$ is convex over the class of positive definite random matrices. It is also a consequence of Corollary 3.1 in Ando [1]. Two alternate proofs of Theorem 1.4 can be given using the lemmas stated above. For details, see Prakasa Rao [17] and Rao [19]. An application of Theorem 1.4 to statistical inference is given in Prakasa Rao [17].

1.4 General Mean Inequalities

Let x and y be positive real numbers. The arithmetic, geometric, harmonic and logarithmic means are defined by

$$m_a(x, y) = \frac{x + y}{2}, m_g(x, y) = \sqrt{xy}$$

and

$$m_h(x, y) = \frac{2}{x^{-1} + y^{-1}}, m_\ell(x, y) = \frac{x - y}{\log x - \log y}.$$

Suppose X and Y are positive random variables. Linearity of the expectation operator implies

$$E[m_a(X, Y)] = m_a(E(X), E(Y)).$$

Cauchy-Schwartz inequality implies that

$$E[m_g(X, Y)] \leq m_g(E(X), E(Y)).$$

Rao [18, 19] and Prakasa Rao [17] proved that

$$E[m_h(X, Y)] \leq m_h(E(X), E(Y))$$

as explained in the last section. Gibilisco and Hansen [9] investigated the generality of these results including the question of whether such a result holds for the logarithmic mean. We will now give a brief survey of some of their results.

Definition 1.1 A bivariate mean is a function $m : R_+ \times R_+ \to R_+$ such that
(i) $m(x, x) = x$;
(ii) $m(x, y) = m(y, x)$;
(iii) $x < y$ implies that $x < m(x, y) < y$;
(iv) $x < x'$ and $y < y'$ implies that $m(x, y) < m(x', y')$;
(v) the function $m(x, y)$ is continuous; and
(vi) the function $m(x, y)$ is positively homogeneous, that is, $m(tx, ty) = t\, m(x, y)$ for $t > 0$.

The definition of the bivariate mean given above is due to Petz and Temesi [15]. Let $\mathcal{M}$ be the set of all bivariate means as defined above.

Definition 1.2 Let $\mathcal{F}$ denote the class of functions $f : R_+ \to R_+$ such that
(i) f is continuous;
(ii) f is monotone increasing;
(iii) $f(1) = 1$;
(iv) $tf(t^{-1}) = f(t)$ for $t > 0$.

It can be checked that there is a bijection between the classes $\mathcal{M}$ and $\mathcal{F}$ given by the formulae

$$m_f(x, y) = yf(y^{-1}x) \quad \text{and} \quad f_m(t) = m(1, t) \tag{1.23}$$

for positive numbers x, y and t. We now give some examples of means.
(i) The function $f(x) = \frac{1+x}{2}$ corresponds to the arithmetic mean $\frac{x+y}{2}$.
(ii) The function $f(x) = \sqrt{x}$ corresponds to the geometric mean $\sqrt{xy}$.
(iii) The function $f(x) = \frac{2x}{x+1}$ corresponds to the harmonic mean $\frac{2}{x^{-1}+y^{-1}}$.
(iv) The function $f(x) = \frac{x-1}{\log x}$ corresponds to the logarithmic mean $\frac{x-y}{\log x - \log y}$.
(v) the function $f(x) = \frac{x^\beta + x^{1-\beta}}{2}$, $0 < \beta < 1$ corresponds to the mean $\frac{x^\beta y^{1-\beta} + x^{1-\beta} y^\beta}{2}$.

Observe that all the functions described in (i)–(v) given above belong to the class $\mathcal{F}$ and they are concave. Gibilisco and Hansen [9] proved the following result.

Theorem 1.5 *Let f be a function in the class $\mathcal{F}$. The inequality*

$$E[m_f(X, Y)] \leq m_f(E(X), E(Y)) \tag{1.24}$$

holds for arbitrary positive random variables X and Y if and only if f is concave.

For a proof of Theorem 1.5, see Gibilisco and Hansen [9]. In particular, the result holds for the logarithmic mean for positive random variables. They extended Theorem 1.5 for operator means in the sense of Kubo and Ando [20]. We refer the reader to Gibilisco and Hansen [9] for details.

A n-mean on a set X is a n-ary operation $m : X^n \to X$ which satisfies the condition

$$m(x, x, \ldots, x) = x, x \in X.$$

It is said to be symmetric if

$$m(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = m(x_1, \ldots, x_n)$$

for $\sigma \in S_n$ where S_n is the symmetric group of order n.

Open Problems: (1) In a recent paper, Gibilisco [8] posed the following problem: suppose f is a function in $\mathcal{F}$ as given by Definition 2 and $X_1, \ldots, X_n$ are positive random variables defined on a probability space $(\Omega, \mathcal{G}, P)$. Is it true that Jensen inequality holds for n-means, that is, if f is concave, does it hold that

$$E(m_f(X_1, \ldots, X_n)) \leq m_f(E(X_1), \ldots, E(X_n))$$

for all positive random variables $X_1, \ldots, X_n$? Jensen's inequality for bivariate means has been proved for operator means in Kubo-Ando [20] (cf. Gibilisco and Hansen [9], Prakasa Rao [17] and Rao [19]). Can it be generalized to n-means in the non-commutative setting?

(2) Given a set of positive numbers $x_1, \ldots, x_n$ and a positive integer $k \leq n$, define

$$M_n^{(k)} = [\frac{\Sigma_{1 \leq i_1, \ldots, i_k \leq n} x_{i_1} \ldots x_{i_k}}{(n_{C_k})}]^{1/k}.$$

It is known that

$$M_n^{(k-1)}(x_1, \ldots, x_n) M_n^{(k+1)}(x_1, \ldots, x_n) \leq [M_n^{(k)}(x_1, \ldots, x_n)]^2, 2 \leq k \leq n-1$$

(cf. Hardy et al. [21]) which in turn implies that

$$M_n^{(1)}(x_1, \ldots, x_n) \geq M_n^{(2)}(x_1, \ldots, x_n) \geq \cdots \geq M_n^{(n)}(x_1, \ldots, x_n).$$

Note that $M_n^{(1)}$ is the arithmetic mean and $M_n^{(n)}$ is the geometric mean. Let I be a nondegenerate interval of R and $n \geq 1$. A function $M : I^n \to R$ is called n-variable mean on I if

$$\min(x_1, \ldots, x_n) \leq M(x_1, \ldots, x_n) \leq \max(x_1, \ldots, x_n), x_1, \ldots, x_n \in I.$$

Let $f : I \to R$ be a continuous and strictly increasing function. The n-variable quasi-arithmetic mean $A_n^f : I^n \to I$ is defined by

$$A_n^f(x_1, \ldots, x_n) = f^{-1}(\frac{1}{n}\sum_{i=1}^{n} f(x_i)), x_1, \ldots, x_n \in I$$

where f^{-1} is the inverse of the function f. The function f is called the generator of the quasi-arithmetic mean A_n^f. The arithmetic mean, the geometric mean and the harmonic mean are quasi-arithmetic means corresponding to the functions $f : R \to R$, $f(x) = x, x \in R$; $f : (0, \infty) \to R$, $f(x) = \log x, x > 0$; and $f : (0, \infty) \to R$, $f(x) = x^{-1}, x > 0$, respectively.

Let I be a compact non-degenerate interval of R and $M_n : I^n \to R, n \geq 1$ be a sequence of functions. It is known that the sequence of functions $M_n, n \geq 1$ is quasi-arithmetic, that is, there exists a continuous strictly monotone function $f : I \to R$ such that

$$M_n(x_1, \ldots, x_n) = A_n^f(x_1, \ldots, x_n), x_1, \ldots, x_n \in I, n \geq 1$$

if and only if the sequence $M_n, n \geq 1$ satisfies the following properties:
(i) M_n is continuous and strictly increasing in each variable for every $n \geq 1$;
(ii) M_n is symmetric for each $n \geq 1$, that is, $M_n(x_1, \ldots, x_n) = M_n(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$ for every permutation $\sigma(1), \ldots, \sigma(n)$ of $1, \ldots, n$;
(iii) $M_n(x, \ldots, x) = x, x \in I$; and
(iv) $M_{n+m}(x_1, \ldots, x_n; y_1, \ldots, y_m) = M_{n+m}(\bar{x}_n, \ldots, \bar{x}_n; y_1, \ldots, y_m)$ for every $n \geq 1, m \geq 1, x_1, \ldots, x_n, y_1, \ldots, y_m \in I^{n+m}$ where $\bar{x}_n = M_n(x_1, \ldots, x_n)$
(cf. Kolmogorov [12], Nagumo [14] and de Finneti [7]).

Bajraktarevic [3] generalized the notion of the quasi-arithmetic mean. Let $n \geq 1$ and I be a non-degenerate interval of R. Let $f : I \to R$ be a continuous and strictly monotone function and $p(.) : I \to (0, \infty)$ be a weight function. Then the Bajraktarevic mean $B_n^{f,p} : I^n \to I$ is given by

$$B_n^{f,p}(x_1, \ldots, x_n) = f^{-1}(\frac{\Sigma_{i=1}^n p(x_i) f(x_i)}{\Sigma_{i=1}^n p(x_i)}), x_1, \ldots, x_n \in I.$$

The pair (f, p) is called the generator of $B_n^{f,p}$. It would be interesting to find out whether the Jensen inequality can be extended to Bajraktarevic means and other types of quasi-arithmetic means defined above for positive random variables.

Acknowledgements This work was supported under the scheme "INSA Honorary scientist" at the CR Rao Advanced Institute of Mathematics, Statistics and Computer Science, Hyderabad, India.

References

1. Ando T (1979) Concavity of certain maps on positive matrices and applications to hadamard products. Linear Alg Appl 26:203–241
2. Ando T (1983) On the arithmetic-geometric-harmonic mean inequalities for positive definite matrices. Linear Alg Appl 52(53):31–37

3. Bajraktarevic M (1958) Sur une equation fonctionnelle aux valeurs moyennes. Period Math-Phys Astron II Ser 13:243–247
4. Bobrovsky BZ, Mayer-Wolf E, Zakai M (1987) Some classes of global cramer-rao bounds. Ann Stat 15:1401–1420
5. Davis C (1963) Notions generalizing convexity for functions defined on spaces of matrices. In: Proceeding of Symposium in Pure Mathematics, vol 8, Convexity, pp 187–201
6. Dey A, Hande S, Tiku ML (1994) Statistical proofs of some matrix results. Linear Multilinear Algebra 38:109–116
7. de Finneti B (1931) Sul concetto di media. G 1st Ital Attuari 2:369–396
8. Gibilisco P (2022) About the jensen inequality for numerical n-means. Int J Modern Phys A. https://doi.org/10.1142/S0217751X22430102
9. Gibilisco P, Hensen S (2017) An inequality for expectation of means of positive random variables. Ann Funct Anal 8:142–151
10. Ibragimov IA, Hasminskii RZ (1981) Statistical estimation : asymptotic theory. Springer
11. Kimeldorf G, Sampson A (1973) A class of covariance inequalities. J Am Statist Assoc 68:228–230
12. Kolmogorov A (1930) Sur la notion de la moyenne. Atti Acad Naz Lincei, Rend, VI Ser 12:388–391
13. Mitra SK (1973) Statistical proofs of some propositions on nonnegative definite matrices. Bull Inter Statist Inst XLV, pp 206–211
14. Nagumo M (1931) On mean values. Tokyo Baturigakto-zassi 40:520–527
15. Petz D, Temesi R (2005) Means of positive numbers. SIAM J Matrix Anal Appl 27:712–720
16. Prakasa Rao BLS (1990) On a matrix version of a covariance inequality. Gujarat Stat Rev 17A:150–159
17. Prakasa Rao BLS (2000) Inequality for random matrices with an application to statistical inference. Student 3:198–202
18. Rao CR (1996) Seven inequalities in statistical estimation theory. Student 1:149–158
19. Rao CR (2000) Statistical proofs of some matrix inequalities. Linear Algebra Appl 321:307–320
20. Kubo F, Ando T (1979) Means of positive linear operators. Math Ann 246:205–224
21. Hardy GH, Littlewood J, Polya G (1952) Inequalities, 2nd ed. Cambridge University Press, Cambridge

Chapter 2
The Impact of Prof. C. R. Rao's Research Used in Solving Problems in Applied Probability



Jeffrey J. Hunter

Mathematics Subject Classification (2010) 15A09 · 15A10 · 15B51 · 60J10 · 60J22 · 60K15

Outline
We commence this article with a summary of the key results developed by Prof. Calyampudi Radhakrishna Rao related to his research on generalized matrix inverses. This is followed by a personal chronology of how the author became exposed to Prof. Rao's fundamental research on generalized matrix inverses. We then present a wide variety of results developed by the author in applying Prof. Rao's results to the field of Applied Probability. Detailed proofs are not provided for all the results, but full references are given. A description of the formal links that Prof. Hunter has had with Prof. Rao is also included.

2.1 Professor Rao's Research on Generalized Matrix Inverses

There were three key publications that set the scene for Rao's work on generalized matrix inverses (see the references [28–30]). In addition, two other related publications by Rao on generalized matrix inverses have been utilized in some of the applied probability applications: [31, 32].

Let us comment briefly on the key results contained in these aforementioned works.

J. J. Hunter (✉)
Auckland University of Technology, Auckland, New Zealand
e-mail: jeffrey.hunter@aut.ac.nz

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_2

Rao [29] **Analysis of Dispersion for Multiply Classified Data with unequal numbers in cells**, ***Sankhya: The Indian Journal of Statistics***, **15(3), 253–280.**

In this paper, Rao introduces the concept of a "pseudo-inverse" of a singular matrix in an appendix to the paper concerning the computational aspects of the general theory of least squares. In matrix notation, the normal equations are expressed as $AT = Q$ which is equivalent to $CAT = CQ = \Delta C^{-1T}T$ where $CA = \Delta C^{-1T}$ implying $T = C^T \nabla CQ$ by formally treating ∇ as the inverse of Δ, where Δ is a diagonal matrix containing the diagonal elements of CA. Thus, $C^T \nabla C$ behaves as the inverse of $A = C^{-1}\Delta C^{-1T}$.

Rao [30] **A note on a generalized inverse of a matrix with applications to problems in Mathematical Statistics**, ***Journal of the Royal Statistical Society, Series B (Methodological)***, **24(1), 152–158.**

In this paper, Rao defines a generalized inverse (g-inverse) of a matrix A, of order $m \times n$, as a matrix of order $n \times m$, denoted as A^- such that for any $\mathbf{y}$ for which $A\mathbf{x} = \mathbf{y}$ is consistent, $\mathbf{x} = A^-\mathbf{y}$ is a solution.

Lemma 2.1 *If A^- is a g-inverse of A, then $AA^-A = A$ and conversely.*

A general solution to $A\mathbf{x} = \mathbf{y}$, when consistent, is $\mathbf{x} = A^-\mathbf{y} + (A^-A - I)\mathbf{z}$ where $\mathbf{z}$ is arbitrary. The computation of a g-inverse, by the sweep-out process, is presented and applied to the theory of least squares.

Rao [31] ***Linear Statistical Inference and its Applications***, **Wiley, New York.**

In this book, Rao builds on the previous paper and introduces the 4−condition Moore-Penrose g-inverse, A^+ (to distinguish it from a 1−condition g-inverse, A^-) satisfying the four properties: (1) AA^+A, (2) $A^+AA^+ = A^+$, (3) $(AA^+)^T = AA^+$, (4) $(A^+A)^T = A^+A$. Such an inverse exists and is unique. This well-known g-inverse was credited by Rao to E. H. Moore in 1935 [26] and R. Penrose in 1955 [28] although some of Moore's results appeared earlier in 1920 [25] as an abstract of his talk at an AMS meeting.

Rao [32] **Generalized inverse for matrices and its applications in mathematical statistics**, ***Research Papers in Statistics, Festschrift for J Neyman***, **Wiley, New York.**

In this paper, extending his earlier results, we have explicitly the following.

Lemma 2.2 *The necessary and sufficient condition that the equation $AX = Y$ is consistent is that $AA^-Y = Y$.*

This leads to the following lemma.

Lemma 2.3 *A necessary and sufficient condition for the equation $AXB = C$ to have a solution is $AA^-CB^-B = C$ where A^- is any g-inverse of A and B^- is any g-inverse of B in which case the general solution is $X = A^-CB^- + Z - A^-AZBB^-$ where Z is arbitrary.*

Rao [33] **Calculus of generalized inverses of matrices, Part 1: General Theory**, ***Sankhya: The Indian Journal of Statistics, Series A*, 29(3), 317–342.**

Finally, in this paper a theorem, providing a representation of all g-inverses, is provided.

Theorem 2.1 *The general solution to a g-inverse of a given $m \times n$ matrix A is $A^- + U - A^-AUAA^-$ where A^- is a particular g-inverse and U is an arbitrary $n \times m$ matrix.*

Regarding multi-condition g-inverses, the two following lemmas are established.

Lemma 2.4 *The necessary and sufficient conditions that $X = GY$ is a minimum norm solution of the consistent equation $AX = Y$ is (i) $AGA = A$ (ii) $(GA)^T = GA$. (i.e. an $A\{1, 4\}$ g-inverse).*

Lemma 2.5 *The necessary and sufficient conditions that $X = GY$ is a least square solution of the equation $AX = Y$ is (i) $AGA = A$ (ii) $(AG)^T = AG$. (i.e. an $A\{1, 3\}$ g-inverse).*

As we shall see, the importance of Rao's work in applied probability lies in the introduction of generalized matrix inverses and the application of Lemmas 2.1 and 2.3 to a variety of situations where linear equations arise naturally in stochastic situations requiring unique solutions. However, to apply such techniques the identification of particular g-inverses is required. Other than assisting in the classification of g-inverses, these last two lemmas are of more relevance to statistical applications rather than to applied probability.

A number of authors have explored the applications of matrix theory to problems in applied probability but specific references to Rao's contributions utilizing generalized inverses and the solution of linear equations are infrequent.

2.2 Professor Hunter Is Exposed to Prof. Rao's Research

In February 1965, the author was in his second semester of coursework toward a Ph.D. in Statistics from the Department of Statistics, University of North Carolina at Chapel Hill. Although he was intending to do research in applied probability, one of the graduate-level courses, designed to enable a student to undertake the preliminary doctoral written examinations, was Statistics 150 "Analysis of Variance with Applications to Experimental Design" taught by Prof. Indra Mohan Chakravarti, using notes prepared by Prof. Raj Chandra Bose. In this course, the notion of a "*conditional inverse*" was introduced. The following are extracts from Bose's notes (with the numbering changed for this article):

> 1. Conditional inverse of a matrix.
> The inverse of a non-singular matrix has already been defined. We shall now consider a generalization of the notion of inverse which we shall find useful in certain circumstances.

Let $A = A(m \times n)$ be any matrix. Then $A^* = A^*(n \times m)$ will be defined to be a conditional inverse of A, if

$$AA^*A = A. \tag{2.1}$$

When A is square and non-singular, $A^{-1}AA^*AA^{-1} = A^{-1}AA^{-1}$. Hence $A^* = A^{-1}$ and is uniquely determined. We shall show that in the general case A^* exists but is not uniquely determined by (2.1), when A is given. We first prove the following theorem.

(a) If $A^* = A^*(n \times m)$ is a conditional inverse of $A = A(m \times n)$ and if the equations

$$A\mathbf{x} = \mathbf{y} \tag{2.2}$$

are consistent, then $\mathbf{x}_1$ is a solution of (2.2) if $\mathbf{x}_1 = A^*y$.

(b) If $\mathbf{x}_1 = A^*y$ is a solution of Eq. (2.2) for every y for which (2.2) are consistent, then A^* must be a conditional inverse of A.

It is interesting to reflect on how the notion of a *conditional inverse* was used in the lecture notes for this course, rather than the now common terminology of a *generalized inverse*, or the simpler form, *a g-inverse.*.

It is worth pointing out that there had been strong interactions between all three aforementioned academics. Firstly, Prof. Raj Chandra Bose joined the University of Calcutta in 1940 where C. R. Rao was in the first group of students that he taught. Professor Bose became Head of the Department of Statistics in 1945 at the University of Calcutta but moved to the Department of Statistics at the University of North Carolina at Chapel Hill in 1949.

Secondly, Prof. Indra M. Chakravarti, who taught the aforementioned graduate-level course, obtained his Ph.D. from the Indian Statistical Institute at Kolkata in 1958 having been supervised by Prof. Rao. Indra joined the Department of Statistics at the University of North Carolina at Chapel Hill in 1964.

Thirdly, the connection between all three was strengthened in 1969 when all three edited the book "Essays in Probability and Statistics", published in the Monograph Series in Probability and Statistics, by the University of North Carolina Press.

The author had the opportunity to study various different areas of statistics, prior to deciding on a research topic, as part of the coursework requirements for a Ph.D. in the U.S. At the Dept. of Statistics at UNC Chapel Hill, he had the opportunity to take courses from a number of leading figures in their fields including Norman Johnson, Wassily Hoeffding, Walter Smith and Ross Leadbetter.

The author's first exposure to Prof. Rao's research was as an undergraduate in Mathematics at the University of Auckland learning of some of his pioneering classical results, in particular the Cramer-Rao lower bound. Besides meeting with Prof. Rao, he also had the opportunity to meet with Prof. Harold Cramer during one of his visits to the Department of Statistics at the University of North Carolina at Chapel Hill while he was writing a book with Ross Leadbetter.

2.3 Professor Hunter's Applications of Prof. Rao's Research in the Field of Applied Probability

Professor Hunter pursued his Ph.D. research in the general area of applied probability under the supervision of Walter Smith completing a thesis "On the necessary and sufficient conditions for the convergence of the renewal density matrix of a semi-Markov process". His early research, following his Ph.D., focused on Markov renewal processes and their embedded Markov chains [2, 3, 7, 8].

Four papers set the scene for his research utilizing generalized matrix inverses in applied probability. Prior to discussing the main results of those papers, we introduce some notation that we shall use throughout this paper.

Let $\{X_n\}$, $(n \geq 0)$, be a discrete time Markov chain $(M.C.)$ with state space $S = \{1, 2, \ldots, m\}$, transition probabilities $p_{ij} = P\{X_{n+1} = j \mid X_n = i\}$, $(i, j) \in S \times S$ and transition matrix $P = [p_{ij}]$. Let $\Pi = \mathbf{e}\boldsymbol{\pi}^T$ where $\mathbf{e}^T = (1, 1, \ldots, 1)$ and $\boldsymbol{\pi}^T$ is the stationary probability vector of the M. C. Let $E = [1]$ and if $A = [a_{ij}]$ define $A_d = [\delta_{ij} a_{ij}]$ (where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise), the diagonal matrix of diagonal elements of A.

Let $\{(X_n, T_n)\}$ $(n \geq 0)$ be a Markov Renewal Process (M. R. P.) with semi-Markov kernel $Q(t) = \left[Q_{ij}(t)\right]$ where X_n denotes the state of the system following the n-th transition, after the initial transition (X_0 being the initial state) in a finite state process with state space S and T_n is the time of the n-th transition. Let $Q_{ij}(t) = P\{X_{n+1} = j,\ T_{n+1} - T_n \leq t \mid X_n = i\}$, $(i, j) \in S$. Further, $P = \left[p_{ij}\right]$ where $p_{ij} = Q_{ij}(+\infty)$, so that P is the transition matrix of the embedded M. C. $\{X_n\}$. Let T_{ij} be the time for the first passage from state i to j in the M. R. P., and let $G_{ij}(t)$ be the distribution function of T_{ij}.

Associated with the M. R. P. is the semi-Markov Process (S. M. P.) $Y_t = X_n$ for $T_n \leq t < T_{n+1}$ that records the state that M. R. P. is in at each time t.

Further let $m_{ij}^{(k)} = \int_0^\infty t^k dG_{ij}(t)$ and $\mu_{ij}^{(k)} = \int_0^\infty t^k dQ_{ij}(t)$ with $m_{ij} = m_{ij}^{(1)}$ and $\mu_{ij} = \mu_{ij}^{(1)}$, so that the $m_{ij}^{(k)}$ is the k-th moment of the first passage from state i to state j.

Define $M^{(r)} = \left[m_{ij}^{(r)}\right]$ and $P^{(r)} = \left[\mu_{ij}^{(r)}\right]$ with $M = M^{(1)}$ and $\boldsymbol{\mu} = P^{(1)}\mathbf{e}$.

Let $N_j(t)$ be the number of visits the M. R. P. makes to state j, up to and including time t, $M(t) = \left[M_{ij}(t)\right]$ where $M_{ij}(t) = E\{N_j(t) \mid X_0 = i\}$, $t \geq 0$, i.e. $M_{ij}(t)$ is a general renewal function of the process, the expected number of visits the M. R. P. makes to state j up to time t, with $X_0 = i$.

In the special case of a M. C. in discrete time, $T_n = n$, $\mu_{ij} = p_{ij}$, and $G_{ij}(t) = \begin{cases} 1, & t \geq 1, \\ 0, & t < 1; \end{cases}$ and the $M_{ij}^{(n)}$ are the number of k $(0 \leq k \leq n)$ such that $X_k = j$ given $X_0 = i$, i.e. the number of visits the M. C. makes to state j, starting in state i, while $N_{ij}^{(n)}$ are the number of k $(1 \leq k \leq n)$, so that $M_{ij}^{(n)} = N_{ij}^{(n)} + \delta_{ij}$.

For a M. C. in continuous time, the elements of the semi-Markov kernel have the form $Q_{ij}(t) = p_{ij}\left[1 - e^{-\lambda_k t}\right]$.

Hunter [2] **On the moments of Markov Renewal Processes**, ***Advances in Applied Probability***, **1(2), 188–210.**

In this first paper, the well-known fundamental matrix of regular Markov chains, $Z = [I - P + \Pi]^{-1}$, as introduced by Kemeny and Snell, was shown to be a 1−condition g-inverse of $I - P$ where P is the transition matrix of the Markov chain. The proof of this result used the well-known result ([23], p. 100) that $(I - P)Z = I - \Pi$.

This matrix Z was then applied to solving a system of linear equations that arose in exploring the properties of a M. R. P. and the special case of a M. C.

It was established that, for $r \geq 1$,

$$m_{ij}^{(r)} = \sum_{k \neq j} p_{ik} m_{kj}^{(r)} + \sum_{s=1}^{r-1} \binom{r}{s} \left\{ \sum_{k \neq j} \mu_{ik}^{(r-s)} m_{kj}^{(s)} \right\} + \sum_{s=1}^{m} \mu_{is}^{(r)},$$

leading to

$$(1 - P)M^{(r)} = \sum_{s=1}^{r-1} \binom{r}{s} P^{(r-s)} \left\{ M^{(s)} - M_d^{(s)} \right\} + P^{(r)} E - P M_d^{(r)}.$$

In particular, when $r = 1$, $(I - P)M = P^{(1)} E - P M_d$.

The paper discusses the solution of these matrix equations, for M. R. P.'s, by solving them directly using generalized matrix techniques, and using Z as a particular g-inverse of $I - P$. In particular, the second main result was an expression for the matrix of the mean first passage times,

$$M = \left[\frac{1}{\lambda_1} \left\{ Z P^{(1)} \Pi - E(Z P^{(1)} \Pi)_d \right\} + I - Z + E Z_d \right] D,$$

where $\lambda_1 = \boldsymbol{\pi}^T \boldsymbol{\mu}$, $D = M_d = \lambda_1 (\Pi_d)^{-1}$.

The third main result of this paper provides an expression for the asymptotic values of the moments of the M. R. P., $M(t)$, in terms of Z,

$$M(t) = \frac{t}{\lambda_1} \Pi + \frac{\lambda_2}{2\lambda_1^2} \Pi + \left[\frac{1}{\lambda_1} \Pi P^{(1)} - I \right] Z \left[\frac{1}{\lambda_1} P^{(1)} \Pi - I \right] - I + o(1) E.$$

Hunter [3] **Generalized Inverses and their application to Applied Probability Problems**, ***Linear Algebra & its Applications***, **45, 157–198.**

The first key result in this paper was the derivation of more general g-inverses of $I - P$, where P is the transition matrix of a finite irreducible M. C. with stationary probability vector $\boldsymbol{\pi}^T$. In particular, if $\boldsymbol{u}$ and $\boldsymbol{t}$ are such that $\boldsymbol{u}^T \boldsymbol{e} \neq 0$ and $\boldsymbol{\pi}^T \boldsymbol{t} \neq 0$, then $I - P + \boldsymbol{t}\boldsymbol{u}^T$ is non-singular and $\left[I - P + \boldsymbol{t}\boldsymbol{u}^T \right]^{-1}$ is a g-inverse of $I - P$.

The results of Rao [32] were utilized to show that any g-inverse of $I - P$ has the form $\left[I - P + \boldsymbol{t}\boldsymbol{u}^T \right]^{-1} + \boldsymbol{e}\boldsymbol{f}^T + \boldsymbol{g}\boldsymbol{\pi}^T$.

In the paper, general procedures were established for finding stationary distributions, moments of first passage time distributions and asymptotic forms of the occupation time random variables for not only M. C.'s but more generally M. R. P.'s and S. M. P.'s including M. C.'s in continuous time. In particular, some specific special results for discrete time M.C.'s are as follows: Let G be any g-inverse of $I - P$.

If $A = I - (I - P)\,G$ then $\boldsymbol{\pi}^T = \dfrac{\boldsymbol{v}^T A}{\boldsymbol{v}^T A \boldsymbol{e}}$ for any $\boldsymbol{v}^T$ such that $\boldsymbol{v}^T A \boldsymbol{e} \neq 0$.

$M = [G\Pi - E(G\Pi)_d + I - G + EG_d]\,D$ where $D = (\Pi_d)^{-1}$.

$\left[EN_{ij}^{(n)}\right] = (n+1)\Pi - I + (I - \Pi)G(I - \Pi) + o(1)E.$

Hunter [7] **Characterizations of Generalized Inverses associated with Markovian kernels**, ***Linear Algebra & its Applications***, **102, 121–142.**

This paper derives characterizations of generalized inverses associated with Markovian kernels $A = I - P$. We consider the standard 4-conditions of the Moore-Penrose g-inverse together with a 5th condition, applicable for square matrices. We let $A\{i, \ldots, k\}$ denote the family of all g-inverses X of the matrix A that satisfy the conditions $i, \ldots, k$ from the following numbered conditions (1) $AXA = A$, (2) $XAX = X$, (3) $(AX)^T = AX$, (4) $(XA)^T = XA$, (5). For square matrices $AX = XA$.

Characterizations of $A\{1, 2\}$, $A\{1, 3\}$, $A\{1, 4\}$, $A\{1, 5\}$, $A\{1, j, k\}$ are all given.

Some special cases are considered.

The group inverse $A\{1, 2, 5\} = \left[I - P + \boldsymbol{e}\boldsymbol{\pi}^T\right]^{-1} - \boldsymbol{e}\boldsymbol{\pi}^T = (1 - \boldsymbol{e}\boldsymbol{\pi}^T)G(1 - \boldsymbol{e}\boldsymbol{\pi}^T)$ for any g-inverse G of $I - P$. The use of this particular g-inverse in relation to M. C.'s was explored extensively by Meyer [24].

The Moore-Penrose g-inverse $A\{1, 2, 3, 4\} = \left[I - P + \alpha\boldsymbol{\pi}\boldsymbol{e}^T\right]^{-1} - \alpha\boldsymbol{e}\boldsymbol{\pi}^T$, where $\alpha = (m\boldsymbol{\pi}^T\boldsymbol{\pi})^{-1/2}$ was given by Paige, Styan and Wachter [27].

It was also shown that one can transform from one g-inverse to another by using the following result.

If, for $i = 1, 2$, $\boldsymbol{u}_i$ and $\boldsymbol{t}_i$ are such that $\boldsymbol{u}_i^T\boldsymbol{e} \neq 0$ and $\boldsymbol{\pi}^T\boldsymbol{t}_i \neq 0$ then

$$\left[I - P + \boldsymbol{t}_2\boldsymbol{u}_2^T\right]^{-1} = (I - U_2)\left[I - P + \boldsymbol{t}_1\boldsymbol{u}_1^T\right]^{-1}(I - T_2) + \beta_{22}\boldsymbol{e}\boldsymbol{\pi}^T$$

where $U_2 = \dfrac{\boldsymbol{e}\boldsymbol{u}_2^T}{\boldsymbol{u}_2^T\boldsymbol{e}}$, $T_2 = \dfrac{\boldsymbol{t}_2\boldsymbol{\pi}^T}{\boldsymbol{\pi}^T\boldsymbol{t}_2}$ and $\beta_{22} = \dfrac{1}{\left(\boldsymbol{\pi}^T\boldsymbol{t}_2\right)\left(\boldsymbol{u}_2^T\boldsymbol{e}\right)}$.

Another interesting result that helps in reformulating different forms of g-inverses of $I - P$, in particular the Moore-Penrose g-inverse, was presented: For any $\boldsymbol{t}$ and $\boldsymbol{u}$ such that $\boldsymbol{\pi}^T\boldsymbol{t} \neq 0$ and $\boldsymbol{u}^T\boldsymbol{e} \neq 0$, $A_\delta \equiv \left[I - P + \delta\boldsymbol{t}\boldsymbol{u}^T\right]^{-1} - \dfrac{\boldsymbol{e}\boldsymbol{\pi}^T}{\delta\left(\boldsymbol{\pi}^T\boldsymbol{t}\right)\left(\boldsymbol{u}^T\boldsymbol{e}\right)}$ does not depend on $\delta \neq 0$.

A range of applications using different g-inverse of $I - P$ in M. C.'s, used by a number of other researchers, is given in the list of references of this paper, [7]. In particular, Decell and Odell [1] used the Moore-Penrose g-inverse in the computation

of the stationary distribution of an ergodic M. C. Rising [34], also explored applications of generalized inverses to M. C.'s, focusing on the derivation of the stationary distribution and mean first passage times.

Hunter [8] **Parametric Forms for Generalized Inverses of Markovian Kernels & their Applications**, ***Linear Algebra & its Applications***, **127, 71–84.**

Since g-inverses can appear in different forms, in this paper parametric forms for generalized inverses of the Markovian kernel $I - P$ were derived. The main result follows.

Given any g-inverse, G, of $I - P$, there exists unique parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and γ such that

$$G = \left[I - P + \boldsymbol{\alpha}\boldsymbol{\beta}^T\right]^{-1} + \gamma \boldsymbol{e}\boldsymbol{\pi}^T.$$

Let $A = I - (I - P)G,\ B = I - G(I - P)$, then $\boldsymbol{\alpha} = A\boldsymbol{e},\ \boldsymbol{\beta}^T = \boldsymbol{\pi}^T B,\ \gamma + 1 = \boldsymbol{\beta}^T G\boldsymbol{\alpha}$.

Note also that $\gamma + 1 = \boldsymbol{\pi}^T G\boldsymbol{\alpha} = \boldsymbol{\beta}^T G\boldsymbol{e},\ \boldsymbol{\pi}^T\boldsymbol{\alpha} = 1,\ \boldsymbol{\beta}^T\boldsymbol{e} = 1$.

These parameters uniquely determine different classes of g-inverses of $I - P$:

$$\begin{aligned}
G \in A\{1,2\} &\Leftrightarrow \gamma = -1,\\
G \in A\{1,3\} &\Leftrightarrow \boldsymbol{\alpha} = \boldsymbol{\pi}/\boldsymbol{\pi}^T\boldsymbol{\pi},\\
G \in A\{1,4\} &\Leftrightarrow \boldsymbol{\beta} = \boldsymbol{e}/\boldsymbol{e}^T\boldsymbol{e},\\
G \in A\{1,5\} &\Leftrightarrow \boldsymbol{\alpha} = \boldsymbol{e},\ \boldsymbol{\beta} = \boldsymbol{\pi}.
\end{aligned}$$

Thus, G is a Moore-Penrose g-inverse if $\boldsymbol{\alpha} = \boldsymbol{e}$, $\boldsymbol{\beta} = \boldsymbol{e}/\boldsymbol{e}^T\boldsymbol{e}$ and $\gamma = -1$, and G is a Group inverse if $\boldsymbol{\alpha} = \boldsymbol{e}$, $\boldsymbol{\beta} = \boldsymbol{\pi}$ and $\gamma = -1$.

The early material of the author, [2, 3], utilizing generalized inverses and applying them to solving for stationary distributions and mean first passage times in M. C.'s is also included in his two books, [4, 5], and the e-book [19].

Professor Hunter has published a number of research papers that utilize the results of the four papers described above—all utilizing generalized matrix inverses and extending their applications in a number of applied probability areas. We summarize the significant results but don't go into full details, and the readers are referred to the individual papers for more information.

Hunter [6] **Stationary distributions of perturbed Markov chains**, ***Linear Algebra & its Applications***, **82, 201–214.**

Techniques for updating the stationary distribution of a finite irreducible M. C., following a rank one perturbation of its transition matrix were discussed in this paper.

Let $P^{(2)} = P^{(1)} + \boldsymbol{a}\boldsymbol{b}^T$, with $\boldsymbol{b}^T\boldsymbol{e} = 0$, be the transition matrix of the perturbed $P^{(1)}$ transition matrix. Choose $\boldsymbol{u}$ and $\boldsymbol{t}$ (with $\boldsymbol{u}^T\boldsymbol{e} \neq 0,\ \boldsymbol{\pi}^{(1)T}\boldsymbol{t} \neq 0$).

Let $\boldsymbol{\alpha}^T = \boldsymbol{u}^T \left[I - P^{(1)} + \boldsymbol{t}\boldsymbol{u}^T\right]^{-1}$ and $\boldsymbol{\beta}^T = \boldsymbol{b}^T \left[I - P^{(1)} + \boldsymbol{t}\boldsymbol{u}^T\right]^{-1}$.

Then $\boldsymbol{\pi}^{(1)T} = \dfrac{\boldsymbol{\alpha}^T}{\boldsymbol{\alpha}^T\boldsymbol{e}}$ and $\boldsymbol{\pi}^{(2)T} = \dfrac{\left(\boldsymbol{\alpha}^T\boldsymbol{a}\right)\boldsymbol{\beta}^T + \left(1 - \boldsymbol{\beta}^T\boldsymbol{a}\right)\boldsymbol{\alpha}^T}{\left(\boldsymbol{\alpha}^T\boldsymbol{a}\right)\boldsymbol{\beta}^T\boldsymbol{e} + \left(1 - \boldsymbol{\beta}^T\boldsymbol{a}\right)\boldsymbol{\alpha}^T\boldsymbol{e}}$.

A variety of situations where such perturbations arise are presented together with suitable procedures for deriving the stationary distributions.

Some subsidiary results relating the fundamental matrices of the two transition matrices are also derived, using obvious notation,

$$Z^{(2)} = \left[I - \frac{\left(\boldsymbol{\pi}^{(1)T}\boldsymbol{a}\right)\boldsymbol{e}\boldsymbol{b}^T Z^{(1)}}{1 - \boldsymbol{b}^T Z^{(1)}\boldsymbol{a}}\right] Z^{(1)} \left[I + \frac{\boldsymbol{a}\boldsymbol{b}^T Z^{(1)}}{1 - \boldsymbol{b}^T Z^{(1)}\boldsymbol{a}}\right].$$

Hunter [9] **The computation of stationary distributions of Markov chains through perturbations**, ***Journal Applied Mathematics & Stochastic Analysis***, **4 (1), 29-46.**

An algorithmic procedure for the determination of the stationary distribution of a finite, $m-$state, irreducible M. C., that does not require the traditional use of methods for solving linear equations, is given in this paper. The construction of the proof revolves around the results derived for perturbed M. C.'s.

The basic idea is simple. Start with the trivial doubly stochastic matrix P_0 with each element having the value $1/m$. The stationary probability vector for this M. C. is simply $\boldsymbol{\pi}^{(0)T} = \boldsymbol{e}^T/m$. Now successively perturb this transition matrix P_0 by successively replacing, for $i = 1, 2, \ldots, m$, its i-th row by the i-th row of the given transition matrix so that at the m-th stage the resulting probability vector is the required stationary probability vector. The procedure can be expressed as follows.

Let $A_0 = I,\ \boldsymbol{u}_0^T = \boldsymbol{e}^T/m$. For $i = 1, 2, \ldots, m$, let

$\boldsymbol{p}_i^T = \boldsymbol{e}_i^T P,$

$\boldsymbol{u}_i^T = \boldsymbol{u}_{i-1}^T + \boldsymbol{p}_i^T - \boldsymbol{e}^T/m,$

$A_i = A_{i-1} + A_{i-1}(\boldsymbol{e}_{i-1} - \boldsymbol{e}_i)(\boldsymbol{u}_{i-1}^T A_{i-1}/\boldsymbol{u}_{i-1}^T A_{i-1}\boldsymbol{e}_i).$

At $i = m,\ \boldsymbol{\pi}^T (= \boldsymbol{\pi}_m^T) = \boldsymbol{u}_m^T A_m/\boldsymbol{u}_m^T A_m \boldsymbol{e}.$

The paper includes some modifications and provides as an example an illustration of the procedure.

Hunter [10] **Stationary distributions and mean first passage times in Markov chains using generalized inverses**, ***Asia-Pacific Journal of Operational Research*** **(2), 145–153.**

In this paper, we consider the joint derivation of stationary distributions and mean first passage times of M. C.'s using g-inverses of $I - P$ (with some minor errors in the initial derivation, [10], corrected).

(1) Compute $G = \left[g_{ij}\right]$, a g-inverse of $I - P$.

(2) Compute sequentially rows $1, 2, \ldots, r(\leq m)$ of $A = I - (I - P)G = [a_{ij}]$, until $\sum_{j=1}^m a_{1j} \neq 0$, $(r = 1)$ or $\sum_{j=1}^m a_{1j} = \sum_{j=1}^m a_{2j} = \cdots = \sum_{j=1}^m a_{r-1\,j} = 0$, $\sum_{j=1}^m a_{rj} \neq 0$, $(r \neq 1)$, where $a_{ij} = \delta_{ij} - g_{ij} + \sum_{k=1}^m p_{ik} g_{kj}$.

(3) Compute the stationary probabilities $\pi_j = a_{rj} / \sum_{k=1}^m a_{rk}$, $j = 1, 2, \ldots, m$.

(4) Set $m_{jj} = \sum_{k=1}^{m} a_{rk}/a_{rj}, \ \left(= 1/\pi_j\right), \ j = 1, 2, \ldots, m.$

(5) Compute, for $i \neq j, \quad i, j = 1, 2, \ldots, m, \quad m_{ij} = \dfrac{\left(g_{jj} - g_{ij}\right)\left(\sum_{k=1}^{m} a_{rk}\right)}{a_{rj}} + \sum_{k=1}^{m} \left(g_{ik} - g_{jk}\right).$

In general, not all the elements of A need to be obtained and, in most cases, only the elements of the first row $a_{11}, a_{12}, \ldots, a_{1m}$ suffice. For such a scenario, all the mean first passage times can be expressed in terms of m^2 elements of the g-inverse $G = [g_{ij}]$ and the m elements $\{a_{1k}\}(k = 1, 2, \ldots, m)$ of the first row of A.

Hunter [11] **Stationary distributions and mean first passage times of perturbed Markov chains**, *Linear Algebra* **& *its Applications***, **410, 217–243.**

Continuing in the theme of the earlier results of perturbed irreducible M. C.'s, it is shown that the mean first passage times play an important role in determining the differences between the stationary probabilities in the perturbed and unperturbed situations. The derivation of the interconnection, under the updating procedure, is explored through the use of generalized matrix inverses.

Let $P = [p_{ij}]$ be the transition matrix of a finite irreducible, m-state M. C. Let $\tilde{P} = [\tilde{p_{ij}}] = P + \boldsymbol{E}$ be the transition matrix of the perturbed M. C. where $\boldsymbol{E} = [\varepsilon_{ij}]$ is the matrix of perturbations. Let $\boldsymbol{\pi}^T$ and $\tilde{\boldsymbol{\pi}}^T$ be the associated stationary probability vectors. Define $\Pi = \boldsymbol{e}\boldsymbol{\pi}^T$ and let G be any g-inverse of $I - P$. If $H \equiv G(I - \Pi)$ then, for any general perturbation $\boldsymbol{E}$, $\tilde{\boldsymbol{\pi}}^T - \boldsymbol{\pi}^T = \tilde{\boldsymbol{\pi}}^T \boldsymbol{E} H$.

The paper also provides a re-expression of the above result in terms of the mean first passage time matrix, M as follows: $\tilde{\boldsymbol{\pi}}^T - \boldsymbol{\pi}^T = \tilde{\boldsymbol{\pi}}^T \boldsymbol{E}(M - M_d)(M_d)^{-1}$.

Further, if $N = [n_{ij}] = \left[(1 - \delta_{ij})m_{ij}/m_{jj}\right] = \left[(1 - \delta_{ij})m_{ij}\pi_j\right]$, then $\tilde{\boldsymbol{\pi}}^T - \boldsymbol{\pi}^T = -\tilde{\boldsymbol{\pi}}^T \boldsymbol{E} N$.

New improved bounds for the relative and absolute differences between the stationary distributions under general perturbations were derived.

In particular, for each fixed $j, \ 1 \leq j \leq m$,

$$|\pi_j - \tilde{\pi}_j| \leq \frac{||E||_\infty}{2} \max_{i \neq j}\{n_{ij}\}, \quad \text{where} ||E||_\infty = \max_{1 \leq k \leq m} \sum_{l=1}^{m} |\varepsilon_{kl}|.$$

Some interesting qualitative results regarding the nature of the relative and absolute changes to the stationary probabilities are also obtained. Similar procedures are used to establish an updating procedure for mean first passage times under perturbations.

Hunter [12] **Mixing times with applications to perturbed Markov chains**, ***Linear Algebra* & *its Applications***, **417, 108–123.**

In this paper, a measure of the "mixing time" or "time to stationarity" in a finite irreducible discrete time M. C. is introduced. For this paper, and the sequel [14], the "*time to mixing*" in a M. C. $\{X_n\}$ can be defined as follows. First, sample from

the stationary distribution $\{\pi_j\}$ to determine a value of the random variable Y, say $Y = j$, the state thus sampled. Then observe the M. C., starting at a given state i. Then "mixing occurs at time $T = n$" when $X_n = j$ for the first such $n \geq 1$, i.e. conditional upon $Y = j$, $T = T_{ij}$, is the first passage time from state i to state j (or return to state j when $i = j$). The finite state space restriction, under irreducibility conditions, ensures the finiteness of the "mixing time T" (a.s), with a finite expectation.

The statistic $\eta_i = \sum_{j=1}^{m} m_{ij}\pi_j$ is shown to be independent of the state i that the chain starts in (i.e. $\eta_i = \eta$ for all i), is minimal in the case of a periodic chain yet can be arbitrarily large in a variety of situations.

Although not identified in this paper, η is now known as Kemeny's constant, (see [20]), which was indirectly established by Kemeny and Snell, Theorem 4.4.10 in [23].

It is shown that η can be derived from any g-inverse of $I - P$, $G = [g_{ij}]$ and the stationary probabilities $\{\pi_j\}$. In particular, $\eta = 1 + \sum_{j=1}^{m}(g_{jj} - g_j\pi_j)$. As a special case, $\eta = \text{tr}(Z)$, where Z is the fundamental matrix of the M. C.

For any irreducible m-state M. C., $\eta \geq \dfrac{m+1}{2}$.

For all irreducible $m-$state M. C.'s undergoing a general perturbation $\boldsymbol{E} = [\varepsilon_{ij}]$, with associated stationary probability vectors $\boldsymbol{\pi}^{(1)}$ and $\boldsymbol{\pi}^{(2)}$, it was shown that

$$||\boldsymbol{\pi}^{(1)} - \boldsymbol{\pi}^{(2)}||_1 \leq (\eta - 1)||E||_\infty$$

where the $1-$norm $||\boldsymbol{v}||_1$ of a vector $\boldsymbol{v}$ is the absolute entry sum.

A variety of special cases are considered.

Hunter [13] **Simple procedures for finding mean first passage times in Markov chains**, ***Asia-Pacific Journal of Operational Research*, 24 (6), 813–829.**

This paper provides elegant new results for finding mean first passage times based upon using a selection of simple g-inverses of $I - P$ starting from the basic form of a g-inverse as $\left[I - P + \boldsymbol{t}\boldsymbol{u}^T\right]^{-1}$ and then choosing $\boldsymbol{t}$ and $\boldsymbol{u}$, or the parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and γ.

We explore a variety of row and column properties of these g-inverses and explore the joint computation of both the stationary distributions and the mean first passage times. A special result deserves highlighting.

If $G_{eb} = \left[I - P + \boldsymbol{e}\boldsymbol{e}_b^T\right] = [g_{ij}]$ where $\boldsymbol{e}_b^T = (0, 0, \ldots, 1, 0, \ldots, 0)$ where 1 is in the b-th position, then $\pi_j = g_{bj}$, $j = 1, 2, \ldots, m$, and $m_{ij} = \begin{cases} 1/g_{bj}, & i = j, \\ (g_{jj} - g_{ij})/g_{bj}, & i \neq j. \end{cases}$

Hunter [14] **Variance of first passage times and applications to mixing times in Markov chains**, *Linear Algebra* ***& its Applications*, 429, 1135–1162.**

In this paper, we explore the variance of the mixing time. We omit many of the technicalities of establishing expressions but a brief synopsis is as follows.

Define $\eta_i^{(k)} = E\left[T^k | X_0 = i\right]$ as the k-th moment of the mixing time, starting in state i, and let $m_{ij}^{(k)}$ be the k-th moment of the first passage time from state i to state j in the M. C. For all i, $\eta_i^{(k)} = \sum_{j=1}^{m} m_{ij}^{(k)}\pi_j$.

The mean mixing time starting in state i, $\eta_i^{(1)} = \eta_i = \sum_{j=1}^{m} m_{ij}\pi_j = \eta$, a constant (*Kemeny's constant*) independent of i, the starting state.

Now, the variance of the mixing time when the M. C. starts in state i, $v_i = \text{var}[T|X_0 = i] = \eta_i^{(2)} - \eta^2 = \sum_{j=1}^{m} m_{ij}^{(2)}\pi_j - \eta^2$, thus we first need to explore the derivation of the first two moments of the first passage times.

By deriving expressions for the matrices of the first and second moments of the first passage times from state i to state j in an irreducible finite M. C. with transition matrix P, we establish that, if $G = [g_{ij}]$ is any g-inverse of $I - P$,

$$\boldsymbol{v} = 2[I - G + EG_d]\boldsymbol{\alpha} + \left[2\eta G - 2\text{tr}(GL) - \eta - \eta^2\right]\boldsymbol{e}$$

where $\boldsymbol{\alpha} = (\boldsymbol{\pi}^T G\boldsymbol{e})\boldsymbol{e} - (GE)_d\boldsymbol{e} + \boldsymbol{e} - (\Pi G)_d D\boldsymbol{e} + G_d D\boldsymbol{e} = (\Pi M)_d\boldsymbol{e}$, $L = G\Pi - E(G\Pi)_d + I - G + EG_d$ and $\eta = 1 + \sum_{j=1}^{m}(g_{jj} - g_{j\bullet}\pi_j)$ with $g_{j\bullet} = \sum_{k=1}^{m} g_{jk}$.

The variances v_i are shown to depend on i and an exploration of recommended starting states, given knowledge of the transition probabilities, is considered.

Hunter [16] **Some stochastic properties of semi-magic and magic Markov chains**, ***Linear Algebra & its Applications***, **433, 893–907.**

Magic squares are arrays of positive integers that have four main properties (1) equal row sums, (2) equal column sums, (3) the main diagonal sum and (4) the main skew-diagonal, all equal to the same sum, the "magic sum". By dividing each element in the array by the magic sum, we obtain the transition matrix of a M. C., a "magic" M. C. This transition matrix is a "super-stochastic" matrix. If we allow one or both of the properties (3) or (4) to be relaxed, we have a "semi-magic" M. C., still with a doubly stochastic transition matrix.

The paper explored the main stochastic properties of "magic" and "semi-magic" M. C.'s (and their diagonal and skew-diagonal variants). A number of different types of g-inverses are explored—some of which have constant row sums and some are semi-magic. We also explored multi-condition g-inverses of the Markovian kernels under doubly stochasticity as well as the properties of the diagonal sums and skew-diagonal sums. We do not go into any details, but an interested reader can consult the paper, [16], which also explores the stationary distribution, the mean first passage times, the variances of the first passage times and the expected times to mixing. Some general results were developed, some conjectures were presented and some special cases, involving three and four states, were explored in detail.

Hunter [17] **Markov chain properties in terms of column sums of the transition matrix, Acta et Commentationes Universitatis Tartuensis de Mathematica. 16, (1), 33–51.**

In this paper, questions are posed regarding the influence that the column sums of the transition probabilities of a stochastic matrix (with row sums all one) have on the stationary distribution, the mean first passage times and the Kemeny constant of the associated irreducible discrete time M. C. Some new relationships, including some inequalities, and partial answers to the questions are given.

Let $P = [p_{ij}]$ be the transition matrix of a finite irreducible, discrete time M. C. $\{X_n\}, n \geq 0$, with state spaces $S = \{1, 2, \ldots, m\}$. The stochastic nature of P implies that the row sums of P are all one. Let $c_j = \sum_{i=1}^{m} p_{ij}$, for all $j = 1, 2, \ldots, m$, be the respective column sums of the transition matrix.

We pose the following questions: What influence does the sequence $\{c_j\}$ have on the stationary distribution $\{\pi_j\}$? What influence does the sequence $\{c_j\}$ have on the mean first passage times $\{m_{ij}\}$? Are there relationships connecting the $\{c_j\}$, the $\{\pi_j\}$ and the $\{m_{ij}\}$? Can we deduce bounds on the $\{\pi_j\}$ and the $\{m_{ij}\}$ involving $\{c_j\}$? What effect does the $\{c_j\}$ have on Kemeny's constant $K = \sum_{j=1}^{m} \pi_j m_{ij}$, which is in fact independent of $i \in \{1, 2, \ldots, m\}$.

To explore these questions, we use a special generalized matrix inverse that had not previously been considered in the literature on M. C.'s. In particular, we define $H \equiv \left[I - P + \boldsymbol{e}\boldsymbol{c}^T\right]^{-1}$, where $\boldsymbol{e}$ is the column vector of ones and $\boldsymbol{c}^T = (c_1, c_2, \ldots, c_m)$. We show that Kemeny and Snell's fundamental matrix, Z, can be expressed in terms of H.

Some simple results:

(1) $\boldsymbol{c}^T H = \boldsymbol{\pi}^T$ implying $\pi_j = \sum_{i=1}^{m} c_i h_{ij}$.
(2) $H\boldsymbol{e} = \boldsymbol{e}/m$ implying $h_{i\bullet} \equiv \sum_{j=1}^{m} h_{ij} = 1/m$.
(3) $\boldsymbol{c}^T H\boldsymbol{e} = 1$, with $\boldsymbol{c}^T\boldsymbol{e} = \sum_{i=1}^{m} c_i = m$.
(4) $h_{\bullet j} \equiv \sum_{i=1}^{m} h_{ij}$ yielding $h_{\bullet j} = 1 - (m-1)\pi_j$.
(5) Kemeny and Snell's fundamental matrix $Z = [I - P + \Pi]^{-1} = H + \Pi - \Pi H$.
(6) The mean first passage times,

$$m_{ij} = \begin{cases} \dfrac{1}{\pi_j} = \dfrac{1}{\sum_{i=1}^{m} c_i h_{ij}}, & i = j, \\ \dfrac{h_{jj} - h_{ij}}{\pi_j} = \dfrac{h_{jj} - h_{ij}}{\sum_{i=1}^{m} c_i h_{ij}}, & i \neq j. \end{cases}$$

(7) Kemeny's constant, $K = 1 - (1/m) + \mathrm{tr}(H)$.

Hunter [18] **Generalized inverses of Markovian kernels in terms of properties of the Markov chain**, ***Linear Algebra & its Applications***, **447, 38–55.**

Generalized matrix inverses (g-inverses) of $I - P$ are typically used to solve systems of linear equations including a variety of the properties of the M. C. In particular, the $\{\pi_j\}$ and the $\{m_{ij}\}$ can be found in terms of g-inverses, either in matrix form or in terms of the elements of the g-inverse. What was not known was that the elements of every g-inverse of $I - P$ can be expressed in terms of the stationary probabilities $\{\pi_j\}$ and the mean first passage times $\{m_{ij}\}$ of the associated M. C. The key thrust to this paper is to first identify the parameters that characterize different sub-families of g-inverses of $I - P$. Then to assign to each sub-family, thus characterized in terms of their parameters $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$ and γ, explicit expressions for the elements of the g-inverses in terms of the $\{\pi_j\}$ and the $\{m_{ij}\}$.

The main result of the paper is the following.

Let $G = G(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma)$ be any g-inverse of $I - P$ then the elements of $G = [g_{ij}]$ can be expressed in terms of the parameters $\{\alpha_j\}$, $\{\beta_j\}$, γ, the stationary probabilities $\{\pi_j\}$, and the mean first passage times $\{m_{ij}\}$, of the Markov chain, with $\delta_j \equiv \sum_{k\neq j}^{m} \beta_k m_{kj}$ as

$$g_{ij} = \begin{cases} \left(1 + \gamma + \delta_j - m_{ij} + \sum_{k\neq i} \pi_k \alpha_k m_{ik} - \sum_{k=1}^{m} \pi_k \alpha_k \delta_k\right) \pi_j, & i \neq j, \\ \left(1 + \gamma + \delta_j + \sum_{k\neq j} \pi_k \alpha_k m_{jk} - \sum_{k=1}^{m} \pi_k \alpha_k \delta_k\right) \pi_j, & i = j. \end{cases}$$

Further, for all g-inverses $G = G(\boldsymbol{\alpha}, \boldsymbol{\beta}, \gamma)$ of $I - P$, with $\delta_j \equiv \sum_{k\neq j}^{m} \beta_k m_{kj}$ then

$$\sum_{k=1}^{m} \pi_k \delta_k = K - 1,$$

where $K = \sum_{j=1}^{m} \pi_j m_{ij}$, a constant (Kemeny's constant) for all $i = 1, 2, \ldots, m$.

The author has also written other papers where generalized matrix inverses have been utilized, and these are included in the references (in particular [15, 20–22]), but in these papers the g-inverses are used primarily as well-known tools for deriving relevant properties of Markov chains.

2.4 Professor Hunter's Links with Prof. Rao

Professor Hunter has had the opportunity to meet personally with Prof. Rao and was involved in inviting and hosting Prof. Rao during his visit to New Zealand in 2005. We were delighted to have him participate in an extensive programme, especially onerous for someone of his age.

The organization of his visit required considerable coordination and a number of individuals and organizations were involved. He started his visit to the South Island with a succession of seminars, firstly at the University of Otago, Dunedin (14 March), then the University of Canterbury, Christchurch (15 March), across to the North Island with seminars at Victoria University of Wellington (17 March) and Massey University, Palmerston North (18 March). Moving up the island, Prof. Nye John organized a CR Rao Workshop on Data Scrutiny and Data Mining at the Waikato Centre of Applied Statistics at University of Waikato, Hamilton (22 March) followed by a seminar at the University of Auckland (23 March). The final activity was a Public Lecture as the Keynote Speaker at the opening of the International Workshop on Matrices and Statistics (IWMS 2005) at Massey University, Albany, Auckland (29 March).

The NZ Statistical Association hosted him as its Visiting Lecturer. Professor Hunter organized him being appointed as a Distinguished Visitor to Massey University as well as hosting him at IWMS 2005 and Dr. Simo Puntanen organized his appointment as the Nokia Lecturer at IWMS 2005.

The following year, Prof. Hunter was invited by Prof. Rao to be a Keynote speaker at a two-day "Workshop on Matrix Theory and in Physical, Biological and Social Sciences" at Penn State University, University Park, PA, USA, July 27–28, 2006.

The invited speakers were G. H. Golub (Stanford Univ), T. Kailath (Stanford Univ), J. Hunter (Massey Univ) and M. B. Rao (Univ Cincinnati). During this time, Prof. Hunter also had the opportunity to be hosted by Prof. Rao and his wife, Bhargavi, at their home at University Park.

2.5 Final Comments

While the significance of Prof. Rao's contributions to a number of statistical fields is well known, the author is proud to have been able to apply his outstanding contributions on generalized matrix inverses and their use in solving systems of linear equations to a large number of problems in applied probability. He also values the opportunity to have met him and wishes to convey to him his congratulations and best wishes for attaining such a milestone birthday.

References

1. Decell HP, Odell PL (1967) On the fixed point probability vector of regular or ergodic transition matrices. J Am Stat Assoc 62:600
2. Hunter JJ (1969) On the moments of Markov Renewal Processes. Adv Appl Probab 1(2):188–210
3. Hunter JJ (1982) Generalized inverses and their application to applied probability problems. Linear Algeb Appl 45:157–198
4. Hunter JJ (1983) Mathematical techniques of applied probability, vol 1. Discrete Time Models: Basic Theory. Academic Press, New York
5. Hunter JJ (1983) Mathematical techniques of applied probability, vol 2. Discrete Time Models: Techniques and Applications. Academic Press, New York
6. Hunter JJ (1986) Stationary distributions of perturbed Markov chains. Linear Algebra Appl 82:201–214
7. Hunter JJ (1988) Characterizations of generalized inverses associated with Markovian kernels. Linear Algebra Appl 102:121–142
8. Hunter JJ (1990) Parametric forms for generalized inverses of Markovian kernels & their applications. Linear Algebra Appl 127:71–84
9. Hunter JJ (1991) The computation of stationary distributions of Markov chains through perturbations. J Appl Math Stoch Anal 4(1):29–46
10. Hunter JJ (1992) Stationary distributions and mean first passage times in Markov chains using generalised inverses. Asia-Pac J Oper Res 2:145–153
11. Hunter JJ (2005) Stationary distributions and mean first passage times of perturbed Markov chains. Linear Algebra Appl 410:217–243
12. Hunter JJ (2006) Mixing times with applications to perturbed Markov chains. Linear Algebra Appl 417:108–123
13. Hunter JJ (2007) Simple procedures for finding mean first passage times in Markov chains. Asia-Pac J Oper Res 24(6):813–829
14. Hunter JJ (2008) Variance of first passage times and applications to mixing times in markov chains. Linear Algebra Appl 429:1135–1162
15. Hunter JJ (2009) Coupling and mixing times in a Markov chain. Linear Algebra Appl 430:2607–2621

16. Hunter JJ (2010) Some stochastic properties of semi-magic and magic Markov chains. Linear Algebra Appl 433:893–907
17. Hunter JJ (2012) Markov chain properties in terms of column sums of the transition matrix. Acta et Commentationes Universitatis Tartuensis de Mathematica 16(1):33–51
18. Hunter JJ (2014) Generalized inverses of Markovian kernels in terms of properties of the Markov chain. Linear Algebra Appl 447:38–55
19. Hunter JJ (2014) Mathematical techniques of applied probability, 1st edn. Discrete Time Models: Basic Theory. Academic Press, Elsevier Science, eBook
20. Hunter JJ (2014) The role of Kemeny's constant in properties of Markov chains. Commun Stat—Theory Methods 43:1–13
21. Hunter JJ (2016) The computation of the key properties of Markov chains via perturbations. Linear Algebra Appl 511:176–202
22. Hunter JJ (2018) The computation of the mean first passage times for Markov chains. Linear Algebra Appl 549:100–122
23. Kemeny JG, Snell JL (1960) Finite Markov Chains. Van Nostrand, Princeton, NJ
24. Meyer CD Jr (1975) The role of the group generalized inverse in the theory of finite Markov chains. SIAM Rev 17(3):443–464
25. Moore EH (1920) On the reciprocal of the general algebraic matrices. Bull Am Math Soc 26:394–395
26. Moore EH (1935) General analysis. American Philosophical Society, Philadelphia
27. Paige CC, Styan GPH, Wachter PG (1975) Computation of the stationary distribution of a Markov chain. J Stat Comput Simul 4(173–186)
28. Penrose R (1955) A generalized inverses for matrices. Math Proc Camb Phil Soc 51(3):406–413
29. Rao CR (1955) Analysis of dispersion for multiply classified data with unequal numbers in cells. Sankhya: Indian J Stat 15(3):253–280
30. Rao CR (1962) A note on a generalized inverse of a matrix with applications to problems in mathematical statistics. J R Stat Soc Ser B (Methodological) 24(1):152–158
31. Rao CR (1965) Linear statistical inference and its applications. Wiley, New York
32. Rao CR (1966) Generalized inverse for matrices and its applications in mathematical statistics. Research Papers in Statistics, Festschrift for J Neyman, Wiley, New York
33. Rao CR (1967) Calculus of generalized inverses of matrices, Part 1: general theory. Sankhya: Indian J Stat Ser A 29(3):317–342
34. Rising W (1991) Applications of generalized inverses to Markov chains. Adv Appl Probab 23(2):293–302

Chapter 3
Upper Bounds for the Euclidean Distances Between the BLUEs Under the Partitioned Linear Fixed Model and the Corresponding Mixed Model



Stephen J. Haslett, Jarkko Isotalo, Augustyn Markiewicz, and Simo Puntanen

2010 Mathematics Subject Classification 62J05 · 62J10

3.1 Introduction

We will consider the partitioned linear *fixed* effects model

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}\,, \quad \mathrm{cov}(\boldsymbol{\varepsilon}) = \mathbf{V},\ \mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}\,,$$

or shortly denoted as

$$\mathscr{F} = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2,\ \mathbf{V}\} = \{\mathbf{y},\ \mathbf{X}\boldsymbol{\beta},\ \mathbf{V}\}\,,$$

S. J. Haslett
Research School of Finance, Actuarial Studies and Statistics, The Australian National University, Canberra, Australia
e-mail: s.j.haslett@massey.ac.nz; stephen.haslett@anu.edu.au

National Institute for Applied Statistics Research Australia (NIASRA), University of Wollongong, Wollongong, NSW 2522, Australia

Centre for Public Health Research, Massey University, Wellington, New Zealand

School of Fundamental Sciences, Massey University, Palmerston North, New Zealand

J. Isotalo · S. Puntanen (✉)
Faculty of Information Technology and Communication Sciences, Tampere University, Tampere 33014, Finland
e-mail: simo.puntanen@tuni.fi

J. Isotalo
e-mail: jarkko.isotalo@tuni.fi

A. Markiewicz
Department of Mathematical and Statistical Methods, Poznań University of Life Sciences, Wojska Polskiego 28, 60637 Poznań, Poland
e-mail: amark@up.poznan.pl

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_3

where $\mathbf{X}_1$ and $\mathbf{X}_2$ are known matrices of order $n \times p_i$, respectively, $p_1 + p_2 = p$, $\boldsymbol{\beta}_1 \in \mathbb{R}^{p_1}$, and $\boldsymbol{\beta}_2 \in \mathbb{R}^{p_2}$ are vectors of unknown (but fixed) effects and $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2')'$; here $'$ stands for the transpose. The covariance matrix $\mathbf{V}$ of the random error vector $\boldsymbol{\varepsilon}$ is assumed to be known.

Consider then the linear *mixed* model $\mathscr{M}$ which is obtained from $\mathscr{F}$ by replacing the fixed vector $\boldsymbol{\beta}_2$ with the random effect vector $\mathbf{u}$:

$$\mathscr{M}: \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\mathbf{u} + \boldsymbol{\varepsilon}, \quad \mathrm{cov}(\boldsymbol{\varepsilon}) = \mathbf{V},\ \mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}\,,$$

where $\mathbf{X}_1$, $\mathbf{X}_2$, $\boldsymbol{\beta}_1$, and $\boldsymbol{\varepsilon}$ are as in $\mathscr{F}$, $\mathbf{u}$ is an unobservable vector of random effects with $\mathrm{E}(\mathbf{u}) = \mathbf{0}$, $\mathrm{cov}(\mathbf{u}) = \mathbf{D}$, $\mathrm{cov}(\boldsymbol{\varepsilon}, \mathbf{u}) = \mathbf{0}$; $\mathbf{V}$ and $\mathbf{D}$ are assumed to be known. Notice that under $\mathscr{F}$ we have $\mathrm{cov}(\mathbf{y}) = \mathbf{V}$ but under $\mathscr{M}$,

$$\mathrm{cov}(\mathbf{y}) = \mathrm{cov}(\mathbf{X}_2\mathbf{u} + \boldsymbol{\varepsilon}) = \mathbf{X}_2\mathbf{D}\mathbf{X}_2' + \mathbf{V} = \boldsymbol{\Sigma}\,.$$

As for notation, $\mathrm{r}(\mathbf{A})$, $\mathbf{A}^-$, $\mathbf{A}^+$, $\mathscr{C}(\mathbf{A})$, and $\mathscr{C}(\mathbf{A})^\perp$, denote, respectively, the rank, a generalized inverse, the (unique) Moore–Penrose inverse, the column space, and the orthogonal complement of $\mathscr{C}(\mathbf{A})$. By $\mathbf{A}^\perp$ we denote any matrix satisfying $\mathscr{C}(\mathbf{A}^\perp) = \mathscr{C}(\mathbf{A})^\perp$. Furthermore, we will write $\mathbf{P}_\mathbf{A} = \mathbf{A}\mathbf{A}^+ = \mathbf{A}(\mathbf{A}'\mathbf{A})^-\mathbf{A}'$ to denote the orthogonal projector onto $\mathscr{C}(\mathbf{A})$. The orthogonal projector onto $\mathscr{C}(\mathbf{A})^\perp$ is denoted as $\mathbf{Q}_\mathbf{A} = \mathbf{I}_a - \mathbf{P}_\mathbf{A}$, where $\mathbf{I}_a$ refers to the $a \times a$ identity matrix and a is the number of rows of $\mathbf{A}$. We use the short notations

$$\mathbf{M} = \mathbf{I}_n - \mathbf{P}_\mathbf{X} \in \{\mathbf{X}^\perp\}\,, \quad \mathbf{M}_i = \mathbf{I}_n - \mathbf{P}_{\mathbf{X}_i} \in \{\mathbf{X}_i^\perp\}\,,\ i = 1,\ 2\,.$$

If we are interested in comparing the BLUEs of $\mathbf{K}_1\boldsymbol{\beta}_1$ under $\mathscr{F}$ and $\mathscr{M}$, then $\mathbf{K}_1\boldsymbol{\beta}_1$ has to be estimable in both models. It is well known that$\mathbf{K}_1\boldsymbol{\beta}_1$ is estimable under $\mathscr{F}$ if and only if $\mathscr{C}(\mathbf{K}_1') \subseteq \mathscr{C}(\mathbf{X}_1'\mathbf{M}_2)$, i.e., $\mathbf{K}_1 = \mathbf{L}\mathbf{M}_2\mathbf{X}_1$ for some matrix $\mathbf{L}$. In other words, the benefit of concentrating on estimating $\boldsymbol{\theta}_1 = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1$ is that the properties obtained are valid for all parametric functions of the type $\mathbf{K}_1\boldsymbol{\beta}_1$ that are estimable under the model $\mathscr{F}$.

Putting $\mathbf{K}_1 = \mathbf{X}_1$ we immediately observe that $\boldsymbol{\mu}_1 = \mathbf{X}_1\boldsymbol{\beta}_1$ is estimable if and only if $\mathscr{C}(\mathbf{X}_1') \subseteq \mathscr{C}(\mathbf{X}_1'\mathbf{M}_2)$, i.e., $\mathscr{C}(\mathbf{X}_1') = \mathscr{C}(\mathbf{X}_1'\mathbf{M}_2)$, so that

$$\boldsymbol{\mu}_1 = \mathbf{X}_1\boldsymbol{\beta}_1 \text{ is estimable under } \mathscr{F} \iff \mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{X}_2) = \{\mathbf{0}\}\,.$$

For Lemma 1, characterizing the BLUE, see, e.g., Rao [17, p. 282].

Lemma 1 *Consider the models $\mathscr{F}$ and $\mathscr{M}$ and denote $\boldsymbol{\Sigma} = \boldsymbol{X}_2\boldsymbol{D}\boldsymbol{X}_2' + \boldsymbol{V}$. Then the following statements hold:*

1. **Ay** *is the* BLUE *for estimable $\mathbf{K}\boldsymbol{\beta} = (\mathbf{K}_1 : \mathbf{K}_2)\boldsymbol{\beta}$ in $\mathscr{F}$ if and only if*

$$\mathbf{A}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V}\mathbf{M}) = (\mathbf{K}_1 : \mathbf{K}_2 : \mathbf{0})\,,\ \textit{i.e.,}\ \mathbf{A} \in \mathscr{P}_{\mathbf{K}\boldsymbol{\beta}\,|\,\mathscr{F}}\,.$$

2. ***By*** *is the* BLUE *for estimable $\boldsymbol{K}_1\boldsymbol{\beta}_1$ under $\mathscr{M}$ if and only if*

$$\boldsymbol{B}(\boldsymbol{X}_1 : \boldsymbol{\Sigma M}_1) = (\boldsymbol{K}_1 : \boldsymbol{0}) \,, \;\; \textit{i.e.,} \; \boldsymbol{B} \in \mathscr{P}_{\boldsymbol{K}_1\boldsymbol{\beta}_1 \,|\, \mathscr{M}} \,.$$

Notice that the notation $\mathscr{P}_{\mathbf{K}\boldsymbol{\beta}\,|\,\mathscr{F}}$ refers to the set of *all solutions* (for $\mathbf{A}$) in equation $\mathbf{A}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{VM}) = (\mathbf{K}_1 : \mathbf{K}_2 : \mathbf{0})$, i.e., it refers to set of matrices $\mathbf{A}$ such that $\mathbf{Ay}$ is the BLUE for estimable $\mathbf{K}\boldsymbol{\beta} = (\mathbf{K}_1 : \mathbf{K}_2)\boldsymbol{\beta}$ in model $\mathscr{F}$.

It is well known, see, e.g., Rao [17], that the matrix

$$\mathbf{G}_{\boldsymbol{\mu}\,|\,\mathscr{F}} = \mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+}, \tag{3.1}$$

where

$$\mathbf{W} = \mathbf{X}_1\mathbf{X}_1' + \mathbf{X}_2\mathbf{X}_2' + \mathbf{V} = \mathbf{X}\mathbf{X}' + \mathbf{V}, \tag{3.2}$$

is one solution for $\mathbf{A}$ in equation $\mathbf{A}(\mathbf{X} : \mathbf{VM}) = (\mathbf{X} : \mathbf{0})$, and thus

$$\mathbf{G}_{\boldsymbol{\mu}\,|\,\mathscr{F}}\mathbf{y} = \mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{F}) \,, \quad \mathbf{G}_{\boldsymbol{\mu}\,|\,\mathscr{F}} \in \mathscr{P}_{\boldsymbol{\mu}\,|\,\mathscr{F}} \,, \;\; \text{where } \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta} \,.$$

It is not necessary to choose $\mathbf{W}$ as in (3.2). For example, $\mathbf{W}$ could be replaced with any matrix from the class $\mathscr{W}(\mathbf{V})$, where

$$\mathscr{W}(\mathbf{V}) = \{\mathbf{W} \in \mathbb{R}^{n\times n} : \mathbf{W} = \mathbf{V} + \mathbf{XUX}', \; \mathscr{C}(\mathbf{W}) = \mathscr{C}(\mathbf{X} : \mathbf{V})\} \,. \tag{3.3}$$

In (3.3) $\mathbf{U}$ can be any $p \times p$ matrix as long as $\mathscr{C}(\mathbf{W}) = \mathscr{C}(\mathbf{X} : \mathbf{V})$ is satisfied; see, e.g., Puntanen et al. [16, Sect. 12.3]. For example, $\mathbf{U}$ could always be chosen as $\mathbf{I}_p$ and in particular, when $\mathscr{C}(\mathbf{X}) \subseteq \mathscr{C}(\mathbf{V})$, $\mathbf{U}$ could be chosen as $\mathbf{0}$. It is noteworthy that the matrix $\mathbf{G}_{\boldsymbol{\mu}\,|\,\mathscr{F}}$ in (3.1) is unique with respect to choices of generalized inverses denoted as "$^-$". This is based on the fact that for nonnull matrices $\mathbf{A}$ and $\mathbf{C}$, the product $\mathbf{AB}^{-}\mathbf{C}$ is invariant for the choice of $\mathbf{B}^{-}$ if and only if

$$\mathscr{C}(\mathbf{C}) \subseteq \mathscr{C}(\mathbf{B}) \quad \text{and} \quad \mathscr{C}(\mathbf{A}') \subseteq \mathscr{C}(\mathbf{B}') \,;$$

see, e.g., Rao and Mitra [18, Lemma 2.2.4]. This invariance property is frequently used in this paper. Notice also that for *all* generalized inverses involved, we have

$$\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{-} \in \mathscr{P}_{\boldsymbol{\mu}\,|\,\mathscr{F}} \,.$$

Below are some solutions providing BLUEs for $\boldsymbol{\theta}_1 = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1$ under $\mathscr{F}$ and $\mathscr{M}$, see, e.g., Puntanen et al. [16, Chap. 10]:

$$\begin{aligned}
\mathbf{G}_{\boldsymbol{\theta}_1\,|\,\mathscr{F}} &= \mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-}\mathbf{X}_1'\mathbf{M}_2 \qquad \in \mathscr{P}_{\boldsymbol{\theta}_1\,|\,\mathscr{F}} \,, \\
\mathbf{G}_{\boldsymbol{\theta}_1\,|\,\mathscr{M}} &= \mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{W}_m^{-}\mathbf{X}_1)^{-}\mathbf{X}_1'\mathbf{W}_m^{+} \qquad \in \mathscr{P}_{\boldsymbol{\theta}_1\,|\,\mathscr{M}} \,,
\end{aligned}$$

i.e., $\mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y}$ and $\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{y}$ are BLUEs for $\boldsymbol{\theta}_1$ under $\mathscr{F}$ and $\mathscr{M}$, respectively; above

$$\mathbf{W}_m = \mathbf{X}_1\mathbf{X}_1' + \boldsymbol{\Sigma} = \mathbf{X}_1\mathbf{X}_1' + \mathbf{X}_2\mathbf{D}\mathbf{X}_2' + \mathbf{V},$$

and the matrix $\mathbf{M}_2$ is defined as

$$\mathbf{M}_2 = \mathbf{M}_2(\mathbf{M}_2\mathbf{W}\mathbf{M}_2)^+\mathbf{M}_2\,.$$

Moreover, see, e.g., Puntanen et al. [16, Chap. 15], $\mathbf{M}_2$ satisfies

$$\mathbf{M}_2 = \mathbf{M}_2(\mathbf{M}_2\mathbf{W}\mathbf{M}_2)^+\mathbf{M}_2 = \mathbf{M}_2(\mathbf{M}_2\mathbf{W}\mathbf{M}_2)^+ = (\mathbf{M}_2\mathbf{W}\mathbf{M}_2)^+.$$

Trivially $\mathbf{V}\mathbf{M} = \mathbf{\Sigma}\mathbf{M}$ and denoting $\mathbf{W}_1 = \mathbf{X}_1\mathbf{X}_1' + \mathbf{V}$, we have

$$\mathbf{M}_2\mathbf{W} = \mathbf{M}_2\mathbf{W}_1 = \mathbf{M}_2\mathbf{W}_m\,, \quad \mathbf{M}_1\mathbf{W}_m = \mathbf{M}_1\mathbf{\Sigma}\,.$$

The solutions to equations in Lemma 1 dealing with $\mathscr{F}$ are unique if and only if $\mathscr{C}(\mathbf{W}) = \mathbb{R}^n$ while those dealing with $\mathscr{M}$ are unique if and only if $\mathscr{C}(\mathbf{W}_m) = \mathbb{R}^n$. The *general* solution for $\mathbf{A}$ in

$$\mathbf{A}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V}\mathbf{M}) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0} : \mathbf{0})$$

can be expressed, e.g., as

$$\mathbf{A}_0 = \mathbf{G}_{\boldsymbol{\theta}_1|\mathscr{F}} + \mathbf{E}\mathbf{Q}_\mathbf{W} = \mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^-\mathbf{X}_1'\mathbf{M}_2 + \mathbf{E}\mathbf{Q}_\mathbf{W}\,,$$

where $\mathbf{E} \in \mathbb{R}^{n\times n}$ is free to vary. By the *consistency* of the model $\mathscr{F}$ it is meant that $\mathbf{y}$ lies in $\mathscr{C}(\mathbf{W})$ with probability 1. Thus under the consistent model $\mathscr{F}$ the vector $\mathbf{A}_0\mathbf{y}$ itself is unique once $\mathbf{y}$ has been observed. The consistency in $\mathscr{M}$ means that $\mathbf{y}$ belongs to $\mathscr{C}(\mathbf{W}_m)$. Notice that

$$\mathscr{C}(\mathbf{W}_m) = \mathscr{C}(\mathbf{X}_1 : \mathbf{X}_2\mathbf{D} : \mathbf{V}) \subseteq \mathscr{C}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V}) = \mathscr{C}(\mathbf{W})\,,$$

with equality holding if and only if $\mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{W}_m) = \mathscr{C}(\mathbf{X}_1 : \mathbf{\Sigma}\mathbf{M}_1)$.

In the consistent linear model $\mathscr{F}$, the estimators $\mathbf{A}\mathbf{y}$ and $\mathbf{B}\mathbf{y}$ are said to be equal (with probability 1) if

$$\mathbf{A}\mathbf{y} = \mathbf{B}\mathbf{y} \text{ for all } \mathbf{y} \in \mathscr{C}(\mathbf{W}) = \mathscr{C}(\mathbf{X} : \mathbf{V}) = \mathscr{C}(\mathbf{X} : \mathbf{V}\mathbf{M})\,, \tag{3.5}$$

while similarly under the model $\mathscr{M}$,

$$\mathbf{A}\mathbf{y} = \mathbf{B}\mathbf{y} \text{ for all } \mathbf{y} \in \mathscr{C}(\mathbf{W}_m) = \mathscr{C}(\mathbf{X}_1 : \mathbf{\Sigma}) = \mathscr{C}(\mathbf{X}_1 : \mathbf{\Sigma}\mathbf{M}_1)\,.$$

In (3.5) we are dealing with the "statistical" equality of the estimators $\mathbf{A}\mathbf{y}$ and $\mathbf{B}\mathbf{y}$. In (3.5) $\mathbf{y}$ refers to a vector in $\mathbb{R}^n$, while in notation $\mathrm{cov}(\mathbf{A}\mathbf{y})$ we understand $\mathbf{y}$ as a *random* vector. For a review of the equality of two linear estimators, see, e.g., Groß and Trenkler [4].

Isotalo et al. [10] found conditions under which an arbitrary representation of the BLUE of $\boldsymbol{\theta}_1 = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1$ under the fixed model $\mathscr{F}$ remains the BLUE for $\boldsymbol{\theta}_1$ under the mixed model $\mathscr{M}$. This kind of property can be denoted shortly as

$$\{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})\} \subseteq \{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})\}\,,\ \text{i.e.,}\ \mathscr{P}_{\theta_1|\mathscr{F}} \subseteq \mathscr{P}_{\theta_1|\mathscr{M}}\,,$$

where the sets $\mathscr{P}_{\theta_1|\mathscr{F}}$ and $\mathscr{P}_{\theta_1|\mathscr{M}}$ are defined as in Lemma 1:

$$\begin{aligned}\mathbf{A} \in \mathscr{P}_{\theta_1|\mathscr{F}} &\iff \mathbf{A}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V}\mathbf{M}) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0} : \mathbf{0})\,,\\ \mathbf{B} \in \mathscr{P}_{\theta_1|\mathscr{M}} &\iff \mathbf{B}(\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0})\,.\end{aligned}$$

Recall that $\boldsymbol{\theta}_1$ generates all estimable functions in both models in the sense that each estimable function in both models can be represented as $\mathbf{p}'\boldsymbol{\theta}_1$, where $\mathbf{p}$ is an arbitrary vector of proper dimension. Isotalo et al. [10, Sect. 2] proved the following result:

Lemma 2 *The following statements hold:*

1. *An arbitrary* BLUE *for* $\boldsymbol{\theta}_1 = \boldsymbol{M}_2\boldsymbol{X}_1\boldsymbol{\beta}_1$ *under* $\mathscr{F}$ *provides also the* BLUE *for* $\boldsymbol{\theta}_1$ *under the mixed model* $\mathscr{M}$*, i.e.,*

$$\{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})\} \subseteq \{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})\}\,,$$

   i.e., $\mathscr{P}_{\theta_1|\mathscr{F}} \subseteq \mathscr{P}_{\theta_1|\mathscr{M}}$*, holds if and only if*

$$\mathscr{C}(\boldsymbol{\Sigma}\boldsymbol{M}_1) \subseteq \mathscr{C}(\boldsymbol{X}_2 : \boldsymbol{\Sigma}\boldsymbol{M})\,. \tag{3.7}$$

2. *The reverse relation* $\{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})\} \subseteq \{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})\}$*, i.e.,* $\mathscr{P}_{\theta_1|\mathscr{M}} \subseteq \mathscr{P}_{\theta_1|\mathscr{F}}$*, holds if and only if*

$$\mathscr{C}(\boldsymbol{X}_2) \subseteq \mathscr{C}(\boldsymbol{X}_{1\cap 2} : \boldsymbol{\Sigma}\boldsymbol{M}_1)\,, \tag{3.8}$$

   where the matrix $\boldsymbol{X}_{1\cap 2}$ *has property* $\mathscr{C}(\boldsymbol{X}_{1\cap 2}) = \mathscr{C}(\boldsymbol{X}_1) \cap \mathscr{C}(\boldsymbol{X}_2)$.

Actually, the matrix $\mathbf{X}_{1\cap 2}$ in (b) was erroneously missing in Isotalo et al. [10]. The inclusion (3.7) is obviously equivalent to $\mathscr{C}(\mathbf{X}_{1\cap 2} : \boldsymbol{\Sigma}\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{X}_2 : \boldsymbol{\Sigma}\mathbf{M})$ and (3.8) is equivalent to $\mathscr{C}(\mathbf{X}_2 : \boldsymbol{\Sigma}\mathbf{M}) \subseteq \mathscr{C}(\mathbf{X}_{1\cap 2} : \boldsymbol{\Sigma}\mathbf{M}_1)$ and thereby $\mathscr{P}_{\theta_1|\mathscr{M}} = \mathscr{P}_{\theta_1|\mathscr{F}}$ holds if and only if

$$\mathscr{C}(\mathbf{X}_{1\cap 2} : \boldsymbol{\Sigma}\mathbf{M}_1) = \mathscr{C}(\mathbf{X}_2 : \boldsymbol{\Sigma}\mathbf{M})\,.$$

For further equalities arising from models $\mathscr{F}$ and $\mathscr{M}$ see Haslett et al. [7].

The following lemma appears useful for our considerations.

Lemma 3 *Using the earlier notation, the following statements hold:*

1. $\boldsymbol{M} = \boldsymbol{I}_n - \boldsymbol{P}_{(\boldsymbol{X}_1:\boldsymbol{X}_2)} = \boldsymbol{I}_n - (\boldsymbol{P}_{\boldsymbol{X}_2} + \boldsymbol{P}_{\boldsymbol{M}_2\boldsymbol{X}_1}) = \boldsymbol{M}_2\boldsymbol{Q}_{\boldsymbol{M}_2\boldsymbol{X}_1} = \boldsymbol{Q}_{\boldsymbol{M}_2\boldsymbol{X}_1}\boldsymbol{M}_2$.
2. $\mathrm{r}(\boldsymbol{M}_2\boldsymbol{X}_1) = \mathrm{r}(\boldsymbol{X}_1) - \dim \mathscr{C}(\boldsymbol{X}_1) \cap \mathscr{C}(\boldsymbol{X}_2)$.
3. $\mathscr{C}(\boldsymbol{W}_m^{+}\boldsymbol{X}_1)^{\perp} = \mathscr{C}(\boldsymbol{W}_m\boldsymbol{M}_1 : \boldsymbol{Q}_{\boldsymbol{W}_m}) = \mathscr{C}(\boldsymbol{\Sigma}\boldsymbol{M}_1 : \boldsymbol{Q}_{\boldsymbol{W}_m})$.
4. $\mathscr{C}(\boldsymbol{X}_2 : \boldsymbol{\Sigma}\boldsymbol{M}) = \mathscr{C}(\boldsymbol{M}_2\boldsymbol{X}_1 : \boldsymbol{Q}_{\boldsymbol{W}})^{\perp}$.
5. $\mathscr{C}[\boldsymbol{A}(\boldsymbol{A}'\boldsymbol{B}^{\perp})^{\perp}] = \mathscr{C}(\boldsymbol{A}) \cap \mathscr{C}(\boldsymbol{B})$.
6. *For* $\boldsymbol{W} \in \mathscr{W}(\boldsymbol{V})$*, we have*

$$\boldsymbol{G}_{\boldsymbol{\mu}\,|\mathscr{F}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{W}^{-}\boldsymbol{X})^{-}\boldsymbol{X}'\boldsymbol{W}^{+} = \boldsymbol{P}_{\boldsymbol{W}} - \boldsymbol{V}\boldsymbol{M}(\boldsymbol{M}\boldsymbol{V}\boldsymbol{M})^{+}\boldsymbol{M}\,. \tag{3.9}$$

For part (b), see, e.g., Marsaglia and Styan [14, Corollary 6.2]. For (c), see, e.g., Rao [17, Sect. 2]. For (d), see Isotalo et al. [10, Lemma, p. 72], for (e), see Rao and Mitra [18, Compl. 7, p. 118], and for (a) and (f), see Puntanen et al. [16, Theorem 8, Proposition 15.2]. Notice from (3.9) that the actual choice of $\mathbf{W}$ does not matter—as long as $\mathbf{W} \in \mathcal{W}(\mathbf{V})$.

Our main goal in this paper is to introduce, in different situations, upper bounds for the Euclidean norm of the distance between the BLUEs of $\boldsymbol{\theta}_1$ under $\mathscr{F}$ and under $\mathscr{M}$:

$$||\mathbf{G}_{\boldsymbol{\theta}_1\,|\mathscr{F}}\mathbf{y} - \mathbf{G}_{\boldsymbol{\theta}_1\,|\mathscr{M}}\mathbf{y}||_2\,, \tag{3.10}$$

where $||\mathbf{x}||_2 = \sqrt{\mathbf{x}'\mathbf{x}}$, when $\mathbf{x} \in \mathbb{R}^n$. The distance in (3.10) depends on the value of $\mathbf{y}$ and we will consider two cases:

$$\mathbf{y} \in \mathscr{C}(\mathbf{W}) = \mathscr{C}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V})\,, \quad \mathbf{y} \in \mathscr{C}(\mathbf{W}_m) = \mathscr{C}(\mathbf{X}_1 : \mathbf{X}_2\mathbf{D} : \mathbf{V})\,.$$

3.2 Upper Bounds for the BLUEs

Let $\mathrm{ch}_1(\cdot)$ denote the largest eigenvalue of the matrix argument. Now the matrix norm corresponding to the Euclidean vector norm is defined as a spectral norm, see, e.g., Horn and Johnson [9, Sect. 5.6],

$$|||\mathbf{A}||| = \sqrt{\mathrm{ch}_1(\mathbf{A}'\mathbf{A})} = \max_{||\mathbf{x}||=1} ||\mathbf{A}\mathbf{x}||_2$$

or

$$|||\mathbf{A}||| = \max_{||\mathbf{x}||\neq\mathbf{0}} ||\mathbf{A}\mathbf{x}||/||\mathbf{x}||\,.$$

Then it is easy to see that

$$||\mathbf{A}\mathbf{x}||_2 \le |||\mathbf{A}||| \cdot ||\mathbf{x}||_2 \tag{3.11}$$

holds. Notice that for a (nonzero) nonnegative definite matrix $\mathbf{A}$ we have $|||\mathbf{A}^{+}||| = 1/\lambda$, where λ is the smallest nonzero eigenvalue of $\mathbf{A}$.

Theorem 1 *Consider the models $\mathscr{F}$ and $\mathscr{M}$. Then for every $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{a} + \boldsymbol{X}_2\boldsymbol{b} + \boldsymbol{V}\boldsymbol{M}\boldsymbol{c} \in \mathscr{C}(\boldsymbol{W})$ we have*

$$\begin{aligned} ||\mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{y} - \mathscr{G}_{\theta_1|\mathscr{M}}\boldsymbol{y}||_2^2 &\le |||\boldsymbol{A}|||^2\ |||\boldsymbol{B}|||^2\ |||\boldsymbol{C}|||^2\ ||\boldsymbol{b}||_2^2 \\ &= \frac{a\cdot c}{b^2}\,\boldsymbol{b}'\boldsymbol{b} =: \alpha\,, \end{aligned}$$

where $\boldsymbol{A} = \boldsymbol{M}_2\boldsymbol{X}_1$, $\boldsymbol{B} = (\boldsymbol{X}_1'\boldsymbol{W}_m^{+}\boldsymbol{X}_1)^{+}$, $\boldsymbol{C} = \boldsymbol{X}_1'\boldsymbol{W}_m^{+}\boldsymbol{X}_2$, $a = \mathrm{ch}_1(\boldsymbol{A}'\boldsymbol{A})$, $c = \mathrm{ch}_1(\boldsymbol{C}'\boldsymbol{C})$, and b is the smallest nonzero eigenvalue of $\boldsymbol{X}_1'\boldsymbol{W}_m^{+}\boldsymbol{X}_1$.

The upper bound α is zero for all $\boldsymbol{b} \in \mathbb{R}^{p_2}$ if and only if a or c is zero:

1. $a = 0 \iff \mathscr{C}(\boldsymbol{X}_1) \subseteq \mathscr{C}(\boldsymbol{X}_2)$,
2. $c = 0 \iff \mathscr{C}(\boldsymbol{X}_2) \subseteq \mathscr{C}(\boldsymbol{\Sigma M}_1 : \boldsymbol{Q}_{\mathbf{W}_m})$.

Moreover, the following three statements are equivalent:

(i) $\alpha = 0$ *for all* $\boldsymbol{b} \in \mathbb{R}^{p_2}$,
(ii) $\mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{W} = \mathscr{G}_{\theta_1|\mathscr{M}}\boldsymbol{W}$,
(iii) $\mathscr{G}_{\theta_1|\mathscr{M}}\boldsymbol{y} = \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})$, *i.e.,* $\mathscr{G}_{\theta_1|\mathscr{M}} \in \mathscr{P}_{\theta_1|\mathscr{F}}$.

Proof Let $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \mathbf{X}_2\mathbf{b} + \mathbf{VMc} = (\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{VM})\mathbf{t}$ for some $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ so that $\mathbf{t} = (\mathbf{a}', \mathbf{b}', \mathbf{c}')'$. Then

$$\begin{aligned}
\mathbf{d} &= \mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y} - \mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{y} \\
&= (\mathscr{G}_{\theta_1|\mathscr{F}} - \mathscr{G}_{\theta_1|\mathscr{M}})(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{VM})\mathbf{t} \\
&= \big[\mathbf{0} : (\mathscr{G}_{\theta_1|\mathscr{F}} - \mathscr{G}_{\theta_1|\mathscr{M}})\mathbf{X}_2 : \mathbf{0}\big]\mathbf{t} \\
&= \mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{X}_2\mathbf{b} \\
&= \mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1)^+\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_2\mathbf{b}\,.
\end{aligned}$$

Thus for the squared norm we get

$$\begin{aligned}
||\mathbf{d}||_2^2 &= ||\mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1)^+\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_2\mathbf{b}||_2^2 \\
&\le |||\mathbf{M}_2\mathbf{X}_1|||^2\ |||(\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1)^+|||^2\ |||\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_2|||^2\ ||\mathbf{b}||_2^2 \\
&= \frac{a \cdot c}{b^2}\,\mathbf{b}'\mathbf{b} = \alpha\,,
\end{aligned} \tag{3.12}$$

where $\mathbf{A} = \mathbf{M}_2\mathbf{X}_1$, $\mathbf{B} = (\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1)^+$, $\mathbf{C} = \mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_2$, $a = \mathrm{ch}_1(\mathbf{A}'\mathbf{A})$, $c = \mathrm{ch}_1(\mathbf{C}'\mathbf{C})$, and b is the smallest nonzero eigenvalue of $\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1$. The inequality in (3.12) follows from the consistency and multiplicativity of the matrix norm $||\cdot||_2$; see (3.11) and Ben-Israel and Greville [3, pp. 19–20].

The upper bound α is zero for all $\mathbf{b}$ if and only if a or c is zero. Obviously

$$a = 0 \iff \mathbf{M}_2\mathbf{X}_1 = \mathbf{0} \iff \mathscr{C}(\mathbf{X}_1) \subseteq \mathscr{C}(\mathbf{X}_2)\,.$$

The equality $\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_2 = \mathbf{0}$ is equivalent to

$$\mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{W}_m^+\mathbf{X}_1)^\perp = \mathscr{C}(\mathbf{W}_m\mathbf{M}_1 : \mathbf{Q}_{\mathbf{W}_m}) = \mathscr{C}(\mathbf{\Sigma M}_1 : \mathbf{Q}_{\mathbf{W}_m})\,,$$

where we have used part (c) of Lemma 3. Hence

$$c = 0 \iff \mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{\Sigma M}_1 : \mathbf{Q}_{\mathbf{W}_m})\,.$$

The equivalence of (i), (ii), and (iii) is clear as (iii) holds if and only if

$$\mathscr{G}_{\theta_1|\mathscr{M}}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{VM}) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0} : \mathbf{0})\,,$$

which further is equivalent to $\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{X}_2 = \mathbf{0}$. This completes the proof. □

Theorem 2 *Consider the models $\mathscr{F}$ and $\mathscr{M}$. Then for every $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{a} + \boldsymbol{\Sigma}\boldsymbol{M}_1\boldsymbol{b} \in \mathscr{C}(\boldsymbol{W}_m)$ we have*

$$||\mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{y} - \mathscr{G}_{\theta_1|\mathscr{M}}\boldsymbol{y}||_2^2 \leq |||\boldsymbol{A}|||^2\,|||\boldsymbol{B}|||^2\,|||\boldsymbol{C}|||^2\,\boldsymbol{b}'\boldsymbol{M}_1\boldsymbol{b} = \frac{a\cdot c}{b^2}\,\boldsymbol{b}'\boldsymbol{M}_1\boldsymbol{b} =: \gamma\,,$$

where $\boldsymbol{A} = \boldsymbol{M}_2\boldsymbol{X}_1$, $\boldsymbol{B} = (\boldsymbol{X}_1'\boldsymbol{M}_2\boldsymbol{X}_1)^+$, $\boldsymbol{C} = \boldsymbol{X}_1'\boldsymbol{M}_2\boldsymbol{\Sigma}\boldsymbol{M}_1$, and $a = \mathrm{ch}_1(\boldsymbol{A}'\boldsymbol{A})$, $c = \mathrm{ch}_1(\boldsymbol{C}'\boldsymbol{C})$, and b is the smallest nonzero eigenvalue of $\boldsymbol{X}_1'\boldsymbol{M}_2\boldsymbol{X}_1$.

The upper bound γ is zero for all $\boldsymbol{b} \notin \mathscr{C}(\boldsymbol{X}_1)$ if and only if any of the following equivalent conditions holds:

1. $\mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{W}_m = \mathscr{G}_{\theta_1|\mathscr{M}}\boldsymbol{W}_m$,
2. $\boldsymbol{X}_1'\boldsymbol{M}_2\boldsymbol{\Sigma}\boldsymbol{M}_1 = \boldsymbol{0}$,
3. $\mathscr{C}(\boldsymbol{\Sigma}\boldsymbol{M}_1) \subseteq \mathscr{C}(\boldsymbol{X}_2 : \boldsymbol{\Sigma}\boldsymbol{M})$,
4. $\{\mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})\} \subseteq \{\mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})\}$.

Proof If $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}$ for some $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{t} = (\mathbf{a}', \mathbf{b}')'$, then

$$\begin{aligned}\mathbf{d} &= (\mathscr{G}_{\theta_1|\mathscr{F}} - \mathscr{G}_{\theta_1|\mathscr{M}})(\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1)\mathbf{t}\\ &= (\mathbf{0} : \mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{\Sigma}\mathbf{M}_1)\mathbf{t} = \mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}\\ &= \mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^+\mathbf{X}_1'\mathbf{M}_2\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}\,.\end{aligned}$$

Hence

$$\begin{aligned}||\mathbf{d}||_2^2 &= ||\mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y} - \mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{y}||_2^2\\ &\leq |||\mathbf{M}_2\mathbf{X}_1|||^2\,|||(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^+|||^2\,|||\mathbf{X}_1'\mathbf{M}_2\boldsymbol{\Sigma}\mathbf{M}_1|||^2\,\mathbf{b}'\mathbf{M}_1\mathbf{b}\\ &= |||\mathbf{A}|||^2\,|||\mathbf{B}|||^2\,|||\mathbf{C}|||^2\,\mathbf{b}'\mathbf{M}_1\mathbf{b}\\ &= \frac{a\cdot c}{b^2}\,\mathbf{b}'\mathbf{M}_1\mathbf{b} = \gamma\,,\end{aligned}$$

where

$$\mathbf{A} = \mathbf{M}_2\mathbf{X}_1\,,\quad \mathbf{B} = (\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^+,\quad \mathbf{C} = \mathbf{X}_1'\mathbf{M}_2\boldsymbol{\Sigma}\mathbf{M}_1\,,$$

and $a = \mathrm{ch}_1(\mathbf{A}'\mathbf{A})$, $c = \mathrm{ch}_1(\mathbf{C}'\mathbf{C})$, and b is the smallest nonzero eigenvalue of $\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1$.

The upper bound γ is zero for all $\mathbf{b} \notin \mathscr{C}(\mathbf{X}_1)$ if and only if $\mathbf{A} = \mathbf{0}$ or $\mathbf{C} = \mathbf{0}$. We observe that $\mathbf{A} = \mathbf{0} \iff \mathscr{C}(\mathbf{X}_1) \subseteq \mathscr{C}(\mathbf{X}_2)$. Thus it is clear that $\mathbf{A} = \mathbf{0} \implies \mathbf{C} = \mathbf{0}$. Now $\mathbf{C} = \mathbf{0}$ if and only if $\mathscr{C}(\boldsymbol{\Sigma}\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{M}_2\mathbf{X}_1)^\perp$, which, in view of $\mathscr{C}(\boldsymbol{\Sigma}\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{W})$, is equivalent to

$$\mathscr{C}(\boldsymbol{\Sigma}\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{M}_2\mathbf{X}_1)^\perp \cap \mathscr{C}(\mathbf{W}) = \mathscr{C}(\mathbf{M}_2\mathbf{X}_1 : \mathbf{Q}_\mathbf{W})^\perp. \tag{3.13}$$

In light of part (d) of Lemma 3, (3.13) can be written as

$$\mathscr{C}(\mathbf{\Sigma M}_1) \subseteq \mathscr{C}(\mathbf{X}_2 : \mathbf{\Sigma M}) \,. \tag{3.14}$$

Moreover, by part (a) of Lemma 2 the inclusion (3.14) is equivalent to

$$\{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})\} \subseteq \{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})\} \,.$$

So the proof is completed. □

Theorem 3 *Let* $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{a} + \boldsymbol{X}_2\boldsymbol{b} + \boldsymbol{V}\boldsymbol{M}\boldsymbol{c}$ *for some* $\boldsymbol{a}$, $\boldsymbol{b}$, *and* $\boldsymbol{c}$, *and define*

$$\boldsymbol{d} = \mathscr{G}_{\theta_1|\mathscr{F}}\boldsymbol{y} - (\mathscr{G}_{\theta_1|\mathscr{M}} + \boldsymbol{E}\boldsymbol{Q}_{\boldsymbol{W}_m})\boldsymbol{y} \,,$$

where $\boldsymbol{E} \in \mathbb{R}^{n\times n}$ *is an arbitrary matrix.*

1. *The norm* $||\boldsymbol{d}||_2$ *has no upper bound unless*

$$\mathscr{C}(\boldsymbol{X}_2) \subseteq \mathscr{C}(\boldsymbol{W}_m) \,, \quad \textit{i.e.,} \quad \boldsymbol{X}_2 = \boldsymbol{X}_1\boldsymbol{A} + \mathbf{\Sigma}\boldsymbol{M}_1\boldsymbol{B} \tag{3.15}$$

   for some $\boldsymbol{A}$ *and* $\boldsymbol{B}$.
2. *Supposing that* (3.15) *holds, for every* $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{a} + \boldsymbol{X}_2\boldsymbol{b} + \boldsymbol{V}\boldsymbol{M}\boldsymbol{c}$ *we have*

$$||\boldsymbol{d}||_2^2 \le |||\boldsymbol{M}_2\boldsymbol{X}_1\boldsymbol{A}|||^2 \, ||\boldsymbol{b}||_2^2 = a \, ||\boldsymbol{b}||_2^2 =: \delta \,,$$

   where $a = \text{ch}_1(\boldsymbol{A}'\boldsymbol{X}_1'\boldsymbol{M}_2\boldsymbol{X}_1\boldsymbol{A})$. *The upper bound* δ *for* $||\boldsymbol{d}||$ *is zero for all* $\boldsymbol{b} \in \mathbb{R}^{p_2}$ *if and only if*

$$\mathscr{C}(\boldsymbol{X}_2) \subseteq \mathscr{C}(\boldsymbol{X}_{1\cap 2} : \mathbf{\Sigma}\boldsymbol{M}_1) \,,$$

   where $\mathscr{C}(\boldsymbol{X}_{1\cap 2}) = \mathscr{C}(\boldsymbol{X}_1) \cap \mathscr{C}(\boldsymbol{X}_2)$, *i.e.,*

$$\{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})\} \subseteq \{\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})\} \,.$$

Proof Let $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \mathbf{X}_2\mathbf{b} + \mathbf{V}\mathbf{M}\mathbf{c} = (\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V}\mathbf{M})\mathbf{t}$ for some $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ so that $\mathbf{t} = (\mathbf{a}', \mathbf{b}', \mathbf{c}')'$. Then, in view of the triangle inequality,

$$\begin{aligned}
||\mathbf{d}||_2 &= ||\mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y} - (\mathscr{G}_{\theta_1|\mathscr{M}} + \mathbf{E}\mathbf{Q}_{\mathbf{W}_m})\mathbf{y}||_2 \\
&= ||\big[\mathscr{G}_{\theta_1|\mathscr{F}} - (\mathscr{G}_{\theta_1|\mathscr{M}} + \mathbf{E}\mathbf{Q}_{\mathbf{W}_m})\big](\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{V}\mathbf{M})\,\mathbf{t}||_2 \\
&= ||\big[\mathbf{0} : (\mathscr{G}_{\theta_1|\mathscr{M}} + \mathbf{E}\mathbf{Q}_{\mathbf{W}_m})\mathbf{X}_2 : \mathbf{0}\big]\,\mathbf{t}||_2 \\
&= ||(\mathscr{G}_{\theta_1|\mathscr{M}} + \mathbf{E}\mathbf{Q}_{\mathbf{W}_m})\mathbf{X}_2\mathbf{b}||_2 \\
&\le ||\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{X}_2\mathbf{b}||_2 + ||\mathbf{E}\mathbf{Q}_{\mathbf{W}_m}\mathbf{X}_2\mathbf{b}||_2 \\
&\le |||\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{X}_2||| \; ||\mathbf{b}||_2 + |||\mathbf{E}||| \; |||\mathbf{Q}_{\mathbf{W}_m}\mathbf{X}_2||| \; ||\mathbf{b}||_2 \,.
\end{aligned} \tag{3.16}$$

Thus $||\mathbf{d}||_2$ has no upper bound unless $\mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{W}_m)$, i.e.,

$$\mathbf{X}_2 = \mathbf{X}_1\mathbf{A} + \boldsymbol{\Sigma}\mathbf{M}_1\mathbf{B} = \mathbf{X}_1\mathbf{A} + \mathbf{W}_m\mathbf{M}_1\mathbf{B} \tag{3.17}$$

for some $\mathbf{A}$ and $\mathbf{B}$. Substituting (3.17) into (3.16) gives

$$\begin{aligned} ||\mathbf{d}||_2^2 &= ||\mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y} - (\mathscr{G}_{\theta_1|\mathscr{M}} + \mathbf{E}\mathbf{Q}_{\mathbf{W}_m})\mathbf{y}||_2^2 \\ &= ||\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{X}_2\mathbf{b}||_2^2 \\ &= ||\mathbf{M}_2\mathbf{X}_1(\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1)^+\mathbf{X}_1'\mathbf{W}_m^+\mathbf{X}_1\mathbf{A}\mathbf{b}||_2^2 \\ &= ||\mathbf{M}_2\mathbf{X}_1\mathbf{A}\mathbf{b}||_2^2 \\ &\le |||\mathbf{M}_2\mathbf{X}_1\mathbf{A}|||^2\ ||\mathbf{b}||_2^2 = a\ ||\mathbf{b}||_2^2 = \gamma\,, \end{aligned}$$

where $a = \mathrm{ch}_1(\mathbf{A}'\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1\mathbf{A})$. Thus, supposing that (3.17) holds, $||\mathbf{d}||_2^2$ has the upper bound γ, which is zero for all $\mathbf{b}$ if and only if $\mathbf{M}_2\mathbf{X}_1\mathbf{A} = \mathbf{0}$, i.e.,

$$\mathbf{A} = \mathbf{Q}_{\mathbf{X}_1'\mathbf{M}_2}\mathbf{Z} \quad \text{for some } \mathbf{Z}\,. \tag{3.18}$$

Combining (3.17) and (3.18) and noting by part (e) of Lemma 3 that

$$\mathscr{C}(\mathbf{X}_1\mathbf{Q}_{\mathbf{X}_1'\mathbf{M}_2}) = \mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{X}_2)\,,$$

yields

$$\mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{X}_{1\cap 2} : \boldsymbol{\Sigma}\mathbf{M}_1)\,, \quad \text{where } \mathscr{C}(\mathbf{X}_{1\cap 2}) = \mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{X}_2)\,,$$

which by part (b) of Lemma 2 is equivalent to $\mathscr{P}_{\theta_1|\mathscr{M}} \subseteq \mathscr{P}_{\theta_1|\mathscr{F}}$. □

3.3 Upper Bounds Related to OLSE

The ordinary least squares estimators, OLSEs, for $\boldsymbol{\theta}_1 = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1$ under $\mathscr{F}$ and $\mathscr{M}$, respectively, are defined as

$$\begin{aligned} \mathbf{A}\mathbf{y} = \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) &\iff \mathbf{A}(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{M}) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0} : \mathbf{0})\,, \\ \mathbf{B}\mathbf{y} = \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) &\iff \mathbf{B}(\mathbf{X}_1 : \mathbf{M}_1) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0})\,. \end{aligned}$$

Thus, denoting $\mathbf{H}_1 = \mathbf{P}_{\mathbf{X}_1}$, we obviously have

$$\begin{aligned} \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{y} &= \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) = \hat{\boldsymbol{\theta}}_1(\mathscr{F})\,, \\ \mathbf{M}_2\mathbf{H}_1\mathbf{y} &= \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) = \hat{\boldsymbol{\theta}}_1(\mathscr{M})\,. \end{aligned}$$

What is the Euclidean distance between the OLSEs under $\mathscr{M}$ and $\mathscr{F}$? We should note that there is just one single matrix providing OLSE in each model, these matrices

being $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}$ and $\mathbf{M}_2\mathbf{H}_1$. These matrices are unique, just like in the BLUE-case the matrices $\mathscr{G}_{\theta_1|\mathscr{F}}$ and $\mathscr{G}_{\theta_1|\mathscr{M}}$ are unique. Yet, we can try to find a condition under which

$$\text{(a) } \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{W}_m = \mathbf{M}_2\mathbf{H}_1\mathbf{W}_m \quad \text{or} \quad \text{(b) } \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{W} = \mathbf{M}_2\mathbf{H}_1\mathbf{W}.$$

What about the upper bound for

$$||(\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\mathbf{y}||_2 \quad \text{when } \mathbf{y} \in \mathscr{C}(\mathbf{W}_m)\,? \tag{3.21}$$

In addition to the upper bound for (3.21) we will consider upper bounds for

$$||\hat{\boldsymbol{\theta}}_1(\mathscr{F}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{F})||_2 \quad \text{and} \quad ||\hat{\boldsymbol{\theta}}_1(\mathscr{M}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{M})||_2\,,$$

where $\tilde{\boldsymbol{\theta}}_1(\mathscr{M}) = \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})$ and $\tilde{\boldsymbol{\theta}}_1(\mathscr{F}) = \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})$.

Theorem 4 *Consider the models $\mathscr{F}$ and $\mathscr{M}$. Then for every $\mathbf{y} = \boldsymbol{X}_1\boldsymbol{a} + \boldsymbol{\Sigma}\boldsymbol{M}_1\boldsymbol{b} \in \mathscr{C}(\boldsymbol{W}_m)$ we have*

$$\begin{aligned} ||\text{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) - \text{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M})||_2^2 &= ||(\boldsymbol{P}_{\boldsymbol{M}_2\boldsymbol{X}_1} - \boldsymbol{M}_2\boldsymbol{H}_1)\boldsymbol{y}||_2^2 \\ &\le |||\boldsymbol{A} - \boldsymbol{B}|||^2\, ||\boldsymbol{M}_1\boldsymbol{b}||_2^2 \\ &= a\, ||\boldsymbol{M}_1\boldsymbol{b}||_2^2 =: \xi\,, \end{aligned}$$

where $\boldsymbol{A} = \boldsymbol{P}_{\boldsymbol{M}_2\boldsymbol{X}_1}\boldsymbol{\Sigma}\boldsymbol{M}_1$, $\boldsymbol{B} = \boldsymbol{M}_2\boldsymbol{H}_1\boldsymbol{\Sigma}\boldsymbol{M}_1$, and $a = \text{ch}_1[(\boldsymbol{A} - \boldsymbol{B})(\boldsymbol{A} - \boldsymbol{B})']$. The upper bound ξ is zero for all $\boldsymbol{b} \notin \mathscr{C}(\boldsymbol{X}_1)$ if and only if $\boldsymbol{A} = \boldsymbol{B}$, i.e., $\boldsymbol{P}_{\boldsymbol{M}_2\boldsymbol{X}_1}\boldsymbol{W}_m = \boldsymbol{M}_2\boldsymbol{H}_1\boldsymbol{W}_m$.

Proof When $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b} = (\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1)\mathbf{t}$, where $\mathbf{t} = (\mathbf{a}', \mathbf{b}')'$, we have

$$\begin{aligned} \mathbf{d} &= (\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\mathbf{y} \\ &= (\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)(\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1)\mathbf{t} \\ &= [\mathbf{0} : (\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\boldsymbol{\Sigma}\mathbf{M}_1]\mathbf{t} \\ &= (\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}\,, \end{aligned}$$

and hence

$$\begin{aligned} ||\mathbf{d}||_2^2 &= ||(\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\mathbf{y}||_2^2 \\ &\le |||\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}_1 - \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1|||^2\, ||\mathbf{M}_1\mathbf{b}||_2^2 \\ &= |||\mathbf{A} - \mathbf{B}|||^2\, ||\mathbf{M}_1\mathbf{b}||_2^2 \\ &= a\, ||\mathbf{M}_1\mathbf{b}||_2^2 = \xi\,, \end{aligned}$$

where $\mathbf{A} = \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}_1$, $\mathbf{B} = \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1$, and $a = \text{ch}_1[(\mathbf{A} - \mathbf{B})(\mathbf{A} - \mathbf{B})']$. The upper bound ξ is zero for all $\mathbf{b} \notin \mathscr{C}(\mathbf{X}_1)$ if and only if $\mathbf{A} - \mathbf{B} = \mathbf{0}$, i.e., $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{W}_m = \mathbf{M}_2\mathbf{H}_1\mathbf{W}_m$. Thus the proof is completed. □

Remark 1 It is straightforward to confirm that $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{W} = \mathbf{M}_2\mathbf{H}_1\mathbf{W}$ if and only if

$$\mathbf{M}_2\mathbf{H}_1\mathbf{X}_2 = \mathbf{0} \quad \text{and} \quad \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M} = \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}\,.$$

The upper bound corresponding to Theorem 4 for every $\mathbf{y} \in \mathscr{C}(\mathbf{W})$ can be obtained (skipped here).

Remark 2 We further observe that

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{y} = \text{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) = \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})$$

if and only if $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}(\mathbf{X}_1 : \mathbf{X}_2 : \boldsymbol{\Sigma}\mathbf{M}) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0} : \mathbf{0})$, i.e.,

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M} = \mathbf{0}\,.$$

Similarly $\text{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) = \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})$ if and only if $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}_1 = \mathbf{0}$. Thus we can conclude that

$$\hat{\boldsymbol{\theta}}_1(\mathscr{F}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{M}) \implies \hat{\boldsymbol{\theta}}_1(\mathscr{F}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{F})\,.$$

Remark 3 Suppose that $\mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{W}_m)$, so that $\mathscr{C}(\mathbf{W}_m) = \mathscr{C}(\mathbf{W})$. Then $\mathscr{C}(\mathbf{M}_2\mathbf{X}_1) \subseteq \mathscr{C}(\mathbf{W}_m)$, and thereby

$$(\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\mathbf{Q}_{\mathbf{W}_m} = \mathbf{0}\,.$$

Hence, if $\mathscr{C}(\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{W}_m)$, the statement

$$(\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\mathbf{W}_m = \mathbf{0}$$

is actually equivalent to $(\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{M}_2\mathbf{H}_1)\mathbf{y} = \mathbf{0}$ for all $\mathbf{y} \in \mathbb{R}^n$, so that $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} = \mathbf{M}_2\mathbf{H}_1$.

Theorem 5 *Consider the model $\mathscr{M}$. Then for every $\boldsymbol{y} = \boldsymbol{X}_1\boldsymbol{a} + \boldsymbol{\Sigma}\boldsymbol{M}_1\boldsymbol{b}$ we have*

$$\begin{aligned} ||\hat{\boldsymbol{\theta}}_1(\mathscr{M}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{M})||_2^2 &= ||\text{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) - \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})||_2^2 \\ &\le |||\boldsymbol{A}|||^2\, \boldsymbol{b}'\boldsymbol{M}_1\boldsymbol{b} \\ &= a\, \boldsymbol{b}'\boldsymbol{M}_1\boldsymbol{b} =: \phi\,, \end{aligned}$$

where $a = \text{ch}_1(\boldsymbol{A}'\boldsymbol{A})$,$\boldsymbol{A} = \boldsymbol{M}_2\boldsymbol{H}_1\boldsymbol{\Sigma}\boldsymbol{M}_1$. The upper bound ϕ is zero for all $\boldsymbol{b} \notin \mathscr{C}(\boldsymbol{X}_1)$ if and only if $\boldsymbol{M}_2\boldsymbol{H}_1\boldsymbol{\Sigma}\boldsymbol{M}_1 = \boldsymbol{0}$, i.e., $\hat{\boldsymbol{\theta}}_1(\mathscr{M}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{M})$.

Proof Applying part (f) of Lemma 3 we have

$$\mathbf{X}_1(\mathbf{X}_1'\mathbf{W}_m^-\mathbf{X}_1)^-\mathbf{X}_1'\mathbf{W}_m^+ = \mathbf{P}_{\mathbf{W}_m} - \boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^+\mathbf{M}_1\,. \tag{3.22}$$

Premultiplying (3.22) by $\mathbf{M}_2\mathbf{H}_1$ we obtain

$$\begin{aligned}\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{y} &= \mathbf{M}_2\mathbf{H}_1\mathbf{y} - \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^{+}\mathbf{M}_1\mathbf{y} \\ &= \hat{\boldsymbol{\theta}}_1(\mathscr{M}) - \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^{+}\mathbf{M}_1\mathbf{y} \\ &= \tilde{\boldsymbol{\theta}}_1(\mathscr{M})\,.\end{aligned}$$

Thus we have

$$\begin{aligned}\hat{\boldsymbol{\theta}}_1(\mathscr{M}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{M}) &= \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) - \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) \\ &= \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^{+}\mathbf{M}_1\mathbf{y}\,,\end{aligned}$$

and for all $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}$,

$$\begin{aligned}||\hat{\boldsymbol{\theta}}_1(\mathscr{M}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{M})||_2^2 &= ||\mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) - \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M})||_2^2 \\ &= ||\mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^{+}\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}||_2^2 \\ &\le |||\mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}||_2^2 \\ &\le |||\mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1|||^2\ \mathbf{b}'\mathbf{M}_1\mathbf{b} \\ &= a\ \mathbf{b}'\mathbf{M}_1\mathbf{b} = \phi\,,\end{aligned}$$

where $a = \mathrm{ch}_1(\mathbf{A}'\mathbf{A})$, $\mathbf{A} = \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1$. The upper bound ϕ is zero for all $\mathbf{b} \notin \mathscr{C}(\mathbf{X}_1)$ if and only if $\mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1 = \mathbf{0}$, i.e., $\hat{\boldsymbol{\theta}}_1(\mathscr{M}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{M})$. □

Theorem 6 *Consider the model $\mathscr{F}$. Then for every $\boldsymbol{y} = \boldsymbol{Xa} + \boldsymbol{VMb}$,*

$$\begin{aligned}||\hat{\boldsymbol{\theta}}_1(\mathscr{F}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{F})||_2^2 &= ||\mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) - \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})||_2^2 \\ &\le |||\boldsymbol{B}|||^2\ \boldsymbol{b}'\boldsymbol{M}\boldsymbol{b} \\ &= b\ \boldsymbol{b}'\boldsymbol{M}\boldsymbol{b} =: \varrho\,,\end{aligned}$$

where $b = \mathrm{ch}_1(\boldsymbol{B}'\boldsymbol{B})$, $\boldsymbol{B} = \boldsymbol{P}_{\boldsymbol{M}_2\boldsymbol{X}_1}\boldsymbol{\Sigma}\boldsymbol{M}$. The upper bound ϱ is zero for all $\boldsymbol{b} \notin \mathscr{C}(\boldsymbol{X})$ if and only if $\boldsymbol{P}_{\boldsymbol{M}_2\boldsymbol{X}_1}\boldsymbol{\Sigma}\boldsymbol{M} = \boldsymbol{0}$, i.e., $\hat{\boldsymbol{\theta}}_1(\mathscr{F}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{F})$.

Proof It is clear that premultiplying the fundamental BLUE-equation

$$\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+}(\mathbf{X}_1 : \mathbf{X}_2 : \boldsymbol{\Sigma}\mathbf{M}) = (\mathbf{X}_1 : \mathbf{0} : \mathbf{0})$$

by $\mathbf{M}_2$ yields

$$\mathbf{M}_2\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+}(\mathbf{X}_1 : \mathbf{X}_2 : \boldsymbol{\Sigma}\mathbf{M}) = (\mathbf{M}_2\mathbf{X}_1 : \mathbf{0} : \mathbf{0})\,,$$

and so $\mathbf{C}\mathbf{y} = \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})$, where $\mathbf{C} = \mathbf{M}_2\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+}$. Moreover, it can be shown that

$$\mathscr{G}_{\theta_1|\mathscr{F}} = \mathbf{M}_2\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+}. \tag{3.23}$$

By part (f) of Lemma 3 we have

$$\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+} = \mathbf{P}_{\mathbf{W}} - \boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\,. \tag{3.24}$$

Premultiplying (3.24) by $\mathbf{M}_2\mathbf{P}_{\mathbf{X}}$ and using (3.23) gives

$$\begin{aligned}
\mathcal{G}_{\theta_1|\mathscr{F}} &= \mathbf{M}_2\mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+} \\
&= \mathbf{M}_2[\mathbf{P}_{\mathbf{X}} - \mathbf{P}_{\mathbf{X}}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}] \\
&= \mathbf{M}_{\mathbf{X}_2}[\mathbf{P}_{\mathbf{X}} - \mathbf{P}_{\mathbf{X}}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}] \\
&= \mathbf{M}_2[(\mathbf{P}_{\mathbf{X}_2} + \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}) - \mathbf{P}_{\mathbf{X}}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}] \\
&= \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1} - \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\,.
\end{aligned}$$

Thus

$$\begin{aligned}
\mathcal{G}_{\theta_1|\mathscr{F}}\mathbf{y} &= \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{y} - \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\mathbf{y} \\
&= \hat{\boldsymbol{\theta}}_1(\mathscr{F}) - \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\mathbf{y},
\end{aligned}$$

and

$$\begin{aligned}
\hat{\boldsymbol{\theta}}_1(\mathscr{F}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{F}) &= \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) - \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) \\
&= \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\mathbf{y}.
\end{aligned}$$

Now for all $\mathbf{y} = \mathbf{X}\mathbf{a} + \mathbf{V}\mathbf{M}\mathbf{b} = \mathbf{X}\mathbf{a} + \boldsymbol{\Sigma}\mathbf{M}\mathbf{b}$,

$$\begin{aligned}
||\hat{\boldsymbol{\theta}}_1(\mathscr{F}) - \tilde{\boldsymbol{\theta}}_1(\mathscr{F})||_2^2 &= ||\mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) - \mathrm{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})||_2^2 \\
&= ||\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\boldsymbol{\Sigma}\mathbf{M}\mathbf{b}||_2^2 \\
&\le ||\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}\mathbf{b}||_2^2 \\
&\le |||\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}|||^2\, \mathbf{b}'\mathbf{M}\mathbf{b} \\
&= b\, \mathbf{b}'\mathbf{M}\mathbf{b} = \varrho\,,
\end{aligned}$$

where $b = \mathrm{ch}_1(\mathbf{B}'\mathbf{B})$, $\mathbf{B} = \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}$. The upper bound ϱ is zero for all $\mathbf{b} \notin \mathscr{C}(\mathbf{X})$ if and only if $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M} = \mathbf{0}$, i.e., $\hat{\boldsymbol{\theta}}_1(\mathscr{F}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{F})$. □

Suppose that

$$\mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) = \mathrm{OLSE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) \quad \text{for all } \mathbf{y} \in \mathscr{C}(\mathbf{W}_m)$$

holds, i.e.,

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}(\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1) = \mathbf{M}_2\mathbf{H}_1(\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1)\,,$$

which, in view of $\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{X}_1 = \mathbf{M}_2\mathbf{H}_1\mathbf{X}_1 = \mathbf{M}_2\mathbf{X}_1$, is equivalent to

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}_1 = \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1 =: \mathbf{S}\,. \tag{3.25}$$

We may now ask under which condition does the following hold:

$$\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) = \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F}) \quad \text{for all } \mathbf{y} \in \mathscr{C}(\mathbf{W}_m)\,.$$

We know that

$$\begin{aligned}\mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y} &= \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\mathbf{y} - \mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\mathbf{y},\\ \mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{y} &= \mathbf{M}_2\mathbf{H}_1\mathbf{y} - \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^{+}\mathbf{M}_1\mathbf{y}.\end{aligned}$$

Thereby, assuming that (3.25) holds, $\text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{M}) = \text{BLUE}(\boldsymbol{\theta}_1 \mid \mathscr{F})$ for all $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b}$ if and only if

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\mathbf{y} = \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1(\mathbf{M}_1\boldsymbol{\Sigma}\mathbf{M}_1)^{+}\mathbf{M}_1\mathbf{y},$$

i.e.,

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b} = \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{b} \quad \text{for all } \mathbf{b} \in \mathbb{R}^n,$$

which by $\mathbf{M} = \mathbf{M}_1\mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}$ and (3.25) can be expressed as

$$\mathbf{P}_{\mathbf{M}_2\mathbf{X}_1}\boldsymbol{\Sigma}\mathbf{M}_1\mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}(\mathbf{M}\boldsymbol{\Sigma}\mathbf{M})^{+}\mathbf{M}\boldsymbol{\Sigma}\mathbf{M}_1 = \mathbf{M}_2\mathbf{H}_1\boldsymbol{\Sigma}\mathbf{M}_1\,. \tag{3.27}$$

On the basis of (3.25) and (3.27), we can conclude the following result.

Corollary 1 *Consider the models $\mathscr{F}$ and $\mathscr{M}$ and assume that $\hat{\boldsymbol{\theta}}_1(\mathscr{M}) = \hat{\boldsymbol{\theta}}_1(\mathscr{F})$ for all $\mathbf{y} \in \mathscr{C}(\boldsymbol{W}_m)$, i.e., (3.25) holds. Then for all $\boldsymbol{y} \in \mathscr{C}(\boldsymbol{W}_m)$,*

1. $\hat{\boldsymbol{\theta}}_1(\mathscr{M}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{M}) \iff \hat{\boldsymbol{\theta}}_1(\mathscr{F}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{F})\,.$
2. $\tilde{\boldsymbol{\theta}}_1(\mathscr{M}) = \tilde{\boldsymbol{\theta}}_1(\mathscr{F}) \iff \boldsymbol{S}\boldsymbol{Q}_{\boldsymbol{M}_1\boldsymbol{X}_2}(\boldsymbol{M}\boldsymbol{\Sigma}\boldsymbol{M})^{+}\boldsymbol{M}\boldsymbol{\Sigma}\boldsymbol{M}_1 = \boldsymbol{S}.$

3.4 Conclusions

In this paper, we consider the partitioned fixed linear model $\mathscr{F}: \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ and the corresponding mixed model $\mathscr{M}: \mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\mathbf{u} + \boldsymbol{\varepsilon}$, where $\boldsymbol{\varepsilon}$ is random error vector and $\mathbf{u}$ is a random effect vector, in other words, $\mathscr{M}$ is obtained from $\mathscr{F}$ by replacing the fixed effects $\boldsymbol{\beta}_2$ with the random effects $\mathbf{u}$. In both models the covariance matrix of $\boldsymbol{\varepsilon}$ is $\mathbf{V}$ while the covariance matrix of $\mathbf{u}$ is $\mathbf{D}$. Isotalo et al. [10] found conditions under which an arbitrary representation of the BLUE of an estimable parametric function of $\boldsymbol{\beta}_1$ in the fixed model $\mathscr{F}$ remains BLUE in the mixed model $\mathscr{M}$. The considerations were done by studying the properties of the BLUE of $\boldsymbol{\theta}_1 = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1$, where $\mathbf{M}_2 = \mathbf{I}_n - \mathbf{P}_{\mathbf{X}_2}$.

In this paper, we establish upper bounds for the Euclidean norm of the difference between the BLUEs of $\boldsymbol{\theta}_1 = \mathbf{M}_2\mathbf{X}_1\boldsymbol{\beta}_1$ under models $\mathscr{F}$ and $\mathscr{M}$. Some corresponding bounds are considered also for the difference between the ordinary least squares

estimators, OLSEs of $\boldsymbol{\theta}_1$. It is a bit tricky to consider the distance between the BLUEs under $\mathscr{F}$ and $\mathscr{M}$ because under $\mathscr{F}$ the response vector $\mathbf{y}$ is supposed to belong to $\mathscr{C}(\mathbf{W}) = \mathscr{C}(\mathbf{X} : \mathbf{V})$ while in $\mathbf{M}$ it belongs to $\mathscr{C}(\mathbf{W}_m) = \mathscr{C}(\mathbf{X}_1 : \boldsymbol{\Sigma}\mathbf{M}_1)$, where $\boldsymbol{\Sigma} = \mathbf{V} + \mathbf{X}_2\mathbf{D}\mathbf{X}_2'$. We pick up particular representations for the BLUEs, $\mathscr{G}_{\theta_1|\mathscr{F}}\mathbf{y}$ and $\mathscr{G}_{\theta_1|\mathscr{M}}\mathbf{y}$, and study the upper bounds for the norms of their Euclidean distances when $\mathbf{y}$ belongs to $\mathscr{C}(\mathbf{W}_m)$ or $\mathscr{C}(\mathbf{W})$.

As for related literature, Baksalary and Kala [2, p. 680] provided an upper bound for the Euclidean distance of the OLSE and the BLUE of $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ under $\{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{V}\}$; see also Haberman [5], Baksalary and Kala [1], and Trenkler [19, p. 261]. The corresponding considerations for the BLUEs under two linear models $\{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{V}_1\}$ and $\{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{V}_2\}$, have been made by Hauke et al. [8]. Upper bounds arising from linear models with new future observations have been considered by Haslett et al. [6, Sect. 3], Markiewicz and Puntanen [12, 13]. See also Kala et al. [11, Sect. 6], who considered the distance between the BLUEs under the original and the transformed models, and Pordzik [15], who studied the distance between restricted and unrestricted estimators in the general linear model.

In practical applications, the paper provides guidance on the effect on the remaining fixed parameters of treating some of the parameters in a linear model as fixed rather than random, or vice versa. When computing estimates using statistical software, this is usually a very simple programming change with unknown consequences, so it is rather too easy to misspecify the correct model. Our paper provides useful guidance and upper bounds on the consequences of such misspecification.

Acknowledgements Thanks to the anonymous referee for constructive comments.

References

1. Baksalary JK, Kala R (1978) A bound for the Euclidean norm of the difference between the least squares and the best linear unbiased estimators. Ann Stat 6:1390–1393. http://dx.doi.org/10.1214/aos/1176344383
2. Baksalary JK, Kala R (1980) A new bound for the Euclidean norm of the difference between the least squares and the best linear unbiased estimators. Ann Stat 8:679–681. http://dx.doi.org/10.1214/aos/1176345018
3. Ben-Israel A, Greville TNE (2003) Generalized inverses: theory and applications. Springer, Heidelberg. http://dx.doi.org/10.1007/b97366
4. Groß J, Trenkler G (1998) On the equality of linear statistics in general Gauss-Markov Model. In: Mukherjee SP, Basu SK, Sinha BK (eds) Front Probab Stat. Narosa Publishing House, New Delhi, pp 189–194
5. Haberman SJ (1975) How much do Gauss–Markov and least square estimates differ? A coordinate-free approach. Ann Stat 3:982–990. http://dx.doi.org/10.1214/aos/1176343201
6. Haslett SJ, Isotalo J, Liu Y, Puntanen S (2014) Equalities between OLSE, BLUE and BLUP in the linear model. Stat Pap 55:543–561. http://dx.doi.org/10.1007/s00362-013-0500-7
7. Haslett SJ, Isotalo J, Puntanen S (2021) Equalities between the BLUEs and BLUPs under the partitioned linear fixed model and the corresponding mixed model. Acta Comment Univ Tartu Math 25:239–257. https://doi.org/10.12697/ACUTM.2021.25.16

8. Hauke J, Markiewicz A, Puntanen S (2012) Comparing the BLUEs under two linear models. Commun Stat Theory Methods 41:2405–2418. http://dx.doi.org/10.1080/03610926.2011.594541
9. Horn RA, Johnson CR (2012) Matrix analysis, 2nd ed. Cambridge University Press. https://doi.org/10.1017/CBO9780511810817
10. Isotalo J, Möls M, Puntanen S (2006) Invariance of the BLUE under the linear fixed and mixed effects models. Acta Comment Univ Tartu Math 10:69–76. https://doi.org/10.12697/ACUTM.2006.10.06
11. Kala R, Markiewicz A, Puntanen S (2017) Some further remarks on the linear sufficiency in the linear model. In: Bebiano N (ed) Applied and computational matrix analysis: MatTriad, Coimbra, Portugal, Sept 2015, Selected, Revised Contributions. Springer, Cham, pp 275–294. http://dx.doi.org/10.1007/978-3-319-49984-0_19
12. Markiewicz A, Puntanen S (2018) Further properties of linear prediction sufficiency and the BLUPs in the linear model with new observations. Afrika Statistika 13:1511–1530. https://www.ajol.info/index.php/afst/issue/view/16938
13. Markiewicz A, Puntanen S (2018) Upper bounds for the Euclidean distances between the BLUPs. Special Matr 6:249–261. https://doi.org/10.1515/spma-2018-0020
14. Marsaglia G, Styan GPH (1974) Equalities and inequalities for ranks of matrices. Linear Multilinear Algeb 2:269–292. https://doi.org/10.1080/03081087408817070
15. Pordzik PR (2012) A bound for the Euclidean distance between restricted and unrestricted estimators of parametric functions in the general linear model. Stat Pap 53:299–304. http://dx.doi.org/10.1007/s00362-010-0336-3
16. Puntanen S, Styan GPH, Isotalo J (2011) Matrix tricks for linear statistical models: our personal top twenty. Springer, Heidelberg. http://dx.doi.org/10.1007/978-3-642-10473-2
17. Rao CR (1973) Representations of best linear estimators in the Gauss–Markoff model with a singular dispersion matrix. J Multivar Anal 3:276–292. http://dx.doi.org/10.1016/0047-259X(73)90042-0
18. Rao CR, Mitra SK (1971) Gen Inverse Matr Appl. Wiley, New York
19. Trenkler G (1994) Characterizations of oblique and orthogonal projectors. In: Caliński T, Kala R (eds) Proceedings of the international conference on linear statistical inference LINSTAT '93. Poznań, Kluwer, Dordrecht, pp 255–270. http://dx.doi.org/10.1007/978-94-011-1004-4

Chapter 4
Nucleolus Computation for Some Structured TU Games via Graph Theory and Linear Algebra

T. E. S. Raghavan

2010 Mathematics Subject Classification 91A30 · 91A68 · 91A80

4.1 TU Cooperative Game

A transferable utility cooperative game is defined by

(i) Player set $N = \{1, 2, \ldots, n\}$.
(ii) Coalitions: $S \subset N$.
(iii) Worth of a coalition $S = v(S) \in \mathbf{R}$.
(iv) Worth of the empty coalition $= v(\phi) = 0$.

A preimputation is any real vector $x = (x_1, x_2, \ldots, x_n)$ with $x(S) = \sum_{i \in S} x_i$ and $x(N) = v(N)$.
An imputation is any real vector $x = (x_1, x_2, \ldots, x_n)$ with $x_i \geq v(\{i\})$, $i = 1, 2, .., n$ and $x(N) = v(N)$.
The satisfaction to a coalition S with imputation x is given by $f(x, S) = x(S) - v(S)$.

Example 1

$N = \{1, 2, 3\}$, $v(\{1\}) = v(\{2\}) = v(\{3\}) = 0$, $v(1, 2) = 9$, and $v(1, 3) = 12$, $v(2, 3) = 15$, $v(1, 2, 3) = 20$. The satisfaction for imputation $x = (7, 9, 4)$ and $y = (2, 8, 10)$ at $S = \{2, 3\}$ are respectively $f(x, S) = (9 + 4) - 15 = -2$ and $f(y, S) = 8 + 10 - 15 = 3$.

T. E. S. Raghavan (✉)
University of Illinois at Chicago, Chicago, IL 60607-7045, USA
e-mail: terctu@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_4

We can write down the satisfactions as a vector in 2^3 space. For our example above, we have the following satisfactions for each imputations.

Coalitions	$x(S)$	$v(S)$	$f(x, S) = x(S) - v(S)$	$y(S)$	$f(y, S)$
ϕ	0	0	0	0	0
1	7	0	7	2	2
2	9	0	9	8	8
3	4	0	4	10	10
1 2	16	9	7	10	1
1 3	11	12	-1	12	0
2 3	13	15	-2	18	3
1 2 3	20	20	0	20	0

One can rearrange the satisfactions in an increasing order for each imputation. Let θ denote such an arrangement; we have

$$\theta(x) = (-2, -1, 0, 0, 4, 7, 7, 9) \text{ and } \theta(y) = (0, 0, 0, 1, 2, 3, 8, 10).$$

We say $y \underset{L}{>} x$ (imputation y is lexicographically better than x when $\theta(y) \underset{L}{>} \theta(x)$). Here, $\theta(y) \underset{L}{>} \theta(x)$. Thus, $x \underset{L}{<} y$ if and only if $\theta(x) \leq \theta(y)$ lexicographically.

Let X be either the set of preimputations or imputations. Then

Definition 4.1 (Schmeidler [25]) The nucleolus of the TU game (v, N) over the set X is $\nu(X) = \{x \in X;$ if $y \in X$ then $\theta(y) \underset{L}{\leq} \theta(x)\}$.

For X one may also consider other sets such as $\left\{x : \sum_{i \in T_1} x_i = v(T_1), \sum_{i \in T_2} x_i = v(T_2), \ldots \sum_{i \in T_k} x_i = v(T_k) : \{T_1, T_2, ..T_k\} \text{ a partition of} \{1, 2, \ldots, n\} = N\right\}$.

Theorem 4.1 *If X is non-empty and compact, then $\nu(X)$ is also non-empty and compact.*

Proof Let $X' = \{x : x \in X, \theta_1(y) \leq \theta_1(x)\ \forall y \in X\}$. Since θ_1 is a continuous function of x, the set X' is non-empty and compact as a continuous image of X. By induction, $X^{(k)} = \{x : x \in X^{(k-1)} \quad \theta_k(y) \leq \theta_k(x) \forall y \in X^{(k-1)}\}$ is continuous and compact for $k = 1, 2, \ldots, 2^n$.

We claim $\nu(X) = X^{(2^n)}$. Suppose $\nu(X) \neq X^{(2^n)}$. This means we have a $y \in X - X^{(2^n)}$, $x \in X^{(2^n)}$ such that $\theta(y) \underset{L}{\geq} \theta(x)$. Let k be the smallest integer such that $y \notin X^{(k)}$. Then $\theta_l(y) = \theta_l(x)$ if $l < k$. However, $\theta(y) \underset{L}{>} \theta(x)$ and this contradiction shows $\nu(X) = X^{(2^n)}$. □

The interesting aspect of $v(X)$ is the following:

Theorem 4.2 *If X is the imputation set, then $v(X)$ is a unique imputation.*

The following lemma is needed for the theorem.

Lemma 4.1 *Let $\theta(x) = \theta(y),\ x \neq y$, then for any $0 < \lambda < 1$ θ is quasi-concave, i.e. $\theta(\lambda x + (1-\lambda y)) \underset{L}{\succ} \theta(x)$.*

Proof Let $z = \lambda x + (1-\lambda)y$. We have

$$\theta(x) = (f(x, S_1), f(x, S_2), \ldots, f(x, S_{2^n}))$$
$$\theta(y) = (f(y, T_1), f(y, T_2), \ldots, f(y, T_{2^n})).$$

Here, $S_1, S_2, \ldots, S_{2^n}$ are a permutation of all possible coalitions in N, with $f(x, S_1) \leq f(x, S_2) \cdots \leq f(x, S_{2^n})$ and similarly $T_1, T_2, \ldots, T_{2^n}$ is yet another permutation of all possible coalitions in N with $f(y, T_1) \leq f(y, T_2) \cdots \leq f(y, T_{2^n})$. We will assume that in case $f(x, S_k) = f(x, S_{k+1})$, their naming satisfies $f(y, S_k) \leq f(y, S_{k+1})$.

Since $x \neq y$ we have $\sum_{\in S} x_i \neq \sum_{i \in S} y_i$ for same coalition S and hence $f(x, S) \neq f(y, S)$. Let $S = S_c$ be the index of S according to $\theta(x)$ and we have $f(x, S_c) \neq f(y, S_c)$. We can find such an S_c with c as the first index with this property.

Let $f(x, S_c) = q$. For any coalition S, and $z = \lambda x + (1-\lambda)y$

$$\begin{aligned} f(z, S) &= \lambda x(S) + (1-\lambda)y(S) - v(S) \\ &= \lambda f(x, S) + (1-\lambda) f(y, S). \end{aligned}$$

For any $k < c$ we have $f(z, S_k) = f(x, S_k)$. Consider any $k \geq c$. We claim that $f(y, S_k) \geq f(x, S_c)$. Suppose not. Then $f(y, S_k) < f(x, S_c)$. In fact if $f(x, S) < f(x, S_c) = q$, then $S = S_k$ and $k < c$ and by assumption $f(y, S) = f(y, S_k) = f(x, S_k) < q$, while in particular $f(y, S_c) \geq f(x, S_c) = q$. Thus for $k \geq c$ either $f(x, S_k) > q$, and $f(y, S_k) \geq q$ or $f(x, S_k) \geq q$ and $f(y, S_k) > q$. Thus $f(z, S_k) > q$. We have $f(z, S_k) = f(x, S_k)$ for $k \leq c$ and $f(z, S_k) > q$:

$$\begin{aligned} \text{i.e.}\quad & \theta_k(z) = \theta_k(x) \ \text{ for } 1 \leq k \leq c-1 \\ & \theta_c(z) > \theta_c(x) \text{ i.e. } \theta(z) \underset{L}{\succ} \theta(x). \end{aligned}$$

When $x \neq y$ and $0 < \lambda < 1$ for $z = \lambda x + (1-\lambda)y$, $\theta(z) \underset{L}{\succ} \theta(x)$. This proves that $v(x)$ is a unique point in X. □

Some special cases where one can explicitly express the nucleolus as a fixed form of the data defining the game are worth investigating. We will start with simple examples.

Example 2

Let v be a 3-person simple majority game. We have

$$X = \{(x_1, x_2, x_3) : \sum x_i = 1,\ x_i \geq 0,\ i = 1, 2, 3\}.$$

We will check $x^* = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is the nucleolus.

Let $y = (y_1, y_2, ..., y_n) \neq x^*$ be any imputation. W.l.g. $y_1 = \frac{1}{3} + \epsilon_1,\ \ y_2 = \frac{1}{3} + \epsilon_2,\ \ y_3 = \frac{1}{3} - \epsilon_1 - \epsilon_2$ where $\epsilon_1, \epsilon_2 > 0$. Similarly, let $z_1 = \frac{1}{3} - \delta_1\ z_2 = \frac{1}{3} - \delta_2\ z_3 = \frac{1}{3} + \delta_1 + \delta_2$. In either case the least entry in $\theta(y)$ or $\theta(z)$ is $< \frac{1}{3}$. Thus $(\frac{1}{3},\ \frac{1}{3},\ \frac{1}{3})$ is the nucleolus.

Example 3 [7]

A landlord who is handicapped owns all the lands in the village. He can hire n-(homogeneous) landless laborers and harvest $f(n)$ bags of rice. Suppose $f(x)$ is an increasing concave function. Then the TU game that we can define is

$$v(S) = \begin{cases} f(|S| - 1) & \text{if } S \ni \text{landlord} \\ 0 & \text{if } S \not\ni \text{landlord.} \end{cases}$$

Find the nucleolus for this game with $n + 1$ players

Lemma 4.2 *The game has a non-empty core, where the core of a game* (v, N) *is defined as*

$$\left\{x : x_i \geq v(i)\ \ \forall\, i, \sum x_i = v(N)\right\} \cap \left\{x : \sum_{i \in S} x_i \geq v(S) \quad S \subset N\right\}.$$

(In general core could be empty.)

Proof Consider the preimputation

$$\phi(w_j) = \frac{f(n) - f(n-1)}{2}, \quad j = 1, 2, \ldots, n$$

$$\phi(L) = f(n) - n\left[\frac{f(n) - f(n-1)}{2}\right].$$

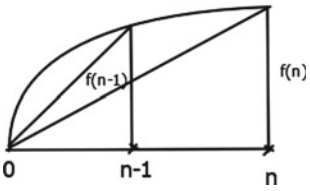


Fig. 4.1 Bags of rice vs number of laborers

We will check that it is an imputation. Since $\frac{f(n)-f(n-1)}{2} \geq 0$, it is enough to check that $\phi(L) = f(n) - n\left[\frac{f(n)-f(n-1)}{2}\right] \geq 0$. We will actually show that $\phi(L) = f(n) - n\left[\frac{f(n)-f(n-1)}{2}\right] \geq \frac{f(n)-f(n-1)}{2} \geq 0$. That is, landlord gets at least what laborers are offered. Since $f(0) = 0$ and $f(n)$ is concave, we have from Fig. 4.1,

$$\frac{f(n-1)}{n-1} \geq \frac{f(n)}{n} \geq \frac{f(n)}{n+1}$$
$$\text{i.e. } (n+1)f(n-1) \geq (n-1)f(n) = [(n+1)-2]f(n).$$

Thus $2f(n) \geq (n+1)[f(n) - f(n-1)]$.
i.e. $f(n) \geq (n+1)\left[\frac{f(n)-f(n-1)}{2}\right] \implies f(n) - n\left[\frac{f(n)-f(n-1)}{2}\right] \geq \frac{f(n)-f(n-1)}{2}$
≥ 0. □

Remark 4.1 What happens when workers are highly productive and the output is an increasing convex function? In this case, one can show that the landlord would check whether what remains after the payment of half the last workers' marginal contribution is feasible leaving him at least that level of rent comparable to the wage of the workers. When it works he will take it. Otherwise, the nucleolus rent and wage will be $\frac{f(n)}{n+1}$, namely possession of land has the same value as the worker's wage.

One can also analyze the case where there are two lame brothers who own all the land in the village and are trying to compete with each other for the n homogeneous landless laborers. When the production function $f(x)$ is concave, they basically have to compete for sharing the workers. Since the marginal contribution is a decreasing function when f is concave, the sharing of workers at the nucleolus price will increase the wage close to $\frac{f(m)-f(m-1)}{2}$ where $2m = n$(say) where the landowners employ equal amount of workers. For finer details locating the nucleolus, see Chetty et al. [7].

4.2 Nucleolus for the Talmud Game [2]

An orthodox Jew borrowed \$1000, \$2000 and \$3000 from 3 banks and spent \$5000 for personal use and died suddenly. His orthodox wife finds just \$1000 in his wallet. The 3 banks want to get the entire \$1000 for themselves. She seeks the advice of her lawyer who is also committed to Jewish law. According to him, the problem has to be resolved as enunciated in the Talmud. The Talmud (Baba Metzia) talks about the following example.

Contested Garment Principle: Two sisters show in court (Jewish), the wills. Elder one says that she is entitled to her mom's marriage cloth (imagine it as an expensive silk saree from Kancheepuram). The younger sister shows her mother's will that the two sisters share the saree equally.

The underlying Jewish principle is that no one can fight for what one is not entitled to. The judge says that the second daughter is not entitled to more than $\frac{1}{2}$ according to the will. Thus, the judge says that the elder daughter gets at least $\frac{1}{2}$ the saree, and the debate is only over the other half on which both daughters' wills say they are the owners. Thus, the judge says a 50 : 50 split of the other half is the most unbiased, giving $\frac{3}{4}$th the saree to elder daughter and $\frac{1}{4}$th the saree to the younger one.

Using this principle, the judge wants to resolve the bank loan problem. Think of the banks as players 1, 2, 3 and let $v(S)$ be what would remain in case each coalition S is entitled to $\max(1000 - \sum_{i \notin S} d_i, 0)$. That is each coalition of bankers can fight over what would remain after closing the debts outside the coalition. In this case, $v(1) = \max(1000 - (2000 + 3000), 0) = 0$. Then $v(1) = v(2) = v(3) = v(1\ 2) = v(1\ 3) = v(2\ 3) = 0$, $v(1\ 2\ 3\ ) = 1000$.

In this case, the judge suggested equal division of \$1000 among the 3 banks. Incidentally, this is also the nucleolus for this game v with $N = \{1, 2, 3\}$.

Suppose the man who borrowed money from the 3 banks spent \$4000 leaving \$2000 in his wallet. We can write down

$$
\begin{aligned}
v(1) &= \max\{2000 - d_2 - d_3, 0\} = 0 \ \text{ where } d_1 = 1000\ d_2 = 2000\ d_3 = 3000.\\
v(2) &= \max\{2000 - d_1 - d_3, 0\} = 0\\
v(3) &= \max\{2000 - d_1 - d_2, 0\} = 0\\
v(1\ 2) = 0,\ v(1\ 3) &= 0,\ v(2\ 3) = 1000,\ v(1\ 2\ 3) = 2000.
\end{aligned}
$$

How could we use contested garment principle to resolve the problem with three persons?

Let us revert to a two-creditor case. Suppose E is the money in the wallet of the diseased person and suppose d_1, d_2 are the loans from the two banks. The amount each bank i concedes to the other bank j is $\max\{E - d_i, 0\}$. Let $\theta_i^+ = \max\{E - d_i, 0\}\}$. Thus, bank 1 concedes to bank 2, an amount θ_1^+. Bank 2 concedes to bank 1 an amount θ_2^+. The amount at issue is $E - \theta_1^+ - \theta_2^+$. It is shared equally between the 2 bankers and each bank receives in addition what the opponent bank has conceded. Thus, the amount is

$$x_1 = \frac{E - \theta_1^+ - \theta_2^+}{2} + \theta_2^+ \; ; \; x_2 = \frac{E - \theta_1^+ - \theta_2^+}{2} + \theta_1^+$$

for banks 1 and 2, respectively.

We will say that this division of E for claims d_1, d_2 is prescribed by the contested garment principle.

If we view this solution as a function of E, for small values of $E, \theta_1^+, \theta_2^+$ will be 0 and $x_1 = x_2 = \frac{E}{2}$. Let $d_1 \leq d_2$ and suppose $d_1 < E < d_2$

$$\theta_1^+ = E - d_1 \quad \text{and} \quad \theta_2^+ = 0.$$

Thus

$$\begin{aligned} x_1 &= \frac{E - (E - d_1) - 0}{2} = \frac{d_1}{2} \\ x_2 &= \frac{E - (E - d_1) - 0}{2} + (E - d_1) \\ &= (E - d_1) + \frac{d_1}{2} \\ &= E - \frac{d_1}{2}. \end{aligned}$$

Thus, bank 1 gets $\frac{1}{2}$, the amount the bank lent and the other bank (bank 2) gets the remaining.

Let us continue to analyze when $d_1 < E, \; d_2 < E$ but $d_1 + d_2 > E$.

We get $\theta_1^+ = (E - d_1)$. $\theta_2^+ = (E - d_2)$. But what remains after conceding by each is $E - \theta_1^+ - \theta_2^+$. This is split evenly by the 2 banks and in addition, each bank receives what the opponent has conceded. Consider the following case.

Let $d_1 = 1000, \; d_2 = 2000, \; E = 2400$ say. The CG (Contested garment principle) will offer

$$\begin{aligned} x_1 &= \frac{2400 - 1400 - 400}{2} + 400 = 700 \\ x_2 &= \frac{2400 - 1400 - 400}{2} + 1400 = 1700. \end{aligned}$$

(Here $\theta_1^+ = \max(2400 - 1000, 0), \; \theta_2^+ = \max(2400 - 2000, 0)$).

Still, this does not resolve the issue for the 3 person case. Our next step is to search for solutions that coincide with the CG principle for each pair for any division of E.

Let $E \leq d_1 + \ldots d_n$ (n banks have lent) $d_1 > d_2 > \ldots d_n > 0$ and the estate E has no money to pay in full, the money borrowed.

We call a solution $(x_1, x_2, \ldots x_n)$ pairwise consistent if for all $i \neq j$ with just the banks i, j that have lent d_i and d_j, the solution says x_i, x_j as though the total estate was only $x_i + x_j$. We call the solution *consistent* if for each pair $i \neq j$ the amounts x_i, x_j correspond to CG division of $x_i + x_j$.

A key observation is that for fixed d_i, d_j, the two awards are monotonic in E.

Theorem 4.3 *Each bankruptcy problem with n lenders lending $d_1, d_2 \ldots d_n$ to a person whose estate is worth $E < \sum_1^n d_j$ has a unique consistent solution.*

Proof We first show there cannot be two distinct consistent solutions: Suppose x and y are two consistent solutions and $x \neq y$. We can find two lenders i, j receiving x_i, x_j according to x and y_i, y_j according to y. Suppose $x_i + x_j \leq y_i + y_j$ yet $x_i < y_i$ and $x_j > y_j$. The CG principle awards y_j to j if $E = y_i + y_j$ and x_j when it is $E = x_i + x_j$. But $y_i + y_j \geq x_i + x_j$, and the monotonicity of CG principle demands $y_j \geq x_j$, a contradiction to $y_j < x_j$.

We will now show how we can build a consistent solution. We assume E is slowly growing for fixed loans $d_1, d_2, \ldots d_n$. For small E the lenders share equally. Suppose $d_1 \leq d_2 \leq \cdots \leq d_n$ then at some stage everyone would have received $\frac{d_1}{2}$. Then temporarily the lender d_1 receives no more payments. The remaining amount $E - n\frac{d_1}{2}$ is now being distributed equally till the 2nd lender gets $\frac{d_2}{2}$. Now both lender 1 and lender 2 are stopped receiving any more. We continue this procedure till each lender i has received $\frac{d_i}{2}$ $i = 1, 2, ..., n$. This will happen when $\frac{d_1+\cdots+d_n}{2} = E$.

In case $E > \frac{D}{2} = \frac{1}{2}$ (Total debt), the process is to think of how much each lender has to lose. If $d_1 - x_1, d_2 - x_2, ...d_n - x_n$ is what they lose, then when $(D - E)$ is small, it is shared equally. Thus, in this case the lender i with d_i receives $d_i = \frac{(D-E)}{n}$. When it hits say $\frac{d_1}{2}$ for the first time, lender 1 is not allowed to lose any more than $\frac{1}{2}$ of what he lent. The procedure continues with the rest of the lenders 2, 3, ..., n losing the same amount till it hits $\frac{d_2}{2}$. Now his losses are distributed among 3, 4..., n, the same way, etc. We have chosen to dig the tunnel from the other end and have met in the middle. □

Example 4

Let $d_1 = 1000$ $d_2 = 2000$ $d_3 = 3000$, $E = 4200$, $D > E > \frac{D}{2}$. Thus, we choose to find awards that fall short of what one has lent.

In round 1, say 500, 500, 500 is the amount all lose. Now it has hit $\frac{d_1}{2} = 500$. Thus, lender 1 has been met with $\frac{1}{2}$ the amount borrower took from lender 1. The estate is now left with $E' = 2700$. Now $d_2' = 2000 - 500 = 1500$ $d_3' = 3000 - 500 = 2500$ $E' > \frac{d_2'+d_3'}{2}$. Thus, suppose 500 more is paid to 2*and* 3 we have got $d_1 - x_1 = 500$, $d_2' - x_2' = 1500 - x_2'$, $d_3' - x_3' = 2500 - x_3'$. If $x_2' = 150$ $x_3' = 150$, then the losses are equally split between banks 2 and 3 and yet each one gets more than $\frac{1}{2}$ their lent amount. Thus $x^* = (500, 1350, 2350)$.

Remark 4.2 When the estate $E < \frac{D}{2} (= \frac{1}{2}$ the total loan), the individual loaners feel happy if they could receive at least half of what they loaned. When $E > \frac{D}{2}$, the loaners aim to get as much as they can and at least would continue till the actual losses are no more than the actual losses of loaners who loaned more.

The consistent rule we used has two properties.

Definition 4.2 A solution x to (E, d) with $d = (d_1 \ldots d_n)$ $x = (x_1 \ldots x_n)$ is *order preserving* if $0 \leq x_1 \leq x_2 \cdots \leq x_n$ and $0 \leq d_1 - x_1 \leq d_2 - x_2 \cdots \leq d_n - x_n$.

Thus, people with a higher claim than another get an award that is no smaller than and at the same time suffer a loss that is no smaller.

Depending on E and d, we can treat the n person problem in one of the following three ways:

(i) Divide E between loaner 1 and the $(n-1)$ persons $\{2, 3, \ldots, n\}$ as a coalition using CG principle where the coalition $\{2, \ldots n\}$ divides the amount by induction.
(ii) Assign equal award to all loaners.
(iii) Award equal losses to all loaners.

Specifically, (i) is used whenever it yields an order preserving result. This happens when $n\frac{d_1}{2} \leq E \leq D - \frac{nd_1}{2}$. We apply (ii) when $E \leq \frac{nd_1}{2}$ and (iii) when $E \geq D - \frac{nd_1}{2}$. Here $d = (d_1, \ldots d_n)$, $D = \sum_1^n d_j$.

Example 5

Banker $1, 2, \ldots 5$ lent 100, 200, 300, 400, and 500 to a person with $E = 510$. Suppose the coalition $\{2, 3, 4, 5\}$ forms. The CG rule offers 50 to 1 and decides the remaining amount 460 to be split among $\{2, 3, 4, 5\}$ with a joint claim of 1400. Suppose this splits into $\{2\}$ and $\{3, 4, 5\}$. Again applying CG rule gives 100 to $\{2\}$ and decides to split 360 among $\{3, 4, 5\}$ with a joint claim of 1200. Now any further split of this coalition as $\{3\}$ and $\{4, 5\}$ applying CG rule will offer 150 to $\{3\}$ and have to share 210 between loaners 4 and 5 and at least one will get less than 150 violating order preservation. Thus, they divide equally giving the final solution as (50, 100, 120, 120 120), which is order preserving. In fact we have the following.

Theorem 4.4 *The above coalitional procedure yields order preserving solution to all bankruptcy problems. Further it is the consistent solution.*

Remark 4.3 To arrive at such a solution, we cannot expect success if we do not choose the right kind of coalitions to apply the CG principle.

A natural cooperative game induced by the bankruptcy problem is through the TU game $\underset{E,d}{v}(S) = \{E - d(N\backslash S), 0\}$. This is a game where each coalition S is willing to forego the maximal demand of others and is willing to share whatever in leftover in the estate if any. This way they try to settle their claims out of court. The following is the main.

Theorem 4.5 *The consistent solution of a bankruptcy problem coincides with the nucleolus of the game corresponding to* $v(S) = (E - d(N\backslash S))_+ = max(E - d(N\backslash S), 0)$.

Proof See Aumann and Machler [2] □

Remark 4.4 The formal proof of the above theorem is based on the notion of reduced game property of the nucleolus. Suppose a coalition S and payoff $\{x_i,\ i \in S\}$ are given. The reduced game on the player space S is defined by

$$\begin{aligned} v^{S,x}(T) &= x(T) \text{ if } T = S \text{ or } T = \phi \\ &= \max\{v(Q \cup T) - x(Q) : Q \subset N \backslash S\} \text{ if } \phi \subsetneqq T \subsetneqq S. \end{aligned}$$

The intuitive idea is that the players in S can try to form new coalitions by trying to hire outside members of S, like $Q \subset N\backslash S$, but paying exactly the wage $x(Q)$ for joining them. Each non-empty coalition $T \neq S$ can look for the right coalition $Q \subset N\backslash S$ which will give maximum net value after the fixed wage $x(Q)$ is paid to them. One demands that if x is the solution to the bankruptcy problem (E, d) with $0 \leq x_i \leq d_i,\ \forall\, i$, then for any coalition S,

$$v_{E,d}^{S,x} = v_{x(S);d\backslash S}.$$

In other words the reduced bankruptcy game coincides with the game corresponding to the reduced bankruptcy problem. We leave its technical proof to [2].

4.3 The Core and Balanced Collection

Given a TU game (v, N), the core $C(v, N) = \{(x_1 \ldots x_n) :\ x_i \geq v(\{i\}),\ \sum_{v \in s} x_i \geq v(S),\ \sum_{i \in N} x_i = v(N)\}$.

In general the core could be empty. For example if we consider the simple game with at least 3 players given by

$$v(S) = \begin{cases} 1 & \text{if } |S| \geq 2 \\ 0 & \text{if } |S| \leq 1. \end{cases}$$

Clearly, if $(x_1, x_2 \ldots, x_n)$ is any imputation, then it is a core element if and only if $x_i + x_j \geq 1\ \forall\, i \neq j$ and $x_1 + \cdots + x_n = 1 \implies x_1 = x_2 = \cdots = x_n = 0$, a contradiction. Thus the game has no core element.

Horse Market: Two sellers and 3 buyers are in a horse market. Each seller has one horse to sell. Each buyer wants to buy one horse. Sellers 1 and 2 are expecting at least 200 K and $300K$ for their horses. The 3 buyers value the two identically looking horses for a price $\leq$350, $\leq$250 and $\leq$200 K, respectively. A TU game with the 5 players $\{S_1, S_2, B_1, B_2, B_3\}$ can be formed. Suppose a price p is announced and 200 K $< p <$ 350 K. Then we can define $V(S_1, B_2)$ as the sum of the gains for the two players $= p - 200$ for the seller and $250 - p$ for the buyer and $v(S_1, B_2) =$

$p - 200 + 250 - p = 50$. We see $v(S_1, B_3) = 0$. Since $p \geq 200 \implies$ the gain is 0 for the coalition, $v(S_1\ S_2\ B_1\ B_2\ B_3) = v(S_1, B_1) + v(S_2\ B_2) = 150 + 0 = 150$.

Any core imputation $(x_1\ x_2\ : y_1\ y_2\ y_3)$ where $x's$ represent sellers reward and $y's$ represent buyers reward satisfies

$$\begin{aligned}
x_1 + y_1 &\geq v(S_1\ B_1) = 150\\
x_1 + y_2 &\geq v(S_1\ B_2) = 50\\
x_1 + y_3 &\geq v(S_1\ B_3) = 0\\
x_2 + y_1 &\geq v(S_2\ B_1) = 350 - 300 = 50\\
x_2 + y_2 &\geq v(S_2\ B_2) = 300 - 250 = 50\\
x_2 + y_3 &\geq v(S_2\ B_3) = 300 - 200 = 100.
\end{aligned}$$

Since coalitions of more than 2 participants split into an optimal coalitional split, $v(S_1, S_2, B_2) = \max\{v(S_1, B_2), v(S_2, B_2)\} = \max\{50, 0\} = 50$.

Here, S_2 expects price $p \geq 300$ K and B_2 expects a price $\leq$250 K. Thus, only the optimal combination of buyer-seller pairing alone is used in defining the coalitional worth. Since coalitions of just sellers or just buyers have no merit, we are looking for a feasible solution to $x_i + y_j \geq a_{ij}$ where a_{ij} is the seller i, buyer j's coalitional gain. Adding dummy buyers or sellers, we can assume the problem to have n buyers and n sellers and with $a_{ij} = 0$ for all dummy buyer-seller combinations. The real pairing will satisfy $x_i + y_j = a_{ij}$ if (i, j) is part of the optimal pairing.

Thus, our problem reduces to finding an optimal pairing $i, \sigma(i)$ of sellers and buyers such that

$$\sum_i a_{i\sigma(i)} \geq \sum_i a_{i\tau(i)} \ \ \forall \text{ pairing } \tau.$$

This is the famous optimal assignment problem and is equivalent to

$$\max \sum_i \sum_j a_{ij} x_{ij}$$

such that

$$\begin{aligned}
\sum_j x_{ij} &= 1\ \forall\ i = 1, 2, \ldots, n\\
\sum_i x_{ij} &= 1\ \forall\ j = 1, 2, \ldots, n\\
x_{ij} &\geq 0\ \forall i, j = 1, 2, \ldots n.
\end{aligned}$$

Since the (x_{ij}) matrix is doubly stochastic, permutation matrices are extreme points of this convex set of doubly stochastic matrices (see Bapat and Raghavan [4]). Our problem becomes a linear programming problem with an optimal permutation matrix

as the solution. By the duality theorem of linear programming, we can consider the dual

$$\min \sum_i u_i + \sum_j v_j$$

such that

$$u_i + v_j \geq a_{ij} \quad \forall\, i, j.$$

At any optimal solution (x^*_{ij}) to the primal, we have by complementary slackness

$$u^*_i + v^*_j = a_{ij} \quad \text{if } x^*_{ij} > 0.$$

Further $\sum_i u^*_i + \sum_j v^*_j$ is the worth of the grand coalition.

Indeed, the core imputations are precisely any optimal solution $(u_1, ..., u_n; v_1, ..., v_n)$ for the seller-buyer pairs of the above dual LP and is therefore non-empty.

Example 6

Find a suitable core imputation for the horse market with 6 sellers and 5 buyers where sellers' base prices and buyers' ceiling prices are

$S_1 \geq 200$	$B_1 \leq 350$
$S_2 \geq 230$	$B_2 \leq 300$
$S_3 \geq 240$	$B_3 \leq 260$
$S_4 \geq 245$	$B_4 \leq 250$
$S_5 \geq 255$	$B_5 \leq 240$
$S_6 \geq 270.$	

We first look for a price p at which demand = supply.

Trial 1: $p = 210 \implies$ exactly 1 seller is interested and 5 buyers are interested at this price. Thus since supply is less than demand, price increases. Say $p = 250$. Exactly 4 sellers are interested, namely S_1, S_2, S_3, S_4. At this price 4 buyers are interested namely B_1, B_2, B_3, B_4. One buyer B_4 has to have no gain at this price. It favors buyers. When $p = 245$, again S_1, S_2, S_3, S_4 are willing and B_1, B_2, B_3, B_4 are willing at this price. Thus, we have 2 prices $p_* = 245$ and $p^* = 250$ and in this range $245 \leq p \leq 250$, we have matching demand and supply. Here, price $p = 245$ favors all buyers and 250 favors all sellers. With $p = 245$ favoring all buyers, the arrangement is say S_1B_1; S_2B_2; S_3B_3; S_4B_4 as the match.

Let $(u_1, u_2...u_6; v_1, v_2...v_5)$ be a dual optimal solution. We have

$$\begin{aligned} u_1 + v_1 &= 350 - 200 \\ u_2 + v_2 &= 300 - 230 \\ u_3 + v_3 &= 260 - 240 \\ u_4 + v_4 &= 250 - 245 \end{aligned}$$

$$\begin{aligned} \text{At } p = 245 : S_1 &\text{ gets a gain of } 245 - 200 = 45 = u_1 \\ : B_1 &\text{ gets a gain of } 350 - 245 = 105 = v_1 \\ : S_2 &\text{ gets a gain of } 245 - 230 = 15 = u_2 \\ : B_2 &\text{ gets a gain of } 300 - 245 = 55 = v_2 \\ : S_3 &\text{ gets a gain of } 245 - 240 = 5 = u_3 \\ : B_3 &\text{ gets a gain of } 260 - 245 = 15 = v_3 \\ : S_4 &\text{ gets a gain of } 245 - 245 = 0 = u_4 \\ : B_4 &\text{ gets a gain of } 250 - 245 = 5 = v_4 \end{aligned}$$

$u_5 = 0 \;\; v_5 = 0 \;\; u_6 = 0.$

The imputation (45, 15, 5, 0, 0, 0; 105, 55, 15, 5, 0) is uniformly good for all buyers. The nucleolus is the midpoint of these extreme ends of the core with p=245 and p=250. It is (47.5, 17.5, 7.5, 2.5, 0, 0; 102.5, 52.5.12.5, 2.5, 0).

The following theorem gives a necessary and sufficient condition for any TU game (v, N) to possess a non-empty core.

Theorem 4.6 (Bondareva and Shapley) *The core of a TU game* (v, N) *is non-empty if and only if the linear program*

$$\begin{aligned} \textit{min} \sum_{i \in N} x_i \\ \textit{subject to} \sum_{i \in S} x_i \geq v(S) \;\; \forall \;\; S \subset N \end{aligned}$$

has a minimum $z^* \leq v(N)$.

Proof Let x^* be any minimizing element. Clearly x^* is in the core. Suppose x is any core element. Then x satisfies $\sum_{i \in S} x_i \geq v(S)$ and $\sum_{i \in N} x_i = v(N)$. Thus the minimum must be $z^* \leq v(N)$. □

Consider the dual to the above linear program. It is given by

$$\begin{aligned} \max \sum_{S \subset N} \lambda_S v(S) \\ \text{subject to} \sum_{S \ni i} \lambda_S = 1 \;\; \forall \;\; i \in N \\ \lambda_S \geq 0 \;\; \forall \;\; S \subset N. \end{aligned}$$

When the core is non-empty, we have an optimal solution to the primal and hence by the duality theorem we have an optimal solution to the above dual program. Further at any optimal solution λ_S^* of the dual, we have

$$\sum \lambda_S^* v(S) \leq v(N).$$

While the theorem per se is only of theoretical interest, the dual problem's extreme solutions have a certain elegant combinatorial structure.

Definition 4.3 Let $S_1, S_2 ... S_m$ be a collection of non-empty subsets of $N = \{1, 2, ..., n\}$. We call them a *balanced collection* if for some $y_1, y_2 ... y_m > 0$

$$\sum_{S_j \ni i} y_j = 1 \quad \text{for each} \quad i \in N.$$

Intuitively, if we think of $S_1, S_2, ..., S_n$ as coalitions and consider only those among them that include player i, he allocates his unit time among these coalitions devoting y_j proportion of his available time to be part of coalition S_j. In a sense it is a generalized partition of N.

Example 7

$N = \{1, 2, 3, 4\}$ $S_1 = (1, 2)$ $S_2 = (1, 3)$ $S_3 = (1, 4)$ $S_4 = (2, 3, 4)$. This is a balanced collection with $y_1 = \frac{1}{3}$ $y_2 = \frac{1}{3}$ $y_3 = \frac{1}{3}$ $y_4 = \frac{2}{3}$.

We call a balanced collection minimal if it has no proper subcollection which is balanced.

Theorem 4.7 *The extreme points of the polyhedra defined by*

$$\begin{gathered} \sum_{S \ni i} \lambda_S = 1 \quad \forall\, i \in \{1, 2, ..., n\} \\ \lambda_S \geq 0 \quad \forall \ S \subset N \end{gathered} \tag{4.1}$$

are the balancing vectors of the minimal balanced collections.

Proof Clearly, $\mathscr{S} = \{S : \lambda_S > 0\}$ is a balancing vector for the collection $\mathscr{S}$. Suppose $\mathscr{S}$ is not minimal, then we have a subcollection $\mathscr{B}$ with balancing vector $\{\mu_S : S \subset N\}$. We know $\mu_S > 0 \implies \lambda_S > 0$. Thus, for ϵ small $\omega = (1 - \epsilon)\lambda + \epsilon\mu$, $\omega' = (1 + \epsilon)\lambda - \epsilon\mu$ satisfy the constraints (4.1). But $\omega \neq \omega'$ as $\omega_S < \omega'_S$ when $S \notin$ subcollection $\mathscr{B}$. Now $\lambda = \frac{\omega + \omega'}{2}$ and λ is not an extreme point.

Conversely let $\mathscr{S}$ be a minimal balanced collection. In case λ is not an extreme point, we can find a $\omega \neq \omega'$ and both satisfy (4.1). Clearly, if $\lambda_S = 0 \implies \omega_S = \omega'_S = 0$ (as ω'_S are ≥ 0). Let $\mathscr{B} = \{S : \omega_S > 0\}$ and $\mathscr{B}' = \{S : \omega'_S > 0\}$. They are

both subcollections of $\mathscr{S}$ and since $\mathscr{S}$ is minimal balanced, $\mathscr{B} = \mathscr{B}' \implies \omega = \omega' = \lambda$. This contradiction shows that λ must be extreme to (4.1). □

Remark 4.5 One can check that at most n of the λ_S's are positive.

Theorem 4.8 *A TU game (v, N) has non-empty core if and only if for each minimal balanced collection $\mathscr{S} = \{S_1, \ldots S_m\}$ with balancing vector $(\lambda_1, \ldots, \lambda_m)$ satisfies*

$$\sum_j \lambda_j v(S_j) \leq v(N).$$

Proof Since the associated LP

$$\begin{aligned} &\max \sum_{S \subset N} \lambda_S v(S) \\ &\text{subject to} \sum_{S \geq 0} \lambda_S = 1 \quad i \in N \\ &\lambda_S \geq 0 \end{aligned}$$

attains its optimal value at an extreme point, we have the required result. □

4.4 The Lexicographic Center [20]

Given a TU game (v, N) and the imputation set X^o, the satisfaction $f(x, S) = x(S) - v(S)$ for any coalition S may vary depending on the imputation x. However, $f(x, \phi) \equiv 0$ and $f(x, N) \equiv 0$ for $x \in X^o$. Let $\Sigma^o =$ all coalitions of N other than the empty coalition ϕ and the grand coalition, N. Consider on X^o the function

$$\theta(x) = \min_{S \in \Sigma^o} f(x, S).$$

Since $f(x, S)$ is affine linear, $\theta(x)$ as a minimum of affine linear functions is a concave function and also $\theta(x)$ is continuous on X^o. Let $\max_{x \in X^o} \theta(x) = \alpha'$. Now consider the set $X' \subset X^o$ where

$$X' = \{x \in X^o : \min_{S \in \Sigma^o} f(x, S) = \alpha'\}.$$

We can check the following. Suppose $x^*, x^o \in X'$. We know $\theta(x^*) = \min_{S \in \Sigma^o} f(x, S) = \alpha'$. Similarly $\theta(x^o) = \alpha'$. Being a concave function on X^o, $\forall\ 0 \leq \lambda \leq 1$, $\theta(\lambda x^* + (1 - \lambda)x^o) \geq \lambda\alpha' + (1 - \lambda)\alpha'$. However, $\max_{x \in X^o} \theta(x) = \alpha'$ and so the imputation $\lambda x^* + (1 - \lambda)x^o$ is also in X'. That is, X' is a closed convex set. Let $\Sigma_1 = \{S \in \Sigma^o : f(S, x) = \alpha'\ \forall\ x \in X'\}$. We claim $\Sigma_1 \neq \phi$. Suppose not. Then given any $S_j \in \Sigma^o$

there exists an $x_j \in X'$ such that $f(x_j, S_j) \neq \alpha'$. Since $x_j \in X'$, $\theta(x_j) = \alpha'$ and so $f(x_j\ S_j) \geq \theta(x_j) = \alpha'$ and so $f(x_j\ S_j) > \alpha'$.

If we define $\bar{x} = \frac{1}{m}\sum x_j$, then since X' is a convex set and since $x_j \in X'$, $\bar{x} \in X'$. Thus, we should have a coalition $\tilde{S}$ such that $f(\bar{x}, \tilde{S})) = \theta(\bar{x}) = \alpha'$. However, $\tilde{S} \in \Sigma^o$ and since $\tilde{S}$ is one of the S_j's in Σ^o, for some x_j, we have $f(x_j, \tilde{S}) > \alpha'$ while $f(x_k, \tilde{S}) \geq \alpha' \implies f(\bar{x}, \tilde{S}) > \alpha'$ a contradiction. Hence, $X' \subset X^o$ and $\Sigma' = \Sigma^o \backslash \Sigma_1 \neq \phi$ and $\Sigma^o \supset \Sigma'$.

We can inductively define for each $k = 2 \ldots k^*$:

$$X^k = \{x \in X^{k-1} : \min_{S \in \Sigma'} f(S, x) = \alpha^k\}$$
$$\Sigma_k = \{S \in \Sigma^{k-1} : f(S, x) = \alpha^k \ \forall\ x \in X^k\}$$
$$\Sigma^k = \Sigma^{k-1} \backslash \Sigma_k$$

and k^* is the first value of k for which $\Sigma^k = \phi$.

We claim that $\alpha' < \alpha^2 < \cdots < \alpha^k < \alpha^{k+1}$. Using $\bar{x} \in X^k$ we have $f(\bar{x}, S) > \alpha^k\ \forall\ S \in \Sigma^k$. Thus, $\alpha^{k+1} \geq \min_S f(\bar{x}, S) > \alpha^k$.

Theorem 4.9 *The lexicographic center of a game consists of a unique point, and it is the nucleolus.*

Proof We know that the lexicographic center $X^{k^*} \neq \phi$. The satisfaction of each coalition S is constant in X^k if $S \in \Sigma_k$ and hence constant in X^{k^*}. As this holds for singleton coalitions, we see that X^{k^*} can't have more than one point. We will show that the lexicographic center coincides with the nucleolus. To prove this we need the following. □

Lemma 4.3 *For any $1 \leq k \leq k^*$, if $x \in X^k$ and if $y \in X^{k-1} \backslash X^k$, then $\theta(y) \underset{L}{\prec} \theta(x)$.*

Proof Consider $\Sigma_1,\ \Sigma_2 \ldots \Sigma_{k-1}, \Sigma^{k-1}, \{\phi, N\}$. They constitute a partition of 2^N. Ignore ϕ *and* N as $\theta(x)$ and $\theta(y)$ have the same value zero for these coordinates. Since $x \in X^k$ and $y \in X^{k-1} \backslash X^k$, we have

$$f(x, S) \geq \alpha^k \text{ and } f(y, S) \geq \alpha^{k-1}.$$

For all S in Σ_n ; $n = 1, 2, k-1$. Further $\forall\ S \in \Sigma^{k-1}$, we have $f(x, S) \geq \alpha^k > \alpha^{k-1}$ and $f(y, S) \geq \alpha^{k-1}$. However, since $y \notin X^k$ we have $\alpha^k > f(R, y) \geq \alpha^{k-1}$ for some $R \in \Sigma^{k-1}$. Hence $\theta(x) \underset{L}{\succ} \theta(y)$. □

Example 8

Find the nucleolus for the following TU game $v(1) = v(2) = v(3) = 0,\ v(1, 2) = v(1, 3) = 30,\ v(2, 3) = 80,\ v(1, 2, 3) = 100$.

We will consider $\alpha' = \max_x \min_{\substack{S \neq \phi \\ \{1,2,3\}}} f(x, S)$. Let $\theta(x) = \min_S f(x, S)$, then $\alpha' = \max \theta(x)$. Thus it reduces to the following LP:

$$\begin{aligned} &\max \theta \\ x_1 \geq \theta, x_2 \geq \theta,\ x_3 \geq \theta,\ x_1 + x_2 - 30 \geq \theta,\ x_1 + x_3 - 30 \geq \theta \\ &\text{and}\ \ x_2 + x_3 - 80 \geq \theta,\ x_1 + x_2 + x_3 = 100. \end{aligned}$$

Solving the LP, one gets $\theta = 10$ and an optimal solution is $x_1^* = 10, x_2^* = 60, x_3^* = 30$. However, the constraints $x_1^* \geq 10,\ x_2^* + x_3^* \geq 90,\ x_1^* + x_2^* + x_3^* = 100 \implies x_1^* = 10,\ x_2^* + x_3^* = 90$. It is so for any optimal solution as it should satisfy the constraints for $\theta = 10$.

Thus $\Sigma_1 = (\{1\}, \{2, 3\})$. Our next LP corresponds to

$$\begin{aligned} &\max \theta \\ x_1 = 10, x_2 + x_3 = 90,\ x_2 \geq \theta,\ x_3 \geq \theta,\ x_1 + x_2 - 30 \geq \theta,\ x_1 + x_3 - 30 \geq \theta. \end{aligned}$$

Let α^2 be the next optimal value. We can check that $\alpha^2 = 25$. Now our constraints give

$$x_1 = 10,\ x_1 + x_2 \geq 30 + \alpha^2 = 55,\ x_1 + x_3 \geq 30 + \alpha^2 = 55.$$

Thus $x_2 \geq 45,\ \ x_3 \geq 45$. This shows that since $x_2 + x_3 = 90,\ \ x_1 = 10,\ \ x_2 = 45,\ x_3 = 45$. We get $X^2 = X^3 = \{(10, 45, 45)\}$, $\Sigma_2 = (\{1, 2\}, \{1, 3\})$ and $\Sigma_3 = (\{2\}, \{3\})$ and we have reached the nucleolus $= X^3 = (10, 45, 45)$.

Example 9

Consider the game with $v(1) = v(2) = v(3) = 0,\ v(1,\ 2) = v(1,\ 3) = 0,\ v(2,\ 3) = 10,\ v(1, 2, 3) = 6$. Find the nucleolus.

Remark 4.6 We notice $v(1\ 2\ 3) < v(2\ 3)$. In fact this game has empty core. We can solve the LP

$$\begin{aligned} &\max \theta(x) \\ x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_1 \geq \theta, x_2 \geq \theta, x_3 \geq \theta \\ x_1 + x_2 \geq \theta, x_1 + x_3 \geq \theta \text{ and } x_2 + x_3 - \theta \geq 10, x_1 + x_2 + x_3 = 6 \end{aligned}$$

giving optimal $\theta = -4$. Thus, the first LP's optimal value $\alpha' = -4$ and $\Sigma_1 = \{(2, 3)\}$ $\Sigma_2 = \{1\}$ and $\Sigma_3 = \{2\}, \{3\}, \{1\ 2\}, \{1\ 3\}$ giving $X' = X^2 = \{x : x_1 = 0\} \cap I$ and $X^3 = \{(0, 3, 3)\}$. This is the nucleolus.

The problem involves solving a sequence of linear programs and each iteration leads to at least one new coalition that has constant satisfaction on the convex subset of the imputation set. The iterations may grow exponentially where even the LP's considered each time will have an exponential number of linear inequality constraints of the type $x(S) - v(S) \geq \theta$ where S's vary over a subcollection of coalitions. In general the problem becomes NP-hard.

We will concentrate on special classes of TU games (v, N) where the data defining the game is a polynomial in the number of players. For example, we saw in the horse market, the only coalitions that matter are buyer-seller pairs.

We are ready to introduce the so-called assignment games.

4.5 Assignment Games [27]

Assignment games with side payments are models of certain two-sided markets. One such example was already discussed in the form of a horse market where all horses are identical except for price expectation by sellers and ceiling prices by buyers. A clear motivation can be given by the so-called real estate game.

House owners H_1 and H_2 are keen on finding buyers who would pay at least $\$200K$ and $\$230K$ for their houses. A real estate agent is approached by them, and they are willing to give a commission proportional to the price gain. Two buyers who want to buy just one house also approach the same agent. The agent wants to know their budgets. The buyers say that it will depend on the house. The agent shows the two houses and buyer 1 values H_1 as at most $\$290K$ and values H_2 as at most $\$260K$. Buyer 2 values H_1 as at most $\$300K$ and values house H_2 as at most $\$280K$.

They are willing to give commission proportional to price savings. Thus, if H_1 is sold to B_2 at a price p with $200 < p < 300$, seller H_1 gains $p - 200$ and buyer B_2 gains $300 - p$. We will assume the commission to the agent is $k(p - 200) + k(300 - p) = k(300 - 200)$. W.l.g. $k = 1$ and the sale fetches a commission $= 300 - 200$ for the real estate agent. Thus, the following matrix summarizes these as coalitional worths:

$$\begin{array}{c} \\ Seller\ H_1 \\ Seller\ H_2 \end{array} \begin{array}{c} \begin{array}{cc} Buyer\ B_1 & Buyer\ B_2 \end{array} \\ \begin{pmatrix} 90 & 100 \\ 20 & 50 \end{pmatrix} \end{array}.$$

We can represent

$$\begin{aligned} v(H_1, B_1) &= 90 \quad v(H_1, B_2) = 100 \\ v(H_2, B_1) &= 20 \quad v(H_2, B_2) = 50. \end{aligned}$$

Clearly, any other coalition like $v(H_1,\ H_2) = v(B_1,\ B_2) = v(H_1) = v(H_2) = v(B_1) = v(B_2) = 0$.

Clearly, we can define $v(H_1,\ H_2,\ B_1) = \max(90, 20) = \max(v(H_1\ B_1), v(H_2\ B_1))$.

Similarly, $v(B_1\ B_2\ H_2) = \max(v(H_2\ B_1), v(H_2\ B_2)) = \max(20, 50) = 50$.

Finally, $v(H_1,\ H_2\ B_1,\ B_2) =$ the optimal assignment which in this case is $v(H_1,\ B_1) + v(H_2,\ B_2) = 140$.

Suppose we denote the single seller and single buyer coalitions as the columns (seller) and rows (buyer) of a matrix and the satisfaction at any imputation for seller-buyer pair as the matrix entries. For example, consider

$$\begin{array}{c} \\ u_1 \\ u_2 \end{array} \begin{array}{c} \begin{array}{cc} v_1 & v_2 \end{array} \\ \begin{pmatrix} 90 & 100 \\ 20 & 50 \end{pmatrix} \end{array}.$$

We restrict u's and v's to satisfy $u_1 \geq 0\ u_2 \geq 0\ v_1 \geq 0\ v_2 \geq 0\ (u_1 + v_1) + (u_2 + v_2) = 140$. Since $v(H_1) = v(H_2) = v(B_1) = v(B_2) = 0$, the singleton coalition satisfactions are also same as $u_1,\ u_2,\ v_1,\ v_2$, etc. However $f(H_1,\ B_1) = u_1 + v_1 - v(H_1,\ B_1) = u_1 + v_1 - 90$. Similarly, $f(H_1,\ B_2) = u_1 + v_2 - 100,\ f(H_2,\ B_1) = u_2 + v_1 - 20,\ f(H_2,\ B_2) = u_2 + v_2 - 50$. For example, we can reward all the coalitional worth to sellers, after $H_1\ B_1$ and $H_2\ B_2$ are paired. We get say

$$\begin{array}{c} \\ 90 \\ 50 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 0 \end{array} \\ \begin{pmatrix} 0 & -10 \\ 30 & 0 \end{pmatrix} \end{array}.$$

Thus, we notice that the coalitional satisfaction for singleton and seller-buyer pair are given above.

Theorem 4.10 *The nucleolus is determined only by singleton satisfaction and buyer-seller coalitional satisfaction. (For a proof, see Solymosi and Raghavan [29])*

The imputation (90, 50; 0, 0) has a satisfaction vector for all singleton and buyer-seller coalitions given by $=$ (90, 50, 0, 0, 0, −10, 30, 0). The rest of the satisfactions are irrelevant to nail down the nucleolus. Rearranging the vector above in increasing order, we get the vector (−10, 0, 0, 0, 0, 30, 50, 90). Thus, the worst hit coalition is $(H_1,\ B_2)$. The worth of the grand coalition is the worth of the optimal assignment $(H_1\ B_1,\ H_2\ B_2)$. We know that since the game has a non-empty core and since the nucleolus is in the core, satisfaction will not change for any imputation in the core for not only the grand coalition but also for those which satisfy $f(x, H_1B_1) = 0,\ \ f(x, H_2B_2) = 0$. We call them settled coalitions where satisfaction is constant over the core and it is currently 0.

We compute

$$\min_{S \neq (H_1,\ B_1)(H_2,\ B_2)} f(u_1, v_1, u_2, v_2, S) = -10.$$

The next intuitive step is to move from the current imputation to a new imputation in a chosen direction that improves the satisfaction of the worst-hit coalition.

Suppose we chose the direction $d = (0, 1; 0, -1)$. For $\beta = -10$, this gives a new imputation

$$(90, 50; 0, 0) - 10(0, 1; 0, -1) = (90, 40; 0, 10).$$

Observe that satisfactions of singleton and buyer-seller coalitions are non-negative. It is also a core imputation. In reality we have reached an extreme corner of the core. Suppose we choose the new direction $(1, 2; -1, -2)$. Again with $\beta = -10$, we get a new imputation $(90, 40; 0, 10) - 10(1, 2; -1, -2) = (80, 20; 10, 30)$. We can check that this new imputation gives a satisfaction vector for singletons and buyer-seller coalition vector $(80, 20; 10; 30, 0, 0, 10, 10)$ which when arranged in increasing order gives $(0, 0, 10, 10, 10, 20, 30, 80)$.

Now moving in the direction $(1, 1; -1, -1)$ with $\beta = -10$ gives $(80, 20, 10, 30) - 10(1, 1; -1, -1) = (70, 10, 20, 40)$, and this imputation gives satisfaction vector for singleton and buyer-seller coalitions $(70, 10; 20, 40, 0, 0, 10, 10)$ which when arranged in increasing order gives

$$(0, 0, 10, 10, 10, 20, 40, 70) \succ (0, 0, 10, 10, 10, 20, 30, 50).$$

In fact we have reached the nucleolus. To claim that this is the nucleolus has to do with the nature of the core in assignment games.

We chose directions with no real explanations. We also chose the new directions with no real explanations. We terminated our calculation with no reason as to why the algorithm terminates now.

For assignment games, a method based on the linear programming technique is not well suited since the combinatorial structure of the characteristic function cannot be effectively translated into a continuous problem formulation. When solving the optimal assignment problem, the ordinary primal simplex method encounters difficulties caused by degeneracy. It is outperformed by well-known algorithms like Kuhn's Hungarian method of (see [4]).

In an assignment game, the first step is to define the worth of the grand coalition. Intuitively, it corresponds to the optimal pairing of the sellers and buyers by the real estate agent that will fetch him the maximal commission. This occurs for the value of the optimal assignment. Suppose we associate a graph with vertices as the optimal assignment. It is quite possible that the assignment could either keep some houses unsold or some buyers unable to buy any house. By introducing another dummy vertex, we can associate with any m seller, n buyer assignment game with $k = \min(m, n)$, a graph with $k + 1$ vertices where when $k < m$ a seller is unable to sell his house or when $k < n$ a buyer is unable to buy any house.

In such a situation, we introduce a dummy vertex 0 such that we have $k + 1$ vertices $0, 1, 2, \ldots, k$. If we have different optimal assignments, then the player common to two different optimal assignments gives us an option to connect two distinct assignments with an edge. Such a graph can be decomposed into several connected components (i.e. maximal complete subgraphs). Having formed such connected components, *the key problem is to find more binding constraints* between players related to different components that would introduce new edges to the graph and reduce the number of components.

Without loss of generality, we can assume that there are enough buyers to buy all the houses in the market. That is, $m = |\bar{M}| \leq |\bar{N}| = n$ where $\bar{M}$ is the set of real house sellers and $\bar{N}$ is the set of real house buyers. It will prove convenient to introduce a fictitious seller and a fictitious buyer who are present in any coalition. Thus $M = \bar{M} \cup \{0\}$ and $N = \bar{N} \cup \{0\}$. We can think of any coalition (S, T) as $S \subset M, \ \ 0 \in S$ and $T \subset N, \ \ 0 \in T$. With this notation, we write $(0, 0)$ for the empty coalitions, $(i, 0)$ for row player $i \in \bar{M}$ and $(0, j)$ for the column player $j \in \bar{N}$. We assume $a_{00} = a_{i0} = a_{0j} = 0$ as the coalitional worth $\forall \ i \in \bar{M}, \ \ j \in \bar{N}$ in the augmented profit matrix $A = (a_{ij})_{(M\ N)}$. Also the payoff is fictitious too with $u_0 = v_0 = 0$ in any payoff vector $(u, v) = (u_0, u_1, \ldots u_m; v_0, v_1, \ldots, v_n)$.

By an (S, T) matching μ, we mean a correspondence between S and T that induces a matching on $(\bar{S}, \bar{T})$ where $\bar{S} = S \backslash \{0\}$ and $\bar{T} = T \backslash \{0\}$ and $(0, 0) \in \mu$. Any player who is not matched to a real player is tied to a fictitious player.

Thus, the characteristic function of an assignment game with augmented profit matrix $A = (a_{ij})_{M,N}$ is given by

$$v(S, T) = \max \left\{ \sum_{(i,j) \in \mu} a_{ij} \ \ \forall \ (S, T) \text{ matching } \mu \right\}.$$

Since $a_{ij} \geq 0 \ \forall \, i, \, j$, we can assume that μ is a full (S, T) matching. That is $\bar{\mu}$ (match between real players in S, T exhausts either $\bar{S}$ or $\bar{T}$ of real players).

Formally, let $f_{ij}(u, v)$ be the satisfaction to coalition (i, j) when players are paid according to (u, v) (here $i =$ seller i, $j =$buyer j). That is $f_{ij}(u, v) = u_i + v_j - a_{ij}$.

Theorem 4.11 *The core C of an assignment game $A(M, N)$ is given by*

$$C = \{(u, v) : \sum_{\substack{i \in M \\ j \in N}} f_{ij}(u, v) = 0, \ \ f_{ij}(u, v) \geq 0 \ \forall \ (i, j) \in (M, N)\}.$$

Shapley and Shubik [27] showed that every element (u, v) in C is an *optimal solution* to the dual of the linear program

$$\begin{aligned} &\max \sum_i \sum_j a_{ij} x_{ij} \\ &\text{subject to } \sum_{j \in \bar{N}} x_{ij} \leq 1; \ \ \sum_{i \in \bar{M}} x_{ij} \leq 1 \ \ \forall \ \ (i, j) \in \bar{M} \times \bar{N} \\ &\qquad x_{ij} \geq 0 \ \ \forall \ \ (i, j). \end{aligned}$$

At the same time, the dual to this primal is

$$\min \sum u_i + \sum v_j$$
$$\text{such that } u_i + v_j \geq a_{ij} \quad \forall\, i \in \bar{M},\ j \in \bar{N}$$
$$u_i, v_j \geq 0 \qquad \forall i, j.$$

The maximum for the primal is also the same as $\max_{\sigma} \sum_i a_{i\sigma(i)}$ where σ is any permutation. This is a consequence of the following theorem of Birkhoff-Von Neumann [4].

Theorem 4.12 *For the convex set X of all $n \times n$ doubly stochastic matrices, i.e.*

$$X = \{(x_{ij}) \ :\ x_{ij} \geq 0 \ \sum_j x_{ij} = \sum_i x_{ij} = 1\}$$

, its set of extreme points are the set of all permutation matrices

$$\Omega_n = \{p^{\sigma} : p^{\sigma}_{ij} = 1 \text{ if } j \in \sigma(i)$$
$$= 0 \text{ if } j \notin \sigma(i)\}$$

where $\sigma = (1, 2..., n) \to (\sigma(1), \sigma(2), ..., \sigma(n))$ is a permutation of $\{1, 2, ..., n\}$.

By complementary slackness, this says that optimally matched pairs share the exact coalitional worth in any core allocation. That is, for an optimal (u^*, v^*) of the dual, $u^*_i + v^*_j = a_{ij}$ when $j = \sigma^*(i)$ and p^{σ^*} is optimal for the primal. But unmatched pairs receive nothing in any core allocation.

The lexicographic center for assignment games gives a truly efficient iterative process to reach the nucleolus in polynomial time. The reasons are many:

(i) The combinatorial nature of assignments
(ii) An easy way to choose a direction in each iteration
(iii) Strictly shrinking set of imputations in each iteration with the dimension of the convex subset of imputation getting reduced by at least one after each iteration
(iv) The lattice structure of the core.

We can essentially ignore all but the singleton and buyer-seller coalitions for applying Maschler-Peleg-Shapley's iteration process as follows: see Mashler et al. [20], Solymosi and Raghavan [29].

Let σ be a fixed optimal (M, N) matching for an assignment game $A(M, N)$. We construct a sequence $(\Delta^o, \Sigma^o), (\Delta^1, \Sigma^1), \ldots, (\Delta^{\rho+1}, \Sigma^{\rho+1})$ that are partitions of (M, N) such that $\Sigma^o \supset \Sigma^1 \supset \cdots \supset \Sigma^{\rho+1} = \phi$ and a nested sequence $X^o \supset X^1 \supset \cdots \supset X^{\rho+1}$ of sets of payoff vectors iteratively defined as follows: Let $\Delta^o = \sigma$ and $\Sigma^o = (M, N) \backslash \sigma$. Let us start with imputation (u^o, v^o) where $u^o_i = a_{i\sigma(i)}\ \forall\, i \in M,\quad v^o_j = 0,\ \ j \in N$. Let $\alpha^o = \min_{(i\ j) \in \Sigma^o} f_{ij}(u^o, v^o) = \min_{(i\ j) \in \Sigma^o} u^o_i + v^o_j - a_{ij}$. To start with $X^o = \{(u, v) : (u, v) \geq (0, 0),\ \ u_i + v_j - a_{ij} = 0 \ \text{ if } (i\ \ j) \in \Delta^o,\ \ u_i + v_j - a_{ij} \geq \alpha^o\ \forall\ (i\ \ j) \in \Sigma^o\}$.

We define recursively for $r = 0, 1, 2, ..\rho$.

(i) $\alpha^{r+1} = \max_{(u,v)\in X^r} \min_{(i\ j)\in\Sigma^r} (u_i + v_j - a_{ij})$

(ii) $X^{r+1} = \{(u, v) : (u, v) \in X^r,\ \min_{(i\ j)\in\Sigma^r} (u_i + v_j - a_{ij}) = \alpha^{r+1}\}$

(iii) $\Sigma_{r+1} = \{(i, j) \in \Sigma^r :\ u_i + v_j - a_{ij}$ is a constant on $X^{r+1}\}$

(iv) $\Sigma^{r+1} = \Sigma^r \backslash \Sigma_{r+1}\ ;\ \Delta^{r+1} = \Delta^r \cup \Sigma_{r+1}$.

Here ρ is the last index r for which $\Sigma^r \neq \phi$. We call $X^{\rho+1}$ the lexicographic center of $A(M, N)$.

Remark 4.7 The lexicographic center is independent of the choice of optimal assignment. In fact if $\alpha^o \leq 0$, then the core $\subset X^o$. In case $\alpha^o \geq 0$, we have $X^o \subset$ core. Further $\alpha^1 \geq 0$ and that $X^1 \subset$ core. Also when $\alpha_1^1 = 0$, $X^1 = C$ (for proof details, see Solymosi and Raghavan [29]). Thus at the end of the first iteration, we always reach the core.

Lemma 4.4 *For $0 \leq r \leq \rho$ if $(u, v) \in X^{r+1}$ and $(y, z) \notin X^{r+1}$ and $(y, z) \in X^r$, then lexicographically $(u, v) \succ (y, z)$.*

Proof Same as Lemma 4.3 for this special case. □

Theorem 4.13 *The lexicographic center of an assignment game coincides with the nucleolus.*

Proof Let $X^{\rho+1} = \{(\overline{u}, \overline{v})\}$. We have $(\overline{u}, \overline{v}) \succ (y, z) \in I$ where $(\overline{u}, \overline{v}) \neq (y, z)$. Therefore the nucleolus $\nu = (\overline{u}, \overline{v})$. □

At any stage of the iteration, we have a particular $\Delta^r = \Delta$, $\Sigma^r = \Sigma$ of (M, N) and set $X^r = X$, the current feasible payoff set. We call Δ *settled coalitions*, and Σ *unsettled coalitions*. The reason that a coalition $(i, j) \in \Delta$ is settled is because its satisfaction is constant over the entire feasible set X. Coalition $(i, j) \in \Sigma$ is called unsettled because the satisfaction of $(i, j) \in \Sigma$ is not a constant over the entire set X.

Suppose $(i, j) \in \Delta$ and also suppose $(i, \sigma(k)) \in \Delta$ where k is a seller and $\sigma(k)$ is the optimal buyer of house k. If so, we have the following important feature of the settled-unsettled partition.

Lemma 4.5 *(i)* $(i, \sigma(k)) \in \Delta \implies (k, \sigma(i)) \in \Delta$

(ii) $(i, \sigma(k)) \in \Delta, (k, \sigma(j)) \in \Delta \implies (i, \sigma(j)) \in \Delta$ *for all $i, j, k \in M$(sellers).*

Proof If $i \in \overline{M}$ (a real seller), there is a unique partner $\sigma(i) \in \overline{N}$ (a real buyer). For $0 \in M$,$\sigma(0) = \{0\} \cup \overline{N}\backslash\sigma(\overline{M})$ and so each column is partner of a row. We have also assumed $|M| \leq |N|$.

To show (i), observe $f_{i\sigma(k)} = u_i + v_{\sigma(k)} - a_{i\sigma(k)} \equiv$ constant over X (by assumption). Also $f_{i\sigma(i)} = u_i + v_{\sigma(i)} - a_{i\sigma(i)} \equiv 0$ on $X^o \supset X$. Thus

$$\begin{aligned} f_{i\sigma(i)} + f_{k\sigma(k)} - f_{i\sigma(k)} = & \\ & [u_i + v_{\sigma(i)} - a_{i\sigma(i)}] + [u_k + v_{\sigma(k)} - a_{k\sigma(k)}] - [u_i + v_{\sigma(k)} - a_{i\sigma(k)}] \\ & \equiv const\ C_1 \quad + \quad const\ C_2 \quad + \quad const\ C_3 \\ & \equiv u_k + v_{\sigma(i)} - [a_{i\sigma(i)} + a_{k\sigma(k)} - a_{i\sigma(k)}] \end{aligned}$$

over the whole set X. Thus $f_{k\sigma(i)} \equiv u_k + v_{\sigma(i)} - a_{k\sigma(i)} \equiv$ const on the whole set X. That is, $(k, \sigma(i)) \in \Delta$ when $(i, \sigma(k)) \in \Delta$.

(ii) The argument is similar. □

Remark 4.8 We notice that (i) and (ii) induce the *equivalence relation* $\sim$ where $i \sim k$ if and only if $(i, \sigma(k)) \in \Delta$). Thus, the rows i and k (sellers i *and* k) are tied by such constraints over the entire set X. (We did not check $i \sim i$ as it is the case by definition.)

Let us decompose the sellers and buyers into equivalence classes $M_o, M_1, \ldots, M_d$ and the equivalence class of partners $N_o, N_1, \ldots, N_d$ where $N_p = \sigma(M_p)$, $0 \leq p \leq d$. We stipulate $0 \in M_o$ and so $0 \in N_o$. We say $i \sim k$ if and only if $\sigma(i) \sim \sigma(k)$. Thus, such a partition of M and N induces the partition of (M, N) with blocks, namely $(M, N) = \underset{0 \leq p,q \leq d}{\cup} (M_p\ N_q)$. The following is the key theorem crucial for our algorithm.

Theorem 4.14 *The settled-unsettled partition* (Δ, Σ) *satisfies*

(i) $\Delta = \underset{0 \leq p \leq d}{\cup} (M_p, N_p)$ *and hence* $\Sigma = \underset{0 \leq p \neq q \leq d}{\cup} (M_p, N_q)$

(ii) If $(i, j), (k, l) \in (M_p, N_q)$ *then* $f_{ij} - f_{kl}$ *is the difference in their satisfaction which is a constant over the whole set* X.

Proof (i) If $(i, j) \in (M_p, N_p) \implies \exists\, k \in M_p$ such that $j = \sigma(k)$. Since $i \sim k$ we have $(i, j) = (i, \sigma(k)) \in \Delta$, hence $\Delta = \underset{0 \leq p \leq d}{\cup} (M_p, N_p)$.

To prove (ii), we claim when $i \sim k$ and $i, k \in M \implies u_i - u_k \equiv$ constant on X.

If $i \sim k \implies (i\ \sigma(k)) \in \Delta$, i.e. $f_{i\sigma(k)} = u_i + v_{\sigma(k)} - a_{i\sigma(k)} \equiv$ const on X. By subtracting $f_{k\sigma(k)}$ we get $u_i - u_k \equiv$ const on X. Similarly, if column j and l are tied then $v_j - v_l \equiv$ constant on X. Thus, if $(i, j), (k, l) \in (M_p\ N_q)$, $i \sim k$ and $j \sim l$. Then $(f_{ij} - f_{kl})(u, v) = (u_i - u_k) + (v_j - v_l) - (a_{ij} - a_{kl}) \equiv$const on X. As a consequence if $\Delta = \Delta^r = (M_o\ N_o) \cup \ldots (M_d, N_d)$ and $r \geq 1$, then $\dim X^r = d$. In fact, linear functions of the form $(f_{i0} - f_{k0})(u, v) = u_i - u_k,\ i, k \in M_p$ are constant on $X = X^r$. Thus for each p we can choose $|M_p| - 1$ linearly independent ones. In the settled class M_o, the function $u_i = u_i - u_0$ are constant on X^r. Thus we have $|M_o| + \sum_{p=1}^{d} (|M_p| - 1)$ linearly independent linear functions that are constant on $X^r = X$. Thus $\dim(X^r) \leq d$;

we claim $\dim(X^r) \geq d$. To see this: the function for $(r \geq 1)$ $f_{i0}(u, v) = u_i, i \in \underset{1 \leq p \leq d}{\cup} M_p$ are non-constant on X^r.

Thus all we can say is that the satisfaction of tied players must change by the same amount. However, there is no binding relationship between players of different classes. Thus, we have d linearly independent vectors in X^r and $\dim X^r \geq d$.

□

The following lemma helps us to locate the coordinates of the nucleolus one by one in the iterative process. This is due to X^r possessing lattice type properties.

Lemma 4.6 *Let $f_{ij}(v)$ be the satisfaction for coalition (i, j) at the nucleolus imputation v. Let $f_{ij}(v) = c_{ij}\ \forall\ (i, j) \in (M, N)$. Then for $r \geq 1$*

$$X^r = Y^r = \left\{(u, v) : f_{ij}(u, v) = c_{ij}\ \ \forall\ (i, j) \in \Delta^r,\ \ f_{ij}(u, v) \geq \alpha^r\ \forall (i, j) \in \Sigma^r\right\}.$$

See [29] for a proof based on induction.

We can extend the definition of Y^r to the case $r = 0$ by $X^0 = Y^0 \cap \{(u, v) : (u, v) \geq (0, 0)\}$. Thus for $0 \leq r \leq \rho$ and $d^r \leq \alpha \leq d^{r+1}$, we can define $Y(r, \alpha) = \left\{(u, v) : f_{ij}(u, v) = c_{ij}\, \forall\, (i, j) \in \Delta^r,\ f_{ij}(u, v) \geq \alpha\, \forall\, (i, j) \in \Sigma^r\right\}$.

As α increases $Y(r, \alpha)$ shrinks. For the maximum value d^{r+1} for α, $Y(r, \alpha) \neq \phi$. Even the sets $Y(r, \alpha)$ possess the following lattice type property, namely if $(u', v'), (u^2, v^2) \in Y(r, \alpha)$ so does $(u' \vee u^2, v' \wedge v^2)$. Similarly $(u' \wedge u^2, v' \vee v^2) \in Y(r, \alpha)$. The proof is easy and is omitted.

Thus for each α, we have two special vertices $\bar{u}_i = \bar{u}_i(r, \alpha)$ as the maximum payoff for each $v \in M$ and $\underline{v}_j = \underline{v}_j(r, \alpha)$ as the minimum payoff for $j \in N$. These two special vertices are the only ones we concentrate on for our algorithm.

Definition 4.4 Fix any $0 \leq r \leq \rho$ and $\alpha^r \leq \alpha \leq \alpha^{r+1}$. Let $\Sigma = \Sigma^r$. Choose the u-best corner of $X(r, \alpha)$. Among unsettled coalitions in Σ

$$\Sigma_{=} = \Sigma_r, \alpha = \{(i, j) \in \Sigma_{=}(r, \alpha) =\ \ f_{ij}(\bar{u}, \underline{v}) = \alpha\}$$

are called active unsettled coalitions. Those in $\Sigma_{>} = \Sigma_{>}(r, \alpha)$ with $f_{ij}(\bar{u}, \underline{v}) > \alpha$ are called passive coalitions.

The directed graph $G = G(r, d)$: Fix any $0 \leq r \leq p$ and $\alpha^r \leq \alpha \leq \alpha^{r+1}$. Let $d = d(r)$, where we have $d + 1$ equivalence classes $0, 1, 2, \ldots, d$. On these nodes, we connect node p and node q by an edge

$$E = \left\{(p, q);\ 0 \leq p \neq q \leq d,\ (M_p\ N_q) \cap \Sigma_{=}(r, \alpha) \neq \phi\right\}.$$

The graph is proper if it is acyclic and has no incoming arcs to 0, and is called improper otherwise.

The following is a fundamental result for the algorithm.

Theorem 4.15 *G is proper if and only if $d^r \leq \alpha \leq d^{r+1} \implies \alpha < d^{r+1}$.*

Proof The proof relies on moving inside the current set of feasible solutions with one end at the u-best v-worst corner and choosing the direction defined by the longest path ending in a node p. Let $l(p)$ be the length of the longest path. Let $s_i = -l(p)$ if $i \in M_p\ \ \forall\ i \in M$ and $t_j = l(q)$ if $j \in N_q\ \ \forall\ j \in N$. The payoff $(u', v') = (\bar{u}, \underline{v}) + \beta(s, t)$ gives satisfaction $f_{ij}(u', v') = \bar{u}_i + \underline{v}_j + \beta(s_i + t_j) - a_{ij}$ which is unaltered if $(i, j) \in \Delta$ as $i \in M_p\ j \in N_p$ (Notice $s_i + t_j = 0$ if $(i, j) \in \Delta$). If $(i,\ j) \in \Sigma_{=}$ then $p \neq q$ and $l(p) + 1 \leq l(q)$ and so $-l(p) + l(q) \geq 1$. Thus for small β, $f_{ij}(u', v') \geq \alpha + \beta \implies (u', v') \in X$ and $d^{r+1} > \alpha$.

Suppose G is improper. This could be due to either an arc from p to 0 or due to a cycle in the graph. Suppose we have an arc $(p, 0)$. Thus $\exists\ i \in M_p \ \ \& \ \ j \in N_o$ such that $\bar{u}_i + \underline{v}_j - a_{ij} = \alpha$. Since $(0, j) \in (M_o, N_o) \subset \Delta \quad f_{ij}(u, v) = v_j = c_{oj} \ \forall\ (u, v) \in X \implies \underline{v}_j = c_{oj}$.

On the other hand, $\bar{u}_i \geq u \implies f_{ij}(u, v) \leq f_{ij}(\bar{u}, \underline{v}) = \alpha \ \forall (u, v) \in X$. Thus $d^{r+1} \leq \alpha$. In case of a cycle $(p, q), (q, r), (r, p)$ (say), we have $f_{i_p j_q}(\bar{u}, \underline{v}) = \alpha$, $f_{i_q j_r}(\bar{u}, \underline{v}) = \alpha$ and $f_{i_k j_p}(\bar{u}\ \underline{v}) = \alpha$. When we sum them, we can replace this sum with a constant since each is at least α at any (u, v) and we get $3\alpha = \sum\limits_{Cycle} f_{ij}(\bar{u}, \underline{v}) = \sum\limits_{Cycle} f_{ij}(u, v) \geq 3\alpha \implies f_{i_p j_q}(u, v) = \alpha \ \forall\ (u, v) \in X$ and $d^{r+1} \leq \alpha$. □

As a consequence of the above theorem, if

$$\beta(r, \alpha) = \max\left\{r : \min_{(i,j)\in\Sigma} f_{ij}(\overline{u}, \underline{v}) + r(s, t) = \alpha + r\right\},$$

then $(\overline{u}, \underline{v}) + \beta(s, t)$ is the u-best corner of $X(r, \alpha + \beta)$.

The algorithm: We construct a finite sequence of payoff vectors where each one is the u-best corner of the current feasible set and the last one is the nucleolus. Finding the next payoff is translated to finding the longest path ending in each node which summarizes the interconnection between unsettled coalitions. It reveals certain relations which prevent any further improvement in their guaranteed satisfaction.

Imagine a federal government distributing a part of the collected GST between states. While individual states can grumble with what they get, a coalition of states say ruled by a party may raise their voices as discriminated. The approach of the federal government is to find the worst hit coalition with current share and redo distribution to make sure that those coalitions who are next in line as the second worst are not uniformly pulled down to be below the worst hit coalition's satisfaction due to the redistribution. Here is how the algorithm works:

(i) While $\Sigma \neq \phi$, do (2) − (12) (iteration 2),
(ii) Build the graph $G := G(r, \alpha)$
(iii) While G is proper, do (4) to (9) (step r, α)
(iv) Find direction (s, t)
(v) Find step size $\beta := \beta(r, \alpha)$
(vi) Update arcs in G
(vii) Update payoff $(u, v) : (u, v) + \beta(s, t)$
(viii) Update satisfactions $f_{ij},\ \forall\ (i, j) \in \Sigma$
(ix) Update guaranteed satisfaction level $\alpha := \alpha + \beta$
(x) Find to be settled coalitions $\overline{\Sigma}$; Σ_{r+1}
(xi) Update partition $\Sigma := \Sigma \backslash \overline{\Sigma},\ \Delta := \Delta \cup \overline{\Sigma}$
(xii) Set $r = r + 1$.

An illustrative example: There are 4 sellers (represented by rows $i = 1, 2, 3, 4$). There are 5 buyers (represented by columns $j = 1, 2, 3, 4, 5$). We introduce a dummy

buyer 0 as well as a dummy seller 0. Thus, any singleton coalition is identified as a buyer-seller coalition with a dummy seller or dummy buyer tied to them. Thus, we have the following augmented matrix with 5 rows and 6 columns. Since the number of equivalence classes is $p = 0, 1, 2, 3, 4$, we also indicate the equivalence class to which each buyer j or seller i belongs and we have the following matrix. The grand coalitional worth is the worth of the optimal assignment and they correspond to the boxed pairings

$$\begin{array}{cccc} \text{p} \rightarrow & & 0 & 0\ 1\ 2\ 3\ 4 \\ \downarrow & \begin{matrix} j: \\ i: \end{matrix} & 0 & 1\ 2\ 3\ 4\ 5 \\ 0\ 0 & & 0 & \left[\boxed{0}\ 0\ 0\ 0\ 0\right] \end{array}$$

$$\begin{array}{cc} 1 & 1 \\ 2 & 2 \\ 3 & 3 \\ 4 & 4 \end{array} \begin{bmatrix} 7 \\ 7 \\ 6 \\ 8 \end{bmatrix} \begin{bmatrix} 6 & \boxed{7} & 4 & 5 & 9 \\ 4 & 3 & \boxed{7} & 8 & 3 \\ 0 & 1 & 3 & \boxed{6} & 4 \\ 2 & 2 & 5 & 7 & \boxed{8} \end{bmatrix} = (a_{ij})$$

We initiate our algorithm with all the coalitional worth given to the sellers and nothing to the buyers. Thus, outside the real coalitional matrix we have sellers $i = 1, 2, 3, 4$ receiving

$$\overline{u} = \begin{bmatrix} 7 \\ 7 \\ 6 \\ 8 \end{bmatrix}$$

and buyers receiving $\underline{v} = \begin{bmatrix} 0\ 0\ 0\ 0\ 0 \end{bmatrix}$.

The optimal assignment is $(1, 2), (2, 3), (3, 4), (4, 5)$ between real seller-buyer pairing. They are the boxed entries. We have buyer $j = 1$ tied to player (seller 0) getting boxed $\boxed{0}$. In our next step, we write down the satisfaction for each coalition of seller-buyers. For example, the satisfaction for the coalition seller 1, buyer 1 is $\overline{u}_1 + \underline{v}_1 - a_{11} = 7 + 0 - 6 = 1$. We write down the satisfaction of seller 1 alone as $\overline{u}_1 + \underline{v}_0 - a_{10} = 7 + 0 - 0 = 7$. The following matrix summarizes the satisfaction details for $\overline{u}$ and $\underline{v}$ given as a blue column vector and blue row vector.

$$\underline{v}$$

$$\begin{array}{lll} & \begin{bmatrix} \boxed{0} & 0 & 0 & 0 & 0 \end{bmatrix} & s_i = \\ \overline{u} \quad \begin{bmatrix} 7 \\ 7 \\ 6 \\ 8 \end{bmatrix} & \begin{bmatrix} 1 & \boxed{0} & 3 & 2 & -2 \\ 3 & 4 & \boxed{0} & -1 & 4 \\ 6 & 5 & 3 & \boxed{0} & 2 \\ 6 & 6 & 3 & 1 & \boxed{0} \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} \quad \beta = 1 \\ t_j = & \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \end{bmatrix} & \end{array}$$

Boxed coalitions are the settled coalitions consisting of (seller 0, buyer 1), (seller 1, buyer 2), (seller 2, buyer 3), (seller 3, buyer 4) and (seller 4, buyer 5). These entries will not change anymore over the entire iterations. The most unhappy coalition is seller 1, buyer 5. We have equivalence classes $p = 0, 1, 2, 3, 4$. $j = 1$ is in N_o, $j = 2$ is in N_1, $i = 1$ is in M_1, etc. Introducing the graph with vertices

$$p = \boxed{0} \quad 1 \; 2 \; 3 \; 4$$

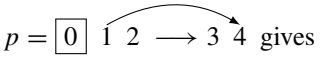
$i = 1,\ j = 5$ is in the equivalence class M_1, N_4 and so this active unsettled coalition gets an edge joining node 1 to node 4 as indicated above. The next worst satisfied passive coalition is $i = 2,\ j = 4$ and in terms of $[p, q]$ it is equal to $(2, 3)$. Here, $t_j = l(q)$ if $j \in N_q \implies t = (0, 0, 0, 0, 1)$ and $i \in M_p \implies s_i = -l(p)$ and $s = (0, 0, 0, -1)$.

We update the data with $\beta = 1$ which will improve the satisfaction of the coalition $i = 1,\ j = 5$ from -2 to -1 and we will have both $(i, j) = (1, 5)$ and $(i, j) = (2, 4)$ as active coalitions when updated as represented below (see * entries).

$$u'_i = \overline{u}_i - \beta s_i \; ; \; v'_j = \overline{v}_j + \beta t_j \text{ giving } u' = (7, 7, 6, 7)^T \text{ and } v' = (0, 0, 0, 0, 1).$$

$$\begin{array}{lll} & \begin{bmatrix} \boxed{0} & 0^{\diamond} & 0^{\diamond} & 0^{\diamond} & 1 \end{bmatrix} & s \\ \begin{bmatrix} 7 \\ 7 \\ 6 \\ 7 \end{bmatrix} & \begin{bmatrix} 1 & \boxed{0} & 3 & 2 & -1^{*} \\ 3 & 4 & \boxed{0} & -1^{*} & 5 \\ 6 & 5 & 3 & \boxed{0} & 3 \\ 5 & 5 & 2 & 0^{\diamond} & \boxed{0} \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ -1 \\ -1 \end{bmatrix} \\ t = & \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \end{bmatrix} & \end{array}$$

Here, the new worst hit coalitions after the initial improvement are $(i, j) = (1, 5), (2, 4)$. We should note that $j = 5$ belongs to N_4 and $j = 4$ belongs to N_3 and $i = 1, 2$ belong to M_1, M_2. Thus $p = 1,\ q = 4,\ p = 2,\ q = 3$ are the two directed arcs introduced.

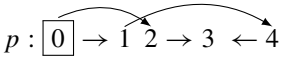


$$t = (0.0, 0, 1, 1)$$
$$s = (0, 0, -1, -1).$$

Here, $*$ entries correspond to the worst satisfied coalitions and $\diamond$ entries mark the next worst satisfied in line but passive coalitions. Thus for $\beta = 1$, the $*$ entries are improved to the level of $\diamond$ entries $= 0$. We update the table entries using the current s and t. We set the new $u = (7, 7, 5, 6)^T$ and the new v is $(0, 0, 0, 1, 2)$. With this, we update the satisfaction matrix and we set the updated matrix that is given below:

$$\begin{array}{cc} & \begin{bmatrix} \boxed{0} & 0^* & 0^* & 1 & 2 \end{bmatrix} \\ \begin{bmatrix} 7 \\ 7 \\ 5 \\ 6 \end{bmatrix} & \begin{bmatrix} 1^\diamond & \boxed{0} & 3 & 3 & 0^* \\ 3 & 4 & \boxed{0} & 0^* & 6 \\ 5 & 4 & 2 & \boxed{0} & 3 \\ 4 & 4 & 1^\diamond & 0^* & \boxed{0} \end{bmatrix} \end{array} \quad s = \begin{bmatrix} -1 \\ -1 \\ -3 \\ -2 \end{bmatrix}$$
$$t = \begin{bmatrix} 0 & 1 & 1 & 3 & 2 \end{bmatrix}$$

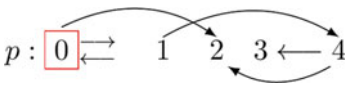
The related graph is still proper, and it is given by (based on * entries)



and this gives $l(p) = t = [0\ 1\ 1\ 3\ 2]^T$ and $s = -t$. For $\beta = \frac{1}{2}$, the worst hit $*$ entries increase to $\frac{1}{2}$ while the second worst hit entries f_{11} and f_{43} both reach the value $\frac{1}{2}$ and the new u and v are given by $\overline{u} = [6\frac{1}{2}, 6\frac{1}{2}, 3\frac{1}{2}, 5]^T$ and $\underline{v} = [0, \frac{1}{2}, \frac{1}{2}, \frac{5}{2}, 3]$ and the new satisfaction at these $(\overline{u}, \underline{v})$ is given by

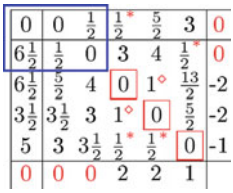
$$\begin{array}{cc} 0 & \begin{bmatrix} \boxed{0} & \frac{1}{2}^* & \frac{1}{2}^* & \frac{5}{2} & 3 \end{bmatrix} \\ \begin{bmatrix} 6\frac{1}{2} \\ 6\frac{1}{2} \\ 3\frac{1}{2} \\ 5 \end{bmatrix} & \begin{bmatrix} \frac{1}{2}^* & \boxed{0} & 3 & 4 & \frac{1}{2}^* \\ \frac{5}{2} & 4 & \boxed{0} & 1 & \frac{1}{2} \\ \frac{7}{2} & 3 & 1 & \boxed{0} & \frac{5}{2} \\ 3 & 3\frac{1}{2} & \frac{1}{2}^* & \frac{1}{2}^* & \boxed{0} \end{bmatrix} \end{array}$$

Currently, the most unhappy coalitions have satisfaction $\frac{1}{2}$, and they are at the $*$ locations. We get the graph corresponding to $M_o \to N_1$, $M_1 \to N_o$, $M_o \to N_2$, $M_1 \to N_4$, $M_4 \to N_2$, $M_4 \to N_3$ given by

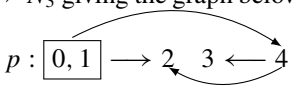


The graph is improper as we have the cycle $0 \leftrightarrow 1$. We melt node 1 to node 0. Thus, all coalitions of block $(0, 1)$ and $(1, 0)$ are settled and hence are left out from any further calculation. They are in a blue block as given below.

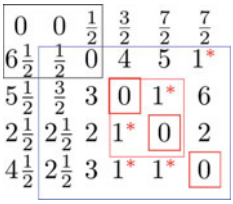
Next one below is proper; we put the block $(0, 1)$ $(1, 0)$ and so with $(0, 0)$ and we are starting with the top singleton row



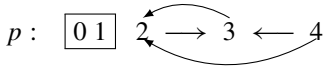
The active to-be-settled coalition is outside the boxed entries, and we have $M_o \to N_2$, $M_1 \to N_4$, $M_4 \to N_2$, $M_4 \to N_3$ giving the graph below since 0 and 1 are identified as same. We have



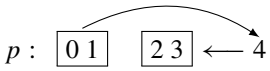
$l(p) = (0\ 2\ 2\ 1) \implies t = (0\ 2\ 2\ 1)$ $s = -t$. With $\beta = \frac{1}{2}$, at least one worst hit active coalitions $*$ (here, say seller-buyer 5) reach the level of the second worst hit coalition in terms of satisfaction denoted by $\diamond$: (seller 3 buyer 3) and also (seller 2 buyer 3). However, at $\beta = \frac{1}{2}$ (seller 0 buyer 3) who was also the worst hit goes above the second worst hit coalition's satisfaction. We get the new $\overline{u}$, $\underline{v}$ and satisfaction matrix at $\overline{u}$, $\underline{v}$ as



The worst hit coalitions are seller 2 buyer 4, seller 3, buyer 3, seller 4, buyer 3 and seller 4, buyer 4. Thus, we have $M_2 \to N_3$, $M_3 \to N_2$, $M_4 \to N_2$, $M_4 \to N_3$ giving the graph



Since $2 \leftrightarrow 3$, node 3 is melted with 2 keeping all arcs. After this the graph is proper. The block $M_2 \to N_3$ and $M_3 \to N_2$ gets settled. After eliminating the above cycle, we get the following proper graph:



$$l(p) \quad 0 \quad 2 \quad 1$$

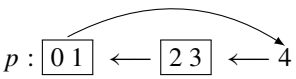
Thus $l(p) = (0, 2, 1)$ for $p = [0\ 1]$, $p = [2\ 3]$, $p = 4$. The step size $\beta = \frac{1}{6}$ will be used with $t = (0,\ 2,\ 1)$ and $s = -t$.

$$\begin{bmatrix} 0 & 0 & \frac{1}{2} & \frac{3}{2} & \frac{7}{2} & \frac{7}{2} \\ 6\frac{1}{2} & \frac{1}{2} & 0 & 4 & 5 & 1^* \\ 5\frac{1}{2} & \frac{3}{2}^{\diamond} & 3 & 0 & 1 & 6 \\ 2\frac{1}{2} & 2\frac{1}{2} & 2 & 1 & 0 & 2 \\ 4\frac{1}{2} & 2\frac{1}{2} & 3 & 1^* & 1^* & 0 \end{bmatrix} \begin{matrix} 0 \\ 0 \\ -2 \\ -2 \\ -1 \end{matrix}$$
$$0 \quad 0 \quad 0 \quad 2 \quad 2 \quad 1$$

Updating with new u, v and satisfaction $f_{ij}(u, v)$, we get

Buyer / Seller	0	1	2	3	4	5
0	0	0	$\frac{1}{2}$	$\frac{11}{6}$	$\frac{23}{6}$	$\frac{22}{6}$
1	$6\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{13}{3}$	$\frac{16}{3}$	$\frac{7}{6}$
2	$\frac{31}{6}$	$\frac{7}{6}$	$\frac{8}{3}$	0	1	$\frac{35}{6}$
3	$\frac{13}{6}$	$\frac{13}{6}$	$\frac{5}{3}$	1	0	$\frac{11}{6}$
4	$\frac{13}{3}$	$\frac{13}{6}$	$\frac{17}{6}$	$\frac{7}{6}$	$\frac{7}{6}$	0

The least happy coalitional satisfaction is $\frac{7}{6}$. The graph for this is



We now have a cycle $[0, 1] \longrightarrow 4 \longrightarrow [2\ 3] \longrightarrow [0\ 1]$. Thus all are melted to 0. The algorithm stops with iteration 3 and $\alpha^3 = \frac{7}{6}$. The final matrix containing the nucleolus is the one in the first column and top row of this matrix above

$$\overline{u} = \left(\frac{13}{2}, \frac{31}{6}, \frac{13}{6}, \frac{13}{3}\right)$$

for real sellers 1, 2, 3, 4 and $\underline{v} = \left(0, \frac{1}{2}.\frac{11}{6}, \frac{23}{6}.\frac{11}{3}\right)$ for real buyers 1, 2, 3, 4, 5.

We illustrate with one more example.

Example 10

Find the nucleolus for an assignment game with 3 sellers and 3 buyers with the coalitional worth of seller-buyer pair given by

$$\begin{bmatrix} 6 & 7 & 7 \\ 0 & 5 & 6 \\ 2 & 5 & 8 \end{bmatrix}$$

where the rows represent sellers and columns represent buyers. The nucleolus is determined by the lexicographic ranking of the satisfactions of singleton and buyer-seller coalitions arranged in a non-decreasing order. Since the singleton players can also be considered as a buyer-seller coalition with a dummy seller 0 or dummy buyer 0, the problem is treated as simply a buyer-seller satisfaction in the 4 players by adjoining dummy seller and dummy buyer. The augmented matrix is

	buyer			
buyer $j=$ / seller $i=$	0	1	2	3
0	0	0	0	0
seller 1	0	6	7	7
2	0	0	5	6
3	0	2	5	8

$= (a_{ij})$

= coalitional worth matrix. The optimal assignment in this case is

$$0 \to 0 \;\; 1 \to 1 \;\; 2 \to 2 \;\; 3 \to 3 \;\; i.e. \sigma(i) = i \;\; i = 0, 1, 2, 3.$$

We initiate the algorithm with all coalitional worth given away to the seller and nothing to the buyer. Let (u', v') be the u-best v-worst imputation $(0, 6, 5, 8\,;\; 0, 0, 0, 0)$. We can write down the satisfaction matrix at the above-given imputation as $f_{ij}(u, v)$ (Here $u = (0, 6, 5, 8)^T$, $v = (0, 0, 0, 0)$) given by

$$\begin{bmatrix} \boxed{0} & 0^{\diamond} & 0^{\diamond} & 0^{\diamond} \\ 6 & \boxed{0} & -1^{*} & -1^{*} \\ 5 & 5 & \boxed{0} & -1^{*} \\ 8 & 6 & 3 & \boxed{0} \end{bmatrix}$$

The most unhappy coalitions are (1, 2), (1, 3) and (2, 3) (denoted by *). The next worst after the most unhappy coalitions are the pairs (0, 1), (0, 2), (0, 3) among all these unsettled coalitions (denoted by $\diamond$). The equivalence classes are

$$p: \quad \boxed{0} \quad 1 \quad 2 \quad 3$$

and the induced graph with arcs are formed by $M_1 \to N_2$ $M_1 \to N_3$ $M_2 \to N_3$ and the graph is

$$\begin{aligned} p: &\ \boxed{0} \ \overset{\frown}{1 \longrightarrow 2 \longrightarrow 3} \\ l(p): &\ 0 \quad 0 \quad 1 \quad 2 \\ t &= (0 \quad 0 \quad 1 \quad 2) \\ s &= -t^T. \end{aligned}$$

We get the augmented satisfaction matrix with augmented s, t values as

$$\begin{array}{c} \\ \\ \\ \\ \text{t:} \end{array} \begin{array}{c} s = \\ \left[\begin{array}{cccc} \boxed{0} & 0^{\diamond} & 0^{\diamond} & 0^{\diamond} \\ 6 & \boxed{0} & -1^{*} & -1^{*} \\ 5 & 5 & \boxed{0} & -1^{*} \\ 8 & 6 & 3 & \boxed{0} \end{array}\right] \\ \begin{array}{cccc} 0 & 0 & 1 & 2 \end{array} \end{array} \begin{array}{c} \\ 0 \\ 0 \\ -1 \\ -2 \end{array}$$

Let us find a β such that for $(u, v) + \beta(s, t)$ for at least one $*$ entry matches a $\diamond$ entry. The satisfaction matrix is

$$\left[\begin{array}{cccc} \boxed{0} & 0 & \beta & 2\beta \\ 6 & \boxed{0} & -1+\beta & -1+2\beta \\ 5-\beta & 5-\beta & \boxed{0} & -1+\beta \\ 8-2\beta & 6-2\beta & 3-\beta & \boxed{0} \end{array}\right].$$

For $\beta = 1$, the $*$ entry $(i = 1, j = 2)$ becomes 0, while the $\diamond$ entry $(i = 0, j = 1)$ is still 0, and we have the updated matrix

$$\begin{array}{c} \\ \\ \\ \\ \text{t:} \end{array} \begin{array}{c} s \\ \left[\begin{array}{cccc} \boxed{0} & 0^{*} & 1 & 2 \\ 6 & \boxed{0} & 0^{*} & 1 \\ 4 & 4 & \boxed{0} & 0^{*} \\ 6 & 4 & 2 & \boxed{0} \end{array}\right] \\ \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \end{array} \begin{array}{c} \\ 0 \\ -1 \\ -2 \\ -3 \end{array}$$

while same $*$ entries like $i = 0$ $j = 2$ became passive and next in line the entries $(i = 0,\ j = 1)$, $(i = 1,\ j = 2)$, $(i = 2,\ j = 3)$ are the worst hit ones and we form the graph

$$\begin{array}{l} p: \ \overset{\frown}{\boxed{0} \ 1 \ 2 \ 3} \\ l(p): \ (0 \ 1 \ 2 \ 3) \end{array} \Longrightarrow t = (0,\ 1,\ 2,\ 3);\ s = -t^T.$$

Again with this new $(u'.v')$, we look for suitable β. Let us write down the satisfaction updates with β. We get

$$\begin{bmatrix} 0 & \beta & 1+2\beta & 2+3\beta \\ 6-\beta & 0 & \beta & 1+2\beta \\ 4-2\beta & 4-\beta & 0 & \beta \\ 6-3\beta & 4-2\beta & 2^{\diamond}-\beta & 0 \end{bmatrix}.$$

Here $f_{32} = 2-\beta$ is the first to reach the increasing minimum level $0+\beta$ at $\beta = 1$. Thus we get the updated matrix as

$$\begin{array}{c} \\ \\ \\ \\ \text{t:} \end{array} \begin{bmatrix} \boxed{0} & 1^{*} & 3 & 5 \\ 5 & \boxed{0} & 1^{*} & 3 \\ 2^{\diamond} & 3 & \boxed{0} & 1^{*} \\ 3 & 2^{\diamond} & 1^{*} & \boxed{0} \\ 0 & 1 & 2 & 2 \end{bmatrix} \begin{array}{c} s \\ \begin{bmatrix} 0 \\ -1 \\ -2 \\ -2 \end{bmatrix} \end{array}$$

The graph for the matrix is $p:\ \boxed{0} \longrightarrow 1 \longrightarrow 2 \leftrightarrow 3$.

Let us update with a suitable β to be determined. We have

$$\begin{bmatrix} \boxed{0} & (1+\beta)^{*} & 3+2\beta & 5+2\beta \\ 5-\beta & \boxed{0} & (1+\beta)^{*} & 3+\beta \\ (2-2\beta)^{*} & 3-\beta & \boxed{0} & 1 \\ 3-2\beta & 2-\beta & 1 & \boxed{0} \end{bmatrix}$$

We find $2-2\beta \downarrow$ and $1+\beta \uparrow$ and they reach equally for $\beta = \frac{1}{3}$. Thus the next one is

$$\begin{array}{c} \\ \\ \\ \\ \end{array} \begin{array}{c} 0 \quad 1 \quad 2 \quad 3 \\ \begin{bmatrix} \boxed{0} & \frac{4}{3}^{*} & \frac{11}{3} & \frac{17}{3} \\ \frac{14}{3} & \boxed{0} & \frac{4}{3}^{*} & \frac{10}{3} \\ \frac{4}{3}^{*} & \frac{8}{3} & \boxed{0} & 1 \\ \frac{7}{3} & \frac{5}{3} & 1 & \boxed{0} \end{bmatrix} \end{array}$$

$$\textit{The graph}: p: \quad \boxed{0} \longrightarrow 1 \longrightarrow \boxed{2\ 3}$$

It is improper and we can melt nodes 1, [2 3] in to 0 and the algorithm terminates. The nucleolus is simply the first column and top row given by

$$\tilde{u} = \left(0,\ \frac{14}{3},\ \frac{4}{3},\ \frac{4}{3}\right) \text{ and } \underline{v} = \left(0.\ \frac{4}{3},\ \frac{11}{3},\ \frac{17}{3}\right).$$

4.6 Domination Power Between Married Couple

Suppose we have 3 boys and 3 girls. Each boy can show interest to marry some of the girls. Each girl can also show interest to marry someone among these boys. Let A, B, C be boys and D, E, F be girls. A marriage could take place if both sides like each other as mates. Let $a_{ij} = 1$ if boy i and girl j like each other and let $a_{ij} = 0$ otherwise. Suppose for our case let

$$\begin{array}{c} \\ (A) \\ (B) \\ (C) \end{array} \begin{array}{c} \begin{array}{ccc} (D) & (E) & (F) \end{array} \\ \begin{bmatrix} \boxed{1} & 1 & 0 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & \boxed{1} \end{bmatrix} \end{array}$$

represent mate possibilities. Clearly, AD, BE, CF coalitions give the maximum number of marriages. Of course, if we view the entries as total power available, we can view any core allocation (which exists as we consider only mates of the opposite sex only) expressing a split of relative strength between a married boy with his mate.

Assumption: The nucleolus split of 1 unit between married couple expresses their relative domination power.

We can show that the nucleolus split is $\frac{3}{4}$ and $\frac{1}{4}$ between A and D, the split is $\frac{1}{2}$ and $\frac{1}{2}$ between B and E and $\frac{1}{4}$ and $\frac{3}{4}$ between C and F. Our initial intuition that C and D will have the least power is also confirmed by the nucleolus solution.

We can also represent the current situation via the following graph. Let each married couple be represented as a vertex and an edge between 2 couples is introduced if the boy of the first couple could also marry the girl of the second couple.

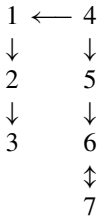


We can think of this flow as a chain reaction that would result if one couple were to separate and both tried to find other mates. As each new marriage would break up another one, we can see the nucleolus values for the men decrease by a constant rate as we move along the path: $\frac{3}{4}$, $\frac{1}{2}$, $\frac{1}{4}$, respectively.

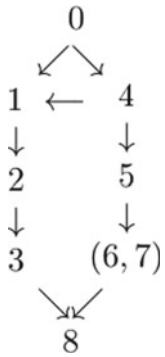
A second example: Let 7 boys and 7 girls induce the following marriage market where rows represent boys and columns represent girls;

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

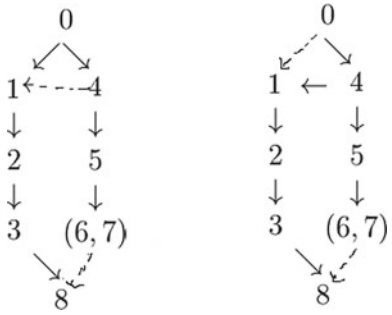
As in the earlier algorithm, the so-called settled coalition $(i\ i') :\ i = 1, 2 \ldots 7$ is treated as the vertex of a graph. Each off diagonal 1 is represented by a directed edge. The graph we get is the following:



We melt any cycle into a single vertex. Given such an acyclic graph, we introduce a dummy vertex called the source and another dummy vertex called the sink. We call them 0 and 8. If a vertex is not having any incoming edges, then we insert an edge from source, to this vertex by an edge. Any vertex with no outgoing edge from this vertex will receive an edge to sink 8. This is shown below



Now we have to find the longest path in this digraph. An algorithm like that of Noltmeier [22] can be used. To perform this algorithm, we first build a tree using depth-first search, starting at the source and going as far as possible, say we go via $0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 8$. We backtrack till we find another vertex. Here, we must go all the way back to 0 to go to $4 \longrightarrow 5 \longrightarrow (6\ 7)$. Now that we have traveled along all vertices, we have an initial tree. We look for any edge we can add to create a longer path. Checking with the edge (67) to 8 we have no way to increase the length of the path. We retrace and find if we can we replace edge $0 \rightarrow 1$ by $4 \rightarrow 1$ and we can increase the length of the path. These two stages are shown in the following 2 graphs.



We have found the longest path as $0 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 8$. Along this path from source to sink, we define a value w for each vertex going from 1 to 0 and decreasing uniformly. In this case, the uniform rate is $\frac{1}{5}$. These vertices are considered *settled* in that their values remain fixed for the remainder of the algorithm. We will show later that these values are the nucleolus values for boys. We then find another path to settle, a path which starts and ends with settled vertices with everything in between unsettled. We have only one such path 4, 5, (67), 8. Since the end points 4 and 8 have values $\frac{4}{5}$ and 0, the uniform rate is $\frac{1}{3}(\frac{4}{5} - 0) = \frac{4}{15}$. This way all the vertices are settled and we have the nucleolus value for each player. Every boy receives the w value and the mates $1 - w$ value. Since (67) are collapsed into a single vertex, those pairs will receive the same split.

The nucleolus solution : Boys Girls

$$\left\{(\tfrac{3}{5}, \tfrac{2}{5}, \tfrac{1}{5}, \tfrac{4}{5}, \tfrac{8}{15}, \tfrac{4}{15}, \tfrac{4}{15}), (\tfrac{2}{5}, \tfrac{3}{5}, \tfrac{4}{5}, \tfrac{1}{5}, \tfrac{7}{15}, \tfrac{11}{15}, \tfrac{11}{15})\right\}$$

Hardwick in his Ph.D. thesis [14] considers the special class of assignment games where each two-person coalition alone matters and they are worth either 1 or 0. The algorithm has two specified methods to compute the nucleolus. The first is a further specialization of the graph theoretic algorithm of Solymosi and Raghavan [29] which itself is a specialization of Maschler-Peleg-Shapley's iterative algorithm [20] in arriving at the lexicographic geometric center. The critical step is to find the longest paths ending in each vertex. The algorithm of Noltmeier [22] is quite essential.

Hardwick's second method is much simpler to implement. At each step, we find the longest path between two settled vertices; the path with the smallest average satisfaction is settled as our average gives us max min δ value. This follows from the following theoretical argument.

It is known that in any maximal matching with $(i\ i')$, the players i and i' split the worth of their coalition exactly, i.e. $w(i) + w(i') = a_{ii}$. If we assume full matching then $a_{ii} = 1 \forall i$ and so $w(i) = 1 - w(i')$ or $w(i') = 1 - w(i)$.

We know for assignment games only one- and two-person coalitions are essential. In fact, we show later only coalitions that correspond to edges of our graph will be essential. For example, if (i, j') is the coalition, the satisfaction at any imputation w is $s_{ij'}(w) = w(i) + w(j') - 1$ when $(i\ j)$ is an edge in G. It is $w(i) - w(j)$ as $(j\ j')$ is part of an optimal assignment.

Let us think of $w(i) - w(j) = s_{ij'}(w)$ as the length of the edge. If w is a core imputation then the length $L(i, j) = w(i) - w(j)$. Thus finding the nucleolus amounts to produce the max min values of edges. For any path $v_0, v_1, \ldots, v_k$, the sum of edge lengths $= L(v_0, v_1) + L(v_1, v_2) + \cdots + L(v_{k-1}, v_k) = w(v_0) - w(v_k)$. In case $v_0, v_1, \ldots, v_k$ is a cycle, we will have $v_k = v_0$ and that the cycle length is $w(v_0) - w(v_k) = 0$. However, since the nucleolus is a core allocation, $L(i, j) \geq 0 \; \forall \; i, j \in V$. In any cycle since cycle length is 0, each edge length $L(i, j)$ is also 0, i.e. $w(i) = w(j)$.

Let G' be a new graph made by collapsing any cycle in G with a single vertex. For any path in G', the sum of the edge lengths will be the same as in G.

Now consider the longest possible path in $G' : 0, v_1 \ldots, v_{k-1}, n + 1$. This path contains k edges (here $0, n + 1$ are dummy vertices). The average length is $\frac{w(0)-w(n+1)}{k}$. By arbitrarily setting $w(0)=1$ $w(n + 1)=0$, the minimum edge length $\leq \frac{w(0)-w(n+1)}{k}$ and nucleolus should maximize the minimum edge length on this longest path. Thus, all edges in this path have the same length $\frac{1}{k}$. If our assumption is indeed true, these values must get fixed. Thus, we can settle each vertex on these parts, so that edge lengths will not change.

To show this formally, it is enough to show that in the algorithm of Solymosi-Raghavan the first iteration gives $\delta_1 = \frac{1}{k}$.

In each iteration, we define the triple (X, Δ, Σ) where $X \subset$ core imputations. Here, Δ is a set of settled coalitions, namely the satisfaction in any coalition belonging to Δ is a constant over the entire set X. That is

$$\Delta = \left\{ (i, j') \in M \times N \; : \; f_{ij'}(x) = s_{ij'}(x) \equiv c_{ij'} \; \forall \; x \in X \right\}$$

where $M = \{0, 1, 2, ..., n\}$ $N = \{1', 2', ..., n + 1'\}$ and $M \times N$ is the set of all 2 player coalitions (or one player coalition), since 0 and $n + 1'$ are dummy players. Even though we should consider all coalitions, it is sufficient to consider only essential coalitions for which $s_{ij}(x)$ is not a constant over all of X. Namely, these are the coalitions outside Δ. We initiate with the core coalitions $X^o = \{x :$ imputation with $0 \leq x(i) \leq 1$ and where $E =$ edge set $= \{(i, j) : a_{ij} = 1, \; x(i) - x(j) \geq 0\}$. We assume that if besides (i, i') if (i, j) is a matched pair where $i \in M$ $j \in N$, $x(j) = 1 - x(i)$. If X^o is the core then $x(i) + x(j') \geq a_{ij} \; \forall \; i, j'$, where $i \in M$, $j' \in N$. That is $x(i) + (1 - x(j)) \geq a_{ij}$. But $(i, j) \in E \implies a_{ij} = 1$. Thus $x(i) - x(j) \geq 0$.

Also on the core when $\Delta^o = \{(i, i') : E^o \in M \backslash 0\}$, and $\Sigma^o = M \times N \backslash \Delta^o$ as well as δ^o, we have for step $r = 1, 2,$ etc.

$$\begin{aligned}
\delta^{r+1} &= \max_{x \in X^r} \min_{(i,j') \in \Sigma^r} s_{ij'}(x) \\
X^{r+1} &= \{x : x \in X^r : \min s_{ij'}(x) = \delta^{r+1}\} \\
\Delta^{r+1} &= \Delta^r \cup \{(i, j) \in \Sigma^r : s_{ij'}(x) = \delta^{r+1}\} \\
\Sigma^{r+1} &= \Sigma^r \backslash \{(i, j) \in \Sigma^r : s_{ij'}(x) = \delta^{r+1} \forall x \in X^{r+1}\}.
\end{aligned}$$

The algorithm terminates with $\Delta = M \times N$, $\Sigma = \phi$, and X consists of a single imputation, the nucleolus.

We will show $\delta' = \frac{1}{k}$. In the first iteration, we choose the longest path from 0 to $(n+1)'$, and we define based on the path length $w(0) - w(n+1') = 1$, $w(i) = 1 - \frac{d(i)}{k}$ where $d(i)$ is the length of vertex i from 0. This particular imputation gives for exact pair, the satisfaction

$$\begin{aligned} s_{ij'} &= w(i) + w(j') - a_{ij'} \\ &= w(i) - w(j) + 1 - a_{ij} \\ &= \frac{d(j) - d(i)}{k} + 1 - a_{ij}. \end{aligned}$$

When $a_{ij} = 1$, we know $(i, j) \in E$ and that $d(j) - d(i) = 1$, i.e. $s_{ij'}(w) \geq \frac{1}{k}$. When $a_{ij} = 0$, $(i, j) \notin E$, but $d(i) - d(j) \leq k - 1 \; \forall (i, j) \in M \times N$, since the length of the longest path is k. Thus $s_{ij'}(w) \geq w(i) - w(j) + 1 - a_{ij} \geq \frac{1-k}{k} + 1$(when $a_{ij} = 0$) $\implies s_{ij'}(w) \geq \frac{1}{k}$.

Thus, we have found an imputation w in X^o for which $\min\limits_{(i,j') \in \Sigma^o} s_{ij'}(w) \geq \frac{1}{k}$. But the longest path average edge length $= \frac{1}{k}$ and so the min edge length of this path $\leq \frac{1}{k}$. Thus, we get

$$\max_{x \in X^o} \min_{(i,j') \in \Sigma^o} s_{ij'}(x) = \frac{1}{k}.$$

Thus, all the vertices along the maximal path are settled and for any $x \in X^o$ all those edges will satisfy $s_{ij'}(x) = \frac{1}{k}$ along that path. We now adjoin the path P with Δ^o and let $\Delta' = \Delta^o \cup P$ and $\Sigma' = \Sigma^o \backslash P$ and set $X' = \{x \in X^o : s_{ij'}(x) = \frac{1}{k} \; \forall \; (i, j) \in P$ and $s_{ij'}(x) \geq \frac{1}{k} \; \forall (i, j) \notin P\}$. But this one is achievable if for all i with $(i, j) \in P$, we fix $x(i) = w(i)$. We can define $V_P = \{i : (i, j) \in P\}$ and thus $X' = \{x \in X^o : x(i) = w(i) \, \forall \, i \in V_P, x(i) \leq w(i) \, \forall \, i \notin V_P\}$. Instead of just fixing satisfactions for coalitions, we are actually fixing payoff for each couple along the path.

Let us assume at some iteration r, we have defined $(X^r, \Delta^r, \Sigma^r)$ and also δ^r. With it, we define an imputation w^r which is the southwest corner of X^r giving the highest possible value of $x(i)$ for all $i \in M$ and $x \in X^r$. Suppose we have also defined a forest $F^r \subset \Sigma^r$ with the following properties: for all $(i, j) \in F^r \; s_{ij'}(w^r) = \delta^r$, and for all j with $(i, j) \in \Sigma^r$ there is actually an edge $(i, j) \in F^r$ for exactly one i. (We can check that $F' \cup P$ is the tree constructed in the first iteration by the Noltmeier algorithm).

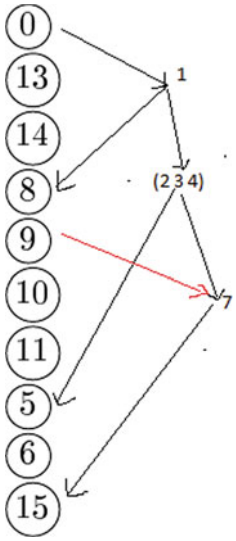
Here we will show how we can apply Hardwick's algorithm to solve the following 14×14 assignment game. Here rows represent males and columns represent females and $a_{ij} = 1$ when male i and female j like each other, $a_{ij} = 0$ otherwise.

$$A = \begin{bmatrix}
1&1&0&0&0&0&0&1&0&0&0&0&0&0\\
0&1&1&0&0&0&0&0&0&0&0&0&0&0\\
0&0&1&1&0&0&1&0&0&0&0&0&0&0\\
0&1&0&1&1&0&0&0&0&0&0&0&0&0\\
0&0&0&0&1&1&0&0&0&0&0&0&0&0\\
0&0&0&0&0&1&0&0&0&0&0&0&0&0\\
0&0&0&0&0&0&1&0&0&0&0&0&0&0\\
0&0&0&0&0&0&0&1&1&0&0&0&0&0\\
0&0&0&0&0&0&1&0&1&1&0&0&0&0\\
0&0&0&0&0&0&0&0&0&1&1&0&0&0\\
0&0&0&0&0&0&0&0&0&0&1&1&0&0\\
0&0&0&0&1&0&0&0&0&0&0&1&0&0\\
0&0&0&0&0&0&0&0&0&0&0&0&1&1\\
0&0&0&0&0&0&0&1&0&0&0&0&0&1
\end{bmatrix}$$

The graph is constructed as follows. We create one vertex for each row/column, plus an extra two dummy vertices, which we call 0 and $n + 1$. Since there are 14 rows and columns in this example, we will name the vertices as 0 through 15. Next, for each off diagonal 1, we create a directed edge from the row number to the column number. We also create edges from 0 to any vertex whose corresponding column index has a 1 only in the diagonal position, and from any vertex whose corresponding column index has a 1 only in the diagonal position, and from any vertex whose row index has a 1 only in the diagonal position to $n + 1$. We do not bother creating dummy edges for the other vertices, as they will be irrelevant since we know there will be a longer path on the graph. The resulting graph for this example is found in Fig. 4.2.

Next, we use Noltemeier's algorithm to build a tree, initially using a depth first search on the graph, resulting in Fig. 4.4, then finally at 4.4. Solid edges represent edges in the tree, and dashed edges represent those in graph, but not the tree.

By deleting the edges along the longest path from 0 to 15, but keeping the rest we generate a forest with settled vertices as the base for each tree in the forest. The initial path had 0 and 15 as the only settled vertices and w' was the equal split for the path length. Thus, we work on each tree of the forest for similar settled end pairs, with unsettled vertices in between. We can choose the path with the smallest average satisfaction settled as that average gives us the next max min δ value. The new graph is (settled vertices circled)

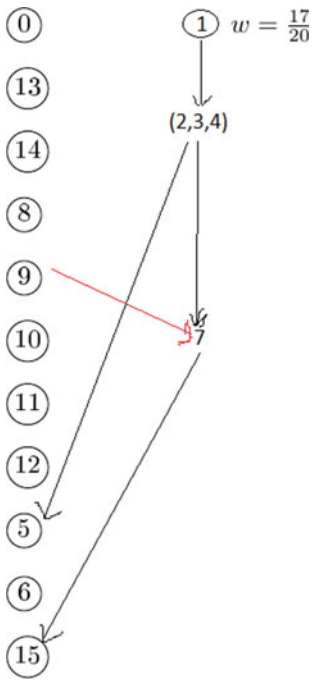


We have paths $0 \to 1 \to 8$ with ends as settled vertices 0 and 8.
We have path $0 \to 1 \to (2\ 3\ 4) \to 5$ with ends as settled vertices 0, 5.
We have path $0 \to 1 \to (2\ 3\ 4) \to 7 \to 15$, with ends as 0, 15 as settled vertices.
The δ value for $0 \to 1 \to 8$ is $\frac{w(0)-w(8)}{2} = \frac{1-\frac{7}{10}}{2} = \frac{3}{20}$.
The δ value for $0 \to 1 \to (2\ 3\ 4) \to 5$ is $\frac{w(0)-w(5)}{2} = \frac{1-\frac{1}{5}}{2} = \frac{4}{15}$.
The δ value for $0 \to 1 \to (2\ 3\ 4) \to 7 \to 15$ is $\frac{1-0}{4} = \frac{1}{4}$.
The δ value for $9 \to 7 \to 15$ is $\frac{\frac{6}{10}-0}{2} = \frac{3}{10}$.
The least value δ is $\frac{3}{20}$.
Thus, the next step is to settle vertex 1 with $w(1) = 1 - \frac{3}{20} = \frac{17}{20}$.

Notice that in each step we build no more than n trees giving $O(n^3)$ calculation per each step as Noltemeier [22] algorithm uses $O(n^2)$ calculations for each one. Since at least one vertex is settled in each step, the total number of steps is $\leq n$. Thus we have $O(n^4)$ calculations.

We will solve our 14×14 game[1] with its attached graph at the end of the first iteration using Noltemier's algorithm. The longest path is $0 \to 13 \to 14 \to 8 \to 9 \to 10 \to 11 \to 12 \to 5 \to 6 \to 15$ giving $w(0) = 1$, $w(13) = \frac{9}{10}$, $w(14) = \frac{8}{10}$, $w(8) = \frac{7}{10}$, $w(9) = \frac{6}{10}$, $w(10) = \frac{5}{10}$, $w(11) = \frac{4}{10}$, $w(12) = \frac{3}{10}$, $w(5) = \frac{2}{10}$, $w(6) = \frac{1}{10}$, $w(15) = 0$. Since all these vertices are settled, we consider the induced forest by deleting the edges connecting any two settled vertices. We now remove the edges joining 1 and 8 and keeping the other paths as below

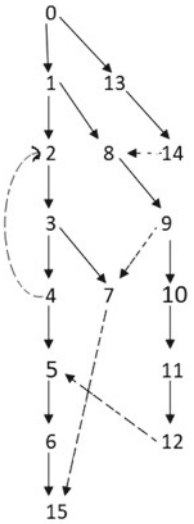
[1] These are found in page 23, 24, 25 and page 33 of John Hardwik's Ph.D. thesis [14].



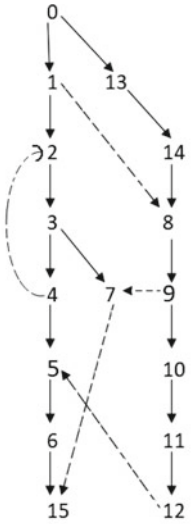
$1 \to (2,3,4) \to 5$ gives $\delta = \frac{\frac{17}{20}-0}{2} = \frac{13}{40}$.
$1 \to (2,3,4) \to 7 \to 15$ gives $\delta = \frac{\frac{17}{20}-0}{3} = \frac{17}{60}$.
These are the paths ending in settled vertices. The least is $\frac{17}{60}$ attained for the path $1 \to (2,3,4) \to 7 \to 15$. Thus $w(2,3,4) = \frac{17}{20} - \frac{17}{60} = \frac{17}{30}$. Now we are left with just one path $(2,3,4) \to 7 \to 4$ and $\frac{\frac{17}{30}-0}{2} = \frac{17}{60} = w(7)$. We have determined nucleolus for all vertices. It is $w(0) = 1$, $w(1) = \frac{17}{20}$, $w(2) = \frac{17}{30}$, $w(3) = \frac{17}{30}$, $w(4) = \frac{17}{30}$, $w(5) = \frac{2}{10}$, $w(6) = \frac{1}{10}$, $w(7) = \frac{17}{60}$, $w(8) = \frac{7}{10}$, $w(9) = \frac{6}{10}$, $w(10) = \frac{5}{10}$, $w(11) = \frac{4}{10}$, $w(12) = \frac{3}{10}$, $w(13) = \frac{9}{10}$, $w(14) = \frac{8}{10}$, $w(15) = 0$ (Fig. 4.2).

4.6.1 The Kernel/Nucleolus of a Standard Tree Game [13]

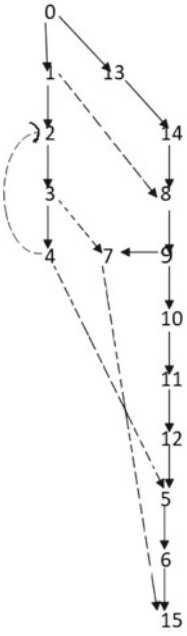
Let T be a rooted tree with root o and with exactly one edge emanating from o. Let V be the vertex set and E be the directed edge set. If N is the player set, then each player occupies exactly one vertex and $|V| \leq |N|$. Players occupying the same vertex are called neighbors. The root vertex o has no occupant and is called a public vertex. Every other vertex $v \in V$ is occupied by at least one player. Each edge $e \in E$ carries with it a cost $c(e) \geq 0$.



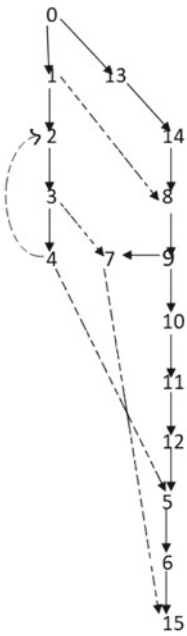
(2a)



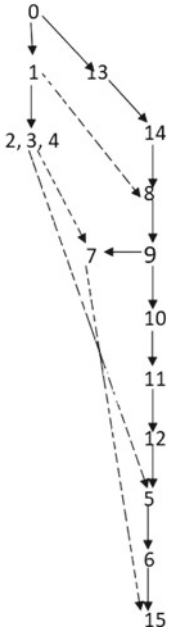
(2b)



(2c)



(2d)



(2e)

Fig. 4.2 We construct a tree and then iteratively update it to find the maximum depth of each vertex, and to collapse any cycles

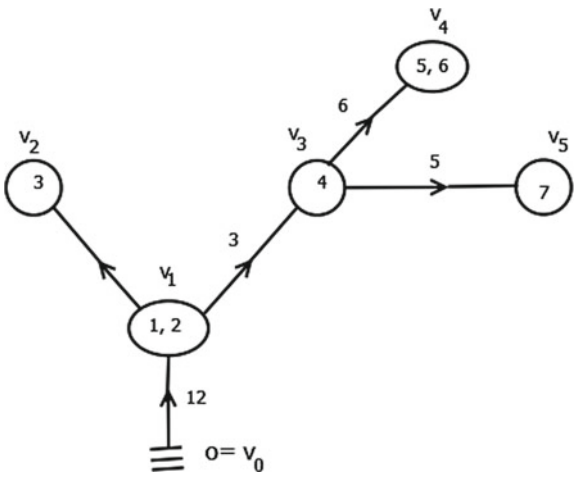


Fig. 4.3 An example for standard tree game

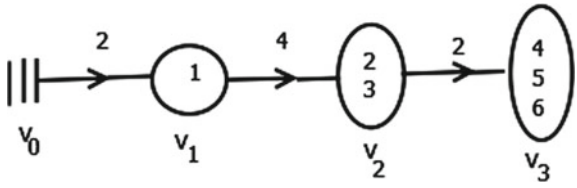


Fig. 4.4 An example of standard tree game

The construction of underground water pipes to supply Lake Michigan water to villages in Cook and Du Page countries in Illinois state can be modeled as a standard tree game.

Sharing the cost of runway construction at an airport by airline companies possessing planes of varying lengths and volumes can be modeled as a tree enterprise which is also a chain.

The key observations are (1) The TU game is a cost game where for each coalition S, the coalitioned cost $c(S)$ is the minimal cost needed to join all members of S to the root o via a connected subgraph of (V, E). (2) The kernel is also the nucleolus. (3) The game is convex in the sense of [28].

The algorithm of [13] initially computes the protonucleolus, an approximation that is refined by remaining bad arcs one by one till one arrives at the nucleolus. An arc $e = (v_i, v_j)$ is bad when $x_i > x_j$ for the protonucleolus x (Figs. 4.3 and 4.4).

Example (Granot et al. [13])

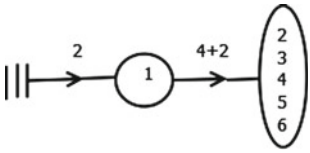
All players in a vertex are charged the same amount. Let $dv_k = \#$ residents at $v_k +$ # edges that leave v_k. Thus $dv_1 = 4,\ dv_2 = 1,\ dv_3 = 3,\ dv_4 = 2,\ dv_5 = 1$. If $e = (v_h, v_k)$, and if $x_h =$ protonucleolus at v_h, then protonucleolus at $v_k = [x_h + c(e)]/dv_k$. We can see $(3, 3, 4, 2, 4, 7)$ as the protonucleolus (here $x_{v_0} = 0$). Here,

player 4 who is further away from the root compared to players 1 and 2 is charged less and (v_1, v_3) is a bad arc.

A chain example (Figure 4.5)

For this chain example, $x_1 = 1, x_2 = x_3 = 5/3, x_4 = x_5 = x_6 = 11/9$. Thus (v_2, v_3) is a bad edge. Trim it and add the cost with (v_1, v_2) cost and put 4, 5, 6 also with 2, 3 in v_2. Trimming bad edges again and again gives the nucleolus as (1,1.4,1.4,1.4,1.4,1.4). This corresponds to the final trimmed chain (Fig. 4.5).

Fig. 4.5 A chain example



4.6.2 Balanced Connected Games [30]

A common feature among routing, sequencing and scheduling situations modeled as TU games share an added feature that all the essential coalitions (breaking them up are disadvantages to the players) are connected with respect to a fixed order of the players. These games are often balanced but connected coalitions could still be of formidable size. If Γ is a minimal balanced collection among essential coalitions, then Γ is just a proper interval partition of N [10]. Like the southwest corner extreme vertex in assignment games, we have the lex min vertex $\hat{x}$ of the core $\mathcal{C} = \{x : x \in R^n : x(S) \geq v(S), \forall$ intervals S and singletons$\}$ defined recursively by $\hat{x}_1 = \min\{x_1 : x \in \mathcal{C}\}, \hat{x}_2 = \{x_2 : x \in \mathcal{C}, x_1 = \hat{x}_1\}, \ldots, \hat{x}_n = \min\{x_n : x \in \mathcal{C}, x_1 = \hat{x}_1, x_2 = \hat{x}_2, x_{n-1} = \hat{x}_{n-1}\}$. Here $\hat{x}_j = \max\{v[i\ j] - \hat{x}[i\ j - 1] : 1 \leq i \leq j\}$. Thus $\hat{x}[1\ j]$ is the value of the maximal interval partition of coalition $[i\ j]$. As in assignment games, one can apply the Maschler-Peleg-Shapely algorithm to the collection of interval coalitions partitioned into $\Delta^0 = \{N\}$ and $\sum^0 = \{$Set of all interval coalitions$\} - \{N\}$. Each time some element I of $\sum^0$ is removed, we start with $\Delta^1 = \{N, I\}, \sum^1 = \sum^0 - I_1$ on the narrowed subset of the core and reach its lex min vertex. The algorithm terminates in $O(n^4)$, operations with the nucleolus.

4.6.3 Nucleolus for Cyclic Permutation Games [31, 33]

Suppose there are n customers forming a queue in front of a counter. Each customer needs exactly the same amount of service time. Due to their professional urgencies, the waiting time costs could vary from person to person. If side payments are allowed, any coalition S of customers can permute among themselves their mutual positions without affecting the waiting times of each member of complement coalition S^c. Let $a_{ij} \geq 0$ be the gain to player i swapping his/her position with player j due to side payment. We have a TU game $v(S) = \max_{\sigma} \sum a_{i\sigma(i)}$ (here σ is any S-permutation).

By the duality theorem, an optimal permutation in Π of $N = \{1, 2, \ldots, n\}$ will give a pair of n-vectors $u \geq 0$, $v \geq 0$ with $\mathcal{D}^+ = \{(u, v) : u \geq 0, v \geq 0, a_{ij} - u_i - v_j = 0, \forall (i, j) \in \Pi$ and $a_{ij} - u_i - v_j \leq 0, \forall (i, j) \notin \Pi\}$.

The game has a non-empty core [33]. Further the core $\mathcal{C} = m(\mathcal{D})$ where $m(u, v) = u + v$ (see [8, 24]).

Here $\mathcal{D}^+$ is the core of the assignment game induced by the matrix A.

Unfortunately, this embedding cannot help to recover the nucleolus of the permutation game from the nucleolus of the above assignment game with matrix A. While the essential coalitions are of size $O(n^2)$ for assignment games, they could be of order $O(2^n)$ for general permutation games. In general, the grand condition is inessential for assignment games, while, say for cyclic permutation games (games where $v(N) = \sum_i a_{i\Pi(i)}$ where Π is a cyclic permutation), the grand coalition is essential. Optimal permutation Π defines as intervals $[i\ j]$ where $[i\ j] = \{i, \Pi(i), \Pi^2(i), \Pi^k(i) = j\}$ and $[i\ j] = \{i\}$ when $i = j$. We can for any permutation σ which maps i to j get a cyclic $[j\ i]$ permutation if $\sigma(h) = \Pi(h)$ for $h \in [j\ i] - \{i\}$.

The assignment game algorithm [29] can be used to locate the nucleolus for cyclic permutation games, but we should start with only the least satisfied coalitions of the buyer-seller type. If we made the main diagonal as the optimal assignment by swapping columns, we should re-swap the columns to their initial positions and use the lexicographic u, v solution. The nucleolus for the permutations game will be the vector $m(u, v) = u + v$.

For the cyclic permutation game A with $\Pi(1) = 2, \Pi(2) = 3, \Pi(3) = 1$, $A = \begin{bmatrix} 2 & \textcircled{4} & 4 \\ 0 & 1 & \textcircled{3} \\ \textcircled{5} & 2 & \end{bmatrix}$. The nucleolus is $(9/2, 2, 11/2)$ based on pair nucleolus $(4, 2, 9/2; 1/2, 0, 1)$, while the nucleolus for the assignment game is $(3, 1, 3; 2, 1, 2)$. All it gives when merging u, v by $u + v$ is only the core allocation $(5, 2, 5)$ for the permutation game.

Remark 4.9 Consider the permutation game with matrix $A = \begin{bmatrix} 0 & \textcircled{8} & 0 & 9 \\ \textcircled{2} & 0 & 0 & 0 \\ 0 & 1 & 0 & \textcircled{4} \\ 0 & 0 & \textcircled{2} & 0 \end{bmatrix}$.

For this game, the image under the map m of the pair nucleolus $(8, 2, 2, 2; 0, 0, 0, 2)$ gives only the core element $x^* = u + v = (8, 2, 2, 4)$. However, the nucleolus for

this permutation game is $(23/3, 7/3, 7/3, 11/3)$. We can easily check that the second imputation is lexicographically better than x^*.

It is an open problem to find an efficient algorithm for the nucleolus of non-cyclic permutation games.

4.6.4 Brief Remarks on Some References Included

Suzuki and Nakayama [32]: They model water resource development as a cooperative venture and propose the nucleolus solution as a fair solution to venturing participants with capital costs and reaping benefits.

Bird [5]: Every game generated from a minimum cost spanning tree with an immovable source admits a non-empty core.

Huberman [15]: A coalition S is essential if and only if $v(S) > \sum_{T \in \Pi} v(T)$ where Π is any non-trivial partition of S. Any TU game v, N with non-empty core is determined just by the essential coalitions.

Muto et al. [21]: In a market with n persons, if only one among them possesses the patent for a new product, the game admits a non-empty core and a stable set.

Arin and Feltkamp [1]: A game (v, N) is veto-rich when one player is critical for all positive coalitional worth. The nucleolus can be computed avoiding any linear program. When all players other than the veto-rich player are homogeneous, this is the same as the Ricardean model considered by Chetty et al. [7]. They give an explicit formula for the nucleolus.

Brânzei et al. [6]: Here, they introduce the notion of strongly essential coalitions. In any game, the coalition of strongly essential coalitions contains all the coalitions which determine the core and when the core is non-empty, the nucleolus. They consider a special class of games called peer group games. They have at most $2n - 1$ strongly essential coalitions. However, for the same class, essential coalitions could be as much as 2^{n-1}. Here, they compute the nucleolus in $O(n^2)$ time directly from the data.

Kleppe et al. [16]: Here, they study weighted nucleolus and per capital nucleolus. They show modifications needed to approximate weighted nucleoli.

References

1. Arin J, Feltkamp V (1997) The nucleolus and kernel of veto-rich transferable utility games. Int J Game Theory 26(1):61–73
2. Aumann R, Machler M (1985) Game theoretic analysis of a bankruptcy problem from the Talmud. J Econ Theory 36:195–213
3. Aumann R, Maschler M, (2020) The bargaining set for cooperative games. Advances in game theory, 443–477 (1964). Princeton University Press, Princeton, pp 140–169
4. Bapat RB, Raghavan TES (2009) Nonnegative matrices and applications. Cambridge University Press
5. Bird CG (1976) On cost allocation for a spanning tree: a game theoretic approach. Networks 6(4):335–350
6. Brânzei R, Solymosi T, Tijs S (2005) Strongly essential coalitions and the nucleolus of peer group games. Int J Game Theory 33(3):447–460
7. Chetty VK, Dasgupta D, Raghavan TES (1976) Power and distribution of profits. Discussion paper no 139. Indian Statistical Institute, Delhi Centre
8. Curiel I, Tijs SH (1986) Assignment games and permutation games. Technical report, Tilburg University, School of Economics and Management
9. Davis M, Maschler M (1965) The kernel of a cooperative game. Naval Res Logist Quart 12(3):223–259
10. Derks J, Kuipers J (1997) On the core of routing games. Int J Game Theory 26(2):193–205
11. Edmonds J (1965) Paths, trees & flowers. Canadian J Math 17:449–467
12. Gillies DB (1953) Some theorems on n-person games. Princeton University
13. Granot D, Maschler M, Owen G, Zhu WR (1996) The kernel/nucleolus of a standard tree game. Int J Game Theory 25(2):219–244
14. Hardwick J (2017) Graphical algorithms for finding the nucleolus of binary-valued matching games. PhD thesis, University of Illinois at Chicago
15. Huberman G (1980) The nucleolus and the essential coalitions. In: Bennousson L (ed) Analysis and optimization of systems, vol 28. Springer, pp 416–422
16. Kleppe J, Reijnierse H, Sudhölter P (2016) Axiomatizations of symmetrically weighted solutions. Ann Oper Res 243(1):37–53
17. Kohlberg E (1971) On the nucleolus of a characteristic function game. SIAM J Appl Math 20(1):62–66
18. Kopelowitz A (1967) Computation of the kernels of simple games and the nucleolus of n-person games. Hebrew University Jerusalem (ISRAEL) Department of mathematics, Technical report
19. Lovasz L, Plummer MD (1986) Matching theory. North-Holland mathematics studies, vol 121
20. Mashler M, Peleg B, Shapley LS (1979) Geometric properties of the kernel, nucleolus and related solution concepts. Math Oper Res 4:303–338
21. Muto S, Potters J, Tijs S (1989) Information market games. Int J Game Theory 18(2):209–226
22. Noltemeier H (1975) An algorithm for the determination of longest distances in a graph. Math Program 9(1):350–357
23. Peleg B (1963) Existence theorem for the bargaining set $\mathcal{M}_1^{(i)}$. Bull Am Math Soc 69(1):109–110
24. Quint T (1996) On one-sided versus two-sided matching games. Games Econ Behav 16(1):124–134
25. Schmeidler D (1969) The nucleolus of a characteristic function game. SIAM J Appl Math 17:1563–1170
26. Schmeidler D (1969) The nucleolus of a characteristic function game. SIAM J Appl Math 17(6):1163–1170
27. Shapely LS, Shubik M (1971) The assignment game in the core. Int J Game Theory 1(1):111–130
28. Shapley LS (1971) Cores of convex games. Int J Game Theory 1(1):11–26
29. Solymosi T, Raghavan TES (1994) An algorithm for finding the nucleolus of assignment games. Int J Game Theory 23:119–143

30. Solymosi T, Aarts H, Driessen T (1998) On computing the nucleolus of a balanced connected game. Math Oper Res 23(4):983–1009
31. Solymosi T, Raghavan TE, Tijs S (2005) Computing the nucleolus of cyclic permutation games. Eur J Oper Res 162(1):270–280
32. Suzuki M, Nakayama M (1976) The cost assignment of the cooperative water resource development: a game theoretical approach. Manag Sci 22(10):1081–1086
33. Tijs S, Parthasarathy T, Potters J, Prasad VR (1984) Permutation games: another class of totally balanced games. Oper-Res-Spektrum 6(2):119–123

Chapter 5
From Linear System of Equations to Artificial Intelligence—The Evolution Journey of Computer Tomographic Image Reconstruction Algorithms

Rekha, Santosh Singh, and Samir Kumar Neogy

Mathematics Subject Classification (2010) 68T01 · 68T07

5.1 Introduction

In the modern world, computed tomography (CT) is one of the most successful and indispensable diagnostic tools in the domain of medical imaging. There are many reports worldwide, which directly/indirectly reflect the success story of the increase in number of the annual CT examinations Geyer et al. [39]. In the early 1970s, the first tomographic imaging modality Ambrose [2], Hounsfield [49] was introduced; since then, the speed of rapid development of hardware of CT machines is amazing Flohr et al. [37], Hsiao et al. [50], Zhang et al. [105]. The rapid development of CT machines since its first clinical scan took place can be imagined by the time taken for scanning the patient then and now. The first clinical scan took approximately 5 min for scanning the patient, whereas the modern machine finishes the rotation in approximately a quarter of a second Geyer et al. [39]. Also, significant success and development in terms of the speed of the reconstruction algorithm, quality of reconstructed images and resolution of reconstructed images. The development in CT scanner is not only from scanning/rotation and reconstruction algorithm speed

Rekha · S. Singh (✉)
Department of Mathematics, Shiv Nadar University, Gautam Budha Nagar, India
e-mail: santosh.singh@snu.edu.in

Rekha
e-mail: re824@snu.edu.in

S. K. Neogy
Statistical Quality Control and Operation Research Unit, Indian Statistical Institute, New Delhi, India
e-mail: skn@isid.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_5

but also in modality as a whole, such as dual source, dual energy, detector size and many more developments Flohr et al. [37].

From a commercial/business point of view, the worldwide increase in the sale of CT scanners is a joy for manufacturers and many minor providers of such a niche medical imaging diagnostic tool. But at the same time, it has always been a worry and problematic situation for the user community due to radiation exposure Abbas et al. [1]. The radiation exposure to the users has significantly increased since the introduction of CT imaging. Due to such hazardous nature of the CT imaging modality, the scientific community is working hard to tackle the health risks involved, and such efforts have shown effective and promising results in the significant reduction in CT dose. There are many ways to look at CT dose reduction. The first one is to reduce CT radiation exposure by using Kalra et al. [57], Slegel et al. [87] technique only when the benefits outweigh the risk as well as the cost. The other way to look at it is the CT dose reduction by the development of sophisticated algorithms as well as the hardware which can produce high-quality reconstructed images. The efforts in this direction have shown significant improvements Fleischmann and Boas [36]. Multiple dose reduction methods were introduced, including tube current modulation, organ-specific care, beam shaping filters and optimization of CT parameters, and following CT protocols, essential parameters such as tube current (mA), tube voltage (kV), pitch voxel size, slice-thickness, reconstruction filters and the number of rotations Kalra et al. [51], Slegel et al. [87]. The different combination of parameters enables significantly different image qualities while delivering the same radiation dose to the patient. In the clinical routine, radiation exposure is frequently controlled by adjusting the tube current. Due to the decrease in the tube current, there is a proportional increase in the image noise. To counter such compromise in the image quality, there is a need for continuous development in the domain of the reconstruction filters and different algorithms, which controls the quality of the reconstructed images Brenner and Hall [11], Murphy et al. [73].

In this article, we will navigate through different reconstruction algorithms since the first CT reconstruction algorithm was introduced Willemink and Noel [97]. Our aim is to analyze the CT reconstruction algorithms with an emphasis on the sampling of projection data. Broadly speaking, there are two categories of CT reconstruction algorithms, one is Filtered Back Projection (FBP) and the other category is dominated by iterative reconstruction techniques. The first and original CT image reconstruction algorithm was based on the iterative method and is popularly known as the algebraic reconstruction technique (ART) Gordon et al. [41]. Due to a lack of computational power, ART was quickly replaced by the filtered back projection technique Kak and Slaney [56]. The reconstruction speed in terms of computational complexity was the major factor for FBP to be widely used. In fact for decades, FBP enjoyed being the only choice for commercial CT scanners Pan et al. [79]. In 2009, the first iterative reconstruction technique was clinically introduced Hsieh [52]. Since then all major CT vendors introduced iterative algorithms for clinical use Nuyts et al. [78]. Interestingly, there was a big overlap between the hype of machine learning/artificial intelligence and iterative algorithms for CT image reconstruction. While navigating through the development of iterative reconstruction algorithms, subsequently our

aim is to understand the direction of CT image reconstruction algorithms in the domain of artificial intelligence. Our belief raises a big question mark, have we done enough to develop a CT image reconstruction algorithm following the non-uniform sampling? The reason to pose this question is meant to provoke mathematicians and CT image reconstruction experts to consider closer collaboration with engineers who design tomographic machines. The scientific/research and design community of tomographic systems have to keep the central objective of dose reduction as the main focus, otherwise argument for the replacement of filtered back projection will be just an academic discussion without product development reality.

5.2 From Filtered Back Projection to Iterative Reconstruction Technique to Clinical Necessity

In December 1970, Gorden et al. presented the first iterative reconstruction algorithm still popularly known as Algebraic Reconstruction Technique (ART), which was initially applied to reconstruct cross-sectional images. It was practically impossible to perform image reconstruction in a reasonable time using the ART algorithm, and the main reason was a lack of computational power. In 1972, Hounsfield's invented an X-ray computed tomographic scanner for which he received the Nobel prize along with Allan Cormack who independently discovered some of the algorithms Hounsfield [49]. Initially, the convolution back projection and then filtered back projection algorithms took off the pace mainly with the advantage of simplicity and reconstruction speed. Hence for a long time, the algorithm used for CT image reconstruction in commercial scanners was filtered back projection and then largely modified version of FBP algorithm, which was developed originally for approximate image reconstruction from circular cone beam data by Feldkamp, Davis and Kress (FDK) in 1984 Feldkamp et al. [34]. As far as the diagnostic image quality of CT reconstructed images is concerned, Feldkamp et al.'s algorithm played a major role with certain minor changes over the years. On the other side, the biggest disadvantage of FDK's algorithm is the direct proportional relation between image noise and radiation exposure as well as following the Nyquist sampling rate for the acquisition of the projection data for perfect reconstruction Herman [48], Mersereau and Oppenheim [69].

While commercial scanners operated with Feldkamp et al.'s algorithm, the CT image reconstruction's scientific community invested significant efforts and time into the development and advancement of the iterative algorithms Cheng et al. [23], Hara et al. [45], Lin et al. [65], Yu et al. [102]. For such advancement, the main focus was to enable low-dose CT algorithms with high diagnostic image quality.

Broadly speaking, these advancements will fall mainly into two categories: (i) iterative algorithms and (ii) hybrid algorithms. The different algorithmic approaches in these categories mainly focused on dose reduction with the improvement in the diagnostic image quality and taking care of computational complexity as well. The

iterative algorithms mainly solve the linear system of equations with a combination of data terms along with a regularization term (or prior term). The data term deals with the observed projection data, whereas the regularization term deals with the non-uniformities of the CT system. There are different statistical based iteration algorithms such as maximum likelihood, least squares and maximum a posteriori estimates. Such methods provide great flexibility in dealing with noise and artifacts. In the case of hybrid algorithms, both the analytical and iterative methods are combined. Mainly analytical methods are used to generate the initial images and then as a second step iterative methods are used to optimize the image characteristic B and Veo [5], Leipsic et al. [63], Thibault et al. [90], Winklehuer et al. [98].

The main progress in the computational complexity has been achieved by using Graphical Processing Units (GPUs) for accelerated CT image reconstruction algorithms. GPUs and high processing computational power allowed the successful implementation of sequential algorithms into parallel processing algorithms and such advancement in the implementation of algorithms played a major trigger for the medical device industry to try alternative algorithms Eklund et al. [32], Kim et al. [59], Yan et al. [99].

In 2009, the first iterative reconstruction-based algorithm (Iterative Reconstruction in Image Space IRIS, Siemens Healthineers) received FDA approval Grant and Flohr [42]. In a way, the algorithm is a hybrid of FBP and IR. In this algorithm, only a single backward projection step is applied for the creation of a cross-sectional image, using the raw data. The reconstructed image quality and other details can be seen in Gordic et al. [40], Moscariello et al. [72].

5.3 Reconstruction Algorithms as Inverse Problems

The CT image reconstruction process can be modeled as the following linear system of equations:

$$y = Ax \tag{5.1}$$

where y denotes the raw data vector (say of size M) that expands the data function, x is the reconstructed image vector (say of size N) that expands the under image function and A is systems matrix of size $M \times N$, which models the discrete X-ray transform. The system matrix depends upon the choice of basis functions for expanding data and image functions. There are different techniques for computing the discrete X-ray transform, in other words the estimation of matrix A plays a vital role and at the same time has many different techniques. The primitive and intuitive way to estimate the elements of the system matrix A is by calculating the intersection length of an X-ray with an image voxel. The measured data y denotes either simulated data or real data for the development of the reconstruction algorithms. The complete details can be found in Kak and Slaney [56].

Consider the following equation:

$$D(x) = |Ax - y|/M \tag{5.2}$$

i.e., the average Euclidean data divergence per detector pixel between the measured data and imaging model. The data divergence term, i.e., $D(x)$ can be zero only in a simulation study in which 'data' is generated by the use of an imaging model. This is one way to generate the simulated data, whereas another way to simulate the data is by estimating the line integral of a phantom. The data divergence term in Eq. 5.2 will necessarily be non-zero in two scenarios, first one is when data vector y is a real data set or it is simulated using the line integral of the phantom. The standard techniques such as SVD for analyzing the properties of systems matrix A are unfortunately impractical because of the sheer size of the matrix, which in general is of size $10^{10} \times 10^{10}$ for a typical cone beam CT-image reconstruction problem Han et al. [44].

It was Gorden et al. in 1970 who presented the first iterative algorithm ART to solve such a linear system of equations, which is modeled as a CT-image reconstruction problem. After that, Simultaneous Algebraic Reconstruction Technique (SART) was developed, which was a modified version of ART in terms of the reduced computational cost Anderson and Kak [3]. After these two algorithms, there is a chain of optimization-based CT-image reconstruction algorithms. For the development of such algorithms, the design of objective function and optimization strategy constitutes the critical step in which the addition of data constraints and prior conditions are also devised and included Han et al. [44].

In this section, we would like to set up a base of optimization algorithms used for CT image reconstruction. Once this base is set, then it will become easy to connect the concept of optimization-based algorithms with the concept of compressed sensing. The main reason to understand this connection is because compressed sensing provides a very sophisticated mathematical framework for exact signal recovery from undersampled data measurements. In the case of CT image reconstruction, undersampled data measurement can be directly related to less number of projections, which probably can be translated in terms of dose reduction.

Coming back to optimization methods, considering Eq. 5.2, data divergence and image positivity, the fundamental optimization-based CT image reconstruction algorithm can be represented as follows Han et al. [44], Jørgensen et al. [55]:

$$D(x) \leq \epsilon \quad \text{and} \quad x_j \geq 0 \tag{5.3}$$

where $\epsilon > 0$ is a selected parameter to incorporate the inconsistencies between the data and the imaging model. x_j represents the value of voxel j of reconstructed image x and $j = 1, 2, ..., N$. This optimization algorithm will provide many possible solutions due to less number of measured data. To deal with such a situation, mathematically we can add an image sparsity constrained to Eq. 5.2, i.e.,

$$D(x) \leq \epsilon, \; x_j \geq 0 \quad \text{and} \quad ||x||_0 \leq s \tag{5.4}$$

where $||x||_0$ denotes the l_0-norm of the image, and s denotes the sparsity parameter, which is the number of non-zero voxels imposed on the image. Now, Eq. 5.4 is a non-convex optimization problem with much tighter constraints compared to Eq. 5.3. To handle such tight constraints, the concept of total variation (TV) has been introduced. The use of the total variation technique is very much in practice for different image processing techniques. Incorporating the TV constraint in Eq. 5.4,

$$D(x) \leq \epsilon,\ ||x||_{TV} \leq t_0,\ x_j \geq \quad \text{and} \quad c_\alpha(x) \leq \gamma \tag{5.5}$$

where $||x||_{TV}$ denotes the image TV, whereas t_0 is a parameter which constrains the TV of the reconstructed image. γ is a negative number larger than 1 and $c_\alpha(x)$ is define as

$$c_\alpha(x) = \frac{d_{TV} \cdot d_{data}}{|d_{TV}||d_{data}|} \tag{5.6}$$

where $d_{TV} = \nabla_x ||x||_{TV}$ and $d_{data} = \nabla_x D^2(x)$. The ∇_x indicates the gradient operator that excluded zero elements of x.

Equation 5.5 brings non-convexity due to the c_α term. Another variation of the above algorithm can be written as follows:

$$D(x) \leq \epsilon,\ ||x||_{TV} \leq t_0,\ x_j \geq 0,\ ||x||_0 \leq s \quad \text{and} \quad c_\alpha(x) \leq \gamma \tag{5.7}$$

The above method is non-convex due to multiple factors.

All the methods explained above in a broader sense sum up the optimization techniques used for CT image reconstruction. In literature Ritschl et al. [83], Sidky and Pan [85], Sidky et al. [86], Tang et al. [89], most of the optimization techniques will be variations of the above-mentioned methods. These explained optimization techniques for CT image reconstruction look like depending upon the explicit parameters on the right-hand side of the inequalities, i.e., ϵ, t_0, s and γ, whereas the basis set spanning the image and data spaces depends on the choice of system matrix A. Although there are a number of research papers on how to choose the system matrix A, the authors of this article plan to use it judiciously in such a way that we can achieve perfect CT image reconstruction.

Once the stage is set with different types of optimization techniques, now the designing of different algorithms for reconstruction comes into play Stiller [88]. The role of these algorithms is to solve the optimization techniques for reconstruction as explained above under the specified constraints. To solve such an optimization technique, the main technique is gradient descent. In the literature, one can find different variations of the gradient descent technique in the form of projection onto convex sets (POCS). These POCS methods are further categorized as non-adaptive and adaptive POCS methods Boyd and Vandenberghe [10].

In the next sections, our aim is to explore the compressed sensing-based techniques with a focus on understanding the advantages of non-uniform sampling versus Nyquist sampling.

5.4 Compressed Sensing-Based CT Image Reconstruction Algorithms

The area of compressed sensing was initiated in 2006 by groundbreaking papers Candes and Romberg [14], Candes and Tao [15], Candes et al. [16, 17], Donoho et al. [29], Donoho and Tsaig [30]. The key idea of compressed sensing is to recover a sparse signal from very few linear measurements by using optimization techniques.

To understand the compressed sensing problem, the aim is to recover the signal of interest, i.e., $x = (x_i)_{i=1}^{n} \in \mathbb{R}^n$. We assume that x is sparse, i.e., $||x||_0 = no.\{i : x_i \neq 0\}$ very few non-zero coefficients or there exists an orthogonal basis or a frame Φ be the elements of the orthonormal basis or the frames as column vectors.

Now mathematically the compressed sensing problem can be formulated as follows.

Estimate x from the given system of linear equations

$$y = Ax \tag{5.8}$$

or estimate c from

$$y = A\Phi c \tag{5.9}$$

In both cases, the linear system of equations is under-determined with sparsity as a priori information.

The mathematical problem leads to the following questions:

- What is a suitable sensing matrix?
- Which signals for recovery are suitable signals?
- Can we always do an exact recovery of the signal?

In this section, our aim is to focus on the connection of the compressed sensing technique with CT image reconstruction. Since 2006, there has been a lot of research done on the theoretical aspect of compressed sensing, and the scientific community has tried to solve varied real-life applications. There have been attempts to use compressed sensing tools and techniques to build CT image reconstruction algorithms with varying number of projections. We always focus the new developments in CT image reconstruction from the viewpoint of reducing the X-ray dosage and maintaining the image quality, although it has been a debatable point about the real success in solving such real applications using the concept of compressed sensing. The reason is to find an appropriate sensing matrix A or $A\Phi$, or the assumption of following too restrictive conditions of exact sparsity for the recovery of signal x.

Before getting into the details related to the concept of sparsity in the development of compressed sensing algorithms, it will be best that we should talk about what we do mean by sparsity in the application of CT image reconstruction. Our aim is to bring sparsity in terms of less acquisition of projections. By doing so, we actually are focusing on fewer projections or in a true sense non-uniform sampling. Mainly, there are three different ways to acquire fewer projections. In the standard practice

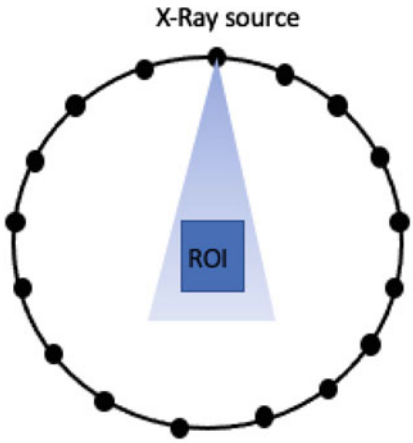
of filtered back projection (FBP) algorithm, data acquisition has to be uniformly sampled and also densely sampled. For example, in Cone Beam CT image reconstruction, sparsity in data acquisition can be brought in different ways as shown in the following Fig. 5.1.

All the cases shown in Fig. 5.1a–c can be considered as fewer acquisitions of projections, whereas in Fig. 5.1d, we have fewer projections but all sampled uniformly and densely. Breast scanning is one such cases where the scanning can be done only through limited views [1]. Such schemes where the data acquisition is sparse or fewer views have been thoroughly investigated using the concept of Compressed sensing. It is much easier to design algorithms with such kinds of sampling patterns, but there will always be a challenge for hardware design or to manipulate hardware accordingly, especially for diagnostic CT machines. Our aim of this paper allows us not to get into the hardware challenges related to sparse sampling.

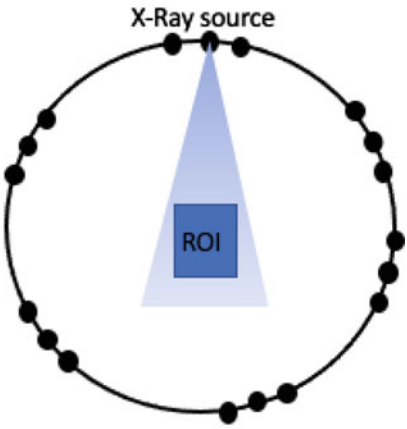
While discussing the iterative/compressed sensing-based algorithms for CT image reconstruction, we need to prepare the setup for such algorithms. First, we need to discretize the image space into pixels or voxels, depending upon whether it's 2D or 3D reconstruction. For simplicity, we consider a 2D CT image reconstruction scenario, where image space is discretized into a 2D square array with 256×256 and a detector having 1120 detector pixels. Now, the discretized image model for CT image reconstruction can be represented as Eq. 5.8, i.e., a system of linear equations $y = Ax$, where y represents a vector of size M corresponding to the total number of measurements, and x is vectorized image of size N. For example, we have discretized image space into a 2D square array of size 256×256, which means $N = 65,536$. Also, if the total number of projections acquired is say 10 and each projection is of size 1120, then $M = 11200$. Accordingly, the size of system matrix A will be $M \times N$. The geometrical representation of fan beam or cone beam discretization can be very well understood in any standard text such as Kak and Slaney [56].

The above-explained CT image reconstruction model fits very well with the concept of compressed sensing. In a compressed sensing scheme, the interest is when $M << N$ and the aim is to recover x based on the measurements, i.e., y. If we don't have any assumption on x, then the problem is an ill-posed problem. However, it has been recognized that if x is sparse and A is designed appropriately, then x may be recovered accurately Devore [27], Haupt and Nowak [47], Indyk [53], Rauhut et al. [82], Zhang [107]. There have been lots of research papers which talk about how the measurements should be carried, but have limited general theoretical developments. Such developments have as a result of statistical analysis Carin et al. [20], Certoft et al. [21], Ralasic et al. [81], Yu and Sapiro [101], Zang and Amin [103] and have shown promising results in array processing.

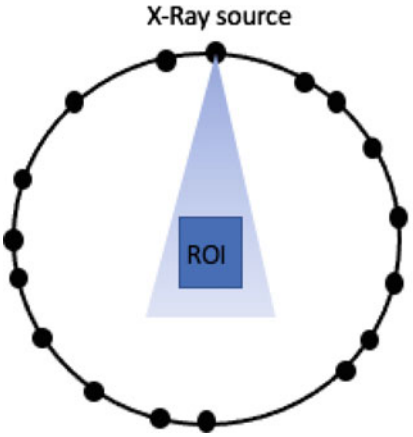
There are two key observations in the theory of compressed sensing for perfect reconstruction, which we would like to go into detail. First is the concept of data coherence, and the second one is the concept of restricted isometry property (RIP). Both these concepts have direct relations on the performance of the compressed sensing inversion algorithm for the estimation of x Bourgain et al. [9], Cai et al. [13], Candes and Tao [19], Dai and Milenkovic [25], DeVore [28], Nguyen and Shin [76], Tropp and Gilbert [94]. The system of linear equations, i.e., $y = Ax$ in most



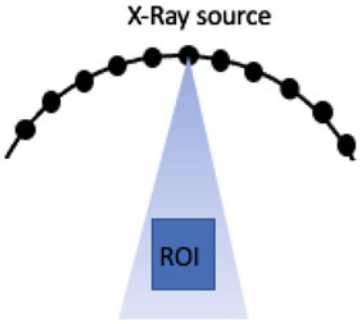
(a) Fanbeam Equiangular Scan



(b) Batch Scanning



(c) Random Scaning Angle



(d) Limited View Scaning Angle

Fig. 5.1 Four different ways for scanning an object for CT scanner

literature has noise component as well. In our discussion, we will not consider the noise term explicitly.

Now, let's talk about the data coherence of A, i.e., the coherence of the columns of A places requirements on the needed sparseness accurately (even perfectly) based on measuring y. The requirements on the level of sparseness are related to the properties of A. For sufficiently sparse x and appropriate choice of A, the components are recovered accurately. To achieve the accurate/perfect recovery of the signal x, the orthogonal matching pursuit (OMP) algorithm is very popular among the image

reconstruction algorithms Foucart and Rauhut [38], Needell and Tropp [74], Tropp and Gilbert [94]. In simple ways, the exact recovery of x using OMP if A, i.e., the sensing/system matrix and x satisfy the following inequality:

$$\mu_A < \frac{1}{2S_x - 1} \tag{5.10}$$

where S = sparsity of x, and μ = mutual coherence of column vectors of A and is defined as follows. Let $A \in \mathbb{C}^{m \times N}$ be a matrix with l_2-normalized columns $\mathbf{a}_1, \ldots, \mathbf{a}_N$, i.e., $||\mathbf{a}_i||_2 = 1$ for all $1 \leq i \leq N$. The coherence μ_A of the matrix A is given as

$$\mu_A = \max_{1 \leq i \neq j \leq N} |\langle \mathbf{a}_i, \mathbf{a}_j \rangle| \tag{5.11}$$

There are limitations on how small the coherence can be, and these limitations depend upon the type of the matrix A. For example, $\mu_A = 0$ if and only if the columns of A form an orthonormal system, and this happens if and only if A is a unitary matrix Foucart and Rauhut [38]. But our focus is on the situation where matrix $A \in \mathbb{C}^{m \times N}$ with $m \ll N$, i.e., in the framework of compressive sensing. In such a situation, our aim is to find how small the coherence can be. The advantage of small coherence implies that columns of sub-matrices of moderate size are well-conditioned. Let A_S denote the matrix formed by the columns of $A \in \mathbb{C}^{m \times N}$ indexed by the subset S of $[N]$. There are many results which set up the conditions on A along with the sparsity on the vector $x \in \mathbb{C}^N$. One of the initial theorem statements below with l_2-normalized columns and sparse vector x with condition on μ_A being less than 1 and the invertibility on A_S.

Theorem 5.1 *Let $A \in \mathbb{C}^{m \times N}$ be a matrix with l_2-normalized columns, and let $1 \leq S \leq N$. For all s-sparse vectors $x \in \mathbb{C}^N$,*

$$(1 - \mu(s-1))||x||_2^2 \leq ||Ax||_2^2 \leq (1 + \mu(s-1))||x||_2^2 \tag{5.12}$$

The above inequality essentially implies that for each set $\subseteq [N]$ with $card(S) \leq s$, the eigenvalues of the matrix $A_S^* A_S$ lie in the interval $[1 - \mu(s-1), 1 + \mu(s-1)]$, equivalently, if $\mu(s-1) < 1$ then $A_S^* A_S$ is invertible. The detailed proof is explained in Foucart and Rauhut [38]. There is a rich theoretical literature available Donoho et al. [31], Liu and Temlyakov [66], Wang and Shim [95] which tries to simplify the conditions mentioned in Eq. 5.12. To handle such conditions, the concepts of frames and tight frames have been used as well.

The analysis of orthogonal matching pursuit explains the performance of sparse recovery algorithms where the coherence is small. There are many theoretical results which guarantee the exact recovery of every s-sparse vector $x \in \mathbb{C}^N$ via orthogonal matching pursuit and via basis pursuit when the measurement matrix has a coherence $\mu_A < \frac{1}{2S_x - 1}$ Mohri and Talwalkar [71].

Before getting into the details of orthogonal matching pursuit (OMP) and its different variations, let's have some light on another strong and interesting property popularly known as Restricted Isometry Property (RIP). RIP facilitates the perfect reconstruction and very well fits into the framework of compressive sensing.

The coherence is a simple and useful measure of the quality of a system/sensing matrix and to achieve the desired quality; the lower bound on the coherence, i.e., $\mu_A \geq \sqrt{N - m/m(N-1)}$ for matrix $A \in \mathbb{C}^{m \times N}$ is one of the major bottlenecks to achieve the desired quality of the signal in practical applications. Restricted Isometry Property (RIP) also known as the uniform uncertainty principle provides a finer measure of the quality for a system/sensing matrix A. Below is the definition of RIP Foucart and Rauhut [38].

Definition 5.1 (*RIP*) The sth restricted isometry constant $\delta_s = \delta_s(A)$ of a matrix $A \in \mathbb{C}^{m \times N}$ is the smallest $\delta \geq 0$ such that

$$(1 - \delta)||x||_2^2 \leq ||Ax||_2^2 \leq (1 + \delta)||x||_2^2 \tag{5.13}$$

for all $s-$ sparse vectors $x \in \mathbb{C}^N$. Equivalently, it is given by

$$\delta_s = \max_{S \subseteq [N], card(S) \leq s} ||A_S^* A_S||_{2 \to 2} \tag{5.14}$$

A satisfies the RIP if δ_s is small for reasonably large S.

The aim is always to find the bounds so that the RIP property is satisfied. Hence, just like the coherence, it is important to know how small the sth RIP property constant of a matrix $A \in \mathbb{C}^{m \times N}$ can be so that it is relevant in the context of compressive sensing. Below is the theorem statement which talks about such a bound.

Theorem 5.2 *For $A \in \mathbb{C}^{m \times N}$ and $2 \leq S \leq N$, one has*

$$m \geq C \frac{S}{\delta_s^2} \tag{5.15}$$

provided $N \geq C_m$ and $\delta_s \leq \delta_*$, where the constants c, C and δ_* depend on each other.

In the literature, a detailed discussion and results are available to ease out the bounds as well as compare the bounds of coherence. There are also mathematical results which try to bridge the gaps between these different bounds. Once the bounds are well studied, then there are efforts to construct the matrices satisfying the RIP of optimal order Candes [18], Blanchard et al. [7], Zhang [108], Baraniuk et al. [6], Tilmann and Pfetsch [91]. There are two ways for the construction of such matrices, deterministic and probabilistic. Both approaches have their own set of challenges, but the probabilistic approach is more popular than the deterministic approach.

As mentioned earlier, orthogonal matching pursuit (OMP) provides methods for perfect reconstruction of the signal. In a similar fashion, Basis Pursuit (BP) establishes the success of the sparse recovery of the signal for the system/sensing matrix with a small restricted isometry constant. Below is a theorem which provides a result for the success of the sparse recovery.

Theorem 5.3 *Suppose that the 2sth restricted isometry constant of the matrix* $A \in \mathbb{C}^{m \times N}$ *satisfies*

$$\delta_{2s} < \frac{1}{3} \tag{5.16}$$

Then every s-sparse vector $x \in \mathbb{C}^N$ *is the unique solution of*

$$\min_{z \in \mathbb{C}^N} ||z||_1$$

subject to

$$Az = Ax$$

There are other simplified mathematical results, which provide simplified bounds and constructing matrices which allow s-sparse recovery whose sth order restricted isometry constant is arbitrarily close to 1.

Now it's time to talk about two main algorithms OMP and BP, which are the theoretical success story for all the concepts established in coherence and RIP. The literature has many different variants Foucart and Rauhut [38], Wang et al. [96] of these algorithms. These algorithms come under the category of greedy algorithms Li et al. [64], Tropp [93]. The pseudo-code described in algorithm 1 explains all the steps for orthogonal matching pursuit Foucart and Rauhut [38]. The pseudo-code described in algorithm 2 are the steps for implementing basis pursuit as a l_1 minimization problem Blumensath and Davis [8]. There are different ways to implement the same Cai and Wang [12], Davis et al. [26], Jain et al. [54], Nin et al. [77], Timbshirani [92].

Algorithm 1 OMP (A, y)

Input: A, y
Result: x_k
Initialization: $r_0 = y, \Lambda_0 = \Phi$
for k = 1,2,... do **do**
Step1. $\lambda_k = argmax_{j \notin \Lambda_{k-1}}$
Step2. $\Lambda_k = \Lambda_{k-1} \cup \{\lambda_k\}$
Step3. $x_k\ (i \in \Lambda_k) = argmin_x ||A_{\Lambda_k} - y||_2$ and $x_k\ (i \notin \Lambda_k) = 0$
Step4. $\hat{y}_k = Ax_k$
Step5. $r_k \leftarrow y - \hat{y}_k$
end for

Apart from the popular pseudo-code presented in the paper, there are variations of orthogonal matching pursuit and basis pursuit algorithms. Gradient pursuit and CoSaMP are popular variations for OMP, whereas hard thresholding and soft thresholding are commonly known implementations of basis pursuit. There are model-based implementations as well, such as model-based CoSaMP and model-based iterative hard thresholding Pope [80].

Algorithm 2 Iterative Hard Thresholding

Input: A, y and sparsity level k
Result: $\hat{x}$
Initialization: $\hat{x}_0 = 0$
for i = 1,2,. . . until stopping criterion is met
do
$\hat{x}_i = H_k\left(\hat{x}_{i-1} + A^T\left(y - A\hat{x}_{i-1}\right)\right)$
end for

The literature on compressive sensing of reconstruction of signals is very well established theoretically but as discussed earlier through some well-known theorems and respective bounds, it's always a challenging task to construct the system matrix. A detailed study about compressive sensing in medical imaging has been done in Christian et al. [24]. While going through this detailed review paper, the main focus is on total variation-based optimization methods. For example, in CT image reconstruction finding out the system matrix is like finding a needle in the haystack without any a priori information. Similar is the case with other real-life applications. In a nutshell, it's one of the most challenging tasks to find the system matrix. In the next section, the authors aim to discuss the concept of sampling, with a special focus on non-uniform sampling Landau [61].

5.5 Nyquist Sampling Versus Non-uniform Sampling

The Nyquist-Shannon sampling theorem establishes a sufficient condition for a sample rate that allows a continuous-time signal of finite bandwidth to discretize in such a way that no information is lost while reconstructing the signal. The theorem only applies to a class of mathematical functions whose bandlimited signal, i.e., support of the Fourier transform of those functions is finite. Suppose a function contains no frequencies higher than Bhertz, then according to Nyquist-Shannon sampling theorem, the sufficient sample rate for a guaranteed perfection would be anything larger than $2B$ samples per second, i.e., $f_s > 2$.

As the Nyquist-Shannon theorem is a sufficient condition, it allows that perfect reconstruction may still be possible when the sampling rate criterion is not followed. In some cases when the proper sampling criterion is not followed, it's possible to have almost perfect reconstruction by allowing additional constraints. The fidelity of these constraints can be verified and quantified using Bochner's theorem Nemirovsky and Shimron [75]. Bochner's theorem characterizes the Fourier transform of a positive finite Borel measure on the real line. From a harmonic analysis point of view, Bochner's theorem asserts that under Fourier transform a continuous positive-definite function on a locally compact abelian group corresponds to finite measure on Pontryagin dual group Feng [35], Mishali and Eldar [70], Zayed [104].

In a similar fashion, H. J. Landau's work presented in the paper "Necessary density conditions for sampling and interpolation of certain entire functions" Landau [61] tries to establish the concept of non-uniform sampling.

The main motivation for this section is to emphasize that the use of concepts of compressive sensing or l_1 minimization problems should be in connection with the mathematical and well-established concepts of non-uniform sampling. Especially in the domain of medical imaging, the most important thing is not to miss any smallest possible information, otherwise it can be a fatal mistake for diagnosis or treatment. There are always efforts for the search or design for perfect reconstruction algorithms. Hence, there is a need to exploit the mathematical conditions applied to the class of functions which can provide a sufficient condition for a non-uniform sampling rate for perfect reconstruction Eng [33], Marvast [67], Yaroslavsky et al. [100]. In brevity, optimization algorithms or compressive sensing-based algorithms should hand in hand go with the mathematical concepts of non-uniform sampling literature.

5.6 Current and Future Developments

Machine learning and artificial intelligence are one of the most popular tools and techniques which are spreading in every domain and aspect of life like a wildfire. There are mainly three types of learning models, supervised, semi-supervised and unsupervised learning. There is another very popular technique known as reinforcement learning. Out of all the different methods developed and established in machine learning, currently the most popular and successful are artificial neural network and deep learning. The most basic unit of neural network is perceptron, and it can be compared with the standard statistical method known as regression. A single perceptron is capable of doing predictions as well as two-class linear classification. The performance is based on how well the weights are being adjusted and type of the activation function. From the basic unit, a complete network is constructed in different layers. There are input layers and output layers and in between multiple hidden layers. The number of hidden layers mostly depends on the type of application. All variations mostly depend upon the choice of the activation function. The performance of these models depends upon how well the training has been performed on the network. For training, the final calculated output is compared to the desired output by a loss or error function. Once training is completed then it should match the testing error for a network to be successful for the task to perform in the future. Once the network has been trained and tested to satisfaction, then the model is ready for prediction, classification or designed task LeCun et al. [62].

There have been multiple objectives of artificial neural network (ANN)/deep learning (DL) in radiology McBee et al. [68]. However, the most common task can be assigned to the following applications: classification, detection, segmentation and so on. Classification task in radiology includes concluding whether a lesion is benign or malignant Chartrand et al. [22]. The existence of lesions and their location are the focus of detection tasks. A promising example is the study by Lakhani et al., who

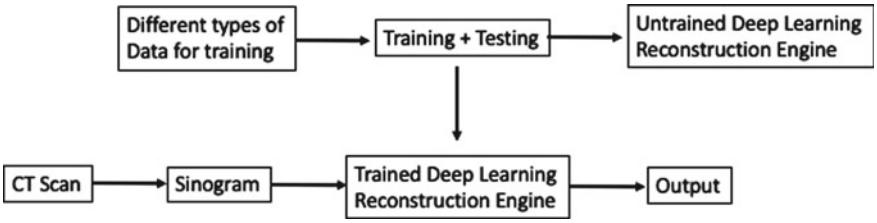


Fig. 5.2 Basic deep learning CT reconstruction workflow

trained two DNNs (Deep Neural Networks) on the detection of tuberculosis Lakhani [60]. The network showed great results with a sensitivity of 97.3% and specificity of 94.7%.

The aim of this review paper is to focus on the CT image reconstruction algorithms. There are huge efforts going on to develop deep learning-based image reconstruction algorithms Zhang and Dong [106] with the objective of improvement of image quality in comparison to existing algorithms, also if possible reduction in the X-ray dosage Greffier et al. [43], Hashemi et al. [46], Kang et al. [58] (Fig. 5.2).

The main assumption to using machine learning, especially the deep learning method against the classical CT image reconstruction algorithms, is that these standard and widely used methods are also prone to be sensitive to noise and incompleteness of measured data as explained in the previous sections Christian et al. [24]. The efforts in the direction of deep learning for CT image reconstruction are to combine classical and well-established methods with deep learning models. Nowadays, the more focus on deep learning is due to its ability to represent complex data as well as end-to-end training capabilities. Hence, combining the classical methods with the deep learning model, mainly two approaches are adopted. The first one is post-processing and the second one is raw to the reconstructed image. In both approaches, a mapping is established. In the post-processing case, a mapping is established between the initially reconstructed low-quality image and its high-quality image. In the second approach, a mapping is established between projection data to the reconstructed image. Our focus is to explore the CT image reconstruction techniques, hence we will only talk about the second approach, i.e., a mapping which is established from projection data to reconstructed image using the deep learning model.

The deep learning model uses the optimization algorithms and unroll the feed-forward networks. Then based on the a priori knowledge of the problem determine which parameters are best to be learned from the data to obtain the final reconstructed image. The main advantage is to have our deep learning architecture more manageable as well as more suitable for less number of parameters to learn Arudt et al. [4], Christian et al. [24], McBee et al. [68].

No matter which algorithm we use, or which hardware technology we use, our main and core objective is to reduce the radiation dose and acquisition time. From an algorithmic point of view, this objective can be achieved mainly by reducing the number of projections. A major problem of reducing the number of projections is the streaking artifacts. These streaking artifacts cover the complete reconstructed image

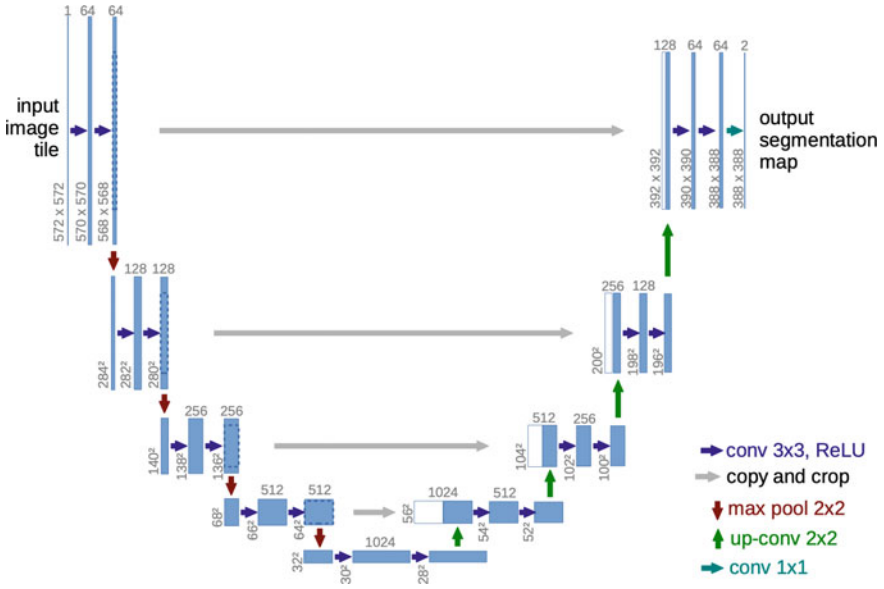


Fig. 5.3 U-Net architecture. Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The X-Y size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrow denote the different operations Ronneberger et al. [84]

but then can be considered simple structures in classical reconstruction methods. Deep learning model with multi-scale architecture allows to capture such global patterns of streaking artifacts. One of the best architectures which provides a better way to reduce such artifacts for sparse view CT image reconstruction is popularly known as U-Net as shown in Fig. 5.3, Chartrand et al. [22], Greffier et al. [43]. There are other well-known deep learning architectures, such as ADMM-Net, Primal-Dual Net and JSR-Net Christian et al. [24].

Recently, deep learning-based models are dominating in medical imaging domain as well, but in the authors' opinion it will be too early to say the success of these models. Also, there are plenty of challenges in deep learning models. These challenges limit the applications and implementations in clinical practices. Also, for supervised learning methods, it's one of the tasks to obtain the labeled data. In the medical imaging domain, privacy issues add up to these challenges. The strict regulations also make life difficult for technical experts to come up with algorithms whose results can solely be used for decision-making by radiologists Zhang and Dong [106].

5.7 Conclusion and Open Question

In this article, the authors tried to give a brief journey of the computer tomography image reconstruction algorithm developments. Apart from all the developments over the years and the recent dominance of machine learning techniques and more importantly deep learning models, still filtered back-projection algorithm with different variations dominates the industrial CT machines for their image reconstruction. The diagnostic decisions in the medical domain heavily depend upon the reconstructed images from different modalities such as MRI, Ultrasound and CT. Due to legal reasons as well as the direct impact on life of human beings, it will always be a challenge for medical experts to accept the probabilistic models. From a scientific/technical perspective, we believe that there should be enough efforts to explore or design such models which have mathematical support in terms of proving sufficient conditions for perfect reconstruction.

Can we use the concepts of sparsity and use the a priori knowledge for satisfying the conditions of restricted isometry property and/or mutual coherence to obtain the perfect reconstruction in the compressive sensing framework? The authors of this paper strongly believe that it's possible to construct a basis set which essentially is sensing matrix A with sufficient a priori knowledge for perfect/almost perfect reconstruction in the framework of a linear system of equations for CT image reconstruction as described above.

References

1. Abbas S, Lee T, Shin S, Lee R, Seungryong (2013) Effects of sparse sampling schemes on image quality in low dose CT. Med Phys 40(11):1–12
2. Ambrose J (1973) Computerized transverse axial scanning (tomography) II. Clinical applications. British J Radiol 552(46):1023–1047
3. Anderson AH, Kak AC (1984) Simultaneous algebraic reconstruction technique (SART: a superior implementation of the ART algorithm. Ultrasound Imaging 6(1):81–94
4. Arudt C, Guttler F, Heinrich A, Burckenmeyer F, Diamantis I, Teichgraber U (2021) Deep learning CT image reconstruction in clinical practice. Fortschr Rontgenstr 193:252–261
5. Thibaul J-B, Veo (2010) CT model-based iterative reconstrruction. GEHealthcare
6. Baraniuk R, Davenport M, DeVorre R, Wakin M (2008) A simple proof of restricted isometry property for random matrices. Construct Approx 28(3):253–263
7. Blanchard J, Cartis C, Tanner J (2010) Compressed sensing: how sharp is the restricted isometry prperty?
8. Blumensath T, Davis ME (2009) Iterative hard thresholding for compressed sensing. Appl Comput Harm Anal 27(13):265–274
9. Bourgain J, Dilworth S, Ford K, Konyagin S, Kutzarova D (2011) Explicit construction of rip matrices and related problems. Duke Math J 159(1):145–185
10. Boyd V (2004) Convex optimization. Cambridge University Press
11. Brenner DJ, Hall EJ (2007) Computed tomography: an increasing source of radiation exposure. N Engl J Med 357:2277–2284
12. Cai TT, Wang L (2011) Orthogonal matching pursuit for sparse signal recovery with noise. IEEE Trans Inf Theory 57(7):4680–4688

13. Cai TT, Wang L, Xu G (2010) New bounds for restricted isometry constants. IEEE Trans Inf Theory 56(9):4388–4394
14. Candes E, Romberg J (2007) Sparsity and incoherence in compressive sensing. Inverse Probl 23(3):969–985
15. Candes E, Tao T (2006) Near optimal signal recovery from random projections: universal encoding strategies. IEEE Trans Inf Theory 52(12):5406–5425
16. Candes E, Romberg J, Tao T (2006) Robust uncertainty principles: exact signal reconstruction from highly incomplete frequency information. IEEE Trans Inf Theory 59(2):489–509
17. Candes E, Romberg J, Tao T (2006) Stable signal recovery from incomplete and inaccurate measurements. Commun Pure Appl Math 59(8):1207–1223
18. Candes EJ (2008) The restricted isomtery property and its implications for compressed sensing. Comptes Rendus Mathematique 346(910):589–592
19. Candes EJ, Tao T (2005) Decoding by linear programming. IEEE Trans Inf Theory 51(12):4203–4215
20. Carin L, Lin D, Guo B (2011) Coherence, compressive sensing and random sensor arrays. IEEE Antennas Propag Mag 53(4):28–39
21. Certoft P, Mechklenbrauker CF, Seong W, Bianco M (2018) Introduction to compressive sensing in acoustics. J Acoust Soc Am 143(6):3731–3736
22. Chartrand G, Cheng PM, Vorontsov E et al (2017) Deep learning: a primer for radiology. Radiographics 37:2113–2131
23. Cheng I, Cheng Y, Fang T, Tyan J (2006) Fast iterative adaptive reconstruction in low-dose CT imaging. Int Conf Imaging Proc 889–892
24. Christian G, Eemil G, Sidky Y (2015) Compressive sensing in medical imaging. Appl Opt 54(8):C23–C44
25. Dai W, Milenkovic O (2009) Subspace pursuit for compressive sensing signal reconstruction. IEEE Trans Inf Theory 55(5):2230–2249
26. Davis G, Mallat S, Avellaneda M (1997) Greedy adaptive approximation. Constr Approx 13:57–98
27. Devore RA (2007) Deterministic construction of compressed sensing matrices. J Complex 22:918–925
28. DeVore RA (2007) Deterministic constructions of compressed sensing matrices. J Complex 23(4):918–925
29. Donoho D (2006) Compressed sensing. IEEE Trans Inf Theory 52(4):1289–1306
30. Donoho D, Tsaig Y (2006) Extensions of compressed sensing. Signal Proc 86(3):533–548
31. Donoho DL, Drori I, Tsaig Y, Strack JL (2012) Sparse solution of under-determined linear equations by stagewise orthogonal matching pursuit. IEEE Trans Inf Theory 58(2):1094–1121
32. Eklund A, Dufort P, Forsberg D, LaConte SM (2013) Medical image processing on the GPU-past, present and future. Med Image Anal 17(6):225–234
33. Eng F (2007) Non-uniform sampling in statistical processing. PhD thesis, Linkopings Universitet, Sweedn
34. Feldkamp LA, Devis LC, Kress JW (1984) Practical cone beam algorithm. J Opt Am A 1:612–619
35. Feng P (1997) Universal minimum-rate sampling and spectrum-blind reconstruction for multiband signals. PhD thesis, University of Illinois at Urbana-Champagin
36. Fleischmann D, Boas FE (2011) Computer tomography—old ideas and new technology. Eur Radiol 21:510–517
37. Flohr TG, McCollough CH, Bruder H et al (2006) First performance evaluation of dual-source CT(DSCT) system. Eur Radiol 16(2):256–268
38. Foucart S, Rauhut H (2012) A mathematical introduction to compressive sensing. Springer
39. Geyer LL, Schepf UJ, Meinel FG, John WNJ, Bastarrika G, Leeipsic JA, Paul NS, Cecco M (2015) State of the art: iterative CT reconstruction technique. Radiology 339–357
40. Gordic S, Morsbach F, Schmidt B et al (2014) Ultra slow dose chest computed tomography for pulmonary nodule detection: first performance evaluation f single energy scanning with spectral shaping. Invest Radiol 49(7):465–473

41. Gordon R, Bender R, Herman GT (1970) Algebraic reconstruction technique (ART) for three dimensional electron microscopy and X-ray photography. J Theor Biol 29(3):471–481
42. Grant K, Flohr T (2010) Iterative reconstruction in image space (IRIS. usasiemenscom/healthcare
43. Greffier J, Hamard A, Pereira F et al (2022) Image quality and dose reduction opportunity of deep learning image reconstruction algorithm for CT: a phantom study. Eur Radiol
44. Han X, Bian J, Ritman EL, Sidly EY, Pan X (2012) Optimization based reconstruction of sparse images from few projections. Phys Med Biol 57:5245–5273
45. Hara A, Paden RG, Silva AC, Kujak JL, Lawder HJ, Pavlicek W (2009) Iterative reconstruction technique for reducing body radiation does at CT: feasibility study. AJR Am J Boentgenol 193(3):764–771
46. Hashemi SM, Beheshti S, Gill PR, Paul NS, Cobbold RSC (2015) Accelerated compressed sensing based CT image reconstruction. Comput Math Methods Med
47. Haupt J, Nowak R (2006) Signal reconstruction from noisy random projections. IEEE Trans Inf Theory 52(9):4036–4048
48. Herman GT (2009) Fundamentals of computerized tomography, 2nd ed. Springer
49. Hounsfield GN (1973) Computerized transverse axial scanning (tomography) I. Description of system. British J Radiol 552(46):1016–1022
50. Hsiao EM, Rybicki FJ, Steagner M (2010) CT coronary angiography: 256 slice and 320 detector row scanners. Curr Cardiol Rep 12(1):68–75
51. Hsich J (2009) Computed tomography principles, design artifacts and recent advances. SPIE Press, Bellingham, WA
52. Hsieh J (2008) Adaptive statistical iterative reconstruction: GE white paper. White Paper
53. Indyk P (2008) Explicit construction of compressed sensing of sparse signals. In: Proceedings of 19th annual ACM-SIAM symposium on discrete algorithms, pp 30-33
54. Jain P, Tewari A, Dhillon IS (2011) Orthogonal matching pursuit with replacement. arXiv
55. Jørgensen JH, Sidky EY, Pan X (2011) Quantifying admissible undersampling for sparsity-exploiting iterative image reconstruction in x-ray CT. https://doi.org/10.48550/ARXIV.1109.0629, arxiv:1109.0629
56. Kak AC, Slaney M (2001) Principles of computerized tomographic imaging. SIAM, Philadelphia
57. Kalra MK, Maher MM, Toth TL, Hamberg LM, Blake MA, Shepard JA, Saini S (2004) Strategies for CT radiation dose optimization. Radiology 230:619–628
58. Kang E, Min J, Ye JC (2017) A deep convolution neural-network using directional wavelets for low-dose X-ray CT reconstruction. Med Phys 44:e360–e375
59. Kim JK, Zhang Z, n JA (2011) Hardware acceleration of iterative image reconstruction for X-ray computed tomography. In: IEEE international conference of acoustics, speech and signal processing, pp 1697–1700
60. Lakhani P (2017) Deep convolution neural networks for endotracheal tube position and X-ray image classification: challenges and opportunities. J Digit Imaging 30:460–468
61. Landau HJ (1967) Necessary density conditions for sampling and interpolation of certain entire functions. Acta Math 117
62. LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521:436–444
63. Leipsic J, Labounty TM, Heilborn B et al (2010) Adaptive statistical iterative reconstruction: assessment of image noise and image quality in coronary CT angiography. AJR Am J Roentgenol 195(3):649–654
64. Li J, Zhaojun W, Feng H, Wang Q (2014) Greedy orthogonal matching pursuit. In: 2014 IEEE international instrumentation and measurement technology conference
65. Lin YJ, Zhu PP, Chen, et al (2007) A new iterative algorithm to reconstruct the referactive index. Phys Med Biol 52(3):764–771
66. Liu E, Temlyakov VN (2012) The orthogonal super greedy algorithm and applications in compressed sensing. IEEE Trans Inf Theory 58:2040–2047
67. Marvast F (2001) Non-uniform sampling: theory and practice. Springer, New York, NY

68. McBee MP, Awan OA, Colucci AT et al (2018) Deep learning in radiology. Acad Radiol 25(25):1472–1480
69. Mersereau R, Oppenheim A (1974) Digital reconstruction of multidimensional signals from projections. Proc IEEE 62(10):1319–1338
70. Mishali M, Eldar YC (2009) Blind multiband signal reconstruction compressed sensing for analog signals. IEEE Trans Signal Proc 57(3):993–1009
71. Mohri M, Talwalkar A (2011) Can matrix coherence be efficient and accurately estimated? In: Proceedings of 14th international conference on artificial intelligence and statistics, FL, USA, vol 15
72. Moscariello A, Takx RA, Schoepf UJ et al (2011) Coronary CT angiography: image quality, diagnostic accuracy and potential for radiation dose reduction using a novel iteration image reconstruction technique comparison with traditional filtered back projection. Eur Radiol 21(10):2130–2138
73. Murphy MJ, Balter J, Balter S, Bencomo JA, Das IJ, Jiang S, Ma CM, Olivera GH, Rodebaug RF, Ruchala KJ, Shirato H, Yin FF (2007) The management of imaging dose during image-guided radiotherapy: report of the AAPM task group 75. Med Phys 34:4041–4063
74. Needell D, Tropp JA (2009) CoSaMP: iterative signal recovery from incomplete and inaccurate samples. Appl Comput Harmon Anal 26(3):301–321
75. Nemirovsky J, Shimron E (2015) Utilizing Bochners theorem for constrained evaluation of missing fourier data. arXiv:1506.03300
76. Nguyen TL, Shin Y (2013) Deterministic sensing matrices in compressive sensing: a survey. Sci World J 2013:1–6
77. Nin S, Gao Y, Bian Z, Huang J, Yu CGW et al (2014) Sparse-view X-ray CT reconstruction via generalized variation regularization. Phys Med Bio 59(12):2997
78. Nuyts J, B DM, Dupont P, Defrise M, Suetens P, Mortelmans I (1998) Iterative reconstruction for helical CT: a simulation study. Phys Med Biol 43(4):729–737
79. Pan X, Sidky EY, Vannier M (2009) Why do commercial CT scanners still employ traditional, filtered back projection for image reconstruction. Inverse Probl 12(25)
80. Pope G (2009) Compressive sensing: a summary of reconstruction algorithms. Masters thesis, Electrical Engineering, ETHZ Zurich
81. Ralasic I, Tafro A, Sersic D (2018) Statistical compressive sensing for efficient signal reconstruction and classification. In: 4th international conference on frontiers of signal processing, pp 44–49
82. Rauhut H, Schnass K, Vandergheynst P (2008) Compressed sensing and redundant dictionaries. IEEE Trans Inf Theory 54(5)
83. Ritschl L, Bergner F, Fleischmann C, Kacelrie M (2011) Improved total variation based CT image reconstruction applied to clinical data. Phys Med Biol 56:1545–1561
84. Ronneberger O, Fischer P, Brox T (2015) U-net: convolutional networks for biomedical image segmentation. arxiv:1505.04597
85. Sidky EY, Pan X (2008) Image reconstruction in circular cone beam computed tomography by constrained total-variation minimization. Phy Med Biol 53:4777–47807
86. Sidky EY, Duchin Y, Pan X, Ullberg C (2011) A constrained, total variation minimization algorithm for low intensity X-ray CT. Med Phy 28:117–125
87. Slegel MJ, Schmidt B, Bradley D, Suess C, Hlldebolt C (2004) Radiation dose and image quality in Pediatrric CT: Effects of technical factors and phantom size and shape. Radiology 233:512–522
88. Stiller W (2018) Basics of iterative reconstruction methods in computed tomography: a vendor independent overview. Euro J Radiol 147–154
89. Tang J, Nelt BE, Chen GH (2009) Performance comparison between total variation TV based compressed sensing and statistical iterative reconstruction algorithm. Med Phys 36:4920–4932
90. Thibault JB, Sauer KD, Bouman CA, Hsieh J (2007) A three-dimensional statistical approach to improved image quality for multiscale helical CT. Med Phys 34(11):4526–4544

91. Tilmann M, Pfetsch ME (2014) The computational complexity of the restricted isometry property, the null space property and related concepts in compressed sensing. IEEE Trans Inf Theory 60(2):1248–1259
92. Timbshirani R (1996) Regression shrinkage and selection via lasso. J R Stat Soc Ser B 58:267–288
93. Tropp JA (2004) Greedy is good: algorithm results for a sparse approximation. IEEE Trans Inf Theory 50:2231–2242
94. Tropp JA, Gilbert AC (2007) Signal recovery from random measurements via orthogonal matching pursuit. IEEE Trans Inf Theory 53(12):4655–4666
95. Wang J, Shim B (2012) On the recovery limit of sparse signals using orthogonal matching pursuit. IEEE Trans Signal Proc 60(9):4973–4976
96. Wang J, Kwon S, Shim B (2012) Generalized orthogonal matching pursuit. IEEE Trans Signal Proc 60(12):6202–6216
97. Willemink MJ, Noel PB (2019) The evolution of image reconstruction for CT—from filtered backprojection to artificial intelligence. Eur Radiol 29:2185–2195
98. Winklehuer A, Karlo C, Puippe G et al (2012) Raw data based iterative reconstruction in body CTA: evaluation of radiation dose saving potential. Euro Radiol 22(12):2597–2606
99. Yan G, Tian J, Shouping Z, Yakang D, Chenghu Q (2008) Fast cone-beam CT image reconstruction using GPU hardware. J X-ray Sci Technol 16(4):225–234
100. Yaroslavsky LP, Shabat G, Salomon BG, Ideses IA, Fishbain B (2008) Non-uniform sampling, image recovery from sparse data and discrete sampling. arXiv
101. Yu G, Sapiro G (2011) Statistical compressed sensing of Gaussian mixture models. IEEE Trans Signal Proc 59(12):5842–5858
102. Yu Z, Thibault JB, Bouman CA, Sauer KD, Hsich J (2011) Fast model-based X-ray CT reconstruction using spatially non-homogeneous ICD optimization. IEEE Trans Image Proc 20(1):161–175
103. Zang YD, Amin MG (2012) Compressive sensing in non-stationary array processing using bilinear transformation. In: IEEE 7th sensor array and multichannel signal processing workshop, pp 349–352
104. Zayed A (1993) Advances in Shannon's sampling theory. Taylor and Francis
105. Zhang D, Li X, Lin B (2011) Objective characterization of GE discovery CT750 HD scanner: gemstone spectral imaging mode. Med Phys 38(3):1178–1188
106. Zhang H, Dong B (2020) A review on deep learning in medical image reconstruction. J Oper Res Soc China 8:311–340
107. Zhang Y (2005) A simple proof for recoverability of l_1- minimization (II): the nonnegative cases. Technical report TR05-10, Rice University, Huston, Texas
108. Zhang Y (2005) When is missing data recoverable? Rice CAAM Department Technical report TR05-15

Chapter 6
Shapley Value and Other Axiomatic Extensions to Shapley Value

T. E. S. Raghavan

6.1 Shapley Value

Cooperative TU (Transferable Utility) game consists of a player set $N = \{1, 2, \ldots, n\}$ and for each subset $S \subset N$ (called a coalition), we associate a real number $v(S)$ called the worth of the coalition S.

Any coalition S can be identified with an n-vector where the ith coordinate is 1 if $i \in S$ and is 0 if $i \notin S$. Thus we associate for $N = \{1, 2, 3, 4\}$, $S = \{1, 3, 4\}$ the 4-tuple $S \leftrightarrow (1, 0, 1, 1)$. With each coalition S, $v(S)$ measures what coalition S on its own can achieve by *full cooperation* of its members. Thus $v : 2^n \to \mathbb{R}$ where $v(\phi) = 0$.

The following examples are just mild variations of the examples in the given references.

Example 1 ([1])

Residents R, D, and U of a house have the responsibility to mow the lawn in front of their house. While R has the strength to start the lawn mower and even mow half the lawn all by himself, ladies U and D need the help of R to start the engine. With his help U can finish mowing the entire lawn or complete mowing unfinished parts. With R's help D can complete $^3/_{4^{th}}$ the lawn. Of course with R's help in starting the engine, U and D can also complete mowing the entire lawn. This game can be represented by 2^3 tuple $v(\phi) = 0,\ v(R) = \frac{1}{2},\ v(U) = 0,\ v(D) = 0,\ v(UD) = 0,\ v(RU) = 1,\ v(RD) = \frac{3}{4},\ v(RUD) = 1$.

We would like to evaluate the relative position and strength of each participant in this cooperative venture.

T. E. S. Raghavan (✉)
University of Illinois at Chicago, Chicago, IL 60607-7045, USA
e-mail: terctu@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_6

Example 2 ([4])

A landlord who is handicapped owns all the lands in a village with n homogeneous landless workers. If he employs k workers and if workers work sincerely he can harvest $f(k)$ bags of rice. Suppose $f(x)$ is a concave increasing function with $f(0) = 0$, then we would like to evaluate a way to share the total output among the landlord and workers.

Example 3 ([16])

An Indian village in Tamil Nadu consists of people belonging to different castes. Suppose the following are the voting population size.

Pillais	300
Nainars	130
Kaunders	400
Chettiars	100
Brahmins	70
	1000

While voters by definition have freedom to make one's own decision on any agenda, in reality it often goes along caste lines en block. Thus we can think of 5 leaders one for each caste playing the game with block votes. Any agenda will pass if the agenda has the support from the majority of voters. But in our case we have 5 players called P, N, K, C, B where B has weight $W_B = 70$. Similarly $W_N = 130$, $W_K = 400$, $W_C = 100$, $W_P = 300$. We can represent this as a weighted majority game with $q = 501$ and a coalition of castes will win if their total weight $\geq q = 501$. We represent this as

$$[q;\ W_B,\ W_C, W_K, W_N, W_P]$$

where for any $S \subset \{B,\ C,\ K,\ N,\ P\}$

$$\begin{aligned} v(S) = 1 \qquad & \text{if} \quad \sum_{i \in S} W_i \geq q \\ = 0 \qquad & \text{Otherwise.} \end{aligned}$$

This is called a *simple* game and is also called *a weighted majority game*.

We want to evaluate the relative strength of each caste in their power to pass successfully an agenda, seeking cooperation from a few castes.

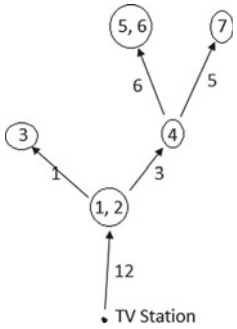
Example 4 (Airport Game) ([14])

The runways in an airport will have to cater to planes of various lengths that require safe landing. The capital cost of building a runway will depend on the longest plane that will land in the airport. Let $C_o < C_1 < \cdots < C_k$ be the capital costs for constructing runways of planes of increasing length. If m_i planes of length l_i are to be using this airport, then what should be the way to split the capital cost among various planes assuming the capital cost as $\max_i C_{l_i}$.

If only $l_{i_1} < l_{i_2} \cdots < l_{i_s}$ are the lengths of planes to use the airport and if r_k planes of lengths l_{i_k} will be using the facility then, the capital cost to be shared is C_{i_s} where $l_{i_1} < l_{i_2} \cdots < l_{i_s}$ and the number of planes using is $\sum_{i=1}^{s} r_{l_i}$. Again we would like some scheme to evaluate the portion of the total capital cost that is to be collected in any fair distribution of costs.

Example 5 (Tree Game) ([10])

There are 7 villages for a TV cable service. Cost of building cables connecting the main TV station to villages 1 and 2 is 12 K. Any cable to connect from villages 1, 2 to village 3 is 1 K. Any cable to connect village 4 from villages 1, 2 is 3 K, villages 5 and 6 can be connected by cable from village 4 for a cost of 6 K and village 7 can be connected to cable service from village 4 for a cost of 5 K. The following tree describes the situation.



If the total cost $= (12 + 1 + 3 + 5 + 6) = 27$ of building the cable network is to be shared among the 7 villages, how should they split the cost among them?

Unanimity Games

Imagine 7 children enjoying a party at my home. My wife prepared a nice cake and is willing to leave it with any set of children to distribute among themselves with one stipulation. The set must include all of my children.

If $N = \{1, 2, 3, 4, 5, 6, 7\}$ and my children are $1, 2, 3$ then any subset S among them can get the whole cake as long as $S \supset \{1, 2, 3\}$. We call this the unanimity game where

$$\begin{aligned} v(S) &= 1 \quad \text{if} \quad S \supset \{1, 2, 3\} \\ &= 0 \quad \text{if} \quad S \not\supset \{1, 2, 3\} \end{aligned}$$

(i) Observe that v treats all of my children *symmetrically*.
(ii) Observe that this characteristic function $v(S) = v(S \cup i)$ for any $i = 4, 5, 6, 7$ and for any $S \not\ni i$. We call them *dummies*.
(iii) When a group of children get the cake, we want to make sure they cannot waste any part but must share the whole cake between them. We call $(x_1, x_2, \ldots, x_7)$ an imputation if $x_i \geq v(\{i\})$ and $\sum_{i=1}^{7} x_i = 1$. (We stipulate *efficiency*)

Since my children are treated symmetrically without any order based on their age, etc., it is only reasonable that any imputation suggested satisfy $x_1 = x_2 = x_3$. Condition (ii) says that neither the presence nor the absence of other children have any influence on what a coalition gets. We call them dummies and it is only fair they are neither punished nor positively rewarded as they are at the mercy of my children in any sharing. Shapley stipulates 0 reward for them. That is $x_i = 0$ if $i \neq 1, 2, 3$. Since no cake is to be wasted, $\sum x_i = 1$ and we get $x_1 + x_2 + x_3 = 1$ and $x_1 = x_2 = x_3$ and that $x_1 = x_2 = x_3 = \frac{1}{3}$.

If we view any TU game (v, N) as a vector in $2^{|N|}$ dimension then we have the following.

Proposition 6.1 *Unanimity games form a basis for the vector space of all TU games.*

Proof We will denote by v^T, the unanimity game

$$\begin{aligned} v^T(S) &= 1 \quad \text{if} \quad S \supset T \\ &= 0 \quad \text{if} \quad S \not\supset T, \end{aligned}$$

we have $2^{|N|}$ unanimity games, one for each $T \subset N$. Suppose they are linearly dependent. Then we have $\sum_{T \subset N} \alpha_T v^T = 0$ for some $\alpha_T \neq 0$.
Let $\alpha_{T_o} \neq 0$ and if $\alpha_T \neq 0$, then let $|T| \geq |T_o|$. From the finitely many we can always find such a T_o. Consider $\sum_T \alpha_T v^T(T_o) = \alpha_{T_o} v^{T_o}(T_o) + \sum_{T \neq T_o} \alpha_T v^T(T_o)$. Let $T \neq T_0$. Here $T \subset T_o \implies \alpha_T = 0$. Thus the only term that remains on the left side of the equation

$$\sum_{T \subset N} \alpha_T v^T = 0$$

is $\alpha_{T_o} v^{T_o}(T_o) = \alpha_{T_o} \neq 0$ a contradiction. □

Shapley value ([15])

With each unanimity game (v^T, N) we can associate the unique imputation $\phi(v^T)$ called the Shapley value given by

$$\phi_i(v^T) = \begin{cases} \frac{1}{t} & \text{if } i \in T \text{ (Here } |T| = t) \\ 0 & \text{if } i \notin T. \end{cases}$$

Further, it trivially satisfies

1. $\phi_\pi(\pi \circ v^T) = \phi(v^T)$ (Here π is any permutation of N) (*Symmetry*).
2. $\sum_{i \in N} \phi_i(v^T) = 1$ (*Efficiency*.)
3. $\phi_i(v^T) = 0$ if i is a *dummy* player.

Theorem 6.1 *There exists a unique imputation ϕ called the Shapley value for all TU games (v, N) with $\hat{N}$ fixed satisfying*

iv $\phi_i(v + w) = \phi_i(v) + \phi_i(w)$, *where* $(v, N), (w, N), (v + w, N)$ *are TU games (Linearity).*

Proof Since unanimity games form a basis of all N person TU games, we can write, $v = \sum_{T \subset N} \alpha_T v^T$ for a unique α_T's. In fact, $\alpha_T v^T$ are themselves TU games with $\alpha_T v^T(N) = \alpha_T$ and linearity determines $\phi_i(v) = \sum_{T \subset N} \alpha_T \phi_i(v^T) = \sum_{T \subset N} \phi_i(\alpha_T v^T)$.

Lemma 6.1 *If $v = \sum_{T \subset N} \alpha_T v^T$, then*

$$\alpha_T = \sum_{\substack{S \subset T \\ |T|=t, |S|=s}} (-1)^{t-s} v(S)$$

Proof Let U be any arbitrary coalition. We will check $v(U) = \sum_{T \subset N} \alpha_T v^T(U)$

$$\sum_{T \subset N} v^T(U) \sum_{\substack{S \subset T \\ |T|=t, |S|=s}} (-1)^{t-s} v(S) = \sum_{T \subset U} v(U) \sum_{\substack{S \subset T, T \subset U \\ |T|=t, |S|=s}} (-1)^{t-s} v(S) \qquad (6.1)$$

Now changing the order of summation since $v^T(U) = 1$ when $U \supset T$, by changing the order of the summation in L H S of step (6.1) we get

$$\sum_{S\subset U} v(S) \sum_{\substack{S\subset T\subset U \\ |S|=s,|T|=t}} (-1)^{t-s}$$

$$= \sum_{S\subset U} v(S) \sum_{t=s}^{t=u} \binom{u-s}{t-s}(-1)^{t-s}$$

$$= \sum_{S\subset U} v(S) \sum_{t-s=0}^{u-s} \binom{u-s}{t-s}(-1)^{t-s}$$

$$= \sum_{S\subset U} v(S)(1-1)^{u-s} \text{(Here if } s < u, \text{ then terms vanish)}$$

$$= v(U) \text{ as} (1-1)^0 = 1 \text{ by convention, when } S = U.$$

□

Theorem 6.2 *For any arbitrary TU game v, Shapley value ϕ is unique and is given by*

$$\phi_i(v) = \sum_{\substack{S\subset N \\ S\ni i \\ |S|=s}} \frac{(s-1)!(n-s)!}{n!}[v(S) - v(S\backslash i)].$$

Proof Since $v = \sum_{T\subset N}, \alpha_T v^T$

$$\phi_i(v) = \sum_{T\subset N} \alpha_T \phi_i(v^T)$$

$$= \sum_{\substack{T\subset N \\ T\ni i \\ |T|=t}} \alpha_T . \frac{1}{t}$$

$$= \sum_{\substack{S\subset T \\ T\ni i}} (-1)^{t-s} v(S) . \frac{1}{t}.$$

Let

$$h_i(S) = \sum_{\substack{T\supset S \\ T\ni i \\ |S|=s}} (-1)^{t-s} . \frac{1}{t},$$

where $|T| = t$. If $S \ni i$, then $h_i(S\backslash i) = -h_i(S)$. Thus $\phi_i(v) = \sum_{\substack{S\subset N \\ i\in S}} h_i(S)[v(S) - v(S\backslash i)]$. We have $\binom{n-s}{t-s}$ coalitions T with t elements such that $S \subset T$.

$$\begin{aligned}
\text{Thus } h_i(S) &= \sum_{t=s}^{n}(-1)^{t-s}\binom{n-s}{t-s}\frac{1}{t} \\
&= \sum_{t=s}^{n}(-1)^{t-s}\binom{n-s}{t-s}\int_0^1 x^{t-1}dx \\
&= \int_0^1 \sum_{t=s}^{n}(-1)^{t-s}\binom{n-s}{t-s}x^{t-1}dx \\
&= \int_0^1 x^{s-1}\sum_{t=s}^{n}(-1)^{t-s}\binom{n-s}{t-s}x^{t-s}dx \\
&= \int_0^1 x^{s-1}(1-x)^{n-s}dx = \frac{(s-1)\,!(n-s)\,!}{n\,!}.
\end{aligned}$$

Hence the value of $\phi_i(v)$ where $i \in S$, $|S| = s$, is given by

$$\phi_i(v) = \sum_{\substack{S\subset N \\ S\ni i}} \frac{(s-1)\,!(n-s)\,!}{n\,!}[v(S) - v(S\backslash i)].$$

□

The formula has certain intuitive appeal. Any player i can arrive in a random order and if he is the last one to become a member of a coalition S, then his marginal contribution to the existing coalition $S\backslash i$ is $v(S) - v(S\backslash i)$. The chance he is the last one to join S in random formation of the grand coalition is given by $\frac{(s-1)\,!(n-s)\,!}{n\,!}$.

Thus the Shapley value is simply the expected marginal contribution by any player in case the coalitions form one by one to reach the grand coalition in a random order.

We will work out the Shapley value for some of our examples above.

Lawn Mowing Problem

We are given as 3 players R, U, and D with $v(R) = \frac{1}{2}$, $v(RU) = 1, v(RD) = \frac{3}{4}$, $v(U) = v(D) = v(UD) = 0, v(RUD) = 1$.

Coalition formation: Say $U\ D\ R$. Where U is first, then D joins and then R joins. By assumption $v(U) = v(\phi) = 0$, $v(UD) = v(U) = 0$, $v(UDR) = v(UD) = 1$. Thus the players U, D, R collect their respective marginal contribution 0, 0, 1.

Thus we can write down the following table of each one's marginal contribution as follows:

				Marginal contribution of U			
				U	D	R	
U	D	R	$\longrightarrow$	0	0	1	(U enters first, then D, then R)
U	R	D	$\longrightarrow$	0	0	1	
D	U	R	$\longrightarrow$	0	0	1	
D	R	U	$\longrightarrow$	$\frac{1}{4}$	0	$\frac{3}{4}$	(D enters first, then R, then U)
R	D	U	$\longrightarrow$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$	
R	U	D	$\longrightarrow$	$\frac{1}{2}$	0	$\frac{1}{2}$	

Thus

$$\begin{array}{lllllll} U & \text{gets} & \frac{\frac{1}{4}+\frac{1}{4}+\frac{1}{2}}{6} & = & \phi(U) & = & \frac{4}{24} \\ D & \text{gets} & \frac{\frac{1}{4}}{6} & = & \phi(D) & = & \frac{1}{24} \\ R & \text{gets} & \frac{1+1+1+\frac{3}{4}+\frac{1}{2}+\frac{1}{2}}{6} & = & \phi(U) & = & \frac{19}{24} \end{array}$$

Landlord Versus Workers

Suppose the landlord hires all the n workers in the village who are landless farmers. There are $(n+1)!$ possible arrangements.
When the landlord is the first to join the ϕ coalition and the rest follow in a random order, his marginal contribution is 0. There are $n!$ cases and out of $(n+1)!$ possible permutations. L's marginal contribution if L is the ith to enter is $f(i), i = 1, 2 \ldots n+1$. Thus his expected marginal contribution is $\frac{1}{n+1}\sum_0^n f(i)$. Thus

$$\phi(L) = \frac{1}{n+1}\sum_0^n f(i).$$

Since farmers are homogeneous $\frac{1}{n}[f(n) - \frac{1}{n+1}\sum_0^n f(i)]$ is the Shapley value for each farmer.

Village Election

Assume that Brahmins en block lost any interest in voting. Now in this case we have only 930 voters in all and simple majority needs 466 votes. Thus the competition is only among the 4 castes with first letters P, N, K, C. Let us enumerate all the 24 possible random arrangement as follows

P N $\underline{\text{K}}$ C	K $\underline{\text{N}}$ P C
P N $\underline{\text{C}}$ K	K $\underline{\text{N}}$ C P
P $\underline{\text{K}}$ N C	K $\underline{\text{P}}$ N C
P $\underline{\text{K}}$ C N	K $\underline{\text{P}}$ C N
P C $\underline{\text{N}}$ K	K $\underline{\text{C}}$ N P
P C $\underline{\text{K}}$ N	K $\underline{\text{C}}$ P N
N P $\underline{\text{K}}$ C	C P $\underline{\text{N}}$ K
N P $\underline{\text{C}}$ K	C P $\underline{\text{K}}$ N
N C $\underline{\text{K}}$ P	C $\underline{\text{K}}$ P N
N C $\underline{\text{P}}$ K	C $\underline{\text{K}}$ N P
N $\underline{\text{K}}$ P C	C N $\underline{\text{P}}$ K
N $\underline{\text{K}}$ C P	C N $\underline{\text{K}}$ P

A player is *pivotal* if the majority forms by the player entering and not before [3]. We have underlined pivotal players in this election. We notice that

K is pivotal in 12 cases
P is pivotal in 4 cases
C is pivotal in 4 cases
N is pivotal in 4 cases.

Thus the Shapley value is

$$\phi(K) = \frac{12}{24}, \phi(P) = \phi(C) = \phi(N) = \frac{4}{24}$$

Airport Game [14]

Let N_j be landings by planes of length l_j and let there be n_j landings by such planes. Let $N = \cup N_j$ be the set of all landings. Given any $S \subset N$, let $\mathrm{j}(S) = \max\{j : S \cap N_j \neq \phi\}$ and the cost to cover all landings in S is $C_{\mathrm{j}(S)}$ and $C(\phi) = 0$.

Let

$$R_k = \cup_{j=k}^{m} N_j \; r_k = \sum_{j=k}^{m} n_j.$$

Let $v_1, \; v_2, \; \ldots v_m$ be TU games defined by

$$\begin{aligned} v_k(S) &= 0 \text{ if } S \cap R_k = \phi \\ &= C_{k-1} - C_k \text{if } S \cap R_k \neq \phi. \end{aligned}$$

If $k \leq \mathrm{j}(S)$ then $S \cap R_k \neq \phi$.
If $k > \mathrm{j}(S)$ then $S \cap R_k = \phi$.

Thus

$$\sum_{k=1}^{m} v_k(S) = \sum_{k=1}^{\mathrm{j}(S)} (C_{k-1} - C_k) = C_o - C_{\mathrm{j}(S)} = -C_{\mathrm{j}(S)}.$$

Thus $v = \sum_{k=1}^{m} v_k$ and $\phi(v) = \sum \varphi(v_k)$ (i.e., $v_k(R_k) = v_k(N)$)
v_k is a game with carrier R_k. Namely $v_k(R_k \cap S) = v_k(N \cap S)$ for all $S \subset N$

$$\begin{aligned} \phi_i(v_k) &= 0 \text{ if } i \notin R_k \\ &= \frac{C_{k-1} - C_k}{r_k} \quad \text{if } i \in R_k \end{aligned}$$

If $i \in N_j$ then $i \in R_k$ if $k \leq j$. Thus the Shapley value is

$$\phi_i(v) = \sum_{k=1}^{j} \frac{C_{k-1} - C_k}{r_k} \text{ if } i \in N_j.$$

6.2 Multi-choice Shapley Value ([12])

In many third world countries, the landless agricultural laborers find it difficult to form unions to fight out rich landlords. Since they have no other skills to sell, they cannot quit the village and often they are stuck in villages to survive with the so-called strategy of absenteeism [4].

Suppose a landlord offers half of the marginal contribution of the last worker as wage for all workers treating them as homogeneous workers. The output is a function of their full cooperation. When the landlord hires one sincere worker, they harvest 20 bags of rice. When landlord hires two sincere workers, they harvest 24 bags of rice. This amounts to $v(L) = 0,\ v(W_1) = v(W_2) = v(W_1\ W_2) = 0,\ v(LW_1) = 20,\ v(LW_2) = 20,\ v(LW_1W_2) = 24$. When the worker gets just 2 bags as half of the last worker's marginal contribution, they have all the reason to show greater marginal contribution by absenting themselves under various pretexts.

Assume they pretend to be working sincerely every day but decide to absent $1/4^{th}$ of cultivating days. As an approximation to the expected output they satisfy $v(L) = 0\ v(W_1) = v(W_2) = v(W_1W_2) = 0\ v(LW_1) = v(LW_2) = (20)\left(\frac{3}{4}\right) + 0\left(\frac{1}{4}\right) = 15$, and to $v(LW_1W_2) = (24)\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) + (20)\left(\frac{3}{4}\right)\left(\frac{1}{4}\right).2 + 0.\left(\frac{1}{4}\right)^2 = \frac{216+120}{16} = \frac{336}{16} = 21$. Thus they show higher marginal contribution by absenteeism.

Motivated by this we introduce, the notion of multi-choice games.

Example 6 ([12])

Players 1, 2, 3, and 4 jointly own a piece of land. They have 3 possible actions to take

σ_o : Don't work
σ_1 : work halfheartedly
σ_2 : work whole heartedly.

Then let the final output of corn in bushels be given by

$$\begin{aligned} v(x_1\ x_2\ x_3\ x_4) &= 0 \quad \text{if } |\{x_i : x_i = \sigma_2\}| = 0 \text{ i.e., no one works whole heartedly.} \\ &= 15 + 3|\{x_i : x_i = \sigma_1\}| \quad \text{if } |\{x_i : x_i = \sigma_2\}| = 1 \\ &= 25 \qquad\qquad \text{if } |\{x_i : x_i = \sigma_2\}| = 2 \\ &= 30 \qquad\qquad \text{if } |\{x_i : x_i = \sigma_2\}| \geq 3. \end{aligned}$$

The main question is what is the relative value for one who works halfheartedly or remain lazy or work whole heartedly.

Even voting games have such action choices as the following example illustrates.

Example 7 ([12])

In a mathematics department with 50 faculty members 10 professors are distinguished. They have a meeting to vote on promoting a junior colleague. The bylaws stipulate that to attend the meeting, one should read the promotion papers in the department secretary's room ahead of time. On the average it takes one hour to go through promotion papers. Those who want to strongly support will need another 3 h to convince colleagues with queries. Marginal supporters spend two hours to follow on others' presentations. Weakly approving colleagues try to attend the meeting by just satisfying minimally the bylaws. Any change of mind occurs only before the meeting. Final voting is open.

A candidate can be promoted only under one of the following scenarios:

1. At least 40 professors show their marginal or strong support and at least 2 distinguished professors are present at the meeting.
2. One distinguished professor and 25 professors in all strongly support the candidate.

Here are possible actions

x_o : Skip the meeting
x_1 : Attend but abstain from voting
x_2 : Support marginally the candidate for promotion

x_3 : Support strongly

Let $z = (z_1, z_2, \ldots z_{50})$ represent the action tuple of professors and the first 10 coordinates represent the actions of distinguished professors. Then let

$$\begin{aligned} v(z) &= 1 \quad \text{if the candidate is promoted} \\ &= 0 \quad \text{otherwise} \end{aligned}$$

For example $v(z) = 1$ if $|\{z_j : z_j = x_1 \text{ or } x_2 \text{ or } x_3,\ \ 1 \le j \le 10\}| \ge 2$ and $|\{z_j : z_j = x_2 \text{ or } x_3,\ 1 \le j \le 50\}| \ge 40$
This is a game with 50 players and 4 actions for each player. How do we measure a professor's influence in the final outcome for various possible actions?

Definition 6.1 A multi-choice cooperative game in characteristic function form is a pair $(\boldsymbol{\beta}^n, v)$ defined by $v = \boldsymbol{\beta}^n \to R$ such that for $\boldsymbol{\beta} = \{0, 1, 2.., m\}$, $v(0, 0, .., 0) = 0$.

Definition 6.2 A multi-choice cooperative game is non-decreasing, if $\boldsymbol{x} \ge \boldsymbol{y} \implies v(\boldsymbol{x}) \ge v(\boldsymbol{y})$.

We call the matrix

$$\phi(v) = \left(\phi_{ij}(v)\right)_{\substack{i=1,2..m \\ j=1,2..n}}$$

the power index where ϕ_{ij} is the power of influence of player j when he/she adopts action i. We assign $\phi_{oj}(v) = 0\ \forall\ j \in N$. We associate weights $w_o, w_1, \ldots, w_m$ with actions 0, 1, 2, ..., m such that $0 = w_0 \le w_1 \le ... \le w_m$.

Axiom 1: Let v be of the form

$$v^x(\boldsymbol{y}) = \begin{cases} C > 0 & \text{if} \quad \boldsymbol{y} \ge \boldsymbol{x} \\ 0 & \text{otherwise} \end{cases}$$

Then $\phi_{x_i,i}$ is proportional to $w(x_i)$.

Axiom 1: States that for binary valued (0 or C) games that stipulate a minimal exertion level from players, the reward for a player is proportional to the weight of the player's minimal exertion level.

Definition 6.3 A vector $\boldsymbol{x}^* \in \boldsymbol{\beta}^n$ is called a carrier of v if $v(\boldsymbol{x}^* \wedge \boldsymbol{x}) = v(\boldsymbol{x})\ \forall \boldsymbol{x} \in \boldsymbol{\beta}^n$. We call $\boldsymbol{x^o}$ a minimal carrier of v if $\sum x_i^o =$ min $\{\sum x_i : x$ a carrier of $v\}$.

Definition 6.4 We call a player dummy if

$$v(\boldsymbol{x} \backslash x_i = k) = v(\boldsymbol{x} \backslash x_i = 0)\ \ \forall\ \boldsymbol{x} \in \boldsymbol{\beta}^n\ \ \&\ \text{ for all } k \in \boldsymbol{\beta}.$$

Axiom 2: If x^* is a carrier of v then for $\boldsymbol{m} = (m, m....m)$

$$\sum_{x_i^*} \phi_{x_i^* i}(v) = v(\boldsymbol{m})$$

Axiom 3: $\phi(v^1 + v^2) = \phi(v^1) + \phi(v^2)$ where $(v^1 + v^2)(\boldsymbol{x}) = v^1(\boldsymbol{x}) + v^2(\boldsymbol{x})$.

Axiom 4:Given $\boldsymbol{x}^o \in \boldsymbol{\beta}^n$ if $v(\boldsymbol{x}) = 0$ when $\boldsymbol{x} \not\geq \boldsymbol{x}^o$ then for each $i \in N, \phi_{k,i}(v) = 0$ for all $k < x_i^o$.

This axiom stipulates that people can't be rewarded when they do not meet even the minimal exertion.

Theorem 6.3 *If $w(0) = 0 \leq w(1) \cdots \leq w(m)$ are given, then there exists a unique function $\phi : G \to M_{m \times n}$ satisfying the axioms 1, 2, 3 and 4.*

Definition 6.5 Given $\boldsymbol{\beta}^n$ and $w_o = 0, w_1, \ldots w_m$, let

$$\|\boldsymbol{x}\|_w = \sum_{r=1}^{m} w(x_r).$$

Theorem 6.4 *The multi-choice value function $\phi : G \to M_{m \times n}$ that satisfies axioms* 1, 2, 3, 4 *and* 5 *is given by*

$$\phi_{ij}(v) = \sum_{k=1}^{i} \sum_{\substack{x_j = k \\ x \neq 0 \\ x \in \beta^n}} \left[\sum_{T \subset M_j(x)} (-1)^{|T|} \frac{w(x_j)}{\|x\|_w + \sum_{r \in T}[w(x_{r+1}) - w(x_r)]} \right] \left[v(\boldsymbol{x}) - v(x - \boldsymbol{b}(j)) \right]$$

(Here $\boldsymbol{b}(j)$ is the unit vector with 1 at the jth coordinate).

The complicated formula has the usual interpretation of a suitable expectation with respect to what we call a Shapley distribution. For the proof details see [12].

6.3 Potential Function and Shapley Value ([11])

In an elegant paper, Hart and Mascolel introduced the notion of a potential function and showed its intimate connection to the Shapley value.

Given a TU game (v, N) and given the set G of all its subgames, let a function $P : G \to R$ associate a real number $P(N, v)$ to every game (N, v) and to every subgame $(N \backslash i, v)$. The marginal contribution of a player in the game is defined by

$$D^i P(N, v) = P(N, v) - P(N \backslash i, v).$$

Here $(N \backslash i, v)$ is the subgame by restricting the game to only sub-coalitions of $N \backslash i$.

Definition 6.6 A function $P : G \to R$ with $P(\phi, v) = 0$ is called a *potential function* if it satisfies the following condition

$$\sum_{i \in N} D^i P(N, v) = v(N). \tag{6.2}$$

for all games v, on the fixed player set N.

Thus the allocation of marginal contributions according to potential function adds up exactly to the worth of the grand coalition.

Theorem 6.5 *There exists a unique potential function P with the resulting pay off vector $(D^i P(N, v))_{i \in N}$. Consider the Shapley value $\phi_i(N, v)$ of the game. Then $D^i P(N, v) = \phi_i(N, v)$ for every game (N, v) and each player $i \in N$.*

Proof We can rewrite (6.2) as

$$P(N, v) = \frac{1}{|N|}\Big[v(N) + P(N \backslash i, v)\Big] \tag{6.3}$$

If we start with $P(\phi, v) = 0$, it determines $P(N, v)$ recursively. Thus P exists and is uniquely determined by (6.2) and (6.3) applied to $(S, v) \ \forall \ S \subset N$. Next we can express

$$v = \sum_{T \subset N} \alpha_T v^T$$

where v^T are unanimity games. Let $d_T = \frac{\alpha_T}{|T|}$ and define

$$P(N, v) = \sum_T d_T = \sum \frac{\alpha_T}{|T|} \tag{6.4}$$

($\frac{d_T}{|T|}$ is often called Harsanyi's dividend to the members of the coalition T).
Now for this $P(N, v)$ we can check

$$\sum_i \big[P(N, v) - P(N \backslash i, v)\big]$$

$$= \sum_{i \in N} \Bigg[\sum_{T \subset N} \frac{\alpha_T}{|T|} - \sum_{T \subset N \backslash i} \frac{\alpha_T}{|T|} \Bigg]$$

$$= \sum_{i \in N} \sum_{\substack{T \subset N \\ T \ni i}} \frac{\alpha_T}{|T|} = v(N).$$

Hence (6.4) defines the unique potential. However

$$\phi_i(N, v) = \sum_{T \ni i} \frac{\alpha_T}{|T|}$$

and we are done. □

We have yet another way to understand the potential function. Suppose from $\{1, 2, ..., n\}$ a random s is chosen. Now you restrict only to coalitions s that are of size $|S| = s$ and one of them is chosen at random with probability $\frac{1}{\binom{n}{s}}$.

Proposition 6.2

$$P(N, v) = E\left[\frac{|N|}{|S|}, v(S)\right]$$

for every game (N, v).

Proof We have to show

$$P(N, v) = \sum \frac{(s-1)!(n-s)!}{n!} v(S)$$

Since the marginal contribution of $D^i P$ is also the Shapley value the above is the potential function. While Shapley value is known to be an expected marginal contribution, we obtain it as a marginal contribution to an expectation (potential). □

Remark 6.1 Even though we deal just with one game and its subgames instead of all games with the number of players fixed, we still get the Shapley value. The data we collect is for any one game and its subgames. To fix uniquely the Shapley value axiomatically, we do need the axioms to run over every possible game with N players. For further extensions and refinements and to multi-choice games see [13].

References

1. Bapat R, Raghavan TES (2009) Nonnegative matrices and applications, vol 64. Cambridge University Press
2. Calvo E (2021) Redistribution of tax resources: a cooperative game theory approach. SERIEs 12(04):633–686
3. Cesari AEMSG, Nepomuceno J (2018) An application of the shapley value to the analysis of co-expression networks. Appl Netw Sci 3(01):1–21
4. Chetty V, Raghavan TES (1976) Absenteeism. Efficiency and income distribution, Discussion Paper No, p 7601, Indian Statistical Institute, New Delhi
5. Ciardiello GAF, Simpson A (2020) A unified cooperative model for environmental costs in supply chains: the shapley value for the linear case. Ann Oper Res 290(01):421–437
6. Conklin PKM, Lipovetsky S (2004) Customer satisfaction analysis: identification of key drivers. Eur J Oper Res 154(03):819–827
7. Ferey S, Dehez P (2016) Multiple causation, apportionment, and the shapley value. J Legal Stud 45(01):143–171
8. Ghorbani A, Zou J (2019) Data shapley: equitable valuation of data for machine learning. In: International conference on machine learning. PMLR, pp 2242–2251
9. Gonzalez P, Herreno G (2004) Optimal sharing of surgical costs in the presence of queues. Math Meth Oper Res 59:435–446
10. Granot MMOGD, Zhu W (1996) The kernel/nucleolus of a standard tree game. Int J Game Theory 25(02):219–244

11. Hart S, Mas-Colell A (1988) The potential of the shapley value. In: the Shapley value. Cambridge University Press, pp 127–137
12. Hsiao C, Raghavan T (1993) Shapley value for multichoice cooperative games, I. Games Econ Behav 5(02):240–256
13. Hsiao YY CR, Mo J (1994) The potential of multi-choice cooperative games. Munich Personal RepEC Archive
14. Owen G (1995) Game theory, 3rd edn. Academic Press, New York
15. Shapley L (1953) Contributions to the theory of games, chapter a value for n-person games. J Int Technol Inf Manag 28:307–317
16. Straffin PD Jr (1993) Game theory and strategy, vol 36. MAA

Chapter 7
An Accelerated Block Randomized Kaczmarz Method



Anjie Qiao, Xuezhong Wang, and Yimin Wei

Mathematics Subject Classification (2010) 15A09 · 65F10

7.1 Introduction

The Kaczmarz method is a kind of row iterative method for solving large-scale linear systems. The randomized version of the Kaczmarz method was presented by Strohmer and Vershynin with the exponential convergence [23].

Let A be a real matrix and b be a real vector. The linear system raised by A and b is

$$Ax = b, \text{ with } A \in \mathbb{R}^{m\times n} \text{ and } b \in \mathbb{R}^m, \tag{7.1}$$

and the rows of the matrix A and the vector b are partitioned,

$$A = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}. \tag{7.2}$$

A. Qiao
School of Mathematical Sciences, Fudan University, Shanghai, People's Republic of China
e-mail: 18300180022@fudan.edu.cn

X. Wang
School of Mathematics and Statistics, Hexi University, Zhangye 734000, People's Republic of China
e-mail: xuezhongwang77@126.com

Y. Wei (✉)
School of Mathematical Sciences and Key Laboratory of Mathematics for Nonlinear Sciences, Fudan University, Shanghai, People's Republic of China
e-mail: ymwei@fudan.edu.cn

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_7

On each step, the randomized Kaczmarz (RK) updates (7.3) and requires only $O(n)$ storage,

$$x_{k+1} = x_k + \frac{b_i - a_i x_k}{\|a_i\|_2^2} a_i^\top, \tag{7.3}$$

where the index i is randomly selected from $1, 2, \ldots, m$.

Updating (7.3) actually conducts an orthogonal projection from the current solution to the hyperplane paralleling the kernel of the selected row. Equation (7.4) is the block version of the Randomized Kaczmarz method (BRK) Needell and Tropp [20] with the block partitions of the matrix A and the vector b,

$$A = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_\tau \end{bmatrix}, \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_\tau \end{bmatrix},$$

and

$$x_{k+1} = x_k + A_\tau^\dagger (b_\tau - A_\tau x_k), \tag{7.4}$$

where τ select a set of rows from $1, 2, \ldots, m$ and $A_\tau^\dagger$ is the Moore-Penrose inverse Wang et al. [26] of A_τ.

If the size of τ is bigger, then the update is similar to a randomized linear system solver with an appropriate selecting approach.

Since RK is a greedy random optimization method, it can be combined with the Nestrov acceleration in real conditions. The accelerated randomized Kaczmarz method (ARK) converges faster than the original RK and in some cases even faster than the conjugate gradient (CG) according to Liu and Wright [13]. Because the randomized Kaczmarz method only uses one sample at a time, it is easy to fall into the local optimal solution, so the convergence performance is not good. Using one batch for a block version of ARK (BARK) at a time can greatly reduce the number of iterations required for convergence and at the same time can make the convergence result closer to the effect of gradient descent. The motivation of this work is that we obtain a BARK with a certain selecting method of τ in (7.4). The convergence of BARK is also proved to be faster than the one of BRK.

The method can also be used on real tensor t-product problems, which is described in Sect. 7.5. Ma and Molitor [16] gave two t-RK methods in 2021, one with the Fourier domain and the other computes matricized version of tensor equations Ma and Molitor [16]. We evolve the two methods to t-BARK and illustrate the performance of the algorithms on some numerical examples in the end.

7.2 Randomized Kaczmarz Method

The RK method for solving (7.1) with (7.3) is shown as Algorithm 1.

Algorithm 1 (Strohmer and Vershynin [23]) (Randomized Kaczmarz)

Input: A, b, x_0 and l

1: for $k = 0, 1, ..., l$

2: Select i_k from {1,2,...,m}

3: Set $x_{k+1} = x_k + \frac{b_i - a_i x_k}{\|a_i\|_2^2} a_i^\top$.

4: end for

output: x_l

There are two common methods to select i_k for each step. By the first one, we select it with the probability of $\frac{\|a_{i_k}\|^2}{\|A\|_F^2}$, and by the second one, we normalize each row of $a_i = \frac{a_i}{\|a_i\|_2}$ before the iteration and select with the same probability. Both of the two selecting methods lead to the exponential convergence.

Theorem 7.1 (Strohmer and Vershynin [23, Theorem 2]) *Let x be the solution of equation (7.1). Algorithm 1 returns x_k in expectation with the average error*

$$\mathbb{E}\|x_k - x\|_2^2 \leq [1 - \kappa(A)^{-2}]^k \, \|x_0 - x\|_2^2, \tag{7.5}$$

where the condition number (Cucker et al. [8]) is

$$\kappa(A) = \|A\|_F \|A^\dagger\|_2. \tag{7.6}$$

The following equations show how (7.3) is established. We select a row of A as a_i. Greedily, we find a γ to optimize

$$\min_{\gamma} \|A(x_k + a_i^\top \gamma) - b\|_2.$$

The solution of the above sub-question is $\gamma = \frac{b_i - a_i x_k}{\|a_i\|_2}$, which obtains (7.3). For block version, we replace a_i with A_τ, where $|\tau| = l$. The sub-question turns to

$$\min_{\gamma \in \mathbb{R}^l} \|A_\tau(x_k + A_\tau^\top \gamma) - b\|_2.$$

The solution is

$$A_\tau^\top \gamma = A_\tau^\dagger (b_\tau - A_\tau x_k).$$

Here we obtain the Block Randomized Kaczmarz method (Algorithm 2).

Algorithm 2 (Needell and Tropp [20]) (Block Randomized Kaczmarz)

Input: A, b, x_0, h and l
1: for $k = 0, 1, \ldots, l$
2: Select $\tau_k \in \mathbb{R}^h$ from $\{1, 2, \ldots, m\}$
3: Set $x_{k+1} = x_k + A_{\tau_k}^\dagger (b_{\tau_k} - A_{\tau_k} x_k)$
4: end for
output: x_l

The convergence of BRK is influenced by how the index set τ is selected.

Definition 7.1 (Needell and Tropp [20]) (Row Paving). An h-row paving of matrix A is a partition $T = \{\tau_1, \tau_2, \ldots, \tau_h\}$, where

$$\alpha \leq \lambda_{\min}(A_\tau A_\tau^*) \text{ and } \lambda_{\max}(A_\tau A_\tau^*) \leq \beta \text{ for each } \tau \in T.$$

The number h is the size of the paving and α and β are the lower and upper paving bounds. Notice that the ratio $\frac{\beta}{\alpha}$ is the bound of the condition number $\kappa^2(A_\tau)$ for each $\tau \in T$.

Theorem 7.2 (Needell and Tropp [20, Theorem 1.2]) *Let x be the solution of linear equations (7.1) and x_k be the result of Algorithm 2. It holds*

$$\mathbb{E}\|x - x_k\|_2^2 \leq \left(1 - \frac{\sigma_{\min}^2(A)}{h\beta}\right)^k \|x_0 - x\|_2^2. \tag{7.7}$$

Every step of Algorithm 2 conducts a basic randomized linear system solver (rNLA) with random row sampling Drineas and Mahoney [9]. Algorithm 2 can also be regarded as an iterative version of the basic randomized linear system solver, which implies its convergence and another usage—to estimate the error of results given by other algorithms at a low cost.

Algorithm 3 (Liu and Wright[13]) (Accelerated Randomized Kaczmarz)

Input: A, b, x_0, l and γ

1: Normalize A as $a_i = \frac{a_i}{\|a_i\|^2}$.

2: Check that $\gamma \in [0, \lambda_{\min}]$;

3: Set $v_0 = x_0$, $\gamma_{-1} = 0$;

4: for $k = 0, 1, \ldots, l$;

5: Set γ_k the larger root of $\gamma_k^2 - \frac{\gamma_k}{m} = (1 - \frac{\gamma_k \lambda}{m}\gamma_{k-1}^2)$;

6: Set $\alpha_k = \frac{m - \gamma_k \lambda}{\gamma_k (m^2 - \gamma)}$, $\beta_k = 1 - \frac{\gamma_k \lambda}{m}$;

7: Select i randomly from $\{1, 2, \ldots, m\}$;

8: Set y_k, x_{k+1}, v_{k+1} as:

$$\begin{cases} y_k = \alpha_k v_k + (1 - \alpha_k) x_k, \\ x_{k+1} = y_k - a_i^\top (a_i y_k - b_i), \\ v_{k+1} = \beta_k v_k + (1 - \beta_k) y_k - \gamma_k a_i^\top (a_i y_k - b_i); \end{cases}$$

9: end for.

output: x_l

7.3 Block Accelerated RK

The update of x_k in step 3 of Algorithm 1 can be written as $x_{k+1} = x_k - \theta_k \nabla f(x_k)$, where θ is the stepsize and $\nabla f(x)$ is the objective gradient. Nestrov's accelerated procedure Nesterov [21] introduces sequences y_k and v_k as

$$\begin{cases} y_k = \alpha_k v_k + (1 - \alpha_k) x_k, \\ x_{k+1} = y_k - \theta_k \nabla f(y_k), \\ v_{k+1} = \beta_k v_k + (1 - \beta_k) y_k - \gamma_k \nabla f(y_k). \end{cases}$$

The procedure gives a better convergence with appropriate α_k, β_k and γ_k. Scalars α_k, β_k and γ_k in Algorithm 3 guarantee a better convergence than Algorithm 1 with an appropriate choice of γ Liu and Wright [13].

In step 8, we can calculate $a_i^\top (a_i y_k - b_i)$ only once to make the update less expensive. The total cost of step 8 is $9n$. If parallel computation is possible, the total cost can be reduced to $4n$ by calculating

$$\begin{cases} s_k = a_i y_k - b_i, \\ g_k = s_k a_i, \\ g'_k = (1 - \alpha_{k+1} + \alpha_{k+1}\gamma_k) s_k a_i, \\ y'_{k+1} = (1 - m\gamma_k)\alpha_{k+1} x_k + (1 - \alpha_{k+1} + m\alpha_{k+1}\gamma_k) y_k, \\ y_{k+1} = y'_{k+1} - g'_k, \\ x_{k+1} = y_k - g_k. \end{cases}$$

We obtain the block version of ARK as follows by combining Algorithms 2 and 2. The following algorithm (orthogonalization with the Gram-Schmidt process) is called before the iterations of BARK (Block Accelerated Random Kaczmarz).

Algorithm 4 (Orthogonalization)

Input: A, b and row paving T

1: for $\tau \in T$

2: for i in each τ

3: $a_i = a_i - \sum_{j \le i} \langle a_i, a_j \rangle a_j, \quad b_i = b_i - \sum_{j \le i} \langle a_i, a_j \rangle b_j$

4: $a_i = \frac{a_i}{\|a_i\|_2}, \quad b_i = \frac{b_i}{\|b_i\|_2}$

5: end for

6: end for

output: A, b

Now we obtain BARK as follows.

Algorithm 5 (Block Accelerated Randomized Kaczmarz)

Input: A, b, x_0, l, T and γ

1: Orthogonalize A with row paving T by Algorithm 4.

2: Check that $\lambda \in [0, \lambda_{\min}]$;

3: Set $v_0 = x_0$, $\gamma_{-1} = 0$;

4: for $k = 0, 1, \ldots, l$;

5: Set γ_k the larger root of $\gamma_k^2 - \frac{\gamma_k}{h} = (1 - \frac{\gamma_k \lambda}{h}\gamma_{k-1}^2)$;
where h is the size of each $\tau \in T$

6: Set $\alpha_k = \frac{h - \gamma_k \lambda}{\gamma_k (h^2 - \gamma)}$, $\beta_k = 1 - \frac{\gamma_k \lambda}{h}$;

7: Select τ from T;

8: Set y_k, x_{k+1}, v_{k+1} as:

$$\begin{cases} y_k = \alpha_k v_k + (1 - \alpha_k) x_k, \\ x_{k+1} = y_k - A_\tau^\dagger (b_\tau - A_\tau x_k), \\ v_{k+1} = \beta_k v_k + (1 - \beta_k) y_k - \gamma_k A_\tau^\dagger (b_\tau - A_\tau x_k); \end{cases}$$

9: end for.

output: x_l

7.4 Convergence and Error

Let us discuss the convergence of Algorithm 4. To give the convergence analysis, we make the following definition.

Definition 7.2 (Shi et al.[22], Zouzias and Freris [31]) Let $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^m$. The energy norm $\|\cdot\|_{(A^\top A)^\dagger}$ is defined as

$$\|x\|^2_{(A^\top A)^\dagger} = \langle x, (A^\top A)^\dagger x\rangle = x^\top (A^\top A)^\dagger x. \tag{7.8}$$

It is trivial that (7.8) yields an energy norm of a vector x.

For BARK, we have the following result. The matrix A in our analysis is already been orthogonalized with the row paving T according to Algorithm 4.

Theorem 7.3 *Let x be the solution of (7.2) and x_k, v_k be the result of Algorithm 4 with the sequence λ and row paving (Definition 7.1) satisfying $\lambda \in [0, \frac{\lambda_{\min}}{\beta}]$. Define $\sigma_1 = 1 + \frac{\sqrt{\lambda}}{2h}$ and $\sigma_2 = 1 - \frac{\sqrt{\lambda}}{2h}$, then*

$$\mathbb{E}\left(\|v_k - x\|^2_{(A^\top A)^\dagger}\right) \le \frac{4\|x_0 - x\|^2_{(A^\top A)^\dagger}}{(\sigma_1^k + \sigma_2^k)^2}, \tag{7.9}$$

and

$$\mathbb{E}\left(\|x_k - x\|_2^2\right) \le \frac{4\lambda\|x_0 - x\|^2_{(A^\top A)^\dagger}}{(\sigma_1^k - \sigma_2^k)^2}. \tag{7.10}$$

We start out proof with two useful lemmas.

Lemma 7.1 *For any $y \in \mathbb{R}^n$, we have*

$$\mathbb{E}_\tau\left(\|A_\tau^\dagger(b_\tau - A_\tau y)\|^2_{(A^\top A)^\dagger}\right) \le \frac{1}{h}\|Ay - b\|_2^2. \tag{7.11}$$

Proof Since τ is chosen at random from T, we have

$$\begin{aligned}&\mathbb{E}_\tau\|A_\tau^\dagger(A_\tau y - b_\tau)\|_{(A^\top A)^\dagger}\\ &= \frac{1}{h}\sum_1^h \left\langle (A^\top A)^\dagger A_\tau^\dagger(A_\tau y - b_\tau), A_\tau^\dagger(A_\tau y - b_\tau)\right\rangle.\end{aligned} \tag{7.12}$$

Since A is orthogonalized with row paving T, we have $A_\tau^\dagger = A_\tau^\top$, (7.12) turns to

$$
\begin{aligned}
&\mathbb{E}_\tau \|A_\tau^\dagger (A_\tau y - b_\tau)\|_{(A^\top A)^\dagger} \\
&= \frac{1}{h} \sum_1^h \left\langle (A^\top A)^\dagger A_\tau^\top (A_\tau y - b_\tau),\, A_\tau^\top (A_\tau y - b_\tau) \right\rangle \\
&= \frac{1}{h} \text{trace} \left((A^\top A)^\dagger \sum_1^h A_\tau^\top (A_\tau y - b_\tau)(A_\tau y - b_\tau)^\top A_\tau \right) \\
&= \frac{1}{h} \text{trace} \left((A^\top A)^\dagger A^\top X A \right) = \frac{1}{h} \text{trace} \left(A^\dagger X A \right),
\end{aligned}
$$

where $X = \text{diag}(A_\tau y - b_\tau)(A_\tau y - b_\tau)^\top$ for all τ in T.

We utilize the singular value decomposition (SVD) of A as $U \Sigma V^\top$ and $(A^\top A)^\dagger = V \Sigma^{-2} V^\top$, where U and V are orthogonal matrices, respectively. Thus

$$
\begin{aligned}
&\mathbb{E}_\tau \|A_\tau^\dagger (A_\tau y - b_\tau)\|_{(A^\top A)^\dagger} \\
&= \frac{1}{h} \text{trace} \left(V \Sigma^{-1} U^\top X U \Sigma V^\top \right) \\
&= \frac{1}{h} \text{trace} \left(U^\top X U \right) = \frac{1}{h} \text{trace}(X) \\
&\le \frac{1}{h} \sum_T \|A_\tau y - b_\tau\|_2^2 = \frac{1}{h} \|Ay - b\|_2^2.
\end{aligned}
$$

□

Lemma 7.2 *For any solution x to (7.2) and any $y \in \mathbb{R}^n$, we have*

$$
\mathbb{E}_\tau \left(\|y + A_\tau^\dagger (b_\tau - A_\tau y) - x\|_2^2 \right) = \|y - x\|_2^2 - \frac{1}{h} \|Ay - b\|_2^2, \tag{7.13}
$$

where τ is selected randomly from rows.

Proof We have

$$
\begin{aligned}
&\mathbb{E}_\tau \left(\|y + A_\tau^\dagger (b_\tau - A_\tau y) - x\|_2^2 \right) \\
&= \|y - x\|_2^2 + \mathbb{E}_\tau (\|A_\tau^\dagger (b_\tau - A_\tau y)\|_2^2) + 2\mathbb{E}_\tau \left\langle y - x,\, A_\tau^\dagger (b_\tau - A_\tau y) \right\rangle \\
&= \|y - x\|_2^2 + \mathbb{E}_\tau (\|A_\tau^\dagger A_\tau (x - y)\|_2^2) + 2\mathbb{E}_\tau \left\langle y - x,\, A_\tau^\dagger A_\tau (x - y) \right\rangle \\
&= \|y - x\|_2^2 - \mathbb{E}_\tau (\|A_\tau^\dagger A_\tau (x - y)\|_2^2) \\
&= \|y - x\|_2^2 - \frac{1}{h} \sum_T \|A_\tau^\top (b_\tau - A_\tau y)\|_2^2 \\
&= \|y - x\|_2^2 - \frac{1}{h} \|Ay - b\|_2^2.
\end{aligned}
$$

We finish the proof. □

Now we prove Theorem 7.3.

Proof Define

$$t(\gamma) := \gamma^2 - \frac{\gamma}{h}\left(1 - \lambda\gamma_{k-1}^2\right) - \gamma_{k-1}^2. \tag{7.14}$$

For $\lambda \leq \frac{\lambda_{\min}}{\sqrt{\beta}} \leq \frac{1}{h}$, $\gamma_{-1} = 0$, we have

$$t(0) = -\gamma_{k-1}^2 \leq 0, \quad t\left(\frac{1}{h}\right) = \gamma_{k-1}^2\left(\frac{\lambda h}{m^2} - 1\right) \leq 0. \tag{7.15}$$

If $\gamma_{k-1} \leq \frac{1}{\sqrt{\lambda}}$, then we have $t\left(\gamma_{k-1}\right) = -\frac{h\gamma_{k-1}}{m}\left(1 - \lambda\gamma_{k-1}^2\right) \leq 0$, and

$$\begin{aligned} t\left(\frac{1}{\sqrt{\lambda}}\right) &= \frac{1}{\lambda} - \frac{h}{m\sqrt{\lambda}}\left(1 - \lambda\gamma_{k-1}^2\right) - \gamma_{k-1}^2 \\ &= \frac{1}{\lambda} - \frac{h}{m\sqrt{\lambda}} + \gamma_{k-1}^2\left(\frac{h\sqrt{\lambda}}{m} - 1\right) \\ &\geq \frac{1}{\lambda} - \frac{h}{m\sqrt{\lambda}} + \frac{1}{\lambda}\left(\frac{h\sqrt{\lambda}}{m} - 1\right) = 0. \end{aligned} \tag{7.16}$$

Equations (7.15) and (7.16) indicate that $\gamma_k \in \left[\gamma_{k-1}, \frac{1}{\sqrt{\lambda}}\right]$. From how α, β and γ yield in out algorithm, there are

$$\begin{cases} \frac{1-\alpha_k}{\alpha_k} = \frac{h^2\gamma_k - h}{h - \gamma_k\lambda} = \frac{h}{\gamma_k}\frac{h\gamma_k^2 - \gamma_k}{h - \gamma_k\lambda h} = \frac{h\gamma_{k-1}^2}{\gamma_k}, \\ \gamma_k^2 - \frac{\gamma_k}{h} - \beta_k\gamma_{k-1}^2 = 0. \end{cases}$$

Let r be the error of the algorithm

$$r_k := \|v_k - x\|_{\left(A^\top A\right)^\dagger}. \tag{7.17}$$

We have

$$\begin{aligned} r_{k+1}^2 &= \left\|v_{k+1} - x\right\|_{\left(A^\top A\right)^\dagger}^2 \\ &= \left\|\beta_k v_k + (1-\beta_k)\, y_k - \gamma_k A_\tau^\dagger\left(A_\tau y_k - b_\tau\right) - x\right\|_{\left(A^\top A\right)^\dagger}^2 \\ &= \left\|\beta_k v_k + (1-\beta_k)\, y_k - x\right\|_{\left(A^\top A\right)^\dagger}^2 + \gamma_k^2\left\|A_\tau^\dagger\left(A_\tau y_k - b_\tau\right)\right\|_{\left(A^\top A\right)^\dagger}^2 \\ &\quad - 2\gamma_k\left\langle\beta_k v_k + (1-\beta_k)\, y_k - x, \left(A^\top A\right)^\dagger A_\tau^\dagger\left(A_\tau y_k - b_\tau\right)\right\rangle \\ &= \left\|\beta_k v_k + (1-\beta_k)\, y_k - x\right\|_{\left(A^\top A\right)^\dagger}^2 + \gamma_k^2\left\|A_\tau^\dagger\left(A_\tau y_k - b_\tau\right)\right\|_{\left(A^\top A\right)^\dagger}^2 \end{aligned}$$

$$
\begin{aligned}
&-2\gamma_k\left\langle\beta_k\left(\frac{1}{\alpha_k}y_k-\frac{1-\alpha_k}{\alpha_k}x_k\right)+(1-\beta_k)\,y_k-x,\left(A^\top A\right)^\dagger A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right\rangle\\
&=\left\|\beta_k v_k+(1-\beta_k)\,y_k-x\right\|^2_{\left(A^\top A\right)^\dagger}+\gamma_k^2\left\|A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right\|^2_{\left(A^\top A\right)^\dagger}\\
&\quad+2\gamma_k\left\langle x-y_k+\frac{1-\alpha_k}{\alpha_k}\beta_k\left(x_k-y_k\right),\left(A^\top A\right)^\dagger A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right\rangle.
\end{aligned}
\tag{7.18}
$$

The first item in (7.18) can be bouded as follows.

$$
\begin{aligned}
&\left\|\beta_k v_k+(1-\beta_k)\,y_k-x\right\|^2_{\left(A^\top A\right)^\dagger}\\
&\le\beta_k\left\|v_k-x\right\|^2_{\left(A^\top A\right)^\dagger}+(1-\beta_k)\left\|y_k-x\right\|^2_{\left(A^\top A\right)^\dagger}\\
&=\beta_k\left\|v_k-x\right\|^2_{\left(A^\top A\right)^\dagger}+\frac{\gamma_k\lambda}{h}\left\|y_k-x\right\|^2_{\left(A^\top A\right)^\dagger}\\
&\le\beta_k\left\|v_k-x\right\|^2_{\left(A^\top A\right)^\dagger}+\frac{\gamma_k}{h}\left\|y_k-x\right\|_2^2.
\end{aligned}
\tag{7.19}
$$

For the second item, recalling Lemmas 7.1 and 7.2, we have

$$
\begin{aligned}
&\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right\|^2_{\left(A^\top A\right)^\dagger}\right)\\
\le&\frac{1}{h}\left\|Ay_k-b\right\|_2^2=\left\|y_k-x\right\|_2^2-\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|x_{k+1}-x\right\|_2^2\right).
\end{aligned}
\tag{7.20}
$$

For the third one, we have

$$
\begin{aligned}
&\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left\langle x-y_k+\frac{1-\alpha_k}{\alpha_k}\beta_k\left(x_k-y_k\right),\left(A^\top A\right)^\dagger\left(A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right)\right\rangle\\
&=\left\langle x-y_k+\frac{1-\alpha_k}{\alpha_k}\beta_k\left(x_k-y_k\right),\left(A^\top A\right)^\dagger\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right)\right\rangle\\
&=\frac{1}{h}\left\langle x-y_k+\frac{1-\alpha_k}{\alpha_k}\beta_k\left(x_k-y_k\right),\left(A^\top A\right)^\dagger\sum_i A_\tau^\dagger\left(A_\tau y_k-b_\tau\right)\right\rangle\\
&=\frac{1}{h}\left\langle x-y_k+\frac{1-\alpha_k}{\alpha_k}\beta_k\left(x_k-y_k\right),\left(A^\top A\right)^\dagger A^\top A\left(y_k-x\right)\right\rangle\\
&\le\frac{1}{h}\left\langle x-y_k+\frac{1-\alpha_k}{\alpha_k}\beta_k\left(x_k-y_k\right),y_k-x\right\rangle\\
&=\frac{1}{h}\left(-\left\|y_k-x\right\|^2+\frac{1-\alpha_k}{\alpha_k}\beta_k\left\langle x_k-y_k,y_k-x\right\rangle\right)\\
&=\frac{1}{h}\left(-\left\|y_k-x\right\|_2^2+\frac{1-\alpha_k}{2\alpha_k}\beta_k\left(\left\|x_k-x\right\|_2^2-\left\|y_k-x\right\|_2^2-\left\|x_k-y_k\right\|_2^2\right)\right)\\
&=\frac{1}{h}\left(-\left(1+\frac{1-\alpha_k}{2\alpha_k}\beta_k\right)\left\|y_k-x\right\|_2^2+\frac{1-\alpha_k}{2\alpha_k}\beta_k\left(\left\|x_k-x\right\|_2^2-\left\|x_k-y_k\right\|_2^2\right)\right)\\
&=-\left(\frac{1}{h}+\frac{\beta_k\gamma_{k-1}^2}{2\gamma_k}\right)\left\|y_k-x\right\|_2^2+\frac{\beta_k\gamma_{k-1}^2}{2\gamma_k}\left(\left\|x_k-x\right\|_2^2-\left\|x_k-y_k\right\|_2^2\right)
\end{aligned}
$$

$$\leq -\left(\frac{1}{h}+\frac{\beta_k\gamma_{k-1}^2}{2\gamma_k}\right)\|y_k-x\|_2^2+\frac{\beta_k\gamma_{k-1}^2}{2\gamma_k}\|x_k-x\|_2^2. \tag{7.21}$$

Combining (7.19), (7.20) and (7.21), we obtain

$$\begin{aligned}
&\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(r_{k+1}^2\right)\\
\leq&\beta_k\left\|v_k-x^*\right\|_{(A^\top A)^\dagger}+\frac{\gamma_k}{h}\left\|y_k-x^*\right\|_2^2\\
&+\gamma_k^2\left(\left\|y_k-x^*\right\|_2^2-\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|x_{k+1}-x^*\right\|_2^2\right)\right)\\
&-\left(\frac{2\gamma_k}{h}+\beta_k\gamma_{k-1}^2\right)\left\|y_k-x^*\right\|_2^2+\beta_k\gamma_{k-1}^2\left\|x_k-x^*\right\|_2^2\\
\leq&\beta_k\left\|v_k-x^*\right\|_{(A^\top A)^\dagger}-\gamma_k^2\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|x_{k+1}-x^*\right\|_2^2\right)+\beta_k\gamma_{k-1}^2\left\|x_k-x^*\right\|_2^2\\
&+\left(\gamma_k^2-\frac{\gamma_k}{h}-\beta_k\gamma_{k-1}^2\right)\left\|y_k-x^*\right\|_2^2\\
=&\beta_k\left\|v_k-x^*\right\|_{(A^T A)^\dagger}-\gamma_k^2\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|x_{k+1}-x^*\right\|_2^2\right)+\beta_k\gamma_{k-1}^2\left\|x_k-x^*\right\|_2^2.
\end{aligned} \tag{7.22}$$

Then we define sequences $\{A_k\}$ and $\{B_k\}$ as follows:

$$A_k\geq 0,\ B_k\geq 0,\ B_0\neq 0,\ B_{k+1}^2=\frac{B_k^2}{\beta_k},\ A_{k+1}^2=\gamma_k^2B_{k+1}^2. \tag{7.23}$$

Let $A_0=0$. Since $\beta_k\in(0,1]$, $B_{k+1}>B_k$. From the definition of β and γ, we have

$$A_{k+1}^2=\frac{B_k^2\gamma_k^2}{\beta_k}=\frac{\gamma_k^2A_k^2}{\beta_k\gamma_{k-1}^2}=\frac{A_k^2\gamma_k^2}{\gamma_k^2-\gamma_k/h}. \tag{7.24}$$

Now we know that $\{A_k\}$ is increasing. Multiplying (7.22) by B_{k+1}^2 on each side, using Algorithm 5 and (7.24), we obtain

$$\begin{aligned}
B_{k+1}^2\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(r_{k+1}^2\right)+A_{k+1}^2\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|x_{k+1}-x^*\right\|_2^2\right)\\
\leq B_k^2r_k^2+A_k^2\left\|x_k-x^*\right\|_2^2,
\end{aligned}$$

and

$$\begin{aligned}
&\mathbb{E}_{\tau(0)\tau(1)\ldots\tau(k)}\left(B_{k+1}^2r_{k+1}^2+A_{k+1}^2\left(\left\|x_{k+1}-x^*\right\|_2^2\right)\right)\\
&=\mathbb{E}_{\tau(0)\tau(1)\ldots\tau(k-1)}\left(B_{k+1}^2\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(r_{k+1}^2\right)\right.\\
&\left.+A_{k+1}^2\mathbb{E}_{\tau(k)|\tau(0),\tau(1),\ldots,\tau(k-1)}\left(\left\|x_{k+1}-x^*\right\|_2^2\right)\right)\\
&\leq\mathbb{E}_{\tau(0)\tau(1)\ldots\tau(k-1)}\left(B_k^2r_k^2+A_k^2\left\|x_k-x^*\right\|_2^2\right).
\end{aligned}$$

Recursively, we obtain

$$\begin{aligned}
&\mathbb{E}_{\tau(0)\tau(1)\ldots\tau(k)}\left(B_{k+1}^2 r_{k+1}^2 + A_{k+1}^2\left(\left\|x_{k+1}-x^*\right\|_2^2\right)\right)\\
&\leq \mathbb{E}_{\tau(0)}\left(B_1^2 r_1^2 + A_1^2\left\|x_1-x^*\right\|_2^2\right)\\
&\leq B_0^2 r_0^2 + A_0^2\left\|x_0-x^*\right\|_2^2 = B_0^2 r_0^2.
\end{aligned}$$

We drop the last term since $A_0 = 0$. We achieve the bounds,

$$\mathbb{E}\left(r_{k+1}^2\right) \leq \frac{B_0^2}{B_{k+1}^2} r_0^2 \quad \text{and} \quad \mathbb{E}\left(\left\|x_{k+1}-x^*\right\|_2^2\right) \leq \frac{B_0^2}{A_{k+1}^2} r_0^2.$$

Now we try to estimate the growth of A_k and B_k. Here we follow Nestrov's proof [21] and we have $B_k^2 = B_{k+1}^2\beta_k = \left(1-\frac{\lambda}{h}\gamma_k\right)B_{k+1}^2 = \left(1-\frac{\lambda A_{k+1}}{hB_{k+1}}\right)B_{k+1}^2$, which implies that

$$\frac{\lambda}{h} A_{k+1}B_{k+1} = B_{k+1}^2 - B_k^2 = \left(B_k + B_{k+1}\right)\left(B_{k+1} - B_k\right).$$

Reminding that $B_{k+1} \geq B_k$, we have

$$B_{k+1} \geq B_k + \frac{\lambda}{2h} A_{k+1}. \tag{7.25}$$

Using the definition of A, B, β and γ, we obtain

$$\begin{aligned}
\frac{A_{k+1}^2}{B_{k+1}^2} - \frac{A_{k+1}}{B_{k+1}h} &= \gamma_k^2 - \frac{\gamma_k}{h}\\
&= \left(1-\frac{\gamma_k\lambda}{h}\right)\gamma_{k-1}^2\\
&= \frac{\beta_k A_k^2}{B_k^2} = \frac{A_k^2}{B_{k+1}^2}.
\end{aligned}$$

Recalling $A_{k+1} > A_k$, we get

$$\frac{1}{m} A_{k+1}B_{k+1} = A_{k+1}^2 - A_k^2 = \left(A_{k+1}+A_k\right)\left(A_{k+1}-A_k\right) \leq 2A_{k+1}\left(A_{k+1}-A_k\right).$$

Thus,

$$A_{k+1} \geq A_k + \frac{B_{k+1}}{2h} \geq A_k + \frac{B_k}{2h}. \tag{7.26}$$

By combining (7.25) and (7.26), we obtain the following estimation:

$$\begin{bmatrix} A_{k+1} \\ B_{k+1} \end{bmatrix} \geq \begin{bmatrix} 1 & \frac{1}{2h} \\ \frac{\lambda}{2h} & 1 \end{bmatrix}^{k+1} \begin{bmatrix} A_0 \\ B_0 \end{bmatrix}. \tag{7.27}$$

The Jordan canonical form of the matrix in (7.27) becomes

$$\begin{bmatrix} \frac{1}{\lambda} & \frac{1}{2h} \\ \frac{\lambda}{2h} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \sqrt{\lambda} & -\sqrt{\lambda} \end{bmatrix}^{-1} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \sqrt{\lambda} & -\sqrt{\lambda} \end{bmatrix},$$

with $\sigma_1 = 1 + \frac{\sqrt{\lambda}}{2h}$, $\sigma_2 = 1 - \frac{\sqrt{\lambda}}{2h}$. Thus, we have

$$\begin{aligned} \begin{bmatrix} 1 & \frac{11}{2h} \\ \frac{\lambda}{2h} & 1 \end{bmatrix}^{k+1} &= \frac{1}{2} \begin{bmatrix} 1 & \frac{1}{\sqrt{\lambda}} \\ 1 & -\frac{1}{\sqrt{\lambda}} \end{bmatrix} \begin{bmatrix} \sigma_1^{k+1} & 0 \\ 0 & \sigma_2^{k+1} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \sqrt{\lambda} & -\sqrt{\lambda} \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} \sigma_1^{k+1} + \sigma_2^{k+1} & \left(\sigma_1^{k+1} - \sigma_2^{k+1}\right)/\sqrt{\lambda} \\ \left(\sigma_1^{k+1} - \sigma_2^{k+1}\right)\sqrt{\lambda} & \sigma_1^{k+1} + \sigma_2^{k+1} \end{bmatrix}. \end{aligned}$$

We get

$$A_{k+1} \geq B_0 \left(\sigma_1^{k+1} - \sigma_2^{k+1}\right)/(2\sqrt{\lambda}), \tag{7.28}$$

and

$$B_{k+1} \geq \left(\sigma_1^{k+1} + \sigma_2^{k+1}\right) B_0/2. \tag{7.29}$$

Applying (7.28) and (7.29) to (7.22), we obtain

$$\begin{cases} \mathbb{E}\left(r_{k+1}^2\right) = \mathbb{E}\left(\|v_{k+1} - x^*\|_{(A^\top A)^\dagger}^2\right) \leq \frac{B_0^2}{B_{k+1}^2} r_0^2 \leq \frac{4\|x_0 - x^*\|_{(A^\top A)^\dagger}^2}{\left(\sigma_1^{k+1} + \sigma_2^{k+1}\right)^2}, \\ \mathbb{E}\left(\|x_{k+1} - x^*\|_2^2\right) \leq \frac{B_0^2}{A_{k+1}^2} r_0^2 \leq \frac{4\lambda\|x_0 - x^*\|_{(A^\top A)^\dagger}^2}{\left(\sigma_1^{k+1} - \sigma_2^{k+1}\right)^2}, \end{cases}$$

and we complete the proof. □

One difficulty of computing with Algorithm 4 is how to initialize λ. We can set $\lambda = 0$ to guarantee the convergence, but it converges slowly. The best λ we are looking for is its upper bound $\frac{\lambda_{\min}}{\beta}$. Theorem 7.3 indicates that we can perform several steps of BRK before BARK to approximate the best λ. From Theorem 3, we give an estimation of λ as

$$\lambda = h\left(1 - \left(\frac{\|Ax_{k_1} - b\|_2^2}{\|Ax_{k_2} - b\|_2^2}\right)^{\frac{1}{k_2 - k_1}}\right),$$

where k_1 and k_2 are the iterations of the original BRK method. However, in our experiment, the result given by the above equation is often larger than what we want. We also give two conservative estimations.

$$\lambda = \left(1 - \left(\frac{\|Ax_{k_1} - b\|_2^2}{\|Ax_{k_2} - b\|_2^2}\right)^{\frac{1}{k_2 - k_1}}\right),$$

and

$$\lambda = \sqrt{h}\left(1 - \left(\frac{\|Ax_{k_1} - b\|_2^2}{\|Ax_{k_2} - b\|_2^2}\right)^{\frac{1}{k_2 - k_1}}\right).$$

The second equation gives us a λ less than $\frac{\lambda_{\min}}{\beta}$. The third one is the previous square average of the two estimations.

7.5 Applications to Tensor Equations

Third tensor is a three-dimensional matrix, which can be regarded as a series of matrices of the same size. When we process tensors, such as images and videos, tensor decompositions or computational methods Che and Wei [3], Wei and Ding [30] are quite different from matrix methods due to the extra dimensions.

The tensor t-product and tensor operations under the t-product Chang and Wei [1, 2], Che and Wei [4], Che et al. [5], Chen et al. [6], Chen and Qin [7], Lu et al. [14], Lund [15], Miao et al. [18, 19], Tarzanagh and Michailidis [25], Wang et al. [27–29] have been presented, which build a connection between the tensor decomposition and the matrix case. These tools inspire us to tackle the tensor by the matrix approach.

Definition 7.3 (Kilmer and Martin [12], Martin et al. [17]) Let $\mathscr{A} \in \mathbb{R}^{m\times n\times k}$ and $\mathscr{B} \in \mathbb{R}^{n\times t\times k}$. Then t-product is defined as

$$\mathscr{C} = \mathscr{A} \star \mathscr{B} = \text{fold}(\text{bcirc}(\mathscr{A}) * \text{unfold}(\mathscr{B})),$$

where

$$\text{bcirc}(\mathscr{A}) = \begin{pmatrix} \mathscr{A}_{::1} & \mathscr{A}_{::k} & \mathscr{A}_{::k-1} & \cdots & \mathscr{A}_{::2} \\ \mathscr{A}_{::2} & \mathscr{A}_{::1} & \mathscr{A}_{::k} & \cdots & \mathscr{A}_{::3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathscr{A}_{::k} & \mathscr{A}_{::k-1} & \mathscr{A}_{::k-2} & \cdots & \mathscr{A}_{::1} \end{pmatrix}, \quad \text{unfold}(\mathscr{B}) = \begin{pmatrix} \mathscr{B}_{::1} \\ \mathscr{B}_{::2} \\ \vdots \\ \mathscr{B}_{::k} \end{pmatrix}$$

and fold is the inverse operation of unfold.

We now use the BARK to solve the tensor equations Du and Sun [10], Ma and Molitor [16], Tang et al. [24],

$$\mathscr{A} \star x = b, \quad \mathscr{A} \in \mathbb{R}^{m\times n\times k}, \quad b \in \mathbb{R}^{n\times 1\times k}. \tag{7.30}$$

7.5.1 Tensor BARK

To solve the tensor equation (7.30), we only need to deal with the matrix equation

$$\text{bcirc}(\mathscr{A}) * \text{unfold}(x) = \text{unfold}(b). \tag{7.31}$$

Then we have the following algorithm.

Since bcirc($\mathscr{A}$) is circulant, we only need to orthogonalize an $m \times nk$ matrix in step 1, and in each iteration, it takes $O(nk)$.

7.5.2 Tensor BARK in the Fourier Domain

The t-product can be computed by means of the discrete Fourier transform (DFT). Firstly, a block circulant matrix Jin et al. [11], Kilmer and Martin [12], Martin et al. [17] can be block-diagonalized by the DFT, i.e., $(F_k \otimes I_m)\text{bcirc}(\mathscr{A})(F_k^* \otimes I_n)$ is a block diagonal matrix.

Algorithm 6 (t-BARK)

Input: $\mathscr{A}$, b, x_0, l, T and γ

1: Orthogonalize $\text{bcirc}(\mathscr{A})_{1:m,:}$ with row paving T by Algorithm 4.
2: Check that $\lambda \in [0, \lambda_{\min}]$;
3: Set $v_0 = \text{unfold}(x_0)$, $\gamma_{-1} = 0$;
4: for $t = 0, 1, ..., l$;
5: Set γ_t the larger root of $\gamma_t^2 - \frac{\gamma_t}{hk} = (1 - \frac{\gamma_t \lambda}{hk}\gamma_{t-1}^2)$;
where h is the size of each $\tau \in T$
6: Set $\alpha_t = \frac{hk - \gamma_t \lambda}{\gamma_t((hk)^2 - \gamma)}$, $\beta_t = 1 - \frac{\gamma_t \lambda}{hk}$;
7: Select τ from T and s from $\{0, 1, ..., k-1\}$;
8: Set y_t, x_{t+1}, v_{t+1} as:

$$\begin{cases} y_t = \alpha_t v_t + (1 - \alpha_t)x_t, \\ x_{t+1} = y_t - A_{\tau,s}^{\dagger}(b_{\tau,1,s} - A_{\tau,s}x_t), \\ v_{t+1} = \beta_t v_t + (1 - \beta_k)y_t - \gamma_t A_{\tau,s}^{\dagger}(b_{\tau,1,s} - A_{\tau,s}x_t), \end{cases}$$

where $A_{\tau,s} = [A_\tau(:, ns+1 : nk), A_\tau(:, 1 : ns)]$;
9: end for.
output: x_l

Definition 7.4 Let $\mathscr{A} \in \mathbb{R}^{m \times n \times k}$. Then

$$\text{bdiag}(\mathscr{A}) := (F_k \otimes I_m)\text{bcirc}(\mathscr{A})(F_k^* \otimes I_n) = \begin{pmatrix} \widehat{A_1} & & & \\ & \widehat{A_2} & & \\ & & \ddots & \\ & & & \widehat{A_k} \end{pmatrix}, \tag{7.32}$$

where F_k is a $k \times k$ the DFT matrix, i.e.,

$$F_k = \left(f_{pq}\right), \ f_{pq} = \omega_k^{pq}, \ \omega_k = \mathrm{e}^{-\frac{2\pi\iota}{k}}, \ \iota = \sqrt{-1}.$$

Then (7.30) can be written as matrix equations

$$\text{bdiag}(\mathscr{A}) * \text{bdiag}(x) = \text{bdiag}(b). \tag{7.33}$$

Here, bdiag($\mathscr{A}$) is a block diagonal matrix, we can solve $\widehat{A}_i * \widehat{x}_i = \widehat{b}_i$, respectively.

Algoritm 7 (t-BARK in the Fourier domain)

Input: $\mathscr{A}$, b, x_0, l, T and γ
1. A = fft(A, [], 3), b = fft(b, [], 3);
2. for $i = 1, 2, \ldots, k$
3. solve $A_i * x_i = b_i$ with Algorithm 5;
4. end for
5. x = ifft(x, [], 3);

output: x

7.6 Numerical Examples

In this section, we perform numerical examples over BARK, BRK and ARK. In the test examples, the algorithms terminate when reach the max iterations we set.

Comparison between BRK and BARK.

First, we concentrate on the nonsingular equations. We generate the test matrix by initializing its SVD. The test matrices have different scales, condition numbers and ranks.

Example 1

The testing matrices are generated by SVD ($A = U\Sigma V^\top$) with random U, Σ and V.

From Fig. 7.1, we conclude that BARK(opt) performs much better than BRK. BARK(auto) can reach a similar convergence rate as λ is calculated. In some cases, BARK(0) is also competitive compared with BRK. Notice that when $\lambda \leq \frac{\lambda_{\min}}{\beta}$, the line shows a periodic feature, which is distinct in Fig. 7.1c. We explain this by Algorithm 3 that gives a periodic list of α_k and β_k, which leads to the periodic performance of convergence. We also note that when the matrix is extremely nice, which means it has a low condition number, BARK is better than the evolved methods.

RK method can also be used to solve the singular equations.

Example 2

We generate the singular testing matrices as in Example 1.

The performance is similar to those on the nonsingular equations (Fig. 7.2).

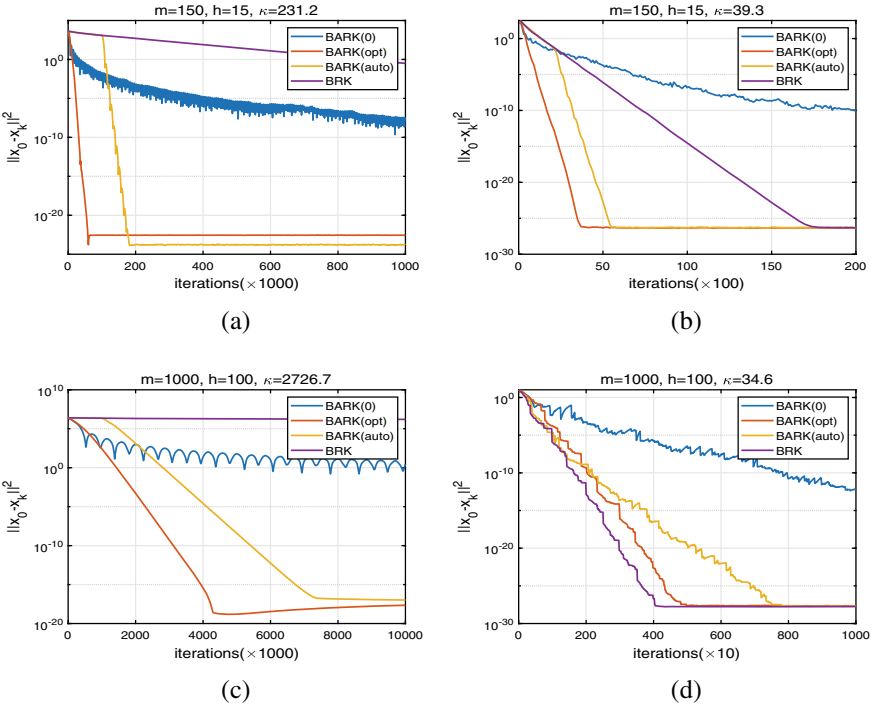


Fig. 7.1 Performance of considered methods for the nonsingular equations. Here m is the size of the matrix, h is the block number and κ is defined in (7.6)

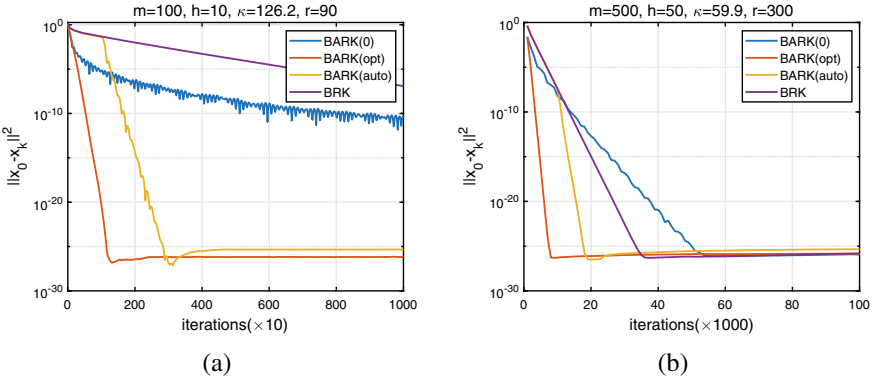


Fig. 7.2 Performance of considered methods for singular equations. r is the rank of the matrix

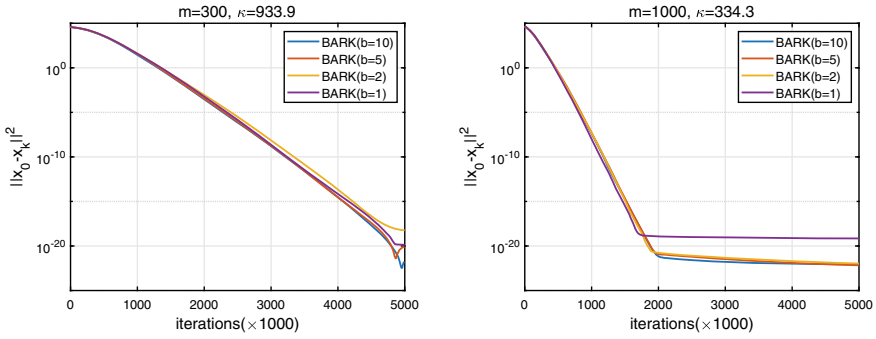


Fig. 7.3 Comparison results between ARK and BARK

7.6.1 Comparison Between ARK and BARK

We compare the two methods by conducting BARK with different block heights (written as b) for the matrices in Example 1.

To balance iterations and complexity, the scale on x-axis is inverse ratio according to b. Thus, points on the vertical line have the same computational complexity. In both cases, all methods with different b have almost the same performance. The greater b is, the more expensive the algorithm cost and the less step it takes. The difference in complexity is mainly generated at the step of orthogonalization (Algorithm 4) before iteration. With parallel calculation, BARK will show its advantage (Fig. 7.3).

7.6.2 T-BARK Performance

The following figure shows that BARK performs on the tensor equations as well as it does on matrix case. The λ estimation also works well in the tensor situation.

Example 3

We consider the testing tensor with each slice of the tensor generated by the same method as in Example 1.

Then we compare t-BARK with and without the FFT (Fig. 7.4).

In these three figures, the x-axis of t-BARK(fft) is its iteration number times the third size of tensor in order to make the complexity the same as the other algorithm. As illustrated above, both convergence rate of the algorithms mainly depends on its λ. The two methods have different λ for the same question. In Fig. 7.5a, t-BARK(fft)

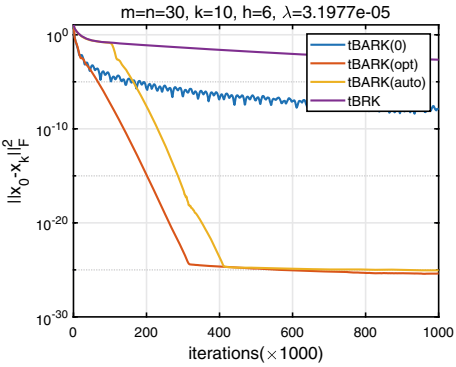


Fig. 7.4 Performance of consider methods for tensor equations

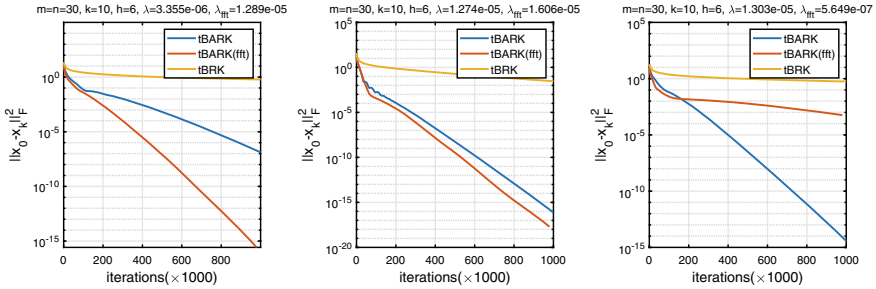


Fig. 7.5 Comparison of t-BARK with and without the FFT. The testing tensors are uniform random tensors

converges faster than t-BARK, and in Fig. 7.5c, it is the contrary case. The experiment shows that the method with a larger λ converges faster. However, t-BARK with the FFT can be computed parallelly and on well-conditioned slices, it converges quickly, which means sometimes it has lower complexity to reach a certain error compared with direct t-BARK.

Acknowledgements The authors would like to thank Professor Manjunatha Prasad Karantha and two referees for their detailed comments. A. Qiao is supported by Shanghai Municipal Science and Technology Commission under grant 23WZ2501400 and Innovation Program of Shanghai Municipal Education Commission. X. Wang is supported by the National Natural Science Foundation of China under grant 12061032; Start-Up Fund of Doctoral Research, Hexi University. Partial work was finished when this author visited Key Laboratory of Mathematics for Nonlinear Sciences at Fudan University. Y. Wei is supported by the National Natural Science Foundation of China under grant 12271108 and Innovation Program of Shanghai Municipal Education Commission.

References

1. Chang SY, Wei Y (2022) T-square tensors-Part I: inequalities. Comput Appl Math 41(1):1–27
2. Chang SY, Wei Y (2022) T-square tensors-part II: tail bounds for sums of random T-product tensors. Comput Appl Math 41(3):1–32
3. Che M, Wei Y (2020) Theory and computation of complex tensors and its applications. Springer
4. Che M, Wei Y (2022) An efficient algorithm for computing the approximate t-URV and its applications. J Sci Comput 92(3):1–27
5. Che M, Wang X, Wei Y, Zhao X (2022) Fast randomized tensor singular value thresholding for low-rank tensor optimization. Numer Linear Algebra Appl e2444
6. Chen J, Wei Y, Xu Y (2022) Tensor CUR decomposition under T-product and its perturbation. Numer Funct Anal Optim 43:698–722
7. Chen X, Qin J (2021) Regularized Kaczmarz algorithms for tensor recovery. SIAM J Image Sci 14(4):1439–1471
8. Cucker F, Diao H, Wei Y (2006) Smoothed analysis of some condition numbers. Numer Linear Algebra Appl 13(1):71–84
9. Drineas P, Mahoney MW (2016) RandNLA: randomized numerical linear algebra. Commun ACM 59(6):80–90
10. Du K, Sun XH (2021) Randomized regularized extended Kaczmarz algorithms for tensor recovery. arXiv:2112.08566
11. Jin X, Wei Y, Zhao Z (2015) Numerical linear algebra and its applications, 2nd edn. Science Press, Beijing
12. Kilmer ME, Martin CD (2011) Factorization strategies for third-order tensors. Linear Algebra Appl 435(3):641–658
13. Liu J, Wright S (2016) An accelerated randomized Kaczmarz algorithm. Math Comput 85(297):153–178
14. Lu C, Feng J, Chen Y, Liu W, Lin Z, Yan S (2019) Tensor robust principal component analysis with a new tensor nuclear norm. IEEE Trans Pattern Anal Mach Intell 42(4):925–938
15. Lund K (2020) The tensor t-function: a definition for functions of third-order tensors. Numer Linear Algebra Appl 27(3):e2288
16. Ma A, Molitor D (2022) Randomized kaczmarz for tensor linear systems. BIT Numer Math 62(1):171–194
17. Martin CD, Shafer R, LaRue B (2013) An order-p tensor factorization with applications in imaging. SIAM J Sci Comput 35(1):A474–A490
18. Miao Y, Qi L, Wei Y (2020) Generalized tensor function via the tensor singular value decomposition based on the T-product. Linear Algebra Appl 590:258–303
19. Miao Y, Qi L, Wei Y (2021) T-Jordan canonical form and T-Drazin inverse based on the T-product. Commun Appl Math Comput 3(2):201–220
20. Needell D, Tropp JA (2014) Paved with good intentions: analysis of a randomized block Kaczmarz method. Linear Algebra Appl 441:199–221
21. Nesterov Y (2003) Introductory lectures on convex optimization: a basic course. Springer Science & Business Media
22. Shi X, Wei Y, Zhang W (2011) Convergence of general nonstationary iterative methods for solving singular linear equations. SIAM J Matrix Anal Appl 32(1):72–89
23. Strohmer T, Vershynin R (2009) A randomized Kaczmarz algorithm with exponential convergence. J Four Anal Appl 15(2):262–278
24. Tang L, Yu Y, Zhang Y, Li H (2022) Sketch-and-project methods for tensor linear systems. Numer Linear Algebra Appl e2470. https://doi.org/10.1002/nla.2470
25. Tarzanagh DA, Michailidis G (2018) Fast randomized algorithms for t-product based tensor operations and decompositions with applications to imaging data. SIAM J Imaging Sci 11(4):2629–2664
26. Wang G, Wei Y, Qiao S (2018) Generalized inverses: theory and computations, 2nd edn. Springer, Singapore and Science Press

27. Wang X, Che M, Wei Y (2020) Tensor neural network models for tensor singular value decompositions. Comput Optim Appl 75(3):753–777
28. Wang X, Che M, Wei Y (2022) Randomized Kaczmarz methods for tensor complementarity problems. Comput Optim Appl 82(3):595–615
29. Wang X, Che M, Mo C, Wei Y (2023) Solving the system of nonsingular tensor equations via randomized Kaczmarz-like method. J Comput Appl Math 421:114856
30. Wei Y, Ding W (2016) Theory and computation of tensors: multi-dimensional arrays. Elsevier/Academic Press, London
31. Zouzias A, Freris NM (2013) Randomized extended Kaczmarz for solving least squares. SIAM J Matrix Anal Appl 34(2):773–793

Chapter 8
Nullity of Graphs—A Survey and Some New Results



S. Arumugam, K. Arathi Bhat, Ivan Gutman, Manjunatha Prasad Karantha, and Raksha Poojary

Mathematics Subject Classification (2010) 05C50 · 05C90

8.1 Basic Results on Nullity

Let $G = (V, E)$ be a graph of order n and let $V(G) = \{v_1, v_2, \ldots, v_n\}$. Since the adjacency matrix $A(G) = (a_{ij})$ is a symmetric binary matrix, its eigenvalues are real numbers. The set of all eigenvalues of G is the spectrum of G and is denoted by $Sp(G)$. The nullity $\eta(G)$ is the algebraic multiplicity of the number zero in the spectrum of G.

S. Arumugam (✉)
Department of Computer Science and Engineering, Ramco Institute of Technology, Rajapalayam 626117, Tamil Nadu, India
e-mail: s.arumugam.klu@gmail.com

K. Arathi Bhat
Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India
e-mail: arathi.bhat@manipal.edu

I. Gutman
Faculty of Science, University of Kragujevac, Kragujevac, Serbia
e-mail: gutman@kg.ac.rs

M. P. Karantha
Department of Data Science, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal, India
e-mail: km.prasad@manipal.edu; kmprasad63@gmail.com

Centre for Advanced Research in Applied Mathematics and Statistics, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India

R. Poojary
Department of Data Science, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_8

In chemistry, in the context of the Hückel molecular orbital theory Coulson et al. [6], the nullity of molecular graphs of so-called "alternant conjugated hydrocarbons" (which necessarily are bipartite graphs) plays an outstanding role. Namely, η is an indicator of the stability of the respective chemical compound: $\eta = 0$ is a necessary (but not sufficient) condition for chemical stability. If $\eta > 0$, then the respective chemical species is extremely reactive and usually does not exist. For further details on this, we refer to Cvetković and Gutman [8], Graovac et al. [13], Longuet-Higgins [23], Yates [34].

If v is an isolated vertex of G, then the row corresponding to v in $A(G)$ is the zero vector and hence $\eta(G) > 0$. Hence, we assume that G is a graph without isolated vertices. There are some classes of graphs for which the spectrum is known and hence the nullity is determined.

Theorem 8.1 (Cvetković and Gutman [7]) *The spectrum of the complete graph K_n consists of $n-1$ and -1 with multiplicity $n-1$.*

Corollary 8.1 *$\eta(K_n) = 0$ for all $n \geq 2$.*

Theorem 8.2 (Cvetković and Gutman [7]) *The spectrum of the P_n is given by*

$$Sp(P_n) = \left\{ 2\cos\left(\frac{\pi r}{n+1}\right) \ : \ 1 \leq r \leq n \right\}.$$

Corollary 8.2 (Cvetković and Gutman [7]) $\eta(P_n) = \begin{cases} 1 \text{ if } n \text{ is odd} \\ 0 \text{ if } n \text{ is even.} \end{cases}$

Theorem 8.3 (Cvetković and Gutman [7]) *The spectrum of the cycle C_n is given by*

$$Sp(C_n) = \left\{ 2\cos\left(\frac{2\pi r}{n}\right) \ : \ 0 \leq r \leq n-1 \right\}.$$

Corollary 8.3 (Cvetković and Gutman [7]) $\eta(C_n) = \begin{cases} 2 \text{ if } n \equiv 0 \pmod 4 \\ 0 \text{ otherwise.} \end{cases}$

Let $\Phi(G, \lambda) = |A - \lambda I| = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n$ be the characteristic polynomial of G. Sachs' theorem gives a formula for the coefficients a_i from the structure of the graph. A subgraph H of G is called an elementary subgraph if every component of H is either K_2 or a cycle. Sachs called such subgraphs basic figures.

Theorem 8.4 (Sachs [28]) *Let $\Phi(G, \lambda) = \sum_{i=0}^{n} a_i\lambda^{n-i}$ be the characteristic polynomial of G. Then*

$$a_i = \sum_{s \in S_i} (-1)^{c(s)} 2^{r(s)} \tag{8.1}$$

where S_i is the set of all elementary subgraphs s of order i, $c(s)$ is the number of components in s and $r(s)$ is the number of cycles in s.

If G is a bipartite graph, then every cycle in G is an even cycle and hence there is no elementary subgraph of odd order. Hence, $a_i = 0$ if i is odd, or equivalently the characteristic polynomial of G contains only even powers of λ. Hence, we have the following theorem, which is called the pairing theorem.

Theorem 8.5 (Cvetković and Gutman [7]) *Let G be a bipartite graph. Then $\lambda \in Sp(G) \implies -\lambda \in Sp(G)$.*

Since the trace of the adjacency matrix is zero, we have $\sum_{i=1}^{n} \lambda_i = 0$. Hence, Theorem 8.5 gives the following corollary.

Corollary 8.4 (Cvetković and Gutman [7]) *Let G be a bipartite graph of odd order. Then $\eta(G) > 0$.*

Theorem 8.6 (Cvetković and Gutman [7]) *Let T be a tree on $n \geq 1$ vertices and let m be the size of its maximum matching. Then $\eta(T) = n - 2m$.*

Proof Since any elementary spanning subgraph of T is a matching in T, it follows that there is no elementary spanning subgraph of order k for any $k > 2m$. Hence, the result follows. □

Theorem 8.6 has been generalized as given below.

Theorem 8.7 (Cvetković and Gutman [7]) *If a bipartite graph G with $n \geq 1$ vertices does not contain any cycle of length $4s$ $(s = 1, 2, \ldots)$, then $\eta(G) = n - 2m$, where m is the size of its maximum matching.*

The vertices of a bipartite graph may be numbered so that the adjacency matrix has the following form:

$$A = \begin{bmatrix} 0 & B \\ B^T & 0 \end{bmatrix}$$

The matrix B is the incidence matrix between the two sets X and Y of vertices of the bipartite graph.

Theorem 8.8 (Gong and Xu [7]) *For the bipartite graph G with n vertices and incidence matrix B, $\eta(G) = n - 2r(B)$, where $r(B)$ is the rank of B.*

The following theorem gives how nullity changes on the removal of a vertex.

Theorem 8.9 (Gong and Xu [12]) *Let v be any vertex of a graph G with an order of at least 2. Then*

$$\eta(G) - 1 \leq \eta(G - v) \leq \eta(G) + 1.$$

It follows from Theorem 8.9 that $\eta(G - v) = \eta(G) - 1$ or $\eta(G)$ or $\eta(G) + 1$. Examples of all three types are given in Fig. 8.1.

The vertex set V of the graph G can be partitioned into three subsets V^-, V^0, V^+ depending on how the nullity changes. Let $V^- = \{v : \eta(G - v) = \eta(G) - 1\}$, $V^0 = \{v : \eta(G - v) = \eta(G)\}$ and $V^+ = \{v : \eta(G - v) = \eta(G) + 1\}$.

We observe that for the complete graph K_n, $V = V^0$, for the even path P_n, $V = V^+$, for cycle C_n, where $n \equiv 0 \pmod 4$, $V = V^-$.

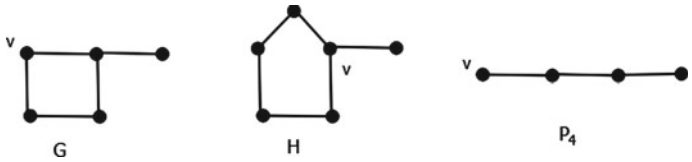


Fig. 8.1 The graphs G, H, P_4 with $\eta(G - v) = \eta(G)$, $\eta(H - v) = \eta(H) - 1$ and $\eta(P_4 - v) = \eta(P_4) + 1$

Problem 8.1 *Characterize graphs for which*

1. $V = V^-$
2. $V = V^0$
3. $V = V^+$.

In Gong and Xu [12], the authors considered the effect of the removal of cut vertices on the nullity and proved the following results.

Theorem 8.10 (Gong and Xu [12]) *Let v be a cut vertex of a graph G of order n and $G_1, G_2, \ldots,$*
G_s be the set of all components of $G - v$. If there exists a component, say G_1, among $G_1, G_2, \ldots, G_s$ such that $\eta(G_1) = \eta(G_1 + v) + 1$, then

$$\eta(G) = \eta(G - v) - 1 = \sum_{i=1}^{s} \eta(G_i) - 1.$$

Theorem 8.11 (Gong and Xu [12]) *Let v be a cut vertex of a graph G of order n and G_1 be a component of $G - v$. If $\eta(G_1) = \eta(G_1 + v) - 1$, then*

$$\eta(G) = \eta(G_1) + \eta(G - G_1).$$

The authors Mohiaddin and Sharaf [25] considered the effect of identifying two vertices on the nullity of the graph. Two vertices u and v are said to be identified if they are replaced by a vertex w such that $N(w) = N(u) \cup N(v)$, where $N(v)$ denotes the open neighborhood of the vertex v. The authors have proved that in general, identifying any pair of distinct vertices cannot increase or decrease the nullity by more than 2.

8.2 Bounds for the Nullity of a Graph

In this section, we present bounds for the nullity of G in terms of other graph theoretic parameters such as clique number, girth and diameter. The following observation gives a tool for obtaining such bounds.

Observation 8.1 Let $G = (V, E)$ be a graph of order n. Let H be an induced subgraph of G. Clearly $r(H) \leq r(G)$, where $r(G)$ is the rank of the adjacency matrix of G. Since $\eta(G) + r(G) = n$ and $\eta(H) + r(H) = |V(H)|$, it follows that

$$\eta(G) \leq n + \eta(H) - |V(H)|. \tag{8.2}$$

Hence if G contains an induced subgraph H whose nullity is known, then it can be used to get an upper bound for $\eta(G)$.

A clique in G is a complete subgraph of G. The clique number $\omega(G)$ is the maximum order of a clique in G. Hence if K_p is a subgraph of G, then $p \leq \omega(G)$.

Theorem 8.12 (Cheng and Liu [4]) *Let G be a graph of order n and let K_p be a subgraph of G with $2 \leq p \leq n$. Then $\eta(G) \leq n - p$.*

Proof The result follows from Corollary 8.1 and Observation 8.1. □

Corollary 8.5 (Cheng and Liu [4]) *Let G be a graph of order n. Then $\eta(G) \leq n - \omega(G)$.*

The girth of G, denoted by $gir(G)$, is the length of a shortest cycle in G. The diameter of G, denoted by $diam(G)$, is the maximum distance between any two vertices. Hence, a cycle of length $gir(G)$ and a diametral path are induced subgraphs of G. Also, the nullity of a path and cycle are known (Corollary 8.2 and Corollary 8.3). Hence, using (8.2), we get the following bounds for $\eta(G)$.

Theorem 8.13 (Cheng and Liu [4]) *Let G be a graph of order n and let the cycle C_p be an induced subgraph of G, where $3 \leq p \leq n$. Then*

$$\eta(G) \leq \begin{cases} n - p + 2 & \text{if } p \equiv 0 \pmod 4 \\ n - p & \text{otherwise.} \end{cases}$$

Corollary 8.6 (Cheng and Liu [4]) *Let G be a graph of order n having at least one cycle. Then*

$$\eta(G) \leq \begin{cases} n - gir(G) + 2 & \text{if } gir(G) \equiv 0 \pmod 4 \\ n - gir(G) & \text{otherwise.} \end{cases}$$

Theorem 8.14 (Cheng and Liu [4]) *Let G be a graph of order n and let the path P_p be an induced subgraph of G. Then*

$$\eta(G) \leq \begin{cases} n - p + 1 & \text{if } p \text{ is odd} \\ n - p & \text{otherwise.} \end{cases}$$

Corollary 8.7 (Cheng and Liu [4]) *Let G be a connected graph of order n. Then*

$$\eta(G) \leq \begin{cases} n - diam(G) + 1 & \text{if } diam(G) \text{ is odd} \\ n - diam(G) & \text{otherwise.} \end{cases}$$

The authors Mustapha and Hansen [26] have given a necessary condition for graphs to satisfy $\eta(G) = n - diam(G)$.

The bounds given in Corollaries 8.6 and 8.7 can be improved. Let $P_k = (v_1, v_2, \ldots, v_k)$ be a path in G. An edge $v_i v_j$ where $|i - j| \geq 2$ is called a chord of the path. A path having no chords is called a monophonic path. Thus, a monophonic path is simply a chordless path or equivalently an induced path in a graph. The monophonic distance $d_m(u, v)$ is the length of a longest $u - v$ monophonic path in G Santhakumaran and Titus [29]. The monophonic diameter $mdiam(G)$ is defined by $mdiam(G) = \max\{d_m(u, v) : u, v \in V\}$. The maximum length of an induced cycle in G is called the upper girth of G and is denoted by $ugir(G)$. Clearly, $mdiam(G) \geq diam(G)$ and $ugir(G) \geq gir(G)$ and the difference between these parameters may be arbitrarily large as shown in the following example. Let G be the graph obtained from the cycle $C_n = (v_1, v_2, \ldots, v_n, v_1)$ by adding the edge $v_1 v_3$ where $n \geq 5$. Then $gir(G) = 3$, $ugir(G) = n - 1$, $diam(G) = \lfloor \frac{n}{2} \rfloor$ and $mdiam(G) = n - 2$.

Observation 8.1 leads to the following new bounds for $\eta(G)$, which is an improvement of the bounds given in Corollaries 8.6 and 8.7.

Theorem 8.15 *Let G be a connected graph of order n, having at least one cycle. Then*

$$\eta(G) \leq \begin{cases} n - ugir(G) + 2 & \text{if } ugir(G) \equiv 0 \pmod 4 \\ n - ugir(G) & \text{otherwise.} \end{cases}$$

Theorem 8.16 *Let G be a connected graph of order n. Then*

$$\eta(G) \leq \begin{cases} n - mdiam(G) + 2 & \text{if } mdiam(G) \text{ is odd} \\ n - mdiam(G) & \text{otherwise.} \end{cases}$$

There are several bounds for nullity in terms of the maximum degree Δ.

Theorem 8.17 (Fiorini [11]) *For a tree T with order n and maximum degree Δ, $\eta(T) \leq n - 2\left\lceil \frac{n-1}{\Delta} \right\rceil$, and the upper bound can be attained.*

A result analogous to Theorem 8.17 has been proved for bipartite graphs.

Theorem 8.18 (Omidi [27]) *Let G be a bipartite graph with n vertices, m edges and maximum vertex degree Δ. If G does not have as subgraph any cycle whose size is divisible by 4, then $\eta(G) \leq n - 2\left\lceil \frac{m}{2} \right\rceil$.*

However, the problem of establishing an upper bound in terms of order and maximum degree for an arbitrary graph was left open for about ten years. Recently, this problem was solved by Zhou et al. [35].

Theorem 8.19 (Zhou et al. [35]) *Let G be a graph with order n and maximum degree $\Delta \geq 1$. If G has no isolated vertex, then $\eta(G) \leq \left(\frac{\Delta - 1}{\Delta}\right) n$, the equality holds if and only if G is the disjoint union of some copies of $K_{\Delta,\Delta}$.*

The authors Zhou et al. [35] posed a conjecture for further study.

Conjecture 8.1 *If G is assumed to be connected, then the upper bound of $\eta(G)$ can be improved to $\frac{(\Delta-2)n+2}{\Delta-1}$, and the upper bound is attained if and only if G is a cycle C_n with n divisible by 4 or a complete bipartite graph with equal size of chromatic sets.*

The authors Wang and Geng [32] have proved the above conjecture.

Characterization of graphs which attain the various bounds given in this section is an open problem.

8.3 Energy and Nullity

In the 1930s, Hückel put forward a method for finding approximate solutions of a class of organic molecules, namely, unsaturated conjugated hydrocarbons. Within this theory, the concept of total electron energy (called "total π-electron energy") was elaborated Coulson et al. [6], Graovac et al. [13], Yates [34], which plays an important role in chemistry. This motivated Gutman to introduce the concept of energy of a graph Gutman [14], which in the general case differs from Hückel's total π-electron energy, but coincides with it for majority of chemically relevant graphs. From a mathematical point of view, Gutman's graph energy has a much simpler form than Hückel's total energy, and it can be straightforwardly applied to all graphs.

As a consequence, the concept of energy of a graph was put forward in Gutman [14].

Definition 8.1 Let $S_p(G) = \{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ be the spectrum of a graph G of order n. Then $\mathcal{E}(G) = \sum_{j=1}^{n} |\lambda_j|$ is called the *energy* of G.

Thus, the energy of a graph G is the sum of the absolute values of the eigenvalues of G. A long-standing open problem in the theory of graph energy is to determine the relation between the energy $\mathcal{E}(G)$ and nullity $\eta(G)$. From a quantum chemical point of view, the greater is $\eta(G)$, the weaker should be the bonding in the underlying molecule, and consequently, the smaller should $\mathcal{E}(G)$ be. Thus, the following has been suggested by arguments from quantum chemistry.

Conjecture 8.2 *$\mathcal{E}(G)$ is a decreasing function of $\eta(G)$.*

Gutman and Triantafillou [18] suggested one formulation of the above conjectured property. If $\eta(G_1) > \eta(G_2)$, then $\mathcal{E}(G_1) < \mathcal{E}(G_2)$, where it must be assumed that all other structural features of the graphs G_1 and G_2 influencing the values of their energies are same or differ negligibly. The energy $\mathcal{E}(G)$ depends mainly on the number of vertices, number of edges and the number, type and mutual arrangements of cycles. Hence, Gutman and Triantafillou [18] considered a family of banana trees of order n.

Definition 8.2 Let $\{K_{1,n_1}, K_{1,n_2}, \ldots, K_{1,n_p}\}$ be a set of p stars where $p \geq 1$ and $n_i \geq 3$. The tree obtained by adding a new vertex v and joining v to one leaf of each star is called a *banana tree*.

If $n_i = k$ for each i, then the tree has $pk + 1$ vertices and is denoted by $B(p, k)$. Let $\mathcal{B}(n) = \{B(p, k) : p \geq 1 \text{ and } k \geq 3\}$. Thus $\mathcal{B}(n)$ is a family of banana trees of order n constructed from isomorphic stars.

Theorem 8.20 (Gutman and Triantafillou [18]) *For a fixed number of vertices n, the energies of the banana trees $B \in \mathcal{B}(n)$ decrease as their nullities increase.*

It has been pointed out by Gutman and Triantafillou [18] that if the stars in the construction of a banana tree are not isomorphic, then the analytical treatment of the nullity-dependence of graph energy as given in Theorem 8.20 is not feasible.

Another work Gutman et al. [19] investigated the validity of the above conjectured property of dependence of $\mathcal{E}(G)$ on $\eta(G)$ by means of computer-aided numerical calculations, as applied to a suitably constructed class of graphs.

Recently, a method for approaching this problem has been put forward in Gutman [15, 16]. In Gutman [15] pairs of bipartite graphs G_1, G_2 have been constructed, called "siblings", such that $\eta(G_1) = 0$ and $\eta(G_2) = 2$ and the difference between their characteristic polynomials is constant. The difference between the energy of siblings is just the energy-effect of nullity. For non-sibling graphs, the above conjectured property in general remains open and the proof of its validity to any family of chemical graphs would be of significant value for theoretical chemistry.

8.4 Graphs with Nullity 0 and 1

In the context of the Hučkel Molecular Orbital model, if $\eta(G) > 0$ for the molecular graph G, then the corresponding chemical compound is highly reactive and unstable. Hence, characterization of graphs for which $\eta(G) = 0$ or equivalently $\eta(G) > 0$ is a significant problem and it was first posed in 1957 by Von and Sinogowitz [31]. This problem still remains unsolved. In this section, we present some of the significant results on graphs with $\eta(G) = 0$ or 1.

Observation 8.2 From Corollaries 8.1, 8.2 and 8.3, we get

(i) $\eta(K_n) = 0$ for all $n \geq 2$.
(ii) $\eta(P_n) = 0$ if and only if n is even.
(iii) $\eta(C_n) = 0$ if and only if $n \not\equiv 0 \pmod 4$.

Ashraf and Bamdad [1] have given infinite families of graphs with nullity zero.

Theorem 8.21 (Ashraf and Bamdad [1]) *Let $n \geq 4$ and $2 \leq t \leq n - 1$. Let $G = K_n - E(K_{1,t})$. Then $\eta(G) = 0$.*

Theorem 8.22 (Ashraf and Bamdad [1]) *Let n and m be positive integers such that $n \geq 5$, $\lceil \frac{n}{2} \rceil \leq m \leq \binom{n}{2}$, $m \neq \binom{n}{2} - 1$ and $m \neq \lceil \frac{n}{2} \rceil$ for odd n. Then there exists a graph G of order n and size m with $\eta(G) = 0$.*

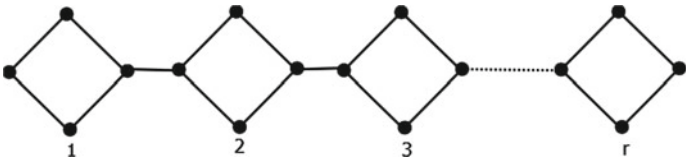


Fig. 8.2 The graph G_r with $\eta(L(G_r)) = r + 1$.

Sciriha [30] has given a method for constructing families of graphs with $\eta(G) = 1$. Consideration of line graphs of trees leads to another family of graphs with nullity one.

Definition 8.3 Let $G = (V, E)$ be graph. Then the graph $L(G)$ with vertex set E in which two elements $e_1, e_2 \in E$ are adjacent in $L(G)$ if they are adjacent edges in G is called the *line graph* of G.

In general, the nullity of line graphs may assume any positive integer. For example, if $G = mK_2$, then $L(G) = \overline{K}_m$ and hence $\eta(L(G)) = m$. Even if we restrict ourselves to connected graphs, the nullity of the line graph can still be any positive integer. Consider the graph G_r given in Fig. 8.2. Then $\eta(L(G_r)) = r + 1$.

Gutman and Sciriha [17] proved the following theorem.

Theorem 8.23 *If T is a tree, then $\eta(L(T)) = 0$ or 1.*

We observe that if $T = P_n$, then $L(T) = P_{n-1}$ and hence $\eta(L(T)) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 1 & \text{if } n \text{ is even.} \end{cases}$

Problem 8.2 *[1.] Characterize trees T for which $\eta(L(T)) = 0$ or equivalently $\eta(L(T)) = 1$.*

More generally, the characterization of graphs with nullities 0 and 1 is a fundamental unsolved problem.

8.5 Graph Operations Preserving Nullity

Three transformations which preserve nullity have been reported in Cvetković and Gutman [7] and Ma et al. [24].

T_1: Let G be any graph and suppose G contains $P_4 = (v_1, v_2, v_3, v_4)$ such that $\deg(v_2) = \deg(v_3) = 2$ in G. Let H be the graph obtained from G by removing v_2 and v_3 and joining v_2 and v_4. Then $\eta(G) = \eta(H)$.

T_2: Suppose G contains a cycle $C_4 = (v_1, v_2, v_3, v_4, v_1)$ such that $\deg(v_2) = \deg(v_3) = 2$. Let H be the graph obtained from G by removing the vertices v_2, v_3 and the edge v_1v_4. Then $\eta(G) = \eta(H)$.

T_3: Let G be a graph with $\delta(G) = 1$. Let $\deg(v) = 1$ and let u be the unique neighbor of v. Let $H = G - \{u, v\}$. Then $\eta(G) = \eta(H)$.

Using the above transformations, we compute the nullity of some classes of graphs.

The Cartesian product $G\Box H$ of two graphs G and H is the graph with vertex set $V(G) \times V(H)$ and two vertices (u_1, v_1) and (u_2, v_2) are adjacent if $u_1 = u_2$ and $v_1v_2 \in E(H)$ or $v_1 = v_2$ and $u_1u_2 \in E(G)$.

Theorem 8.24 *Let $G = P_2\Box P_n$ where $n \geq 2$. Then* $\eta(G) = \begin{cases} 2 & \text{if } n \equiv 2 \pmod 3 \\ 0 & \text{otherwise.} \end{cases}$

Proof If $n = 2$, then $G = C_4$ and $\eta(G) = 2$. If $n = 3$, the graph obtained by using T_2 is P_4. Hence $\eta(P_2\Box P_3) = \eta(P_4) = 0$. If $n = 4$, the graph obtained by using T_2, T_3 is P_4. Hence $\eta(P_2\Box P_4) = \eta(P_4) = 0$.

We claim that for $n \geq 5$, $\eta(P_2\Box P_n) = \eta(P_2\Box P_{n-3})$. In fact if $G = P_2\Box P_n$, let G_1 be the graph obtained from G by applying T_2. The graph G_1 has 2 vertices of degree 1 and we apply T_3 for both the vertices of degree 1. The resulting graph is $P_2\Box P_{n-3}$. Since the transformations preserve nullity, it follows that $\eta(P_2\Box P_n) = \eta(P_2\Box P_{n-3})$.

Hence the result follows. □

The corona of two vertex disjoint graphs G_1 and G_2 is the graph $G = G_1 \circ G_2$ obtained from one copy of G_1 and $|V(G_1)|$ copies of G_2 by joining the ith vertex of G_1 to all the vertices in the ith copy of G_2.

Theorem 8.25 *$\eta(G \circ K_1) = 0$ for any connected graph G.*

Proof Let $H = G \circ K_1$. The graph obtained from H using T_3 for any pendant vertex u with neighbor v is the graph $(G - v) \circ K_1$.

By repeatedly applying the above transformation $(n - 2)$ times, we get the graph $P_2 \circ K_1 = P_4$. Therefore, $\eta(G \circ K_1) = \eta(P_4) = 0$. □

Remark 8.1 Theorem 8.25 shows that every graph can be embedded in a graph with nullity zero. Hence there is no forbidden subgraph characterization for graphs with nullity zero. Also since the corona of a disconnected graph with K_1 is the disjoint union of the graphs obtained as corona of each component with K_1 and the adjacency matrix of a disconnected graph is the direct sum of the adjacency matrices of the components, it follows that Theorem 8.25 is true for disconnected graphs as well.

Theorem 8.26 *Let G be a graph with $\Delta(G) = n - 1$. Let k denote the number of vertices with degree $n - 1$. Let H be a graph obtained from G by attaching one pendant vertex at each vertex v with a degree less than $n - 1$. Then* $\eta(H) = \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$

Proof By applying T_3 for each pendant vertex, we obtain the complete graph K_k. Hence the result follows. □

Definition 8.4 A *threshold graph* is a graph constructed from a one vertex graph by repeated applications of the following two operations:

1. Addition of a single isolated vertex to the graph.
2. Addition of a single dominating vertex to the graph, i.e., a single vertex that is adjacent to all other vertices.

Theorem 8.27 *Let G_n be a connected threshold graph in which v_i is added as an isolated vertex if i is odd and dominating if i is even. Then $\eta(G) = 0$.*

Proof Let $V(G_n) = \{v_1, v_2, \ldots, v_n\}$ where the vertices are listed in the order in which they were added. Therefore, $\deg(v_n) = n - 1$ and $\deg(v_{n-1}) = 1$. Applying T_3 for G_n, at the vertex v_{n-1}, we obtain G_{n-2}. Repeated application of T_3 gives K_2. Since, $\eta(K_2) = 0$, the result follows. □

Definition 8.5 A *half graph* G_n is a graph with $2n$ vertices $v_1, v_2 \ldots v_n$ and $u_1, u_2, \ldots u_n$ obtained by joining v_i with u_j by an edge whenever $i \leq j$.

Theorem 8.28 *Let G_n be a half graph of order $2n$. Then $\eta(G_n) = 0$.*

Proof Let $V(G_n) = V \cup U$, where $V = \{v_1, v_2, \ldots, v_n\}$, $U = \{u_1, u_2, \ldots, u_n\}$ and $N(u_i) = \{v_i, v_{i+1}, \ldots, v_n\}$. Applying T_3 to G_n, at the pendant vertex v_1, we obtain G_{n-1}. Repeated application of T_3 gives K_2. Hence $\eta(G_n) = 0$. □

We now define a family of graphs $\mathcal{F} = \{G_1, G_2, \ldots, G_n, \ldots\}$ starting from the graphs G_1 and H given in Fig. 8.3.

For $n \geq 2$, the graph G_n is obtained from G_{n-1} by attaching the right-most vertical side of H to the vertical side of G_{n-1} on the left side. The graph G_2 is given in Fig. 8.4.

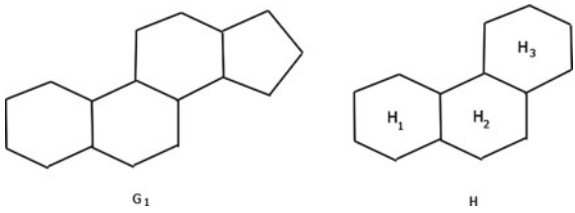


Fig. 8.3 The graphs G_1 and H

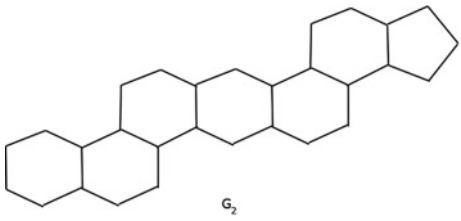


Fig. 8.4 The graph G_2

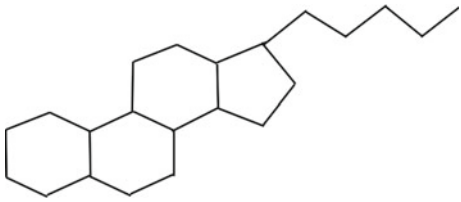


Fig. 8.5 Molecular graph of cholesterol compound

Theorem 8.29 *The nullity of G_n is 0 for all $G_n \in \mathcal{F}$.*

Proof It follows from the definition of $\mathcal{F}$ that the number of cycles in G_n is $3k + 1$.

The proof is by induction on n.

When $n = 1$, by applying the transformations $T_1, T_2, T_3, T_3, T_3, T_3$ and T_1, to G_1, we obtain the graph K_3. Hence $\eta(G_1) = 0$.

We now assume that the result is true for $n - 1$. Now by applying $T_1, T_2, T_3, T_3, T_3, T_3$ to the three consecutive hexagons H_1, H_2, H_3 in G_n, we get G_{n-1}. Hence by induction the proof is complete. □

The graph G obtained from G_1 in $\mathcal{F}$ by attaching a path P_6 at a vertex of degree 2 in C_5 is the molecular graph corresponding to the cholesterol compound. This graph is given in Fig. 8.5. The following theorem shows that the cholesterol compound is chemically stable.

Theorem 8.30 *Let G be the graph obtained from $G_n \in \mathcal{F}$, by attaching a path P_r at a vertex of degree 2 in C_5. Then $\eta(G) = 0$.*

Proof If r is odd, then by applying $\frac{r-1}{2}$ times T_3 on the attached path, we get G_n. If r is even, then by applying the transformations given in Theorem 8.29, to G_n, we get the graph isomorphic to C_3 together with the path P_r attached to C_3. Now by applying $\frac{r}{2}$ times T_3 to the path P_r we obtain K_2. Hence $\eta(G) = 0$. □

We now define another family of graphs $\mathcal{B} = \{G_1, G_2, \ldots, G_n, \ldots\}$ starting from the graph G_1 given in Figure 8.6 and the hexagon C_6. For $n \geq 2$, the graph G_n is obtained by identifying the vertical edge of the last hexagon in G_{n-1} and the vertical edge of C_6.

The graphs G_2 and G_3 in $\mathcal{B}$ are given in Fig. 8.7.

The graphs given in the family $\mathcal{B}$ represent the benzoacene monoradical series given in Dias [9, Fig. 8].

Theorem 8.31 *Let $G_n \in \mathcal{B}$. Then $\eta(G_n) = 1$.*

Proof The proof is by induction on n. Applying three times the transformations T_1 for G_1, we obtain the graph H given in Fig. 8.8.

Since H is a bipartite graph of odd order, it follows that $\eta(H) > 0$ and hence $\eta(G_1) > 0$. Now we shall compute the coefficient a_6 in the characteristic polynomial,

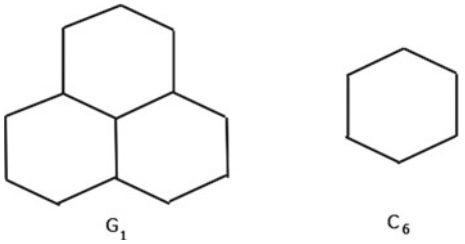


Fig. 8.6 The graphs G_1 and C_6

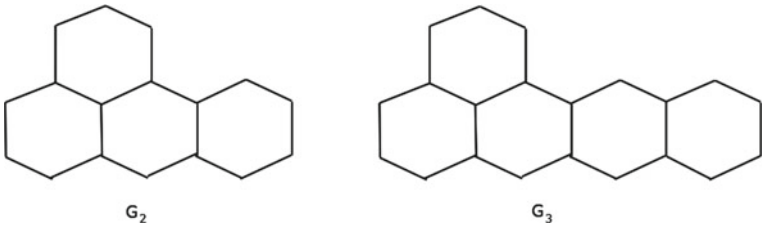
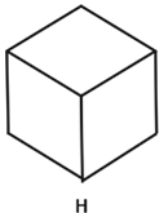


Fig. 8.7 The graphs G_2 and G_3

Fig. 8.8 The graph H



$\sum_{i=0}^{7} a_i \lambda^i$ using Sachs' theorem. Any elementary subgraph of order 6 is either $C_4 \cup K_2$ or $3K_2$ or C_6. The number of elementary subgraphs of type $C_4 \cup K_2$ is 6, the number of elementary subgraphs of type $3K_2$ is 11 and the number of elementary subgraphs of type C_6 is 4. Therefore, $a_6 = 12 - 11 - 8 = -7$. Hence $\eta(H) = 1$. Hence it follows that $\eta(G_1) = 1$.

For G_2, by applying the transformations T_1, T_2 and T_3, we obtain K_1 and hence $\eta(G_2) = 1$.

For $n \geq 3$, by applying T_1, T_2, T_3, T_3 on the right side of G_n, we obtain G_{n-2}. Hence by induction, it follows that $\eta(G_n) = 1$. □

Theorem 8.32 *Let G_n^{2k} be the graph obtained from the path $P_n = (v_1, v_2, \ldots, v_n)$ by replacing every edge of P_n by an even cycle of length C_{2k}. Then $\eta(G_n^{2k}) = n$.*

Proof The proof is by induction on k. For $k = 2$, G_n^4 is the graph obtained from P_n by replacing every edge by C_4. There are $(n - 1)$ copies of C_4 and by applying the transformation T_2 for each cycle, we obtain $\overline{K}_n$. Therefore, $\eta(G_n^4) = n$.

We now assume that the result is true for $k-1$. Consider G_n^{2k}. For each cycle C_{2k} in G_n^{2k} we apply the transformation T_1. This gives the graph $G_n^{2(k-1)}$. Hence by induction, $\eta(G_n^{2k}) = n$. □

Theorem 8.33 *Let G_n be the graph obtained from the path $P_n = (v_1, v_2, \ldots, v_n)$ with $n \geq 3$, by adding $n-2$ vertices $u_1, u_2, \ldots, u_{n-2}$ and joining u_i with v_i and v_{i+2} by edges. Then* $\eta(G_n) = \begin{cases} 0 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd.} \end{cases}$

Proof By applying the transformation T_2 and T_3 to G_n, we obtain G_{n-2}. Hence $\eta(G_n) = \eta(G_{n-2})$. Now $G_3 = C_4$ and hence $\eta(G_3) = 2$. For the graph G_4 by applying the transformation T_2, we get P_4. Hence $\eta(G_4) = 0$.

Hence $\eta(G_n) = \begin{cases} 0 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd.} \end{cases}$ □

8.6 The Nullity Set of a Family of Graphs

Let $\mathcal{F}$ be a family of graphs. The nullity set $N(\mathcal{F})$ of $\mathcal{F}$ is defined by $N(\mathcal{F}) = \{\eta(G) : G \in \mathcal{F}\}$.

Example 1

Let $\mathcal{F}$ be the family of all line graphs of a tree. It follows from Theorem 8.23 that $N(\mathcal{F}) = \{0, 1\}$. Thus for an infinite family of graphs, the nullity set may be a finite set.

Example 2

Let $\mathcal{F}$ be the family of all line graphs. If $G = pK_2$, then $L(G) = \overline{K_p}$ and hence $\eta(L(G)) = p$. Thus for any positive integer p, there exists a graph G such that $\eta(L(G)) = p$. Hence $N(\mathcal{F}) = \{0, 1, 2, 3, \ldots, n, \ldots\}$.

Example 3

Let $\mathcal{F} = \{L(G) : G \text{ is a connected graph}\}$. Consider the graph G_r consisting of r cycles of length 4 and edges joining vertices of cycles as shown in Fig. 8.9.

It can be easily verified that $\eta(L(G_r)) = r + 1$. Hence $n(\mathcal{F}) = \{0, 1, 2, \ldots, n, \ldots\}$.

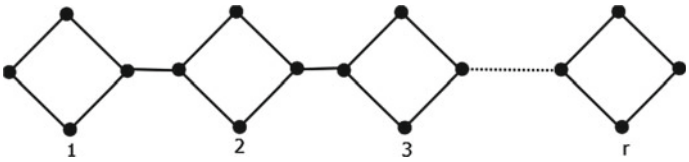


Fig. 8.9 The graph G_r with $\eta(L(G_r)) = r + 1$.

Fig. 8.10 Trees with nullity $n - 4$

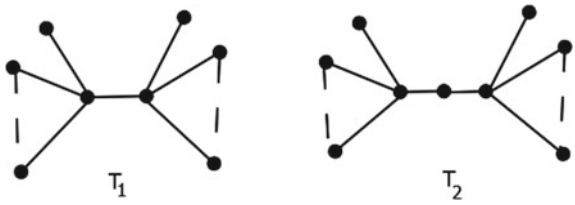
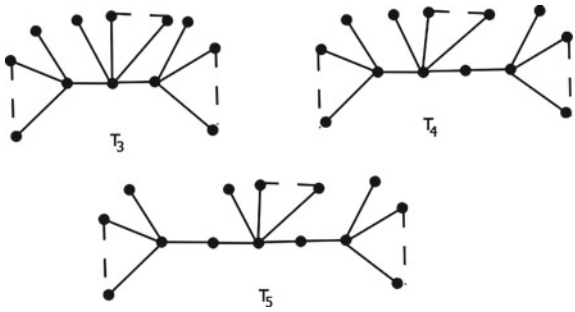


Fig. 8.11 Trees with nullity $n - 6$



Let $\mathcal{T}_n$ be the set of all trees of order n. We now proceed to determine $N(\mathcal{T}_n)$. We need the following theorems.

Theorem 8.34 (Ellingham [10]) *Let $T \in \mathcal{T}_n$. Then $\eta(T) \leq n - 2$ and equality holds if and only if T is isomorphic to the star $S_n = K_{1,n-1}$.*

Theorem 8.35 (Li [22]) *Let $T \in \mathcal{T}_n - \{S_n\}$. Then $\eta(T) \leq n - 4$ and equality holds if and only if T is isomorphic to one of the trees T_1 or T_2 given in Fig. 8.10.*

Theorem 8.36 (Li [22]) *Let $T \in \mathcal{T} - \{S_n, T_1, T_2\}$. Then $\eta(T) \leq n - 6$ and equality holds if and only if T is isomorphic to one of the trees T_3, T_4 or T_5 given in Fig. 8.11.*

In general, we can use trees in $\mathcal{T}_n$ whose nullity is $n - 6$ to determine trees of order n with nullity $n - 8$ Li [22]. Hence, we have the following.

Theorem 8.37 *The nullity set of T_n is $\{0, 2, 4, \ldots, n - 4, n - 2, n\}$ if n is even and $\{1, 3, 5, \ldots, n - 4, n - 2\}$, otherwise.*

Let $\mathcal{U}_n$ denote the set of all unicycle graphs of order n and let $\mathcal{B}_n$ denote the set of all bicyclic graphs of order n. Xuezhong and Liu [33] have determined the nullity set of $\mathcal{U}_n$ where $n \geq 5$.

Theorem 8.38 (Xuezhong and Liu [33]) *The nullity set of* $\mathcal{U}_n$, $n \geq 5$, *is* $\{0, 1, 2, \ldots, n-4\}$.

Hu et al. [20] and Li [22] determined the nullity set of $\mathcal{B}_n$.

Theorem 8.39 (Hu et al. [20], Li [22]) *The nullity of* $\mathcal{B}_n$ *is* $\{0, 1, 2, \ldots, n-2\}$.

8.7 Graphs with Maximum Nullity

If G is a graph of order n, then $\eta(G) = n$ if and only if $G = \overline{K_n}$. Since $\sum_{i=1}^{n} \lambda_i = 0$, it follows that there is no graph G with $\eta(G) = n-1$. Thus if G is a graph of order n and $G \neq \overline{K_n}$, then $0 \leq \eta(G) \leq n-2$. Hence the natural problem is to characterize graphs with maximum nullity $n-2$, second maximum nullity $n-3$ and so on. Substantial progress has been made in this direction. Since any isolated vertex contributes 1 to $\eta(G)$, throughout this section we assume that G is a graph of order n having no isolated vertices.

Cheng and Liu [4] obtained a characterization of graphs with nullity $n-2$ and nullity $n-3$.

Theorem 8.40 (Cheng and Liu [4]) *Let G be a graph of order n having no isolated vertices. Then*

1. $\eta(G) = n-2$ *if and only if G is isomorphic to a complete bipartite graph* K_{n_1,n_2} *where* $n_1 > 0, n_2 > 0$ *and* $n_1 + n_2 = n$.
2. $\eta(G) = n-3$ *if and only if G is isomorphic to a complete tripartite graph* K_{n_1,n_2,n_3} *where* $n_1, n_2, n_3 > 0$ *and* $n_1 + n_2 + n_3 = n$.

There are several results on graphs with nullity $n-4$. Li [22] obtained a characterization of graphs of order n with $\delta = 1$ and nullity $n-4$. Let G_1^* be a graph of order n obtained from $K_{r,s}$ and $K_{1,t}$ by identifying a vertex of $K_{r,s}$ with the center of $K_{1,t}$, where $r, s, t \geq 1$ and $r + s + t = n$. Let G_2 be the graph obtained from the complete tripartite graph $K_{1,l,m}$ and the star $K_{1,p}$ by identifying a vertex v of maximum degree in $K_{1,l,m}$ with the center of $K_{1,p}$, where $l, m, p \geq 1$ and $l + m + p + 1 = n$. The graphs G_1^* and G_2 are shown in Fig. 8.12.

Theorem 8.41 (Li [22]) *Let G be a connected graph of order n with minimum degree $\delta = 1$. Then $\eta(G) = n-4$ if and only G is isomorphic to the graph G_1^* or G_2^*, where G_1^* is given in Fig. 8.12 and G_2^* is a connected spanning subgraph of G_2 given in Fig. 8.12, which contains $K_{l,m}$ as its subgraph.*

Corollary 8.8 (Li [22]) *Let T be a tree of order n and $T \neq K_{1,n-1}$. Then $\eta(T) = n-4$ if and only if T is isomorphic to one of the trees T_1 or T_2 given in Fig. 8.13.*

Chang et al. [3] obtained a characterization of all graphs with rank 4. Since the sum of the rank and nullity is n, this gives a characterization of all graphs with nullity $n-4$.

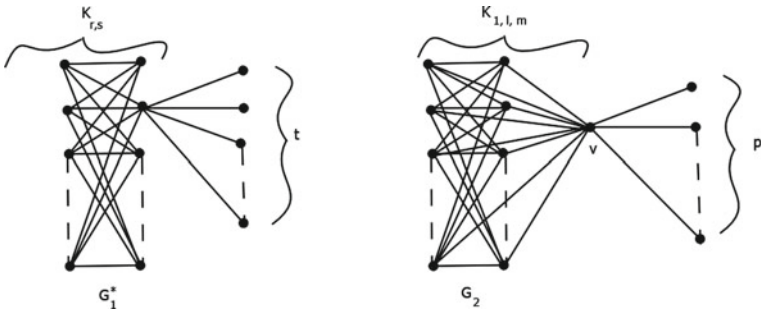


Fig. 8.12 The graphs G_1^* and G_2

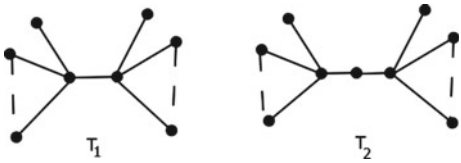


Fig. 8.13 Trees with nullity $n-4$

Definition 8.6 Let G be a graph with $V(G) = \{v_1, v_2, \ldots, v_n\}$. Let $\mathbf{m} = (m_1, m_2, \ldots, m_n)$ be a vector of positive integers. The graph obtained from G by replacing each vertex v_i with an independent set $\{v_i^1, v_i^2, \ldots, v_i^{m_i}\}$ and joining v_i^s with v_j^t if and only if $v_i v_j \in E(G)$ is denoted by $G \circ \mathbf{m}$. This graph is said to be obtained from G by multiplication of vertices.

Theorem 8.42 (Chang et al. [3]) *Let G be a connected graph of order n. Then $r(G) = 4$ (or equivalently $\eta(G) = n - 4$) if and only if G can be obtained from one of the graphs given in Fig. 8.14 by multiplication of vertices.*

Xuezhong and Liu [33] obtained a characterization of unicyclic graphs G of order n with $\eta(G) = n - 4$. This result is also included in Borovicanin and Gutman [2].

Theorem 8.43 (Xuezhong and Liu [33]) *Let G be a unicyclic graph of order n with $n \geq 5$. Then $\eta(G) = n - 4$ if and only if G is isomorphic to one of the graphs U_1, U_2, U_3, U_4 or U_5 given in Fig. 8.15.*

Li et al. [21] and Hu et al. [20] independently characterized bicyclic graphs of order n with $\eta(G) = n - 4$. This result has also been reported in the survey paper Borovicanin and Gutman [2].

Let G be a connected graph of order n and size m. The graph G is called a bicyclic graph if $m = n + 1$. Let $\mathcal{B}_n$ denote the set of all bicyclic graphs of order n. The graph obtained from two cycles C_k and C_l by joining a vertex of C_k to a vertex of C_l by a path of length $q - 1$ is a bicyclic graph and is denoted by $B(k, q, l)$. Such a graph is called an ∞-graph. A graph consisting of three internally disjoint u-v paths of lengths l, p and q where $l, p, q \geq 1$ and at most one of them is 1 is called a θ-graph and is denoted by $P(l, p, q)$. There are three types of bicyclic graphs. The first type,

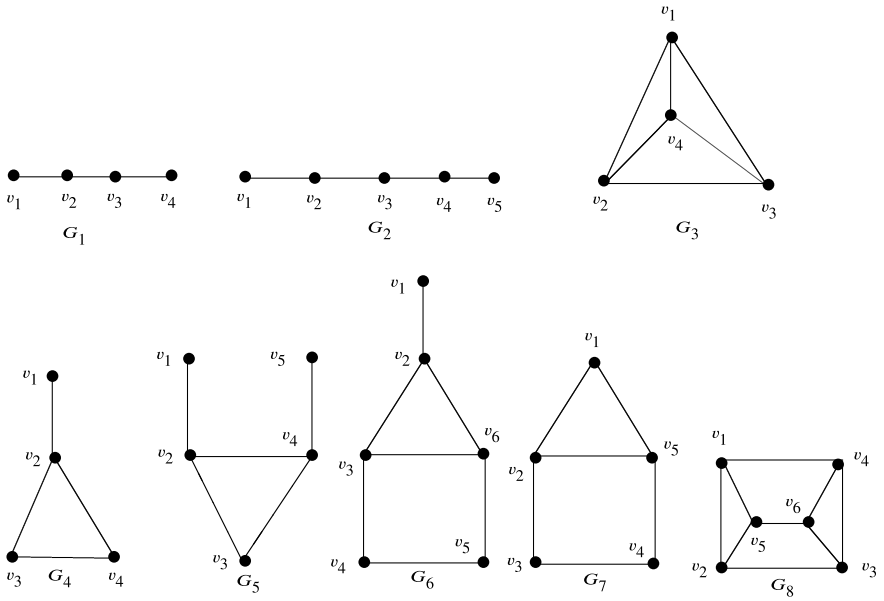


Fig. 8.14 Graphs for Theorem 8.42

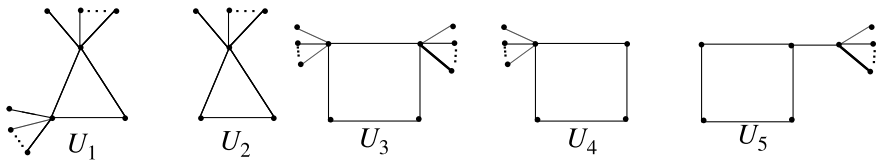


Fig. 8.15 Unicyclic graphs with $\eta(G) = n - 4$.

denoted by $\mathcal{B}_n^+$, is the set of all graphs obtained from an ∞-graph with trees attached when $q = 1$. The second type $\mathcal{B}_n^{++}$ is the set of all graphs obtained from an ∞-graph with trees attached when $q > 1$.

The third type, denoted by θ_n, is the set of all graphs obtained from a θ-graph with trees attached. Then $\mathcal{B}_n = \mathcal{B}_n^+ \cup \mathcal{B}_n^{++} \cup \theta_n$.

Theorem 8.44 (Li et al. [21]) *Let G be a bicyclic graph of order n in θ_n. Then $\eta(G) = n - 4$ if and only if G is isomorphic to one of the graphs $\mathcal{B}_i$, $1 \le i \le 6$, given in Fig. 8.16.*

Another characterization of graphs with $\eta(G) = n - 4$ is given in Cheng and Liu [5].

Let G be an r-partite graph with partite sets $X_1, X_2, \ldots, X_r$ where $r \ge 2$. Let $|X_i| = n_i$.

Definition 8.7 A *chain-like r-partite graph* is a simple r-partite graph with partition $(X_1, X_2, \ldots, X_r)$ in which each vertex of X_i is joined to each vertex of X_{i+1} for

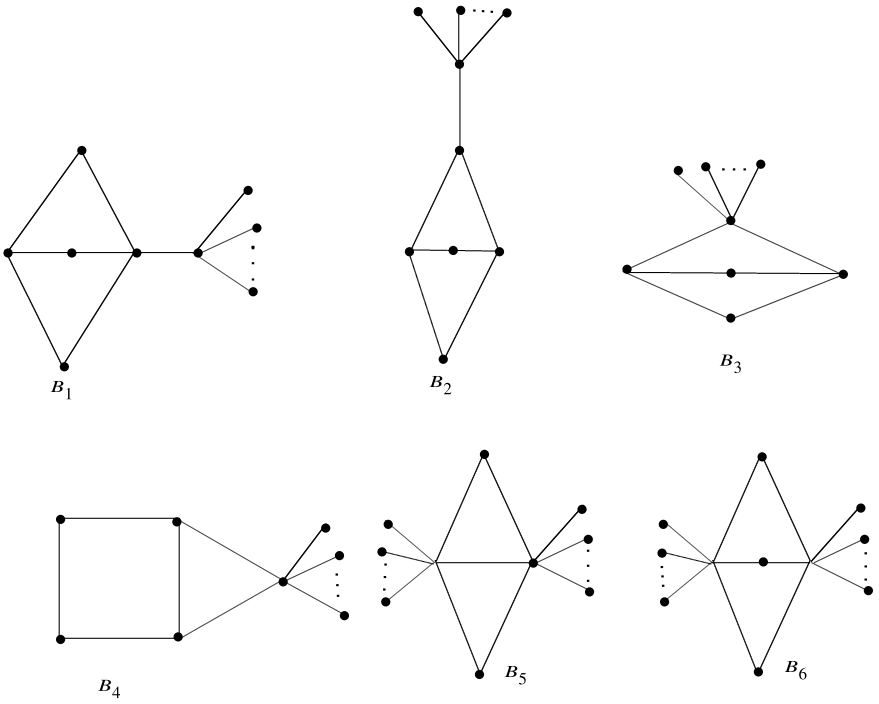


Fig. 8.16 Bicyclic graphs with $\eta = n - 4$.

$i = 1, 2, \ldots, r-1$. This graph is denoted by $P_{n_1,n_2,\ldots,n_r}$ where $|X_i| = n - i$. For $r \geq 4$, $P^{(1)}_{n_1,n_2,\ldots,n_r}$ is the graph obtained from $P_{n_1,n_2,\ldots,n_r}$ by joining each vertex of X_2 to each vertex of X_4 by an edge. For $r \geq 5$, the graph $P^{(2)}_{n_1,n_2,\ldots,n_r}$ is the graph obtained from $P_{n_1,n_2,\ldots,n_r}$ by joining each vertex of X_1 to each vertex of X_4 and X_5 by an edge. For $r \geq 6$, the graph $P^{(3)}_{n_1,n_2,\ldots,n_r}$ is the graph obtained from $P^{(2)}_{n_1,n_2,\ldots,n_r}$ by joining each vertex of X_6 to each vertex of X_2 and X_3 by an edge.

Theorem 8.45 (Cheng and Liu [5]) *Let G be a connected graph of order n with $n \geq 4$. Then $\eta(G) = n - 4$ if and only if G is isomorphic to $K_{n_1,n_2,\ldots,n_r} (r = 4)$, $P_{n_1,n_2,\ldots,n_r} (r = 4 \text{ or } 5)$, $P^{(1)}_{n_1,n_2,\ldots,n_r} (r = 4 \text{ or } 5)$, $P^{(2)}_{n_1,n_2,\ldots,n_r} (r = 5 \text{ or } 6)$ or $P^{(3)}_{n_1,n_2,\ldots,n_r}$ $(r = 6)$, where $n_1 + n_2 + \cdots + n_r = n$ and $n_i > 0$ for all $i = 1, 2, \ldots, r$.*

Li [22] obtained a characterization of all graphs with nullity $n - 5$.

Let G^*_3 be a graph of order n obtained from the complete tripartite graph $K_{r,s,t}$, and $K_{1,q}$ by identifying a vertex of $K_{r,s,t}$ with the center of $K_{1,q}$ where $r, s, t, q > 0$ and $r + s + t + q = n$.

Let G_4 be the graph obtained from $K_{1,l,m,p}$ and $K_{1,d}$ by identifying the vertex v of maximum degree in $K_{1,l,m,p}$ with the center of $K_{1,d}$, where $l, m, p, d > 0$ and $l + m + p + d + 1 = n$.

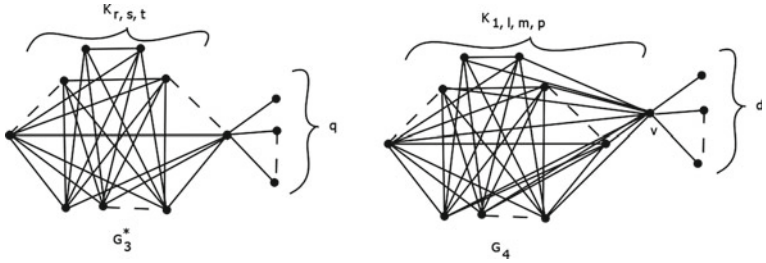


Fig. 8.17 Graphs G_3^* and G_4

Theorem 8.46 (Li [22]) *Let G be a connected graph of order n. Then $\eta(G) = n - 5$ if and only if G is isomorphic to the graph G_3^* or G_4^* where G_3^* is given in Fig. 8.17 and G_4^* is a connected spanning subgraph of G_4 given in Fig. 8.17 such that G_4^* contains $K_{l,m,p}$ as a subgraph.*

8.8 Conclusion

In this paper, we have presented a survey of general mathematical results on the nullity of graphs along with chemically relevant aspects and some new results. Open problems for further investigation have also been included. Though we have covered most of the significant results on nullity, this survey is not exhaustive.

Acknowledgements Among the authors, Manjunatha Prasad Karantha wishes to acknowledge the funding support by Science and Engineering Research Board (DST), India, through the projects CRG/2019/000238 and MTR/2018/000156.

References

1. Ashraf F, Bamdad H (2008) A note on graphs with zero nullity. MATCH- Commun Math Comput Chem 60(1):15–19
2. Borovicanin B, Gutman I (2008) Nullity of graphs. Zborruk Radova 13(21):107–122
3. Chang GJ, Huang L, Yeh H (2011) A characterization of graphs with rank 4. Linear Algebra Appl 434(8):1793–1798
4. Cheng B, Liu B (2007) On the nullity of graphs. Electron J Linear Algebra 16:60–67
5. Cheng B, Liu B (2011) On the nullity of tricyclic graphs. Linear Algebra Appl 434(8):1799–1810
6. Coulson CA, O'Leary B, BMallion R (1978) Hückel theory for organic chemists. Academic Press, London
7. Cvetković DM, Gutman I (1972) The algebraic multiplicity of the number zero in the spectrum of a bipartite graph. Matematički Vesnik 9(24):141–150

8. Cvetković D, Gutman I, Trinajstić N (1975) Graphical studies on the relations between the structure and reactivity of conjugated systems: the role of non-bonding molecular orbitals. J Mol Struct 28:289–303
9. Dias J (1999) Resonance structures of benzenoid conjugated radicals. Phys Chem Chem Phys 1(22):5081–5086
10. Ellingham MN (1993) Basic subgraphs and graph spectra. Austral J Comb 8:247–266
11. Fiorini S, Gutman I, Sciriha I (2005) Trees with maximum nullity. Linear Algebra Appl 397:245–251
12. Gong S, Xu GH (2012) On the nullity of a graph with cut-points. Linear Algebra Appl 436(1):135–142
13. Graovac A, Gutman I, Trinajstić N (1977) Topological approach to the chemistry of conjugated molecules. Springer, Berlin
14. Gutman I (1978) The energy of a graph. Ber Math-Statist Sekt Forshungsz Graz 103:1–22
15. Gutman I (2021) Computing the dependence of graph energy on nullity: the method of siblings. Discr Math Lett 7:30–33
16. Gutman I (2021) Graph energy and nullity. Open J Discret Appl Math 4:25–28
17. Gutman I, Sciriha I (2001) On the nullity of line graphs of trees. Discrete Math 232(1–3):35–45
18. Gutman I, Triantafillou T (2016) Dependence of graph energy and nullity, a case study. MATCH- Commun Math Comput Chem 76:761–769
19. Gutman I, Cmiljanović N, Milosavljević S, Radenković S (2004) Effect of nonbonding molecular orbitals on total π-electron energy. Chem Phys Lett 383(1–2):171–175
20. Hu S, Xuezhong T, Liu B (2008) On the nullity of bicyclic graphs. Linear Algebra Appl 429(7):1387–1391
21. Li J, Chang A, Shiu WC (2008) On the nullity of bicyclic graphs. MATH Commun Math Compt Chem 60:21–36
22. Li S (2008) On the nullity of graphs with pendant vertices. Linear Algebra Appl 429(7):1619–1628
23. Longuet-Higgins HC (1950) Some studies in molecular orbital theory I. Resonance structure and MO in unsaturated hydrocarbons. J Chem Phys 18(3):265–274
24. Ma H, Li D, Xie C (2020) Non-singular trees, unicyclic graphs and bicyclic graphs. Appl Math 11:1–7
25. Mohiaddin GH, Sharaf KR (2021) Change of nullity of a graph under two operations. Kuwait J Sci 48(4)
26. Mustapha A, Hansen P (2017) On the nullity number of graphs. Electron J Graph Theory Appl 2(5):335–346
27. Omidi GR (2009) On the nullity of bipartite graphs. Graphs Combin 25:111–114
28. Sachs H (1964) Beziehungen zwischen den in einem graphen enthaltenen kreisenund seinem charakteristischen polynom. Publ Math (Debrecen) 11:119–134
29. Santhakumaran AP, Titus P (2017) Monophonic distance in graphs. Graph theory—advanced algorithms and applications InTech
30. Sciriha I (1998) On the construction of graphs of nullity one. Discrete Math 181(1–3):193–211
31. Von CL, Sinogowitz U (1957) Spectras of finite graphs. Abh Math Sem Univ Hamburg 21(1):63–77
32. Wang L, Geng X (2020) Proof of a conjecture on the nullity of a graph. J Graph Theory 95(4):586–593
33. Xuezhong T, Liu B (2005) On the nullity of unicyclic graphs. Linear Algebra Appl 408:212–220
34. Yates K (1978) Hučkel Molecular Orbital Theory. Academic Press, New York
35. Zhou Q, Wong D, Sun D (2018) An upper bound of the nullity of a graph in terms of order and maximum degree. Linear Algebra Appl 555:314–320

Chapter 9
Some Observations on Algebraic Connectivity of Graphs



Ravindra B. Bapat, A. K. Lal, and S. Pati

Mathematics Subject Classification (2010) 05C50 · 05C05 · 15A18

9.1 Introduction

Let $G = (V, E)$ be a connected simple graph on the vertex set $V = \{1, 2, \ldots, n\}$ and the edge set E. For any two vertices i and j, $i \neq j$, by $i \sim j$, we will mean that i is *adjacent* to j, that is, the edge $[i, j] \in E$. The *adjacency matrix* $A(G)$ of G is an $n \times n$ matrix with the (i, j)-th entry 1, if $i \sim j$ and 0, otherwise. The *Laplacian matrix* $L(G)$, in short L, is defined as $D(G) - A(G)$, where $D(G)$ is the diagonal degree matrix of G. It is well-known that the Laplacian matrix is a positive semi-definite matrix with 0 as the smallest eigenvalue and $\nVdash$, the vector of all ones, as a corresponding eigenvector. Fiedler [7] observed that the second smallest eigenvalue of $L(G)$ is positive if and only if G is connected. This eigenvalue is viewed as an algebraic measure of the connectivity of G and known as the *algebraic connectivity* of G. It is denoted by $a(G)$.

R B Bapat and S Pati dedicate this article to late Professor A K Lal.

R. B. Bapat
Indian Statistical Institute, New Delhi, India
e-mail: rbb@isid.ac.in

A. K. Lal
IIT Kanpur, Kanpur, India
e-mail: arlal@iitk.ac.in

S. Pati (✉)
I I T Guwahati, Guwahati, India
e-mail: pati@iitg.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_9

The eigenvectors of $L(G)$ corresponding to the algebraic connectivity are popularly known as the *Fiedler vectors* of G. The literature is full of interesting research pieces characterizing graphs that have the smallest (or largest) algebraic connectivities among certain classes of graphs, see, for example, Fallat et al. [6], Fallat and Kirkland [5], Guo [8], Kirkland and Fallat [9], Kirkland et al. [11], Merris [13], Patra and Lal [14], Shao et al. [15].

The first such results were obtained by Fiedler himself. It follows from results 4.1, 4.3, and 4.4 of Fiedler [7] that *among all trees on $n > 1$ vertices, the path has the smallest algebraic connectivity and the star has the largest algebraic connectivity*.

Let G be a connected graph and $\mathbf{y}$ be a Fiedler vector. A vertex v is called a *characteristic vertex* of G with respect to $\mathbf{y}$ if $\mathbf{y}(v) = 0$ and v has a neighbor w such that $\mathbf{y}(w) \neq 0$. An edge $[u, v]$ is called a *characteristic edge* with respect to $\mathbf{y}$ if $\mathbf{y}(u)\mathbf{y}(v) < 0$. The collection of all characteristic elements of G with respect to $\mathbf{y}$ is called the *characteristic set* of G. Following Bapati and Pati [1], we denote it by $\mathcal{C}(G, \mathbf{y})$.

There are many interesting facts known about the characteristic vertices, edges and set, see Fiedler [7], Merris [12, 13], Kirkland et al. [11], Kirkland and Neumann [10], Kirkland and Fallat [9], Fallat and Kirkland [5], Bapati and Pati [1], Bapati et al. [2]. For example, for trees it is known that $|\mathcal{C}(T, \mathbf{y})| = 1$. A general result about the cardinality of the characteristic set can be found in Bapati and Pati [1]. It is known that, if for some Fiedler vector $\mathbf{y}$, we get $\mathcal{C}(G, \mathbf{y}) = \{u\}$, then for every Fiedler vector $\mathbf{z}$ of G, we will have $\mathcal{C}(G, \mathbf{z}) = \{u\}$, see Bapat et al. [2]. For a connected graph G, if $\mathcal{C}(G, \mathbf{y})$ is not a singleton vertex, then it always lies in the same block (a maximal induced subgraph without a cut vertex in it) of G, independent of the different Fiedler vectors, see Bapat et al. [2]. In Bapat et al. [2], the authors have shown that for any connected graph G and any Fiedler vector $\mathbf{y}$, if $\mathcal{C}(G, \mathbf{y})$ is an edge then, $a(G)$ is simple.

One of the key techniques in understanding the algebraic connectivity of trees is that of a *Perron branch*, a concept introduced by Kirkland et al. [11]. Let G be connected and $W \subsetneq V(G)$. Then, a branch of G at W is any connected component of $G - W$. Let B be a branch of G at W. Denote by $\hat{L}(B)$ the principal submatrix of $L(G)$ corresponding to B. It can be argued that $\hat{L}(B)$ is a nonsingular M-matrix and hence has an entrywise positive inverse. Let us denote $\hat{L}(B)^{-1}$ by $\mathbf{B}[B]$ and the Perron value of $\mathbf{B}[B]$ by $\rho(B)$.

Let G be a connected graph and u be any cut vertex. A branch B at u is called a *Perron branch* if $\rho(B)$ is the largest among all branches at u. Note that this is equivalent to saying that $\rho(B) \geq \frac{1}{a(G)}$ (see Bapat et al. [2]). If T is a tree and B is a branch at u then, an explicit description of $\mathbf{B}[B]$ is known (see Fallat and Kirkland [5], Kirkland and Neumann [10], Kirkland et al. [11]). For an unicyclic graph, a corresponding description can be found in Bapat et al. [2, 3].

Using the concept of Perron components the following result was proved. Let T be a tree and B be a non-Perron branch at u. (In this case, it is known that B cannot contain the characteristic set.). Let T' be obtained by moving a leaf in B away from u. Then, $a(T') \leq a(T)$ (see Kirkland and Fallat [9]), that is, as we move an edge away from the characteristic set the algebraic connectivity tends to decrease, see Fig. 9.1.



Fig. 9.1 $a(T') \leq a(T)$

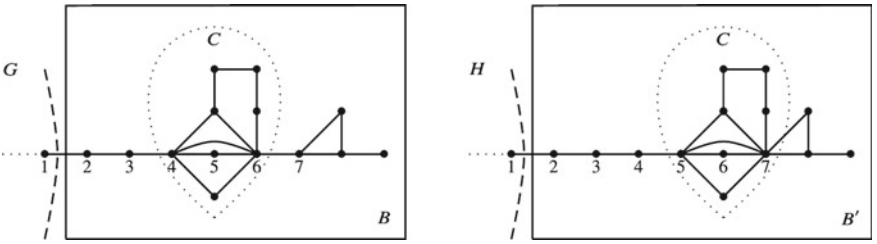


Fig. 9.2 H is obtained from G by sliding C along P one unit away from $u = 1$

This happens because the bottleneck matrix $\mathbf{B}[B'] \geq \mathbf{B}[B]$ (entrywise domination), thereby increasing the Perron value. Notice that, in the above example, we are sliding the tree T'' along a path, away from u.

A more general view of the above operation is 'sliding a connected graph, not necessarily a tree, along a path'. Here, we allow the path to contain more than one vertex of the graph. We give a formal definition below.

Definition 9.1 Let G be a connected graph, u be a cut vertex and B be a non-Perron branch at u. Let u_1 be the only vertex in B adjacent to u and let $P = [u_1, \ldots, u_\ell]$ be a path in B. Let C be a connected subgraph of B such that $V(C) \cap V(P) = \{u_{k+1}, u_{k+2}, \ldots, u_{k+r}\}$. Here, we assume $k + r < \ell$ and the edges of P that are not in C are bridges of P. Replace each edge of $E(C) - E(P)$ of the form $[x, u_i]$ with $[x, u_{i+1}]$, if x is not on P. Replace each edge of $E(C) - E(P)$ of the form $[u_i, u_j]$ with $[u_{i+1}, u_{j+1}]$. We call this operation as 'sliding C along the path P by one unit away from u'.

Example 1

(i) Consider the graph G in Fig. 9.2. It may be having more vertices on the left side. Take $u = 1$. Then, B is the component at 1 that contains 2. Here $P = [2, 3, 4, 5, 6, 7]$, $u_1 = 2$, $u_{k+1} = 4$, $u_{k+r} = 6$, and $u_\ell = 7$. The graph C is the one inside the dotted curve. The graph H is now obtained by moving C one unit away from 1 along the path P.

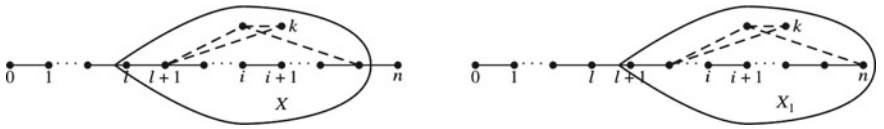


Fig. 9.3 Left side graph: $P(X)$. Right side graph: $P(X)_1$

(ii) In general, consider a path $P = [0, 1, \ldots, n]$ and a connected graph X on vertices $l, l+1, \ldots, n-1, n+1, \ldots, m$ (where $l \geq 1$) such that X contains the path $[l, l+1, \ldots, n-1]$. Let $P(X)$ denote the graph $P \cup X$ and $P(X)_1$ denote the graph obtained by sliding X along P one unit away from 0 (see Fig. 9.3). Also, let us denote the isomorphic copy of the subgraph X in $P(X)_1$ by X_1. Note that the isomorphism ϕ from X to X_1 is given by $\phi(i) = i + 1$, whenever $l \leq i \leq n-1$ and $\phi(i) = i$ for $i \geq n+1$.

Recall that in Fig. 9.1, we had $a(T') \leq a(T)$ owing to the fact that $\mathbf{B}[B'] \geq \mathbf{B}[B]$ entrywise. We need to make this domination work in general case. This is our main result.

Theorem 9.1 *Let the graphs $X, G = P(X), X_1$ and $H = P(X)_1$ be as shown in Example 1 (ii). Then, $\boldsymbol{B}[G - 0]$ is entrywise dominated by $\boldsymbol{B}[H - 0]$.*

In order to do the above, we first give a less known combinatorial description of the bottleneck matrix and some of its immediate useful implications. This is done in Sect. 9.2. In Sect. 9.3, we prove our main result and use it to obtain comparisons between the algebraic connectivity of a few graphs.

9.2 Describing the Bottleneck Matrix

For a square matrix A, by $A(i_1, \ldots, i_k | j_1, \ldots, j_k)$, we mean the submatrix obtained by deleting the rows $i_1, \ldots, i_k$ and the columns $j_1, \ldots, j_k$ of A. Recall that the matrix-tree theorem says that $\det L(i|j) = (-1)^{i+j}\kappa(G)$, where $\kappa(G)$ is the number of spanning trees in G. Let $s, t \geq 1$. Then, by a forest of G of the type $(i_1, \ldots, i_s || j_1, \ldots, j_t)$, we mean a spanning forest of G having two trees, one containing the vertices $i_1, \ldots, i_s$ (and may be some more vertices) and the other containing the vertices $j_1, \ldots, j_t$ (and may be some more vertices). By $N_{G(i_1,\ldots,i_s||j_1,\ldots,j_t)}$, we denote the number of spanning forests of G of the type $(i_1, \ldots, i_s || j_1, \ldots, j_t)$. With the above notations, we have our first result which is a direct application of Chaiken's result, see Chaiken [4].

Lemma 9.1 *Let G be a connected graph on vertices $0, 1, \ldots, n$ and $0 < i, j$ (we allow $i = j$). Then,* $\det L(0, i|0, j) = (-1)^{i+j} N_{G(0||i,j)}$.

Proof Form a complete loop-less directed graph H from G by replacing each undirected edge $[i, j]$ in G with two directed oppositely oriented edges (i, j) and (j, i) of weights 1 each. For each pair $\{i, j\}$ which is not an edge in G, draw the edges (i, j) and (j, i) with weights 0.

Take $S = \{0, 1, \ldots, n\} = V(H)$. Then, the matrix A as defined in Eq. (1) of Chaiken [4] is nothing but $L(G)$.

Now, take $W = \{0, i\}$ and $U = \{0, j\}$. Then, $\epsilon(W, S) = (-1)^{i-1}$ and $\epsilon(U, S) = (-1)^{j-1}$, see the lines below Eq. (3) on page 321 of Chaiken [4].

Now, take $A = \bar{W} = \{1, \ldots, i-1, i+1, \ldots, n\}$ and $B = \bar{U} = \{1, \ldots, j-1, j+1, \ldots, n\}$. Then, for any matching $\pi : A \to B$, we get a path from j to i and this is the only path defined by π (see the first paragraph of Section 2 of Chaiken [4]). Thus, in our case, the bijection $\pi^* : \{0, i\} \to \{0, j\}$ is $\pi^*(0) = 0$ and $\pi^*(i) = j$. Note that π^* has no inversions. Hence $\epsilon(\pi^*) = 1$.

Note that in our notation $A(W|U)$ is nothing but $A(\bar{W}|\bar{U})$ in Chaiken [4]. Thus, from the all minors matrix-tree theorem in Chaiken [4], we have

$$\det A(W|U) = \epsilon(W, S)\epsilon(U, S) \sum_F \epsilon(\pi^*) a_F,$$

where the sum is over all spanning forests F in H such that the following holds:

1. F contains exactly $|W| = |U| = 2$ trees.
2. Each tree in F contains exactly one vertex in U and exactly one vertex in W. That is, one tree T_0 contains the vertex 0 and the other tree, say T_1, contains the vertices i and j.
3. Each arc in F is directed away from the vertex in U of the tree containing that arc. This means that T_0 is rooted at 0 and all its arcs are directed away from 0 and hence it may be viewed as an undirected tree. The same holds for the tree T_1.
4. a_F is the product of the weights of the edges in F.

In our case, note that $a_F = 1$ if F corresponds to a spanning forest in G, otherwise $a_F = 0$. Therefore,

$$\det L(0, i|0, j) = \det A(W|U) = (-1)^{i+j-2} \sum_F a_F = (-1)^{i+j} N_{G(0|i,j)}.$$

The proof is complete.

We remark here that, one could directly give a proof of the previous result using the properties of the vertex edge incidence matrix. We have chosen the above proof as it is a direct application of the theorem in Chaiken [4].

As an immediate corollary, we present a description of the (i, j)-entry of a bottleneck matrix.

Corollary 9.1 *Let G be a connected graph on vertices $0, 1, \ldots, n$ and $0 < i, j$ (we allow $i = j$). Then,*

$$\boldsymbol{B}[G-0](i,j)=\frac{N_{G(0||i,j)}}{\kappa(G)}.$$

Proof Note that $L(0|0)$ is symmetric and $\mathbf{B}[G-0]=L(0|0)^{-1}$. Hence,

$$\mathbf{B}[G-0](i,j)=(-1)^{i+j}\frac{\det L(0,i|0,j)}{\det L(0|0)}=\frac{N_{G(0|i,j)}}{\kappa(G)},$$

as claimed.

We use Corollary 9.1 to obtain quite a few well-known results. To proceed with our first result, let G be a connected graph and let B be a branch at a vertex u. As the matrix $\hat{L}(B)$ is a diagonally dominant nonsingular M-matrix, its inverse has all positive entries and moreover $(\hat{L}(B))_{ii}\geq(\hat{L}(B))_{ij}$, for all $i\neq j$. A combinatorial explanation of this is stated next.

Corollary 9.2 *Let G be a connected graph on vertices $0,1,\ldots,n$ and let $i>0$. Then, $\boldsymbol{B}[G-0](i,i)\geq\boldsymbol{B}[G-0](i,j)$, for each $j>0$ with $j\neq i$. The inequality is strict if there is a forest of the type $(0,j||i)$.*

Proof Using Corollary 9.1, we have

$$\kappa(G)\mathbf{B}[G-0](i,i)-\kappa(G)\mathbf{B}[G-0](i,j)=N_{G(0||i)}-N_{G(0||i,j)}.$$

Now, notice that each forest of the type $(0||i,j)$ is also a forest of the type $(0||i)$. Hence $N_{G(0||i)}-N_{G(0||i,j)}\geq 0$. If there is at least one forest of the type $(0,j||i)$ then, its contribution to $N_{G(0||i)}-N_{G(0||i,j)}$ is at least 1 and hence, in this case, the inequality is strict.

Our second corollary is the famous result in Kirkland et al. [11] that gives the description of the bottleneck matrix of a tree in terms of common edges.

Corollary 9.3 (Kirkland et al. [11]) *Let T be a tree on vertices $0,1,\ldots,n$. Then, $\boldsymbol{B}[T-0](i,j)$ is the number of edges common to the 0-i-path and the 0-j-path.*

Proof Note that for a tree T, a spanning forest of the type $(0||i,j)$ can only be obtained by deleting a common edge of the 0-i-path and the 0-j-path. Hence, $N_{T(0||i,j)}$ equals the numbers of such common edges.

As our next corollary, we obtain a description of the bottleneck matrix of a cycle.

Corollary 9.4 (Bapat et al. [2]) *Let G be the cycle $[0,1,2,\ldots,n,0]$ and $0<i\leq j$. Then,*

$$\boldsymbol{B}[G-0](i,j)=\frac{i(n+1-j)}{(n+1)}.$$

Proof A spanning forest of the type $(0||i, j)$ can be obtained only by deleting an edge from the 0-i-path that does not contain j and by deleting an edge from the 0-j-path that does not contain i. Thus, there are only $i(n+1-j)$ such spanning forests. As $\kappa(G) = n+1$, the required result follows.

To state and prove the next corollary, we need the following notation. By $A(:, i)$ (respectively, $A(i, :)$), we mean the i-th column (respectively, row) of A. By J, we mean the matrix of all ones of an appropriate size.

Corollary 9.5 (Bapat et al. [3], Fallat and Kirkland [5]) *Let G and H be two connected graphs on vertex sets $0, 1, \ldots, n$ and $n, n+1, \ldots, k$, respectively. Put $\delta = \boldsymbol{B}[G-0](n, n)$. Then,*

$$\boldsymbol{B}[G \cup H - 0] = \left[\begin{array}{c|c} \boldsymbol{B}[G-0] & \boldsymbol{B}[G-0](:, n)\not\Vdash^t \\ \hline \not\Vdash \boldsymbol{B}[G-0](n, :) & \delta J + \boldsymbol{B}[H-n] \end{array}\right].$$

Proof Note that $V(G) \cap V(H) = \{n\}$. Hence

$$N_{G\cup H(0||i,j)} = N_{G(0||i,j)}\kappa(H), \qquad \text{for } 1 \le i, j \le n.$$

So,

$$\mathbf{B}[G \cup H - 0](i, j) = \frac{N_{G\cup H(0||i,j)}}{\kappa(G \cup H)} = \frac{N_{G(0||i,j)} \cdot \kappa(H)}{\kappa(G) \cdot \kappa(H)} = \frac{N_{G(0||i,j)}}{\kappa(G)} = \mathbf{B}[G-0](i, j).$$

If $1 \le i \le n$ and $j > n$ then, a spanning forest of $G \cup H$ of the type $(0||i, j)$ uniquely gives rise to a spanning forest of G of the type $(0||i, n)$ and a spanning tree of H. Hence,

$$\mathbf{B}[G \cup H - 0](i, j) = \frac{N_{G\cup H(0||i,j)}}{\kappa(G \cup H)} = \frac{N_{G(0||i,n)} \cdot \kappa(H)}{\kappa(G) \cdot \kappa(H)} = \frac{N_{G(0||i,n)}}{\kappa(G)} = \mathbf{B}[G-0](i, n).$$

Now, let $j \ge i > n$. Then, a spanning forest of $G \cup H$ of the type $(0||i, j)$ gives rise to

1. either a spanning tree of G and a spanning forest of H of the type $(n|i, j)$,
2. or a spanning forest of G of the type $(0|n)$ and a spanning tree of H.

Thus,

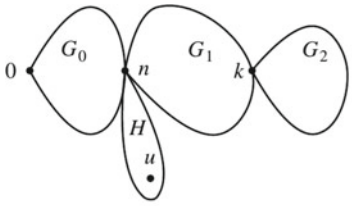
$$\begin{aligned}\mathbf{B}[G \cup H - 0](i, j) &= \frac{N_{G\cup H(0||i,j)}}{\kappa(G \cup H)} = \frac{\kappa(G)N_{H(n||i,j)} + N_{G(0||n)}\kappa(H)}{\kappa(G)\kappa(H)} \\ &= \mathbf{B}[G-0](n, n) + \mathbf{B}[H-n](i, j) = \delta + \mathbf{B}[H-n](i, j).\end{aligned}$$

This completes the proof.

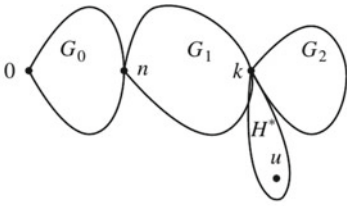
We are now ready to state and prove the next corollary.

Corollary 9.6 (Bapat et al. [3] (Shifting)) *Let G_0, G_1, G_2, H be connected graphs on vertices $\{0, 1, \ldots, n\}$, $\{n, n+1, \ldots, k\}$, $\{k, k+1, \ldots, r\}$, and $\{n, r+1, \cdots, s\}$, respectively. Let H^* be obtained from H by relabeling the vertex n to k (see the figure below). Then*

$$\boldsymbol{B}[G_0 \cup G_1 \cup G_2 \cup H - 0] \leq \boldsymbol{B}[G_0 \cup G_1 \cup G_2 \cup H^* - 0].$$



$G = G_0 \cup G_1 \cup G_2 \cup H$



$G^* = G_0 \cup G_1 \cup G_2 \cup H^*$

Proof Put $G = G_0 \cup G_1 \cup G_2 \cup H$ and $G^* = G_0 \cup G_1 \cup G_2 \cup H^*$. As $\kappa(G) = \kappa(G^*)$, using Corollary 9.1, we have

$$\kappa(G)\mathbf{B}[G - 0](i, j) - \kappa(G)\mathbf{B}[G^* - 0](i, j) = N_{G(0||i,j)} - N_{G^*(0||i,j)}.$$

Using Corollary 9.5, we see that

$$\mathbf{B}[G_0 \cup G_1 \cup G_2 - 0](n, n) < \mathbf{B}[G_0 \cup G_1 \cup G_2 - 0](k, k).$$

Thus, again using Corollary 9.5, we only need to show that $N_{G(0||i,j)} \leq N_{G^*(0||i,j)}$, whenever i is a vertex of $G_0 \cup G_1 \cup G_2$) and $\underline{j \; is \; a \; vertex \; of \; H}$.

We do so as follows:

Case 1. Assume that $i \in \{1, 2, \ldots, n\}$. Then, a spanning forest of G of the type $(0||i, j)$ is a spanning forest of G_0 of the type $(0||i, n)$ and spanning trees of H, G_1 and G_2. A similar argument holds for $N_{G^*(0||i,j)}$. Hence, in this case,

$$N_{G(0||i,j)} = N_{G(0||i,n)}\kappa(H)\kappa(G_1)\kappa(G_2) = N_{G^*(0||i,j)}.$$

Case 2. Assume that $i \in \{n+1, n+2, \ldots, k\}$. Then, a spanning forest of G of the type $(0||i, j)$ is a spanning forest of G_0 of the type $(0||n)$ and spanning trees of H, G_1 and G_2. So,

$$N_{G(0||i,j)} = N_{G_0(0||n)}\kappa(H)\kappa(G_1)\kappa(G_2).$$

Now, observe that a spanning forest of G^* of the type $(0||i, j)$ gives rise to either

1. a spanning forest of G_1 of the type $(n||i, k)$ and spanning trees of G_0, G_2 and H^*, or
2. a spanning forest of G_0 of the type $(0||n)$ and spanning trees of G_1, G_2 and H^*.

Thus, in this case, we have

$$\begin{aligned} N_{G^*(0|i,j)} &= \kappa(G_0)N_{G_1(n|i,k)}\kappa(H)\kappa(G_2) + N_{G_0(0|n)}\kappa(H)\kappa(G_1)\kappa(G_2) \\ &\geq N_{G_0(0|n)}\kappa(H)\kappa(G_1)\kappa(G_2) = N_{G(0|i,j)}. \end{aligned}$$

Case 3. Assume that $i \in \{k+1, k+2, \ldots, r\}$. The argument is similar to Case 2 and hence is omitted.

Therefore, combining the three cases given above, the required result follows.

Some new observations related to bottleneck matrix that will be used in the sequel are given next. The first one implies that if i and j are adjacent then, the entrywise difference of row i and row j is bounded above by 1. Their proofs are given by applying Corollary 9.1.

Proposition 9.1 *Let G be a connected graph on vertices $0, 1, \ldots, n$, $0 < i, j$ such that $i \sim j$ and $k > 0$. Also, let N_1 be the number of spanning trees of G that have paths of the form $[0, \ldots, j, i, \ldots, k]$ and N_2 be the number of spanning trees of G that have paths of the form $[0, \ldots, i, j, \ldots, k]$. Then,*

$$\boldsymbol{B}[G-0](i,k) - \boldsymbol{B}[G-0](j,k) = \frac{N_{G(0,j||i,k)} - N_{G(0,i||j,k)}}{\kappa(G)} = \frac{N_1 - N_2}{\kappa(G)}.$$

In particular, $\left|\boldsymbol{B}[G-0](i,k) - \boldsymbol{B}[G-0](j,k)\right| \leq 1$. Moreover, equality holds if and only if either $N_1 = \kappa(G)$ and $N_2 = 0$ or $N_1 = 0$ and $N_2 = \kappa(G)$. Or equivalently, equality holds if and only if $[i, j]$ is a bridge that separates the vertices 0 and k.

Proof Using Corollary 9.1, we have

$$\kappa(G)\mathbf{B}[G-0](i,k) - \kappa(G)\mathbf{B}[G-0](j,k) = N_{G(0||i,k)} - N_{G(0||j,k)}.$$

We first note that only the forests of the type $(0, j||i, k)$ contribute 1 to $N_{G(0||i,k)} - N_{G(0||j,k)}$. Similarly, we see that the forests of the type $(0, i||j, k)$ contribute -1 to $N_{G(0||i,k)} - N_{G(0||j,k)}$.

Hence, $N_{G(0||i,k)} - N_{G(0||j,k)} = N_{G(0,j||i,k)} - N_{G(0,i||j,k)}$. Thus, we only need to show that the above number is $\leq \kappa(G)$.

Toward that, we will show that $N_{G(0,j||i,k)}$ is the number of spanning trees of G that have paths of the form $[0, \ldots, j, i, \ldots, k]$. For that, take a forest of the type $(0, j||i, k)$ and add the edge $[i, j]$ to get a spanning tree having a path of the form $[0, \ldots j, i, \ldots, k]$. Conversely, if we take any spanning tree that contains a path of the form $[0, \ldots, j, i, \ldots, k]$ then, the forest obtained by deleting the edge $[i, j]$ is a forest of the type $(0, j||i, k)$.

Similarly, $N_{G(0,i||j,k)}$ is the number of spanning trees of G that have paths of the form $[0, \ldots, i, j, \ldots, k]$. The conclusion follows.

Proposition 9.2 *Let G be a connected graph on vertices $0, 1, \ldots, n$ and $0 < i, j, u, v$ such that $i \sim j$ and $u \sim v$ (some of the vertices i, j, u, v need not be distinct). Then,*

$$\boldsymbol{B}[G-0](i,u) - \boldsymbol{B}[G-0](j,v) = \frac{N_{G(0,v||i,u)} + N_{G(0,j||i,u,v)} - N_{G(0,i||j,v,u)} - N_{G(0,u||j,v)}}{\kappa(G)}.$$

In particular, $\left|\boldsymbol{B}[G-0](i,u) - \boldsymbol{B}[G-0](j,v)\right| \leq 1$.

Proof By Corollary 9.1,

$$\kappa(G)\mathbf{B}[G-0](i,u) - \kappa(G)\mathbf{B}[G-0](j,v) = N_{G(0||i,u)} - N_{G(0||j,v)}. \tag{9.1}$$

Note that a forest of G of the type $(0||i,u)$ is

1. either a forest of the type $(0, v||i,u)$,
2. or a forest of the type $(0, j||i,u,v)$,
3. or a forest of the type $(0||i,u,j,v)$.

The forests of G of the type $(0|j,v)$ can be partitioned similarly. So, Eq. (9.1) reduces to

$$\kappa(G)\big(\mathbf{B}[G-0](i,u) - \mathbf{B}[G-0](j,v)\big) = N_{G(0,v||i,u)} + N_{G(0,j||i,u,v)} - N_{G(0,u|j,v)} - N_{G(0,i|j,v,u)}.$$

This completes the proof of the first part.

For the proof of the second part, note that only the forests of the type $(0, j||i,u)$ or $(0, v||i,u)$ can contribute 1 to $N_{G(0||i,u)} - N_{G(0||j,v)}$. Hence, we consider two cases depending on whether we have forests of the type $(0, j||i,u)$ or $(0, v||i,u)$.

Case 1. If $i = u$ and $j = v$, then by adding the edge $[i, j]$ to a forest of the type $(0, j||i)$, we obtain a spanning tree of G. We are done in this case.
Case 2.1 Either $i \neq u$ or $j \neq v$. Suppose that we have forest of the type $(0, j||i,u)$. By adding the edge $[i, j]$, we obtain a spanning tree of G. Note that in this tree, the 0-u-path contains both the vertices i and j.
Case 2.2 Either $i \neq u$ or $j \neq v$. Suppose that we have forest of the type $(0, v||i,u)$. By adding the edge $[u, v]$, we obtain a spanning tree of G. Note that in this tree, the 0-u-path cannot contain both i and j, unless $i = u$ and $j = v$. So, these trees are different from the ones obtained in the previous case.

Thus, $\left|N_{G(0|i,u)} - N_{G(0|j,v)}\right| \leq \kappa(G)$ and hence using Eq. (9.1), the conclusion follows.

As a last observation, we have the following result.

Corollary 9.7 *Let G be any connected graph on $0, 1, \ldots, n$. Assume that $0 \sim 1$ and $j \geq 1$. Then, $\boldsymbol{B}[G-0](1, j) \leq 1$. Equality holds if and only if every path from j to 0 passes through 1.*

Proof Note that by Corollary 9.1, $\kappa(G)\mathbf{B}[G-0](1,j) = N_{G(0||1,j)}$. By inserting the edge [0, 1], we see that, $N_{G(0||1,j)} \leq \kappa(G)$. Hence, we get the first part.

Suppose that every path from j to 0 passes through 1. Then, the edge [0, 1] is a bridge and $N_{G(0||1)} = N_{G(0||1,j)}$. But $N_{G(0||1)} = \kappa(G)$ and hence,

$$\mathbf{B}[G-0](1,j) = \frac{N_{G(0||1,j)}}{\kappa(G)} = \frac{N_{G(0||1)}}{\kappa(G)} = 1.$$

Now, let us assume that $\mathbf{B}[G-0](1,j) = 1$. Hence, $\kappa(G) = N_{G(0||1,j)}$. We also note that from every forest of the type $(0||1,j)$, we can create a unique spanning tree by adding the edge [0, 1]. As $\kappa(G) = N_{G(0||1,j)}$, we see that there cannot be any other spanning tree. In other words, every path from j to 0 passes through 1.

9.3 The Sliding Principle and its Application

We are ready to prove our main result which is stated again.

Theorem 9.2 *Let the graphs X, $G = P(X)$, X_1 and $H = P(X)_1$ be as shown in Example 1. Then, $\boldsymbol{B}[G-0]$ is entrywise dominated by $\boldsymbol{B}[H-0]$.*

Proof Note that we need to show that $\mathbf{B}[G-0](i,j) \leq \mathbf{B}[H-0](i,j)$, for all $i, j \geq 1$. To do so, we divide the proof into five cases depending on the location of the vertices i, j.

Case 1. Let $i, j \leq l$. Then, by Corollary 9.5, we have $\mathbf{B}[G-0](i,j) = \mathbf{B}[H-0](i,j)$.

Case 2. Let $i \leq l$ and $j > l$. Then, by Corollary 9.5,

$$\mathbf{B}[G-0](i,j) = \mathbf{B}[G-0](i,l) = \mathbf{B}[H-0](i,l) = \mathbf{B}[H-0](i,j).$$

Case 3. Let $i = n$. We divide this case into three subcases.

Subcase 1. Let $j = n$. Then, using Corollary 9.5, we have

$$\begin{aligned}\mathbf{B}[G-0](n,n) &= l + \mathbf{B}[X-l](n-1,n-1) + 1\\ &= l + 1 + \mathbf{B}[X_1 - (l+1)](n,n) = \mathbf{B}[H-0](n,n).\end{aligned}$$

Subcase 2. Let $j > n$. Then, using Corollary 9.5, we have

$$\begin{aligned}\mathbf{B}[G-0](n,j) &= l + \mathbf{B}[X-l](n-1,j)\\ &< l + \mathbf{B}[X-l](n-1,j) + 1\\ &= l + 1 + \mathbf{B}[X_1 - (l+1)](n,j)\\ &= \mathbf{B}[H-0](n,j).\end{aligned}$$

Subcase 3. Let $l < j < n$. Then, note that $j \sim j-1$ and $j \geq 2$. Thus, using Corollary 9.5 and Proposition 9.1, we have

$$\begin{aligned}\mathbf{B}[G-0](n,j) &= l + \mathbf{B}[X-l](n-1,j)\\ &\leq l + \mathbf{B}[X-l](n-1,j-1) + 1\\ &= l+1+\mathbf{B}[X_1-(l+1)](n,j) = \mathbf{B}[H-0](n,j).\end{aligned}$$

Case 4. Let $l < i < n$, $j > l$ and $j \neq n$. Then, note that $\mathbf{B}[G-0](i,j) = l + \mathbf{B}[X-l](i,j)$.

Subcase 1. Let $i = l+1$. Then, by Corollary 9.7, $\mathbf{B}[X-l](l+1,j) \leq 1$ and by Corollary 9.5

$$\mathbf{B}[H-0](l+1,j) = l+1 \geq l + \mathbf{B}[X-l](l+1,j) = \mathbf{B}[G-0](l+1,j).$$

Subcase 2. Let $l+1 < i < n$, $j > l+1$ and $j \neq n$. First, let $j > n$. By Proposition 9.1, $\mathbf{B}[X-l](i,j) - \mathbf{B}[X-l](i-1,j) \leq 1$. Thus

$$\begin{aligned}\mathbf{B}[H-0](i,j) &= l+1+\mathbf{B}[X_1-(l+1)](i,j)\\ &= l+1+\mathbf{B}[X-l](i-1,j)\\ &\geq l + \mathbf{B}[X-l](i,j) = \mathbf{B}[G-0](i,j).\end{aligned}$$

Now, for $l+1 < i \leq j < n$, we have $\mathbf{B}[X-l](i,j) - \mathbf{B}[X-l](i-1,j-1) \leq 1$ (using Proposition 9.2). Thus,

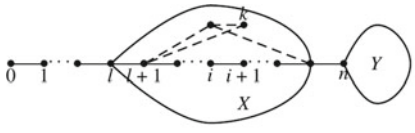
$$\begin{aligned}\mathbf{B}[H-0](i,j) &= l+1+\mathbf{B}[X_1-(l+1)](i,j)\\ &= l+1+\mathbf{B}[X-l](i-1,j-1)\\ &\geq l + \mathbf{B}[X-l](i,j) = \mathbf{B}[G-0](i,j).\end{aligned}$$

Case 5. Let $n < i \leq j$, that is, both i, j are outside P. Then, using Corollary 9.5, we see that $\mathbf{B}[H-0](i,j) = l+1+\mathbf{B}[X_1-(l+1)](i,j) = l+1+\mathbf{B}[X-l](i,j) = \mathbf{B}[G-0](i,j)+1$.

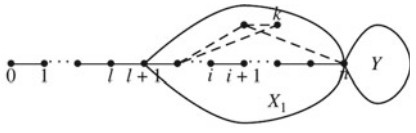
This completes the proof of our main result.

As direct applications of Theorem 9.1, we state a few important results without proof.

Corollary 9.8 *Let G and H be graphs as shown in the following picture. Then, $\boldsymbol{B}[G-0]$ is entrywise dominated by $\boldsymbol{B}[H-0]$.*

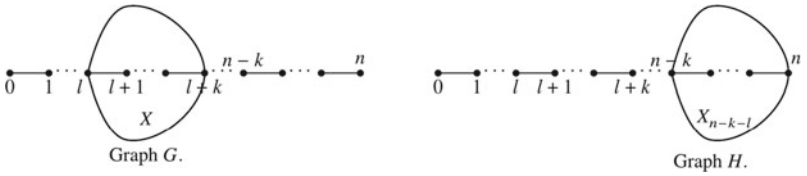


Graph G: Y is a connected subgraph.



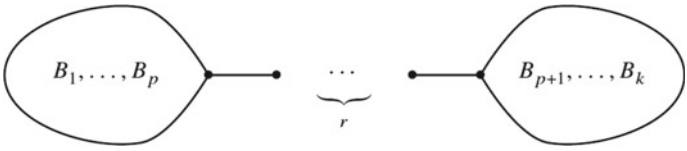
Graph H: Y is a connected subgraph.

Corollary 9.9 *Let G and H be graphs as shown in the following picture, where X_{n-k-l} is an isomorphic copy of X obtained by sliding X exactly $n-k-l$ times. Then, $\boldsymbol{B}[G-0]$ is entrywise dominated by $\boldsymbol{B}[H-0]$.*



As mentioned in Sect. refsec:1, it was proved in Bapat et al. [2] that if G is a connected graph and $\mathbf{y}$ is a Fiedler vector with $|\mathcal{C}(G,\mathbf{y})| = 1$ (the size of the characteristic set) then, either $\mathcal{C}(G,\mathbf{y}) = \{u\}$, for a cut vertex u or $\mathcal{C}(G,\mathbf{y}) = \{[u,v]\}$, for a bridge $[u,v]$. Then, a repeated application of Theorem 9.1 (or that of the above two corollaries) gives us the following result.

Theorem 9.3 *Let G be a connected graph and let $\mathbf{y}$ be a Fiedler vector with $|\mathcal{C}(G,\mathbf{y})| = 1$. Also, assume that G has r trivial blocks (K_2) and k nontrivial blocks $B_1, \ldots B_k$. Then, there is a graph H which has an induced path of length r and whose nontrivial blocks are $B_1, \ldots, B_k$ such that $a(H) \leq a(G)$.*



By definition, a tree has no nontrivial block and hence as a final corollary to Theorem 9.1, we obtain the following well-known result which was first stated and proved by Fiedler.

Corollary 9.10 ([7]) *Let T_n be a tree on n vertices. Then, $a(T_n) \geq a(P_n)$.*

References

1. Bapat R, Pati S (1998) Algebraic connectivity and the characteristic set of a graph. Linear Multilinear Algebra 45(2–3):247–273
2. Bapat R, Kirkland S, Pati S (2001) The perturbed laplacian matrix of a graph. Linear Multilinear Algebra 49(3):219–242
3. Bapat R, Lal A, Pati S (2012) On algebraic connectivity of graphs with at most two points of articulation in each block. Linear Multilinear Algebra 60(4):415–432
4. Chaiken S (1982) A combinatorial proof of the all minors matrix tree theorem. SIAM J Algebra Discret Methods 3(3):319–329
5. Fallat S, Kirkland S (1998) Extremizing algebraic connectivity subject to graph theoretic constraints. Electron J Linear Algebra 3:48–74
6. Fallat SM, Kirkland S, Pati S (2002) Minimizing algebraic connectivity over connected graphs with fixed girth. Discret Math 254(1–3):115–142

7. Fiedler M (1973) Algebraic connectivity of graphs. Czechoslovak Math J 23(2):298–305
8. Guo JM (2008) A conjecture on the algebraic connectivity of connected graphs with fixed girth. Discret Math 308(23):5702–5711
9. Kirkland S, Fallat S (1998) Perron components and algebraic connectivity for weighted graphs. Linear Multilinear Algebra 44(2):131–148
10. Kirkland S, Neumann M (1997) Algebraic connectivity of weighted trees under perturbation. Linear Multilinear Algebra 42(3):187–203
11. Kirkland S, Neumann M, Shader BL (1996) Characteristic vertices of weighted trees via perron values. Linear Multilinear Algebra 40(4):311–325
12. Merris R (1987) Characteristic vertices of trees. Linear Multilinear Algebra 22(2):115–131
13. Merris R (1994) Laplacian matrices of graphs: a survey. Linear Algebra Appl 197:143–176
14. Patra K, Lal A (2008) The effect on the algebraic connectivity of a tree by grafting or collapsing of edges. Linear Algebra Appl 428(4):855–864
15. Shao JY, Guo JM, Shan HY (2008) The ordering of trees and connected graphs by algebraic connectivity. Linear Algebra Appl 428(7):1421–1438

Chapter 10
Orthogonality for Biadjoints of Operators



T. S. S. R. K. Rao

10.1 Introduction

Let X, Y be, real, non-reflexive Banach spaces. An important problem in operator theory is to consider standard Banach space theoretic operations on the space of operators, $\mathcal{L}(X, Y)$ or its subspaces, like the space of compact operators, weakly compact operators, etc., and see if the resulting structure is again (or in) a space of operators, perhaps between different Banach spaces. For example, one important property that has attracted a lot of attention recently (and also has some bearing on the current investigation), is the quotient space problem. For Banach spaces X, Y, Z with $Z \subset Y$, in the space of operators $\mathcal{L}(X, Y)$, when one performs the 'quotient operation' with the subspace $\mathcal{L}(X, Z)$, a natural question is to ask, when is the quotient space $\mathcal{L}(X, Y)/\mathcal{L}(X, Z)$, surjective isometric to a space of operators? Here a natural target space is $\mathcal{L}(X, Y/Z)$. If $\pi : Y \to Y/Z$ denotes the quotient map, then a possible isometry between the spaces involved is, $[T] \to \pi \circ T$ where the equivalence class, $[T] \in \mathcal{L}(X, Y)/\mathcal{L}(X, Z)$. See Monika et al. [7], Botelho et al. [3] and Rao [10].

In this paper we investigate similar aspects for the Banach space theoretic property of Birkhoff-James orthogonality and the operator theoretic property of taking biadjoints of an operator. Let $Z \subset Y$ be a closed subspace, let $y \in Y$. We recall that $y \perp Z$ in the sense of Birkhoff-James, if $\|y + z\| \geq \|y\|$ all $z \in Z$. Deviating from standard notation, we denote the set of all such y by $Z^{\dagger}$. For a closed subspace $M \subset \mathcal{L}(X, Y)$, with $T \perp M$, we are interested in the question, when is T^{**} orthogonal to the smallest weak*-closed subspace of $\mathcal{L}(X^{**}, Y^{**})$, containing $\{S^{**} : S \in M\}$.

T. S. S. R. K. Rao (✉)
Department of Mathematics, Shiv Nadar Institution of Eminence, Goutam Buddha Nagar 201314, UP, India
e-mail: srin@fulbrightmail.org

Manipal Academy of Higher Education, Manipal, India

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_10

Let $Z \subset Y$ be a closed, non-reflexive subspace. We consider the case $M = \mathcal{L}(X, Z)$. We use the canonical embedding of a Banach space X in its bidual X^{**} and also recall the identification of the bidual of the quotient space $(Y/Z)^{**}$ as $Y^{**}/Z^{\perp\perp}$ and Z^{**} as $Z^{\perp\perp} \subset Y^{**}$. Using these embedding, we have, $d(y, Z) = d(y, Z^{\perp\perp})$ for all $y \in Y$. For $T \in \mathcal{L}(X, Y)$, the biadjoint, $T^{**} \in \mathcal{L}(X^{**}, Y^{**})$ and when T is Z-valued, T^{**} takes values in $Z^{\perp\perp}$ (we also note for a future use, when T is compact, T^{**} takes values in Z).

So in this case our problem can be formulated as, if $T \perp M$, when is $T^{**} \perp \mathcal{L}(X^{**}, Z^{\perp\perp})$? We note that $\mathcal{L}(X^{**}, Z^{\perp\perp})$ is closed in the weak*-operator topology of $\mathcal{L}(X^{**}, Y^{**})$, and we show that this is the smallest closed subspace containing M. So that by solving this orthogonality problem, we have a solution to the biadjoint question proposed at the beginning, somewhat analogous to the von Neumann double commutant theorem for C^*-algebras (interpreted as concrete identification of the closure of a space, in the weak-operator topology) . We refer to Diestel and Uhl [4] for standard results on spaces of operators and tensor products.

This paper is an expanded version of my talk given at ICLAA2021, organized by CARAMS, MAHE, Manipal during December 15–17, 2021. I thank the Editors of this special volume for their efficient handling of the MS during the pandemic. Thanks are also due to the referee for the extensive comments.

10.2 Main Results

For a Banach space X, we denote the closed unit ball by X_1 . Following the lead of the well-known Bhatia-Semrl theorem Bhatia and Semrl [2] on point-wise orthogonality of two orthogonal operators (established in the case of finite dimensional Hilbert spaces), one can ask, if $T \perp \mathcal{L}(X, Z)$, does there exist an extreme point $\Lambda \in X_1^{**}$ such that $\|T\| = \|T^{**}\| = \|T^{**}(\Lambda)\|$ and $T^{**}(\Lambda) \perp Z^{\perp\perp}$?

Most generalizations of the Bhatia-Semrl theorem for Hilbert spaces or for operators between Banach spaces (see the survey article Paul and Sain [8]) require T to attain its norm, which is a strong requirement even when X is reflexive. On the other hand, it is well-known that if T is a compact operator, T^{**} attains its norm at an extreme point and also a result of Zizler [12], asserts that $\{T \in \mathcal{L}(X, Y) : T^{**} \textit{ attains its norm}\}$ is norm dense in $\mathcal{L}(X, Y)$. Thus anticipating T^{**} to attain its norm is a more natural condition for non-reflexive Banach spaces. Now we propose a general interpretation of the Bhatia-Semrl theorem as ' when does global behavior on an operator, leads to a similar local behavior at points where the biadjoint attains its norm?' Our generalization here involves orthogonality of an operator to a space of operators. See Rao [11] for another interpretation.

We will first investigate the sufficient condition, leading to very interesting min-imax formulae, see condition (iii) in Proposition 10.1 below. It is well-known and easy to see that $y \perp Z$ if and only if $d(y, Z) = \|y\|$.

Proposition 10.1 *Suppose* $\|T\| = \|T^{**}\| = \|T^{**}(\Lambda)\|$ *for some* $\Lambda \in X_1^{**}$ *and* $T^{**}(\Lambda) \perp Z^{\perp\perp}$. *The following hold.*

(i) $T^{**} \perp \mathcal{L}(X^{**}, Z^{\perp\perp})$ *as well as* $T \perp \mathcal{L}(X, Z)$.

(ii) Now using the distance interpretation of orthogonality for subspaces, in this case we have,

$$\|T\| = \|T^{**}\| = d(T, \mathcal{L}(X, Z)) = d(T^{**}, \mathcal{L}(X^{**}, Z^{\perp\perp})).$$

(iii) $d(T^{**}, \mathcal{L}(X^{**}, Z^{\perp\perp})) = sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} = d(T^{**}(\Lambda), Z^{\perp\perp})$.

Proof To see (i) and (ii), for any $S \in \mathcal{L}(X^{**}, Z^{\perp\perp})$, $\|T^{**} - S\| \geq \|T^{**}(\Lambda) - S(\Lambda)\| \geq \|T^{**}(\Lambda)\| = \|T\|$, since $S(\Lambda) \in Z^{\perp\perp}$. Thus $T^{**} \perp \mathcal{L}(X^{**}, Z^{\perp\perp})$ as well as $T \perp \mathcal{L}(X, Z)$. Now using the distance interpretation of orthogonality for subspaces, in this case we have,

$$\|T\| = \|T^{**}\| = d(T, \mathcal{L}(X, Z)) = d(T^{**}, \mathcal{L}(X^{**}, Z^{\perp\perp})).$$

To see (iii), by the distance interpretation of orthogonality again,

$$d(T^{**}(\Lambda), Z^{\perp\perp}) = \|T^{**}(\Lambda)\| \leq sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\}. \tag{10.1}$$

Fix $S \in \mathcal{L}(X^{**}, Z^{\perp\perp})$ and $\tau \in X_1^{**}$.

$$d(T^{**}(\tau), Z^{\perp\perp}) \leq \|T^{**}(\tau) - S(\tau)\| \leq \|T^{**} - S\|.$$

Therefore $sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} \leq d(T^{**}, Z^{\perp\perp})$. Thus by (10.1) and (ii), we get (iii). □

Remark 10.1 We note that the conclusions of Proposition 10.1 are also valid if we replace bounded operators by compact operators. In Proposition 10.1, suppose T is also a compact operator. Since $T^{**}(\Lambda) \perp Z$, we also have,

$$\|T^{**}(\Lambda)\| = d(T^{**}(\Lambda), Z).$$

Remark 10.2 It is of independent interest to note that, for $T \in \mathcal{L}(X, Y)$, if

$$d(T, \mathcal{L}(X, Z)) = sup_{x \in X_1}\{d(T(x), Z)\}.$$

then again (iii) holds. To see this note that $d(T(x), Z) = d(T^{**}(x), Z^{\perp\perp})$. Now

$$d(T^{**}, \mathcal{L}(X^{**}, Z^{\perp\perp})) \leq d(T, \mathcal{L}(X, Z))$$

and it follows from the proof of (iii),

$$sup_{x \in X_1}\{d(T(x), Z)\} \leq sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} \leq d(T^{**}, Z^{\perp\perp}).$$

Thus (iii) holds.

Our aim in this paper is to partially solve the converse problem, for all domains X and imposing conditions only on the pair (Y, Z).

Since Proposition 10.1 is a solution to an optimization problem (see Rao [9]) on the weak*-compact convex set X_1^{**}, we next show that if there is a solution, then there is an extremal solution. We recall that for a compact convex set K, $E \subset K$ is an extreme set, if for $x, y \in K, \lambda \in (0, 1), \lambda x + (1-\lambda)y \in E$ implies $x, y \in E$. An extreme convex set is called a face. For $x \in E$, if $face(x)$ denotes the smallest face of K, containing x, then clearly $E = \cup\{face(x) : x \in E\}$. See Alfsen [1] page 122. We apply these ideas to $K = X_1^{**}$, equipped with the weak*-topology, along with a Krein-Milman type theorem to get the extremal solution.

Proposition 10.2 *Let $\Lambda \in X_1^{**}$ be such that*

$$\|T^{**}\| = \|T^{**}(\Lambda)\| \text{ and } sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} = d(T^{**}(\Lambda), Z^{\perp\perp}).$$

Then there is an extreme point of X_1^{**}, satisfying the same conditions.

Proof Let $E = \{\tau_0 \in X_1^{**} : \|T^{**}(\tau_0)\| = \|T^{**}\| \text{ and } sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} = d(T^{**}(\tau_o), Z^{\perp\perp})\}$. It is easy to see that being an intersection of two extreme sets, E is an extreme subset of X_1^{**}. Now we use a result of T. Johannesen (Theorem 5.8 in Lima [6]), which says that if the complement of E, E^c in X_1^{**} is a union of weak*-compact convex sets, then E has an extreme point of X_1^{**} . Note that

$$E^c = \cup_n\{\tau' \in X_1^{**} : \|T^{**}(\tau')\| \leq \|T^{**}\| - \frac{1}{n}\}$$

$$\cup_n\{\tau' \in X_1^{**} : d(T^{**}(\tau'), Z^{\perp\perp}) \leq sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} - \frac{1}{n}\}.$$

We next note that if $\pi : Y \to Y/Z$ is the quotient map, then $\pi^{**} : Y^{**} \to Y^{**}/Z^{\perp\perp}$ is the canonical quotient map and $d(T^{**}(\tau), Z^{\perp\perp}) = \|\pi^{**}(\tau)\|$. Thus by the convexity and weak*-lower-semi-continuity of $\|.\|$ and the weak*-continuity of the adjoint map, we see that all the sets in the union are weak*-compact and convex. Therefore E has an extreme point of X_1^{**}. □

We recall that $\mathcal{L}(X, Y^{**})$ is the dual of the projective tensor product space $X \otimes_\pi Y^*$. See Diestel and Uhl [4] Chapter VIII. Our next Lemma is folklore and is included here for the sake of completeness. We use the canonical identification of $Y^*/Z^\perp = Z^*$ and $\pi : Y^* \to Y^*/Z^\perp$ denotes the quotient map. For $\Lambda \in X^{**}$ and $y^* \in Y^*$, $\Lambda \otimes \pi(y^*)$ denotes the (weak*-continuous functional) on $\mathcal{L}(X^{**}, Z^{\perp\perp})$, defined by $(\Lambda \otimes \pi(y^*))(S) = S(\Lambda)(y^*)$ for $S \in \mathcal{L}(X^{**}, Z^{\perp\perp})$. Thus $(\Lambda \otimes \pi(y^*))(T^{**}) = \Lambda(T^*(\pi(y^*)))$, for any $T \in \mathcal{L}(X, Z)$.

Lemma 10.1 *Let $Z \subset Y$ be a closed subspace and let $M = \{T^{**} : T \in \mathcal{L}(X, Z)\}$. Then M is weak*-dense in $\mathcal{L}(X^{**}, Z^{\perp\perp}) = (X^{**} \otimes_\pi Y^*/Z^\perp)^*$.*

Proof Suppose for $\Lambda \in X^{**}$ and $y^* \in Y^*$, $(\Lambda \otimes \pi(y^*))(T^{**})=0$ for all $T \in \mathcal{L}(X, Z)$. For any $z \in Z$ choose y^* such that $\pi(y^*)(z) = 1$. For $x^* \in X^*$ and for the operator $T^* = z \otimes x^*$, we have $\Lambda(T^*(\pi(y^*))) = 0 = \Lambda(x^*)$. Therefore $\Lambda = 0$. Hence by an application of the separation theorem for the weak*-topology, M is weak*-dense in $\mathcal{L}(X^{**}, Z^{\perp\perp})$. □

We recall that if $\pi : Y \to Y/Z$ is the quotient map, then π^* is the inclusion map of $Z^\perp$ in Y^*. In what follows, we use a contractive projection to achieve the required compact 'lifting'. See Rao [10] for other alternate approaches.

Theorem 10.1 *Let X be a Banach space and let $Z \subset Y$ be a closed finite codimensional subspace such that there is a contractive linear projection $P : Y \to Y$ such that $ker(P) = Z$. Let $T \in \mathcal{L}(X, Y)$ and $T \perp \mathcal{L}(X, Z)$. Then there is an extreme point $\Lambda \in X_1^{**}$ such that $\|T\| = \|T^{**}\| = \|T^{**}(\Lambda)\|$ and $T^{**}(\Lambda) \perp Z^{\perp\perp}$.*

Proof Since Z is of finite codimension, $\pi \circ T : X \to Y/Z$ is a compact operator. It is well-known (see Harmand et al. [5] page 226) that there are extreme points $\Lambda \in X_1^{**}$, $y^* \in Z_1^\perp$ such that $\|\pi \circ T\| = \Lambda(T^*(y^*)) = T^{**}(\Lambda)(y^*) \leq d(T^{**}(\Lambda), Z^{\perp\perp}) \leq sup_{\tau \in X_1^{**}}\{d(T^{**}(\tau), Z^{\perp\perp})\} \leq d(T^{**}, \mathcal{L}(X^{**}, Z^{\perp\perp})) \leq d(T, \mathcal{L}(X, Z)) = \|T\|$.

Now let $P' : Y/Z \to Y$ defined by $P'(\pi(y)) = P(y)$ be the canonical embedding. Let $S = P' \circ \pi \circ T$. Then $\|S\| \leq \|\pi \circ T\|$. Also note that $\pi \circ S = \pi \circ T$. To see this note that for any $x \in X$, $S(x) - T(x) = P'(\pi(T(x)) - T(x) = P(T(x)) - T(x) \in Z$. Thus $\|S\| = \|\pi \circ T\|$ and $d(T, \mathcal{L}(X, Z)) = d(S, \mathcal{L}(X, Z)) \leq \|S\| = \|\pi \circ T\|$. Therefore equality holds in the inequalities and we have the required conclusion. □

Remark 10.3 We have used the finite codimensionality to get $\pi \circ T$ compact. The theorem also holds by assuming $\pi \circ T$ is compact. Also note that since P is a contractive projection, $\|(x - P(x)) - P(y)\| \geq \|P(y)\|$ for all $x, y \in X$, so in Theorem 10.1, the range of P is contained in $Z^\dagger$. As noted in Remark 10.1, if one considers these questions only in the class of compact operators, the minimax formulae from Proposition 10.1, are still valid. It was shown recently in Rao [11] that these can be used to compute the directional derivative of the operator norm at these operators.

References

1. Alfsen EM (1971) Compact convex sets and boundary integrals. Springer, New York
2. Bhatia R, Semrl P (1999) Orthogonality of matrices and the distance problem. Linear Algebra Appl 287:77–85
3. Botelho F, Fleming RJ, Rao TSSRK (2022) Proximinality of subspaces and the quotient lifting property. Linear Multilinear Algebra. https://doi.org/10.1080/03081087.2023.2173131
4. Diestel J, Uhl JJ (1977) Vector measures, vol 15. AMS: Mathematical Surveys

5. Harmand P, Werner D, Werner W (1993) M-ideals in Banach spaces and Banach algebras. Springer LNM 1547, Berlin
6. Lima A (1978) Intersection properties of balls in spaces of compact operators. Ann Inst Fourier 28:35–65
7. Monika, Botelho F, Fleming RJ (2021) The existence of linear selection and the quotient lifting property. Indian J Pure Appl Math. https://doi.org/10.1007/s13226-021-00171-z
8. Paul K, Sain D (2019) Birkhoff-james orthogonality and its applications in the study of geometry of Banach spaces. In: Ruzhansky M, Dutta H (eds) Advanced topics in mathematical analysis. CRC Press, Boca Raton, pp 245–284
9. Rao TSSRK (2021) Operators birkhoff-james orthogonal to spaces of operators. Numer Funct Anal Optim 42:1201–1208
10. Rao TSSRK (2022a) Order preserving quotient lifting properties. Positivity 26:37. https://doi.org/10.1007/s11117-022-00907-z
11. Rao TSSRK (2022b) Subdifferential set of an operator. Monatsh Math 199:891–898. https://doi.org/10.1007/s00605-022-01739-5
12. Zizler V (1973) On some extremal problems in Banach spaces. Math Scand 32:214–224

Chapter 11
Permissible Covariance Structures for Simultaneous Retention of BLUEs in Small and Big Linear Models



Stephen J. Haslett, Jarkko Isotalo, Augustyn Markiewicz, and Simo Puntanen

Mathematics Subject Classification (2010) 62J05 · 62J10

11.1 Introduction

In this paper, we consider the partitioned linear model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$, or shortly denoted as

$$\mathscr{M}_{12} = \{\mathbf{y},\ \mathbf{X}\boldsymbol{\beta},\ \mathbf{V}\} = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2,\ \mathbf{V}\}\,. \tag{11.1}$$

S. J. Haslett
Research School of Finance, Actuarial Studies and Statistics, The Australian National University, Canberra, Australia
e-mail: s.j.haslett@massey.ac.nz; stephen.haslett@anu.edu.au

National Institute for Applied Statistics Research Australia (NIASRA), University of Wollongong, NSW 2522 Keiraville, Australia

Centre for Public Health Research, Massey University, Wellington, New Zealand

School of Fundamental Sciences, Massey University, Palmerston North, New Zealand

J. Isotalo · S. Puntanen (✉)
Faculty of Information Technology and Communication Sciences, FI-33014 Tampere University, Tampere, Finland
e-mail: simo.puntanen@tuni.fi

J. Isotalo
e-mail: jarkko.isotalo@tuni.fi

A. Markiewicz
Department of Mathematical and Statistical Methods, Poznań University of Life Sciences, Wojska Polskiego 28, PL-60637 Poznań, Poland
e-mail: amark@up.poznan.pl

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_11

In (11.1), $\mathbf{y}$ is an n-dimensional observable variable, and $\boldsymbol{\varepsilon}$ is an unobservable random error with a known covariance matrix $\mathrm{cov}(\boldsymbol{\varepsilon}) = \mathbf{V} = \mathrm{cov}(\mathbf{y})$ and expectation $\mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$. The matrix $\mathbf{X}$ is a known $n \times p$ matrix, i.e., $\mathbf{X} \in \mathbb{R}^{n\times p}$, partitioned columnwise as $\mathbf{X} = (\mathbf{X}_1 : \mathbf{X}_2)$. Vector $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2')' \in \mathbb{R}^p$ is a vector of fixed (but unknown) parameters; here symbol $'$ stands for the transpose. We will also denote $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta},\ \boldsymbol{\mu}_i = \mathbf{X}_i\boldsymbol{\beta}_i,\ i = 1, 2$.

As for notations, the symbols $\mathrm{r}(\mathbf{A})$, $\mathbf{A}^-$, $\mathbf{A}^+$, $\mathscr{C}(\mathbf{A})$, and $\mathscr{C}(\mathbf{A})^\perp$ denote, respectively, the rank, a generalized inverse, the Moore–Penrose inverse, the column space, and the orthogonal complement of the column space of the matrix $\mathbf{A}$. By $\mathbf{A}^\perp$, we denote any matrix satisfying $\mathscr{C}(\mathbf{A}^\perp) = \mathscr{C}(\mathbf{A})^\perp$. Furthermore, we will write $\mathbf{P}_\mathbf{A} = \mathbf{P}_{\mathscr{C}(\mathbf{A})} = \mathbf{A}\mathbf{A}^+ = \mathbf{A}(\mathbf{A}'\mathbf{A})^-\mathbf{A}'$ to denote the orthogonal projector onto $\mathscr{C}(\mathbf{A})$ and $\mathbf{Q}_\mathbf{A} = \mathbf{I} - \mathbf{P}_\mathbf{A}$ where $\mathbf{I}$ is an identity matrix of conformable order. In particular, we denote

$$\mathbf{M} = \mathbf{I}_n - \mathbf{P}_\mathbf{X}\,, \quad \mathbf{M}_i = \mathbf{I}_n - \mathbf{P}_i\,, \quad \mathbf{P}_i = \mathbf{P}_{\mathbf{X}_i}\,,\ i = 1, 2.$$

In addition to the *full* or *big* model $\mathscr{M}_{12}$, we will consider the *small* model $\mathscr{M}_1 = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1,\ \mathbf{V}\}$ and the *reparametrized* model

$$\mathscr{M}_{12t} = \{\mathbf{y},\ \mathbf{X}_t\boldsymbol{\beta},\ \mathbf{V}\} = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{M}_1\mathbf{X}_2\boldsymbol{\beta}_2,\ \mathbf{V}\}\,,\quad \mathbf{X}_t = (\mathbf{X}_1 : \mathbf{M}_1\mathbf{X}_2)\,.$$

Under the model $\mathscr{M}_{12}$, the statistic $\mathbf{G}\mathbf{y}$, where $\mathbf{G}$ is an $n \times n$ matrix, is the best linear unbiased estimator, BLUE, of $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ if $\mathbf{G}\mathbf{y}$ is unbiased, i.e., $\mathbf{G}\mathbf{X} = \mathbf{X}$, and it has the smallest covariance matrix in the Löwner sense among all unbiased linear estimators of $\mathbf{X}\boldsymbol{\beta}$. For the proof of the following fundamental lemma, see, e.g., Rao [15, p. 282].

Lemma 11.1 *Consider the general linear model* $\mathscr{M}_{12} = \{\boldsymbol{y}, \boldsymbol{X\beta}, \boldsymbol{V}\}$. *Then the statistic* $\boldsymbol{Gy}$ *is the* BLUE *for* $\boldsymbol{\mu} = \boldsymbol{X\beta}$ *if and only if* $\boldsymbol{G}$ *satisfies the equation*

$$\boldsymbol{G}(\boldsymbol{X} : \boldsymbol{V}\boldsymbol{X}^\perp) = (\boldsymbol{X} : \boldsymbol{0})\,. \tag{11.2}$$

One well-known solution for $\mathbf{G}$ in (11.2) (which is always solvable) is

$$\mathbf{G}_{12} = \mathbf{X}(\mathbf{X}'\mathbf{W}^-\mathbf{X})^-\mathbf{X}'\mathbf{W}^+, \quad \text{where}\mathbf{W} = \mathbf{V} + \mathbf{X}\mathbf{X}'. \tag{11.3}$$

It is worth noting, see, e.g., Puntanen et al. [11, Proposition 12.1 and 15.2], that $\mathbf{G}_{12}$, which is invariant for the choices of the generalized inverses denoted by "$-$", can be expressed as

$$\begin{aligned}\mathbf{G}_{12} &= \mathbf{X}(\mathbf{X}'\mathbf{W}^-\mathbf{X})^-\mathbf{X}'\mathbf{W}^+ = \mathbf{P}_\mathbf{W} - \mathbf{V}\mathbf{M}(\mathbf{M}\mathbf{V}\mathbf{M})^-\mathbf{M}\mathbf{P}_\mathbf{W} \\ &= \mathbf{P}_\mathbf{W} - \mathbf{V}\mathbf{M}(\mathbf{M}\mathbf{V}\mathbf{M})^+ = \mathbf{P}_\mathbf{W} - \mathbf{V}\mathbf{M}(\mathbf{M}\mathbf{V}\mathbf{M})^+\mathbf{M}\,.\end{aligned} \tag{11.4}$$

The *general* solution for $\mathbf{G}$ can be expressed, for example, as

$$\mathbf{G}_0 = \mathbf{G}_{12} + \mathbf{N}\mathbf{Q}_{(\mathbf{X}:\mathbf{V})}\,, \quad \text{where } \mathbf{N} \in \mathbb{R}^{n\times n}\text{is free to vary,}$$

and $\mathbf{Q}_{(\mathbf{X}:\mathbf{V})} = \mathbf{I}_n - \mathbf{P}_{(\mathbf{X}:\mathbf{V})}$. However, the equality

$$\mathbf{G}_0\mathbf{y} = (\mathbf{G}_{12} + \mathbf{N}\mathbf{Q}_{(\mathbf{X}:\mathbf{V})})\mathbf{y} \quad \text{for all } \mathbf{y} \in \mathscr{C}(\mathbf{X} : \mathbf{V})$$

holds when the model is *consistent* in the sense that $\mathbf{y} \in \mathscr{C}(\mathbf{X} : \mathbf{V})$ with probability 1. Notice that

$$\mathscr{C}(\mathbf{X} : \mathbf{V}) = \mathscr{C}(\mathbf{X} : \mathbf{V}\mathbf{M}) = \mathscr{C}(\mathbf{X}) \oplus \mathscr{C}(\mathbf{V}\mathbf{M}) = \mathscr{C}(\mathbf{X}) \boxplus \mathscr{C}(\mathbf{M}\mathbf{V}) ,$$

where "$\oplus$" refers to the direct sum and "$\boxplus$" refers to the direct sum of orthogonal subspaces. The solution for $\mathbf{G}$ in (11.2) is unique if and only if $\mathscr{C}(\mathbf{X} : \mathbf{V}) = \mathbb{R}^n$.

The following lemma will be repeatedly used in our considerations.

Lemma 11.2 *Consider* $\boldsymbol{X} = (\boldsymbol{X}_1 : \boldsymbol{X}_2)$ *and let* $\boldsymbol{M}_1 = \boldsymbol{I}_n - \boldsymbol{P}_{X_1}$. *Then*

1. $\mathscr{C}(\boldsymbol{X}_1 : \boldsymbol{X}_2) = \mathscr{C}(\boldsymbol{X}_1 : \boldsymbol{M}_1\boldsymbol{X}_2)$, *i.e.,* $\mathscr{C}(\boldsymbol{X}) = \mathscr{C}(\boldsymbol{X}_1) \boxplus \mathscr{C}(\boldsymbol{M}_1\boldsymbol{X}_2)$,
2. $\boldsymbol{M} = \boldsymbol{I}_n - \boldsymbol{P}_X = \boldsymbol{I}_n - (\boldsymbol{P}_{X_1} + \boldsymbol{P}_{M_1X_2}) = \boldsymbol{M}_1\boldsymbol{Q}_{M_1X_2} = \boldsymbol{Q}_{M_1X_2}\boldsymbol{M}_1$,
3. $\boldsymbol{Q}_{(X:V)} = \boldsymbol{M} - \boldsymbol{P}_{MV} = \boldsymbol{M}(\boldsymbol{I} - \boldsymbol{P}_{MV}) = \boldsymbol{M}\boldsymbol{Q}_{MV} = \boldsymbol{Q}_{MV}\boldsymbol{M}$, *and*
4. $\mathrm{r}(\boldsymbol{M}_1\boldsymbol{X}_2) = \mathrm{r}(\boldsymbol{X}_2) - \dim \mathscr{C}(\boldsymbol{X}_1) \cap \mathscr{C}(\boldsymbol{X}_2)$.

Consider now the following linear models:

$$\mathscr{M}_1(\mathbf{V}_0) = \{\mathbf{y}, \mathbf{X}_1\boldsymbol{\beta}_1, \mathbf{V}_0\}, \quad \mathscr{M}_{12}(\mathbf{V}_0) = \{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{V}_0\},$$
$$\mathscr{M}_1(\mathbf{V}) = \{\mathbf{y}, \mathbf{X}_1\boldsymbol{\beta}_1, \mathbf{V}\}, \quad \mathscr{M}_{12}(\mathbf{V}) = \{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{V}\}.$$

For the property that every representation of the BLUE of $\boldsymbol{\mu}$ under $\mathscr{M}_{12}(\mathbf{V}_0)$ remains BLUE of $\boldsymbol{\mu}$ under $\mathscr{M}_{12}(\mathbf{V})$, we can use the short notation

$$\big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{V}_0))\big\} \subseteq \big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{V}))\big\}. \tag{11.5}$$

The notation $\big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{A})\big\}$ refers to the set of all representations of the BLUE of $\boldsymbol{\mu}$ under the model $\mathscr{A}$, which can be symbolically denoted as

$$\big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{V}))\big\} = \big\{\mathbf{G}\mathbf{y} : \mathbf{G}(\mathbf{X} : \mathbf{V}\mathbf{M}) = (\mathbf{X} : \mathbf{0})\big\}.$$

By Rao [14, Theorem 5.2], the statement (11.5) is equivalent to

$$\mathscr{C}(\mathbf{V}\mathbf{M}) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}). \tag{11.6}$$

By Rao's Theorem 5.3, (11.6) is equivalent to the fact that $\mathbf{V}$ can be expressed as

$$\mathbf{V} = \mathbf{X}\mathbf{A}\mathbf{A}'\mathbf{X}' + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0 \tag{11.7}$$

for some $\mathbf{A}$ and $\mathbf{B}$. According to Rao's Corollary 5.3.1, a further equivalent form is

$$\mathbf{V} = \mathbf{V}_0 + \mathbf{X}\mathbf{C}\mathbf{C}'\mathbf{X}' + \mathbf{V}_0\mathbf{M}\mathbf{D}\mathbf{D}'\mathbf{M}\mathbf{V}_0, \tag{11.8}$$

for some $\mathbf{C}$ and $\mathbf{D}$. As a matter of fact, Rao uses symmetric matrices $\mathbf{\Gamma}$ and $\mathbf{\Delta}$ instead of $\mathbf{AA}'$ and $\mathbf{BB}'$, etc. But it is slightly simpler to use nonnegative definite matrices $\mathbf{AA}'$ and $\mathbf{BB}'$, etc. For further references regarding (11.6)–(11.8), see also Rao [13, Lemma 5], Rao [15, p. 289], and Mitra and Moore [10, Theorem 4.1–4.2].

As noted by Mitra and Moore [10, p. 148] and Rao [15, p. 289], Rao in Theorem 5.3 of his 1971 paper had an unnecessary condition $\mathscr{C}(\mathbf{X} : \mathbf{V}_0\mathbf{M}) = \mathbb{R}^n$ which was not used in the proofs. Notice that $\mathscr{C}(\mathbf{X} : \mathbf{V}_0) = \mathscr{C}(\mathbf{X} : \mathbf{V}_0\mathbf{M})$ and thereby $\mathscr{C}(\mathbf{X} : \mathbf{V}_0\mathbf{M}) = \mathbb{R}^n$ means that the BLUE of $\boldsymbol{\mu}$ has a unique representation.

Let us define the set $\mathscr{V}_{12}$ of nonnegative definite matrices as follows:

$$\mathbf{V} \in \mathscr{V}_{12} \iff \mathbf{V} = \mathbf{V}_0 + \mathbf{XAA}'\mathbf{X}' + \mathbf{V}_0\mathbf{MBB}'\mathbf{MV}_0\,, \tag{11.9}$$

for some $\mathbf{A}$ and $\mathbf{B}$. The corresponding set for the small model is defined as

$$\mathbf{V} \in \mathscr{V}_1 \iff \mathbf{V} = \mathbf{V}_0 + \mathbf{X}_1\mathbf{CC}'\mathbf{X}_1' + \mathbf{V}_0\mathbf{M}_1\mathbf{DD}'\mathbf{M}_1\mathbf{V}_0\,,$$

for some $\mathbf{C}$ and $\mathbf{D}$. In view of $\mathscr{C}(\mathbf{X}_t) = \mathscr{C}(\mathbf{X}_1 : \mathbf{M}_1\mathbf{X}_2) = \mathscr{C}(\mathbf{X})$, it is clear that

$$\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}) = \mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12t})\,.$$

Thereby $\mathscr{V}_{12} = \mathscr{V}_{12t}$, and we can equally well replace (11.9) with

$$\mathbf{V} \in \mathscr{V}_{12} \iff \mathbf{V} = \mathbf{V}_0 + \mathbf{X}_t\mathbf{AA}'\mathbf{X}_t' + \mathbf{V}_0\mathbf{MBB}'\mathbf{MV}_0\,.$$

In Sects. 11.2 and 11.3, we are interested in the relations between the sets $\mathscr{V}_1$ and $\mathscr{V}_{12}$, i.e., we want to find necessary and sufficient conditions for $\mathscr{V}_1 \subseteq \mathscr{V}_{12}$ and for $\mathscr{V}_{12} \subseteq \mathscr{V}_1$.

11.2 The Case when $\mathbf{V}_0$ is the Identity Matrix

Putting $\mathbf{V}_0 = \mathbf{I}$, the identity matrix of order $n \times n$ means that we have the models $\mathscr{M}_{12}(\mathbf{I}) = \{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{I}\}$ and $\mathscr{M}_{12}(\mathbf{V}) = \{\mathbf{y}, \mathbf{X}\boldsymbol{\beta}, \mathbf{V}\}$ and

$$\big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{I}))\big\} = \big\{\mathbf{By} : \mathbf{B}(\mathbf{X} : \mathbf{M}) = (\mathbf{X} : \mathbf{0})\big\} = \big\{\mathbf{P}_{\mathbf{X}}\mathbf{y}\big\}\,,$$

i.e., the unique representation of the BLUE of $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ under $\mathscr{M}_{12}(\mathbf{I})$ is $\mathbf{P}_{\mathbf{X}}\mathbf{y}$, the ordinary least squares estimator, OLSE, of $\boldsymbol{\mu}$. Thus, the inclusion

$$\big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{I}))\big\} \subseteq \big\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{V}))\big\}$$

can be interpreted as the equality

$$\mathrm{OLSE}(\boldsymbol{\mu}) = \mathrm{BLUE}(\boldsymbol{\mu}) \ \text{ under} \mathscr{M}_{12}(\mathbf{V})\,.$$

Using $\mathbf{X}_t = (\mathbf{X}_1 : \mathbf{M}_1\mathbf{X}_2)$ in the characterization we consider the sets

$$\begin{aligned}\mathbf{V} \in \mathscr{V}_{12}(\mathbf{I}) &\iff \mathbf{V} = \mathbf{I} + \mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' + \mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\,,\\ \mathbf{V} \in \mathscr{V}_{1}(\mathbf{I}) &\iff \mathbf{V} = \mathbf{I} + \mathbf{X}_1\mathbf{C}\mathbf{C}'\mathbf{X}_1' + \mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\,.\end{aligned}$$

Notice that if $\mathbf{V}_0 = \mathbf{I}$, then we use notations $\mathscr{V}_1(\mathbf{I})$ and $\mathscr{V}_{12}(\mathbf{I})$. In the general case, we use notations $\mathscr{V}_1$ and $\mathscr{V}_{12}$. Take now an arbitrary

$$\mathbf{V} = \mathbf{I} + \mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' + \mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M} \in \mathscr{V}_{12}(\mathbf{I})\,.$$

Denoting

$$\mathbf{A} = \begin{pmatrix}\mathbf{A}_1\\ \mathbf{A}_2\end{pmatrix}, \quad \mathbf{A}\mathbf{A}' = \begin{pmatrix}\mathbf{A}_1\mathbf{A}_1' & \mathbf{A}_1\mathbf{A}_2'\\ \mathbf{A}_2\mathbf{A}_1' & \mathbf{A}_2\mathbf{A}_2'\end{pmatrix} = \begin{pmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{pmatrix},$$

we have, for conformable partitioning of $\mathbf{A}$,

$$\mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' = \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{21}\mathbf{X}_1'\,.$$

Now $\mathbf{V} = \mathbf{I} + \mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' + \mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M} \in \mathscr{V}_1(\mathbf{I})$ if and only if

$$\mathrm{OLSE}(\boldsymbol{\mu}) = \mathrm{BLUE}(\boldsymbol{\mu}) \ \text{ under} \mathscr{M}_1(\mathbf{V})\,,$$

which holds, see, e.g., Rao [12] and Zyskind [16], if and only if any of the following equivalent statements hold:

$$\mathbf{P}_1\mathbf{V}\mathbf{M}_1 = \mathbf{0}\,, \quad \mathscr{C}(\mathbf{V}\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{M}_1)\,, \quad \mathbf{P}_1\mathbf{V} = \mathbf{V}\mathbf{P}_1\,. \tag{11.10}$$

Applying (11.10) yields

$$\mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2'\mathbf{M}_1 = \mathbf{0}\,.$$

Thus, in view of $\mathbf{M} = \mathbf{M}_1\mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}$, the matrix $\mathbf{V}$ can be written as

$$\begin{aligned}\mathbf{V} &= \mathbf{I} + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\\ &= \mathbf{I} + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1(\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2' + \mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}\mathbf{B}\mathbf{B}'\mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2})\mathbf{M}_1\\ &=: \mathbf{I} + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1\mathbf{S}\mathbf{S}'\mathbf{M}_1\,.\end{aligned} \tag{11.11}$$

What about the reverse relation $\mathscr{V}_1(\mathbf{I}) \subseteq \mathscr{V}_{12}(\mathbf{I})$? Take an arbitrary

$$\mathbf{V} = \mathbf{I} + \mathbf{X}_1\mathbf{C}\mathbf{C}'\mathbf{X}_1' + \mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1 \in \mathscr{V}_1(\mathbf{I})\,. \tag{11.12}$$

Now $\mathbf{V} \in \mathscr{V}_{12}(\mathbf{I})$ if and only if $\mathbf{P}_{\mathbf{X}}\mathbf{V}\mathbf{M} = \mathbf{0}$, which holds if and only if

$$\mathbf{P}_{\mathbf{X}}\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{M} = \mathbf{0}\,, \quad \text{i.e.,} \quad \mathbf{P}_{\mathbf{M}_1\mathbf{X}_2}\mathbf{D}\mathbf{D}'\mathbf{M} = \mathbf{0}\,. \tag{11.13}$$

Requesting that $\mathbf{V}$ has the representation of the type (11.11) and, at the same time, $\mathbf{V}$ has the representation (11.12) with (11.13) holding and observing that $\mathbf{P_X M}_1\mathbf{SS}'\mathbf{M}_1\mathbf{M} = \mathbf{0}$, we can conclude that $\mathscr{V}_{12}(\mathbf{I}) = \mathscr{V}_1(\mathbf{I})$ if and only if $\mathbf{V}$ is of the form (11.11).

Theorem 11.1 *The following statements hold.*

1. $\mathscr{V}_{12}(\boldsymbol{I}) \subseteq \mathscr{V}_1(\boldsymbol{I}) \iff$ *the matrix* $\boldsymbol{V} \in \mathscr{V}_{12}$ *can be expressed as*

$$\boldsymbol{V} = \boldsymbol{I} + \boldsymbol{X}_1\boldsymbol{A}_{11}\boldsymbol{X}_1' + \boldsymbol{M}_1\boldsymbol{X}_2\boldsymbol{A}_{22}\boldsymbol{X}_2'\boldsymbol{M}_1 + \boldsymbol{MBB}'\boldsymbol{M} \tag{11.14}$$

   for some matrices $\boldsymbol{A}_1$, $\boldsymbol{A}_2$ *and* $\boldsymbol{B}$; $\boldsymbol{A}_{ij} = \boldsymbol{A}_i\boldsymbol{A}_j'$, $i, j = 1, 2$.
2. $\mathscr{V}_1(\boldsymbol{I}) \subseteq \mathscr{V}_{12}(\boldsymbol{I}) \iff$ *the matrix* $\boldsymbol{V} \in \mathscr{V}_1$ *can be expressed as*

$$\boldsymbol{V} = \boldsymbol{I} + \boldsymbol{X}_1\boldsymbol{CC}'\boldsymbol{X}_1' + \boldsymbol{M}_1\boldsymbol{DD}'\boldsymbol{M}_1$$

   for some matrices $\boldsymbol{C}$ *and* $\boldsymbol{D}$, *where*

$$\boldsymbol{P_X M}_1\boldsymbol{DD}'\boldsymbol{M}_1\boldsymbol{M} = \boldsymbol{0}\,, \quad \textit{i.e.,} \quad \boldsymbol{P}_{\boldsymbol{M}_1\boldsymbol{X}_2}\boldsymbol{DD}'\boldsymbol{M} = \boldsymbol{0}\,.$$

3. $\boldsymbol{V} \in \mathscr{V}_1 \cap \mathscr{V}_{12}$ *if and only if* $\boldsymbol{V}$ *can be expressed as in* (11.14)*, in other words,* $\mathscr{V}_{12}(\boldsymbol{I}) = \mathscr{V}_1(\boldsymbol{I}) \iff \mathscr{V}_{12}(\boldsymbol{I}) \subseteq \mathscr{V}_1(\boldsymbol{I})$.

Remark 11.1 Let us have a look at the situation when we use $\mathbf{X}$ instead of $\mathbf{X}_t$ in characterizing $\mathscr{V}_{12}$. In this case

$$\mathbf{XAA}'\mathbf{X}' = \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2' + \mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2' + \mathbf{X}_2\mathbf{A}_{21}\mathbf{X}_1'\,. \tag{11.15}$$

Now the inclusion $\mathscr{V}_{12}(\mathbf{I}) \subseteq \mathscr{V}_1(\mathbf{I})$ holds if and only if

$$\mathbf{V} = \mathbf{I} + \mathbf{XAA}'\mathbf{X}' + \mathbf{MBB}'\mathbf{M} \in \mathscr{V}_1(\mathbf{I})\,,$$

i.e., $\mathbf{P}_1\mathbf{VM}_1 = \mathbf{0}$, i.e.,

$$\mathbf{P}_1\mathbf{XAA}'\mathbf{X}'\mathbf{M}_1 = \mathbf{P}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2'\mathbf{M}_1 = \mathbf{0}\,. \tag{11.16}$$

If $\mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{X}_2) = \{\mathbf{0}\}$ then (11.16) simplifies to $\mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2' = -\mathbf{P}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'$, in which case

$$\begin{aligned}\mathbf{XAA}'\mathbf{X}' &= \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2' - \mathbf{P}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2' - \mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{P}_1 \\ &= \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2' - \mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{P}_1\,.\end{aligned}$$

Thus if $\mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{X}_2) = \{\mathbf{0}\}$, then $\mathscr{V}_{12}(\mathbf{I}) \subseteq \mathscr{V}_1(\mathbf{I})$ if and only if $\mathbf{V} \in \mathscr{V}_{12}(\mathbf{I})$ can be expressed as

$$\mathbf{V} = \mathbf{I} + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2' - \mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{P}_1 + \mathbf{MBB}'\mathbf{M}\,.$$

The use of (11.15) seems to yield rather complicated characterizations. Let's consider an alternative formulation. Namely, it is clear, see, e.g., Puntanen et al. [11, p. 219], that $\mathbf{V} \in \mathscr{V}_{12}(\mathbf{I})$ if and only if $\mathbf{V}_d$ can be expressed as

$$\mathbf{V}_d = \mathbf{I} + \mathbf{P}_{\mathbf{X}}\mathbf{A}\mathbf{A}'\mathbf{P}_{\mathbf{X}} + \mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}$$

for some $\mathbf{A}$ and $\mathbf{B}$. Now $\mathbf{V}_d \in \mathscr{V}_1(\mathbf{I})$ if and only if $\mathbf{P}_1\mathbf{V}\mathbf{M}_1 = \mathbf{0}$, which holds if and only if $\mathbf{P}_1\mathbf{A}\mathbf{A}'\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2} = \mathbf{0}$, yielding the expression

$$\mathbf{V}_d = \mathbf{I} + \mathbf{P}_1\mathbf{A}\mathbf{A}'\mathbf{P}_1 + \mathbf{P}_{\mathbf{M}_1\mathbf{X}_2}\mathbf{A}\mathbf{A}'\mathbf{P}_{\mathbf{M}_1\mathbf{X}_2} + \mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\,,$$

which is interestingly related to (11.14). □

11.3 The General Case of $\mathbf{V}_0$

Consider next the case of arbitrary nonnegative definite $\mathbf{V}_0$ and the two sets:

$$\mathbf{V} \in \mathscr{V}_{12} \iff \mathbf{V} = \mathbf{V}_0 + \mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0\,, \tag{11.17a}$$

$$\mathbf{V} \in \mathscr{V}_1 \iff \mathbf{V} = \mathbf{V}_0 + \mathbf{X}_1\mathbf{C}\mathbf{C}'\mathbf{X}_1' + \mathbf{V}_0\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{V}_0\,. \tag{11.17b}$$

Take an arbitrary $\mathbf{V}_e \in \mathscr{V}_{12}$ and ask when does the following hold:

$$\mathbf{V}_e = \mathbf{V}_0 + \mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0 \in \mathscr{V}_1\,? \tag{11.18}$$

In other words, if we have an arbitrary $\mathbf{V}_e$ of the form (11.17a), what is needed that $\mathbf{V}_e$ can be expressed in the form (11.17b) for some $\mathbf{C}$ and $\mathbf{D}$? Shortly denoted, we want to find a necessary and sufficient condition for the inclusion $\mathscr{V}_{12} \subseteq \mathscr{V}_1$. By Rao [14, Theorem 5.2], the condition (11.18) is equivalent to

$$\mathscr{C}(\mathbf{V}_e\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)\,. \tag{11.19}$$

In view of $\mathbf{M} = \mathbf{M}_1\mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}$, we have

$$\mathscr{C}(\mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)$$

and hence the inclusion (11.19) holds if and only if

$$\mathscr{C}(\mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t'\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)\,.$$

Notice that

$$\mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t'\mathbf{M}_1 = \mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1\,.$$

Consider a special situation when

$$\mathscr{C}(\mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1) = \mathscr{C}(\mathbf{M}_1\mathbf{X}_2\mathbf{A}_2) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)\,. \tag{11.20}$$

Requesting (11.20) to hold for any $\mathbf{A}_2$, it becomes

$$\mathscr{C}(\mathbf{M}_1\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)\,. \tag{11.21}$$

Then, in light of $\mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{V}_0\mathbf{M}_1) = \{\mathbf{0}\}$, and assuming that (11.21) holds, the inclusion

$$\mathscr{C}(\mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t'\mathbf{M}_1) = \mathscr{C}(\mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)$$

holds if and only if

$$\mathbf{X}_1\mathbf{A}_{12}\mathbf{X}_2'\mathbf{M}_1 = \mathbf{0}\,,$$

which further is equivalent to

$$\mathbf{X}_t\mathbf{A}\mathbf{A}'\mathbf{X}_t' = \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1' + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1\,.$$

Thus under (11.21) the inclusion $\mathscr{V}_{12} \subseteq \mathscr{V}_1$ holds if and only if $\mathbf{V} \in \mathscr{V}_{12}$ can be expressed as

$$\mathbf{V} = \mathbf{V}_0 + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1 + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0\,.$$

So we have proved the following.

Theorem 11.2 *Let us denote*

$$\boldsymbol{X}_t = (\boldsymbol{X}_1 : \boldsymbol{M}_1\boldsymbol{X}_2)\,, \quad \boldsymbol{A} = (\boldsymbol{A}_1' : \boldsymbol{A}_2')', \quad \boldsymbol{A}_{ij} = \boldsymbol{A}_i\boldsymbol{A}_j',\ i, j = 1, 2.$$

1. *Inclusion $\mathscr{V}_{12} \subseteq \mathscr{V}_1$ holds if and only if the matrix $\boldsymbol{V} \in \mathscr{V}_{12}$ can be expressed as*

$$\boldsymbol{V} = \boldsymbol{V}_0 + \boldsymbol{X}_t\boldsymbol{A}\boldsymbol{A}'\boldsymbol{X}_t + \boldsymbol{V}_0\boldsymbol{M}\boldsymbol{B}\boldsymbol{B}'\boldsymbol{M}\boldsymbol{V}_0\,,$$

   where

$$\mathscr{C}(\boldsymbol{X}_t\boldsymbol{A}\boldsymbol{A}'\boldsymbol{X}_t'\boldsymbol{M}_1) \subseteq \mathscr{C}(\boldsymbol{V}_0\boldsymbol{M}_1)\,,$$

   i.e.,

$$\mathscr{C}(\boldsymbol{X}_1\boldsymbol{A}_{12}\boldsymbol{X}_2'\boldsymbol{M}_1 + \boldsymbol{M}_1\boldsymbol{X}_2\boldsymbol{A}_{22}\boldsymbol{X}_2'\boldsymbol{M}_1) \subseteq \mathscr{C}(\boldsymbol{V}_0\boldsymbol{M}_1)\,,$$

   for some $\boldsymbol{A}$ and $\boldsymbol{B}$.
2. *Suppose that*

$$\mathscr{C}(\boldsymbol{M}_1\boldsymbol{X}_2) \subseteq \mathscr{C}(\boldsymbol{V}_0\boldsymbol{M}_1)\,.$$

Then $\mathscr{V}_{12} \subseteq \mathscr{V}_1$ holds if and only if the matrix $\boldsymbol{V} \in \mathscr{V}_{12}$ can be expressed as

$$\boldsymbol{V} = \boldsymbol{V}_0 + \boldsymbol{X}_1\boldsymbol{A}_{11}\boldsymbol{X}_1 + \boldsymbol{M}_1\boldsymbol{X}_2\boldsymbol{A}_{22}\boldsymbol{X}_2'\boldsymbol{M}_1 + \boldsymbol{V}_0\boldsymbol{M}\boldsymbol{B}\boldsymbol{B}'\boldsymbol{M}\boldsymbol{V}_0\,.$$

What about the reverse relation $\mathscr{V}_1 \subseteq \mathscr{V}_{12}$? Take an arbitrary

$$\mathbf{V}_f = \mathbf{V}_0 + \mathbf{X}_1\mathbf{C}\mathbf{C}'\mathbf{X}_1' + \mathbf{V}_0\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{V}_0 \in \mathscr{V}_1\,.$$

By Rao [14, Theorem 5.2], the inclusion $\mathscr{V}_1 \subseteq \mathscr{V}_{12}$ is equivalent to

$$\mathscr{C}(\mathbf{V}_f\mathbf{M}) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M})\,. \tag{11.22}$$

Now we have

$$\mathbf{V}_f\mathbf{M} = \mathbf{V}_0\mathbf{M} + \mathbf{V}_0\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{V}_0\mathbf{M}\,,$$

and thus (11.22) holds if and only if

$$\mathscr{C}(\mathbf{V}_0\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{V}_0\mathbf{M}) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M})\,. \tag{11.23}$$

Consider the equality $\mathscr{V}_1 = \mathscr{V}_{12}$, i.e., we should characterize the set of matrices $\mathbf{V}$ belonging to $\mathscr{V}_1 \cap \mathscr{V}_{12}$. In other words, we should combine (11.23) and Theorem 11.2. Suppose that $\mathscr{C}(\mathbf{M}_1\mathbf{X}_2) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M}_1)$, so that

$$\mathbf{M}_1\mathbf{X}_2\mathbf{A}_2 = \mathbf{V}_0\mathbf{M}_1\mathbf{T} \quad \text{for some} \mathbf{T}. \tag{11.24}$$

Now our request is that the following two expressions are both holding:

1. $\mathbf{V} = \mathbf{V}_0 + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1 + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0$, and
2. $\mathbf{V} = \mathbf{V}_0 + \mathbf{X}_1\mathbf{C}\mathbf{C}'\mathbf{X}_1' + \mathbf{V}_0\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{V}_0$ subject to (11.23).

In light of (11.24), the matrix $\mathbf{V}$ in 1 can be expressed as

$$\begin{aligned}\mathbf{V} &= \mathbf{V}_0 + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1 + \mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0\\ &= \mathbf{V}_0 + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1 + \mathbf{V}_0\mathbf{M}_1(\mathbf{T}\mathbf{T}' + \mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}\mathbf{B}\mathbf{B}'\mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2})\mathbf{M}_1\mathbf{V}_0\\ &=: \mathbf{V}_0 + \mathbf{X}_1\mathbf{A}_{11}\mathbf{X}_1 + \mathbf{V}_0\mathbf{M}_1\mathbf{R}\mathbf{R}'\mathbf{M}_1\mathbf{V}_0\,,\end{aligned}$$

where $\mathbf{R} = \mathbf{T} + \mathbf{Q}_{\mathbf{M}_1\mathbf{X}_2}\mathbf{B}$. Now

$$\begin{aligned}\mathbf{V}_0\mathbf{M}_1\mathbf{R}\mathbf{R}'\mathbf{M}_1\mathbf{V}_0\mathbf{M} &= (\mathbf{M}_1\mathbf{X}_2\mathbf{A}_{22}\mathbf{X}_2'\mathbf{M}_1 + \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0)\mathbf{M}\\ &= \mathbf{V}_0\mathbf{M}\mathbf{B}\mathbf{B}'\mathbf{M}\mathbf{V}_0\mathbf{M}\,,\end{aligned}$$

where we have used $\mathbf{M}_1\mathbf{M} = \mathbf{M}$, and hence

$$\mathscr{C}(\mathbf{V}_0\mathbf{M}_1\mathbf{R}\mathbf{R}'\mathbf{M}_1\mathbf{V}_0\mathbf{M}) \subseteq \mathscr{C}(\mathbf{V}_0\mathbf{M})\,.$$

Thus, we can conclude that the matrix $\mathbf{V}$ in 1 is also of the form $\mathbf{V}$ in 2, and we have proved the following.

Theorem 11.3 *The following statements hold.*

1. *The inclusion $\mathscr{V}_1 \subseteq \mathscr{V}_{12}$ holds if and only if the matrix $\boldsymbol{V} \in \mathscr{V}_1$ can be expressed as*

$$\boldsymbol{V} = \boldsymbol{V}_0 + \boldsymbol{X}_1\boldsymbol{C}\boldsymbol{C}'\boldsymbol{X}_1' + \boldsymbol{V}_0\boldsymbol{M}_1\boldsymbol{D}\boldsymbol{D}'\boldsymbol{M}_1\boldsymbol{V}_0\,,$$

   for some matrices $\boldsymbol{C}$ and $\boldsymbol{D}$ where

$$\mathscr{C}(\boldsymbol{V}_0\boldsymbol{M}_1\boldsymbol{D}\boldsymbol{D}'\boldsymbol{M}_1\boldsymbol{V}_0\boldsymbol{M}) \subseteq \mathscr{C}(\boldsymbol{V}_0\boldsymbol{M})\,. \tag{11.25}$$

2. *Suppose that $\mathscr{C}(\boldsymbol{M}_1\boldsymbol{X}_2) \subseteq \mathscr{C}(\boldsymbol{V}_0\boldsymbol{M}_1)$. Then $\mathscr{V}_{12} = \mathscr{V}_1$ holds, i.e., $\boldsymbol{V} \in \mathscr{V}_1 \cap \mathscr{V}_{12}$, if and only if the matrix $\boldsymbol{V}$ can be expressed as*

$$\boldsymbol{V} = \boldsymbol{V}_0 + \boldsymbol{X}_1\boldsymbol{A}_{11}\boldsymbol{X}_1 + \boldsymbol{M}_1\boldsymbol{X}_2\boldsymbol{A}_{22}\boldsymbol{X}_2'\boldsymbol{M}_1 + \boldsymbol{V}_0\boldsymbol{M}\boldsymbol{B}\boldsymbol{B}'\boldsymbol{M}\boldsymbol{V}_0\,.$$

Remark 11.2 Putting $\mathbf{V}_0 = \mathbf{I}$ in part 2 of Theorem 11.2 yields claim 1 of Theorem 11.1. Similarly, putting $\mathbf{V}_0 = \mathbf{I}$, condition (11.25) becomes

$$\mathscr{C}(\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{M}) \subseteq \mathscr{C}(\mathbf{M})\,, \quad \text{i.e.,} \quad \mathbf{P}_{\mathbf{X}}\mathbf{M}_1\mathbf{D}\mathbf{D}'\mathbf{M}_1\mathbf{M} = \mathbf{0}\,,$$

which is 2 of Theorem 11.1. Thus, Theorem 11.1 follows from Theorems 11.2 and 11.3. However, we feel that it is instructive to go separately through the proof of Theorem 11.1; for example, it is interesting to notice the role of the conditions for the equality of OLSE and BLUE.

11.4 A Related Sidetrack

For the property that every representation of the BLUE of (estimable) $\boldsymbol{\mu}_1$ under $\mathscr{M}_{12}(\mathbf{V}_0)$ remains BLUE of $\boldsymbol{\mu}_1$ under $\mathscr{M}_{12}(\mathbf{V})$, we can use the notation

$$\big\{\mathrm{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12}(\mathbf{V}_0))\big\} \subseteq \big\{\mathrm{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12}(\mathbf{V}))\big\}\,. \tag{11.26}$$

In view of Haslett and Puntanen [4, Theorem 2.1], (11.26) holds if and only if (see also Mathew and Bhimasankaram [9, Theorem 2.4])

$$\mathscr{C}(\mathbf{M}_2\mathbf{V}\mathbf{M}) \subseteq \mathscr{C}(\mathbf{M}_2\mathbf{V}_0\mathbf{M})\,, \quad \text{i.e.,} \quad \mathscr{C}(\mathbf{V}\mathbf{M}) \subseteq \mathscr{C}(\mathbf{X}_2 : \mathbf{V}_0\mathbf{M})\,.$$

Defining $\mathbf{W}_0 = \mathbf{V}_0 + \mathbf{X}\mathbf{X}'$, we have $\mathscr{C}(\mathbf{X} : \mathbf{V}_0\mathbf{M} : \mathbf{Q}_{\mathbf{W}_0}) = \mathbb{R}^n$. Hence an arbitrary nonnegative definite $\mathbf{V}$ can be expressed as $\mathbf{V} = \mathbf{F}\mathbf{F}'$, where

$$\mathbf{F} = \mathbf{X}_1\mathbf{L}_1 + \mathbf{X}_2\mathbf{L}_2 + \mathbf{V}_0\mathbf{M}\mathbf{L}_3 + \mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\,,$$

for some matrices $\mathbf{L}_1, \ldots, \mathbf{L}_4$ so that

$$\mathbf{V} = \mathbf{F}\mathbf{F}' = \mathbf{X}_1\mathbf{L}_{11}\mathbf{X}_1' + \mathbf{X}_2\mathbf{L}_{22}\mathbf{X}_2' + \mathbf{V}_0\mathbf{M}\mathbf{L}_{33}\mathbf{M}\mathbf{V}_0 + \mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_{44}\mathbf{Q}_{\mathbf{W}_0} + \mathbf{Z} + \mathbf{Z}',$$

where $\mathbf{L}_{ij} = \mathbf{L}_i\mathbf{L}_j'$, $i, j = 1, \ldots, 4$, and

$$\begin{aligned}\mathbf{Z} = {} & \mathbf{X}_1\mathbf{L}_{12}\mathbf{X}_2' + \mathbf{X}_1\mathbf{L}_{13}\mathbf{M}\mathbf{V}_0 + \mathbf{X}_1\mathbf{L}_{14}\mathbf{Q}_{\mathbf{W}_0} \\ & + \mathbf{X}_2\mathbf{L}_{23}\mathbf{M}\mathbf{V}_0 + \mathbf{X}_2\mathbf{L}_{24}\mathbf{Q}_{\mathbf{W}_0} + \mathbf{V}_0\mathbf{M}\mathbf{L}_{34}\mathbf{Q}_{\mathbf{W}_0}.\end{aligned}$$

Moreover, $\mathbf{M}_2\mathbf{F} = \mathbf{M}_2\mathbf{X}_1\mathbf{L}_1 + \mathbf{M}_2\mathbf{V}_0\mathbf{M}\mathbf{L}_3 + \mathbf{M}_2\mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4$, and

$$\mathbf{F}'\mathbf{M} = \mathbf{L}_3'\mathbf{M}\mathbf{V}_0\mathbf{M} + \mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0}\mathbf{M} = \mathbf{L}_3'\mathbf{M}\mathbf{V}_0\mathbf{M} + \mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0} =: \mathbf{K},$$

where we have used $\mathbf{Q}_{\mathbf{W}_0}\mathbf{M} = \mathbf{Q}_{\mathbf{W}_0}$; this is due to $\mathbf{Q}_{\mathbf{W}_0} = \mathbf{Q}_{\mathbf{M}\mathbf{V}_0\mathbf{M}}$. Now

$$\mathbf{M}_2\mathbf{F}\mathbf{F}'\mathbf{M} = \mathbf{M}_2\mathbf{X}_1\mathbf{L}_1\mathbf{K} + \mathbf{M}_2\mathbf{V}_0\mathbf{M}\mathbf{L}_3\mathbf{K} + \mathbf{M}_2\mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{K},$$

and thereby $\mathscr{C}(\mathbf{M}_2\mathbf{F}\mathbf{F}'\mathbf{M}) \subseteq \mathscr{C}(\mathbf{M}_2\mathbf{V}_0\mathbf{M})$ holds if and only if

$$\mathscr{C}(\mathbf{M}_2\mathbf{X}_1\mathbf{L}_1\mathbf{K} + \mathbf{M}_2\mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{K}) \subseteq \mathscr{C}(\mathbf{M}_2\mathbf{V}_0\mathbf{M}) = \mathscr{C}(\mathbf{M}_2\mathbf{W}_0\mathbf{M}),$$

which further is equivalent to

$$\mathscr{C}(\mathbf{X}_1\mathbf{L}_1\mathbf{K} + \mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{K}) \subseteq \mathscr{C}(\mathbf{X}_2 : \mathbf{W}_0\mathbf{M}). \tag{11.27}$$

Premultiplying (11.27) by $\mathbf{Q}_{\mathbf{W}_0} = \mathbf{M}\mathbf{Q}_{\mathbf{M}\mathbf{V}_0}$ yields $\mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{K} = \mathbf{0}$, i.e.,

$$\mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{L}_3'\mathbf{M}\mathbf{V}_0\mathbf{M} + \mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0} = \mathbf{0},$$

which further holds if and only if $\mathbf{Q}_{\mathbf{W}_0}\mathbf{L}_4\mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0} = \mathbf{0}$, i.e., $\mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0} = \mathbf{0}$. By (11.27) we have

$$\mathbf{X}_1\mathbf{L}_1\mathbf{K} = \mathbf{X}_2\mathbf{U}_1 + \mathbf{W}_0\mathbf{M}\mathbf{U}_2 \quad \text{for some} \mathbf{U}_1 \text{and} \mathbf{U}_2. \tag{11.28}$$

It is easy to conclude that (11.28) holds if and only if $\mathbf{X}_1\mathbf{L}_1\mathbf{K} = \mathbf{0}$, i.e.,

$$\mathbf{X}_1\mathbf{L}_1\mathbf{L}_3'\mathbf{M}\mathbf{V}_0\mathbf{M} + \mathbf{X}_1\mathbf{L}_1\mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0} = \mathbf{0},$$

which holds if and only if $\mathbf{X}_1\mathbf{L}_1\mathbf{L}_3'\mathbf{M}\mathbf{V}_0 = \mathbf{0}$, which together with $\mathbf{L}_4'\mathbf{Q}_{\mathbf{W}_0} = \mathbf{0}$ yields

$$\mathbf{Z} = \mathbf{X}_1\mathbf{L}_1\mathbf{L}_2'\mathbf{X}_2' + \mathbf{X}_2\mathbf{L}_2\mathbf{L}_3'\mathbf{M}\mathbf{V}_0. \tag{11.29}$$

Thus we have proved the following.

Theorem 11.4 *The inclusion*

$$\{\text{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12}(\boldsymbol{V}_0))\} \subseteq \{\text{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12}(\boldsymbol{V}))\}$$

holds if and only if the matrix $\boldsymbol{V}$ *can be expressed as*

$$\begin{aligned}\boldsymbol{V} &= \boldsymbol{X}_1\boldsymbol{L}_1\boldsymbol{L}_1'\boldsymbol{X}_1' + \boldsymbol{X}_2\boldsymbol{L}_2\boldsymbol{L}_2'\boldsymbol{X}_2' + \boldsymbol{V}_0\boldsymbol{M}\boldsymbol{L}_3\boldsymbol{L}_3'\boldsymbol{M}\boldsymbol{V}_0 + \boldsymbol{Z} + \boldsymbol{Z}' \\ &= \boldsymbol{X}\boldsymbol{L}\boldsymbol{L}'\boldsymbol{X}' + \boldsymbol{V}_0\boldsymbol{M}\boldsymbol{L}_3\boldsymbol{L}_3'\boldsymbol{M}\boldsymbol{V}_0 + \boldsymbol{X}_2\boldsymbol{L}_2\boldsymbol{L}_3'\boldsymbol{M}\boldsymbol{V}_0 + \boldsymbol{V}_0\boldsymbol{M}\boldsymbol{L}_3\boldsymbol{L}_2'\boldsymbol{X}_2' \qquad (11.30)\end{aligned}$$

for some $\boldsymbol{L}' = (\boldsymbol{L}_1' : \boldsymbol{L}_2')$, $\boldsymbol{L}_3$, *and* $\boldsymbol{Z}$ *is defined in* (11.29).

Notice the striking similarity between (11.7) and (11.30), the only difference concerning the term $\mathbf{X}_2\mathbf{L}_2\mathbf{L}_3'\mathbf{M}\mathbf{V}_0 + \mathbf{V}_0\mathbf{M}\mathbf{L}_3\mathbf{L}_2'\mathbf{X}_2'$. The characterization of $\mathbf{V}$ in Theorem 11.4 can be also expressed as

$$\mathbf{V} = \mathbf{X}\boldsymbol{\Gamma}\mathbf{X}' + \mathbf{V}_0\mathbf{M}\boldsymbol{\Delta}\mathbf{M}\mathbf{V}_0 + \mathbf{X}_2\boldsymbol{\Sigma}\mathbf{M}\mathbf{V}_0 + \mathbf{V}_0\mathbf{M}\boldsymbol{\Sigma}'\mathbf{X}_2'$$

for some matrices $\boldsymbol{\Gamma}$, $\boldsymbol{\Delta}$, and $\boldsymbol{\Sigma}$ such that $\mathbf{V}$ is nonnegative definite.

Isotalo and Puntanen [6, Theorem 1] characterized the set of matrices $\mathbf{V}$ such that the ordinary least squares estimator of estimable $\mathbf{K}\boldsymbol{\beta}$ equals BLUE. Corresponding characterizations for $\mathbf{V}$ retaining the equality of the BLUEs of $\mathbf{K}\boldsymbol{\beta}$ in the spirit of Theorem 11.4 can be done.

11.5 OLSE Equals BLUE Under the Small Model, What About the Big Model?

One can assume that $\text{BLUE}(\boldsymbol{\mu}_1) = \text{OLSE}(\boldsymbol{\mu}_1)$ under the small model and then check when the equality continues when more X-variables are added; here OLSE refers to the ordinary least squares estimator which is

- under $\mathscr{M}_1$: $\hat{\boldsymbol{\mu}}_1(\mathscr{M}_1) = \mathbf{P}_{\mathbf{X}_1}\mathbf{y} = \mathbf{P}_1\mathbf{y} = \mathbf{X}_1\mathbf{X}_1^+\mathbf{y}$,
- under $\mathscr{M}_{12}$: $\hat{\boldsymbol{\mu}}_1(\mathscr{M}_{12}) = \mathbf{P}_{1\#}\mathbf{y} = \mathbf{X}_1(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^-\mathbf{X}_1'\mathbf{M}_2\mathbf{y}$.

Our aim is to give several equivalent characterizations for the continuation of the equality of OLSE and BLUE when more X-variables are added. Some related considerations were made by Isotalo et al. [7]. Haslett and Puntanen [4, Theorem 3.1] considered the effect of adding regressors on the equality of the BLUEs under two linear models with different covariance matrices.

Above, and throughout this section, we assume that $\boldsymbol{\mu}_1 = \mathbf{X}_1\boldsymbol{\beta}_1$ is estimable under $\mathscr{M}_{12}$, i.e., $\mathscr{C}(\mathbf{X}_1) \cap \mathscr{C}(\mathbf{X}_2) = \{\mathbf{0}\}$ and thereby

$$\mathbf{P}_{\mathbf{X}} = \mathbf{P}_{(\mathbf{X}_1:\mathbf{X}_2)} = \mathbf{P}_{1\#} + \mathbf{P}_{2\#}\,,$$

where $\mathbf{P}_{2\#} = \mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^-\mathbf{X}_2'\mathbf{M}_1$. Recall that by (11.3) one expression for the BLUE of $\boldsymbol{\mu}$ under $\mathscr{M}_{12}$ is $\mathbf{G}_{12}\mathbf{y}$ where

$$\mathbf{G}_{12} = \mathbf{X}(\mathbf{X}'\mathbf{W}^{-}\mathbf{X})^{-}\mathbf{X}'\mathbf{W}^{+}, \quad \text{where} \mathbf{W} = \mathbf{V} + \mathbf{X}\mathbf{X}',$$

while the BLUE of $\boldsymbol{\mu}_1$ under $\mathscr{M}_1$ can be expressed as $\mathbf{G}_1\mathbf{y}$ where

$$\mathbf{G}_1 = \mathbf{X}_1(\mathbf{X}_1'\mathbf{W}_1^{-}\mathbf{X}_1)^{-}\mathbf{X}_1'\mathbf{W}_1^{+}, \quad \text{where} \mathbf{W}_1 = \mathbf{V} + \mathbf{X}_1\mathbf{X}_1'.$$

Let us denote

$$\mathbf{G}_{1\#} = \mathbf{X}_1(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1)^{-}\mathbf{X}_1'\mathbf{M}_2\,, \quad \mathbf{G}_{2\#} = \mathbf{X}_2(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2)^{-}\mathbf{X}_2'\mathbf{M}_1\,,$$

where $\mathbf{M}_1$ and $\mathbf{M}_2$ are (unique) matrices defined as

$$\mathbf{M}_1 = \mathbf{M}_1(\mathbf{M}_1\mathbf{W}\mathbf{M}_1)^{+}\mathbf{M}_1\,, \quad \mathbf{M}_2 = \mathbf{M}_2(\mathbf{M}_2\mathbf{W}\mathbf{M}_2)^{+}\mathbf{M}_2\,.$$

Now under the estimability of $\boldsymbol{\mu}_1 = \mathbf{X}_1\boldsymbol{\beta}_1$, we have

$$\text{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12}) = \mathbf{G}_{1\#}\mathbf{y}, \quad \text{BLUE}(\boldsymbol{\mu}_2 \mid \mathscr{M}_{12}) = \mathbf{G}_{2\#}\mathbf{y},$$

and, see Haslett et al. [5, Proposition 3.1],

$$\mathbf{G}_{12} = \mathbf{G}_{1\#} + \mathbf{G}_{2\#}\,.$$

Following Haslett et al. [5, Proposition 4.1], it is straightforward to confirm that $\mathbf{G}_1\mathbf{G}_{1\#} = \mathbf{G}_{1\#}$ and $\mathbf{G}_1\mathbf{G}_{12} = \mathbf{G}_1$ and thereby

$$\mathbf{G}_1 - \mathbf{G}_{1\#} = \mathbf{G}_1(\mathbf{G}_{12} - \mathbf{G}_{1\#}) = \mathbf{G}_1\mathbf{G}_{2\#}$$

and so we have

$$\mathbf{G}_{1\#} = \mathbf{G}_1 - \mathbf{G}_1(\mathbf{G}_{12} - \mathbf{G}_{1\#}) = \mathbf{G}_1 - \mathbf{G}_1\mathbf{G}_{2\#}\,, \tag{11.31a}$$

$$\mathbf{P}_{1\#} = \mathbf{P}_1 - \mathbf{P}_1(\mathbf{P}_{(\mathbf{X}_1:\mathbf{X}_2)} - \mathbf{P}_{1\#}) = \mathbf{P}_1 - \mathbf{P}_1\mathbf{P}_{2\#}\,. \tag{11.31b}$$

Assume that under $\mathscr{M}_1$, we have (with probability 1) $\hat{\boldsymbol{\mu}}_1(\mathscr{M}_1) = \tilde{\boldsymbol{\mu}}_1(\mathscr{M}_1)$, i.e., $\mathbf{P}_1\mathbf{y} = \mathbf{G}_1\mathbf{y}$ for all $\mathbf{y} \in \mathscr{C}(\mathbf{W}_1)$, so that

$$\mathbf{P}_1(\mathbf{X}_1 : \mathbf{V}\mathbf{M}_1) = \mathbf{G}_1(\mathbf{X}_1 : \mathbf{V}\mathbf{M}_1) = (\mathbf{X}_1 : \mathbf{0})\,, \tag{11.32}$$

which obviously holds if and only if $\mathbf{P}_1\mathbf{V}\mathbf{M}_1 = \mathbf{0}$. Notice that (11.32) means that $\mathbf{P}_1 = \mathbf{G}_1$. Suppose that $\mathbf{P}_1 = \mathbf{G}_1$. Then (11.31a) and (11.31b) become

$$\mathbf{G}_{1\#} = \mathbf{P}_1 - \mathbf{P}_1\mathbf{G}_{2\#}\,, \quad \mathbf{P}_{1\#} = \mathbf{P}_1 - \mathbf{P}_1\mathbf{P}_{2\#}\,,$$

and in this situation

$$\mathbf{G}_{1\#} - \mathbf{P}_{1\#} = \mathbf{P}_1\mathbf{G}_{2\#} - \mathbf{P}_1\mathbf{P}_{2\#}\,,$$

and thereby the equality $\mathbf{P}_{1\#}\mathbf{y} = \mathbf{G}_{1\#}\mathbf{y}$ for all $\mathbf{y} \in \mathscr{C}(\mathbf{W})$ can be written as

$$\mathbf{P}_1\mathbf{P}_{2\#}\mathbf{W} = \mathbf{P}_1\mathbf{G}_{2\#}\mathbf{W}. \tag{11.33}$$

Denote $\mathbf{y} = \mathbf{X}_1\mathbf{a} + \mathbf{X}_2\mathbf{b} + \mathbf{VMc} = (\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{VM})\mathbf{t}$ for some $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ so that $\mathbf{t} = (\mathbf{a}', \mathbf{b}', \mathbf{c}')'$. Then

$$\begin{aligned}\mathbf{P}_1\mathbf{P}_{2\#}\mathbf{y} - \mathbf{P}_1\mathbf{G}_{2\#}\mathbf{y} &= \mathbf{P}_1(\mathbf{P}_{2\#} - \mathbf{G}_{2\#})(\mathbf{X}_1 : \mathbf{X}_2 : \mathbf{VM})\mathbf{t} \\ &= (\mathbf{0} : \mathbf{0} : \mathbf{P}_1\mathbf{P}_{2\#}\mathbf{VM})\mathbf{t} = \mathbf{P}_1\mathbf{P}_{2\#}\mathbf{VMc}\,.\end{aligned}$$

Thus (11.33) is equivalent to $\mathbf{P}_1\mathbf{P}_{2\#}\mathbf{VM} = \mathbf{0}$, i.e., $\mathbf{X}_1'\mathbf{P}_{2\#}\mathbf{VM} = \mathbf{0}$.

Actually, (11.33) means the equality

$$\mathbf{P}_1\mathbf{P}_{2\#} = \mathbf{P}_1\mathbf{G}_{2\#}\,, \quad \text{i.e.,} \quad \mathbf{X}_1'\mathbf{P}_{2\#} = \mathbf{X}_1'\mathbf{G}_{2\#}\,.$$

This can be concluded from the fact that $\mathbf{P}_1\mathbf{P}_{2\#}\mathbf{Q}_\mathbf{W} = \mathbf{P}_1\mathbf{G}_{2\#}\mathbf{Q}_\mathbf{W} = \mathbf{0}$. Thus we have proved the following.

Theorem 11.5 *Consider the models $\mathscr{M}_{12}$ and $\mathscr{M}_1$ and suppose that $\boldsymbol{\mu}_1 = \boldsymbol{X}_1\boldsymbol{\beta}_1$ is estimable under $\mathscr{M}_{12}$, and* $\mathrm{OLSE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_1) = \mathrm{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_1)$ *. Then the following statements are equivalent:*

1. $\mathrm{OLSE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12}) = \mathrm{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_{12})$ *, i.e.,* $\boldsymbol{P}_{1\#}\boldsymbol{W} = \boldsymbol{G}_{1\#}\boldsymbol{W}$,
2. $\boldsymbol{X}_1'\boldsymbol{P}_{2\#}\boldsymbol{W} = \boldsymbol{X}_1'\boldsymbol{G}_{2\#}\boldsymbol{W}$,
3. $\boldsymbol{X}_1'\boldsymbol{P}_{2\#} = \boldsymbol{X}_1'\boldsymbol{G}_{2\#}$,
4. $\boldsymbol{X}_1'\boldsymbol{P}_{2\#}\boldsymbol{VM} = \boldsymbol{0}$ *, and*
5. $\boldsymbol{X}_1'\mathrm{OLSE}(\boldsymbol{\mu}_2 \mid \mathscr{M}_{12}) = \boldsymbol{X}_1'\mathrm{BLUE}(\boldsymbol{\mu}_2 \mid \mathscr{M}_{12})$.

Remark 11.3 Consider the following task: Define the subspace $\mathcal{A} \subseteq \mathbb{R}^n$ so that $\mathrm{OLSE}(\boldsymbol{\mu}) = \mathrm{BLUE}(\boldsymbol{\mu})$ under $\mathscr{M}_{12}$ for all $\mathbf{y} \in \mathcal{A}$, i.e.,

$$\mathbf{P}_\mathbf{X}\mathbf{y} = \mathbf{G}_{12}\mathbf{y} \quad \text{for all} \mathbf{y} \in \mathcal{A}\,,$$

where $\mathbf{P}_\mathbf{X} = \mathbf{P}_{(\mathbf{X}_1:\mathbf{X}_2)}$ and by (11.4),

$$\mathbf{G}_{12} = \mathbf{X}(\mathbf{X}'\mathbf{W}^-\mathbf{X}')^-\mathbf{X}'\mathbf{W}^+ = \mathbf{P}_\mathbf{W} - \mathbf{VM}(\mathbf{MVM})^+\mathbf{M}\,, \tag{11.34}$$

where $\mathbf{W} = \mathbf{V} + \mathbf{XX}'$. Premultiplying (11.34) by $\mathbf{P}_\mathbf{X}$ gives

$$\mathbf{G}_{12} = \mathbf{X}(\mathbf{X}'\mathbf{W}^-\mathbf{X}')^-\mathbf{X}'\mathbf{W}^+ = \mathbf{P}_\mathbf{X} - \mathbf{P}_\mathbf{X}\mathbf{VM}(\mathbf{MVM})^+\mathbf{M}\,.$$

Thus we have to solve $\mathcal{A}$ from the condition

$$\mathbf{P}_\mathbf{X}\mathbf{VM}(\mathbf{MVM})^+\mathbf{My} = \mathbf{0} \quad \text{for all} \mathbf{y} \in \mathcal{A}\,,$$

i.e.,

$$\mathcal{A} = \mathcal{N}(\mathbf{P}_{\mathbf{X}}\mathbf{Z}), \quad \text{where } \mathbf{Z} = \mathbf{VM}(\mathbf{MVM})^{+}\mathbf{M},$$

where $\mathcal{N}(\cdot)$ refers to the nullspace of the matrix argument. Krämer [8] considered the corresponding problem assuming that $\mathbf{V}$ is positive definite and $\mathbf{X}$ has full column rank. He showed that in that case

$$\mathcal{A} = \mathcal{C}(\mathbf{X}) \boxplus [\mathcal{C}(\mathbf{M}) \cap \mathcal{C}(\mathbf{VM})]. \tag{11.35}$$

Let us relax from the rank assumptions on $\mathbf{X}$ and $\mathbf{V}$. We utilize the result of Baksalary and Trenkler [1, Theorem 1] stating that for idempotent matrices $\mathbf{R}$ and $\mathbf{S}$ (of the same size), the following holds:

$$\mathcal{N}(\mathbf{RS}) = \mathcal{N}(\mathbf{S}) \oplus [\mathcal{N}(\mathbf{R}) \cap \mathcal{C}(\mathbf{S})]. \tag{11.36}$$

Now $\mathbf{P}_{\mathbf{X}}$ and $\mathbf{Z}$ are idempotent and $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{VM})$, and thus (11.36) gives

$$\mathcal{A} = \mathcal{N}(\mathbf{Z}) \oplus [\mathcal{C}(\mathbf{M}) \cap \mathcal{C}(\mathbf{Z})] = \mathcal{N}(\mathbf{Z}) \oplus [\mathcal{C}(\mathbf{M}) \cap \mathcal{C}(\mathbf{VM})].$$

Clearly, $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{Z})$, and thereby $\mathcal{C}(\mathbf{X}) = \mathcal{N}(\mathbf{Z})$ if and only if

$$\mathrm{r}(\mathbf{X}) = n - \mathrm{r}(\mathbf{Z}) = n - \mathrm{r}(\mathbf{VM}), \quad \text{i.e., } \mathrm{r}(\mathbf{X} : \mathbf{VM}) = \mathrm{r}(\mathbf{X} : \mathbf{V}) = n. \tag{11.37}$$

Thus Krämer's result (11.35) applies if and only if (11.37) holds. Further expressions for $\mathcal{A}$ appear in Baksalary et al. [2].

11.6 Conclusions

In this article, we consider the partitioned linear model $\mathscr{M}_{12}(\mathbf{V}_0) = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2,\ \mathbf{V}_0\}$ and the corresponding small model $\mathscr{M}_1(\mathbf{V}_0) = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1,\ \mathbf{V}_0\}$. Following Rao [14, Sect. 5], we can characterize the sets $\mathscr{V}_{12}$ and $\mathscr{V}_1$ of nonnegative definite matrices so that

$$\begin{aligned}
\mathbf{V} \in \mathscr{V}_1 &\iff \{\mathrm{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_1(\mathbf{V}_0))\} \subseteq \{\mathrm{BLUE}(\boldsymbol{\mu}_1 \mid \mathscr{M}_1(\mathbf{V}))\}, \\
\mathbf{V} \in \mathscr{V}_{12} &\iff \{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{V}_0))\} \subseteq \{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{M}_{12}(\mathbf{V}))\},
\end{aligned}$$

where $\boldsymbol{\mu}_1 = \mathbf{X}_1\boldsymbol{\beta}_1$, $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, and $\mathscr{M}_{12}(\mathbf{V}) = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2,\ \mathbf{V}\}$ and $\mathscr{M}_1(\mathbf{V}) = \{\mathbf{y},\ \mathbf{X}_1\boldsymbol{\beta}_1,\ \mathbf{V}\}$. The notation $\{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{A}\} \subseteq \{\mathrm{BLUE}(\boldsymbol{\mu} \mid \mathscr{B}\}$ means that every representation of the BLUE for $\boldsymbol{\mu}$ under $\mathscr{A}$ remains BLUE for $\boldsymbol{\mu}$ under $\mathscr{B}$. In the first three sections of this article, we focus on characterizing the mutual relations between the sets $\mathscr{V}_1$ and $\mathscr{V}_{12}$.

In Sect. 11.5, we assume that under the small model $\mathscr{M}_1$ the ordinary least squares estimator, OLSE, of $\boldsymbol{\mu}_1$ equals the BLUE and give several characterizations for the continuation of the equality of OLSE and BLUE when more X-variables are added.

Our main questions are somewhat unusual and, we admit, are slightly on the nonpractical side. However, they offer us interesting linear-algebraic problems, and, moreover, sometimes new theoretical considerations may provide surprises for applications. Moreover, as the Referee pointed out: "Models with added covariates those of primary interest occur quite frequently in applications".

Acknowledgements This research has its origin in the first author's online talk, see Haslett [3], considering the related problems from a different perspective, in the 28th International Workshop on Matrices and Statistics, Manipal Academy of Higher Education, India, 13–15 December 2021. Thanks go to the anonymous referee for constructive remarks.

References

1. Baksalary OM, Trenkler G (2013) On column and null spaces of functions of a pair of oblique projectors. Linear Multilinear Algebra 61:1116–1129. https://doi.org/10.1080/03081087.2012.731055
2. Baksalary OM, Trenkler G, Liski E (2013) Let us do the twist again. Stat Pap 54:1109–1119. https://doi.org/10.1007/s00362-013-0512-3
3. Haslett SJ (2021) The role of block diagonal matrices in conditions for equality of BLUEs in linear models. In: Online talk given in the 28th international workshop on matrices and statistics, Manipal Academy of Higher Education, India. https://carams.in/events/iwms2021/
4. Haslett SJ, Puntanen S (2010) Effect of adding regressors on the equality of the BLUEs under two linear models. J Stat Plann Inference 140:104–110. http://dx.doi.org/10.1016/j.jspi.2009.06.010
5. Haslett SJ, Markiewicz A, Puntanen S (2022) Properties of BLUEs in full versus small linear models. Commun Stat Theory Methods. https://doi.org/10.1080/03610926.2022.2052899
6. Isotalo J, Puntanen S (2009) A note on the equality of the OLSE and the BLUE of the parametric function in the general Gauss–Markov model. Stat Pap 50:185–193. http://dx.doi.org/10.1007/s00362-007-0055-6
7. Isotalo J, Puntanen S, Styan GPH (2007) Effect of adding regressors on the equality of the OLSE and BLUE. Int J Stat Sci 6:193–201. http://people.uta.fi/~simo.puntanen/Isotalo-Puntanen-Styan-IJSS-2007.pdf
8. Krämer W (1980) A note on the equality of ordinary least squares and Gauss-Markov estimates in the general linear model. Sankhyā, Ser A 42:130–131. https://www.jstor.org/stable/25050222
9. Mathew T, Bhimasankaram P (1983) On the robustness of LRT in singular linear models. Sankhyā Ser A 45:301–312. https://www.jstor.org/stable/25050442
10. Mitra SK, Moore BJ (1973) Gauss–Markov estimation with an incorrect dispersion matrix. Sankhyā, Ser A 35:139–152. https://www.jstor.org/stable/25049864
11. Puntanen S, Styan GPH, Isotalo J (2011) Matrix tricks for linear statistical models: our personal top twenty. Springer, Heidelberg. http://dx.doi.org/10.1007/978-3-642-10473-2
12. Rao CR (1967) Least squares theory using an estimated dispersion matrix and its application to measurement of signals. In: Lucien M, Cam L, Neyman J (eds) Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability, vol 1, University of California Press, Berkeley, pp 355–372. https://projecteuclid.org/euclid.bsmsp/1200512997
13. Rao CR (1968) A note on a previous lemma in the theory of least squares and some further results. Sankhyā, Ser A 30:259–266. https://www.jstor.org/stable/25041357

14. Rao CR (1971) Unified theory of linear estimation. Sankhyā, Ser A 33:371–394. https://www.jstor.org/stable/25049750
15. Rao CR (1973) Representations of best linear estimators in the Gauss–Markoff model with a singular dispersion matrix. J Multivar Anal 3:276–292. http://dx.doi.org/10.1016/0047-259X(73)90042-0
16. Zyskind G (1967) On canonical forms, non-negative covariance matrices and best and simple least squares linear estimators in linear models. Ann Math Stat 38:1092–1109. https://www.jstor.org/stable/2238829

Chapter 12
On Some Special Matrices and Their Applications in Linear Complementarity Problem



Samir Kumar Neogy and Gambheer Singh

Mathematics Subject Classification (2010) 91A15 · 90C33

12.1 Introduction

The linear complementarity problem (LCP) is a combination of linear and nonlinear equations and inequalities. LCP is crucial in mathematical programming because it gives a unified approach to studying linear programming, linear fractional programming, convex quadratic programming and bimatrix games. It is always a demanding and interesting area for optimization researchers. The linear complementarity problem finds a vector in a finite-dimensional real vector space that satisfies a specific system of inequalities. The problem is described below.

Given a matrix $A \in \mathbb{R}^{n\times n}$ and a vector $q \in \mathbb{R}^n$, LCP is to find a vector $z \in \mathbb{R}^n$ such that

$$q + Az \geq 0,\ z \geq 0, \tag{12.1}$$

$$z^T(q + Az) = 0, \tag{12.2}$$

or to demonstrate that no such vector z exists. This problem is denoted by LCP(q, A).

Let

$$w = q + Az. \tag{12.3}$$

S. K. Neogy (✉)
Indian Statistical Institute, 7, S. J. S Sansanwal Marg, New Delhi 110016, India
e-mail: skn@isid.ac.in

G. Singh
Department of Mathematics, University of Delhi, Delhi 110007, India

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_12

Then LCP (12.1) and (12.2) can be formulated as follows:

$$w = q + Az, \quad w \geq 0,\ z \geq 0, \tag{12.4}$$

$$w^T z = 0. \tag{12.5}$$

Condition (12.5) is frequently substituted in place of (12.2). When talking about algorithms, this method of LCP representation is useful.

For a matrix $A \in \mathbb{R}^{n\times n}$ and a vector $q \in \mathbb{R}^n$, the feasible set and the solution set of LCP(q, A) can be defined by $F(q, A) = \{z \in \mathbb{R}^n \mid z \geq 0,\ Az + q \geq 0\}$ and $S(q, A) = \left\{z \in F(q, A) \mid z^T(Az + q) = 0\right\}$, respectively. If $w + z > 0$, a solution (w, z) of the LCP(q, A) is said to be nondegenerate. If every solution (w, z) of the LCP(q, A) is nondegenerate, then a vector $q \in \mathbb{R}^n$ is said to be nondegenerate with respect to matrix A.

In the literature on mathematical programming, LCP is well studied, and it has several applications in operations research, mathematical economics, geometry and engineering which have been described by Cottle et al. [5] and Murty and Yu [21]. In the field of stochastic games, the linear complementarity problem has recently found several applications (see Mohan et al. [16]). The research of the LCP, particularly the study of the LCP algorithms and the study of matrix classes, has been inspired by these kinds of applications and the possibility of future applications. In fact, the assumption that matrix A belongs to a particular class of matrices is the foundation of most of the linear complementarity theory and algorithms. This chapter discusses recent classes of matrices studied in the last two decades and summarizes some main results pertaining to these classes.

The chapter is organized as follows. Section 12.2 presents some notations, definitions of various matrix classes, the concept of principal pivot transform (PPT) and the basics of the zero-sum matrix game. Section 12.3 introduces a hidden Z-matrix with some examples and properties. Further, it shows the game theory-based characterization of hidden Z-matrices. Section 12.4 is devoted to studying the class of positive subdefinite matrices (PSBD) and indicates that LCP with PSBD matrices of rank ≥ 2 is processable by Lemke's algorithm. Section 12.5 discusses the generalized positive subdefinite matrices (GPSBD) and their properties in the context of LCP. Section 12.6 defines the weak generalized positive subdefinite matrices (WGPSBD). Section 12.7 presents GPSBD matrices of level k and identifies that this class is a subclass of row sufficient matrices. The aim of Sect. 12.8 is to introduce the class of fully copositive matrices and their properties. Section 12.9 presents the singular N_0-matrices with Q-property. In Sect. 12.10, we study the almost type class of matrices with Q-property and obtain a sufficient condition for almost $\bar{N}$-matrices to hold Q-property. Further, we present the generalization of almost type class of matrices with examples. Section 12.11 introduces various classes of matrices that are defined based on principal pivot transforms. Section 12.12 presents a list of open problems.

12.2 Preliminaries

We consider matrices and vectors with real entries. Any vector $x \in \mathbb{R}^n$ is a column vector unless otherwise specified, and x^T denotes the transpose of x. For any matrix $A \in \mathbb{R}^{m\times n}$, a_{ij} denotes its ith row and jth column entry. For any matrix $A \in \mathbb{R}^{m\times n}$, $A_{i.}$ denotes the ith row of A and $A_{.j}$ denotes the jth column of A. For any set α, $|\alpha|$ denotes its cardinality. For any index set $\alpha \subseteq \{1, 2, \ldots, n\}$, $\bar{\alpha}$ denotes its complement in $\{1, 2, \ldots, n\}$. If A is a matrix of order n, $\alpha \subseteq \{1, 2, \ldots, n\}$ and $\beta \subseteq \{1, 2, \ldots, n\}$, then $A_{\alpha\beta}$ represents the submatrix of A consisting of only those rows and columns whose indices are in α and β, respectively. If $\alpha = \beta$, then the submatrix $A_{\alpha\alpha}$ is referred to as a principal submatrix of A and $\det A_{\alpha\alpha}$ is referred to as a principal minor of A. We denote the identity matrix by I and the empty set by $\emptyset$. A vector $x \in \mathbb{R}^n$ is unisigned, if either $x_i \geq 0$ or $x_i \leq 0$ for all $1 \leq i \leq n$.

We now define some matrix classes that will be needed in the subsequent sections, such as positive definite (PD), positive semidefinite (PSD), positive subdefinite (PSBD), merely positive subdefinite (MPSBD), semimonotone (E_0), fully semimonotone (E_0^f), copositive (C_0), strictly copositive (C), copositive-plus $\left(C_0^+\right)$, copositive-star $\left(C_0^*\right)$ and fully copositive (C_0^f).

A matrix $A \in \mathbb{R}^{n\times n}$ is said to be

- Q: if $\forall\ q \in \mathbb{R}^n$, LCP$(q, A)$ has a solution.
- Q_0: if $F(q, A) \neq \emptyset \Rightarrow S(q, A) \neq \emptyset$.
- PD: if $\forall\ 0 \neq x \in \mathbb{R}^n,\ x^T Ax > 0$.
- PSD: if $\forall\ x \in \mathbb{R}^n,\ x^T Ax \geq 0$.
- PSBD: if $\forall\ x \in \mathbb{R}^n,\ x^T Ax < 0$ implies either $A^T x \leq 0$ or $A^T x \geq 0$.
- MPSBD: if A is a PSBD matrix but not a PSD matrix.
- E_0: if $\forall\ 0 \neq x \geq 0$ there exists i such that $x_i > 0$ and $(Ax)_i \geq 0$.
- E_0^f: if every PPT of A is in E_0.
- C_0: if $\forall\ x \geq 0 \Rightarrow x^T Ax \geq 0$.
- C: if $\forall\ 0 \neq x \geq 0 \Rightarrow x^T Ax > 0$.
- C_0^+: if $A \in C_0$ and $\left[x^T Ax = 0,\ x \geq 0\right] \Rightarrow \left[\left(A + A^T\right)x = 0\right]$.
- C_0^*: if $A \in C_0$ and $\left[x^T Ax = 0,\ Ax \geqslant 0,\ x \geqslant 0\right] \Rightarrow A^T x \leqslant 0$.
- C_0^f: if every PPT of A is in C_0.
- Z: if $a_{ij} \leq 0\ \forall\ i, j \in \{1, 2, \ldots, n\}$ such that $i \neq j$.
- N: if all its principal minors are negative.
- N_0: if all its principal minors are nonpositive.
- P: if all its principal minors are positive.
- P_0: if all its principal minors are nonnegative.
- K: if it is a Z-matrix as well as P-matrix.
- S: if there exists a vector $x > 0$ such that $Ax > 0$.
- R: if $\forall\ t \geq 0$, LCP(te, A) has only the trivial solution.
- R_0: if LCP$(0, A)$ has only the trivial solution.
- Column sufficient: if $\forall\ x \in \mathbb{R}^n,\ x_i(Ax)_i \leqslant 0\ \forall\ i \Rightarrow x_i(Ax)_i = 0\ \forall\ i$.

- Row sufficient: if A^T is column sufficient.
- Sufficient: if A and A^T are both column sufficient.

Note that $P \subseteq P_0 \subseteq E_0^f \subseteq E_0$ and $C_0^f \subseteq E_0^f$. A matrix class is said to be complete if all principal submatrices of its elements belong to the same class. We claim that an LCP can be processed by Lemke's algorithm if Lemke's algorithm can either compute a solution to it, if one exists, or demonstrate that there is no solution. (See Cottle et al. [5] for more details on Lemke's algorithm.)

The concept of principal pivot transforms (PPTs) was introduced by Tucker [31]. The PPT of a matrix A with respect to $\alpha \subseteq \{1, ..., n\}$ is defined as the matrix given by

$$A' = \begin{bmatrix} A'_{\alpha\alpha} & A'_{\alpha\bar{\alpha}} \\ A'_{\bar{\alpha}\alpha} & A'_{\bar{\alpha}\bar{\alpha}} \end{bmatrix},$$

where $A'_{\alpha\alpha} = (A_{\alpha\alpha})^{-1}$, $A'_{\alpha\bar{\alpha}} = -(A_{\alpha\alpha})^{-1} A_{\alpha\bar{\alpha}}$, $A'_{\bar{\alpha}\alpha} = A_{\bar{\alpha}\alpha} (A_{\alpha\alpha})^{-1}$, $A'_{\bar{\alpha}\bar{\alpha}} = A_{\bar{\alpha}\bar{\alpha}} - A_{\bar{\alpha}\alpha} (A_{\alpha\alpha})^{-1} A_{\alpha\bar{\alpha}}$. The expression $A_{\bar{\alpha}\bar{\alpha}} - A_{\bar{\alpha}\alpha} (A_{\alpha\alpha})^{-1} A_{\alpha\bar{\alpha}}$ is the Schur complement of $A_{\alpha\alpha}$ in A and is denoted as $(A/A_{\alpha\alpha})$. The PPT of LCP (q, A) with respect to α (obtained by pivoting on $A_{\alpha\alpha}$) is given by LCP (q', A') where $q'_\alpha = -A^{-1}_{\alpha\alpha} q_\alpha$ and $q'_{\bar{\alpha}} = q_{\bar{\alpha}} - A_{\bar{\alpha}\alpha} A^{-1}_{\alpha\alpha} q_\alpha$. We use the notation $\wp_\alpha(A)\ (= A')$ for PPT of A with respect to $\alpha \subseteq \{1, \ldots, n\}$. Note that PPT is only defined with respect to those α for which $\det A_{\alpha\alpha} \neq 0$.

Tucker demonstrated that A is a P-matrix if the diagonal entries for each PPT of A are positive. However, A need not be a P_0-matrix if the diagonal entries for each PPT of A are nonnegative. The diagonal elements of each PPT of the matrix $A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$ can be easily verified to be nonnegative, but $A \notin P_0$.

The following proposition observed by Dubey and Neogy [8] shows that the class of S-matrices is invariant under PPT.

Proposition 12.1 *Let $\wp_\alpha(A) \in \mathbb{R}^{n\times n}$ be a PPT of a given matrix $A \in \mathbb{R}^{n\times n}$ with respect to $\alpha \subseteq \{1, \ldots, n\}$. Then $A \in S$ if and only if $\wp_\alpha(A) \in S$.*

We present some fundamentals of a two-person zero-sum matrix game due to Von Neumann and Morgenstern [32], which will be required in the sequel. A two-player zero-sum matrix game is defined as follows.

Player 1 selects an integer $i\,(i = 1, \ldots, m)$ and player 2 selects an integer $j\,(j = 1, \ldots, n)$ simultaneously. Then player 1 pays player 2 an amount a_{ij} (which can be positive, negative or zero). The game is considered to be zero-sum because player 2's gain is player 1's loss. A strategy for player 1 is a probability vector $x \in \mathbb{R}^m$, the ith component of which represents the probability of selecting an integer i where $x_i \geq 0$ for $i = 1, \ldots, m$ and $\sum_{i=1}^m x_i = 1$. Similarly, a strategy for player 2 is a probability vector $y \in \mathbb{R}^n$.

According to Von Neumann's fundamental minimax theorem, there exist strategies x^*, y^* and a real number v such that

$$\sum_{i=1}^{m} x_i^* a_{ij} \leq v, \quad \forall\, j = 1, \ldots, n.$$

$$\sum_{j=1}^{n} y_j^* a_{ij} \geq v, \quad \forall\, i = 1, \ldots, m.$$

The strategies (x^*, y^*) with $x^* \in \mathbb{R}^m$ and $y^* \in \mathbb{R}^n$ are said to be optimal strategies for player 1 and player 2, respectively, and v is known as the minimax value of the game. The payoff matrix in the game described is $A = \left(a_{ij}\right)$. The value of the game corresponding to the payoff matrix A is denoted by $v(A)$. The minimizer is player 1, and the maximizer is player 2 in the game mentioned above. The value of the game $v(A)$ is positive (nonnegative) if there exists a $0 \neq y \geq 0$ such that $Ay > 0 (Ay \geq 0)$. Similarly, $v(A)$ is negative (nonpositive) if there exists a $0 \neq x \geq 0$ such that $A^T x < 0 \left(A^T x \leq 0\right)$. If $x > 0$, a mixed strategy for player 1 is said to be completely mixed, which means that all of the rows of the payoff matrix A are selected with a positive probability. In the same way, a mixed strategy y for player 2 is defined. A matrix game with a payoff matrix $A \in \mathbb{R}^{n \times n}$ is said to be completely mixed if all optimum mixed strategies x for player 1 and y for player 2 are completely mixed.

12.3 The Class of Hidden Z-Matrices

Mangasarian presented a generalization of the class of Z-matrices for studying the class of linear complementarity problems that can be solved by linear programs, Mangasarian [10–12]. Since many of the properties that hold to the class Z also hold to the class hidden Z, Pang proposed calling this generalization of the class Z the class of hidden Z-matrices. As per Mangasarian [10] and Pang [28], a hidden Z-matrix is as follows.

Definition 12.1 A matrix $A \in \mathbb{R}^{n \times n}$ is said to be a hidden Z-matrix if there exist Z-matrices $X, Y \in \mathbb{R}^{n \times n}$ and $r, s \in \mathbb{R}_+^n$ such that

(i) $AX = Y$,
(ii) $r^T X + s^T Y > 0$.

Hidden Z is the abbreviation for the class of hidden Z-matrices. A necessary and sufficient condition was developed by Pang [28] for a hidden Z-matrix to be a P-matrix. Chu [2] investigated the class of hidden $Z \cap P$-matrices, which is also known as the class of hidden K-matrices.

We have the following example of a hidden Z-matrix.

Example 1

Consider the matrix

$$A = \begin{bmatrix} 0 & -2 & 2 \\ -4 & 12 & -10 \\ 0 & -6 & 4 \end{bmatrix}$$

with

$$X = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & 0 & -2 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 0 & -4 & 0 \\ -8 & 24 & -4 \\ 0 & -12 & 4 \end{bmatrix}.$$

Let $r = \begin{bmatrix} 16 \\ 0 \\ 0 \end{bmatrix}$ and $s = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$, then $r^T X + s^T Y = \begin{bmatrix} 8 & 24 & 4 \end{bmatrix}$.
We see that X and Y are Z-matrices. Also, $AX = Y$ and $r^T X + s^T Y > 0$. Thus A is a hidden Z-matrix.

Now, we study some properties of hidden Z-matrices and discuss the existence of the solution of LCP(q, A) where A is a hidden Z-matrix.

It should be noted that many of the results that apply to the Z class admit an extension to the hidden Z class because the class of hidden Z-matrices generalizes the class of Z-matrices. All proper principal submatrices need not be hidden Z-matrices, even if A is a hidden Z-matrix. Now, we have the following example.

Example 2

From Example 1, we can see that A is a hidden Z-matrix, but for $\alpha = \{1, 3\}$, the principal submatrix $A_{\alpha\alpha} = \begin{bmatrix} 0 & 2 \\ 0 & 4 \end{bmatrix}$ is not a hidden Z-matrix. To prove this, suppose on the contrary that there exists a Z-matrix $\tilde{X} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$ such that $\tilde{Y} = A_{\alpha\alpha}\tilde{X} = \begin{bmatrix} 2x_{21} & 2x_{22} \\ 4x_{21} & 4x_{22} \end{bmatrix} \in Z$. Since $\tilde{X} \in Z$, we have $x_{12} \leq 0$ and $x_{21} \leq 0$. Further, $\tilde{Y} \in Z$ implies $x_{22} \leq 0$. As both entries in the second column of $\tilde{X}$ and $\tilde{Y}$ are less than or equal to zero, therefore for any $r, s \in \mathbb{R}^2_+$, the second component of the row vector $r^T\tilde{X} + s^T\tilde{Y}$ will always be less than or equal to zero. Hence $A_{\alpha\alpha}$ is not a hidden Z-matrix.

Furthermore, it is commonly known that the inverse of a K-matrix is nonnegative. However, unlike the case of a K-matrix, the inverse of a hidden K-matrix is not always nonnegative. The same is demonstrated in the example below.

Example 3 (Dubey and Neogy [8])

Consider the matrix

$$A = \begin{bmatrix} 2 & -3 & -4 \\ -1 & 2 & 3 \\ 2 & -2 & 5 \end{bmatrix}.$$

We can easily see that A is a hidden K-matrix. But

$$A^{-1} = \begin{bmatrix} 2.2857 & 3.2857 & -0.1429 \\ 1.5714 & 2.5714 & -0.2857 \\ -0.2857 & -0.2857 & 0.1429 \end{bmatrix}$$

is not a nonnegative matrix. Indeed, no PPT of A is a nonnegative matrix.

Now, we consider a class of hidden Z-matrices from the LCP's point of view. To solve LCPs as linear programs, Mangasarian introduced the hidden Z class, and he demonstrated the following result.

Theorem 12.1 (Mangasarian [10]) *Let $A \in$ hidden Z and $F(q, A) \neq \emptyset$. Then the LCP(q, A) has a solution which can be obtained by solving the linear program* LP(p, q, A)*:*

$$\begin{aligned} &\textit{Minimize } p^T x \\ &\textit{subject to } q + Ax \geq 0, \\ &x \geq 0, \end{aligned}$$

where $p = r + A^T s$, r and s are as in Definition 12.1.

We state the following lemmas proved by Dubey and Neogy [8], which are useful to prove subsequent theorems.

Lemma 12.1 (Dubey and Neogy [8]) *Let $A \in \mathbb{R}^{n \times n}$ be a hidden Z-matrix. Let $\wp_\alpha(A)$ be a PPT of A with respect to $\alpha \subseteq \{1, \ldots, n\}$. Then $\wp_\alpha(A)$ is a hidden Z-matrix.*

By constructing an equivalent LCP, we demonstrate that an LCP(q, A) involving a hidden Z-matrix can be solved whenever it is feasible. We construct this LCP in the subsequent lemma.

Lemma 12.2 (Dubey and Neogy [8]) *Consider* LCP$(\tilde{q}, \mathcal{A})$ *with* $\mathcal{A} = \begin{bmatrix} 0 & -A^T \\ A & 0 \end{bmatrix}$, $\tilde{q} = \begin{bmatrix} p \\ q \end{bmatrix}$, $A \in \mathbb{R}^{n\times n}$ *is a hidden Z-matrix,* $p,\ q \in \mathbb{R}^n$, *where* $p = r + A^T s$, *and* r *and* s *are as in Definition 12.1. If* $\begin{bmatrix} x \\ y \end{bmatrix} \in F(\tilde{q}, \mathcal{A})$ *then* $\left(I - A^T\right) y + p > 0$.

The relationship between the solution of LCP(q, A) and LCP$(\tilde{q}, \mathcal{A})$ is examined in the following lemma.

Lemma 12.3 (Dubey and Neogy [8]) *Consider the* LCP$(\tilde{q}, \mathcal{A})$ *in Lemma 12.2 and* LCP(q, A). *Then* LCP(q, A) *has a solution if and only if* LCP$(\tilde{q}, \mathcal{A})$ *has a solution.*

Theorem 12.2 (Dubey and Neogy [8]) *Suppose* $A \in$ *hidden* Z. *Then* $A \in Q_0$.

Proof It is noted that the feasibility of LCP(q, A) implies the feasibility of LCP$(\tilde{q}, \mathcal{A})$. Since $\mathcal{A} \in$ PSD, therefore $\mathcal{A} \in Q_0$ implies that LCP$(\tilde{q}, \mathcal{A})$ has a solution. By Lemma 12.3, LCP(q, A) has a solution if and only if LCP$(\tilde{q}, \mathcal{A})$ has a solution. Thus, the feasibility of LCP(q, A) implies solvability of LCP(q, A). Hence $A \in Q_0$. □

Theorem 12.3 (Dubey and Neogy [8]) *Let* $\wp_\alpha(A) \in \mathbb{R}^{n\times n}$ *be a PPT of a given matrix* $A \in \mathbb{R}^{n\times n}$ *for* $\alpha \subseteq \{1, \ldots, n\}$. *Suppose* $\wp_\alpha(A) \in$ *hidden* $Z \cap S$. *Then* $A \in Q$.

Proof Assume that for $\alpha \subseteq \{1, \ldots, n\}$, $\wp_\alpha(A) \in$ hidden $Z \cap S$. It follows from Proposition 12.1 and Lemma 12.1 that $A \in$ hidden $Z \cap S$. Further, by Theorem 12.2, we get $A \in Q_0 \cap S$. Since $Q = Q_0 \cap S$ (see Cottle et al. [5, p. 146]), therefore $A \in Q$. □

It should be noted that it is unknown whether or not Lemke's algorithm can solve LCP(q, A) involving a hidden Z-matrix. However, in the following proposition, we see that the equivalent LCP$(\tilde{q}, \mathcal{A})$ is processable by Lemke's algorithm.

Proposition 12.2 (Dubey and Neogy [8]) LCP$(\tilde{q}, \mathcal{A})$ *is processable by Lemke's algorithm.*

Proof Note that $\mathcal{A} \in$ PSD $\subset P_0 \cap Q_0$. As a result of Aganagic and Cottle's [1] theorem, LCP$(\tilde{q}, \mathcal{A})$ is processable by Lemke's algorithm. □

Neogy and Das [23] proposed an open problem for the complete characterization of the class of matrices with at least one PPT that is a Z-matrix. Dubey and Neogy [8] observed that this class is a subclass of the hidden Z class. Let $\mathcal{Z}$ denote the class of matrices for which at least one PPT is a Z-matrix.

Theorem 12.4 (Dubey and Neogy [8]) $\mathcal{Z} \subset$ *hidden* Z.

Proof Let $A \in \mathcal{Z}$ and for $\alpha = \alpha_0 \subseteq \{1, \ldots, n\}$, $\wp_{\alpha_0}(A) \in Z$. This implies $\wp_{\alpha_0}(A) \in$ hidden Z. From Lemma 12.1, it follows that $A \in$ hidden Z. □

The matrix A in Example 3 demonstrates that $\mathcal{Z}$ does not capture the entire hidden Z class. It is easy to show that $\wp_\alpha(A) \notin Z$ for any $\alpha \subseteq \{1, \ldots, n\}$.

12.3.1 Various Game Theory-Based Characterizations of Hidden Z-Matrices

In this section, we demonstrate how two-player matrix game results from Von Neumann and Morgenstern [32], and Raghavan [30] can be applied to generate interesting results in LCP and related matrix classes.

The value of the matrix game corresponding to the payoff matrix $A \in \mathbb{R}^{n\times n}$ is maintained in all of its PPTs, indicating that the value of the matrix game $v(A) > 0$ if and only if $v(\wp_\alpha(A)) > 0$, Neogy and Das [23]. The existence of solutions to the LCP(q, A) is fundamentally dependent on the study of the matrix classes Q and Q_0. The value of the matrix game $v(A)$ can be used to associate the matrix classes Q and Q_0. Indeed, $A \in Q$ if and only if $A \in Q_0$ for any matrix game with $v(A) > 0$.

The following lemma proved by Raghavan [30] is useful to prove the subsequent theorem.

Lemma 12.4 (Raghavan [30]) *Let $A \in \mathbb{R}^{n\times n}$ be a payoff matrix whose off-diagonal entries are nonpositive. If $v(A) > 0$, then the game is completely mixed.*

Dubey and Neogy [8] have generalized the above lemma in the following theorem.

Theorem 12.5 *Let $A \in \mathcal{Z}$. If $v(A) > 0$ then the game is completely mixed.*

Proof It is worth noting that $A \in \mathcal{Z}$ implies that $\wp_\alpha(A) \in Z$ for some $\alpha \subseteq \{1, \ldots, n\}$. From Theorem 12.2, $A \in Q_0$. In addition, since $v(A) > 0$, $A \in Q$. Since the PPT of a Q-matrix is a Q-matrix, it follows that $\wp_\alpha(A) \in Q$. As a result, $v(\wp_\alpha(A)) > 0$. From Lemma 12.4, the game with payoff matrix $\wp_\alpha(A)$ is completely mixed. We find that the game with payoff matrix $A \in \mathcal{Z}$ is also completely mixed, since the payoff matrix A is a PPT of payoff matrix $\wp_\alpha(A)$. □

The following is another result involving hidden Z-matrices presented by Dubey and Neogy [8].

Theorem 12.6 *Let A be a hidden Z-matrix; there exists a positive diagonal matrix D such that $\left(AD + DA^T\right)$ is positive definite. Then $A \in Q$.*

Proof The matrix A is hidden Z implies that $A \in Q_0$. To prove that $A \in Q$, we need to show that $v(A) > 0$. Suppose to contrary that $v(A) \leq 0$. Then there exists a probability vector $y \in \mathbb{R}^n$ such that $y^T A \leq 0$. Thus for a positive diagonal matrix D, $y^T AD \leq 0$. Now $y^T \left(AD + DA^T\right) y = y^T ADy + y^T DA^T y \leq 0$. This contradicts the positive definiteness of $\left(AD + DA^T\right)$. Therefore $v(A) > 0$. Hence $A \in Q$. □

12.4 The Class of Positive Subdefinite Matrices

In this section, we study positive subdefinite matrices (PSBD) introduced by Crouzeix et al. [7] and show that LCPs with PSBD matrices of rank ≥ 2 are processable by Lemke's algorithm and that a PSBD matrix of rank ≥ 2 belongs to the class of sufficient matrices studied by Cottle et al. [4]. The class of PSBD matrices is a generalization of the class of PSD matrices and is helpful in the study of quadratic programming problems.

Knowing whether a PSD matrix's properties also apply to a PSBD matrix is important, since a PSBD matrix is a natural generalization of a PSD matrix. We may particularly examine whether

(i) A is PSBD if and only if $(A + A^T)$ is PSBD,
(ii) any PPT of a PSBD matrix is a PSBD matrix.

The following examples demonstrate that the above statements are not true.

Example 4

Consider a matrix

$$A = \begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}.$$

Then for any vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $x^T Ax = 2x_1x_2 < 0$ implies x_1 and x_2 are of opposite sign. Clearly, $A \in$ PSBD because $x^T Ax < 0$ and $A^T x = \begin{bmatrix} -2x_2 \\ 4x_1 \end{bmatrix}$ imply either $A^T x \leqslant 0$ or $A^T x \geqslant 0$.

Also, we can easily see that

$$A + A^T = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

is not a PSBD matrix.

Similarly, consider a matrix

$$A = \begin{bmatrix} 0 & -4 \\ 2 & 0 \end{bmatrix} \quad \text{so that} \quad A + A^T = \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}.$$

It can be easily verified that $A + A^T$ is PSBD, but A is not a PSBD matrix.

Example 5

Consider the matrix

$$A = \begin{bmatrix} 0 & 4 \\ -2 & 0 \end{bmatrix}$$

as in Example 4. It should be noted that $A \in$ PSBD, but it is easy to verify that

$$A^{-1} = \begin{bmatrix} 0 & -0.5 \\ 0.25 & 0 \end{bmatrix}$$

is not a PSBD matrix. Since A^{-1} is a PPT of A, therefore any PPT of a PSBD matrix is not a PSBD matrix.

We now know whether PSBD $\subseteq Q_0$ and PSBD matrices are processable by Lemke's algorithm. We need the following results to prove the subsequent theorems.

Theorem 12.7 (Crouzeix et al. [7]) *Let* $A = ab^T$, *where* $a \neq b$; $a, b \in \mathbb{R}^n$. A *is* PSBD *if and only if one of the following conditions holds:*

(i) there exists a $t > 0$ *such that* $b = ta$,
(ii) for all $t > 0$, $b \neq ta$ *and either* $b \geqslant 0$ *or* $b \leqslant 0$.

Furthermore, assume that $A \in$ MPSBD. *Then* $A \in C_0$ *if and only if either* ($a \geqslant 0$ *and* $b \geqslant 0$) *or* ($a \leqslant 0$ *and* $b \leqslant 0$) *and* $A \in C_0^*$ *if and only if* $A \in C_0$ *and* $b_i = 0 \Rightarrow a_i = 0$.

Theorem 12.8 (Crouzeix et al. [7]) *Suppose* $A \in \mathbb{R}^{n\times n}$ *is* PSBD *and* rank$(A) \geqslant 2$. *Then* A^T *is* PSBD *and at least one of the following conditions holds:*

(i) $A \in$ PSD;
(ii) $\left(A + A^T\right) \leqslant 0$;
(iii) $A \in C_0^*$.

Lemma 12.5 (Mohan et al. [14]) *Suppose* $A \in \mathbb{R}^{n\times n}$ *is a* PSBD *matrix with* rank$(A) \geqslant 2$ *and* $A + A^T \leqslant 0$. *If* A *is not a skew-symmetric matrix, then* $A \leqslant 0$.

Theorem 12.9 (Mohan et al. [14]) *Suppose* $A \in \mathbb{R}^{n\times n}$ *is a* PSBD *matrix with rank(A)* ≥ 2. *Then* $A \in Q_0$.

Proof By Theorem 12.8, $A^T \in$ PSBD. Also, by the same theorem, either $A \in$ PSD or $\left(A + A^T\right) \leqslant 0$ or $A \in C_0^*$. If $A \in C_0^*$, then $A \in Q_0$ (see Cottle et al. [5]). Now, if $\left(A + A^T\right) \leqslant 0$, and A is not skew-symmetric, then from Lemma 12.5, we have $A \leqslant 0$. In this case, $A \in Q_0$. However, if A is skew-symmetric, then $A \in$ PSD. Hence $A \in Q_0$. □

Corollary 12.1 (Mohan et al. [14]) *Let A be a* PSBD *matrix with* $\text{rank}(A) \geqslant 2$. *Then Lemke's algorithm can solve* LCP(q, A). *If* $\text{rank}(A) = 1$, *(that is* $A = ab^T$, a, $b \neq 0$*) and A is a copositive matrix, then* LCP(q, A) *can be processed by Lemke's algorithm whenever* $b_i = 0 \Rightarrow a_i = 0$.

Proof Assume $\text{rank}(A) \geqslant 2$. By Theorem 12.8 and from the proof of Theorem 12.9, it follows that A is either a PSD matrix or $A \leqslant 0$ or $A \in C_0^*$. Thus, LCP(q, A) can be processed by Lemke's algorithm (see Cottle et al. [5]). For PSBD $\cap C_0$ matrices of $\text{rank}(A) = 1$, i.e., for $A = ab^T$, a, $b \neq 0$, such that $b_i = 0 \Rightarrow a_i = 0$. By Theorem 12.7, we have $A \in C_0^*$. Hence, LCP(q, A) with such matrices can be processed by Lemke's algorithm. □

Theorem 12.10 (Mohan et al. [14]) *Let* $A \in \mathbb{R}^{n \times n}$ *be a* PSBD $\cap\, C_0$ *matrix with rank(A)* ≥ 2. *Then A is a sufficient matrix.*

The following example demonstrates that PSBD matrices need not be P_0-matrix in general.

Example 6

Consider a matrix

$$A = \begin{bmatrix} 0 & -2 \\ -2 & 0 \end{bmatrix}.$$

Then for any vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$,

$$x^T Ax = -4x_1x_2 < 0$$

implies that x_1 and x_2 are of the same sign. Since $A^T x = \begin{bmatrix} -x_2 \\ -x_1 \end{bmatrix}$ implies either $A^T x \leqslant 0$ or $A^T x \geqslant 0$, A is PSBD matrix. But A is not a P_0-matrix.

The following example studied by Mohan et al. [14] demonstrates that in general PSBD matrices need not be Q_0-matrix.

Example 7

Consider a matrix

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

Then for any vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$,

$$A^T x = \begin{bmatrix} x_1 + x_2 \\ 0 \end{bmatrix}$$

implies either $A^T x \leqslant 0$ or $A^T x \geqslant 0$. Therefore A is a PSBD matrix. Take $q = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$. It is easy to see that LCP(q, A) is feasible but has no solution. Hence $A \notin Q_0$.

12.5 The Class of Generalized Positive Subdefinite Matrices

This section is dedicated to the study of the class of generalized positive subdefinite (GPSBD) matrices introduced by Crouzeix and Komlósi [6]. Further, we study some properties of GPSBD matrices in the context of LCP. Finally, we show that if a matrix A can be written as a sum of a copositive-plus MGPSBD matrix with some additional condition and a copositive matrix and if it satisfies the feasibility condition, then LCP(q, A) is processable by Lemke's algorithm.

Definition 12.2 (Crouzeix and Komlósi [6]) A matrix $A \in \mathbb{R}^{n\times n}$ is said to be a GPSBD matrix if there exist nonnegative multipliers $s_i,\ t_i$ with $s_i + t_i = 1,\ i = 1, 2, \ldots, n$ such that

$$\text{for all } z \in \mathbb{R}^n, \quad z^T Az < 0 \Rightarrow \begin{cases} \text{either} & -s_i z_i + t_i \left(A^T z\right)_i \geq 0 \ \ \forall\, i, \\ \text{or} & -s_i z_i + t_i \left(A^T z\right)_i \leq 0 \ \ \forall\, i. \end{cases}$$

Remark 12.1 It is easy to prove that the matrix A is PSBD if and only if A is GPSBD with multipliers $s_i = 0$ and $t_i = 1$ for all i.

Neogy and Das [24] have introduced an equivalence definition of GPSBD matrices.

Definition 12.3 Let S and T be two nonnegative diagonal matrices with diagonal entries $s_i,\ t_i$, where $s_i + t_i = 1$ for $i = 1, \ldots, n$. A matrix $A \in \mathbb{R}^{n\times n}$ is said to be GPSBD if there exist two nonnegative diagonal matrices S and T with $S + T = I$ such that

$$\text{for all } z \in \mathbb{R}^n,\ z^T Az < 0 \ \Rightarrow \begin{cases} \text{either} & -Sz + TA^T z \geq 0, \\ \text{or} & -Sz + TA^T z \leq 0.. \end{cases}$$

Remark 12.2 Note that if $S = 0$, then GPSBD matrix reduces to PSBD matrix. A is called a merely generalized positive subdefinite (MGPSBD) matrix if A is a GPSBD matrix but not a PSBD matrix.

A nontrivial example of a GPSBD matrix is shown below.

Example 8 (Neogy and Das [24])

Consider a matrix

$$A = \begin{bmatrix} 0 & 2 & 0 \\ -1 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}.$$

It should be noted that $\nu_-\left(A + A^T\right) = 1$. Then for any vector $z = \begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix}^T$, $z^T Az = z_1 z_2 < 0$ implies that z_1 and z_2 have opposite signs. Clearly, $A^T z = \begin{bmatrix} -z_2 & 2z_1 + z_3 & -z_2 \end{bmatrix}^T$ and for $z = \begin{bmatrix} -1 & 1 & 5 \end{bmatrix}^T$, $z^T Az < 0$ does not indicate $A^T z$ is unisigned. Thus, A is not a PSBD matrix. However, by selecting $s_1 = 0,\ s_2 = 1$ and $s_3 = 0$, it is easy to see that A is a GPSBD matrix.

A careful analysis of the GPSBD matrix definition yields the following result.

Proposition 12.3 (Neogy and Das [24]) *A matrix $A \in \mathbb{R}^{n\times n}$ is a* GPSBD *matrix if any one of the following conditions holds:*

(i) $Z_1 \subseteq Z_2$, *where* $Z_1 = \left\{z \mid z^T Az < 0\right\}$ *and* $Z_2 = \left\{z \mid z^T\left(-S + TA^T\right)z < 0\right\}$ *and* $(-S + AT)$ *is* PSBD.
(ii) $t_i = k,\ 0 \le k \le 1,\ \forall\, i$ *and* $(-S + AT)$ *is* PSBD.
(iii) $\bar{Z} = \left\{z \mid z^T Az < 0\right\} \subseteq \mathbb{R}^n_+ \cup -\mathbb{R}^n_+$.
(iv) $a_{ii} \ge 0\ \forall\, i$ *and there exists an* $i = i_0$ *such that* $A_{i_0 \cdot} \ge 0$ *and* $a_{ij} = 0\ \forall\, i, j,\ i \ne i_0,\ j \ne i$.
(v) $a_{ii} \ge 0\ \forall\, i$ *and there exists a* $j = j_0$ *such that* $A_{\cdot j_0} \ne 0$ *and* $a_{ij} = 0\ \forall\, i, j,\ i \ne j,\ j \ne j_0$.

Neogy and Das [24] studied the following result on GPSBD matrices.

Theorem 12.11 *Let A be a* GPSBD *matrix. Then for $\alpha \subseteq \{1, \ldots, n\}$, $A_{\alpha\alpha} \in$* GPSBD.

Proof Let $x_\alpha \in \mathbb{R}^{|\alpha|}$ and for $\alpha \subseteq \{1, \ldots, n\}$,

$$A = \begin{bmatrix} A_{\alpha\alpha} & A_{\alpha\bar{\alpha}} \\ A_{\bar{\alpha}\alpha} & A_{\bar{\alpha}\bar{\alpha}} \end{bmatrix}.$$

Let $x_\alpha^T A_{\alpha\alpha} x_\alpha < 0$. Now define $z \in \mathbb{R}^n$ by taking $z_\alpha = x_\alpha$ and $z_{\bar{\alpha}} = 0$. Then $z^T Az = x_\alpha^T A_{\alpha\alpha} x_\alpha$. Since A is a GPSBD matrix, $z^T Az = x_\alpha^T A_{\alpha\alpha} x_\alpha < 0$ implies either $\left(-S + TA^T\right)z \ge 0 \Rightarrow \left(-S_{\alpha\alpha} + T_{\alpha\alpha} A_{\alpha\alpha}^T\right) x_\alpha \ge 0$, or $\left(-S + TA^T\right)z \le 0 \Rightarrow (-S_{\alpha\alpha} + T_{\alpha\alpha} A_{\alpha\alpha}^T) x_\alpha \le 0$. Therefore $A_{\alpha\alpha} \in$ GPSBD. □

The above theorem shows that GPSBD is a complete class in the sense of Cottle et al. [5]. Crouzeix et al. [7] have shown that if A is a PSBD matrix with rank(A) ≥ 2, then A^T is also a PSBD matrix. However, the following example shows that in general this is not true for GPSBD matrices.

Example 9 (Neogy and Das [24])

Consider a matrix

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 10 & 0 & 1 \end{bmatrix}.$$

Take $z = \begin{bmatrix} 1 & \frac{5}{2} & -1 \end{bmatrix}^T$. Clearly, A and A^T are not PSBD matrices. Now by selecting $s_1 = 1,\ s_2 = \frac{1}{2},\ s_3 = \frac{1}{2}$; it is easy to verify that A is a GPSBD matrix.

Now consider A^T. It should be noted that for $z = \begin{bmatrix} 1 & \frac{5}{2} & -1 \end{bmatrix}^T$,

$$(-S + TA)z = \begin{bmatrix} (1-2s_1)\,z_1 - (1-s_1)\,z_2 \\ -(1-s_2)\,z_1 + (1-2s_2)\,z_2 \\ 10\,(1-s_3)\,z_1 + (1-2s_3)\,z_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}s_1 - \frac{3}{2} \\ -4s_2 + \frac{3}{2} \\ -8s_3 + 9 \end{bmatrix}.$$

It is easy to verify that there does not exist any $s_1,\ s_2,\ s_3$ for which A^T satisfies the definition of a GPSBD matrix.

Mohan et al. [14] have proved that a copositive PSBD matrix of rank ≥ 2 is sufficient. However, the following example demonstrates that a GPSBD matrix is not always sufficient.

Example 10

Consider the GPSBD matrix A from Example 9. We note that A is not a PSBD matrix. It is easy to verify that A is a row sufficient matrix, though A is not a column sufficient matrix.

We study the following result on row sufficiency, proved by Neogy and Das [24].

Theorem 12.12 *Let $A \in \text{MGPSBD} \cap C_0$ with $0 < t_i < 1\ \forall\, i$. Then A is a row sufficient matrix.*

Proof Assume $z_i \left(A^T z\right)_i \leq 0$ for all $i = 1, \ldots, n$. Let $I_1 = \{i : z_i > 0\}$ and $I_2 = \{i : z_i < 0\}$. Now consider three cases.

Case (i): $I_2 = \emptyset$. Then $z^T Az = z^T A^T z = \sum_i z_i \left(A^T z\right)_i \leq 0$. Since $A \in C_0$, $\left[z_i \left(A^T z\right)_i\right] = 0\ \forall\, i$.

Case (ii): $I_1 = \emptyset$. Then $(-z)^T A^T (-z) = z^T A^T z = \sum_i z_i \left(A^T z\right)_i \leq 0$. Since $A \in C_0$, $\left[z_i \left(A^T z\right)_i\right] = 0\ \forall\, i$.

Case (iii): Let us assume that there exists a vector z such that $z_i \left(A^T z\right)_i \leq 0$ for all $i = 1, 2, \ldots, n$ and $z_k \left(A^T z\right)_k < 0$ for at least one $k \in \{1, 2, \ldots, n\}$. Let $I_1 \neq \emptyset$ and $I_2 \neq \emptyset$. $z^T Az = z^T A^T z = \sum_i \left[z_i \left(A^T z\right)_i\right] < 0$, which implies $-s_i z_i + t_i \left(A^T z\right)_i \geq 0 \; \forall \; i$ or $-s_i z_i + t_i \left(A^T z\right)_i \leq 0 \; \forall \; i$.

Without loss of generality, assume that $-s_i z_i + t_i \left(A^T z\right)_i \geq 0 \; \forall \; i$. Then for all $i \in I_1$, $-s_i z_i^2 + t_i z_i \left(A^T z\right)_i \geq 0$, which implies $\left[z_i \left(A^T z\right)_i\right] \geq \frac{s_i}{t_i} z_i^2 > 0 \; \forall \; i \in I_1$. Therefore, $\sum_{i \in I_1} \left[z_i \left(A^T z\right)_i\right] > 0$. Since $z_i \left(A^T z\right)_i \leq 0$ for all $i = 1, \ldots, n$, it results in a contradiction. Thus, $\left[z_i \left(A^T z\right)_i\right] = 0 \; \forall \; i$. Hence A is row sufficient. □

The following example demonstrates that the assumption $0 < t_i < 1 \; \forall \; i$ in the above theorem cannot be relaxed.

Example 11 (Neogy and Das [24])

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & 0 & 0 \end{bmatrix}.$$

It is easy to see that A is copositive but not a row sufficient matrix. For $z = \begin{bmatrix} -1 & -1 & 1 \end{bmatrix}^T$, $z^T Az < 0$. Clearly, A is not PSBD. Now for $z = \begin{bmatrix} -1 & -1 & 1 \end{bmatrix}^T$,

$$\left(-S + TA^T\right) z = \begin{bmatrix} (1 - 2s_1) z_1 + (1 - s_1) 2z_2 + (1 - s_1) 10 z_3 \\ (1 - 2s_2) z_2 \\ -s_3 z_3 \end{bmatrix} = \begin{bmatrix} 7 - 6s_1 \\ 2s_2 - 1 \\ -s_3 \end{bmatrix}.$$

Note that there does not exist $s_1, \; s_2, \; s_3$, where $0 < s_i < 1 \; \forall \; i$, i.e., there does not exist t_1, t_2, t_3 where $0 < t_i < 1 \; \forall \; i$ for which the definition of GPSBD matrix is satisfied. However, by selecting $s_1 = \frac{1}{2}, s_2 = \frac{1}{2}, s_3 = 0$, A is a MGPSBD matrix.

Murthy and Parthasarathy [17] have studied the following characterization on nonnegative Q_0-matrices, which is useful in the subsequent theorem.

Theorem 12.13 (Murthy and Parthasarathy [17]) *Suppose A is nonnegative, where $n \geq 2$. Then, $A \in Q_0$ if and only if for every $i \in \{1, 2, \ldots, n\}$, $A_{i\cdot} \neq 0 \Rightarrow a_{ii} > 0$.*

Theorem 12.14 (Neogy and Das [24]) *Suppose A is a nonnegative* MGPSBD *matrix with $0 < t_i < 1 \; \forall \; i$. Then $A \in C_0^*$.*

Proof From Theorem 12.12, A is row sufficient. Therefore, $A \in Q_0$ Cottle et al. [4, p. 239]. Now from Theorem 12.13, for any nonnegative Q_0-matrix $A_{i\cdot} \neq 0 \Rightarrow a_{ii} > 0$. Let $\alpha = \{i \mid a_{ii} > 0\}$. Then $x^T Ax = 0$ for $x_\alpha = 0$. It is easy to verify that for $(0, x_{\bar{\alpha}})$, $Ax \geq 0 \Rightarrow A^T x \leq 0$. Hence $A \in C_0^*$. □

From Theorem 12.12, we conclude that LCP(q, A) where $A \in C_0 \cap$ MGPSBD matrix with $0 < t_i < 1$ is processable by Lemke's algorithm.

Neogy and Das [24] obtained the following result for solving LCP(q, A) by Lemke's algorithm when A satisfies certain conditions stated in the theorem below.

Theorem 12.15 (Neogy and Das [24]) *Assume $A \in \mathbb{R}^{n\times n}$ can be written as $M + N$, where $M \in$ MGPSBD $\cap\, C_0^+$ is nondegenerate with $0 < t_i < 1\ \forall\ i$, and $N \in C_0$. If the system $q + Mx - N^T y \geq 0,\ y \geq 0$, is feasible, then Lemke's algorithm for* LCP(q, A) *with covering vector $d > 0$ terminates with a solution.*

The class MGPSBD $\cap C_0^+$ is shown nonempty in the following example.

Example 12 (Neogy and Das [24])

Consider the copositive-plus matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 8 & 0 & 1 \end{bmatrix}.$$

Take $z = \begin{bmatrix} -1 & -1 & 1 \end{bmatrix}^T$. Note that A is not MPSBD. However, by selecting $s_i = \frac{1}{2}\ \forall\ i,\ A$ is a MGPSBD matrix.

12.6 The Class of WGPSBD Matrices

In this section, we study the class of weak generalized positive subdefinite matrices (WGPSBD) introduced by Neogy and Das [26]. WGPSBD is a weaker version of the class of GPSBD matrices. Row sufficient matrices, proposed by Cottle et al. [4], also capture this weaker class of matrices.

Definition 12.4 A matrix $A \in \mathbb{R}^{n\times n}$ is called a weak generalized positive subdefinite (WGPSBD) matrix if there exist two nonnegative diagonal matrices S and T with $S + T = I$ such that for all $z \in \mathbb{R}^n,\ z^T Az < 0$ implies at least $(n - 1)$ coordinates of the vector $(-Sz + TA^T z)$ are unisigned.

Note 12.1 WGPSBD contains GPSBD.

Remark 12.3 A matrix A is called a merely weak generalized positive subdefinite (MWGPSBD) matrix if $A \in$ WGPSBD but $A \notin$ GPSBD.

Another version of WGPSBD matrices can be defined as follows.

Definition 12.5 A matrix $A \in \mathbb{R}^{n\times n}$ is said to be a WGPSBD matrix if there exist nonnegative multipliers $s_i,\ t_i$ with $s_i + t_i = 1,\ i = 1, 2, \ldots, n$ such that

$$\text{for all } z \in \mathbb{R}^n, \quad z^T Az < 0 \ \text{ implies}$$

either $\{$at least $(n-1)$ coordinates of $-s_i z_i + t_i \left(A^T z\right)_i \geq 0,\ i \in \mathcal{I} \subseteq \{1, 2, \ldots, n\}\}$ or $\{$at least $(n-1)$ coordinates of $-s_i z_i + t_i \left(A^T z\right)_i \leq 0,\ i \in \mathcal{I} \subseteq \{1, 2, \ldots, n\}\}$.

The below example is a nontrivial example of WGPSBD $\cap\ P_0$.

Example 13 (Neogy and Das [26])

Consider the matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It is easy to verify that A is a copositive matrix. Now we show that $A \notin$ GPSBD.

It should be noted that for $z = \left[1, \frac{1}{2}, -1\right]^T,\ z^T Az < 0.$

$$\left(-S + TA^T\right) z = \begin{bmatrix} (1-2s_1)\, z_1 - (1-s_1)\, z_2 \\ -(1-s_2)\, z_1 + (1-2s_2)\, z_2 \\ 2\,(1-s_3)\, z_1 + (1-2s_3)\, z_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - \frac{3}{2}s_1 \\ -\frac{1}{2} \\ 1 \end{bmatrix}.$$

It is clear that there are no $s_1,\ s_2,\ s_3$ for which A meets the criteria for a GPSBD matrix. However, by definition A is a WGPSBD matrix.

Neogy and Das [26] studied the following result on WGPSBD matrices.

Theorem 12.16 *Let A be a* WGPSBD *matrix. Then $A_{\alpha\alpha} \in$* WGPSBD*, where $\alpha \subseteq \{1, \ldots, n\}$.*

12.7 The Class of GPSBD Matrices of Level *k*

This section introduces the class of generalized positive subdefinite (GPSBD) matrices of level k, studied by Dubey and Neogy [9]. This class is a subclass of row sufficient matrices, which is not a subclass of copositive matrices. Dubey and Neogy [9] relaxed the definition of a GPSBD matrix and defined a GPSBD matrix of level k to identify subclasses of row sufficient matrices.

Definition 12.6 (Dubey and Neogy [9]) A matrix $A \in \mathbb{R}^{n\times n}$ is called a GPSBD matrix of level k (GPSBD$_k$), $1 \le k \le n$, if there exist two nonnegative diagonal matrices S and T with $S + T = I$ such that for all $z \in \mathbb{R}^n$, $z^T Az < 0$ implies exactly k coordinates of the vector $\left(-Sz + TA^T z\right)$ are nonnegative or nonpositive and the remaining $(n - k)$ coordinates of the vector $\left(-Sz + TA^T z\right)$ are negative or positive, respectively.

The class of GPSBD matrices of level k is denoted by GPSBD$_k$. It should be noted that for $k = n$, GPSBD$_n$ = GPSBD and WGPSBD = GPSBD$_{n-1} \cup$ GPSBD. The following example shows that this class is nonempty.

Example 14 (Dubey and Neogy [9])

Let

$$A = \begin{bmatrix} 1 & -1 & 10 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

Note that

$$-Sz + TA^T z = \begin{bmatrix} (1-2s_1)z_1 + (1-s_1)z_2 + (1-s_1)z_3 \\ -(1-s_2)z_1 + (1-2s_2)z_2 \\ 10(1-s_3)z_1 + (1-2s_3)z_3 + (1-s_3)z_4 \\ -s_4 z_4 - (1-s_4)z_3 \\ (1-2s_5)z_5 + (1-s_5)z_6 + (1-s_5)z_7 \\ -(1-s_6)z_5 + (1-2s_6)z_6 \\ -s_7 z_7 - (1-s_7)z_5 \end{bmatrix}.$$

For $z = \begin{bmatrix} 1 & -1 & -1 & -1 & 1 & -1 & -1 \end{bmatrix}^T$, $z^T Az < 0$ and

$$-Sz + TA^T z = \begin{bmatrix} -1 \\ -2 + 3s_2 \\ 8 - 7s_3 \\ 1 \\ -1 \\ -2 + 3s_6 \\ -1 + 2s_7 \end{bmatrix}.$$

It is easy to see that there does not exist any $0 \le s_i \le 1$; therefore, A is not a GPSBD matrix. However, for $s_1 = 1$, $s_2 = \frac{1}{2}$, $s_3 = 1$, $s_4 = 0$, $s_5 = 1$, $s_6 = \frac{1}{2}$, $s_7 = 0$, A satisfies the definition of a GPSBD$_5$ matrix. As we have

$$-Sz + TA^T z = \begin{bmatrix} -z_1 \\ -\frac{1}{2}z_1 \\ -z_3 \\ -z_3 \\ -z_5 \\ -\frac{1}{2}z_5 \\ -z_5 \end{bmatrix}$$

and for $z^T Az = z_1^2 + z_2^2 + z_3^2 + z_5^2 + z_6^2 + 11z_1z_3 < 0$, z_1 and z_3 have to be of opposite sign.

We know that the class of GPSBD matrices is complete. However, for a GPSBD$_k$ matrix, Dubey and Neogy [9] proved a theorem which says that every principal submatrix of a GPSBD matrix of level k is also a GPSBD matrix of some level (say k_α which may or may not be equal to k). This ensures the completeness of the class $\bigcup_{k=1}^{n}$ GPSBD$_k$.

Theorem 12.17 (Dubey and Neogy [9]) *Suppose $A \in \mathbb{R}^{n\times n}$ is a* GPSBD$_k$ *matrix, $1 \le k \le n$. Then $A_{\alpha\alpha} \in$* GPSBD$_{k_\alpha}$ *where $\alpha \subseteq \{1, \ldots, n\}$ and $1 \le k_\alpha \le |\alpha|$.*

12.8 The Class of Fully Copositive Matrices

The aim of this section is to study the class of fully copositive (C_0^f) matrices and their properties. Further, we demonstrate that the fully copositive matrices with positive diagonal entries are column sufficient and fully copositive matrices which belong to Q_0 are sufficient.

Positive semidefinite matrices are known to be sufficient. Murthy et al. [20] demonstrated that $C_0^f \cap Q_0$ matrices are sufficient. Further, Mohan et al. [15] have shown how this result follows from the one that Cottle and Guu [3] proved earlier.

Theorem 12.18 *A matrix $A \in \mathbb{R}^{n\times n}$ is sufficient if and only if every matrix obtained from it by means of a PPT operation is sufficient of order* 2.

We need the following result to prove the subsequent theorem.

Theorem 12.19 (Murthy and Parthasarathy [18]) *If $A \in \mathbb{R}^{2\times 2} \cap C_0^f \cap Q_0$, then A is a* PSD *matrix.*

As a consequence, Mohan et al. [15] studied the following theorem.

Theorem 12.20 *Let $A \in C_0^f \cap Q_0$. Then A is sufficient.*

Proof Since A and all its PPTs are completely Q_0-matrices, therefore, all 2×2 submatrices of A or its PPTs are $C_0^f \cap Q_0$-matrices. From Theorem 12.19, all 2×2 submatrices of A or their PPTs are PSD and so sufficient. Thus, A or every matrix obtained by a PPT operation is sufficient of order 2. Hence from Theorem 12.18, A is sufficient. □

In view of the above result, we observe that if $A \in C_0^f \cap Q_0$, then it is processable by Lemke's algorithm.

Murthy and Parthasarathy [17] studied the following theorem.

Theorem 12.21 *Let $A \in \mathbb{R}^{n \times n}$ be a fully copositive matrix. Assume that $a_{ii} > 0 \forall i \in \{1, 2, \ldots, n\}$. Then $A \in P_0$.*

Contrary to what was stated above, we see that a C_0^f-matrix is a column sufficient matrix under the assumption of positive diagonal entries and that if a matrix A with positive diagonal entries and its transpose are in C_0^f, then such a matrix is in Q_0 and hence it is a completely Q_0-matrix.

Theorem 12.22 (Mohan et al. [15]) *Suppose $A \in \mathbb{R}^{n \times n}$ be a fully copositive matrix. Assume that $a_{ii} > 0 \; \forall \; i \in \{1, 2, \ldots, n\}$. Then $A \in Q_0$.*

(i) A is column sufficient.
(ii) In addition, if $A^T \in \mathbb{R}^{n \times n} \cap C_0^f$, then A is a completely Q_0-matrix.

The following example demonstrates how it is important to assume that A^T is also a fully copositive matrix in the previous theorem in order to get a stronger conclusion in (ii).

Example 15

Consider the matrix

$$A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

It is easy to see that $A \in C_0^f$, but $A \notin Q_0$. For example, the vector

$$q = \begin{bmatrix} -8 \\ -5 \\ 2 \end{bmatrix}$$

is feasible, but LCP(q, A) has no solution. Also, it is easy to see that $A^T \notin C_0^f$.

12.9 The Class of Singular N_0-Matrices

In this section, we study the class of singular N_0-matrices with Q-property. Pye [29] investigated nonsingular N_0-matrices and posed an obvious query on the connection to the class Q. The fact that singular N_0-matrices of orders 1 and 2 are not Q-matrices is easy to see. In his discussion of singular $N_0 \cap Q$-matrices, Pye [29] provides examples and observes that if A is a singular N_0-matrix and all of its principal minors are zero, then $A \notin Q$. Pye offers two examples of two singular N_0-matrices A_1 and A_2 where

$$A_1 = \begin{bmatrix} -1 & 2 & 1 \\ 2 & -1 & -\frac{1}{2} \\ 1 & 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

Here, $A_1 \in Q$, but $A_2 \notin Q$.

The following lemma is useful in the subsequent theorem.

Lemma 12.6 (Neogy and Das [25], p. 815) *Let $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 3)$ be an N_0-matrix. If A is nonsingular, then there exists a $\emptyset \neq \alpha \subseteq \{1, 2, \ldots, n\}$ so that A can be written in the partitioned form (if necessary, after a principal rearrangement of its rows and columns)*

$$A = \begin{bmatrix} A_{\alpha\alpha} & A_{\alpha\bar{\alpha}} \\ A_{\bar{\alpha}\alpha} & A_{\bar{\alpha}\bar{\alpha}} \end{bmatrix}$$

for $\emptyset \neq \alpha \subseteq \{1, 2, \ldots, n\}$ where $A_{\alpha\alpha} \leqslant 0$, $A_{\bar{\alpha}\bar{\alpha}} \leqslant 0$, $A_{\bar{\alpha}\alpha} \geqslant 0$ and $A_{\alpha\bar{\alpha}} \geqslant 0$.

Neogy and Das [25] studied the following result on singular N_0-matrices.

Theorem 12.23 (Neogy and Das [25]) *Let $A \in \mathbb{R}^{n\times n}$ be a singular N_0-matrix $(n \geqslant 3)$ with $v(A) > 0$. Suppose that A has the partitioned form as in Lemma 12.6 and the values of the games, determined by the proper principal submatrices of A of order 2 or more which contain at least one positive entry, are positive. If $A \in R_0$, then $A \in Q$.*

Using the above theorem in the following example, we deduce that the given matrix $A \in Q$.

Example 16 (Neogy and Das [25])

Consider the matrix

$$A = \begin{bmatrix} -4 & 2 & 4 & 5 \\ 1 & 0 & 0 & 0 \\ 2.1 & -1 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{bmatrix}.$$

Clearly, A is a singular N_0-matrix with partitioned form as in Lemma 12.6 and $v(A) > 0$. It is easy to see that $A \in R_0$. Also, the values of proper principal submatrices of

order (≥ 2) with at least one positive entry are positive. Hence by Theorem 12.23, it follows that $A \in Q$.

Theorem 12.24 (Neogy and Das [25]) *Let* $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 3)$. *If* $A \in N_0 \cap Q$, *then* $A \in R_0$.

12.10 Almost Type Classes of Matrices

This section is dedicated to the study of almost type classes of matrices with Q-property. Neogy and Das [22] introduce the class of almost $\bar{N}$-matrices which is a subclass of the almost N_0-matrices and obtained a sufficient condition to hold Q-property. The generalization of almost type class of matrices is called exact order matrices. In particular, the matrices of exact order 1 are called almost type class of matrices.

Definition 12.7 A matrix $A \in \mathbb{R}^{n\times n}$ is called an almost $\bar{N}$-matrix if there exists a sequence $\left\{A^{(k)}\right\}$ where $A^{(k)} = \left[a_{ij}^{(k)}\right]$ are almost N-matrices such that $a_{ij}^{(k)} \to a_{ij}$ for all $i, j \in \{1, 2, \ldots, n\}$.

Example 17 (Neogy and Das [22])

Consider the matrix

$$A = \begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 2 \\ 1 & 1 & -1 \end{bmatrix}.$$

It should be noted that A belongs to almost N_0-matrix. Also, it is easy to verify that A belongs to almost $\bar{N}$, since the matrix A can be obtained as a limit point of the sequence of almost N-matrices

$$A^{(k)} = \begin{bmatrix} -1 & 2 & 2 \\ 1/k & -1/k & 2 \\ 1 & 1 & -1 \end{bmatrix}$$

which converges to A as $k \to \infty$.

The following example demonstrates that the matrix class almost $\bar{N}$ is a proper subclass of the matrix class almost N_0.

Example 18 (Neogy and Das [22])

Consider the matrix

$$A = \begin{bmatrix} 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix}.$$

Note that A is an almost N_0-matrix. However, it is easy to see that A is not an almost $\bar{N}$-matrix because A cannot be obtained as a limit point of a sequence of almost N-matrices.

Now, we take almost N_0-matrices into consideration and propose the following question. Let $A \in$ almost N_0. Then, is it true that

(i) $A \in Q \Rightarrow A \in R_0$?
(ii) $A \in R_0 \Rightarrow A \in Q$?

We partially resolve the above-mentioned problems in the sequel. The following example shows that $A \in$ almost $N_0 \cap Q$, but $A \notin R_0$.

Example 19 (Neogy and Das [22])

Let

$$A = \begin{bmatrix} -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{bmatrix}.$$

Note that $A \in$ almost N_0. Now taking a PPT with respect to $\alpha = \{1, 3\}$, we get

$$\wp_\alpha(A) = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}.$$

Since $\wp_\alpha(A)$, a PPT $A \in Q$. Therefore $A \in Q$. However, $(0, 1, 0, 0)$ solves LCP$(0, A)$ implies $A \notin R_0$.

Olech et al. [27, p. 120] demonstrate that an almost N_0-matrix, even with a positive value, not necessarily belongs to the class Q or R_0.

Example 20

Consider the following matrix A with vector q,

$$A = \begin{bmatrix} -2 & -2 & -2 & 2 \\ -2 & -1 & -3 & 3 \\ -2 & -3 & -1 & 3 \\ 2 & 3 & 3 & 0 \end{bmatrix}, \quad q = \begin{bmatrix} -1001 \\ -500 \\ -500 \\ 500 \end{bmatrix}.$$

It is easy to verify that $A \in$ almost N_0, but $A \notin Q$ even though $v(A) > 0$. Also, $A \notin R_0$.

However, in the following theorem, Neogy and Das [22] proved that if $A \in$ almost $\bar{N} \cap R_0$ and $v(A) > 0$, then $A \in Q$.

Theorem 12.25 *Suppose $A \in$ almost $\bar{N} \cap \mathbb{R}^{n\times n}$, $n \geq 4$ with $v(A) > 0$. Then $A \in Q$ if $A \in R_0$.*

12.10.1 A Generalization of Almost Type Class of Matrices

Neogy and Das [22] introduced the $\bar{N}$-matrix of exact order 2, which is a generalization of almost $\bar{N}$-matrix. This class can be obtained as a limit of a sequence of N-matrices of exact order 2.

Definition 12.8 A matrix $A \in \mathbb{R}^{n\times n}$ is called an $\bar{N}$-matrix of exact order 2 if there exists a sequence $\left\{A^{(k)}\right\}$ where $A^{(k)} = \left[a_{ij}^{(k)}\right]$ are N-matrices of exact order 2 such that $a_{ij}^{(k)} \to a_{ij}$ for all $i, j \in \{1, 2, \ldots, n\}$.

Example 21

Consider the matrix

$$A = \begin{bmatrix} 0 & -90 & -80 & -70 & 0 \\ -90 & -2 & -2 & -2 & 2 \\ -70 & -2 & -1 & -3 & 3 \\ -50 & -2 & -3 & -0.8 & 3 \\ 0 & 2 & 3 & 3 & 0 \end{bmatrix}.$$

Note that A is an N_0-matrix of exact order 2. Further, since A can be obtained as a limit point of the sequence of N-matrices of exact order 2, therefore, A is an $\bar{N}$-matrix of exact order 2:

$$A^{(k)} = \begin{bmatrix} -1/k^2 & -90 & -80 & -70 & 1/k \\ -90 & -2 & -2 & -2 & 2 \\ -70 & -2 & -1 & -3 & 3 \\ -50 & -2 & -3 & -0.8 & 3 \\ 1/k & 2 & 3 & 3 & -1/k^2 \end{bmatrix}$$

which converges to A as $k \to \infty$.

The following example demonstrates that an N_0-matrix of exact order 2 need not be an $\bar{N}$-matrix of exact order 2.

Example 22

Consider the matrix

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & -1 & -1 & 2 \\ 0 & -1 & 0 & -1 & 1 \\ 1 & -1 & -1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

Note that A is an N_0-matrix of exact order 2. However, since we are unable to obtain A as the limit of a sequence of N-matrices of exact order 2, A is not an $\bar{N}$-matrix of exact order 2.

In the following theorem, Neogy and Das [22] proved a sufficient condition for $\bar{N}$-matrix of exact order 2 to hold Q-property.

Theorem 12.26 (Neogy and Das [22]) *Let $A \in \bar{N} \cap \mathbb{R}^{n\times n}$, $n \geq 5$ of exact order 2 with $v(A) > 0$. Suppose there exists at most one nonpositive principal submatrix of order $n-1$ and the values of proper principal submatrices of order (≥ 2), which contains at least one positive entry, are positive. Then $A \in Q$ if $A \in R_0$.*

12.11 Principal Pivot Transforms of some Classes of Matrices

The aim of this section is to study and characterize various classes of matrices that are defined based on the principal pivot transforms (PPTs). Neogy and Das [23] look at a few additional classes that were defined using PPTs. One of these classes has the characteristic that all of its PPTs are either C_0 or almost C_0, with at least one PPT being almost C_0. The other class that is taken into consideration in this study has the property that all of its PPTs are either E_0 or almost C_0, with at least one PPT being almost C_0. Take note that an almost C_0-matrix is not always an E_0-matrix.

By demonstrating that this class is in P_0, Neogy and Das [23] proved that if it also belongs to Q_0, then it is in E_0^f.

Definition 12.9 A matrix $A \in \mathbb{R}^{n\times n}$ is called an almost fully copositive (almost C_0^f)-matrix if its PPTs are either C_0 or almost C_0 and there exists at least one PPT $\wp_\alpha(A)$ of A for some $\alpha \subset \{1, 2, \ldots, n\}$ that is almost C_0.

Example 23 (Neogy and Das [23])

Consider the matrix

$$A = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

It is easy to see that A is an almost fully copositive matrix.

Theorem 12.27 (Neogy and Das [23]) *Let $\wp_\alpha(A) \in \mathbb{R}^{n\times n}$ be a PPT of a given matrix $A \in \mathbb{R}^{n\times n}$. Then $v(A) > 0$ if and only if $v(\wp_\alpha(A)) > 0$.*

Proof It is sufficient to show that $v(A) > 0 \Rightarrow v(\wp_\alpha(A)) > 0$. Assume $v(A) > 0$, then there exists a $z > 0$ such that $Az > 0$.

Let

$$\begin{bmatrix} w_\alpha \\ w_{\bar{\alpha}} \end{bmatrix} = \begin{bmatrix} A_{\alpha\alpha} & A_{\alpha\bar{\alpha}} \\ A_{\bar{\alpha}\alpha} & A_{\bar{\alpha}\bar{\alpha}} \end{bmatrix} \begin{bmatrix} z_\alpha \\ z_{\bar{\alpha}} \end{bmatrix}.$$

On premultiplying by

$$\begin{bmatrix} -A_{\alpha\alpha} & 0 \\ -A_{\bar{\alpha}\alpha} & I_{\bar{\alpha}\bar{\alpha}} \end{bmatrix}^{-1} = \begin{bmatrix} -A_{\alpha\alpha}^{-1} & 0 \\ -A_{\bar{\alpha}\alpha}A_{\alpha\alpha}^{-1} & I_{\bar{\alpha}\bar{\alpha}} \end{bmatrix},$$

and rewriting we get

$$\begin{bmatrix} z_\alpha \\ w_{\bar{\alpha}} \end{bmatrix} = \begin{bmatrix} A'_{\alpha\alpha} & A'_{\alpha\bar{\alpha}} \\ A'_{\bar{\alpha}\alpha} & A'_{\bar{\alpha}\bar{\alpha}} \end{bmatrix} \begin{bmatrix} w_\alpha \\ z_{\bar{\alpha}} \end{bmatrix},$$

where $A'_{\alpha\alpha} = (A_{\alpha\alpha})^{-1}$, $A'_{\alpha\bar{\alpha}} = -(A_{\alpha\alpha})^{-1} A_{\alpha\bar{\alpha}}$, $A'_{\bar{\alpha}\alpha} = A_{\bar{\alpha}\alpha}(A_{\alpha\alpha})^{-1}$, $A'_{\bar{\alpha}\bar{\alpha}} = A_{\bar{\alpha}\bar{\alpha}} - A_{\bar{\alpha}\alpha}(A_{\alpha\alpha})^{-1} A_{\alpha\bar{\alpha}}$.

Since $\begin{bmatrix} z_\alpha \\ w_{\bar{\alpha}} \end{bmatrix} > 0$ and $\begin{bmatrix} w_\alpha \\ z_{\bar{\alpha}} \end{bmatrix} > 0$, it follows that $v(\wp_\alpha(A)) > 0$. □

Murthy et al. [19] proved that if $A \in Q$, then $v(A) > 0$. Since any PPT $\wp_\alpha(A)$ of a Q-matrix is again a Q-matrix. Therefore, for any Q-matrix $v(\wp_\alpha(A)) > 0$ in all its PPTs $\wp_\alpha(A)$. However, it is simple to demonstrate that for any matrix A with $v(A) > 0$, A is a Q-matrix if and only if A is a Q_0-matrix.

Theorem 12.28 (Neogy and Das [23]) *Let* $A \in \mathbb{R}^{n\times n} \cap$ *almost* $C_0^f \cap Q_0$ $(n \geq 3)$. *Then* $A \in P_0$.

Now consider the matrix class whose members have PPTs that are either E_0 or almost C_0 with at least one PPT that is almost C_0. The example below demonstrates that this class is nonempty.

Example 24 (Neogy and Das [23])

Let

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

It is easy to see that all its PPTs are either E_0 or almost C_0. Also, $A \in Q_0$.

Theorem 12.29 (Neogy and Das [23]) *Let* $A \in \mathbb{R}^{n\times n} \cap Q_0$ $(n \geq 3)$ *and the PPTs of* A *be either* E_0 *or almost* C_0 *with at least one PPT almost* C_0. *Then* A *is a* P_0*-matrix.*

12.12 A List of Open Problems

In this section, we present some open problems. Dubey and Neogy [8] noticed that the class of hidden Z-matrices is not complete. However, Pang [28] proved that the subclass hidden K of hidden Z is complete. The below example demonstrates that the subclass of hidden Z-matrices which is complete properly contains the subclass hidden K.

Example 25 (Dubey and Neogy [8])

Consider the matrix

$$A = \begin{bmatrix} 0 & -2 & -2 \\ -1 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$$

with $X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$ and $AX = Y = \begin{bmatrix} 2 & -2 & 0 \\ -2 & 2 & -1 \\ -3 & 0 & 2 \end{bmatrix}$.

Take $r = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ and $s = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $r^T X + s^T Y = \begin{bmatrix} 2 & 6 & 1 \end{bmatrix}$. Hence A is a hidden Z-matrix. It should be noted that A is not a P-matrix. But every principal submatrix of A is a hidden Z-matrix.

Now, we present the following open problems.

Problem 12.1 Characterize the class of hidden $Z \setminus P$-matrices, which is complete.

Problem 12.2 Mohan [13] demonstrated that if $A \in Z$, then LCP(q, A) can be solved by using the simplex method to the problem

$$\begin{aligned} &\text{Minimize} \quad z_0 \\ &\text{subject to} \quad w - Az - e_n z_0 = q \\ &\qquad w,\ z,\ z_0 \geq 0, \end{aligned}$$

where e_n denotes the column vector of all 1's. Indeed, it is shown that the simplex algorithm and Lemke's algorithm provide the same sequence of basic feasible solutions. Can we extend this result to the class of hidden Z-matrices?

Problem 12.3 Characterize the subclass of hidden K-matrices for which the inverse is nonnegative.

Neogy and Das [23] studied and characterized various classes of matrices that are defined based on PPTs. Now we have the following open problem.

Problem 12.4 Give the complete characterization of the class of matrices for which at least one PPT is a Z-matrix.

Acknowledgements The second author wants to thank the Council of Scientific and Industrial Research, Junior Research Fellowship, India (Grant no. 09/045(1709)/2019-EMR-I), for financial support for this research.

References

1. Aganagić M, Cottle RW (1987) A constructive characterization of Q_0-matrices with nonnegative principal minors. Math Program 37(2):223–231
2. Chu TH (2003) On semimonotone matrices with nonnegative principal minors. Linear Algebra Appl 367:147–154
3. Cottle RW, Guu SM (1992) Two characterizations of sufficient matrices. Linear Algebra Appl 170:65–74
4. Cottle RW, Pang JS, Venkateswaran V (1989) Sufficient matrices and the linear complementarity problem. Linear Algebra Appl 114:231–249
5. Cottle RW, Pang JS, Stone RE (2009) The linear complementarity problem. Revised ed of the 1992 original edn. SIAM, Philadelphia, PA
6. Crouzeix JP, Komlósi S (2001) The linear complementarity problem and the class of generalized positive subdefinite matrices. In: Optimization theory. Springer, pp 45–63
7. Crouzeix JP, Hassouni A, Lahlou A, Schaible S (2000) Positive subdefinite matrices, generalized monotonicity, and linear complementarity problems. SIAM J Matrix Anal Appl 22(1):66–85
8. Dubey D, Neogy SK (2016) On hidden Z-matrices and the linear complementarity problem. Linear Algebra Appl 496:81–100

9. Dubey D, Neogy SK (2018) On generalizations of positive subdefinite matrices and the linear complementarity problem. Linear Multilinear Algebra 66(10):2024–2035
10. Mangasarian OL (1976) Linear complementarity problems solvable by a single linear program. Math Program 10(1):263–270
11. Mangasarian OL (1978) Characterization of linear complementarity problems as linear programs. In: Complementarity and fixed point problems. Springer, pp 74–87
12. Mangasarian OL (1979) Simplified characterizations of linear complementarity problems solvable as linear programs. Math Oper Res 4(3):268–273
13. Mohan SR (1976) On the simplex method and a class of linear complementary problems. Linear Algebra Appl 14(1):1–9
14. Mohan SR, Neogy SK, Das AK (2001) More on positive subdefinite matrices and the linear complementarity problem. Linear Algebra Appl 338(1–3):275–285
15. Mohan SR, Neogy SK, Das AK (2001) On the classes of fully copositive and fully semimonotone matrices. Linear Algebra Appl 323(1–3):87–97
16. Mohan SR, Neogy SK, Parthasarathy T (2001c) Pivoting algorithms for some classes of stochastic games: a survey. Int Game Theory Rev 3(02n03):253–281
17. Murthy GSR, Parthasarathy T (1995) Some properties of fully semimonotone, Q_0-matrices. SIAM J Matrix Anal Appl 16(4):1268–1286
18. Murthy GSR, Parthasarathy T (1998) Fully copositive matrices. Math Program 82(3):401–411
19. Murthy GSR, Parthasarathy T, Ravindran G (1995) On copositive, semi-monotone Q-matrices. Math Program 68(1):187–203
20. Murthy GSR, Parthasarathy T, Sriparna B (2000) On the solution sets of linear complementarity problems. SIAM J Matrix Anal Appl 21(4):1229–1235
21. Murty KG, Yu FT (1988) Linear complementarity, linear and nonlinear programming, vol 3. Citeseer
22. Neogy SK, Das AK (2005) On almost type classes of matrices with Q-property. Linear Multilinear Algebra 53(4):243–257
23. Neogy SK, Das AK (2005) Principal pivot transforms of some classes of matrices. Linear Algebra Appl 400:243–252
24. Neogy SK, Das AK (2006) Some properties of generalized positive subdefinite matrices. SIAM J Matrix Anal Appl 27(4):988–995
25. Neogy SK, Das AK (2011) On singular N_0-matrices and the class Q. Linear Algebra Appl 434(3):813–819
26. Neogy SK, Das AK (2013) On weak generalized positive subdefinite matrices and the linear complementarity problem. Linear Multilinear Algebra 61(7):945–953
27. Olech C, Parthasarathy T, Ravindran G (1991) Almost N-matrices and linear complementarity. Linear Algebra Appl 145:107–125
28. Pang JS (1979) Hidden Z-matrices with positive principal minors. Linear Algebra Appl 23:201–215
29. Pye WC (1992) Almost P_0-matrices and the class Q. Math Program 57(1):439–444
30. Raghavan TES (1978) Completely mixed games and M-matrices. Linear Algebra Appl 21(1):35–45
31. Tucker AW (1963) Principal pivotal transforms of square matrices. In: SIAM Review, vol 5, p 305
32. Von Neumann J, Morgenstern O (1944) Theory of games and economic behaviour. Princeton University Press, Princeton, NJ

Chapter 13
On Nearest Matrix with Partially Specified Eigen-Structure



Rafikul Alam

Mathematics Subject Classification (2010) 65F15 · 15A57 · 15A18 · 65F35

13.1 Introduction

Let $\mathbb{C}^{n\times n}$ denote the vector space of all $n \times n$ matrices with entries in $\mathbb{C}$. For $A \in \mathbb{C}^{n\times n}$, let $\mathrm{eig}(A)$ denote the spectrum of A. Let $\lambda \in \mathrm{eig}(A)$. We denote the algebraic multiplicity and the geometric multiplicity of λ by $\mathbf{am}(\lambda, A)$ and $\mathbf{gm}(\lambda, A)$, respectively. If $\Lambda := \{\lambda_1, \ldots, \lambda_\ell\} \subset \mathrm{eig}(A)$ then the total algebraic multiplicity of Λ, denoted by $\mathbf{am}(\Lambda, A)$, is given by

$$\mathbf{am}(\Lambda, A) := \mathbf{am}(\lambda_1, A) + \cdots + \mathbf{am}(\lambda_\ell, A).$$

We denote the singular values of A by $\sigma_1(A) \geq \sigma_2(A) \geq \cdots \geq \sigma_{n-1}(A) \geq \sigma_n(A) \geq 0$ and refer to $\sigma_j(A)$ as the j-th singular value of A.

Eigenvalue Distance Problem: Let $A \in \mathbb{C}^{n\times n}$. Suppose that $\Lambda := \{\lambda_1, \ldots, \lambda_\ell\} \subset \mathbb{C}$ is a given set and $m \in \mathbb{N}$ is such that $\ell \leq m \leq n$. Then solve the minimization problem

$$\delta_m(\Lambda, A) := \inf\{\|X - A\|_2 : X \in \mathbb{C}^{n\times n},\ \Lambda \subset \mathrm{eig}(X) \text{ and } \mathbf{am}(\Lambda, X) \geq m\}. \tag{13.1}$$

R. Alam (✉)
Department of Mathematics, IIT Guwahati, Guwahati 781039, India
e-mail: rafik@iitg.ac.in; rafikul68@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_13

If the infimum is attained at A_{opt} then we have $\delta_m(\Lambda, A) = \|A - A_{\text{opt}}\|_2$. The minimization problem (13.1) is easy to solve when $\Lambda = \{\lambda\}$ and $m = 1$. Indeed, from the SVD of $A - \lambda I$ it follows that

$$\delta_1(\lambda, A) = \sigma_{\min}(A - \lambda I) \quad \text{and} \quad A_{\text{opt}} = A - \sigma_{\min}(A - \lambda I)uv^*,$$

where $\sigma_{\min}(A - \lambda I)$ is the smallest singular value of $A - \lambda I$ and, u and v are left and right singular vectors of $A - \lambda I$ corresponding to $\sigma_{\min}(A - \lambda I)$.

For the special case when $\Lambda = \{\lambda\}$ and $m = 2$, Malyshev [22] proved that

$$\delta_2(\lambda, A) = \max_{\gamma \geq 0} \sigma_{2n-1}\left(\begin{bmatrix} A - \lambda I & \gamma I \\ 0 & A - \lambda I \end{bmatrix}\right) \tag{13.2}$$

and constructed A_{opt} from left and right singular vectors. Mengi [23] considered the case $\Lambda = \{\lambda\}$ and $m \leq n$. Generalizing Malyshev's result, it is proved in Mengi [23] that

$$\delta_m(\lambda, A) \geq \sup\{\sigma_{mn-m+1}(I_m \otimes (A - \lambda I) - N^\top \otimes I_n) : N \in \mathbf{UN}(m, \mathbb{C})\}$$

and that the equality holds under two crucial assumptions, where $\mathbf{UN}(m, \mathbb{C}) \subset \mathbb{C}^{m \times m}$ denotes the space of strictly upper triangular (and hence nilpotent) matrices.

Gracia [10] and Lippert [20] considered the case of two distinct eigenvalues, namely, the case when $\Lambda = \{\lambda_1, \lambda_2\}$ and $m = 2$. It is proved in Gracia [10] that

$$\delta_2(\Lambda, A) = \max_{\gamma \geq 0} \sigma_{2n-1}\left(\begin{bmatrix} A - \lambda_1 I & \gamma I \\ 0 & A - \lambda_2 I \end{bmatrix}\right)$$

provided that the maximum is attained at some $\gamma_0 > 0$. In such a case, A_{opt} is constructed from the associated left and right singular vectors.

The general case of ℓ distinct eigenvalues and $m \geq \ell$, namely, the case when $\Lambda = \{\lambda_1, \ldots, \lambda_\ell\}$ and $\ell \leq m \leq n$ is considered by Lippert [21]. It is proved that

$$\delta_m(\Lambda, A) \geq \sup\{\sigma_{mn-m+1}(I_m \otimes A - B^\top \otimes I_n) : B \in \mathbf{S}(m, \Lambda)\},$$

where $\mathbf{S}(m, \Lambda)$ consists of $m \times m$ matrices B with $\text{eig}(B) = \Lambda$ such that the Weyr characteristic of B satisfies certain constraints; see Lippert [21].

Eigenvalue distance problems of the kind (13.1) have been considered for generalized eigenvalue problems in Kressner et al. [18], for polynomial eigenvalue problems in Karow and Mengi [14], and for nonlinear eigenvalue problems in Karow et al. [15]. See also Papathanasiou and Psarrakos [24], Psarrakos [25], Kokabifar et al. [16, 17] and the references therein. The distance problem (13.1) is an offshoot of the well-known Wilkinson's problem, which has been studied extensively in the literature (Wilkinson [27–32], Ruhe [26], Demmel [7, 8], Kahan [13], Malyshev [22], Alam [1, 2], Alam and Bora [3], Alam et al. [4]).

Wilkinson Problem (1965): Suppose that $A \in \mathbb{C}^{n\times n}$ has n distinct eigenvalues. Let $w(A) := \inf\{\|X - A\|_2 : X \text{ is defective}\}$. Determine $w(A)$ and construct A_{opt} such that A_{opt} is defective and $\|A - A_{\text{opt}}\|_2 = w(A)$.

It follows from (13.2) that $w(A) = \inf\{\delta_2(\lambda, A) : \lambda \in \mathbb{C}\}$. We refer to Alam [1, 2], Alam and Bora [3], Alam et al. [4] for a comprehensive treatment of Wilkinson's problem based on pseudospectra of matrices. An algorithm that provides a solution to Wilkinson's problem can be found in Alam et al. [4].

Now consider a more specialized eigenvalue distance problem. Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$ be such that $\text{eig}(B) = \Lambda$. Then solve the minimization problem

$$\text{d}(A, B) := \inf\left\{\|X - A\|_2 : X \in \mathbb{C}^{n\times n},\ X \simeq \left[\begin{array}{c|c} B & \times \\ \hline 0 & \times \end{array}\right]\right\}, \tag{13.3}$$

where the blocks denoted by $\times$ represent some matrices. Here $X \simeq Y$ means X is similar to Y. If the infimum is attained at A_{opt} then A_{opt} is similar to a block upper triangular dilation of B. Hence the eigen-structure of B is subsumed by that of A_{opt}. Thus A_{opt} has a partially specified eigen-structure given by the eigen-structure of B. Obviously, we have

$$\delta_m(\Lambda, A) = \inf\{\text{d}(A, B) : B \in \mathbb{C}^{m\times m},\ \text{eig}(B) = \Lambda\}.$$

The main purpose of this work is to present a concise framework for analysis of the distance problem (13.3) and undertake an in-depth analysis of various issues that influence the existence of a solution. We also revisit the eigenvalue distance problem (13.1). Our approach is based on extensive use of SVD of the Sylvester operator $\mathbf{T}_{A,B}(X) := AX - XB$ and subdifferential calculus of singular value functions. Analysis of the distance problem (13.3) involves optimization of certain singular value functions which are locally Lipschitz continuous and hence are suitable for nonsmooth analysis. In fact, the distance problem (13.3) leads to optimization of the singular value function given by

$$f(N) := \sigma_m(A, D + N)$$

for all $N \in \mathbf{UN}(m, \mathbb{C})$, where $\sigma_m(A, D + N)$ is the m-th smallest singular value of the Sylvester operator $\mathbf{T}_{A,D+N}$ and $D \in \mathbb{C}^{m\times m}$ is a diagonal matrix such that $\text{eig}(D) = \Lambda$. We show that $f(N)$ has a global maximum and that either $f(N) \equiv 0$ on $\mathbf{UN}(m, \mathbb{C})$ or $f(N) \neq 0$ for all nonderogatory matrices $N \in \mathbf{UN}(m, \mathbb{C})$, the latter set being dense in $\mathbf{UN}(m, \mathbb{C})$.

We determine subdifferentials of locally Lipschitz continuous singular value functions and show that nonsmooth analysis of singular value functions together with SVD of Sylvester operators provides a powerful unified framework for analysing the eigenvalue distance problem (13.3) irrespective of multiplicity of the singular value function $f(N)$ at a critical point. As a byproduct, we derive short and simple proofs of the main results. Also, we prove several new results which provide new insights

into various conditions that lead to a solution of the distance problem (13.3). For instance, if B is nonderogatory with Jordan canonical form J then we show that

$$\mathrm{d}(A, B) = \mathrm{d}(A, J) \geq \max\{\sigma_m(A, D + N) : N \in \mathbf{UN}(m, \mathbb{C})\} = f(N_0)$$

and the equality holds under the assumption that

$$U^*U = V^*V \text{ and } \operatorname{rank}(V) = m, \tag{13.4}$$

where U and V are left and right singular vectors of $\mathbf{T}_{A,D+N_0}$ corresponding to $\sigma_m(A, D + N_0)$ and $D := \operatorname{diag}(J)$. We mention that a solution of (13.1) has been obtained in the literature under the assumption that the conditions in (13.4) hold. However, these crucial conditions have not been analysed in the literature. With a view to gaining insight into these conditions, we characterize the conditions in (13.4) and show that these conditions are related to centralizer of the $m \times m$ matrix $D + N_0$.

The rest of the paper is organized as follows. In Sect. 13.2, we present singular value decomposition of a Sylvester operator and discuss basic results which will be used in the subsequent development. In Sect. 13.3, we reformulate the eigenvalue distance problem and discuss minimal perturbations and their construction from SVD of a Sylvester operator. In Sect. 13.4, we consider singular value functions and determine their subdifferentials. In Sect. 13.5, we present a singular value characterization of the eigenvalue distance problem and provide a solution. Also, for the special case when $m = 2$, we present a short proof of the fact that a solution of the problem (13.1) always exists.

Notation

Let $X \in \mathbb{C}^{m\times n}$. We denote the null space of X by $\ker(X)$ and the column space of X by $\operatorname{span}(X)$. Two $n \times n$ matrices A and B are said to be similar if there is an $n \times n$ nonsingular matrix S such that $S^{-1}AS = B$. We write $A \simeq B$ when A and B are similar matrices. We denote the conjugate transpose (resp., transpose) of a matrix $A \in \mathbb{C}^{m\times n}$ by A^* (resp., $A^\top$). The Kronecker product of two matrices A and B is denoted by $A \otimes B$. The $n \times n$ identity matrix is denoted by I_n. We consider the standard inner product on $\mathbb{C}^n$, that is, $\langle x, y\rangle := y^*x$ for $x, y \in \mathbb{C}^n$, and the norm $\|x\|_2 = \sqrt{\langle x, x\rangle}$. We denote the spectral norm of $A \in \mathbb{C}^{m\times n}$ by $\|A\|_2$ which is defined by $\|A\|_2 := \max\{\|Ax\|_2 : x \in \mathbb{C}^n,\ \|x\|_2 = 1\}$. We consider the Frobenius inner product on $\mathbb{C}^{m\times n}$ given by $\langle X, Y\rangle := \operatorname{Tr}(Y^*X)$ for $X, Y \in \mathbb{C}^{m\times n}$, where $\operatorname{Tr}(A)$ is the trace of $A \in \mathbb{C}^{n\times n}$. Then $\|X\|_F := \sqrt{\langle X, X\rangle}$ is the Frobenius norm of X. We write an $n \times n$ diagonal matrix D with diagonal (scalar) entries $\lambda_1, \ldots, \lambda_n$ as $D = \operatorname{diag}(\lambda_1, \cdots, \lambda_n) = \operatorname{diag}(\lambda_i)$. Similarly, we write a block diagonal matrix A with diagonal blocks A_1 and A_2 as $A = A_1 \oplus A_2$, that is, $A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix} = A_1 \oplus A_2$. On the other hand, if $A \in \mathbb{C}^{m\times m}$ then $\operatorname{diag}(A)$ denotes the diagonal matrix obtained from A by setting the off diagonal entries of A to zero. We denote the Moore-Penrose pseudo-inverse of $A \in \mathbb{C}^{m\times n}$ by A^+. Then $A^+ \in \mathbb{C}^{n\times m}$ is the unique matrix such that AA^+ and A^+A are orthogonal projections and $AA^+A = A$ and $A^+AA^+ = A^+$.

13.2 SVD of Sylvester Operator

Singular value decomposition (SVD) of a Sylvester operator will play a crucial role in analysing the eigenvalue distance problem. We briefly recall singular value decomposition of a linear operator acting on a finite dimensional inner product space. Let H be a finite dimensional Hilbert space and let $\dim H = n$. Let $L(H)$ denote the space of all linear operators on H.

Theorem 13.1 (SVD, Axler [5]) *Let $T \in L(H)$. Then there exist unique real numbers $\sigma_1(T) \geq \cdots \geq \sigma_n(T) \geq 0$ and orthonormal bases $\{u_1, \ldots, u_n\}$ and $\{v_1, \ldots, v_n\}$ of H such that*

$$Tx = \sigma_1(T)\langle x, v_1\rangle u_1 + \cdots + \sigma_n(T)\langle x, v_n\rangle u_n, \quad x \in H.$$

Equivalently, $Tv_j = \sigma_j(T)u_j$ and $T^*u_j = \sigma_j(T)v_j$ for $j = 1 : n$.

The numbers $\sigma_1(T), \ldots, \sigma_n(T)$ are called singular values of T. The vectors u_j and v_j are called left and right singular vectors, respectively, of T corresponding to the singular value $\sigma_j(T)$ for $j = 1 : n$. It follows that

$$\dim \ker(T) \geq \ell \iff \operatorname{rank}(T) \leq n - \ell \iff \sigma_{n-\ell+1}(T) = 0. \tag{13.5}$$

Singular values are well behaved under perturbation in the sense that a small perturbation of the operator results in a small variation in the singular values. More precisely, for any $\Delta T \in L(H)$, we have the following bounds (Horn and Johnson [11])

$$|\sigma_j(T + \Delta T) - \sigma_j(T)| \leq \|\Delta T\|, \quad j = 1 : n. \tag{13.6}$$

Theorem 13.2 (Eckart-Young, Horn and Johnson [11]) *Let $T \in L(H)$. Consider the singular value decomposition of T given by*

$$Tx = \sigma_1(T)\langle x, v_1\rangle u_1 + \cdots + \sigma_n(T)\langle x, v_n\rangle u_n, \quad x \in H.$$

For $\ell < n$, define the linear operator $T_\ell : H \longrightarrow H$ by

$$T_\ell x = \sigma_1(T)\langle x, v_1\rangle u_1 + \cdots + \sigma_\ell(T)\langle x, v_n\rangle u_\ell, \quad x \in H.$$

Then $\min\{\|T - S\| : S \in L(H)$ and $\operatorname{rank}(S) \leq \ell\} = \sigma_{\ell+1}(T) = \|T - T_\ell\|$.

Thus, if $\ell < \operatorname{rank}(T)$ then T_ℓ is the best rank-ℓ approximation of T from $L(H)$.

Sylvester Operator: Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. The Sylvester operator

$$\mathbf{T}_{A,B} : \mathbb{C}^{n\times m} \longrightarrow \mathbb{C}^{n\times m} \text{ is given by } \mathbf{T}_{A,B}(X) := AX - XB.$$

It is well known (Horn and Johnson [11]) that $\ker(\mathbf{T}_{A,B}) = \{0\} \iff \text{eig}(A) \cap \text{eig}(B) = \emptyset$. Consequently, $\ker(\mathbf{T}_{A,B}) \neq \{0\} \iff \text{eig}(A) \cap \text{eig}(B) \neq \emptyset \iff$ the equation $AX - XB = 0$ has a nontrivial solution X. The adjoint of $\mathbf{T}_{A,B}$ is given by

$$\mathbf{T}^*_{A,B}(X) = A^*X - XB^* \text{ for } X \in \mathbb{C}^{n\times m}.$$

For $X := \begin{bmatrix} x_1 \cdots x_m \end{bmatrix} \in \mathbb{C}^{n\times m}$, let $\text{Vec}(X) \in \mathbb{C}^{mn}$ denote the vector in $\mathbb{C}^{mn}$ obtained by stacking the columns of X given by

$$\text{Vec}(X) = \text{Vec}(\begin{bmatrix} x_1 \cdots x_m \end{bmatrix}) := \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} \in \mathbb{C}^{mn}.$$

Then the map $\text{Vec} : \mathbb{C}^{n\times m} \longrightarrow \mathbb{C}^{mn}$, $X \longmapsto \text{Vec}(X)$, defines a linear isometric isomorphism. Further, for $X \in \mathbb{C}^{n\times m}$, we have

$$\text{Vec}(\mathbf{T}_{A,B}(X)) = (I_m \otimes A - B^\top \otimes I_n)\text{Vec}(X).$$

SVD of the Sylvester operator $\mathbf{T}_{A,B}$ will play an important role in solving the eigenvalue distance problem. Since $\mathbf{T}_{A,B} : \mathbb{C}^{n\times m} \longrightarrow \mathbb{C}^{n\times m}$ is a linear operator on the inner product space $\mathbb{C}^{n\times m}$ equipped with the inner product $\langle X, Y\rangle := \text{Tr}(Y^*X)$, by Theorem 13.1 we have the following result.

Theorem 13.3 (SVD of $\mathbf{T}_{A,B}$) *There exist orthonormal bases* $\{U_1, \ldots, U_{mn}\}$ *and* $\{V_1, \ldots, V_{mn}\}$ *of* $\mathbb{C}^{n\times m}$ *and unique real numbers* $\sigma_1(\mathbf{T}_{A,B}) \geq \cdots \geq \sigma_{mn}(\mathbf{T}_{A,B}) \geq 0$ *such that*

$$\mathbf{T}_{A,B}(X) = \sigma_1(\mathbf{T}_{A,B})\langle X, V_1\rangle U_1 + \cdots + \sigma_{mn}(\mathbf{T}_{A,B})\langle X, V_{mn}\rangle U_{mn} \tag{13.7}$$

for all $X \in \mathbb{C}^{n\times n}$. *Equivalently, we have*

$$AV_j - V_jB = \sigma_j(\mathbf{T}_{A,B})U_j \text{ and } A^*U_j - U_jB^* = \sigma_j(\mathbf{T}_{A,B})V_j, \quad j = 1 : mn. \tag{13.8}$$

We mention that the SVD in (13.7) can also be equivalently written as

$$\mathbf{T}_{A,B}(V_j) = \sigma_j(\mathbf{T}_{A,B})U_j \text{ and } \mathbf{T}^*_{A,B}(U_j) = \sigma_j(\mathbf{T}_{A,B})V_j, \quad j = 1 : mn.$$

The matrices U_j and V_j are called left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding the singular value $\sigma_j(\mathbf{T}_{A,B})$ for $j = 1 : mn$. The rank of $\mathbf{T}_{A,B}$ is the number of nonzero singular values of $\mathbf{T}_{A,B}$.

Corollary 13.1 *Consider the SVD of* $\mathbf{T}_{A,B}$ *as given in Theorem 13.3. Set* $u_j := \text{Vec}(U_j)$ *and* $v_j := \text{Vec}(V_j)$ *for* $j = 1 : mn$. *Then the SVD of* $(I_m \otimes A - B^\top \otimes I_n)$ *is given by*

$(I_m \otimes A - B^\top \otimes I_n)v_j = \sigma_j(\mathbf{T}_{A,B})u_j$ *and* $(I_m \otimes A - B^\top \otimes I_n)^* u_j = \sigma_j(\mathbf{T}_{A,B})v_j$

for $j = 1 : mn$. *Further,* $\sigma_j(\mathbf{T}_{A,B}) = \sigma_j(I_m \otimes A - B^\top \otimes I_n)$ *for* $j = 1 : mn$. *Furthermore, we have*

$$\dim \ker(\mathbf{T}_{A,B}) \geq m \iff \operatorname{rank}(\mathbf{T}_{A,B}) \leq mn - m \iff \sigma_{mn-m+1}(\mathbf{T}_{A,B}) = 0.$$

Proof Since $\operatorname{Vec}(\mathbf{T}_{A,B}(X)) = (I_m \otimes A - B^\top \otimes I_n)\operatorname{Vec}(X)$ for all $X \in \mathbb{C}^{n\times m}$ and $\operatorname{Vec} : \mathbb{C}^{n\times m} \longrightarrow \mathbb{C}^{mn}$ is an isometric isomorphism, the SVD of $(I_m \otimes A - B^\top \otimes I_n)$ follows from Theorem 13.3. The assertion on rank of $\mathbf{T}_{A,B}$ follows from (13.5). □

We consider the induced operator norm of $\mathbf{T}_{A,B}$ given by

$$\|\mathbf{T}_{A,B}\| := \sup\{\|\mathbf{T}_{A,B}(X)\|_F : X \in \mathbb{C}^{n\times m} \text{ and } \|X\|_F = 1\}.$$

Then it follows that $\|\mathbf{T}_{A,B}\| = \sigma_1(\mathbf{T}_{A,B}) = \|(I_m \otimes A - B^\top \otimes I_n)\|_2$. For the rest of the article, we set $\sigma_j(A, B) := \sigma_{mn-j+1}(\mathbf{T}_{A,B})$ for $j = 1 : mn$. This counts the singular values of $\mathbf{T}_{A,B}$ in ascending order of magnitude and we have

$$\sigma_1(A, B) \leq \cdots \leq \sigma_{mn}(A, B).$$

Proposition 13.1 *Let* $\Delta A \in \mathbb{C}^{n\times n}$ *and* $\Delta B \in \mathbb{C}^{m\times m}$. *For* $j = 1 : mn$, *we have*

$$\begin{aligned}
|\sigma_j(A + \Delta A, B) - \sigma_j(A, B)| &\leq \|\Delta A\|_2, \\
|\sigma_j(A, B + \Delta B) - \sigma_j(A, B)| &\leq \|\Delta B\|_2, \\
|\sigma_j(A + \Delta A, B + \Delta B) - \sigma_j(A, B)| &\leq \|I_m \otimes \Delta A - (\Delta B)^\top \otimes I_n\|_2.
\end{aligned}$$

Further, for any unitary matrices $U \in \mathbb{C}^{n\times n}$ *and* $V \in \mathbb{C}^{m\times m}$, *we have*

$$\sigma_j(U^*AU,\ V^*BV) = \sigma_j(A, B),\ \ j = 1 : mn.$$

Proof We have $\mathbf{T}_{A+\Delta A,B+\Delta B} = \mathbf{T}_{A,B} + \mathbf{T}_{\Delta A,\Delta B}$. Hence by (13.6), we have

$$\begin{aligned}
|\sigma_j(A + \Delta A, B + \Delta B) - \sigma_j(A, B)| &= |\sigma_{mn-j+1}(\mathbf{T}_{A+\Delta A,B+\Delta B}) - \sigma_{mn-j+1}(\mathbf{T}_{A,B})| \\
&\leq \|\mathbf{T}_{\Delta A,\Delta B}\| = \|I_m \otimes \Delta A - (\Delta B)^\top \otimes I_n\|_2.
\end{aligned}$$

Setting $\Delta B = 0$, we have $\sigma_j(A + \Delta A, B) - \sigma_j(A, B)| \leq \|\Delta A\|_2$. Similarly, setting $\Delta A = 0$, we have $|\sigma_j(A, B + \Delta B) - \sigma_j(A, B)| \leq \|\Delta B\|_2$.

Finally, note that $V^\top \otimes U^*$ and $(V^\top \otimes U^*)^* = (V^\top)^* \otimes U$ are unitary and

$$\begin{aligned}
(V^\top \otimes U^*)(I_m \otimes A - B^\top \otimes I_n)(V^\top \otimes U^*)^* &= I_m \otimes U^*AU - V^\top B(V^\top)^* \otimes I_n \\
&= I_m \otimes U^*AU - (V^*BV)^\top \otimes I_n.
\end{aligned}$$

Hence for $j = 1 : mn$, by Corollary 13.1, we have

$$\begin{aligned}\sigma_j(U^*AU,\ V^*BV) &= \sigma_{mn-j+1}(I_m \otimes U^*AU - (V^*BV)^\top \otimes I_n)\\ &= \sigma_{mn-j+1}((V^\top \otimes U^*)(I_m \otimes A - B^\top \otimes I_n)(V^\top \otimes U^*)^*)\\ &= \sigma_{mn-j+1}(I_m \otimes A - B^\top \otimes I_n) = \sigma_j(A, B).\end{aligned}$$

This completes the proof. □

Observe that $\|I_m \otimes \Delta A - (\Delta B)^\top \otimes I_n\|_2 \le \|\Delta A\|_2 + \|\Delta B\|_2$. Hence we have $|\sigma_j(A + \Delta A, B + \Delta B) - \sigma_j(A, B)| \le \|\Delta A\|_2 + \|\Delta B\|_2$. Next, by Theorem 13.2,

$$\min\{\|\mathbf{T}_{A,B} - \mathbf{S}\| : \mathbf{S} \in L(\mathbb{C}^{n\times m}),\ \mathrm{rank}(\mathbf{S}) \le mn - m\} = \sigma_m(A, B) \tag{13.9}$$

and the minimum is attained at $\mathbf{S}_{\mathrm{opt}}$ given by

$$\mathbf{S}_{\mathrm{opt}}(X) := \sigma_1(\mathbf{T}_{A,B})\langle X, V_1\rangle U_1 + \cdots + \sigma_{mn-m}(\mathbf{T}_{A,B})\langle X, V_{mn-m}\rangle U_{mn-m}$$

for all $X \in \mathbb{C}^{n\times n}$. Hence $\|\mathbf{T}_{A,B} - \mathbf{S}_{\mathrm{opt}}\| = \sigma_{mn-m+1}(\mathbf{T}_{A,B})) = \sigma_m(A, B)$.

We mention that $\mathbf{S}_{\mathrm{opt}} - \mathbf{T}_{A,B}$ may not be of the form $\mathbf{T}_{\Delta A,0}$, that is, there may not exist ΔA such that $\mathbf{S}_{\mathrm{opt}} = \mathbf{T}_{A+\Delta A,B}$. Next, we consider conditions under which we have $\mathbf{S}_{\mathrm{opt}} = \mathbf{T}_{A+\Delta A,B}$. This will play a crucial role in solving the eigenvalue distance problem.

13.3 Minimal Perturbation

Let $A \in \mathbb{C}^{n\times n}$ and $X \in \mathbb{C}^{n\times m}$ be such that $\mathrm{rank}(X) = m$. Then $\mathrm{span}(X)$ is said to be an invariant subspace of A if $A\ \mathrm{span}(X) \subset \mathrm{span}(X)$. Clearly, $\mathrm{span}(X)$ is an invariant subspace of $A \iff AX = XB$ for some $B \in \mathbb{C}^{m\times m}$. In such a case, we have $\mathrm{eig}(B) \subset \mathrm{eig}(A)$.

Definition 13.1 Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. We write $B \subset A$ if $m \le n$ and there exists $X \in \mathbb{C}^{n\times m}$ such that $\mathrm{rank}(X) = m$ and $AX = XB$.

Obviously $B \subset A \Longrightarrow \mathrm{eig}(B) \subset \mathrm{eig}(A)$. Moreover, the spectral structure of B is subsumed by that of A. This is evident from the fact that

$$B \subset A \iff A \simeq \left[\begin{array}{c|c} B & \times \\ \hline 0 & \times \end{array}\right], \tag{13.10}$$

where the blocks denoted by $\times$ represent some matrices. Thus A is similar to a block upper triangular dilation of B.

Let $C(B) := \{X \in \mathbb{C}^{m\times m} : BX = XB\}$ be the centralizer of B. Then $C(B)$ is a subspace of $\mathbb{C}^{m\times m}$ and by Gantmacher [9, Theorem 1, p.219] $\dim C(B)) \ge m$.

Theorem 13.4 *Suppose that $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Then the following results hold.*

(a) $B \subset A \Longrightarrow S^{-1}BS \subset A$ *for any* $m \times m$ *nonsingular matrix* S.
(b) $B \subset A \iff \ker(\mathbf{T}_{A,B})$ *contains a full column rank matrix.*
(c) $B \subset A \Longrightarrow \dim\ker(\mathbf{T}_{A,B}) = \dim\ker(I_m \otimes A - B^\top \otimes I_n) \geq m$.
(d) $B \subset A \Longrightarrow \sigma_m(A, B) = \sigma_{mn-m+1}(I_m \otimes A - B^\top \otimes I_n) = 0$.

Proof The proofs of (a) and (b) are immediate. Suppose that $B \subset A$. Then there exists $X \in \ker(\mathbf{T}_{A,B})$ such that $\operatorname{rank}(X) = m$. Now for any $D \in \mathbb{C}^{m\times m}$, it is easily seen that

$$XD \in \ker(\mathbf{T}_{A,B}) \iff BD = DB \iff D \in C(B).$$

By Gantmacher [9, Theorem 1, p.219] the dimension of $C(B)$ is at least m. Since X has full column rank and $XD \in \ker(\mathbf{T}_{A,B})$ for all $D \in C(B)$, it follows that $\dim\ker(\mathbf{T}_{A,B}) \geq m$. Since

$$X \in \ker(\mathbf{T}_{A,B}) \iff \operatorname{Vec}(X) \in \ker(I_m \otimes A - B^\top \otimes I_n),$$

the proof of (c) follows.

Finally, since $\sigma_m(A, B) = \sigma_{mn-m+1}(\mathbf{T}_{A,B}) = \sigma_{mn-m+1}(I_m \otimes A - B^\top \otimes I_n)$, the proof of (d) follows from Corollary 13.1. □

Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Then, in view of (13.3) and (13.10), we have

$$\mathrm{d}(A, B) = \inf\{\|\Delta A\|_2 : \Delta A \in \mathbb{C}^{n\times n} \text{ and } B \subset A + \Delta A\}. \tag{13.11}$$

By Theorem 13.4(a), $B \subset A + \Delta A \Longrightarrow S^{-1}BS \subset A + \Delta A$ for any $m \times m$ nonsingular matrix S. Hence it follows that

$$\mathrm{d}(A, B) = \mathrm{d}(A,\ S^{-1}BS)$$

for any $m \times m$ nonsingular matrix S. For the rest of the paper, consider the set $\Lambda := \{\lambda_1, \ldots, \lambda_\ell\} \subset \mathbb{C}$ consisting of ℓ distinct real or complex numbers. For any natural number m such that $\ell \leq m \leq n$, consider the set

$$\mathbf{M}_m(\Lambda) := \{X \in \mathbb{C}^{n\times n} : \Lambda \subset \operatorname{eig}(X),\ \mathbf{am}(\Lambda, X) \geq m\}.$$

Then we have $\delta_m(\Lambda, A) = \inf\{\|\Delta A\|_2 : A + \Delta A \in \mathbf{M}_m(\Lambda)\} =: \operatorname{dist}(A, \mathbf{M}_m(\Lambda))$. Our aim is to determine $\delta_m(\Lambda, A)$ and a matrix A_{opt} such that $\|A - A_{\text{opt}}\|_2 = \delta_m(\Lambda, A)$. It is evident that

$$\delta_m(\Lambda, A) = \inf\{\mathrm{d}(A, B) : B \in \mathbb{C}^{m\times m},\ \operatorname{eig}(B) = \Lambda\}. \tag{13.12}$$

Thus, in order to determine $\delta_m(\Lambda, A)$, we need to determine $\mathrm{d}(A, B)$. For this purpose, we need the following result.

Theorem 13.5 *Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Then the following results hold.*

(a) $B \subset A + \Delta A \Longrightarrow \sigma_m(A, B) = \sigma_{mn-m+1}(I_m \otimes A - B^\top \otimes I_n) \leq \|\Delta A\|_2$.
(b) $B \subset A + \Delta A \Longrightarrow \sigma_m(A, S^{-1}BS) \leq \|\Delta A\|_2$ *for any nonsingular matrix* S.
(c) $A + \Delta A \in \mathbf{M}_m(\Lambda) \iff$ *there exists* $B \in \mathbb{C}^{m\times m}$ *such that* $\mathrm{eig}(B) = \Lambda$ *and* $B \subset A + \Delta A$.
(d) $A + \Delta A \in \mathbf{M}_m(\Lambda) \Longrightarrow$ *there exists* $B \in \mathbb{C}^{m\times m}$ *such that* $\mathrm{eig}(B) = \Lambda$ *and* $\sigma_m(A, B) = \sigma_{mn-m+1}(I_m \otimes A - B^\top \otimes I_n) \leq \|\Delta A\|_2$.

Proof By Theorem 13.4(d), $B \subset A + \Delta A \Longrightarrow \sigma_m(A + \Delta A, B) = 0$. Hence by Proposition 13.1, we have $\sigma_m(A, B) = |\sigma_m(A + \Delta A, B) - \sigma(A, B)| \leq \|\Delta A\|_2$. Consequently, we have $\sigma_m(A, B) = \sigma_{mn-m+1}(I_m \otimes A - B^\top \otimes I_n) \leq \|\Delta A\|_2$. This proves (a). Next, by Theorem 13.4(a), $B \subset A + \Delta A \Longrightarrow S^{-1}BS \subset A + \Delta A$ for any nonsingular S. Hence (b) follows from (a).

Now, if $A + \Delta A \in \mathbf{M}_m(\Lambda)$ then $\mathbf{am}(\Lambda, A + \Delta A) \geq m \Longrightarrow$ there exist $X \in \mathbb{C}^{n\times m}$ with $\mathrm{rank}(X) = m$ and $B \in \mathbb{C}^{m\times m}$ such that $(A + \Delta A)X = XB$ and $\mathrm{eig}(B) = \Lambda$. In other words, $B \subset A + \Delta A$ and $\mathrm{eig}(B) = \Lambda$. Conversely, if there exists $B \in \mathbb{C}^{m\times m}$ such that $B \subset A + \Delta A$ and $\mathrm{eig}(B) = \Lambda$ then obviously $A + \Delta A \in \mathbf{M}_m(\Lambda)$. This proves (c). Finally, (d) follows from (a) and (c). □

The next result shows that $\mathrm{d}(A, B) = \sigma_m(A, B)$ provided that appropriate conditions are satisfied.

Theorem 13.6 (Minimal Perturbation) *Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Let $U \in \mathbb{C}^{n\times m}$ and $V \in \mathbb{C}^{n\times m}$ be left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding to the singular value $\sigma_m(A, B)$, respectively. Suppose that* $\mathrm{rank}(V) = m$ *and* $U^*U = V^*V$. *Define*

$$\Delta A := -\sigma_m(A, B)UV^+,$$

where V^+ is the Moore-Penrose pseudo-inverse of V. Then $B \subset A + \Delta A$. Moreover,

$$(A + \Delta A)V = VB \text{ and } \|\Delta A\|_2 = \sigma_m(A, B) = \mathrm{d}(A, B).$$

Define $\mathbb{S}_{\mathrm{opt}} := \mathbf{T}_{A+\Delta A,B}$. *Then* $\dim\ker\left(\mathbf{S}_{\mathrm{opt}}\right) \geq m$ *and*

$$\|\mathbf{S}_{\mathrm{opt}} - \mathbf{T}_{A,B}\| = \min\{\|\mathbf{S} - \mathbf{T}_{A,B}\| : \mathbf{S} \in L(\mathbb{C}^{n\times n}),\ \mathrm{rank}(\mathbf{S}) \leq mn - m\} = \sigma_m(A, B).$$

Proof Note that $V^+V = I_m$. Hence the SVD of $\mathbf{T}_{A,B}$ gives

$$AV - VB = \sigma_m(A, B)U \Longrightarrow (A - \sigma_m(A, B)UV^+)V = VB \Longrightarrow (A + \Delta A)V = VB.$$

This shows that $B \subset A + \Delta A$. Next, since $U^*U = V^*V$ and VV^+ is an orthogonal projection, we have

$$\begin{aligned}\|UV^+\|_2^2 &= \lambda_{\max}((UV^+)^*UV^+) = \lambda_{\max}((V^+)^*U^*UV^+) = \lambda_{\max}((V^+)^*V^*VV^+)\\ &= \lambda_{\max}((VV^+)^*VV^+) = \lambda_{\max}(VV^+) = 1,\end{aligned}$$

where $\lambda_{\max}(Y)$ denotes the largest eigenvalue of a positive semidefinite matrix Y. This proves that $\|\Delta A\|_2 = \sigma_m(A, B)$.

Since $B \subset A + \Delta A$, by Theorem 13.4(c), $\dim \ker(\mathbf{S}_{\text{opt}}) = \dim \ker(\mathbf{T}_{A+\Delta A,B}) \geq m$. Thus $\text{rank}(\mathbf{S}_{\text{opt}}) \leq mn - m$ and $\|\mathbf{S}_{\text{opt}} - \mathbf{T}_{A,B}\| = \|\mathbf{T}_{\Delta A,0}\| = \|\Delta A\|_2 = \sigma_m(A, B)$. Hence by (13.9), it follows that $\mathbf{S}_{\text{opt}}$ is the best approximation of $\mathbf{T}_{A,B}$ such that $\dim \ker(\mathbf{S}_{\text{opt}}) \geq m$.

Finally, since $B \subset A + \Delta A$, by (13.11), $\text{d}(A, B) \leq \|\Delta A\|_2 = \sigma_m(A, B)$. On the other hand, by Theorem 13.5(a), for any ΔZ such that $B \subset A + \Delta Z \Longrightarrow \sigma_m(A, B) \leq \|\Delta Z\|_2$. Now taking infimum over all such ΔZ, we have $\sigma_m(A, B) \leq \text{d}(A, B)$. Hence we have $\text{d}(A, B) = \sigma_m(A, B)$. This completes the proof. □

Remark 13.1 The proof of Theorem 13.6 shows that

$$U^*U = V^*V \Longrightarrow \|UV^+\|_2 = 1$$

irrespective of the rank of V. When V is rank deficient, assume that the condition $UV^+V = U$ is satisfied. Then we have $(A - \sigma_m(A, B)UV^+)V = VB$ and $\|UV^+\|_2 = 1$. This shows that for $\Delta A := -\sigma_m(A, B)UV^+$, we have $\|\Delta A\|_2 = \sigma_m(A, B)$ and that $A + \Delta A$ and B have common eigenvalues and $\text{span}(V)$ is an invariant subspace of $A + \Delta A$ corresponding to the common eigenvalues. Note that the full column rank of V is crucial for ensuring that $B \subset A + \Delta A$.

In view of Remark 13.1, we have the following result.

Corollary 13.2 (Rank Deficient) *Let $A \in \mathbb{C}^{n\times n}$ and $B \in \mathbb{C}^{m\times m}$. Let $U \in \mathbb{C}^{n\times m}$ and $V \in \mathbb{C}^{n\times m}$ be left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding to the singular value $\sigma_m(A, B)$, respectively. Suppose that $UV^+V = U$ and $U^*U = V^*V$, where V^+ is the Moore-Penrose pseudo-inverse of V. Define*

$$\Delta A := -\sigma_m(A, B)UV^+.$$

Then $(A + \Delta A)V = VB$ and $\|\Delta A\|_2 = \sigma_m(A, B)$. Further, $\text{span}(V)$ is an invariant subspace of $A + \Delta A$ corresponding to some eigenvalues in $\text{eig}(A + \Delta A) \cap \text{eig}(B)$.

Next, we characterize the condition $UV^+V = U$ when U and V are left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding to $\sigma_m(A, B)$. We have the following result.

Proposition 13.2 *Let U and V be left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding to $\sigma_m(A, B) \neq 0$, respectively. Then the following conditions are equivalent.*

(a) $B \ker(V) \subset \ker(V)$.
(b) $\ker(V) \subset \ker(U)$.
(c) $UV^+V = U$.

In particular, if $U^*U = V^*V$ then $\ker(V) = \ker(U)$ and $B \ker(V) \subset \ker(V)$.

Proof First, recall that $\mathbf{T}_{A,B}(V) = \sigma_m(A, B)U \Longrightarrow AV - VB = \sigma_m(A, B)U$.

Now, assume that (a) holds. Let $x \in \ker(V)$. Then $Bx \in \ker(V)$. Consequently, we have $\sigma_m(A, B)Ux = AVx - VBx = 0$ which shows that $Ux = 0$. Hence $x \in \ker(U)$. Since x is arbitrary, we have $\ker(V) \subset \ker(U)$. This proves (a)$\Rightarrow$(b).

Next, suppose that (b) holds. Observe that V^+V is an orthogonal projection on $\text{span}(V^*)$ and $V(I_m - V^+V) = V - VV^+V = V - V = 0$. Hence for $x \in \mathbb{C}^m$, we have

$$x = V^+Vx + (I_m - V^+V)x \;\text{ and }\; (I_m - V^+V)x \in \ker(V) \subset \ker(U).$$

Consequently, we have $UV^+Vx = U(V^+Vx + (I_m - V^+V)x) = Ux$ for all $x \in \mathbb{C}^m$. Hence $UV^+V = U$. This proves (b)$\Rightarrow$(c).

Finally, assume that (c) holds. Let $x \in \ker(V)$. Then $U = UV^+V \Longrightarrow Ux = UV^+Vx = 0$. Hence $(AV - VB)x = \sigma_m(A, B)Ux \Longrightarrow VBx = 0 \Longrightarrow Bx \in \ker(V)$. Since $x \in \ker(V)$ is arbitrary, we have $B\ker(V) \subset \ker(V)$. This proves (c)$\Rightarrow$(a).

Note that $U^*U = V^*V \Longrightarrow \ker(V) = \ker(U)$. Next, $AV - VB = \sigma_m(A, B)U$ shows that if $x \in \ker(V)$ then $VBx = 0 \Longrightarrow Bx \in \ker(V)$. Hence $B\ker(V) \subset \ker(V)$. This completes the proof. □

Now, we analyze the conditions $U^*U = V^*V$ and $\text{rank}(V) = m$ and determine sufficient conditions under which these conditions are satisfied.

Theorem 13.7 *Let U and V be left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding to $\sigma_m(A, B)$. Suppose that $\sigma_m(A, B) \neq 0$ and that $C := BU^*V - U^*VB$ is nilpotent. Then $U^*U = V^*V$. If B is nonderogatory then there exists a polynomial p of degree $\leq m - 1$ such that $U^*V = p(B)$. If $p(\lambda) \neq 0$ for $\lambda \in \text{eig}(B)$ then* $\text{rank}(V) = m$.

Proof Since U and V are left and right singular vectors of $\mathbf{T}_{A,B}$, we have

$$AV - VB = \sigma_m(A, B)U \;\text{ and }\; A^*U - UB^* = \sigma_m(A, B)V.$$

This gives $\sigma_m(A, B)(U^*U - V^*V) = BU^*V - U^*VB = C$ which shows that C is Hermitian. Since C is nilpotent, we have $C = 0$. Now $C = 0 \Longrightarrow U^*U = V^*V$ as $\sigma_m(A, B)$ is nonzero.

Finally, since U^*V commutes with B and B is nonderogatory, by Horn and Johnson [11, Theorem 3.2.4.2, p.178], we have $U^*V = p(B)$ for some polynomial p of degree at most $m - 1$. Since $p(B)$ is nonsingular as $p(\lambda) \neq 0$ for $\lambda \in \text{eig}(B)$, we have $U^*V = p(B)$ is nonsingular. This shows that $\text{rank}(V) = m$. □

Next, we characterize the condition $U^*U = V^*V$. These characterizations provide insight into the relation between U, V and B. Recall that

$$C(B) := \{X \in \mathbb{C}^{m\times m} : BX = XB\}$$

is the centralizer of B. We have seen that $\dim C(B)) \geq m$.

Theorem 13.8 *Let U and V be left and right singular vectors of $\mathbf{T}_{A,B}$ corresponding to $\sigma_m(A, B)$. Suppose that $\sigma_m(A, B) \neq 0$. Then the following conditions are equivalent.*

(a) $C := BU^*V - U^*VB$ *is nilpotent.*
(b) $U^*V \in C(B)$.
(c) $U^*U = V^*V$.
(d) $U = WV$ *for some unitary matrix* $W \in \mathbb{C}^{n\times n}$.
(e) *Either* U^*V *or* B *commute with* C.

In particular, if both B and U^*V are upper triangular then $U^*U = V^*V$.

Proof The equivalence of (a), (b) and (c) follow from the proof of Theorem 13.7. By Horn and Johnson [11, Theorem 7.3.11, p.452], $U^*U = V^*V \iff$ there is a unitary matrix $W \in \mathbb{C}^{n\times n}$ such that $U = WV$. This proves the equivalence of (c) and (d).

Now, if (e) holds then by Jacobson's Lemma (Horn and Johnson [11, Exercise 2.4.P12, p.126]), we have $C^m = 0$. Hence C is nilpotent. On the other hand, if C is nilpotent then by Theorem 13.7, we have $C = 0$. Consequently, C commutes with both U^*V and B. This proves equivalence of (a) and (e).

Finally, if U^*V and B are upper triangular then $C := BU^*V - U^*VB$ is strictly upper triangular and hence nilpotent. Consequently, $C = 0$ and hence $U^*V \in C(B)$ which in turn implies that $U^*U = V^*V$. □

Remark 13.2 Without loss of generality we can assume that B is upper triangular. In such a case, if U^*V is upper triangular then $C := BU^*V - U^*VB$ is nilpotent. In fact, $U^*V \in C(B)$ when both B and U^*V are upper triangular. However, it is not necessary that U^*V be upper triangular in order for it to be in $C(B)$.

To summarize: We have seen that $B \subset A + \Delta A \Longrightarrow S^{-1}BS \subset A + \Delta A$ and that $\mathrm{d}(A, B) = \mathrm{d}(A, S^{-1}BS)$ for any $m \times m$ nonsingular matrix S. We have also seen that $B \subset A + \Delta A \Longrightarrow \sigma_m(A, B) \leq \|\Delta A\|_2 \Longrightarrow \sigma_m(A, B) \leq \mathrm{d}(A, B)$ and the equality holds under the assumptions as given in Theorem 13.6. By Proposition 13.1, we have $\sigma_m(A, W^*BW) = \sigma_m(A, B)$ for any $m \times m$ unitary matrix W. Hence by Theorem 13.6, we have

$$\mathrm{d}(A, B) = \sigma_m(A, B) = \sigma_m(A, W^*BW) = \mathrm{d}(A, W^*BW)$$

for any unitary matrix W. Thus, without loss of generality, we can assume that B is an upper triangular matrix. Indeed, by Schur theorem there is a unitary matrix W such that W^*BW is upper triangular.

Our next aim is to derive a singular value characterization of $\mathrm{d}(A, B)$ assuming that B is upper triangular and relate the conditions in Theorem 13.6 with optimization of the singular value function $X \longmapsto \sigma_m(A, X)$. By Proposition 13.1, the singular value function $X \longmapsto \sigma_m(A, X)$ is Lipschitz continuous. Hence Clarke subdifferential of $\sigma_m(A, X)$ can be gainfully utilized for optimization of the function $X \longmapsto \sigma_m(A, X)$. In the next section, we determine Clarke subdifferential of locally Lipschitz singular value functions.

13.4 Clarke Subdifferential of Singular Value Functions

Let H be a finite dimensional Hilbert space over the field $\mathbb{C}$. A function $f : H \longrightarrow \mathbb{R}$ is said to be locally Lipschitz continuous at $x_0 \in H$ if there is a neighbourhood $\mathrm{Nbd}(x_0)$ of x_0 and a constant $\alpha > 0$ such that

$$x, y \in \mathrm{Nbd}(x_0) \Longrightarrow |f(x) - f(y)| \leq \alpha \|x - y\|.$$

We consider Clarke subdifferential of a locally Lipschitz function and refer to it as subdifferential. We denote the real part of a complex number z by $\mathrm{re}(z)$.

Definition 13.2 (Clarke [6]) Let $f : H \to \mathbb{R}$ be a locally Lipschitz continuous function at $x_0 \in H$. The generalized directional derivative of f at x_0 in the direction of $u \in H$ is denoted by $f^\circ(x_0; u)$ and is defined by

$$f^\circ(x_0; u) := \limsup_{x \to x_0 \; t \to 0^+} \frac{f(x + tu) - f(x)}{t}.$$

Definition 13.3 (Clarke [6]) Let $f : H \to \mathbb{R}$ be a locally Lipschitz continuous function at $x_0 \in H$. Then the subdifferential of f at x_0 is denoted by $\partial f(x_0)$ and is defined by

$$\partial f(x_0) := \{x \in H : f^\circ(x_0; u) \geq \mathrm{re}\langle u, x\rangle \text{ for all } u \in H\}.$$

Each $v \in \partial f(x_0)$ is called a subgradient of f at x_0.

Definition 13.4 *(Clarke Stationary Point)* Let $f : H \to \mathbb{R}$ be locally Lipschitz and $x_0 \in H$. Then x_0 is said to be a Clarke stationary point of f if $0 \in \partial f(x_0)$.

We now prove the chain rule for subdifferential of $g \circ f$ when g is Lipschitz and f is locally Lipschitz. This will help us to determine subdifferentials of local Lipschitz singular value functions.

Theorem 13.9 *Let U and V be finite dimensional Hilbert spaces over $\mathbb{C}$. Let $f : U \to V$ be continuously differentiable in a neighbourhood of $x_0 \in U$ and $g : V \to \mathbb{R}$ be Lipschitz continuous. Then the subdifferential of $g \circ f$ at x_0 is given by*

$$\partial(g \circ f)(x_0) = (Df(x_0))^* \partial g(f(x_0)),$$

where $Df(x_0)$ is the (Frechet) derivative of f at x_0, $(Df(x_0))^*$ is the adjoint operator of the linear map $Df(x_0) : U \to V, h \mapsto Df(x_0)h$, and $\partial g(f(x_0))$ is the subdifferential of g at $f(x_0)$.

Proof Since f is continuously differentiable in a neighbourhood of x_0 and g is Lipschitz, it follows that $g \circ f$ is locally Lipschitz at x_0. Hence $\partial(g \circ f)(x_0)$ is well defined. By the Chain rule (Clarke [6]), we have

$$\partial(g \circ f)(x_0) \subset (Df(x_0))^* \partial g(f(x_0)).$$

We now show that the equality holds. Let $y \in (Df(x_0))^* \partial g(f(x_0))$. Then we have $y = (Df(x_0))^* v$ for some $v \in \partial g(f(x_0))$. Consequently, for $x \in U$, we have

$$\langle x, y\rangle = \langle x, (Df(x_0))^* v\rangle = \langle Df(x_0)x, v\rangle. \tag{13.13}$$

Set $h := g \circ f$. We calculate the generalized directional derivative $h^\circ(x_0; u)$ for $u \in U$. Note that $f(x + tu) = f(x) + tDf(x)u + \epsilon(t)$ for some $\epsilon(t)$ such that $\epsilon(t)/t \longrightarrow 0$ as $t \to 0$. Hence $h(x + tu) = g(f(x + tu)) = g(f(x) + tDf(x)u + \epsilon(t))$. Since g is Lipschitz, we have

$$g(f(x) + tDf(x)u) - \epsilon(t) \le g(f(x) + tDf(x)u + \epsilon(t)) \le g(f(x) + tDf(x)u) + \epsilon(t).$$

Consequently, we have $(h(x + tu) - h(x))/t \le (g(f(x) + tDf(x)u) - g(f(x)))/t + \epsilon(t)/t$ and $(h(x + tu) - h(x))/t \ge (g(f(x) + tDf(x)u) - g(f(x)))/t - \epsilon(t)/t$ for $t > 0$. Hence taking limsup as $x \to x_0$ and $t \to 0$, we have $h^\circ(x_0, u) = g^\circ(f(x_0); Df(x_0)u)$ for all $u \in U$. Now by (13.13) and the definition of subdifferential, we have

$$\mathrm{re}\langle x, y\rangle = \mathrm{re}\langle Df(x_0)x, v\rangle \le g^\circ(f(x_0); Df(x_0)x) = h^\circ(x_0; x)$$

for all $x \in U$, which shows that $y \in \partial h(x_0) = \partial(g \circ f)(x_0)$. This completes the proof. □

Let $A \in \mathbb{C}^{m\times n}$. We denote the singular values of A by $\sigma_1(A) \ge \cdots \ge \sigma_p(A) \ge 0$, where $p = \min(m, n)$. Consider the $m \times n$ diagonal matrix $\Sigma := \mathrm{diag}(\sigma_1(A), \ldots, \sigma_p(A))$. Then there exist unitary matrices $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ such that $A = U\Sigma V^*$ is a singular value decomposition (SVD) of A. The columns of U are left singular vectors and columns of V are right singular vectors of A. We write $\sigma(A)$ to denote a singular value of A. If u and v are left and right singular vectors of A corresponding to $\sigma(A)$ then $Av = \sigma(A)u$ and $A^*u = \sigma(A)v$. If ℓ is the multiplicity of $\sigma(A)$ then there exist isometries $U \in \mathbb{C}^{m\times \ell}$ and $V \in \mathbb{C}^{n\times \ell}$ such that $AV = \sigma(A)U$ and $A^*U = \sigma(A)V$.

Definition 13.5 Let $U \subset \mathbb{C}^{p\times q}$ be an open set. A function $f : U \longrightarrow \mathbb{R}$ is said to be a singular value function if f is of the form

$$f(X) = \sigma(F(X)), \quad X \in U,$$

for some function $F : U \longrightarrow \mathbb{C}^{m\times n}$, where $\sigma(F(X))$ is a singular value of the matrix $F(X)$.

If $F(X)$ is continuously differentiable on U then it follows that the singular value function $f(X) := \sigma(F(X))$ is locally Lipschitz continuous on U. We now determine subdifferential of a locally Lipschitz singular value function.

Theorem 13.10 *Let $A : \mathbb{C} \to \mathbb{C}^{m\times n}$ be continuously differentiable in a neighbourhood of $\lambda \in \mathbb{C}$. Let $g : \mathbb{C} \to \mathbb{R}$ be given by $g(z) := \sigma(A(z))$, where σ is a singular*

value of $A(z)$. Suppose that $g(\lambda) \neq 0$ and that the multiplicity of $\sigma(A(\lambda))$ is p. Then the subdifferential of g at λ is given by

$$\partial g(\lambda) = \mathcal{F}(V^*(A'(\lambda))^*U),$$

where $U \in \mathbb{C}^{m\times p}$ and $V \in \mathbb{C}^{n\times p}$ have orthonormal columns such that $A(\lambda)V = g(\lambda)U$ and $(A(\lambda))^*U = g(\lambda)V$. Here $\mathcal{F}(X)$ denotes the Field of values of a square matrix X and $A'(\lambda)$ denotes the derivative of $A(z)$ at λ.

Proof First note that $g(z)$ is locally Lipschitz at λ. Indeed, since $A(z)$ is continuously differentiable in a neighbourhood of λ and $\sigma : \mathbb{C}^{m\times n} \to \mathbb{R}$ is Lipschitz, it follows that the function $g = \sigma \circ A$ is locally Lipschitz at λ. Consequently, by Theorem 13.9, we have

$$\partial g(\lambda) = (DA(\lambda))^* \partial\sigma(A(\lambda)),$$

where $\partial\sigma(A(\lambda))$ is the subdifferential of the map $X \mapsto \sigma(X)$ evaluated at $A(\lambda)$.

Note that the derivative $DA(\lambda) : \mathbb{C} \longrightarrow \mathbb{C}^{m\times n}, z \longmapsto A'(\lambda)z$, is a linear map and its adjoint $(DA(\lambda))^* : \mathbb{C}^{m\times n} \longrightarrow \mathbb{C}$ is given by $(DA(\lambda))^*X = \mathrm{Tr}((A'(\lambda))^*X)$. Indeed, we have

$$\begin{aligned}\langle z, (DA(\lambda))^*X\rangle &= \langle DA(\lambda)z, X\rangle = \langle A'(\lambda)z, X\rangle = \mathrm{Tr}(X^*A'(\lambda)z)\\ &= z\,\mathrm{Tr}(X^*A'(\lambda)) = \langle z, \overline{\mathrm{Tr}(X^*A'(\lambda))}\rangle\end{aligned}$$

which shows that $(DA(\lambda))^*X = \overline{\mathrm{Tr}(X^*A'(\lambda))} = \mathrm{Tr}((A'(\lambda))^*X)$.

Next, by Lewis and Sendov [19, Corollary 6.4] the subdifferential of $X \mapsto \sigma(X)$ at $A(\lambda)$ is given by

$$\partial\sigma(A(\lambda)) = \text{Convex hull}\left\{Uxx^*V^* : x \in \mathbb{C}^p \text{ and } \|x\|_2 = 1\right\}.$$

Consequently, we have

$$\begin{aligned}\partial g(\lambda) &= \text{Convex hull}\left\{\mathrm{Tr}((A'(\lambda))^*Uxx^*V^*) : x \in \mathbb{C}^p \text{ and } \|x\|_2 = 1\right\}\\ &= \text{Convex hull}\left\{x^*(V^*(A'(\lambda))^*U)x : x \in \mathbb{C}^p \text{ and } \|x\|_2 = 1\right\}\\ &= \mathcal{F}(V^*(A'(\lambda))^*U).\end{aligned}$$

This completes the proof. □

Next, we consider partial and directional subdifferentials of singular values. Let $\sigma : \mathbb{C}^{m\times n} \to \mathbb{R}, X \mapsto \sigma(X)$, be a singular value map, that is, $\sigma(X)$ is a singular value of X. Suppose that $\sigma(X)$ is nonzero. We define the directional subdifferential of σ at X in the direction of $E \in \mathbb{C}^{m\times n}$ by

$$\partial_E\sigma(X) := \partial g(0), \tag{13.14}$$

where $g : \mathbb{R} \to \mathbb{R}$ is given by $g(t) := \sigma(X + tE)$. For the special case when $E = e_i e_j^*$, the canonical basis element, we denote the directional subdifferential by $\partial_{ij}\sigma(X)$ and refer to it as the (i, j)-th partial subdifferential of σ at X. By Theorem 13.10 we have

$$\partial_E \sigma(X) = \mathcal{F}(V^* E^* U) \text{ and } \partial_{ij}\sigma(X) = \mathcal{F}(V^* e_j e_i^* U), \tag{13.15}$$

where U and V are isometry such that $XV = \sigma(X)U$ and $X^*U = \sigma(X)V$.

Theorem 13.11 *Let $F : \mathbb{C}^{p\times q} \to \mathbb{C}^{m\times n}$ be continuously differentiable in a neighbourhood of $A \in \mathbb{C}^{p\times q}$. Define $g : \mathbb{C}^{p\times q} \to \mathbb{R}$ by $g(X) := \sigma(F(X))$, where $\sigma : \mathbb{C}^{m\times n} \to \mathbb{R}$, $X \mapsto \sigma(X)$, is a singular value map. Suppose that $g(A) \neq 0$ and that the multiplicity of $g(A)$ is r. Let $U \in \mathbb{C}^{m\times r}$ and $V \in \mathbb{C}^{n\times r}$ be isometry such that $F(A)V = g(A)U$ and $(F(A))^*U = g(A)V$. Then for $E \in \mathbb{C}^{p\times q}$ we have*

$$\partial_E g(A) = \mathcal{F}(V^*(DF(A)E)^*U) \quad \text{and} \quad \partial_{ij} g(A) = \mathcal{F}\left(V^*\left(\frac{\partial F(A)}{\partial X_{ij}}\right)^* U\right)$$

where $DF(A)$ is the derivative of F at A and $\dfrac{\partial F(A)}{\partial X_{ij}}$ is the partial derivative of $F(X)$ with respect to the (i, j)-th entry of X evaluated at A.

Proof For $E \in \mathbb{C}^{p\times q}$, define $g_E : \mathbb{R} \to \mathbb{R}$ by $g_E(t) := g(A + tE)$. Then $\partial_E g(A) = \partial g_E(0)$. Hence by Theorem 13.10, we have $\partial_E g(A) = \mathcal{F}(V^*(DF(A)E)^*U)$. Now by considering $E := E_{ij}$, the matrix with (i, j)-th entry 1 and zero everywhere else, we obtain the desired result for the partial subdifferential. □

For the special case when $r = 1$ (that is, when $g(A)$ is a simple singular value) and $q = 1$, Theorem 13.11 gives the result obtained by Ji-Guang [12].

A special case of Theorem 13.11 will be useful in the subsequent development. Suppose that $F : \mathbb{C}^{p\times q} \to \mathbb{C}^{km\times ln}$ is given by

$$F(X) := \phi(X) \otimes B,$$

where $B \in \mathbb{C}^{m\times n}$ and $\phi : \mathbb{C}^{p\times q} \to \mathbb{C}^{k\times l}$ is continuously differentiable in a neighbourhood of A. Then by Theorem 13.11, we have

$$\partial_E g(A) = \mathcal{F}\left(V^*\left(D\phi(A)E \otimes B\right)^* U\right), \ \partial_{ij} g(A) = \mathcal{F}\left(V^*\left(\frac{\partial \phi(A)}{\partial X_{ij}} \otimes B\right)^* U\right). \tag{13.16}$$

Further, a partial subgradient $\omega \in \partial_{ij} g(A)$ is given by

$$\omega = \mathrm{Tr}\left(V_\omega^* B^* U_\omega \left(\frac{\partial \phi(A)^\top}{\partial X_{ij}}\right)^*\right),$$

where $V_\omega \in \mathbb{C}^{n\times \ell}$ and $U_\omega \in \mathbb{C}^{m\times k}$ are such that $\mathrm{Vec}(V_\omega) = Vx$ and $\mathrm{Vec}(U_\omega) = Ux$ for some $x \in \mathbb{C}^r$. Indeed, we have $\omega = x^*V^*\left(\frac{\partial \phi(A)}{\partial X_{ij}} \otimes B\right)^* Ux$ for some $x \in \mathbb{C}^r$. Then with U_ω and V_ω as defined above, we have

$$\omega = \mathrm{Vec}(V_\omega)^* \mathrm{Vec}\left(B^*U_\omega \left(\frac{\partial \phi(A)^\top}{\partial X_{ij}}\right)^*\right) = \mathrm{Tr}\left(V_\omega^* B^* U_\omega \left(\frac{\partial \phi(A)^\top}{\partial X_{ij}}\right)^*\right)$$

as desired.

13.5 Singular Value Characterization of Eigenvalue Distance Problem

We have seen that $\mathrm{d}(A, B) = \mathrm{d}(A, S^{-1}BS)$ for any $m \times m$ nonsingular matrix S. Thus, if $J := S^{-1}BS$ is the Jordan canonical form of B then $\mathrm{d}(A, B) = \mathrm{d}(A, J)$. In particular, for any unitary matrix W, we have

$$\mathrm{d}(A, B) = \mathrm{d}(A, W^*BW) \geq \sigma_m(A, W^*BW) = \sigma_m(A, B). \tag{13.17}$$

Hence, without loss of generality, we can assume that B is upper triangular.

Recall that $\mathbf{UN}(m, \mathbb{C}) \subset \mathbb{C}^{m\times m}$ is the space of $m \times m$ upper triangular nilpotent matrices. By Schur theorem, there is a unitary matrix $W \in \mathbb{C}^{m\times m}$ such that

$$W^*BW = D + N \ \text{ and } \ \mathrm{eig}(D) = \mathrm{eig}(B) = \Lambda, \tag{13.18}$$

where D is a diagonal matrix and $N \in \mathbf{UN}(m, \mathbb{C})$. The eigenvalues of B can be made to appear in any order in D by choosing an appropriate W. This means that $\sigma_m(A,\ D + N) = \sigma_m(A, B)$ is independent of the order of the eigenvalues that appear in D. Henceforth, we assume that B is upper triangular and is given by $B = D + N$ with $D := \mathrm{diag}(B)$ and $N \in \mathbf{UN}(m, \mathbb{C})$.

It is proved in Gracia [10] that $\sigma_m(A, D + N) \longrightarrow 0$ as $\|N\|_2 \to \infty$ when $m = 2$. We now show that $\sigma_m(A, D + N) \longrightarrow 0$ as $\|N\|_2 \to \infty$ for any $m < n$.

Lemma 13.1 *Define* $f : \mathbf{UN}(m, \mathbb{C}) \longrightarrow \mathbb{R}$ *by* $f(N) := \sigma_m(A, D + N)$. *Then we have* $f(N) \longrightarrow 0$ *as* $\|N\|_2 \to \infty$. *Further, there exists* $N_{\mathrm{opt}} \in \mathbf{UN}(m, \mathbb{C})$ *such that*

$$\sup\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\} = \max\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\} = f(N_{\mathrm{opt}}).$$

Proof We have

$$f(N) = \sigma_{mn-m+1}(I_m \otimes A - (D + N)^\top \otimes I_n) = \sigma_{mn-m+1}(\mathcal{A} - \mathcal{N}),$$

where $\mathcal{A} := I_m \otimes A - D \otimes I_n$ is block diagonal and $\mathcal{N} := N^\top \otimes I_n$ is strictly block lower triangular and hence nilpotent.

If $D = \text{diag}(\mu_1, \cdots, \mu_m)$ then $\mathcal{A} = (A - \mu_1 I_n) \oplus \cdots \oplus (A - \mu_m I_n)$. Thus $\mathcal{A} - \mathcal{N}$ is block lower triangular. Note that $\mathcal{A} - \mathcal{N}$ is invertible $\iff$ $\mathcal{A}$ is invertible. For $i = 1 : m-1$ and $j = 2 : m$, let r_{ij} denote the (i, j)-th entry of the strictly upper triangular matrix N. Then $\|N\|_2 \to \infty \iff |r_{ij}| \to \infty$ for some (i, j).

First, suppose that $\text{eig}(D) \not\subset \text{eig}(A)$. Then $\mathcal{A}$ is invertible and $\mathcal{N}\mathcal{A}^{-1}$ is strictly block lower triangular and hence nilpotent and is given by

$$\begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ r_{12}(A-\mu_1 I_n)^{-1} & 0 & 0 & \cdots & 0 \\ r_{13}(A-\mu_1 I_n)^{-1} & r_{23}(A-\mu_2 I_n)^{-1} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ r_{1m}(A-\mu_1 I_n)^{-1} & r_{2m}(A-\mu_2 I_n)^{-1} & \cdots & r_{m-1,m}(A-\mu_{m-1} I_n)^{-1} & 0 \end{bmatrix}.$$

Note that $r_{12}(A - \mu_1 I_n)^{-1}, r_{23}(A - \mu_2 I_n)^{-1}, \ldots, r_{m-1,m}(A - \mu_{m-1} I_n)^{-1}$ are the first sub-diagonal blocks of $\mathcal{N}\mathcal{A}^{-1}$. Similarly, $(\mathcal{N}\mathcal{A}^{-1})^2$ is block lower triangular whose first sub-diagonals blocks consist of zero blocks and the second sub-diagonal blocks are given by

$$r_{12}r_{23}(A-\mu_1 I_n)^{-1}(A-\mu_2 I_n)^{-1}, r_{23}r_{34}(A-\mu_2 I_n)^{-1}(A-\mu_3 I_n)^{-1}, \ldots,$$
$$r_{m-2,m-1}r_{m-1,m}(A-\mu_{m-2} I_n)^{-1}(A-\mu_{m-1} I_n)^{-1}.$$

More generally, $(\mathcal{N}\mathcal{A}^{-1})^j$ is block lower triangular and the first $j-1$ sub-diagonal blocks consist of zero blocks. Hence $(\mathcal{N}\mathcal{A}^{-1})^{m-1}$ has only one sub-diagonal block (the m-th sub-diagonal block) given by $\prod_{j=1}^{m-1} r_{j,j+1}(A - \mu_j I_n)^{-1}$ and $(\mathcal{N}\mathcal{A}^{-1})^m = 0$. Consequently, we have

$$(\mathcal{A} - \mathcal{N})^{-1} = \mathcal{A}^{-1} + \sum_{j=1}^{m-1} \mathcal{A}^{-1}(\mathcal{N}\mathcal{A}^{-1})^j =: \mathcal{A}^{-1} + F(N).$$

Note that $\text{rank}\left((\mathcal{N}\mathcal{A}^{-1})^j\right) \geq m$ if $(\mathcal{N}\mathcal{A}^{-1})^j \neq 0$. Hence it follows that $F(N)$ is strictly block lower triangular and hence nilpotent and $\text{rank}(F(N)) \geq m$.

If X_{ij} denotes the (i, j)-th block of $F(N)$ then it follows that

$$X_{ij} = (A - \mu_j I)^{-1} r_{ij} (A - \mu_i I)^{-1} + \text{other terms}.$$

Hence $\sigma_m(X_{ij}) \to \infty$ as $|r_{ij}| \to \infty$. It is well known that $\sigma_m(F(N)) \geq \sigma_m(X_{ij})$. Consequently, $\sigma_m(F(N)) \to \infty$ as $|r_{ij}| \to \infty$. This shows that $\sigma_m(F(N)) \to \infty$ as $\|N\|_2 \to \infty$. Next, we have

$$f(N) = \sigma_{mn-m+1}(\mathcal{A} - \mathcal{N}) = \frac{1}{\sigma_m((\mathcal{A} - \mathcal{N})^{-1})} = \frac{1}{\sigma_m(\mathcal{A}^{-1} + F(N))}.$$

Now using the fact that $|\sigma_m(\mathcal{A}^{-1} + F(N)) - \sigma_m(F(N))| \leq \|\mathcal{A}^{-1}\|_2$, we have

$$\sigma_m(\mathcal{A}^{-1} + F(N)) \geq \sigma_m(F(N)) - \|\mathcal{A}^{-1}\|_2 \longrightarrow \infty \text{ as } \|N\|_2 \to \infty$$

because $\sigma_m(F(N)) \to \infty$ as $\|N\|_2 \to \infty$. Hence $f(N) = \dfrac{1}{\sigma_m(\mathcal{A}^{-1} + F(N))} \to 0$ as $\|N\|_2 \to \infty$. This proves the result when $\mathcal{A}$ is nonsingular.

Now suppose that $\mathcal{A}$ is singular. For any $\epsilon > 0$, we choose $\mathcal{A}_\epsilon := A_1 \oplus \cdots \oplus A_m$ such that $\mathcal{A}_\epsilon$ is nonsingular and

$$\|\mathcal{A} - \mathcal{A}_\epsilon\|_2 = \max\{\|A - \mu_j I_n - A_j\|_2 : j = 1 : m\} < \epsilon/2.$$

Then as above we have $\sigma_{mn-m+1}(\mathcal{A}_\epsilon - \mathcal{N}) \longrightarrow 0$ as $\|N\|_2 \to \infty$. Hence there exists $M > 0$ such that $\sigma_{mn-m+1}(\mathcal{A}_\epsilon - \mathcal{N}) < \epsilon/2$ for all $\|N\|_2 \geq M$. Now

$$|\sigma_{mn-m+1}(\mathcal{A} - \mathcal{N}) - \sigma_{mn-m+1}(\mathcal{A}_\epsilon - \mathcal{N})| \leq \|\mathcal{A} - \mathcal{A}_\epsilon\|_2 < \epsilon/2$$

shows that $f(N) = \sigma_{mn-m+1}(\mathcal{A} - \mathcal{N}) < \sigma_{mn-m+1}(\mathcal{A}_\epsilon - \mathcal{N}) + \epsilon/2 < \epsilon$ for all $\|N\|_2 \geq M$. This proves that $f(N) \to 0$ as $\|N\|_2 \to \infty$.

Finally, since $f(N)$ is Lipschitz continuous and $f(N) \to 0$ as $\|N\|_2 \to \infty$, there is a closed bounded (compact) subset $\mathcal{S} \subset \mathbf{UN}(m, \mathbb{C})$ such that

$$\begin{aligned}\sup\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\} &= \sup\{f(N) : N \in \mathcal{S}\} = \max\{f(N) : N \in \mathcal{S}\} \\ &= f(N_{\text{opt}})\end{aligned}$$

for some $N_{\text{opt}} \in \mathcal{S}$ as $\mathcal{S}$ is compact. This completes the proof. □

Consider $\mathbf{SN}(m, \mathbb{C}) := \{N \in \mathbf{UN}(m, \mathbb{C}) : N(j, j+1) \neq 0,\ j = 1 : m-1\}$. The set $\mathbf{SN}(m, \mathbb{C})$ consists of special upper triangular nilpotent matrices in the sense that $N \in \mathbf{SN}(m, \mathbb{C}) \Longrightarrow \text{rank}(N) = m - 1$ and that $\mathbf{SN}(m, \mathbb{C})$ is dense in $\mathbf{UN}(m, \mathbb{C})$. Since f is Lipschitz continuous, we have

$$\sup\{f(N) : N \in \mathbf{SN}(m, \mathbb{C})\} = \max\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\}. \tag{13.19}$$

We need the following result which follows from max-min characterization of singular values. For completeness, we provide a proof.

Lemma 13.2 *Let $A \in \mathbb{C}^{m\times n}$ and $\sigma_1(A) \geq \cdots \geq \sigma_p(A)$ be singular values of A, where $p := \min(m, n)$. Let $V \in \mathbb{C}^{n\times n}$ be nonsingular and $U \in \mathbb{C}^{m\times m}$. Then*

$$\sigma_j(UAV) \leq \|U\|_2\|V\|_2\, \sigma_j(A),\ \ j = 1 : p.$$

Proof By Horn and Johnson [11, Theorem 7.3.8, p.451], for $k = 1 : p$, we have

$$\sigma_k(A) = \max_{S:\dim(S)=k,\ x\in S\setminus\{0\}} \min \frac{\|Ax\|_2}{\|x\|_2},$$

where $S \subset \mathbb{C}^n$ is a subspace. Now,

$$\frac{\|UAVx\|_2}{\|x\|_2} \leq \|U\|_2 \frac{\|AVx\|_2}{\|Vx\|_2} \frac{\|Vx\|_2}{\|x\|_2} \leq \|U\|_2 \|V\|_2 \frac{\|Ay\|_2}{\|y\|_2},$$

where $y := Vx$. Since V is nonsingular, we have $\sigma_k(UAV) \leq \|U\|_2 \|V\|_2 \, \sigma_k(A)$ for $k = 1 : p$. □

Recall that ℓ is the number of distinct eigenvalues of D and $\text{eig}(D) = \{\lambda_1, \ldots, \lambda_\ell\}$.

Theorem 13.12 *Either* $f(N) = \sigma_m(A,\ D + N) \equiv 0$ *on* $\mathbf{UN}(m, \mathbb{C})$ *or* $f(N) \neq 0$ *for all* $N \in \mathbf{SN}(m, \mathbb{C})$. *In particular, if* $m = \ell$ *then either* $f(N) \equiv 0$ *on* $\mathbf{UN}(m, \mathbb{C})$ *or* $f(N) \neq 0$ *for all* $N \in \mathbf{UN}(m, \mathbb{C})$.

Proof First, note that if $N \in \mathbf{SN}(m, \mathbb{C})$ then $\text{rank}(D + N - \lambda I_m) \geq m - 1$ for all $\lambda \in \mathbb{C}$. This shows that $\mathbf{gm}(\lambda_j,\ D + N) = 1,\ \ j = 1 : \ell$, for all $N \in \mathbf{SN}(m, \mathbb{C})$. Consequently, there exists unique natural numbers $m_1 \geq \cdots \geq m_\ell$ such that

$$J := J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_\ell}(\lambda_\ell)$$

is the Jordan canonical form of $D + N$ for all $N \in \mathbf{SN}(m, \mathbb{C})$, where $J_{m_j}(\lambda_j)$ is the Jordan block of size $m_j \times m_j$ with eigenvalue λ_j for $j = 1 : \ell$. Thus $D + N \simeq J$ for all $N \in \mathbf{SN}(m, \mathbb{C})$.

Now, suppose that $f(N_0) = 0$ for some $N_0 \in \mathbf{SN}(m, \mathbb{C})$. Let $N \in \mathbf{SN}(m, \mathbb{C})$. Since $D + N \simeq J \simeq D + N_0$, there is a nonsingular matrix S such that $D + N = S^{-1}(D + N_0)S$. Then we have

$$\begin{aligned} I_m \otimes A - (D + N)^\top \otimes I_n &= I_m \otimes A - (S^{-1}(D + N_0)S)^\top \otimes I_n \\ &= (S^\top \otimes I_n)(I_m \otimes A - (D + N_0)^\top \otimes I_n)(S^\top \otimes I_n)^{-1}. \end{aligned}$$

By Lemma 13.2, we have

$$\begin{aligned} f(N) &= \sigma_{mn-m+1}((S^\top \otimes I_n)(I_m \otimes A - (D + N_0)^\top \otimes I_n)(S^\top \otimes I_n)^{-1}) \\ &\leq \|S^{-1}\|_2 \|S\|_2 f(N_0) = 0. \end{aligned}$$

This shows that $f(N) = 0$ for all $N \in \mathbf{SN}(m, \mathbb{C})$. Consequently, $f(N) = 0$ for all $N \in \mathbf{UN}(m, \mathbb{C})$ as $f(N)$ is Lipschitz continuous and $\mathbf{SN}(m, \mathbb{C})$ is dense in $\mathbf{UN}(m, \mathbb{C})$. On the other hand, if $f(N_0) \neq 0$ then we have $f(N_0) \leq \|S^{-1}\|_2 \|S\|_2 \, f(N)$ showing that $f(N) \neq 0$ for all $N \in \mathbf{SN}(m, \mathbb{C})$. This proves the first assertion.

Finally, if $m = \ell$ then the diagonal entries of D are distinct. Thus $D + N$ is diagonalizable for all $N \in \mathbf{UN}(m, \mathbb{C})$. Hence either $f(N) = 0$ for all $N \in \mathbf{UN}(m, \mathbb{C})$ or $f(N) \neq 0$ for all $N \in \mathbf{UN}(m, \mathbb{C})$. This completes the proof. □

The proof of Theorem 13.12 shows that the following result holds.

Corollary 13.3 *Let* $S \in \mathbb{C}^{m \times m}$ *be nonsingular and let* $\text{cond}(S) := \|S^{-1}\|_2 \|S\|_2$. *Then we have*

$$\frac{\sigma_m(A, B)}{\text{cond}(S)} \le \sigma_m(A, S^{-1}BS) \le \text{cond}(S)\sigma_m(A, B).$$

We denote the Jordan canonical form of a matrix $X \in \mathbb{C}^{m\times m}$ by $\text{JCF}(X)$. Let $\mathbf{JCF}(m, \Lambda)$ denote the set of all $m \times m$ matrices in Jordan canonical form with spectrum Λ. Thus

$$\mathbf{JCF}(m, \Lambda) := \{\text{JCF}(X) : X \in \mathbb{C}^{m\times m},\ \text{eig}(X) = \Lambda\}.$$

Let $\mathbf{Diag}(m, \Lambda)$ denote the set of $m \times m$ diagonal matrices with spectrum Λ, that is,

$$\mathbf{Diag}(m, \Lambda) := \{D \in \mathbb{C}^{m\times m} : D \text{ is diagonal and } \text{eig}(D) = \Lambda\}.$$

Then $\mathbf{Diag}(m, \Lambda) \oplus \mathbf{UN}(m, \mathbb{C})$ consists of all $m \times m$ upper triangular matrices with spectrum Λ and

$$\mathbf{JCF}(m, \Lambda) = \text{JCF}\left(\mathbf{Diag}(m, \Lambda) \oplus \mathbf{UN}(m, \mathbb{C})\right).$$

On the other hand, $\mathbf{Diag}(m, \Lambda) \oplus \mathbf{SN}(m, \mathbb{C})$ consists of $m \times m$ nonderogatory upper triangular matrices with spectrum Λ. Indeed, if $D + N \in \mathbf{Diag}(m, \Lambda) \oplus \mathbf{SN}(m, \mathbb{C})$ then it is easy to see that $\text{rank}(D + N - \lambda I_m) \ge m - 1$ for $\lambda \in \mathbb{C}$. Consequently, we have $\mathbf{gm}(\lambda_j, D + N) = 1$ for $j = 1 : \ell$. Hence there exist unique natural numbers $m_1 \ge \cdots \ge m_\ell$ such that

$$\text{JCF}(D + N) = J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_\ell}(\lambda_\ell),$$

where $J_{m_j}(\lambda_j)$ is the Jordan block of size $m_j \times m_j$ with eigenvalue λ_j for $j = 1 : \ell$. This shows that $D + N$ is nonderogatory. In fact, $\mathbf{gm}(\lambda_j,\ D + N) = 1$ for all $j = 1 : \ell$ and for all $N \in \mathbf{SN}(m, \mathbb{C})$. Hence we have

$$\text{JCF}(D + N) = J_{m_1}(\lambda_1) \oplus \cdots \oplus J_{m_\ell}(\lambda_\ell) \ \text{ for all } \ N \in \mathbf{SN}(m, \mathbb{C}). \tag{13.20}$$

We have seen that $\text{d}(A, B) = \text{d}(A,\ \text{JCF}(B))$. Suppose that B is nonderogatory. Then there exists $D \in \mathbf{Diag}(m, \Lambda)$ such that $\text{JCF}(B) = \text{JCF}(D + N)$ for all $N \in \mathbf{SN}(m, \mathbb{C})$. Consequently, we have

$$\text{d}(A, B) = \text{d}(A,\ \text{JCF}(D + N)) = \text{d}(A, D + N) \ge \sigma_m(A, D + N)$$

for all $N \in \mathbf{SN}(m, \mathbb{C})$. This shows that

$$\text{d}(A, B) \ge \sup\{f(N) : N \in \mathbf{SN}(m, \mathbb{C})\} = \max\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\}.$$

Thus we have the following result. See Mengi [23], Kressner et al. [18], Gracia [10], Malyshev [22], Lippert [20] for similar results.

Theorem 13.13 *Let $B \in \mathbb{C}^{m\times m}$ be nonderogatory and upper triangular such that* $\text{eig}(B) = \Lambda$. *Set* $D := \text{diag}(B)$ *and consider* $f(N) := \sigma_m(A,\ D + N)$. *Then*

$$\text{d}(A, B) \geq \sup\{f(N) : N \in \mathbf{SN}(m, \mathbb{C})\} = \max\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\}.$$

Suppose that the maximum is attained at N_0 and that $f(N_0) \neq 0$. Also suppose that there exist left and right singular vectors U and V of $\mathbf{T}_{A,B}$ corresponding to $f(N_0)$ such that $U^*U = V^*V$ and $\text{rank}(V) = m$. *Define* $\Delta A := -f(N_0)UV^+$. *Then*

$$(A + \Delta A)V = V(D + N_0) \ \text{ and } \ \|\Delta A\|_2 = f(N_0) = \text{d}(A, D + N_0).$$

If $m = \ell$ then $\text{d}(A, B) = f(N_0) = \sigma_m(A, D + N_0)$. *If $m > \ell$ and $N_0 \in \mathbf{SN}(m, \mathbb{C})$ then* $\text{d}(A, B) = f(N_0) = \sigma_m(A, D + N_0)$.

Proof By Theorem 13.6, we have $(A + \Delta A)V = V(D + N_0)$ and $\|\Delta A\|_2 = f(N_0) = \text{d}(A, D + N_0)$. Now, if $m = \ell$, then B is an $\ell \times \ell$ matrix with ℓ distinct eigenvalues. Hence $B \simeq D + N_0$. Consequently, $\text{d}(A, B) = \text{d}(A, D + N_0) = f(N_0) = \sigma_m(A, D + N_0)$. On the other hand, if $N_0 \in \mathbf{SN}(m, \mathbb{C})$ when $m > \ell$, then again $B \simeq D + N_0$ as $D + N_0$ is nonderogatory. Hence $\text{d}(A, B) = \text{d}(A, D + N_0) = f(N_0) = \sigma_m(A, D + N_0)$. This completes the proof. □

Remark 13.3 Theorem 13.13 assumes that either $m = \ell$ or $N_0 \in \mathbf{SN}(m, \mathbb{C})$. Now suppose that $N_0 \in \mathbf{UN}(m, \mathbb{C})$ but not in $\mathbf{SN}(m, \mathbb{C})$. Then $N_0(j, j + 1) = 0$ for some j. For any $\epsilon > 0$, replacing the zero entries in $N_0(j, j + 1)$ with ϵ, we have $N_\epsilon \in \mathbf{SN}(m, \mathbb{C})$ such that $\|N_0 - N_\epsilon\|_2 = \epsilon$. Set $B_\epsilon := D + N_\epsilon$. Now assuming that $\mathbf{T}_{A,B_\epsilon}$ has left and right singular vectors U and V corresponding to $f(N_\epsilon)$ such that $U^*U = V^*V$ and $\text{rank}(V) = m$, we have $\text{d}(A, B_\epsilon) = f(N_\epsilon)$. Then

$$|\text{d}(A, B) - \text{d}(A, B_\epsilon)| = |f(N_0) - f(N_\epsilon)| \leq \|N_0 - N_\epsilon\|_2 = \epsilon.$$

This shows that if $N_0 \notin \mathbf{SN}(m, \mathbb{C})$ then $\text{d}(A, B)$ can be approximated arbitrarily closely by $\text{d}(A, B_\epsilon)$ provided the conditions on the singular vectors are satisfied for $f(N_\epsilon)$.

The main question now is to determine whether or not the conditions on the singular vectors in Theorem 13.13 are satisfied at the point of maximum N_0. It is proved in Malyshev [22] that these conditions are satisfied when $\ell = 1$ and $m = 2$. In that case

$$\delta_2(\lambda, A) = \max\{f(N) : N \in \mathbf{UN}(2, \mathbb{C})\} = f(N_0) = \sigma_2(A, \lambda I_2 + N_0).$$

Note that $N_0 = \begin{bmatrix} 0 & t_0 \\ 0 & 0 \end{bmatrix}$ for some $t_0 \in \mathbb{C}$. For the case when $m = \ell = 2$, it is shown in Gracia [10] that the conditions on the singular vectors are satisfied and that

$$\delta_2(\Lambda, A) = \max\{f(N) : N \in \mathbf{UN}(2, \mathbb{C})\} = f(N_0) = \sigma_2(A, D + N_0)$$

provided that $N_0 \neq 0$, that is, $N_0 \in \mathbf{SN}(2, \mathbb{C})$. We provide a unified simple proof of these results using subdifferential of $f(N_0)$. We show the condition $U^*U = V^*V$ is satisfied for $m > 2$ when the singular value $f(N_0)$ is simple. However, the rank condition ($\text{rank}(V) = m$) may or may not be satisfied. If the rank condition is not satisfied then we have an analogue of Corollary 13.2 which ensures that some elements in Λ are eigenvalues of $A + \Delta A$ with $\|\Delta A\|_2 = f(N_0)$.

Define $F : \mathbf{UN}(m, \mathbb{C}) \longrightarrow \mathbb{C}^{mn \times mn}$ by $F(N) := I_m \otimes A - (D + N)^\top \otimes I_n$. Then recall that $f(N) = \sigma_m(A, D + N) = \sigma_{mn-m+1}(F(N))$. We now show that if $f(N_0)$ is a simple singular value then the associated left and right singular vectors U and V of $\mathbf{T}_{A,D+N}$ satisfy the condition $U^*U = V^*V$. In fact, in such a case U^*V is upper triangular.

Theorem 13.14 *Suppose that* $f(N_0) = \max\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\}$ *and that* $f(N_0) \neq 0$. *Suppose that the multiplicity of* $f(N_0)$ *is* r. *Let* $\mathcal{U} \in \mathbb{C}^{nm \times r}$ *and* $\mathcal{V} \in \mathbb{C}^{nm \times r}$ *be isometry such that* $F(N_0)\mathcal{V} = f(N_0)\mathcal{U}$ *and* $(F(N_0))^*\mathcal{U} = f(N_0)\mathcal{V}$. *Then we have*

$$0 \in \mathcal{F}\left(\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}\right) \text{ for all } E \in \mathbf{UN}(m, \mathbb{C}).$$

For each $E \in \mathbf{UN}(m, \mathbb{C})$, *there exist left and right singular vectors* U *and* V *of* $\mathbf{T}_{A,D+N_0}$ *corresponding to* $f(N_0)$ *such that* $\text{Tr}(U^*VE) = 0$. *In particular, if* $m = 2$ *then* U^*V *is upper triangular and* $U^*U = V^*V$.

Now, suppose that $f(N_0)$ *is simple. Let* U *and* V *be left and right singular vectors of* $\mathbf{T}_{A,D+N_0}$ *corresponding to* $f(N_0)$. *Then* U^*V *is upper triangular and we have* $U^*U = V^*V$.

Proof Note that for any $E \in \mathbf{UN}(m, \mathbb{C})$, the function $g_E(t) := f(N_0 + tE)$ attains a maximum at $t = 0$. Hence $t = 0$ is a Clarke stationary point of $g_E(t)$. Consequently, $0 \in \partial g_E(0) = \partial_E f(N_0)$. By Theorem 13.11, we have $\partial_E f(N_0) = \mathcal{F}\left(\mathcal{V}^*(E^\top \otimes I_n)^*\mathcal{U}\right)$. Hence we have

$$0 \in \mathcal{F}\left(\mathcal{V}^*(E^\top \otimes I_n)^*\mathcal{U}\right) \text{ for all } E \in \mathbf{UN}(m, \mathbb{C}). \tag{13.21}$$

Equivalently, we have $0 \in \mathcal{F}\left(\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}\right)$ for all $E \in \mathbf{UN}(m, \mathbb{C})$. Thus there exists $x \in \mathbb{C}^r$ such that $x^*\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}x = 0$. Let $V \in \mathbb{C}^{n \times r}$ and $U \in \mathbb{C}^{n \times r}$ be such that $\text{Vec}(V) = \mathcal{V}x$ and $\text{Vec}(U) = \mathcal{U}x$. Then we have

$$\text{Tr}(U^*VE) = x^*\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}x = 0.$$

Now, if $m = 2$, then for $E = e_1e_2^\top$, we have $e_2^\top U^*Ve_1 = \text{Tr}(U^*VE) = 0$. This shows that U^*V is upper triangular. Since $D + N_0$ is upper triangular, by Theorem 13.8, we have $U^*U = V^*V$. This proves the results when $f(N_0)$ is multiple.

Now consider the case when $f(N_0)$ is simple. Then $\mathcal{U}$ and $\mathcal{V}$ are vectors in $\mathbb{C}^{nm}$. Consequently, $\mathcal{F}\left(\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}\right) = \{\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}\}$. Hence by (13.21), we have $\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V} = 0$ for all $E \in \mathbf{UN}(m, \mathbb{C})$. Let $U \in \mathbb{C}^{n \times m}$ and $V \in \mathbb{C}^{n \times m}$ be such that $\text{Vec}(U) = \mathcal{U}$ and $\text{Vec}(V) = \mathcal{V}$. Then $\text{Tr}(U^*VE) = \mathcal{U}^*(E^\top \otimes I_n)\mathcal{V} =$

0 for all $E \in \mathbf{UN}(m, \mathbb{C})$. In other words, $\langle U^*V, E^* \rangle = 0$ for all $E \in \mathbf{UN}(m, \mathbb{C})$. Thus U^*V is orthogonal to all strictly lower triangular matrices in $\mathbb{C}^{m\times m}$ and hence U^*V is upper triangular. Since $D + N_0$ is upper triangular, by Theorem 13.8, we have $U^*U = V^*V$. This completes the proof. □

Remark 13.4 If $f(N_0)$ is multiple then Theorem 13.14 does not guarantee existence of left and right singular vectors U and V of $\mathbf{T}_{A,D+N_0}$ such that $U^*U = V^*V$. The existence is guaranteed if there exists $x \in \mathbb{C}^r$ such that

$$0 = x^*\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}x \in \mathcal{F}\left(\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}\right)$$

for all $E \in \mathbf{UN}(m, \mathbb{C})$. In such a case, we choose $U \in \mathbb{C}^{n\times m}$ and $V \in \mathbb{C}^{n\times m}$ such that $\mathrm{Vec}(U) = \mathcal{U}x$ and $\mathrm{Vec}(V) = \mathcal{V}x$. Then $\mathrm{Tr}(U^*VE) = x^*\mathcal{U}^*(E^\top \otimes I_n)\mathcal{V}x = 0$ for all $E \in \mathbf{UN}(m, \mathbb{C})$ showing that U^*V is upper triangular. Since $D + N_0$ is upper triangular, by Theorem 13.8, we have $U^*U = V^*V$. Equivalently, if there exists $x \in \mathbb{C}^r$ such that

$$0 = x^*\mathcal{U}^*(e_je_i^\top \otimes I_n)\mathcal{V}x \in \mathcal{F}\left(\mathcal{U}^*(e_je_i^\top \otimes I_n)\mathcal{V}\right),\ i = 1:m-1,\ j = 2:m,$$

then $U \in \mathbb{C}^{n\times m}$ and $V \in \mathbb{C}^{n\times m}$ such that $\mathrm{Vec}(U) = \mathcal{U}x$ and $\mathrm{Vec}(V) = \mathcal{V}x$ satisfy the condition $U^*U = V^*V$. We have seen that such a vector $x \in \mathbb{C}^m$ exists when $m = 2$. So, does there exist such a vector $x \in \mathbb{C}^m$ when $m > 2$? For $m > 2$, an affirmative answer can be obtained if the joint-field of values (i.e. the joint numerical range) of $\mathcal{U}^*(e_je_i^\top \otimes I_n)\mathcal{V}$, $i = 1 : m - 1$, $j = 2 : m$, contains 0.

We have seen that there exist left and right singular vectors U and V of $\mathbf{T}_{A,D+N_0}$ corresponding to $f(N_0)$ such that $U^*U = V^*V$ when $m = 2$ irrespective of the multiplicity of the singular value $f(N_0) = \sigma_m(A, D + N_0)$. We now show that the condition $\mathrm{rank}(V) = 2$ holds when $m = 2$ irrespective of the multiplicity of $f(N_0)$ provided that $N_0 \neq 0$. Compare our proof with those in Malyshev [22], Gracia [10], Lippert [20].

Theorem 13.15 *Let* $f(N_0) = \max\{\sigma_m(A, D + N) : N \in \mathbf{UN}(m, \mathbb{C})\}$. *Suppose that* $f(N_0) \neq 0$. *If* $m = 2$ *and* $N_0 \neq 0$ *then there exist left and right singular vectors* U *and* V *of* $\mathbf{T}_{A,D+N_0}$ *corresponding to* $f(N_0)$ *such that* $U^*U = V^*V$ *and* $\mathrm{rank}(V) = 2$.

Proof Note that $\mathbf{UN}(2, \mathbb{C}) = \left\{\begin{bmatrix} 0 & t \\ 0 & 0 \end{bmatrix} : t \in \mathbb{C}\right\}$. By Theorem 13.14 there exist left and right singular vectors U and V of $\mathbf{T}_{A,D+N_0}$ corresponding to $f(N_0)$ such that $U^*U = V^*V$.

If possible, suppose that $\mathrm{rank}(V) < 2$. Note that $\ker(V) = \ker(U)$. Since $\mathrm{Tr}(V^*V) = 1$, we have $\dim \ker(V) = 1$. Let $x \in \ker(V)$ be nonzero. Then $AV - V(D + N_0) = f(N_0)U \Longrightarrow V(D + N_0)x = 0 \Longrightarrow (D + N_0)x \in \ker(V)$. Hence x is a eigenvector of $D + N_0$. Similarly, $A^*U - U(D + N_0)^* = f(N_0)V \Longrightarrow U(D + N_0)^*x = 0 \Longrightarrow (D + N_0)^*x \in \ker(U) = \ker(V)$. Hence x is an eigenvector of $(D + N_0)^*$. This shows that x is a common eigenvector of $D + N_0$ and $(D + N_0)^*$.

Note that $N_0 = \begin{bmatrix} 0 & t_0 \\ 0 & 0 \end{bmatrix}$ for some $t_0 \in \mathbb{C}$ and $t_0 \neq 0$. Hence $N_0 \in \mathbf{SN}(2, \mathbb{C})$. Now consider the case when $\ell = 2$. Then $D = \mathrm{diag}(\lambda_1, \lambda_2)$ and $\lambda_1 \neq \lambda_2$. Thus x is a common eigenvector of

$$D + N_0 = \begin{bmatrix} \lambda_1 & t_0 \\ 0 & \lambda_2 \end{bmatrix} \text{ and } (D + N_0)^* = \begin{bmatrix} \bar{\lambda}_1 & 0 \\ \bar{t}_0 & \bar{\lambda}_2 \end{bmatrix}.$$

Since $t_0 \neq 0$, it follows that $x = 0$, which is a contradiction. Hence $\mathrm{rank}(V) = 2$.

On the other hand, if $\ell = 1$ then $\lambda_1 = \lambda_2 = \lambda$ and x is a common eigenvector of

$$D + N_0 = \begin{bmatrix} \lambda & t_0 \\ 0 & \lambda \end{bmatrix} \text{ and } (D + N_0)^* = \begin{bmatrix} \bar{\lambda} & 0 \\ \bar{t}_0 & \bar{\lambda} \end{bmatrix}.$$

Since $t_0 \neq 0$, it follows that $x = 0$, which is a contradiction. Hence $\mathrm{rank}(V) = 2$. □

Remark 13.5 We mention that if $\ell = m = 2$ then it is important that $f(N)$ attains maximum at $N_0 \neq 0$, that is, $N_0 \in \mathbf{SN}(2, \mathbb{C})$. In such a case $D + N_0$ is nonderogatory and hence we have

$$\delta_2(\Lambda, A) = \mathrm{d}(A, D) = \mathrm{d}(A, D + N_0) = f(N_0) = \sigma_2(A, N_0).$$

On the other hand, if $\ell = 1$ and $m = 2$ then it has been proved that $\delta_2(\lambda, A) = f(N_0)$ even when the maximum is attained at $N_0 = 0$. In such a case, a rank-1 perturbation ΔA can be constructed such that λ is a multiple eigenvalue of $A + \Delta A$ and $\|\Delta A\|_2 = f(N_0) = \delta_2(\lambda, A)$; see Malyshev [22].

Remark 13.6 Recall that $\mathrm{d}(A, B) = \mathrm{d}(A, S^{-1}BS)$ for any nonsingular matrix S and

$$\begin{aligned} \delta_m(\Lambda, A) &= \inf\{\mathrm{d}(A, B) : B \in \mathbb{C}^{m\times m},\ \mathrm{eig}(B) = \Lambda\} \\ &= \inf\{\mathrm{d}(A, J) : J \in \mathbf{JCF}(m, \Lambda)\} \\ &= \min\{\mathrm{d}(A, J) : J \in \mathbf{JCF}(m, \Lambda)\}. \end{aligned}$$

Hence, in view of (13.20), if $m = \ell$ then for any $D \in \mathbf{Diag}(m, \Lambda)$, we have

$$\delta_m(\Lambda, A) = \mathrm{d}(A, D) \geq \max\{f(N) : N \in \mathbf{UN}(m, \mathbb{C})\} = f(N_0)$$

and the equality holds when $f(N_0)$ is simple and $\mathrm{rank}(V) = m$. The equality also holds when $U^*U = V^*V$ and $\mathrm{rank}(V) = m$. Here U and V is are left and right singular vectors of $\mathbf{T}_{A,D+N_0}$ corresponding to $f(N_0)$. On the other hand, if $\ell < m$ then $D + N$ will have different JCF for different choices of $D \in \mathbf{Diag}(m, \mathbb{C})$. Then, in view of Theorem 13.13 and Remark 13.3, $\delta_m(\Lambda, A)$ can be approximated arbitrary closely by taking minimum of $\mathrm{d}(A, J)$ over all nonderogatory $J \in \mathbf{JCF}(m, \Lambda)$.

References

1. Alam R (2005) On the construction of nearest defective matrices to a normal matrix. Linear Algebra Appl 395:367–370
2. Alam R (2006) Wilkinson's problem revisited. J Anal 4:176–205
3. Alam R, Bora S (2005) On sensitivity of eigenvalues and eigendecompositions of matrices. Linear Algebra Appl 396:273–301
4. Alam R, Bora S, Byers R, Overton ML (2011) Characterization and construction of the nearest defective matrix via coalescence of pseudospectral components. Linear Algebra Appl 435(3):494–513
5. Axler S (2015) Linear algebra done right, vol 2. Springer
6. Clarke FH (1990) Optimization and nonsmooth analysis. SIAM, Philadelphia
7. Demmel J (1990) Nearest defective matrices and the geometry of ill-conditioning. In: Cox M, Hammarling S (eds) Reliable numerical computation, vol 44, Oxford Sci., pp 35–55
8. Demmel JW (1987) On condition numbers and the distance to the nearest ill-posed problem. Numerische Mathematik 51(3):251–289
9. Gantmacher FR (1959) The theory of matrices. Chelsea 2
10. Gracia JM (2005) Nearest matrix with two prescribed eigenvalues. Linear Algebra Appl 401:277–294
11. Horn RA, Johnson CR (2013) Matrix analysis, 2nd edn. Cambridge University Press
12. Ji-Guang S (1988) A note on simple non-zero singular values. J Comput Math, pp 258–266
13. Kahan W (1981) On neighbouring defective matrices. Notes for lecture at Gatlinburg VIII
14. Karow M, Mengi E (2015) Matrix polynomials with specified eigenvalues. Linear Algebra Appl 466:457–482
15. Karow M, Kressner D, Mengi E (2014) Nonlinear eigenvalue problems with specified eigenvalues. SIAM J Matrix Anal Appl 35(3):819–834
16. Kokabifar E, Loghmani G, Psarrakos P, Karbassi S (2017) On the distance from a matrix polynomial to matrix polynomials with k prescribed distinct eigenvalues. Linear Multilinear Algebra 65(4):658–676
17. Kokabifar E, Psarrakos P, Loghmani GB (2018) On the distance from a matrix polynomial to matrix polynomials with some prescribed eigenvalues. Linear Algebra Appl 544:158–185
18. Kressner D, Mengi E, Nakić I, Truhar N (2014) Generalized eigenvalue problems with specified eigenvalues. IMA J Numer Anal 34(2):480–501
19. Lewis AS, Sendov HS (2005) Nonsmooth analysis of singular values. Part ii: applications. Set-Valued Anal 13(3):243–264
20. Lippert RA (2005) Fixing two eigenvalues by a minimal perturbation. Linear Algebra Appl 406:177–200
21. Lippert RA (2010) Fixing multiple eigenvalues by a minimal perturbation. Linear Algebra Appl 432(7):1785–1817
22. Malyshev AN (1999) A formula for the 2-norm distance from a matrix to the set of matrices with multiple eigenvalues. Numerische Mathematik 83(3):443–454
23. Mengi E (2011) Locating a nearest matrix with an eigenvalue of prespecified algebraic multiplicity. Numerische Mathematik 118(1):109–135
24. Papathanasiou N, Psarrakos P (2008) The distance from a matrix polynomial to matrix polynomials with a prescribed multiple eigenvalue. Linear Algebra Appl 429(7):1453–1477
25. Psarrakos PJ (2012) Distance bounds for prescribed multiple eigenvalues of matrix polynomials. Linear Algebra Appl 436(11):4107–4119
26. Ruhe A (1970) Properties of a matrix with a very ill-conditioned eigenproblem. Numerische Mathematik 15(1):57–60
27. Wilkinson J (1972) Note on matrices with a very ill-conditioned eigenproblem. Numerische Mathematik 19(2):175–178
28. Wilkinson J (1984) On neighbouring matrices with quadratic elementary divisors. Numerische Mathematik 44(1):1–21

29. Wilkinson J (1984) Sensitivity of eigenvalues. Utilitas Mathematica 25:5–76
30. Wilkinson J (1987) Eigenvlue problems. In: A I, Powell M (ed) The state of art in numerical analysis. Claredon Press, Oxford, pp 1–39
31. Wilkinson J (1996) The algebraic eigenvalue problem. Oxford University Press
32. Wilkinson JH (1986) Sensitivity of eigenvalues ii. Util Math 30:243–286

Chapter 14
Equality of BLUEs for Full, Small, and Intermediate Linear Models Under Covariance Change, with Links to Data Confidentiality and Encryption



Stephen J. Haslett and Simo Puntanen

Mathematics Subject Classification (2010) 62J05 · 62J10

14.1 Introduction

Generally, when the error covariance structure in a linear model is changed, the BLUEs of estimable functions of its parameters also change. Rao [18] explored the necessary and sufficient conditions that the change in error covariance results in no change to the BLUEs.

Rao's conditions for allowable changes to a shared error covariance while still retaining BLUEs may be applied separately to two models, one a submodel (the "small model") of the other (the "full model"). The necessary and sufficient condition for BLUEs for the full model applies to all estimable functions of its parameters, not just to estimable functions containing parameters only from the small model. BLUEs

S. J. Haslett (✉)
School of Mathematical and Computational Sciences & Research Centre for Hauora and Health, Massey University, Palmerston North, New Zealand

Research School of Finance, Actuarial Studies and Statistics, The Australian National University, Canberra, Australia

NIASRA, Faculty of Engineering and Information Sciences, University of Wollongong, Wollongong, Australia
e-mail: s.j.haslett@massey.ac.nz; stephen.haslett@anu.edu.au

S. Puntanen
Faculty of Information Technology and Communication Sciences, Tampere University, Tampere, Finland
e-mail: simo.puntanen@tuni.fi

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_14

of estimable functions of parameters that are in both models are not necessarily equal in the full and small models, but for each model its own BLUEs remain unaltered.

We show that if both model and submodel have the same original covariance structure, the permissible changes to the error covariance for retention of BLUEs of all estimable functions of all parameters in both models are similar in block diagonal structure of the added quadratic form, but are nevertheless different to those for the two models considered separately.

Calling the original model the "full model" and the submodel the "small model" allows explicit specification of any "intermediate models" between them, i.e., all models containing the same parameters as the small model plus some but not all the parameters in the full model that are not in the small model. It also permits discussion of restrictions on changes to error covariance so that, in addition to the full and small models, all intermediate models also retain their own BLUEs. The results developed apply regardless of the order in which regressors in a reparametrised version of the full model that are not in the small model (including any possible interaction terms) are added or removed.

Haslett et al. [9] considered properties of BLUEs in full versus small linear models with new observations where the supplementary quadratic form added to the original covariance structure implicitly involves block diagonal matrices based only on functions of the design matrices of the linear models.

Haslett et al. [10] have outlined in detail the permitted error covariance changes that retain BLUEs of estimable functions of parameters for both full and small fixed effect linear models, but the paper does not consider intermediate models or retention of the covariance of the BLUEs.

The key questions addressed in this paper are

1. When two nested fixed effect linear models (the "full" and the "small" models) are each to retain the equality of BLUEs for their own estimable functions of their own parameters, how are the changes in shared error covariance matrix that are permissible linked to block diagonal matrices?
2. What additional restrictions on shared error covariance matrix are needed when two nested fixed effect linear models (the "full" and the "small" model) each must retain the equality of BLUEs as well as the covariance for estimable functions of their own parameters?
3. What further limitations are placed on changes to the original shared error covariance matrix if BLUEs of estimable parameters are retained for all intermediate models as well as for the full and small models, and what further restrictions would allow the covariance of these BLUEs also to remain unchanged?

Having established the necessary core results, the way they are related to data confidentiality and encryption is outlined. For data confidentiality, it is shown that given a pre-specified full model that may include interactions of any order, it is possible to construct any number of datasets that have exactly the same BLUEs and covariances of BLUEs for the estimable function of parameters as the original dataset, for all possible submodels. These methods extend the results of Haslett and Govindaraju [7, 8] on confidentiality and encryption and simplify their implementation. The results

have roots in the data cloning literature, in the wider sense of non-identical datasets that produce identical statistics. Also see, for example, Anscombe [1], Chatterjee and Firat [2] and Govindaraju and Haslett [5]. Replications of the original data are another very simple form of cloning which, via Markov Chain Monte Carlo methods, can be used to obtain maximum likelihood estimators of the parameters in a range of models from linear models to generalised linear mixed models (Lele et al. [13]).

Example 1

Let

$$Y = \begin{pmatrix}1 & 5 & 9 & 23 & 36\end{pmatrix}'$$
$$Y_{new} = \begin{pmatrix}1.132843 & 5.3215964 & 10.559413 & 24.992382 & 34.103914\end{pmatrix}'.$$
$$X = \begin{pmatrix}1 & 1 & 1\\ 1 & 2 & 4\\ 1 & 3 & 9\\ 1 & 4 & 16\\ 1 & 5 & 25\end{pmatrix} \quad \text{and} \quad X_{new} = \begin{pmatrix}1.0298573 & 1.0298573 & 1.0298573\\ 1.0565343 & 2.2798898 & 4.7266007\\ 0.9799919 & 3.0624747 & 9.6284206\\ 0.9825015 & 4.1142250 & 17.377995\\ 0.9473309 & 4.7366547 & 23.683273\end{pmatrix}.$$

The BLUE for the original data is

$$\hat{\beta} = \begin{pmatrix}2.4\\ -3.2\\ 2\end{pmatrix}.$$

For the same data, the BLUE for the submodel with the last column of X deleted is

$$\hat{\beta} = \begin{pmatrix}-11.6\\ 8.8\end{pmatrix},$$

and the BLUE using only the first column of X is $\hat{\beta} = 14.8$.

Using Y_{new} and X_{new} produces exactly the same set of BLUEs, with *exactly* the same covariances for these BLUEs as for the original dataset, Y and X.

Why and how? We return to Example 1 in Sect. 14.7.

14.2 Full Model $\mathcal{M}_{12}$ and Small Model $\mathcal{M}_1$

Consider the two nested linear models, $\mathcal{M}_{12}$ (the full model (1)) and $\mathcal{M}_1$ (the small model (2)), both with fixed parameters:

$$\begin{aligned} Y &= X_1\beta_1 + X_{02}\beta_{02} + e \\ &= X_0\beta_0 + e \qquad (14.1) \\ \text{and} \quad Y &= X_1\beta_1 + e \qquad (14.2) \end{aligned}$$

where $cov(e) = V_0$ is non-negative definite. The subscript " 0" is used to designate this "original" model in (14.1) because we later find it useful to consider reparametrised versions of $\mathcal{M}_{12}$.

The vector Y is an n-dimensional observable response variable, and e is an unobservable random error with a known covariance matrix $cov(e) = V_0 = cov(Y)$ and expectation $\mathrm{E}(e) = \mathbf{0}$, where $\mathbf{0}$ is the zero vector. The matrices X_1 which is $n \times p_1$ and X_{02} which is $n \times p_{02}$ are known with $p_1 + p_{02} = p_0$. So X_0 is $n \times p_0$ where $p_0 < n$, sometimes (though not necessarily) partitioned as $X_0 = (X_1 : X_{02})$. Neither X_1 nor X_{02} are necessarily full (column) rank. The parameter vector is $\beta_0 = (\beta_1' : \beta_{02}')'$ where the symbol $'$ indicates transpose. Let $\mu = X_0\beta_0$.

Let the symbols A^-, A^+, $C(A)$, $C(A)_\perp$ denote a generalised inverse, the (unique) Moore-Penrose inverse, the column space, and the orthogonal complement of the column space of the matrix A. We denote any matrix satisfying $C(A_\perp) = C(A)_\perp$ as $A_\perp$. Let $P_A = P_{C(A)} = AA^+ = A(A'A)^-A'$ denote the orthogonal projector (with respect to the standard inner product) onto $C(A)$. The orthogonal projector onto $C(A)_\perp$ is designated as $Q_A = I_a - P_A$ where I_a is the $a \times a$ identity matrix with rows and columns. When $A = X = (X_1 : X_2)$, set $M = I_n - P_X$, $M_1 = I_n - P_{X_1}$ and $M_2 = I_n - P_{X_2}$. An obvious choice for $X_\perp$ is M.

Following Rao [18, Theorems 5.2 and 5.3], irrespective of the rank of the necessary and sufficient condition on a covariance matrix for a linear model

$$Y = X\beta + e \qquad (14.3)$$

with $cov(e) = V_0$ to continue to have the same BLUE for $X\beta$ is that

$$V = V_0 + XAX' + V_0X_\perp BX_\perp'V_0. \qquad (14.4)$$

Note that X and $X_\perp$ are orthogonal, that A and B are arbitrary, conformable, and hence square matrices of a possibly different dimension, and that (although it will be true when V_0 is of full rank) there is no requirement that $C(V_0X_\perp) \cup C(X) = \mathbb{R}^n$, i.e., it is not necessary that $C(V_{0\perp}) = \{0\}$ (where $C(V_{0\perp})$ denotes the column space orthogonal to V_0), or that the model is weakly singular, i.e., that $C(X) \subset C(V_0)$. Rao [18] specified tighter conditions, but Mitra and Moore [15, p. 148] and Rao [19, p. 289] both noted that the proof in Rao [18] did not require that

$$C(X : V_0) = C(X : V_0M) = \mathbb{R}^n;$$

however if this additional condition holds, then BLUE(μ) is unique.

Although both are square, the requirement that A or B in (14.4) be symmetric can be loosened. If a quadratic form CEC' is itself symmetric, then by taking transposes

$CEC' = CE'C'$ and hence $CEC' = C\{\frac{(E+E')}{2}\}C'$. Because $\frac{(E+E')}{2}$ is symmetric, and because both V and V_0 are symmetric then, unless $XAX' + V_0X_\perp BX'_\perp V_0$ is symmetric but its two components quadratic forms are not, A and B in Eq. (14.4) can be set to be symmetric without loss of generality.

For A and B symmetric, set $A = UU'$ and $B = U_rU'_r$ so that Eq. (14.4) yields

$$V = V_0 + XUU'X' + V_0X_\perp U_rU'_rX'_\perp V_0. \tag{14.5}$$

14.3 Reparametrised Full Model $\mathcal{M}_{12t}$ and Small Model $\mathcal{M}_1$

Recall that $M_1 = I - P_{X_1}$, where $P_{X_1} = X_1(X'_1X_1)^-X'_1$ is the orthogonal projector onto $C(X_1)$. Note that P_{X_1} is invariant to the choice of generalised inverse $(X'_1X_1)^-$. Let $X_{2|1} = P_{X_{2|1}}X_{02} = M_1X_{02}$. Then $P_{X_{2|1}} = P_{M_1X_{02}} = M_1X_{02}(X'_{02}M_1X_{02})^-X'_{02}M_1$ is the orthogonal projector onto $C(X_{1\perp})$ the column space perpendicular to $C(X_1)$. Note that $P_{X_{2|1}}$ is invariant to the choice of $(X'_{02}M_1X_{02})^-$. The transformation of X_{02} to $X_{2|1}$ ensures that X_1 and $X_{2|1}$ are orthogonal.

Noting that generally $\beta_2 \neq \beta_{02}$ define a reparametrised full model $\mathcal{M}_{12t}$ by

$$\begin{aligned} Y &= X_1\beta_1 + P_{X_{2|1}}X_{02}\beta_2 + e \\ &= X_1\beta_1 + M_1X_{02}\beta_2 + e \\ &= X_1\beta_1 + X_{2|1}\beta_2 + e \\ &= X_t\beta_t + e \end{aligned} \tag{14.6}$$

again with $cov(e) = V_0$.

For the BLUEs under $\mathcal{M}_{12}$, because $C(X_t) = C(X_1 : M_1X_{02}) = C(X_0)$,

$$\text{BLUE}(\mu|\mathcal{M}_{12}) = BLUE(\mu|\mathcal{M}_{12t}), \tag{14.7}$$

so that without loss of generality we may set

$$X_{2|1}\beta_2 = X_{02}\beta_{02}. \tag{14.8}$$

If required β_{02} (or its estimate) can be recovered from β_2 (or its estimate) via

$$\begin{aligned} \beta_{02} &= (X'_{02}X_{02})^-X'_{02}X_{2|1}\beta_2 \\ &= (X'_{02}X_{02})^-X'_{02}M_1X_{02}\beta_2 \end{aligned} \tag{14.9}$$

and an appropriate choice of generalised inverse.

The reason for using the reparametrised or transformed model $\mathcal{M}_{12t}$ is that the full model in its original form, $\mathcal{M}_{12}$, does not provide any straightforward way to specify the condition that the sets $\{V\}$ for $\mathcal{M}_{12}$ and $\mathcal{M}_1$ coincide, because it is not

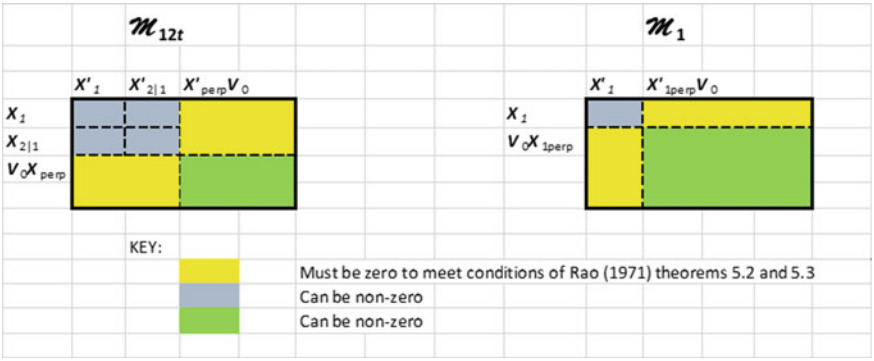


Fig. 14.1 Matrices of form U_0U_0' $(n \times n)$ which when added to V_0 are necessary and sufficient for unchanged BLUEs for reparametrised full model $\mathcal{M}_{12t}$ and for small model $\mathcal{M}_1$, considered separately, under the assumption that $\mathcal{C}(X_{2|1}) \subset \mathcal{C}(V_0X_{1\perp})$

generally possible to distinguish terms that contain both X_1 and X_2 from the quadratic forms in only X_1 or only in X_2. This in turn leads to complications ensuring that in $\mathcal{M}_1$, $\mathcal{C}(V_0X_{12\perp}) \cap \mathcal{C}(X_2)$ does not overlap with part of $\mathcal{C}(X_1)$. Note that, unless $\mathcal{C}(X_1) \cap \mathcal{C}(X_2) = \{0\}$, $\mathcal{C}(X_2 : V_0X_{12\perp}) \cap \mathcal{C}(X_1) \neq \{0\}$.

For $\mathcal{M}_{12t}$, Eq. (14.5) can be written

$$\begin{aligned} V = V_0 &+ X_1U_1U_1'X_1' + X_{2|1}U_{2|1}U_{2|1}'X_{2|1}' \\ &+ X_1U_1U_{2|1}'X_{2|1}' + X_{2|1}U_{2|1}U_1'X_1' \\ &+ V_0X_{12\perp}U_{12r}U_{12r}'X_{12\perp}'V_0. \end{aligned} \tag{14.10}$$

For $\mathcal{M}_1$ the parallel result is that

$$V = V_0 + X_1U_1U_1'X_1' + V_0X_{1\perp}U_{1r}U_{1r}'X_{1\perp}'V_0. \tag{14.11}$$

Both (14.10) and (14.11) can be written in block diagonal form so that all their quadratic forms can be contained in a square matrix of dimension $n \times n$ that we denote by U_0U_0'. However, the internal structure of these $n \times n$ block diagonal matrices is different for full and small models, as illustrated in Fig. 14.1.

Note that in Figs. 14.1, 14.2, 14.3, 14.4, and 14.5, although the block matrix of principal interest U_0U_0' is in usual matrix notation, the collection of row labels and column labels is not. To return the pictorial representation to usual matrix notation, row labels (which form a column in the figures) should be read in sequence as a row vector which pre-multiplies U_0U_0', and the column labels (which form a row in the figures) should be read in sequence as a column vector which post-multiplies U_0U_0'. Here, $V_0X_{\text{perp}} \equiv V_0X_{12\perp}$, and $V_0X_{1\text{perp}} \equiv V_0X_{1\perp}$. When a matrix product involving X_1, $V_0X_{1\perp}$, $X_{2|1}$ and/or $V_0X_{12\perp}$ is zero, this is represented in the figures by setting the relevant part of U_0U_0' to zero.

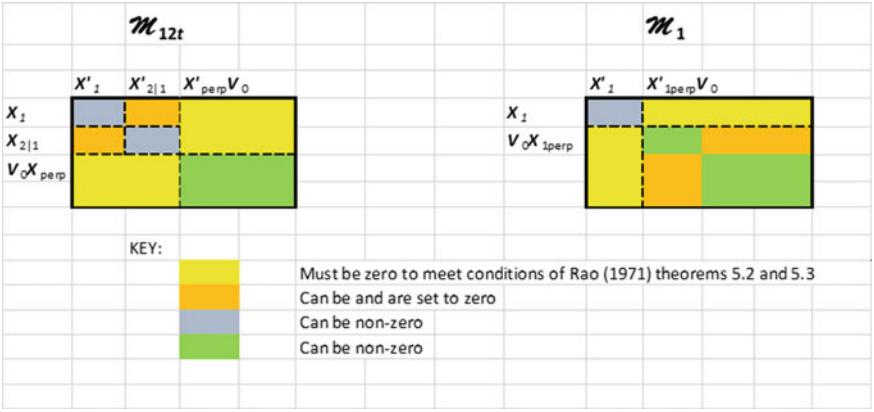


Fig. 14.2 Matrices of form U_0U_0' $(n \times n)$ necessary and sufficient for unchanged BLUEs for reparametrised full model $\mathcal{M}_{12t}$ and for small model $\mathcal{M}_1$, considered simultaneously, under the assumption that $C(X_{2|1}) \subset C(V_0X_{1\perp})$

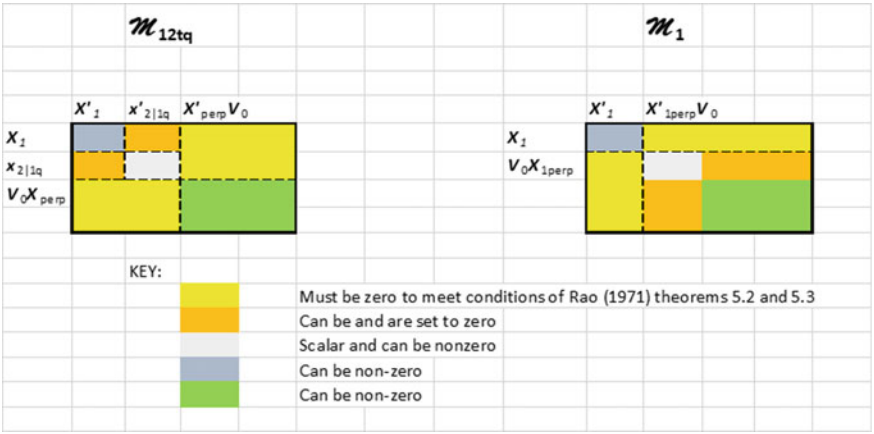


Fig. 14.3 Matrices of form U_0U_0' $(n \times n)$ necessary and sufficient for unchanged BLUEs for the reparametrised full model $\mathcal{M}_{12tq}$ and for the small model $\mathcal{M}_1$, considered simultaneously, where $X_{2|1q} = x_{2|1q}$ a vector, under the assumption that $C(X_{2|1}) \subset C(V_0X_{1\perp})$

Note that although in general $C(X_1) \cup C(X_{2|1}) \cup C(V_0X_{12\perp}) \cup C(V_{0\perp}) = \mathbb{R}^n$ and $C(X_1) \cup C(V_0X_{1\perp}) \cup C(V_{0\perp}) = \mathbb{R}^n$, where $C(V_{0\perp})$ is the column space orthogonal to $C(V_0)$, Fig. 14.1 does not include any terms in $C(V_{0\perp})$. This is not because $C(V_{0\perp}) = \{0\}$ necessarily, but because via Rao [18, Theorems 5.2 and 5.3] all terms (if any) in (14.10) and (14.11) that involve $V_{0\perp}$ must be zero. The assumption inherent in Fig. 14.1 is that $C(X_{2|1}) \subset C(V_0X_{1\perp})$ is not necessary. If it does not hold, then it is because $C(V_{0\perp}) \neq \{0\}$ and $C(X_{2|1}) \cap C(V_{0\perp}) \neq \{0\}$, so that some part of $C(X_{2|1}) \not\subset C(V_0X_{1\perp})$. Then, when considering conditions on V for retention of BLUEs for $\mathcal{M}_1$ and $\mathcal{M}_{12t}$ simultaneously, $U_1U_{2|1}'$ must be zero to match the

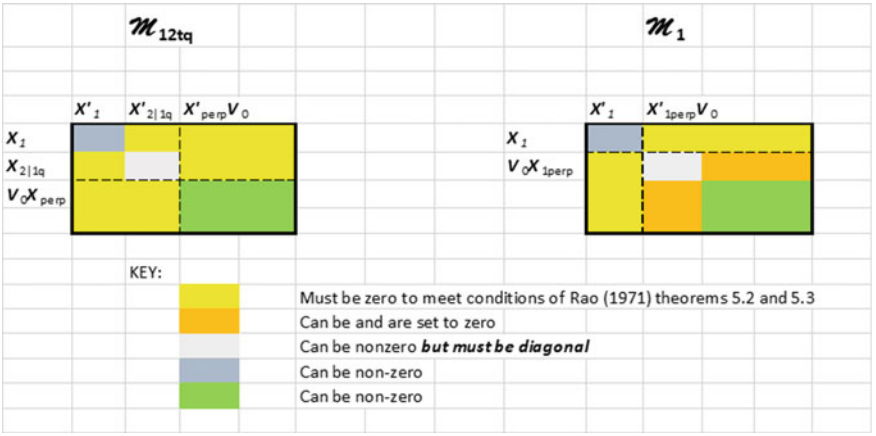


Fig. 14.4 Matrices of form U_0U_0' $(n \times n)$ necessary and sufficient for unchanged BLUEs for reparametrised full model $\mathcal{M}_{12tq}$ and for small model $\mathcal{M}_1$, considered simultaneously, where $X_{2|1q}$ is a matrix, under the assumption that $\mathcal{C}(X_{2|1}) \subset \mathcal{C}(V_0X_{1\perp})$

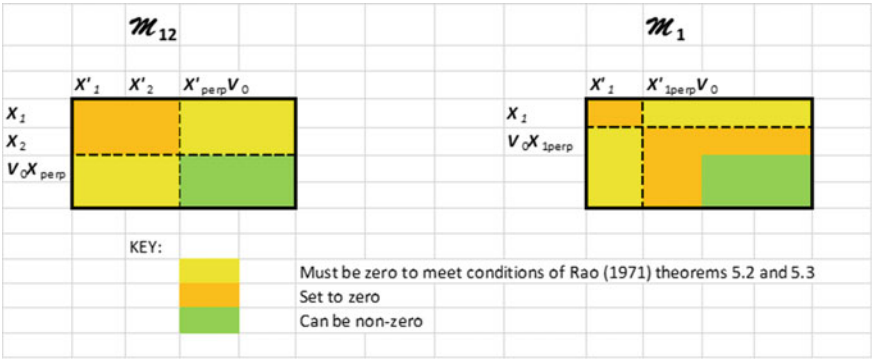


Fig. 14.5 Matrices of form U_0U_0' $(n \times n)$ sufficient for unchanged BLUEs and covariances of functions of estimable parameters for original full model $\mathcal{M}_{12}$ and for all its submodels including the small model $\mathcal{M}_1$, considered simultaneously

result of Rao's [18] theorems. For any intermediate models, exactly which parts of U_0U_0' must be zero is difficult to specify without orthogonalising the columns of $X_{2|1}$. This orthogonalisation is discussed in Sect. 14.4 and comment made there about removing the assumption that $\mathcal{C}(X_{2|1}) \subset \mathcal{C}(V_0X_{1\perp})$, but in the remainder of Sect. 14.3 and the initial part of Sect. 14.4, to simplify the exposition, V_0 is assumed weakly singular (i.e., $\mathcal{C}(X) \subset \mathcal{C}(V_0)$). Although weak singularity is a more familiar condition, it is stronger than $\mathcal{C}(X_{2|1}) \subset \mathcal{C}(V_0X_{1\perp})$; however when in the small model $\mathcal{M}_1$, the design matrix X_1 has only one column (a situation important to data confidentiality and encryption); the two conditions become very similar. As an aside, the condition Rao [18, Theorem 5.3] assumed unnecessarily,

namely $C(X : V_0) = C(X : V_0M) = \mathbb{R}^n$ is not implied by weak singularity, i.e., by $C(X) \subset C(V_0)$, because although under weak singularity $C(X : V_0) = C(V_0)$, it is not necessarily then true that $C(X : V_0) = \mathbb{R}^n$ (i.e., that V_0 is full rank).

Illustrated (by differing blocks of U_0U_0' needing to be zero) in Fig. 14.1 is that the required V for Rao's [18] condition for $\mathcal{M}_{12t}$ (and hence for $\mathcal{M}_{12}$) and for $\mathcal{M}_1$ to each have the same BLUEs are not equivalent, even for weakly singular V_0. Additional restrictions need to be placed on the structure of V for both $\mathcal{M}_{12t}$ and $\mathcal{M}_1$, so that the condition that the BLUEs be unchanged applies to both models simultaneously. Put more formally, the subspaces in Eq. (14.10), namely $C(X_1)$, $C(X_{2|1})$, and $C(V_0X_{12\perp})$, and in Eq. (14.11) namely $C(X_1)$ and $C(V_0X_{1\perp})$, are not the same partition of $\mathbb{R}^n \backslash C(V_{0\perp})$ (i.e., of $\mathbb{R}^n$ excluding $C(V_{0\perp})$).

Essentially, our representational convention for Fig. 14.1 suggests that what is required in specifying V for which BLUEs are retained in both $\mathcal{M}_1$ and $\mathcal{M}_{12t}$ when V_0 is weakly singular is that all submatrices of the entire matrix U_0U_0' that need to be zero in either model when they are considered separately, need to be zero in both models when they are considered together.

Without further loss of generality (based on $C(V_0X_{1\perp})$ and $C(X_{2|1}) \cup C(V_0X_{12\perp})$ having exactly the same column space, since both only exclude $C(X_1)$ from $\mathbb{R}^n \backslash C(V_{0\perp})$), specify the orthogonal basis for $V_0X_{1\perp}$ as being subdivided into two parts, one part in $C(X_{2|1})$ and the other part in $C(V_0X_{12\perp})$.

For $\mathcal{M}_{12t}$, now consider a restricted form of Eq. (14.10)

$$V = V_0 + X_1U_1U_1'X_1' + X_{2|1}U_{2|1}U_{2|1}'X_{2|1}') + V_0X_{12\perp}U_{12r}U_{12r}'X_{12\perp}'V_0. \quad (14.12)$$

All terms in U_1U_{12r}', $U_{2|1}U_{12r}'$ and their transposes must be zero in Eq. (14.12) because $X_{12\perp}'V(V_0X_{12\perp})_\perp = 0$ via Rao [18, Theorem 5.2]. The term in $U_1U_{2|1}'$ and its transpose that appears in Eq. (14.10) are now zero by construction in Eq. (14.12).

For $\mathcal{M}_1$, the parallel result is that

$$V = V_0 + X_1U_1U_1'X_1' + V_0X_{1\perp}U_{1r}U_{1r}'X_{1\perp}'V_0 \quad (14.13)$$

where $V_0X_{1\perp}U_{1r}U_{1r}'X_{1\perp}'V_0$ is restricted so that

$$V_0X_{1\perp}U_{1r}U_{1r}'X_{1\perp}'V_0 = X_{2|1}U_{2|1}U_{2|1}'X_{2|1}' + V_0X_{12\perp}U_{12r}U_{12r}'X_{12\perp}'V_0 \quad (14.14)$$

i.e.,

$$V = V_0 + X_1U_1U_1'X_1' + X_{2|1}U_{2|1}U_{2|1}'X_{2|1}' + V_0X_{12\perp}U_{12r}U_{12r}'X_{12\perp}'V_0 \quad (14.15)$$

which is exactly Eq. (14.12).

The structure for V in Eqs. (14.12) and (14.15) is necessary and sufficient via this extension of [18] because no other structure can ensure that the conditions required for $\mathcal{M}_{12t}$ and $\mathcal{M}_1$ to retain their BLUEs simultaneously are met.

So in Fig. 14.2, only the diagonal blocks of U_0U_0' can be non-zero when specifying V such that Rao's condition applies to $\mathcal{M}_{12t}$ and $\mathcal{M}_1$ simultaneously.

14.4 Models Intermediate Between $\mathcal{M}_{12t}$ and $\mathcal{M}_1$

The question may be extended to considering the requirements on $\{V\}$ for $\mathcal{M}_{12t}$, $\mathcal{M}_1$, and all models intermediate between them (interpretable as parameters being added one at a time to $X_1\beta_1$), each to have the same BLUEs.

However, there is a complication.

Recall that $X_{2|1}$ was constructed to ensure that $X_{2|1}$ was orthogonal to X_1 in order to apply Rao [18, Theorem 5.3]. In the same manner, when considering intermediate models where we need to reassign the column space in sequence, we need to use a transformed version of $X_{2|1}$ in which each column is orthogonal.

This can be achieved, for example, via the QR decomposition of $X_{2|1}$,

$$X_{2|1} = X_{2|1q} R_{2|1q} \tag{14.16}$$

where $X_{2|1q}$ is $n \times p_2$ with orthogonal columns and $R_{2|1q}$ is an upper $p_2 \times p_2$ triangular matrix. Note that $\text{rank}(X_{2|1q} R_{2|1q}) = \text{rank}(X_{2|1})$. Then defining $\beta_{2q} = R_{2|1q}\beta_2$, it will be the columns in $X_{2|1q}$ along with the corresponding parameters in β_{2q} that can be deleted one at a time, rather than the columns of $X_{2|1}$. This new model is designated $\mathcal{M}_{12tq}$. Again by a parallel argument to (14.7)

$$\text{BLUE}(\mu|\mathcal{M}_{12}) = \text{BLUE}(\mu|\mathcal{M}_{12t}) = \text{BLUE}(\mu|\mathcal{M}_{12tq}). \tag{14.17}$$

Because they are orthogonal, the columns of $X_{2|1q}$ can be added or removed in any order so that all intermediate models are included not just those in a particular addition or deletion sequence. There is also no loss of applicability of the result, at least algebraically, if there are interaction terms within X_0 in the original model $\mathcal{M}_{12}$.

Consider first the case where $X_{2|1q} = x_{2|1q}$ a vector, i.e., β_{2q} is scalar, so only one additional parameter and one additional column from $X_{2|1q}$ is added to $\mathcal{M}_1$. See Fig. 14.3.

Because V must still hold for both the full and the small model, the new requirement on V is a special case of Eqs. (14.12) and (14.15) with $X_{2|1q} = x_{2|1q}$:

$$V = V_0 + X_1 U_1 U_1' X_1' + x_{2|1q} u_{2|1q}^2 x_{2|1q}' + V_0 X_{12\perp} U_{12r} U_{12r}' X_{12\perp}' V_0. \tag{14.18}$$

The scalar term $u_{2|1q}^2$ in (14.18) is represented in $U_0 U_0'$ of Fig. 14.3 in light blue.

The extension to a second single parameter from β_{2q} repeats the same requirement, namely the addition of a single further diagonal element. Repeating the process can continue to add only diagonal elements to $U_{2|1q} U_{2|1q}'$.

Hence, the condition on weakly singular V so that $\mathcal{M}_{12tq}$, $\mathcal{M}_1$, and all models intermediate between them have the same BLUEs for $X_{2|1q}\beta_{2q}$ is that

$$V = V_0 + X_1 U_1 U_1' X_1' + X_{2|1q} \text{diag}(U_{2|1q} U_{2|1q}') X_{2|1q}' + V_0 X_{12\perp} U_{12r} U_{12r}' X_{12\perp}' V_0 \tag{14.19}$$

where diag designates a diagonal matrix.

Note that, if $X_{2|1q}\text{diag}(U_{2|1q}U'_{2|1q})X'_{2|1q}$ in Eq. (14.19) is written in terms of X_2 rather than $X_{2|1q}$, then Eq. (14.19) can induce terms of the form $X_2FX'_1$ for some F containing both X_1 and X_2.

Because $X_{2|1q}$ has only orthogonal columns, it is now possible to be explicit on how the assumption that $C(X_{2|1}) \subset C(V_0X_{1\perp})$ can be removed from the results in Sects. 14.3 and 14.4 for full, small, and intermediate models. Each column vector in $X_{2|1q}$ is either in $C(V_0X_{1\perp})$ or in $C(V_{0\perp})$ but cannot be in both, since $C(V_0X_{1\perp}) \cap C(V_{0\perp}) = \{0\}$. So

$$V = V_0 + X_1U_1U'_1X'_1 + X_{2|1q}U_{2|1q}U'_{2|1q}X'_{2|1q} + V_0X_{12\perp}U_{12r}U'_{12r}X'_{12\perp}V_0 \tag{14.20}$$

holds for small and full models, provided that where a column of $X_{2|1q}$ does not belong to $C(V_0X_{1\perp})$ the corresponding row and column in $U_{2|1q}U'_{2|1q}$ contain only zeros, and that if intermediate models are also included then $U_{2|1q}U'_{2|1q}$ is diagonal.

The utility of the results for intermediate models is most obvious for linear models with polynomials as regressors and some types of experimental design. For polynomials, the required transformation can be achieved by fitting instead some class of orthogonal polynomials, and for many balanced experimental designs no transformation to orthogonality is required. For models that do not have orthogonal regressors however, the results are more theoretical than practical. Interpretation is complicated because it is not the original regressors that are being removed one at a time in transiting via intermediate models from full model to small model or vice versa, but particular orthogonal linear combinations of those regressors.

Consequently, since results are more easily interpretable in practical terms, for intermediate models it is more straightforward to look at the subset of $\{V\}$ in (14.20) which avoids the need to consider the structure of all terms involving X_{02}, $X_{2|1}$, $X_{2|1q}$, or X_1 (except in so far as they determine $X_{12\perp}$). It also has another major advantage, which is discussed in the next section.

14.5 BLUEs and Their Covariance

The results in Sects. 14.2–14.4 concern retention of BLUEs under changes of error covariance. Generally, however, covariances of the BLUEs of estimable functions of parameters are not conserved, even for weakly singular V_0. However, it is possible to limit the structure of V in a way that both the BLUEs and their covariances are conserved.

Consider

$$V = V_0 + V_0X_{12\perp}U_{12r}U'_{12r}X'_{12\perp}V_0 \tag{14.21}$$

with V_0 and $U_{12r}U'_{12r}$ full rank.

Then following Puntanen et al. [17, p. 261, Problem 10.6], under V_0

$$cov(\text{BLUE}(X_0\beta_0)) = X_0(X_0'V_0^{-1}X_0)^-X_0' \tag{14.22}$$

and under V

$$cov(\text{BLUE}(X_0\beta_0)) = X_0(X_0'V^{-1}X_0)^-X_0'. \tag{14.23}$$

However, from (14.21), via the Sherman-Morrison-Woodbury formula, attributed to Duncan [3] in Puntanen et al. [17, p. 301],

$$\begin{aligned} V^{-1} &= (V_0 + V_0X_{12\perp}U_{12r}U_{12r}'X_{12\perp}'V_0)^{-1} \\ &= V_0^{-1} + V_0^{-1}V_0X_{12\perp}\{X_{12\perp}'V_0V_0^{-1}V_0X_{12\perp} + (U_{12r}U_{12r}')^{-1}\}^{-1}X_{12\perp}'V_0V_0^{-1} \\ &= V_0^{-1} + X_{12\perp}\{X_{12\perp}'V_0V_0^{-1}V_0X_{12\perp} + (U_{12r}U_{12r}')^{-1}\}^{-1}X_{12\perp}' \end{aligned} \tag{14.24}$$

so that, because $X_0'X_{12\perp} = 0$ (since X_0 and $X_{12\perp}$ are orthogonal),

$$X_0'V^{-1}X_0 = X_0'V_0^{-1}X_0 \tag{14.25}$$

and hence $cov(\text{BLUE}(X_0\beta_0))$ is identical under the change of error covariance from V_0 to V when V is defined by (14.21).

Puntanen et al. [17, p. 261, Problem 10.7] generalise (14.22) and (14.23) by replacing V_0^{-1} by V_0^- and V^{-1} by V^-, i.e.,

$$cov(\text{BLUE}(X_0\beta_0)) = X_0(X_0'V_0^-X_0)^-X_0' \tag{14.26}$$

and under V

$$cov(\text{BLUE}(X_0\beta_0)) = X_0(X_0'V^-X_0)^-X_0'. \tag{14.27}$$

However, the various available parallels to the Sherman-Morrison-Woodbury formula for generalised inverses apply only under restricted conditions. Riedel [20] is not directly relevant because the necessary condition on the two components of the sum that the rank of the sum equals the sum of the ranks is not met. Fill and Fishkind [4] who extended Riedel's results noted the centrality of his rank condition even to their extensions. Hung and Markham [11] had earlier proved a more general formula, but this does not easily simplify in the way required here. Kala and Klaczyński [12, Corollary 2(e)] is useful, but still restrictive as it applies here; their result applied to V is that, given $C(X_{1\perp}'V_0) \subset C(U_{12r}U_{12r}')$ and since $C(V_0X_{1\perp}) \subset C(V_0)$,

$$\begin{aligned} V^+ &= V_0^+ + V_0^+V_0X_{12\perp}\{X_{12\perp}'V_0V_0^+V_0X_{12\perp} + (U_{12r}U_{12r}')^+\}^{(1)}X_{12\perp}'V_0V_0^+ \\ &= V_0^+ + V_0^+V_0X_{12\perp}\{X_{12\perp}'V_0X_{12\perp} + (U_{12r}U_{12r}')^+\}^{(1)}X_{12\perp}'V_0V_0^+, \end{aligned} \tag{14.28}$$

where the superscript (1) indicates a 1-generalised inverse, i.e., that $KK^-K = K$ for $K = \{X_{12\perp}'V_0V_0^+V_0X_{12\perp} + (U_{12r}U_{12r}')^+\}$.

So, provided

$$X_0'V_0^+V_0X_{12\perp} = (X_0'V_0^+)(V_0X_{12\perp}) = X_0'(V_0^+V_0)X_{12\perp} = (X_0'V_0^+V_0)(V_0^+V_0X_{12\perp}) = 0 \tag{14.29}$$

then

$$\begin{aligned} X_0'V^+X_0 &= X_0'V_0^+X_0 + X_0'V_0^+V_0X_{12\perp}(X_{12\perp}'V_0X_{12\perp} + (U_{12r}U_{12r}')^+)^{(1)}X_{12\perp}'V_0V_0^+X_0 \\ &= X_0'V_0^+X_0 \end{aligned} \tag{14.30}$$

which implies the equality of covariance of BLUEs conditions (14.26) and (14.27) since these results are invariant to the choices of generalised inverse.

Now $P_{C(V_0)} = V_0V_0^+$ is the projection operator into $C(V_0)$ and when V_0 is weakly singular (because then by definition $C(X_0) \subset C(V_0)$), the projection of X_0 into $C(V_0)$ is X_0 itself, i.e., $V_0V_0^+X_0 = P_{C(V_0)}X_0 = X_0$ so that

$$X_{12\perp}'V_0V_0^+X_0 = X_{12\perp}'(V_0V_0^+X_0) = X_{12\perp}'(P_{C(V_0)}X_0) = X_{12\perp}'X_0 = 0 \tag{14.31}$$

which implies (14.30) holds when V_0 is weakly singular, given $C(X_{1\perp}'V_0) \subset C(U_{12r}U_{12r}')$.

To summarise, when V_0 is weakly singular the requirement for retention of BLUEs and their covariance for estimable functions of parameters is that $C(X_{1\perp}'V_0) \subset C(U_{12r}U_{12r}')$, a condition that can be simply met, for example by ensuring $U_{12r}U_{12r}'$ is full rank.

As an aside in the explanation of the choice made in specifying V in (14.21), as straightforward algebra shows if Kala and Klaczyński [12, Corollary 2(e)] and its conditions are applied instead to (14.4) with $A \neq 0$, i.e., to $V = V_0 + XAX' + V_0X_\perp BX_\perp'V_0$, or applied to $V = V_0 + XAX'$, although the BLUEs of estimable functions of parameters are unaltered by the change from V_0 to V, the covariance of these BLUEs under V_0 and V will necessarily be different. A construction of this type, namely $V = V_0 + XX'$ is used by [18] to facilitate estimation of BLUEs when V_0 is singular but not weakly singular, i.e., when $C(X) \not\subset C(V_0)$.

14.6 BLUPs for Mixed Models

In principle, the results of Sects. 14.1–14.5 can be extended to mixed models of the form

$$Y = X\beta + Z\gamma + u \tag{14.32}$$

for γ and uncorrelated and $cov(u) = V_{0u}$, $cov(\gamma) = R$ (both non-negative definite), by substitution of

$$X_0 = \begin{pmatrix} X & Z \\ 0 & I \end{pmatrix} \text{ and } V_0 = \begin{pmatrix} V_{0u} & 0 \\ 0 & R \end{pmatrix} \tag{14.33}$$

noting that

$$X_{0\perp} = \begin{pmatrix} I \\ -Z \end{pmatrix} X_{\perp}. \tag{14.34}$$

Considering intermediate models between a specified small mixed model and a full model may involve adding (or deleting) either fixed or random parameters or both. These two types of addition (or deletion) can be considered separately, one additional parameter or set of parameters at a time (as in Sect. 14.4).

14.7 Data Cloning

Let us return to Example 1 to uncover the trick.

The original fit is via ordinary least squares (OLS) so $V_0 = \sigma^2 I$ and in the example (to allow for models to be reduced to their simplest form, e.g., only an intercept), the only term added to V_0 to create V is of the form, in line with (14.21):

$$V_0 X_{12\perp} U_{12r} U'_{12r} X'_{12\perp} V_0 = \sigma^4 X_{12\perp} U_{12r} U'_{12r} X'_{12\perp} \tag{14.35}$$

from (14.12). Use this new full rank V and find its Cholesky or the square root of its eigendecomposition, $V^{1/2}$. Pre-multiplying the original model by $V^{1/2}$ and using different selections of U_{12r} can produce an infinite collection of datasets that have the same regression coefficients for all submodels as the original data as well as the same parameter covariances. For $V_0 = \sigma^2 I$, the transformed model is

$$V^{1/2} Y = V^{1/2} X_1 \beta_1 + V^{1/2} X_{02} \beta_{02} + V^{1/2} e \tag{14.36}$$

and when the parameters in this transformed model are estimated using OLS, it will produce the same BLUEs of estimable functions of parameters and their covariance as for the OLS fit to the original data and, via the results in Sects. 14.1–14.5, the same will be true of all submodels.

For any full rank V_0, the required transformation is

$$(V^{1/2} V_0^{-1/2}) Y = (V^{1/2} V_0^{-1/2}) X_1 \beta_1 + (V^{1/2} V_0^{-1/2}) X_{02} \beta_{02} + (V^{1/2} V_0^{-1/2}) e. \tag{14.37}$$

V_0 can instead be weakly singular rather than $V_0 = \sigma^2 I$, or V_0 and/or V being full-rank, and BLUE$(X_0 \beta_0)$ and its covariance will again be retained by the cloned data. See (14.38). When V_0 singular but not weakly singular, the BLUEs of estimable functions of original and cloned data will coincide, but for the reasons outlined in Sect. 14.5, it is an open question whether the covariances of the BLUEs always will.

For weakly singular V_0 with $\mathcal{C}(X'_{1\perp} V_0) \subset \mathcal{C}(U_{12r} U'_{12r})$ and original data $\{Y : X_0\}$, the cloned data is $\{Y_c : X_{1c} : X_{02c}\} = \{Y_c : X_c\}$ and the model is

$$[V^{1/2} (V_0^+)^{1/2}] Y = [V^{1/2} (V_0^+)^{1/2}] X_1 \beta_1 + [V^{1/2} (V_0^+)^{1/2}] X_{02} \beta_{02} + [V^{1/2} (V_0^+)^{1/2}] e \tag{14.38}$$

i.e.,

$$\begin{aligned} Y_c &= X_{1c}\beta_1 + X_{02c}\beta_{02} + e_c \\ &= X_c\beta_0 + e_c \end{aligned} \tag{14.39}$$

where via Sect. 14.5,

$$cov(\text{BLUE}(X_c\beta_0)) = cov(\text{BLUE}(X_0\beta_0)).$$

So, via (14.21), the model using the cloned data and the model using V_0 will produce the same BLUEs of estimable functions of the parameters and covariances for these BLUEs for all submodels as does the original data fitted using V_0.

Further, because what has been added to V_0 is only $V_0X_{12\perp}U_{12r}U'_{12r}X'_{12\perp}V_0$ with $C(X'_{1\perp}V_0) \subset C(U_{12r}U'_{12r})$, this addition to V_0 will work for all submodels. See Fig. 14.5. Any choice made from $\{U_{12r}U'_{12r}\}$ need not be used for all submodels, but if a single cloned and hence confidentialised dataset is supplied to a user (so as to minimise data transfer, and complexity of analysis) then $\{U_{12r}U'_{12r}\}$ would be fixed for each user.

Because there are no terms in $X_{2|1}U_{2|1}U'_{2|1}X'_{2|1}$ included, there is no need to orthogonalise to form $X_{2|1}$ or $X_{2|1q}$. Of course, orthogonalisation of X would be straightforward for Example 1 by specifying the original model using orthogonal polynomials, but in general orthogonalisation would be an extra unnecessary step and would complicate interpretation. The addition only of $V_0X_{12\perp}U_{12r}U'_{12r}X'_{12\perp}V_0$ focuses changes to V_0 on the column space of the errors and residuals for the full model, which is necessarily contained in the corresponding column spaces for the error and residuals of the small and all the intermediate models. It also ensures retention of the covariances of BLUEs of estimable functions of model parameters for the full model and all its submodels.

14.8 Conclusions

Sections 14.2–14.6 have outlined necessary and sufficient conditions for simultaneous retention of BLUEs of estimable function for full, small, and all intermediate models under changes in error covariance. Generally, however, because the additions allowed to the original error covariance V_0 to form V include terms in X_0, the covariances of these BLUEs are not maintained.

One implication of Sect. 14.7 is that by excluding terms in X_0, all new datasets generated using the results there give the same BLUEs of estimable parameters and these BLUEs also have the same covariances as for the original dataset. This additional BLUE covariance equality property is what makes these results so useful in applications, e.g., for data cloning: data visualisation, smoothing, confidentiality, and encryption. Data confidentiality and encryption are important internationally.

Encryption is important to national statistical agencies and to industry, and both are important to various government agencies.

Haslett and Govindaraju [8] note that for the pivot methods they used, because they did not preserve a sufficiently wide range of statistical properties, data confidentialisation applications were restricted. The additional material here in Sects. 14.2–14.7 broadens the possibilities by outlining how a given dataset can be adjusted via a changed error covariance matrix in a way that preserves BLUEs via fitting a saturated (or some other suitable) linear model. The subset of such possible changes to error covariance outlined in Sect. 14.7 also preserves the covariance of the BLUEs. These new methods consequently provide a wider range of statistical uses than pivot methods.

The new methods would allow statistical agencies to provide confidentialised data that, rather than containing the random error which many confidentialisation methods currently add to datasets, maintained analytic results for linear models using any subset of the original full model via its cloned form. By including vectors of ones for subgroups in the design matrix of the full model fitted by the agency to the un-confidentialised data, because results in Sect. 14.7 show that using cloned data based on (14.21) does not change the estimates of the parameters associated with the columns of this original design matrix, even means of subgroups and overall means can still be estimated from the cloned data without additional error and with the correct variances. The full model initially fitted by the agency might also include interaction terms where relevant. This might be simply extended to confidentialised datasets for fitting fixed effects in mixed linear models by incorporating random effects into V_0.

Methods needed for generalised linear models would be more complicated. The results in Sects. 14.2–14.7 assume that when the error covariance V_0 is known or estimated as V_0 and used in the full model, it is not changed. So there are limitations to confidentialised data when iterative procedures (such as iterated generalised least squares—IGLS) are to be used by external users for analysis. The exceptions are when the data supplied by the agency is from the final iteration of the full model fit (as would be suitable for estimating fixed parameters in mixed linear models) or provides extensive back-computation information (as would be necessary for generalised linear models). This is because to be able to reverse the iterative process from the confidentialised data at the final iteration of IGLS for its underlying linear model requires interim results from the fit that preceded it in the forward (rather than the reverse or backward) direction. See del Pino [16] for further discussion of the role and basis of IGLS.

For full (linear) models fitted by the agency that have error covariance that is a multiple of the identity matrix or that have error covariance specified by incorporating random effects into V_0 (to allow estimation of the fixed effects in a mixed model), there is no complication in principle with using an estimated (and hence known) error covariance $\hat{V}_0$ when fitting the full model to the original data rather than known V_0. The confidentialised data $\{Y_c : X_c\}$ can be provided to data users along with V_0 or $\hat{V}_0$ (simplest and of minimum size when $V_0 = \sigma^2 I$, and containing no more than $\frac{n(n+1)}{2}$ elements if it has internal structure), while the agency retains the key V, which can

be different or the same for different data users without any additional compromise to confidentiality.

The cloning mechanism has focused on a particular dependent variable Y but, subject to conditions outlined below, may nevertheless still produce cloned data with the same BLUEs and covariances for its estimable function for *different* dependent variables using the *same* cloned dataset. This is a potentially important property because the same cloned dataset may then be used for the analysis of several different dependent variables, only one of which has been used to produce the cloned data. It broadens the utility of cloned data when it is used for confidentialisation.

The same transformation $V^{1/2}(V_0^+)^{1/2}$ in (14.38) is applied to both Y and $(X_1 : X_{02}) = X_0$. This gives the impression that, for a given cloned dataset, the roles of Y and one of the columns of X_0 might be interchanged so that the cloned dataset would also provide the correct estimates and standard errors for a different dependent variable. However, the original Y cannot be included in the new design matrix and still retain the required BLUE properties, because it is not a column in the original X_0 and consequently not included in the original cloning process. So, despite (14.38) suggesting otherwise, interchange is not possible. However, whether the new Y is from the columns of X_0 or not, choosing a new Y and having it transformed by $V^{1/2}(V_0^+)^{1/2}$ is permitted, provided the covariance of the error in the new model belongs to the set $\{V\}$. Feasible new error covariance structures include any rescaling of V_0 such as would occur when the internal structure of V_0 is a property of the study design. The possibly marked increase in cloned data utility when the same cloned data can be used for several dependent variables does however have a price, albeit relatively small. Firstly, each suitable transformed new Y must be included in the cloned dataset supplied by the agency. Secondly, the agency needs to verify the required error covariance property holds for each dependent variable of interest to the users. If it does not conform, a separate cloned dataset will be required for the new Y; the advantage is that this additional cloned dataset can easily be set up to include the original Y as a predictor variable.

The methods of Sect. 14.7 provide an alternative to the usual confidentialised unit record files (CURFs) used internationally to release unit record data by national statistical agencies. Unlike CURFs, when analysed statistically, data produced by cloning methods produces the correct answers. Additional research is however warranted if the dataset is from a complex survey, because then even linear model fitting ought not to use standard statistical methods (Lumley [14]).

As also noted by Haslett and Govindaraju [8], an "important use for cloned data may be for encryption of statistical databases, and of relational databases via data cubes, irrespective of whether the "same regression properties are of any interest at all". The methods in Sects. 14.2–14.5, not just those in the current section, improve the results given by Haslett and Govindaraju [8] for encryption, because they do not require the pivots needed there. The encryption key needed to recover the original data is now the particular V and V_0 that has been used. Although V and V_0 do require storage additional to the cloned data, encryption via the methods in Sects. 14.2–14.7 does not increase data volume of the encrypted data itself (since there are still the

same number of variables in total). Encrypted datasets can be stored on site (separate from the encryption key of course) or moved between secure areas.

In a more general context, even when the original data are known, cloned datasets might be generated and amalgamated to increase apparent sample size as an aid to model fitting via residual diagnostics. Whether $U_{12r}U'_{12r}$ would best be constrained in this type of application, for example by setting an upper bound on some form of its norm or to have distributional assumptions imposed (which would introduce a Bayesian element), are open questions. What is central to understand, however, is that any apparent increase in statistical power from combining multiple copies of cloned data is not a free good. It is instead the consequence of making the final model fit no longer conditional on only one particular model but instead conditional on the entire (or possibly some reduced or probability weighted) set of *all* models with the same BLUEs and covariances for estimable parameters of the model parameters.

Acknowledgements This paper originated in the first author's online talk at the 28th International Workshop on Matrices and Statistics, Manipal Academy of Higher Education, India, 13–15 December 2021, see [6], and his previous research on data confidentiality and encryption with Dr. K (Raj) Govindaraju.

References

1. Anscombe FJ (1973) Graphs in statistical analysis. Am Stat 27:17–21
2. Chatterjee S, Firat A (2007) Generating data with identical statistics but dissimilar graphics: a follow up to the Anscombe dataset. Am Stat 61:248–254
3. Duncan WJ (1944) Some devices for the solution of large sets of simultaneous linear equations (with an appendix on the reciprocation of partitioned matrices). Lond, Edinb Dublin Philos Mag J Sci, Seventh Series 35:660–670
4. Fill JA, Fishkind DE (1999) The Moore-Penrose generalized inverse for sums of matrices. SIAM J Matrix Anal Appl 21(2):629–635
5. Govindaraju K, Haslett SJ (2008) Illustration of regression towards the means. Int J Math Educ Sci Technol 39(4):544–550
6. Haslett SJ (2021) The role of block diagonal matrices in conditions for equality of BLUEs in linear models. In: Online talk given in the 28th international workshop on matrices and statistics, Manipal Academy of Higher Education, India, 13–15 December 2021 IWMS 2021
7. Haslett SJ, Govindaraju K (2009) Cloning data: generating datasets with exactly the same multiple linear regression fit. Aust N Z J Stat 51(4):499–503. https://doi.org/10.1111/j.1467-842X.2009.00560.x
8. Haslett SJ, Govindaraju K (2012) Data cloning: data visualisation, smoothing, confidentiality, and encryption. J Stat Plan Inference 142(2):410–422. https://doi.org/10.1016/j.jspi.2011.07.020
9. Haslett SJ, Markiewicz A, Puntanen S (2020) Properties of BLUEs and BLUPs in full versus small linear models with new observations. In: Holgersson T, Singull M (eds) Recent developments in multivariate and random matrix analysis. Springer, Berlin, Chap 8, pp 123–146
10. Haslett SJ, Isotalo J, Markiewicz A, Puntanen S (2023) Permissible covariance structures for simultaneous retention of BLUEs in small and big linear models. In: Springer proceedings of the conferences in honor of C R Rao & A K Lal: Applied Linear Algebra, Probability and Statistics (ALAPS)
11. Hung CH, Markham TL (1977) The Moore-Penrose inverse of a sum of matrices. J Aust Math Soc Ser A 24:385–392

12. Kala R, Klaczyński K (1994) Generalized inverses of a sum of matrices. Sankhyā Ser A 56(3):458–464
13. Lele SR, Nadeem K, Schmuland B (2010) Estimability and likelihood inference for generalized linear mixed models using data cloning. J Am Stat Assoc 105(492):1617–1625
14. Lumley T (2010) Complex surveys: a guide to analysis using R. Wiley
15. Mitra SK, Moore BJ (1973) Gauss-Markov estimation with an incorrect dispersion matrix. Sankhyā Ser A 35:139–152
16. del Pino G (1989) The unifying role of iterative generalized least squares in statistical algorithms. Stat Sci 4(4):394–403
17. Puntanen S, Styan GPH, Isotalo J (2012) Matrix tricks for linear statistical models: our personal top twenty. Springer, Berlin
18. Rao CR (1971) Unified theory of linear estimation (corr: 72v34 p 194; 72v34 p 477). Sankhyä Ser A 33:371–394
19. Rao CR (1973) Representations of best linear estimators in the Gauss-Markov model with a singular dispersion matrix. J Multivar Anal 3:276–292
20. Riedel KS (1992) A Sherman-Morrison-Woodbury identity for rank augmenting matrices with application to centering. SIAM J Matrix Anal Appl 13:659–662

Chapter 15
Statistical Inference for Middle Censored Data with Applications



Sudeep R. Bapat

Mathematics Subject Classification 60E05 · 62N01 · 62N02

15.1 Introduction

Statistical literature is well content with the ideas of different censoring schemes. Some of the widely used ones include *right censoring, left censoring, interval censoring* and *double censoring*. Some of the key applications where censoring is observed include biometry, clinical trials, survival analysis, medical follow-up studies and epidemiology. At the start, we will quickly review some of these censoring ideas, just for completeness.

(i) Right censoring is seen when the subject leaves the study before the event occurs. Hence, if the subject leaves the study at time t_R, the event occurs in (t_R, ∞). For example, consider patients in a clinical trial to study the effect of treatment on a particular disease. The data becomes right censored if the patient does not develop the disease symptoms before the study ends. Further, Type I right censoring occurs when the study ends at a pre-specified time, whereas Type II right censoring is seen if the study ends after a fixed number of failures are observed.
(ii) Left censoring happens when the event of interest has already occurred before the patient enrols for the study. However, such censoring is rarely observed in practice.

One may refer to [25] where a general right censoring scheme is discussed, [26] who discuss a few censoring issues in survival analysis, [28] who present assumptions of right censoring in the presence of left truncation, [15] which gives a particular hybrid censoring scheme, [23] which considers Bayesian inference under progressive

S. R. Bapat (✉)
Indian Institute of Management, Indore, India
e-mail: sudeepb@iimidr.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_15

censoring and [24] which outlines a Type II progressive hybrid censoring scheme, among many others, for more extensions and applications.

This survey paper however focuses on a different kind of a censoring scheme which is relatively new and emerging, called "middle censoring". This is also a less restrictive and more general censoring scheme. Middle censoring occurs if a data point becomes unobservable if it falls inside a random interval. The data point in such situations includes a censorship indicator and a random interval. Clearly, right censoring, left censoring and double censoring happen to be special cases of middle censoring, by taking suitable choices for the censoring intervals. Middle censored observations can be found in a variety of real-life applications. To name a few, in a lifetime study if the subject is temporarily withdrawn, we get an interval of censorship, or in biomedical studies the patient may be absent from the study for a short period, during which the event of interest occurs.

A brief construction of a middle censored setup is as follows: Let there be n items under study, with lifetimes $T_1, T_2, \ldots, T_n$. For each item, there exists a random censoring interval (L_i, R_i), where observations cannot be made. The point to note is that the censoring interval is random, and independent of the lifetimes. The observed data in this case hence happens to be

$$X_i = \begin{cases} T_i & \text{if } T_i \notin [L_i, R_i] \\ (L_i, R_i) & \text{otherwise.} \end{cases} \tag{15.1}$$

Looking back, [20] first introduced this setup. Since then, numerous related papers have been published where the authors have tried to extend the model under different distributional and non-parametric setups. A well-documented work containing overviews of estimation strategies under middle censoring is that by [6], which is also the author's Ph.D. thesis. The author mainly discusses important contributions in middle censoring prior to the year 2011. However, a lot has been done since then, which we will try to review in this paper, along with some interesting applications. A brief outline of the following sections is as follows: We discuss the non-parametric middle censoring estimation approach as proposed by [20] in Sect. 15.2. Section 15.3 consists of an overview of some estimation techniques for middle censored data under some parametric models. We focus mainly on the exponential [16], Weibull [30] and Burr-XII [2] type lifetime distributions. As an extension to the exponential model, we also discuss estimation under middle censored competing risks data following an exponential distribution [4]. We briefly outline both frequentist and Bayesian estimation approaches. Section 15.4 gives a flavour of estimation under discrete middle censored lifetimes. Specifically, we cover situations where the lifetimes follow the geometric distribution [12] and a multinomial distribution [17]. Additionally, we also review estimation for middle censored data under the presence of covariates [19]. In Sect. 15.5, we cover parametric estimation of lifetimes which are middle censored and follow a circular distribution, namely the von Mises distribution [21], while Sect. 15.6 contains similar methodologies for an additive risks regression model [31]. A slightly different approach to estimation, namely the martingale method, is discussed here. Section 15.7 presents brief conclusions and a few final remarks.

15.2 Non-parametric Estimation for Middle Censored Data

Reference [20] introduced a self-consistent estimator (SCE) and a non-parametric maximum likelihood estimator (NPMLE) for middle censored data. A short overview is as follows: Let T_i be a sequence of independent and identically distributed random variables, with unknown distribution function F_0. Further, let $Y_i = (L_i, R_i)$ be a sequence of i.i.d. random vectors, independent of T_i's with an unknown bivariate distribution G such that $P(L_i < R_i) = 1$. We thus observe the following dataset:

$$Z_i = \begin{cases} T_i & \text{if } \delta_i = 1 \\ (L_i, R_i) & \text{if } \delta_i = 0, \end{cases} \tag{15.2}$$

where δ_i is the indicator function which takes the value 1 if $T_i \in (L_i, R_i)$. This is similar to Eq. (15.1) and a typical construction which is found in related middle censoring problems, which we will explore in later sections.

Now if one tries to estimate the distribution function using the EM algorithm, the resulting equation takes the form

$$\hat{F}(t) = E_{\hat{F}}[E_n(t)|\mathbb{Z}], \tag{15.3}$$

where $\mathbb{Z}$ refers to the set from Eq. (15.2), and which is as described by [34], where E_n is the empirical distribution function. This is also referred to as the self-consistency equation in the literature. It so turns out that under a middle censoring case, the SCE satisfies the equation:

$$F_n(t) = \frac{1}{n}\sum_{i=1}^{n}\left\{\delta_i I(T_i \le t) + \bar{\delta}_i I(R_i \le t) + \bar{\delta}_i I[t \in (L_i, R_i)]\frac{F_n(t) - F_n(L_i)}{F_n(R_i-) - F_n(L_i)}\right\}, \tag{15.4}$$

where $\bar{\delta}_i = 1 - \delta_i$. Again, an analytical solution to the above equation does not exist and one has to resort to an EM algorithm setup. An updated estimator using an iterative formula is given by

$$\hat{F}^{(m+1)}(t) = E_{\hat{F}^{(m)}}[E_n(t)|T]. \tag{15.5}$$

The authors further show that in this case, the NPMLE also satisfies the self-consistency equation given in Eq. (15.4). Further, if there exist any uncensored observations, the NPMLE attaches all its mass to them. This is well discussed in the form of the next proposition.

Proposition 2.1

If each observed censored interval (L_i, R_i) contains at least one uncensored observation T_j, $j \neq i$, then any distribution function which satisfies (15.4) attaches all its mass on the uncensored observations.

On the other hand, the authors also discuss a situation where the censoring interval (L_i, R_i) does not contain any uncensored observations. If a similar situation occurs in a right censoring setup, the extra mass is left unassigned. However, under a middle censoring scheme there exists a natural choice where the mass may be assigned to the midpoint of the corresponding interval. We will now look at a basic construction involving a couple of intervals, which will highlight the application of the proposed methodology. Let $n = 5$ and $z_1 = 2$, $z_2 = 4$, $z_3 = 6$, $z_4 = (1, 5)$, $z_5 = (3, 7)$. Let p_1, p_2, p_3 be the masses to be assigned to z_1, z_2 and z_3. The likelihood function hence becomes

$$p_1 p_2 p_3 (p_1 + p_2)(p_2 + p_3).$$

Now assuming that p_i's add up to 1 and one can interchange the roles of p_1 and p_3, one can reduce the problem to that of maximizing $(x^2)(1 - 2x)(1 - x)^2$ with $p_1 = p_3 = x$ and $p_2 = 1 - 2x$, and the solution turns out to be $x = (5 - \sqrt{5})/10$, and $1 - x = 1/\sqrt{5}$.

The authors also prove that the SCE is a uniformly strong consistent estimator, and one may refer to their paper for more information and details. A few other related papers which are worth a mention are by [27] where the authors derive a necessary and sufficient condition for developing the equivalence of the self-consistent estimators and NPMLEs, by [32] who came up with an inverse probability weighted approach to estimate the distribution function under middle censoring, and by [33] who demonstrated that the NPMLE can be obtained by using the [35] EM algorithm or the self-consistent estimating equation given by [20] with an initial estimator which puts more weight on the innermost intervals.

15.3 Middle Censoring Under Parametric Models

This section outlines a few interesting estimation strategies under middle censored data for some parametric models. Specifically, we will focus on exponential, Weibull and Burr-XII lifetime models.

15.3.1 Middle Censoring Under an Exponential Setup

This problem is along the lines of [16]. We assume that the observed data is similar to that of Eq. (15.2) but the sequence $T_1, T_2, \ldots, T_n$ is an i.i.d. sequence following an exponential distribution with mean $1/\theta_0$. The probability density function thus becomes

$$f(x; \theta_0) = \begin{cases} \theta_0 e^{-\theta_0 x}, & \text{if } x > 0, \theta_0 > 0 \\ 0 & otherwise. \end{cases} \tag{15.6}$$

Further, as seen before, let $(L_1, R_1), \ldots, (L_n, R_n)$ be i.i.d. with some unknown bivariate distribution G, and $(L_1, Z_1), \ldots, (L_n, Z_n)$ be i.i.d. sequence where L_i and $Z_i = R_i - L_i$ are independent exponential random variables which are independent of T_i. Also assume that L_i and Z_i have means $1/\alpha$, $1/\beta$, respectively. In short, one assumes that the censoring scheme is independent of the lifetimes. Such assumptions are standard in any middle censoring literature. In the next subsections, we briefly outline the maximum likelihood estimator of θ_0, and a Bayesian estimation setup for estimating the parameter θ.

15.3.1.1 Maximum Likelihood Estimator

Without a loss of generality, let the first n_1 and the rest n_2 observations ($n_1 + n_2 = n$) be uncensored and censored, respectively. Thus, the observed data becomes

$$\{(T_1, 1), \ldots, (T_{n_1}, 1), (L_{n_1+1}, R_{n_1+1}), \ldots, (L_{n_1+n_2}, R_{n_1+n_2})\}.$$

One can hence write down the likelihood function which takes the form,

$$l(\theta) = c\theta^{n_1} e^{-\theta \sum_{i=1}^{n_1} t_i} \prod_{i=n_1+1}^{n_1+n_2} (e^{-\theta l_i} - e^{-\theta r_i}), \tag{15.7}$$

where c is the normalizing constant depending on α and β. Now one has to take the derivative of the above likelihood function, set it equal to 0 and solve, to find the MLE. However, this equation does not give a closed-form solution and one has to resort to an iterative procedure. Let the equation $\frac{\partial l}{\partial \theta} = 0$ be written as $h(\theta) = \theta$, where $h(\theta)$ is a function of the variable and the constants. Thus, a basic iterative procedure can be to start with an initial guess $\theta^{(1)}$, then obtain $\theta^{(2)} = h(\theta^{(1)})$ and so on. The procedure can be stopped if $|\theta^{(i)} - \theta^{(i+1)}| < \epsilon$, where ϵ is a predefined small positive number. One definitely needs an initial guess for θ and one can pick $\theta^{(1)} = n_1 / \sum_{i=1}^{n_1} t_i$. A suitable EM algorithm can also be applied but we will omit further details. One may refer to the source paper. The authors also outline a series of lemmas and propositions which give the convergence of the iterative procedure, the likelihood function and the MLE, along with deriving the consistency of the MLE, and its asymptotic distribution.

15.3.1.2 Bayesian Analysis

Rather than using the conventional MLE for estimating the parameter θ, one can approach the problem through a Bayesian angle. Let the parameter θ have a $Gamma(a, b)$ prior distribution, with the pdf given by

$$\pi(\theta) = \pi(\theta|a,b) = \frac{b^a}{\Gamma(a)}\theta^{a-1}e^{-b\theta}. \tag{15.8}$$

The likelihood function of the observed data hence becomes

$$l(data|\theta) = c\theta^{n_1}e^{-\theta\sum_{i=1}^{n_1}t_i}\prod_{i=n_1+1}^{n_1+n_2}(1-e^{-\theta z_i})e^{-\theta\sum_{i=n_1+1}^{n_1+n_2}l_i}. \tag{15.9}$$

Now based on (15.8) and (15.9), the posterior density is given by

$$\pi(\theta|data) = \frac{l(data|\theta)\pi(\theta)}{\int_0^\infty l(data|\theta)\pi(\theta)d\theta}. \tag{15.10}$$

It turns out that if there are no censored observations, $Gamma(a+n, b+\sum_{i=1}^{n}t_i)$ happens to be posterior. However, if there exist censored observations, a Gibbs sampling technique proves useful to find the estimators. In this case, one can start by looking at a restricted estimator which is given by

$$f_{T|T\in(L,R)}(t|\theta) = \frac{\theta e^{-\theta t}}{e^{-\theta L} - e^{-\theta R}}. \tag{15.11}$$

A rough outline of the Gibbs sampling technique is as follows:

1. Generate $\theta_{(1)}$ from a $Gamma(a+n_1, b+\sum_{i=1}^{n_1}t_i)$ distribution
2. Generate the incomplete data t_i^* from the restricted distribution given in (15.11)
3. Generate $\theta_{(2)}$ from $Gamma(a+n, b+\sum_{i=1}^{n_1}t_i+\sum_{i=n_1+1}^{n}t_i^*)$
4. Go to Step 2 and replace $\theta_{(1)}$ by $\theta_{(2)}$. Repeat Steps 2 and 3 N times.

Finally, the Bayes estimate under squared error loss is given by

$$\hat{\theta}_{Bayes} = \frac{1}{N-M}\sum_{i=M+1}^{N}\theta_{(i)}, \tag{15.12}$$

where M is the burn-in sample.

A related work connecting an exponential distribution with middle censoring is by [4], where the authors also add competing risks data. We will briefly discuss the methodology in the next subsection.

15.3.2 Middle Censored Competing Risks Data Under Exponential Distribution

Reference [4] discusses possible estimation, both under a frequentist and a Bayesian approach, of the exponential parameter under middle censoring and competing risks data, which we will briefly explore for completeness.

Assume that n identical units are put on the test, and T_i is the failure time of the ith unit. An alternative notation for the failure times is

$$T_i = \min\{X_{i1}, X_{i2}, \ldots, X_{is}\},$$

where X_{ij} is the latent failure time of the ith unit under the jth cause of failure, $i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, s$. Now a general notion under competing risks data is that the failures are independent but not observable in nature, whereas T_i, C_i are observable where $C_i = j$ if failure is due to cause j. Further, let X_{ij} follow an exponential distribution with pdf,

$$f_j(t; \theta_j) = \theta_j e^{-\theta_j t}, \quad t, \theta_j > 0. \tag{15.13}$$

Now using the independence of the lifetimes, one can derive the density of the lifetime and the corresponding cause (T_i, C_i), which is given by

$$f_{T_i, C_i}(t, j; \theta) = f_j(t; \theta_j) \prod_{l=1, l\neq j}^{s} [1 - F_l(t; \theta_l)] = \theta_j e^{-\theta_* t}, \tag{15.14}$$

where $\theta = (\theta_1, \ldots, \theta_s)$ and $\theta_* = \sum_{j=1}^{s} \theta_j$. Now similar to the construction of the observed data seen in the last subsection, the data in this case becomes

$$(Y_i, C_i, \delta_i) = \begin{cases} (T_i, C_i, 1), & \text{if } T_i \neq [L_i, R_i] \\ ([L_i, R_i], C_i, 0) & T_i \in [L_i, R_i]. \end{cases} \tag{15.15}$$

Further, we assume that $(L_1, Z_1), \ldots, (L_n, Z_n)$ are i.i.d. where L_i and $Z_i = R_i - L_i$ are independent random variables with exponential distributions having pdfs,

$$f_{L_i}(l; \alpha) = \alpha e^{-\alpha l}$$

and

$$f_{Z_i}(z; \beta) = \beta e^{-\beta z},$$

$l, \alpha, \beta, z > 0$. Note that α, β do not depend on θ and L_i, Z_i are independent of T_i. Now again assuming a simple reordering of the data, the final observed data is given by

$$\{(T_1, C_1, 1), \ldots, (T_{n_1}, C_{n_1}, 1), ([L_{n_1+1}, R_{n_1+1}], C_{n_1+1}, 0), \ldots, ([L_{n_1+n_2}, R_{n_1+n_2}], C_{n_1+n_2}, 0)\}.$$

Based on this data, the likelihood function can be written as

$$L(\theta) \propto \prod_{j=1}^{s} \prod_{i=n_1+1}^{n_1+n_2} \{F(r_i, j; \theta) - F(l_i, j; \theta)\}^{I(C_i=j)} \tag{15.16}$$

$$= \left(\frac{1}{\theta_*}\right)^{n-n_1} \left(\prod_{j=1}^{s} \theta_j^{m_j}\right) e^{-\theta_*(\sum_{i=1}^{n_1} t_i + \sum_{i=n_1+1}^{n_1+n_2} l_i)} \prod_{i=n_1+1}^{n_1+n_2} (1 - e^{-\theta_* z_i}), \tag{15.17}$$

where $z_i = r_i - l_i$ and $m_j = \sum_{i=1}^{n} I(C_i = j)$ is the total number of failures due to cause j. A routine method to find the MLE will be to take the derivative, set it equal to 0 and solve. It can be seen that the MLE of θ_j, say $\hat{\theta}_j$, satisfies the following equation:

$$\hat{\theta}_j = \frac{m_j}{n_1 + n_2} \hat{\theta}_*, \tag{15.18}$$

$j = 1, 2, \ldots, s$, where $\hat{\theta}_*$ is the MLE of θ_*. Now since this equation does not give an analytical solution, one has to resort to an appropriate iterative idea or an EM algorithm. One may refer to the source paper for a suitable methodology and further details. We will now briefly look at a Bayesian estimation angle to estimate the unknown parameter θ_j, $j = 1, 2, \ldots, s$.

15.3.2.1 Bayesian Estimation

A suitable prior distribution in this case will be a $Gamma(a_j, b_j)$ distribution. We assume that all the hyperparameters a_j, b_j are known. Now going by the standard approach, the posterior density can be derived and is seen proportional to

$$\pi(\theta|data) = \prod_{j=1}^{s} \left\{\theta_j^{m_j+\alpha_j-1} e^{-\theta_j(b_j + \sum_{i=1}^{n_1} t_i + \sum_{i=n_1+1}^{n_1+n_2} l_i)}\right\} \prod_{i=n_1+1}^{n_1+n_2} \left\{\frac{1}{\theta_*}(1 - e^{-\theta_* z_i})\right\}. \tag{15.19}$$

The Bayesian estimator of θ_j under a squared error loss function can be thus obtained as $E(\theta_j|data)$. Now again, the Bayesian estimator cannot be obtained analytically and hence one has to resort to some suitable iterative methods. The authors outline two such methods, namely Lindley's approximation and Gibbs sampling to obtain the estimators. A point to note is that if there do not exist any censored observations, the Bayesian estimator of θ_j can be easily evaluated as $(m'_j + a_j)/(b_j + \sum_{i=1}^{n_1} t_i)$, where $m'_j = \sum_{i=1}^{n_1} I(C_i = j)$. We now provide the form of the estimator under Lindley's approximation just for completeness.

Let $\hat{\theta}_j^{SE}$ denote the estimator obtained through Lindley's approximation which is given by

$$\hat{\theta}_j^{SE} = \left[\theta_j + \sum_{j'=1}^{s} \left(\frac{a_{j'} - 1}{\theta_{j'}} - b_{j'} \right) \sigma_{jj'} + \frac{1}{2} \sum_{i'=1}^{s} \sum_{j'=1}^{s} \sum_{k=1}^{s} L_{i'j'k} \sigma_{i'j'} \sigma_{kj} \right]_{\hat{\theta}}, \quad (15.20)$$

where

$$L_{i'j'k} = \frac{\partial^3 \log L(\theta)}{\partial \theta_{i'} \partial \theta_{j'} \partial \theta_k} = \begin{cases} \frac{2mj'}{\theta_{j'}^3} - A, & i' = j' = k, \\ -A & \text{otherwise.}, \end{cases}$$

$$A = \frac{2n_2}{\theta_*^3} + \sum_{i=n_1+1}^{n_1+n_2} \frac{1 + e^{\theta_* z_i}}{(1 - e^{\theta_* z_i})^3} z_i^3 e^{\theta_* z_i},$$

where $\hat{\theta} = (\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_s)$ and $\sigma_{i'j'} = (i', j')^{th}$ elements of the inverse matrix $[-L_{i'j'}]_{s \times s}$, $j = 1, 2, \ldots, s$.

15.3.2.2 Interval Estimation

We will now try to give a flavour of interval estimation of the MLEs. Using the standard assumption of asymptotic normality of the MLEs under some regularity conditions, one can construct suitable confidence intervals for the unknown parameters. On utilizing the asymptotic distribution of the MLEs, the sampling distribution of $(\hat{\theta}_j - \theta_j)/\sqrt{\sigma_{jj}}$, $j = 1, 2, \ldots, s$ can be approximated by a standard normal distribution. Thus for a given $\gamma \in (0, 1)$, a two-sided $100(1 - \gamma)\%$ confidence interval can be constructed as $\hat{\theta}_j \pm z_{\gamma/2}\sqrt{\sigma_{jj}}$, where $z_{\gamma/2}$ is the usual upper percentile point of a standard normal distribution. Now taking the help of the generated samples using the EM algorithm and using the method described in [8], a highest posterior density (HPD) credible interval can be easily constructed.

15.3.2.3 Reconstruction of Censored Failure Times

The authors in their paper have discussed an interesting and needy issue which is that of reconstructing the censored failure times. If in a middle censoring scheme $\delta_i = 0$, one only observes the censoring interval $[L_i, R_i]$. Thus one would argue as to what the actual observations are. A possible reconstruction of the censored failure times is a solution in this case. Several point reconstructors of the censored failure time T_i are obtained such as the unbiased conditional reconstructor, the conditional median and mode reconstructor and the Bayesian reconstructor. We will briefly outline one of them here.

A good choice of a point reconstructor of T_i happens to be $E(T_i|data) = E(T_i|T_i \in [l_i, r_i])$. Thus upon invoking the conditional distribution of T_i given $T_i \in [l_i, r_i]$, one can derive the point reconstructor as

$$\hat{T} = L_i + \frac{1}{\theta_*} - \frac{Z_i e^{-\theta_* Z_i}}{1 - e^{-\theta_* Z_i}}. \tag{15.21}$$

One can also derive the conditional expectation of $\hat{T}$ given $T_i \in [L_i, R_i]$ which is given by

$$E(\hat{T}|T_i \in [L_i, R_i]) = \frac{\alpha + \beta + 2\theta_*}{(\alpha + \theta_*)(\beta + \theta_*)}. \tag{15.22}$$

If the parameter θ_* is unknown, one can replace it with the MLE. Further, the authors also discuss a comparison among the three above-mentioned reconstruction methods using a suitable Pitman's measure of closeness. We will leave out the details for brevity.

We would like to mention a few other recent but related papers for completeness. The first one is by [13], where the authors have developed a dependent middle censoring scheme with a censoring interval of fixed length, where the lifetime and lower bound of the censoring interval are variables with a Marshall-Olkin bivariate exponential distribution. The next is by [36], where the author established point and interval estimation problems for the exponential distribution when the failure data is middle censored with two independent competing failure risks. The next is by [29], where the authors have dealt with the modelling of cumulative incidence function using improper Gompertz distribution based on middle censored competing risks survival data. The last one is by [37], where the authors considered a dependent competing risks model under a Marshall-Olkin bivariate Weibull distribution. A suitable MLE approach, a midpoint approximation estimator and appropriate confidence intervals are designed. Additionally, a Bayesian approach using a Gamma-Dirichlet prior distribution is proposed, where the estimators are obtained through an acceptance-rejection sampling method. One may refer to the source papers for more details.

15.3.3 Middle Censoring Under a Few Other Parametric Models

We will now briefly outline the parametric estimation procedure under a middle censored data where the lifetimes follow a Weibull distribution and Burr-XII distribution, respectively.

15.3.3.1 Estimation Under a Weibull Model

This work is along the lines of [30]. Let T be a non-negative lifetime of a subject under study, which follows a $Weibull(\alpha, \beta)$ distribution having the following density function:

$$f(t) = \alpha\beta^{-1}(\beta^{-1}t)^{\alpha-1}\exp\{-(\beta^{-1}t)^{\alpha}\}; \quad t > 0, \alpha > 0, \beta > 0. \tag{15.23}$$

Further, let (U, V) be the censoring interval which is independent of T and with a bivariate cumulative distribution G. Also assume that Z denotes a set of covariates which maybe continuous or mere indicator variables. Thus, the observed vector is denoted by

$$X = \begin{cases} T, & \text{if } \delta = 1 \\ (U, V) & \text{if } \delta = 0, \end{cases} \tag{15.24}$$

where δ is the usual indicator variable. The authors assume a Cox proportional hazards model under which the conditional density takes the form,

$$f(t|z) = \alpha\gamma^{-1}(\gamma^{-1}t)^{\alpha-1}\exp\{-(\gamma^{-1}t)^{\alpha}\}; \quad t, \alpha, \gamma > 0, \tag{15.25}$$

where $\gamma = \beta e^{-\theta Z/\alpha}$ and $\theta = (\theta_1, \ldots, \theta_p)$ is a vector of regression coefficients. Now let (X_i, δ_i, Z_i) be n i.i.d. copies of (X, δ, Z) corresponding to n subjects under investigation. One can note that the likelihood function in this case becomes

$$L(\psi) \propto \prod_{i=1}^{n} f(x_i|z_i)^{\delta_i}[S(u_i|z_i) - S(v_i|Z_i)]^{1-\delta_i}, \tag{15.26}$$

where ψ is the collective parameter vector given by $(\alpha, \beta, \theta_1, \ldots, \theta_p)'$. Again in this case, finding an analytical solution for the MLEs is not possible and hence one may resort to an EM algorithm, which is outlined in the source paper. We leave out the details for brevity. It is also shown that the derived MLE $\hat{\psi} = (\hat{\alpha}, \hat{\beta}, \hat{\theta})$ follows asymptotically $(p + 2)-$variate normal distribution with mean vector 0 and dispersion matrix I_{ψ}^{-1}, where $I(\psi)$ denotes the Fisher information matrix that can be found out after a set of usual manipulations.

15.3.3.2 Estimation Under a Burr-XII Distribution

The Burr-XII distribution was introduced by [7] and finds its use to model stochastic events and lifetimes of products as it has a non-monotone hazard function. This estimation methodology was proposed by [2]. A random variable is said to follow Burr-XII distribution with parameters a and b if the density function is given by

$$f(t) = abt^{b-1}(1 + t^{b})^{-a-1}, \quad t, a, b > 0. \tag{15.27}$$

Assume that $T_1, \ldots, T_n$ are i.i.d. Burr-XII variables and let $Z_i = R_i - L_i, i = 1, \ldots, n$ denote the length of the censoring interval with an exponential distribution with parameter γ. We also assume that the left bound of the interval L_i also follows an exponential distribution with parameter λ. Moreover, T_i, L_i, Z_i are independent.

We now try to represent the likelihood function under this case by reordering the observed data into uncensored and censored observations. Thus, the observed data becomes

$$\{T_1, \ldots, T_{n_1}, (L_{n_1+1}, R_{n_1+1}), \ldots, (L_{n_1+n_2}, R_{n_1+n_2})\},$$

and the likelihood can be written as

$$L(a, b|t) = c(ab)^{n_1} \prod_{i=1}^{n_1} t_i^{b-1}(1+t_i^b)^{-(a+1)} \prod_{i=n_1+1}^{n_1+n_2} \left[(1+r_i^b)^{-a} - (1+l_i^b)^{-a}\right], \tag{15.28}$$

where c is a normalizing constant that depends on λ and γ. One can follow the usual pattern of finding the log-likelihood and maximizing that to find the MLEs. However, an analytical solution is not possible and one has to resort to an iterative procedure. The authors choose to adopt a Newton-Raphson method to do the same.

15.4 Inference for Discrete Middle Censored Data

Reference [12] in their paper developed a discrete middle censoring problem where the lifetime, lower bound and length of censoring interval are variables with the geometric distribution. A brief outline of the problem is as follows: Let $T_1, T_2, \ldots, T_n$ be a random sample of size n from a geometric distribution with mean $1/\theta_0$. The pdf is given by

$$P(T_i = t_i) = \theta_0 \lambda_0^{t_i}, \tag{15.29}$$

where $\lambda_0 = 1 - \theta_0, t_i = 0, 1, 2, \ldots$ Further, let $(L_1, X_1), \ldots, (L_n, X_n)$ be i.i.d. where L_i and $X_i = R_i - L_i$ are independent geometric random variables with means $1/p_i$ and q_x/p_x, where

$$P(L_i = l_i) = p_l q_l^{l_i}, \quad q_l = 1 - p_l, \quad l_i = 0, 1, 2, \ldots$$

and

$$P(X_i = x_i) = p_x q_x^{x_i - 1}, \quad q_x = 1 - p_x, \quad x_i = 1, 2, \ldots,$$

where p_l, p_x do not depend on θ_0, and T_i is independent of L_i and X_i, $i = 1, 2, \ldots$

15.4.1 Maximum Likelihood Estimator

As seen in the previous section, the MLE of θ_0 does not exist in a closed form, and hence an appropriate EM algorithm has to be applied. Now just to be in line with the notations used in the paper, we first define the following two random variables:

$$U_i = \begin{cases} T_i, & \text{if } T_i \notin [L_i, R_i], \\ [L_i, R_i], & T_i \in [L_i, R_i] \end{cases} \tag{15.30}$$

and

$$V_i = \begin{cases} 1, & T_i \notin [L_i, R_i], \\ 0, & T_i \in [L_i, R_i], \end{cases} \tag{15.31}$$

$i = 1, 2, \ldots, n$. Now further let $W_i = (U_i, V_i)$, $i = 1, 2, \ldots, n$, and $\mathbf{W} = (W_1, \ldots, W_n)$. Now in order to get the density function of W, one has to make use of the variables U, V from Eqs. (15.30) and (15.31). The pmf of W thus becomes

$$P_{\theta_0}(W = w) = P_{\theta_0}(U = u, V = v) = \left[c_1 \lambda^l (1 - \lambda^{x+1})\right]^{1-v} \left[c_2 \theta_0 \lambda^{t_i}\right]^v,$$

where $c_1 = P_{\theta_0}(L = l, R = r)$, $c_2 = P_{\theta_0}(T \notin [L, r])$ and $x = r - L$. Again as assumed earlier, let the first n_1 observations be uncensored, whereas the rest n_2 observations be censored. The final observed data thus becomes

$$\mathbf{W} = [(T_1, 1), \ldots, (T_n, 1), ([L_{n_1+1}, R_{n_1+1}], 0), \ldots, ([L_{n_1+n_2}, R_{n_1+n_2}], 0)].$$

The likelihood function takes the form,

$$L(\theta)_0 = c\theta_0^{n_1}(1 - \theta_0)^{\left(\sum_{i=1}^{n_1} t_i + \sum_{i=n_1+1}^{n_1+n_2} l_i\right)} \prod_{i=n_1+1}^{n_1+n_2} (1 - (1 - \theta_0)^{x_i+1}), \tag{15.32}$$

where $c = c_1^{n_2} c_2^{n_1}$ is the normalizing constant which does not depend on θ_0. Again, one cannot find an explicit solution of the MLE and hence one has to resort to an EM algorithm. Further one can also show that the MLE is a consistent estimator of θ_0. We omit details for brevity and one may refer to the original paper, where the authors have discussed several lemmas in this regard.

15.4.2 Bayesian Inference

In this context, we assume a $Beta(a, b)$ prior distribution for the parameter θ. The pdf takes the form,

$$\pi(\theta) \propto \theta^{a-1}(1 - \theta)^{b-1}, \ 0 < \theta < 1, a, b > 0. \tag{15.33}$$

Now denoting $x_i = x_{n_1+i}$ and $l_i = l_{n_1+i}$, one can rewrite the likelihood function from (15.16) as

$$L(\theta) = L(\mathbf{w}|\theta) = c\theta^{n_1}\lambda^{(\sum_{i=1}^{n_1} t_i + \sum_{i=1}^{n_2} l_i)} \prod_{i=1}^{n_2}(1-\lambda^{x_i+1}). \tag{15.34}$$

The posterior density of θ can be written as

$$\pi(\theta|\mathbf{w}) = \frac{L(\mathbf{w}|\theta)\pi(\theta)}{\int_0^1 L(\mathbf{w}|\theta)\pi(\theta)d\theta}. \tag{15.35}$$

After a few simplifications, one can write the posterior density of θ, and can come up with its Bayes estimator. We omit details for brevity. It turns out that when n_2 is small, the evaluation of $E(\theta|w)$ is straightforward. However, when n_2 is large, it is difficult to obtain a numerical solution. One can apply a suitable Metropolis-Hastings algorithm as per [9]. The authors consider the starting density as a $Beta(3, 3)$ distribution and calculate the estimated risk using the formula,

$$ER(\delta) = \frac{1}{N}\sum_{i=1}^{N}(\delta_i - \theta)^2, \tag{15.36}$$

where δ_i is an estimate of θ in the ith replication and N is the number of replications. One can also easily construct a credible interval for θ using the procedure described in [8].

A somewhat related and extended work is by [19] where the authors discuss the analysis of discrete lifetime data under middle censoring and in the presence of covariates. We will briefly look at the same in the next subsection.

15.4.3 Discrete Lifetime Data Under Middle Censoring and Presence of Covariates

The paper by [19] tries to extend the ideas proposed by [12] by generalizing their setup by bringing in covariates. The proofs and results are discussed by exploiting the relationship between an exponential and a geometric distribution. This connection is used to derive the MLEs under middle censoring and in the presence of covariates, by considering an accelerated failure time model for the geometric distribution. The following proposition gives the connection between exponential and geometric distributions.

Proposition 4.1

If X is an exponentially distributed random variable with parameter λ, then $Y = \lfloor X \rfloor$ where $\lfloor \cdot \rfloor$ stands for the integer part of x, and *is a geometrically distributed random variable with parameter $p = 1 - e^{-\lambda}$.*

Also interestingly, both exponential and geometric distributions satisfy the famous "memoryless" property. Thus, by invoking the memoryless property and the above proposition, one can generate geometric lifetimes from the exponential lifetimes, $T_i \sim Exp(\lambda)$. The geometric lifetimes hence can be derived as $Y_i = \lfloor T_i \rfloor \sim geometric(p)$, where the pmf becomes

$$P(Y_i = y_i) = p(1-p)^{y_i},$$

where $y_i = 0, 1, 2, \ldots$ and $p = 1 - e^{-\lambda}$. Now let the left bound of the censored interval be denoted by $U_i = \lfloor L_i \rfloor \sim geomteric(p_u)$, where $L_i \sim Exp(\alpha)$, $p_u = 1 - e^{-\alpha}$. Further let the length of the interval be denoted by $W_i = \lfloor S_i \rfloor \sim geometric(p_w)$, where $S_i \sim Exp(\beta)$, $S_i = R_i - L_i$, $p_w = 1 - e^{-\beta}$ and $W_i = V_i - U_i$, where V_i is the right bound of the censored interval. It turns out the Y_i, U_i, W_i are independent for all i.

We will now look at a geometric model in the presence of covariates, where the covariates are taken as fixed. Let T_i be the survival time for each patient along with its covariates denoted by Z_i. The authors then propose an accelerated failure time model and argue that it is equivalent to the Cox proportional hazard assumption when the baseline distribution is exponential. The pdf of this particular exponential density is given by

$$f(t|Z_i) = \lambda e^{\theta^\top Z_i} \exp(-\lambda e^{\theta^\top Z_i} t), \tag{15.37}$$

where θ is the effect of each covariate $\mathbf{Z}$, and $\top$ stands for transpose. Hence, the geometric lifetimes become $Y_i = \lfloor T_i \rfloor \sim geometric(p_i)$, where $p_i = 1 - \exp(-\lambda e^{\theta^\top Z_i} t)$. The left endpoint, length and right endpoint of the censored interval is denoted as before as U_i, W_i, V_i, respectively. One can thus write the necessary likelihood function and derive the MLE of p. Again, as discussed before, after a slight reordering of the data, the observed data becomes

$$\{Y_i, Y_2, \ldots, Y_{n_1}, [U_{n_1+1}, V_{n_1+1}], [U_{n_1+2}, V_{n_1+2}], \ldots, [U_{n_1+n_2}, V_{n_1+n_2}]\}.$$

Now similar to the expression seen in [12], the likelihood function can be written down as per Eq. (15.16). Finally, as the analytical solution cannot be obtained even here, one can resort to an EM algorithm appropriately. One may refer to the source paper for more details.

15.4.4 Middle Censoring in the Multinomial Distribution with Applications

Another relevant paper in the discrete distribution area is by [17], where the authors develop estimation strategies under a multinomial setup and middle censoring. An interesting application in this context is that of a "consumer rating" scenario. Imagine

that a group of customers are told to rate a certain product (in terms of stars) which they have bought, maybe through an online shopping portal. A small addition is that the customers are allowed to provide interval ratings say, between 1 and 2 stars or between 2 and 3 stars, etc. Such a situation can be easily modelled using a multinomial distribution, which we will briefly explore here.

Assume that the company contacts n individuals, each of them being asked to rate the product in terms of $\{1, 2, \ldots, k\}$ stars, according to his/her liking for the product. Let f_j stand for the number/frequency of people giving j stars. If we denote the true probabilities of giving $1, 2, \ldots, k$ stars by $p_1, p_2, \ldots, p_k$ respectively, we have the standard multinomial scheme with $\sum_{j=1}^{k} f_j = n$ and $\sum_{j=1}^{k} p_j = 1$. Now further assume that each individual is also allowed to give "interval ratings" for the product. We discuss how the estimated probabilities for each category change. The next subsection gives a general likelihood function in such a case.

15.4.4.1 The Likelihood Under a General Scheme

Let I represent an interval say, j_1 to j_2 of categories with corresponding probability $P_I = \sum_{j \in I} p_j$ for this interval. Now out of n individuals, let m of them provide interval ratings that belong to the intervals $\{I_j;\, j = 1, 2, \ldots, m\}$ with r of these being distinct. The remaining $n - m$ individuals provide specific singleton ratings, of which let there be k. Then the probabilities become

$$\sum_{j=1}^{r} P_{I_j} + \sum_{i=1}^{k} p_i = 1. \tag{15.38}$$

Further assume that the frequency in the interval I_j is F_j and the frequencies in the k individual categories are f_i. Then $\sum_{j=1}^{r} F_j = m$ and $\sum_{i=1}^{k} f_i = n - m$, so that

$$\sum_{j=1}^{r} F_j + \sum_{i=1}^{k} f_i = n. \tag{15.39}$$

One can then easily write down the likelihood function for the vector $\boldsymbol{p}$, which is given by

$$L \propto \prod_{j=1}^{r} P_{I_j}^{F_j} \times \prod_{i=1}^{k} p_i^{f_i}. \tag{15.40}$$

Finally, one aims to find the associated MLEs by either using the usual maximization approach or a suitable iterative method. One has to note that if certain intervals overlap, finding analytical MLE is even more difficult. We now briefly discuss a situation where there exist two disjoint interval ratings and their associated problem formulation.

Let $[i_1, j_1]$ and $[i_2, j_2]$ be two disjoint intervals with frequencies $f_{i_1 j_1}$, $f_{i_2 j_2}$ and respective probabilities $p_{i_1 j_1}$, $p_{i_2 j_2}$. We further have $p_{i_1 j_1} + p_{i_2 j_2} + \sum_{i=1}^{k} p_i = 1$. The likelihood function then becomes

$$p_1^{f_1} \dots p_{i_1}^{f_{i_1}} \dots p_{j_1}^{f_{j_1}} \dots p_{i_2}^{f_{i_2}} \dots p_{j_2}^{f_{j_2}} \dots p_k^{f_k} p_{i_1 j_1}^{f_{i_1 j_1}} p_{i_2 j_2}^{f_{i_2 j_2}}, \tag{15.41}$$

where $1 \le i_1 < j_1 < i_2 < j_2 \le k$ and $p_k = (1 - p_1 - \dots - 2p_{i_1} - \dots - 2p_{j_1} - p_{j_1+1} - \dots - 2p_{i_2} - \dots - 2p_{j_2} - \dots - p_{k-1})$. The MLEs in this case exist in analytical form which are given by

$$\hat{p_1} = \frac{f_1}{n}, \dots, \hat{p}_{i_1-1} = \frac{f_{i_1-1}}{n}, \hat{p}_{j_1+1} = \frac{f_{j_1+1}}{n}, \dots, \hat{p}_{i_2-1} = \frac{f_{i_2-1}}{n}, \hat{p}_{j_2+1} = \frac{f_{j_2+1}}{n}, \dots, \hat{p}_k = \frac{f_k}{n} \tag{15.42}$$

and,

$$\hat{p}_{l_1} = \frac{f_{l_1}}{2n}\left(1 + \frac{f_{i_1 j_1}}{f_{i_1} + \dots + f_{j_1}}\right), \qquad \hat{p}_{l_2} = \frac{f_{l_2}}{2n}\left(1 + \frac{f_{i_2 j_2}}{f_{i_2} + \dots + f_{j_2}}\right), \tag{15.43}$$

where $l_1 = i_1, (i_1 + 1), \dots, j_1$ and $l_2 = i_2, (i_2 + 1), \dots, j_2$. Now if no individual opts for the given interval ratings, the MLEs take a simpler form and are given by $\hat{p}_{l_1} = f_{l_1}/2n$ and $\hat{p}_{l_2} = f_{l_2}/2n$. The authors also provide asymptotic variances of the estimates, but we leave details for brevity.

15.4.4.2 Bayesian Estimation

As seen above, the unknown parameter vector is $\boldsymbol{p} = (P_{I_1}, P_{I_2}, \dots, P_{I_r}, p_1, p_2, \dots, p_k)$. Going by the usual ideology, we will assume a Dirichlet prior to the parameter vector. Hence, the setup is as follows: Let $\boldsymbol{X} = (X_1, X_2, \dots, X_n)$ denote the choices of the n individuals with each X_i taking either an interval or a singleton rating. Hence we can assume

$$\begin{aligned} X|\boldsymbol{p} &\sim Multinomial(\boldsymbol{p}) \\ \boldsymbol{p}|\boldsymbol{\alpha} &\sim \pi(\boldsymbol{p}) \equiv Dir(\boldsymbol{\alpha}), \end{aligned}$$

where $Dir(\boldsymbol{\alpha})$ stands for a Dirichlet distribution with parameter vector $\boldsymbol{\alpha} = (\alpha_1^*, \dots, \alpha_r^*, \alpha_1, \alpha_2, \dots, \alpha_k)$, where $\alpha_i^* (> 0)$ corresponds to the respective prior parameter on each P_{I_j}, $j = 1, 2, \dots, r$. The Dirichlet density is given by

$$Dir(\boldsymbol{p}|\boldsymbol{\alpha}) = \frac{\Gamma(\sum_{j=1}^{r} \alpha_j^* + \sum_{i=1}^{k} \alpha_i)}{\prod_{j=1}^{r} \Gamma(\alpha_j^*) \prod_{i=1}^{k} \Gamma(\alpha_i)} \prod_{j=1}^{r} P_{I_j}^{\alpha_j^* - 1} \times \prod_{i=1}^{k} p_i^{\alpha_i - 1}. \tag{15.44}$$

Now on applying the usual approach, one can get the posterior density which is given by

$$\pi(\boldsymbol{p}|data) = \frac{L(data|\boldsymbol{p})\pi(\boldsymbol{p})}{\int_{\boldsymbol{p}} L(data|\boldsymbol{p})\pi(\boldsymbol{p})d\boldsymbol{p}}. \tag{15.45}$$

A point to note is that one can obtain the Bayes estimate of $\boldsymbol{p}$ under some specific illustrations, which are neatly outlined in the source paper. The authors also include extensive parametric bootstrap and simulation analyses to complement their ideas.

15.4.4.3 A Real Data Example

Now to complement the methodology from the above subsection, we will give a flavour of a real data application. A survey was conducted at the University of California, Santa Barbara, by one of the authors of the paper, which asked a class of students about their overall experience with regard to a software capable of organizing online meetings and live sessions. They could give ratings of 1 through 5 (5 being the highest rating) along with a couple of "interval ratings" consisting of [1, 2] and [4, 5]. The data was collected from a group of $n = 90$ students from the class. This falls perfectly in the paradigm of middle censoring, especially the situation which is discussed in Sect. 4.4.1. The likelihood and log-likelihood functions in this case take the following forms:

$$L = p_1^{f_1} p_2^{f_2} p_3^{f_3} p_4^{f_4} p_5^{f_5} (p_1 + p_2)^{f_{12}} (p_4 + p_5)^{f_{45}},$$

and,

$$\log L = f_1 \log p_1 + \cdots + f_5 \log p_5 + f_{12} \log(p_1 + p_2) + f_{45} \log(p_4 + p_5),$$

where $p_3 = (1 - 2p_1 - 2p_2 - 2p_4 - 2p_5)$. We then obtain the estimated probabilities (MLEs) from Eqs. (15.42) and (15.43). Table 15.1 outlines results for these students' observed frequencies. Note that all these results are from a single run (single question in the survey).

As can be seen from the above table, the intervals [1, 2] and [4, 5] put a slight additional mass on $\hat{p}_1^{MLE}$, $\hat{p}_2^{MLE}$ as well as $\hat{p}_4^{MLE}$, $\hat{p}_5^{MLE}$ compared to the original values of $f_i/n, i = 1, 2, 4, 5$.

Table 15.1 Real data example with $n = 90$

i	1	2	3	4	5	[1, 2]	[4, 5]
f_i	2	8	13	21	6	11	29
f_i/n	0.022	0.088	0.144	0.233	0.066	0.122	0.322
$\hat{p}_i^{MLE}$	0.023	0.093	0.144	0.241	0.069	0.116	0.31

15.5 A General Censoring Scheme for Circular Data

In the previous sections, we have seen a variety of non-parametric and parametric estimation methodologies under middle censored data. However, all the datasets were examples of usual samples (lying either on the entire real line or a subset of it). Reference [21] discusses a general censoring scheme which is similar to a middle censoring scheme for circular data. Circular data is a set of independent and identically distributed (i.i.d.) values in two-dimensional directions. Such variables are also called angular data, as they can be represented on the circumference of a circle. A few examples consist of wind directions, the angles made by sea turtles while walking from the sea to the mainland, or the vanishing angles for a group of birds. Now a middle censored observation arises if an observation falls inside a random interval, whose endpoints may be observed but not the actual data point. A simple example is that of the bird's vanishing angle at the horizon, which may be blocked due to a passing cloud for a short while in between, thus making it middle censored. The authors specifically develop estimation strategies under middle censoring for a von Mises distribution, which we will explore next.

Let $\alpha_1, \ldots, \alpha_n$ be a set of angular measurements following a von Mises distribution. The pdf is given by

$$f(\alpha; \boldsymbol{\theta}) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\alpha - \mu)}, \quad 0 \leq \alpha \leq 2\pi, \tag{15.46}$$

where μ is the mean direction and κ is called as the concentration parameter. Here, $\boldsymbol{\theta} = (\mu, \kappa) \in \Theta = [0, 2\pi) \times [0, \infty)$ and I_v is the modified Bessel function of the first kind and order v, which takes the form,

$$I_v(z) = \frac{1}{2\pi} \int_0^{2\pi} \cos vt e^{z \cos t} dt.$$

Now the goal is to estimate the parameter vector under middle censoring which means when a few observations fall inside a random interval it forms an arc on the circumference. Let these observed arcs be (l_i, r_i). Then the likelihood function becomes

$$L_n(\boldsymbol{\alpha}; \boldsymbol{\theta}) = \frac{1}{(2\pi I_o(\kappa))^n} \exp\left[\kappa \sum_{i=1}^{n} \delta_i \cos(\alpha_i - \mu)\right] \prod_{i=1}^{n} \left[\int_{l_i}^{r_i} \exp[\kappa \cos(t - \mu)] dt\right]^{1-\delta_i}, \tag{15.47}$$

where δ_i is the usual indicator function which takes the value 0 if the observation is censored or 1 otherwise. The authors further provide the log-likelihood function and its derivatives from which the MLE can be found numerically. One can also substitute the MLE in the information matrix to obtain the observed information matrix $\hat{I}$. Moreover, $\sqrt{n}(\hat{\theta}_n - \theta)$ is asymptotically normal with mean 0 and covariance $\hat{I}^{-1}$.

15.5.1 Non-parametric Estimation of Censored Data

We will now briefly look at the self-consistent estimator of the parameter $\boldsymbol{\theta}$. Let A_i be a sequence of i.i.d. circular random variables with an unknown distribution F_0. Let (L_i, R_i) be a sequence of i.i.d. random vectors, independent of $A_i's$. Assume that both these components take values in $[0, 2\pi)$. Interestingly since there is no ordering on a circle, R_i may happen to be less than the corresponding L_i, for example, a censoring arc from 355 to 18. Now following the usual procedure, we will equate the distribution function with the conditional expectation of the empirical distribution given the data. In this case it becomes

$$F(t) = E_F[(E_n)(t)|A_i, (L_i, R_i)], \tag{15.48}$$

where E_n is the empirical distribution function. This equation is referred to as the self-consistency equation according to [14]. Now in the presence of censored data, the above distribution function satisfies the equation,

$$F(t) = \frac{1}{n}\sum_{i=1}^{n}\left\{\delta_i I(A_i \le t) + \bar{\delta}_i I(R_i \le t) + \bar{\delta}_i I[t \in (L_i, R_i)]\frac{F(t) - F(L_i)}{F(R_i-) - F(L_i)}\right\}. \tag{15.49}$$

Again it turns out that the above equation does not have any closed-form solution and one has to resort to an iterative procedure. The authors also discuss a NPMLE for the parameter and show that the maximizer of the non-parametric likelihood function also becomes self-consistent. We skip the details for brevity.

15.6 Additive Risks Regression Model for Middle-Censored Data

Generally in survival studies, covariates are important to represent the differences in the population. As seen in the above subsections, the papers by [30], where the authors discuss a proportional hazard regression model for middle censored data, and [19], who analysed discrete middle censored data in the presence of covariates, discuss ideas in this regard. This work is by [31] which we will briefly discuss here. In the literature, a widely applied hazards model is that by [10]. It is a multiplicative hazards model which takes the form,

$$h(t|z) = h_0(t)\exp(z^\top\theta), \tag{15.50}$$

where T has a baseline hazard function $h_0(t)$, z is a $p \times 1$ vector of covariates, and $\theta = (\theta_1, \ldots, \theta_p)^\top$ is the vector of regression coefficients. However, the authors argue that this PH model does not fit lifetime data well and hence the additive risks model (AR) given by [1] proves to be beneficial, and takes the following form:

$$h(t|z) = h_0(t) + z^\top \theta. \tag{15.51}$$

The following paper discusses the estimation of the unknown baseline survival function $S_0(t)$ which is subject to middle censoring.

Now as seen before, let T be the lifetime variable, (U, V) be the censoring interval having a bivariate cdf given by $G(u, v) = P(U \leq u, V \leq v)$, and one observes the vector (X, δ, z) where δ is the usual indicator function. In the following subsections, we briefly outline an estimation technique which seems useful in this case.

15.6.1 Martingale Method of Estimation

This method consists of developing a partial likelihood function under the model (15.51), and a stochastic integral representation of the score function derived from this partial likelihood function is used to infer the unknown regression coefficient. Let the observed data be (X_i, z_i, δ_i), $1 \leq i \leq n$ which are n i.i.d. replicates of (X, z, p). In the presence of middle censoring, let the counting process corresponding to the ith individual be represented by $N_i(t) = I(X_i \leq t, \delta_i = 1), t \geq 0$, which indicates whether the event occurred at time t or not. One can define the at-risk process $R_i(t) = I(X_i \geq t, \delta_i = 1) + I(U_i \geq t, \delta_i = 0)$ where value 1 indicates whether the ith individual is at risk at time t or not. Further let $\sigma\{N_i(u), R_i(u+), z_i : i = 1, 2, \ldots, n; 0 \leq u \leq t\}$ be the filtration denoted by $\mathscr{F}_t$. Now under model (15.51), the conditional cumulative hazard rate for the ith individual is given by $H(t|z_i) = H_o(t) + z_i^\top \theta t$, where $H_0(t)$ is the baseline hazard function. Model (15.51) also assumes

$$E[N_i(t)|\mathscr{F}_{t-}] = (h_0(t) + \theta^\top z_i) R_i(t) dt, \tag{15.52}$$

and the intensity function corresponding to the counting process $N_i(t)$ can be written as

$$N_i(t) = M_i(t) + \int_0^t R_i(a) dH(a|z_i), \tag{15.53}$$

where $M(\cdot)$ is the local square integrable martingale. One may refer to [5] for more details. Now the derivative of the above equation can be written as

$$dN_i(t) = dM_i(t) + R_i(t) dH(t|z_i), \tag{15.54}$$

such that

$$\sum_{i=1}^{n} dM_i(t) = \sum_{i=1}^{n} [dN_i(t) - R_i(t)(dH_o(t) + \theta^\top z_i dt)] = 0. \tag{15.55}$$

To estimate θ, we will consider the partial likelihood function given by [10, 11], which is defined as

$$L(\theta) = \prod_{i=1}^{k} \frac{h(t_{(i)}|z_{(i)}}{\sum_{i=1}^{n} R_l(t_{(i)})h(t_{(i)}|z_l)}, \tag{15.56}$$

where $t_{(1)}, \ldots, t_{(k)}$ are the k observed exact lifetimes arranged in an increasing order. Now one can go along the usual lines of first obtaining the log-likelihood function, which in terms of the counting process can be defined as

$$C(\theta) = \sum_{i=1}^{n} \int_0^{\infty} \log\left(h_0(s) + z_i^{\top}\theta\right) dN_i(s) - \int_0^{\infty} \log\left(\sum_{l=1}^{n} R_l(s)(h_0(s) + z_i^{\top}\theta)\right) d\bar{N}(s), \tag{15.57}$$

where $\bar{N}(s) = \sum_{i=1}^{n} N_i(s)$. The score function thus becomes the derivative of the above function $C(\theta)$ and let it be denoted by $U(\theta)$. The estimate of the true regression coefficient θ_0 can thus be obtained by slightly simplifying $U(\theta)$, which is given by

$$U(\theta) = \sum_{i=1}^{n} \int_0^{\infty} \{z_i - \bar{z}\}\{dN_i(t) - R_i(t)z_i^{\top}\theta dt\}, \tag{15.58}$$

where $\bar{z} = \sum_{i=1}^{n} z_i R_i(t) / \sum_{i=1}^{n} R_i(t)$. Equation (15.58) assumes that $U(\theta_0)$ is a martingale integral with mean 0. It is also linear in θ, and one can get the estimator as

$$\hat{\theta} = \left[\sum_{i=1}^{n} \int_0^{\infty} [z_i - \bar{z}]^{\otimes 2} R_i(t) dt\right]^{-1} \sum_{i=1}^{n} \int_0^{\infty} [z_i - \bar{z}] dN_i(t), \tag{15.59}$$

where $a^{\otimes 2} = aa^{\top}$. Using this, one can further obtain the estimator of the cumulative hazard function $H_0(t)$ given by

$$\hat{H}_0(\hat{\theta}, t) = \int_0^t \frac{\sum_{i=1}^{n} \{dN_i(a) - R_i(a)z_i^{\top}\hat{\theta} da\}}{\sum_{i=1}^{n} R_i(a)} \tag{15.60}$$

and the estimator of the corresponding conditional survival function which is found as

$$\hat{S}(t|z) = \exp\{-\hat{H}_0(\hat{\theta}, t) - z^{\top}\hat{\theta}t\}. \tag{15.61}$$

The authors in their paper have further considered an iterative procedure to find the estimators of the underlying parameters, but we will leave out the details for brevity as it is quite similar to some of the earlier sections.

15.7 Conclusions

In this paper, a detailed review of estimation strategies under middle censored data is covered. Even though middle censoring is a relatively new censoring idea, which was proposed by [20], a lot of ground has been covered since then, both in terms of non-parametric and parametric distribution models. Some of the key distributions where middle censoring has been applied and well studied include the exponential, Weibull, Burr-XII and other related models. Among the discrete distributions, the geometric, its extensions and the multinomial have been covered. As mentioned in the introduction, the thesis by Nathan Bennett was an attempt to document some of the prominent papers in this area, prior to 2011. This paper focuses on other models, settings and extensions, which were majorly published recently. Some of these ideas include estimation under middle censoring in the presence of covariates, the additive risks regression model, estimation under middle censored competing risks data or parameter estimation for circular data. Due to the page limitation, this paper fails to include some important and interesting ideas pertaining to middle censoring. To mention a few, [18] proposed an approximation for the distribution function through which the estimator and its asymptotic properties are easily established, [3] considered the exponential and Weibull distributions to study the robustness of their parameter estimates under right and middle censoring and [22], where the authors established a semi-parametric regression model for analysis of middle censored lifetime data. Some of the ideas are also complemented through interesting applications, e.g., the "interval rating" problem under the multinomial distribution setup. It is truly believed that this paper will serve as a ready reference for future research in this area.

References

1. Aalen OO (1989) A linear regression model for the analysis of life times. Stat Med 8(8):907–925
2. Abuzaid AH (2015) The estimation of the burr-xii parameters with middle-censored data. Springerplus 4(1):1–10
3. Abuzaid AH, El-Qumsan MA, El-Habil A (2017) On the robustness of right and middle censoring schemes in parametric survival models. Commun Stat-Simul Comput 46(3):1771–1780
4. Ahmadi K, Rezaei M, Yousefzadeh F (2017) Statistical analysis of middle censored competing risks data with exponential distribution. J Stat Comput Simul 87(16):3082–3110
5. Andersen PK, Gill RD (1982) Cox's regression model for counting processes: a large sample study. Ann Stat 1100–1120
6. Bennett NA (2011) Some Contributions to Middle-Censoring. University of California, Santa Barbara
7. Burr IW (1942) Cumulative frequency functions. Ann Math Stat 13(2):215–232
8. Chen MH, Shao QM (1999) Monte Carlo estimation of Bayesian credible and HPD intervals. J Comput Graph Stat 8(1):69–92
9. Chib S, Greenberg E (1995) Understanding the metropolis-hastings algorithm. Am Stat 49(4):327–335
10. Cox DR (1972) Regression models and life-tables. J Roy Stat Soc: Ser B (Methodol) 34(2):187–202

11. Cox DR (1975) Partial likelihood. Biometrika 62(2):269–276
12. Davarzani N, Parsian A (2011) Statistical inference for discrete middle-censored data. J Stat Plan Inference 141(4):1455–1462
13. Davarzani N, Parsian A, Peeters R (2015) Statistical inference on middle-censored data in a dependent setup. J Stat Theory Pract 9(3):646–657
14. Efron B (1967) The two sample problem with censored data. In: Proceedings of the fifth Berkeley symposium on mathematical statistics and probability, vol 4, pp 831–853
15. Gupta RD, Kundu D (1998) Hybrid censoring schemes with exponential failure distribution. Commun Stat-Theory Methods 27(12):3065–3083
16. Iyer SK, Jammalamadaka SR, Kundu D (2008) Analysis of middle-censored data with exponential lifetime distributions. J Stat Plan Inference 138(11):3550–3560
17. Jammalamadaka SR, Bapat SR (2020) Middle censoring in the multinomial distribution with applications. Stat Probab Lett 167:108916
18. Jammalamadaka SR, Iyer SK (2004) Approximate self consistency for middle-censored data. J Stat Plan Inference 124(1):75–86
19. Jammalamadaka SR, Leong E (2015) Analysis of discrete lifetime data under middle-censoring and in the presence of covariates. J Appl Stat 42(4):905–913
20. Jammalamadaka SR, Mangalam V (2003) Nonparametric estimation for middle-censored data. J Nonparametric Stat 15(2):253–265
21. Jammalamadaka SR, Mangalam V (2009) A general censoring scheme for circular data. Stat Methodol 6(3):280–289
22. Jammalamadaka SR, Prasad SN, Sankaran PG (2016) A semi-parametric regression model for analysis of middle censored lifetime data. Statistica (Bologna) 76(1):27–40
23. Kundu D (2008) Bayesian inference and life testing plan for the Weibull distribution in presence of progressive censoring. Technometrics 50(2):144–154
24. Kundu D, Joarder A, Krishna H (2009) On Type-ll progressively hybrid censoring. J Mod Appl Stat Methods 8(2):18
25. Lagakos SW (1979) General right censoring and its impact on the analysis of survival data. Biometrics 139–156
26. Leung KM, Elashoff RM, Afifi AA (1997) Censoring issues in survival analysis. Annu Rev Public Health 18(1):83–104
27. Mangalam V, Nair GM, Zhao Y (2008) On computation of NPMLE for middle-censored data. Stat Probab Lett 78(12):1452–1458
28. Qian J, Betensky RA (2014) Assumptions regarding right censoring in the presence of left truncation. Stat Probab Lett 87:12–17
29. Rehman H, Chandra N (2021) Estimation of cumulative incidence function in the presence of middle censoring using improper Gompertz distribution. Statistica (Bologna) 81(2):163–182
30. Sankaran P, Prasad S (2014) Weibull regression model for analysis of middle-censored lifetime data. J Stat Manag Syst 17(5–6):433–443
31. Sankaran P, Prasad S (2017) An additive risks regression model for middle-censored lifetime data. Stat Transit New Ser 18(3):459–479
32. Shen Ps (2010) An inverse-probability-weighted approach to the estimation of distribution function with middle-censored data. J Stat Plan Inference 140(7):1844–1851
33. Shen PS (2011) The nonparametric maximum likelihood estimator for middle-censored data. J Stat Plan Inference 141(7):2494–2499
34. Tsai WY, Crowley J (1985) A large sample study of generalized maximum likelihood estimators from incomplete data via self-consistency. Ann Stat 1317–1334
35. Turnbull BW (1976) The empirical distribution function with arbitrarily grouped, censored and truncated data. J Roy Stat Soc: Ser B (Methodol) 38(3):290–295
36. Wang L (2016) Estimation for exponential distribution based on competing risk middle censored data. Commun Stat-Theory Methods 45(8):2378–2391
37. Yan W, Yimin S, Min W (2019) Statistical inference for dependence competing risks model under middle censoring. J Syst Eng Electron 30(1):209–222

Chapter 16
Matrix Partial Orders Based on the Secondary-Transpose

Umashankara Kelathaya, Manjunatha Prasad Karantha, and Ravindra B. Bapat

Mathematics Subject Classification (2010) 15A09 · 06A06

16.1 Introduction

The minus partial order and the star partial order on the class of rectangular matrices, and the sharp order on the class of square matrices are among important matrix relations. These relations are studied with the help of different classes of generalized inverses to a great extent. Therefore, these partial orders on the respective classes of matrices are called $\mathcal{G}$-based partial orders. Several authors in the recent past have defined new relations in the class of matrices, considering different types of generalized inverses and outer inverses. For more details, readers are referred to Drazin [5], Hartwig [7], Nambooripad [18], Hartwig and Drazin [8], Mitra [14], Baksalary and Mitra [1], Mitra [15], Mitra and Hartwig [16], Ben-Israel and Greville [4], Baksalary and Trenkler [2], Mitra et al. [17], Wang [23] and the references therein.

The Moore-Penrose inverse with reference to the involution of secondary-transpose (often called s-g inverse) is among a new class of generalized inverses, having several interesting properties. Given a rectangular matrix A of rank r, it has been noted in Eagambaram et al. [6] that a matrix can be written uniquely as a sum of disjoint rank one factors (subject to certain conditions) with reference to star partial

U. Kelathaya · M. P. Karantha (✉)
Department of Data Science, Prasanna School of Public Health, Manipal Academy of Higher Education, Manipal 576 104, Karnataka, India
e-mail: kmprasad63@gmail.com; km.prasad@manipal.edu

U. Kelathaya
e-mail: umashankar.kelathaya@learner.manipal.edu; usksullia@gmail.com

R. B. Bapat
Indian Statistical Institute, Delhi Center, 7, SJS Sansanwal Marg, New Delhi 110 016, India
e-mail: rbb@isid.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_16

order. This decomposition is closely related to singular values of matrix A and the associated eigenvectors of AA^T. Several interesting properties of star partial order studied in the past lead us to introduce a new partial order, in the present paper, based on the secondary-transpose (equivalently, with reference to s-g inverse) and probe its properties.

The secondary-transpose was initially introduced on the class of square matrices by Lee [11]. Following the definition, secondary-transpose on the class of real matrices has several interesting combinatorial properties. Further, more spectral properties associated with the secondary-transpose are studied by Krishnamoorthy and Vijayakumar and introduced the notion of s-g inverse (see Krishnamoorthy and Vijayakumar [9] and Krishnamoorthy and Vijayakumar [10]). However, a few examples which countered the results presented in this literature lead authors of Varkady et al. [20] to reconsider several research problems, and to obtain the results including an analogue to spectral decomposition. Noting that s-g inverse need not always exist for a given matrix, they provided necessary and sufficient conditions for the existence of s-g inverse. Observing that the notion of secondary-transpose could be extended to the class of rectangular matrices, in this article, we introduce new relations on the class of all real rectangular matrices and study their properties analogous to the case of star partial order.

16.2 Preliminaries

In this section, we introduce the notations used, definitions of some important generalized inverses and matrix partial orders. We write $\mathbb{R}^{m\times n}$ to denote the class of $m \times n$ matrices over a real field. Let $A \in \mathbb{R}^{m\times n}$. We denote by $\text{rank}(A)$, $\text{Trace}(A)$, A^T, $C(A)$ and $\mathcal{R}(A)$, the rank, trace, transpose, column space and row space of A, respectively. For matrices X and Y, we write $X \underset{SP}{\simeq} Y$ if $C(X) = C(Y)$ and $\mathcal{R}(X) = \mathcal{R}(Y)$. An $n \times m$ matrix G is called a *generalized inverse* of A if $AGA = A$. An arbitrary generalized inverse of A is denoted by A^-. An $n \times m$ matrix G satisfying $GAG = G$ is called an *outer inverse* of A. A generalized inverse of A which is also an outer inverse is called a *reflexive generalized inverse* of A. The unique $n \times m$ matrix G satisfying the matrix equations, often referred to as Penrose conditions, $AXA = A$, $XAX = X$, $(AX)^T = AX$ and $(XA)^T = XA$ is called the *Moore-Penrose inverse* of A and denoted by $A^\dagger$. For more details on generalized inverses, readers are referred to Rao and Mitra [19] and Ben-Israel and Greville [4].

We refer to Krishnamoorthy and Vijayakumar [9], Krishnamoorthy and Vijayakumar [10], Vijayakumar [21], Vijayakumar [22] and Varkady et al. [20] for the following definitions. Throughout our discussion, V denotes a square matrix of appropriate size (depending on the context), with 1's on the secondary diagonal and 0's elsewhere. Given an $m \times n$ matrix $A = (a_{ij})$ over real field, the *secondary-transpose* A^s of A is an $n \times m$ matrix with (i, j)-th entry given by

$$(A^s)_{ij} = a_{m-j+1,n-i+1}.$$

Evidently, $A^s = V_n A^T V_m$. A square matrix A is called *s-symmetric* if $A^s = A$. An $m \times n$ matrix A is said to be s-cancellable if $A^s AX = A^s AY \implies AX = AY$ and $XAA^s = YAA^s \implies XA = YA$. A matrix P is said to be an *idempotent (projector)* if $P^2 = P$. A projector P is called a *s-symmetric projector* if $P^s = P$. The class of all s-symmetric projectors with $C(P) = \mathcal{M}$ is denoted by $\mathcal{P}(\mathcal{M})$. Though we consider the matrices over a real field, the results discussed in this paper could be extended to the case of matrices over a complex field by replacing A^s with its analogue $A^\theta = V_n A^* V_m$.

Now, we define the s-g inverse of a matrix by considering the Penrose conditions with reference to the involution of secondary-transpose as given in the following.

Definition 16.1 For an $m \times n$ matrix A, consider the following matrix equations:

$$(1)\ AXA = A \quad (2)\ XAX = X \quad (3)\ (AX)^s = AX \quad (4)\ (XA)^s = XA.$$

An $n \times m$ matrix G, if it exists, satisfying all of the above equations is called the Moore-Penrose inverse of A with reference to the secondary-transpose (also called the s-g inverse of A) and denoted by $A^{\dagger_s}$.

It may be noted that the s-g inverse of A, whenever it exists, is the unique reflexive generalized inverse G of A such that $G \underset{SP}{\simeq} A^s$. Note that if $A^{\dagger_s}$ exists, then A is s-cancellable. It is proved in Varkady et al. [20, Theorem 3.1] that the converse is also true.

Theorem 16.1 (Varkady et al. [20]) *Given an $n \times n$ matrix A, $A^{\dagger_s}$ exists if and only if* $\text{rank}(A) = \text{rank}(AA^s) = \text{rank}(A^s A)$ *(equivalently A is s-cancellable). Further, in the case of* $\text{rank}(A) = 1$, *$A^{\dagger_s}$ exists if and only if* $\text{Trace}(A^s A) \neq 0$, *in which case,*

$$A^{\dagger_s} = \frac{1}{\text{Trace}(A^s A)} A^s. \tag{16.1}$$

Whenever $\text{rank}(A) = r > 1$, a determinental formula is given in Varkady et al. [20, Theorem 3.5, Eq. 3.5] to compute s-g inverse of A.

The minus partial order introduced by Hartwig [7] and Nambooripad [18] dominates most of the other partial orders defined on the rectangular matrices, and we provide the definition of the same in the following.

Definition 16.2 (*Minus partial order, Minus partial order, Hartwig* [7]*, Nambooripad* [18]) The relation '$\leq^-$' on the class of real rectangular matrices, defined by $B \leq^- A$ if $BG_B = AG_B$ and $G_B B = G_B A$ for some $G_B \in \{B^-\}$, is a partial order, and the same is called the minus partial order. In the above case, we say that B is less than A under minus partial order.

The star order was initially introduced and studied by Drazin [5] and Hartwig and Drazin [8], respectively, without referring to the Moore-Penrose inverse.

Definition 16.3 (*Star order, Star order, Drazin* [5]) The relation '$\leq^*$' on the class of real rectangular matrices, defined by $B \leq^* A$ if $BB^T = AB^T$ and $B^TB = B^TA$, is a partial order and the same is called the star partial order. In the above case, we say that B is less than A under star partial order.

It was also noted by Drazin [5] that $B \leq^* A \Leftrightarrow BB^\dagger = AB^\dagger$ and $B^\dagger B = B^\dagger A$, replacing '^{T}' by '†'. Therefore, the minus partial order dominates the star order (i.e., $B \leq^* A \implies B \leq^- A$).

For more details on matrix partial orders, we refer the readers to Mitra et al. [17] and Bapat [3]. The following theorem describes several equivalent conditions describing the properties of minus partial order. For the proof, readers are referred to Mitra [13].

Theorem 16.2 (Mitra [13]) *Given the rectangular matrices A, B and C such that $C = A - B$, the following statements are equivalent:*

(i) *B and $A - B$ are disjoint matrices, i.e., $C(B) \cap C(A - B) = \{0\}$ and $\mathcal{R}(B) \cap \mathcal{R}(A - B) = \{0\}$*
(ii) $\text{rank}(B) + \text{rank}(A - B) = \text{rank}(A)$ *(rank additivity)*
(iii) $B \leq^- A$
(iv) $\{A^-\} \subseteq \{B^-\}$
(v) $C(B) \oplus C(A - B) = C(A)$
(vi) $\mathcal{R}(B) \oplus \mathcal{R}(A - B) = \mathcal{R}(A)$.

Later in Eagambaram et al. [6], the star order has been studied with reference to column space decomposition and characterized all the matrices B such that $B \leq^* A$.

Motivated by the study of star order, in Sect. 16.4, we define a new relation called s-order on the class of real rectangular matrices, which is based on secondary-transpose, and a $\mathcal{G}$-based relation called $\dagger_s$-order using the s-g inverse. Though the definition of s-order is analogous to the definition of star order, unlike star order, s-order is not a partial order on the entire class of real rectangular matrices. So, we investigate how the new relation is different or the same as the star order. We identify a subclass of real matrices on which the relations s-order and $\dagger_s$-order are partial orders. Also, in Sect. 16.5, we characterize the column space of factors of given matrices in relation to the relations considered, as similar to the case of singular value decomposition related to star order. In the process, we obtain a necessary and sufficient condition for the existence of rank 1 factor with reference to $\dagger_s$-order for a given matrix. In Sect. 16.6, a new decomposition is studied, establishing a relationship between s-g inverse of a given matrix and s-g inverses of its components.

16.3 Projectors and Generalized Inverses

In this section, we discuss some interesting properties of s-symmetric projectors. It may be noted that every subspace S of $\mathbb{R}^n$ has an orthogonal projector onto itself. But the same is not true in the case of s-symmetric projector. For example, for $S =$

$\{(\alpha, 0, 0)^T \in \mathbb{R}^3 : \alpha \in \mathbb{R}\}$, the $\mathcal{P}(S) = \emptyset$. The reason is described in the following Lemma.

Lemma 16.1 *Given a subspace S of $\mathbb{R}^n$ and any matrix E such that $C(E) = \mathcal{M}$, we have the following:*

(i) If $P, Q \in \mathcal{P}(\mathcal{M})$, then $P = Q$ (s-symmetric projector onto a given space is unique, whenever it exists)
(ii) If $P \in \mathcal{P}(\mathcal{M})$, then $\mathcal{R}(P) = \mathcal{R}(E^s)$
(iii) There exists $P \in \mathcal{P}(\mathcal{M})$ if and only if $\text{rank}(E^s E) = \text{rank}(E)$, in which case, $P = E(E^s E)^- E^s$ for any arbitrary choice of $(E^s E)^-$.

Proof For $P, Q \in \mathcal{P}(\mathcal{M})$, it is trivial that $PQ = Q$ and $QP = P$ as we have $C(P) = C(Q) = \mathcal{M}$. Now, both P, Q are s-symmetric which implies that $P = QP = Q^s P^s = (PQ)^s = Q^s = Q$, proving the part (i) of the lemma.

From the definition of s-symmetric matrix, we have that $P = P^s = VP^T V$ and therefore $\mathcal{R}(P) = \mathcal{R}(P^T V) = C(VP)$. Since $C(E) = C(P)$, we have that $\mathcal{R}(P) = C(VE) = \mathcal{R}(E^T V) = \mathcal{R}(VE^T V) = \mathcal{R}(E^s)$. Thus, part (ii) of the lemma is proved.

It may be noted from Lemma 4.3 of Manjunatha Prasad [12] that given matrices A, X and Y, there exists an outer inverse G such that $C(G) = C(X)$ and $\mathcal{R}(G) = \mathcal{R}(Y)$ if and only if $\text{rank}(X) = \text{rank}(YAX) = \text{rank}(Y)$, in which case $G = X(YAX)^- Y$ for any choice of $(YAX)^-$. Therefore, the part (iii) of the lemma follows from the fact that $P \in \mathcal{P}(\mathcal{M})$ is an outer inverse of the identity matrix I (in other words, an idempotent matrix) with $C(P) = C(E)$ and $\mathcal{R}(P) = \mathcal{R}(E^s)$, as noted in the part (ii).

The following theorem characterizes the matrices having s-g inverse.

Theorem 16.3 *Given $m \times n$ real matrix A, $A^{\dagger_s}$ exists if and only if both $C(A)$ and $\mathcal{R}(A)$ have s-symmetric projectors.*

Proof If $A^{\dagger_s}$ exists, it is clear from its definition that $AA^{\dagger_s}$ and $(A^{\dagger_s} A)^T$ are the s-symmetric projectors onto $C(A)$ and $\mathcal{R}(A)$, respectively.

Conversely, if $C(A)$ has an s-symmetric projector onto itself, then referring to part (iii) of Lemma 16.1, we get that $\text{rank}(A^s A) = \text{rank}(A)$. On similar lines of argument, if $\mathcal{R}(A)(= C(A^T))$ has an s-symmetric projector then we get $\text{rank}(A) = \text{rank}(A^T) = \text{rank}(A^{T^s} A^T) = \text{rank}(AA^s)$. Now, the existence of $A^{\dagger_s}$ follows from Theorem 3.1 of Varkady et al. [20], given in Theorem 16.1.

16.4 The s-Order and $\dagger_s$-Order

We begin this section by defining s-order on the class of real rectangular matrices.

Definition 16.4 The relation '$\leq^s$' on the class of real rectangular matrices defined by $B \leq^s A$ if $B^s B = B^s A$ and $BB^s = AB^s$ is called s-order.

In case $B^{\dagger_s}$ exists, we have

$$\begin{aligned} B^sB = B^sA &\implies B^{\dagger_s}(B^{\dagger_s})^sB^sB = B^{\dagger_s}(B^{\dagger_s})^sB^sA \\ &\implies (B^{\dagger_s}BB^{\dagger_s})B = (B^{\dagger_s}BB^{\dagger_s})A \ \because\ BB^{\dagger_s} \text{ is } s\text{-symmetric} \\ &\implies B^{\dagger_s}B = B^{\dagger_s}A \\ &\implies (B^sB)B^{\dagger_s}B = (B^sB)B^{\dagger_s}A \\ &\implies B^sB = B^sA \ \because\ B^{\dagger_s} \underset{SP}{\simeq} B^s. \end{aligned}$$

So, we have $B^sB = B^sA$ if and only if $B^{\dagger_s}B = B^{\dagger_s}A$. On similar lines, we prove $BB^s = AB^s$ if and only if $BB^{\dagger_s} = AB^{\dagger_s}$, whenever $B^{\dagger_s}$ exists. Therefore,

$$B \leq^s A \Leftrightarrow B^{\dagger_s}B = B^{\dagger_s}A \ \text{ and } \ BB^{\dagger_s} = AB^{\dagger_s}, \tag{16.2}$$

for all matrices B for which $B^{\dagger_s}$ exists.

The above Eq. 16.2 leads us to define a $\mathcal{G}$-based relation $\leq^{\dagger_s}$ (called $\dagger_s$-order) on the class of rectangular matrices defined by $B \leq^{\dagger_s} A$ if $B^{\dagger_s}$ exists and

$$B^{\dagger_s}B = B^{\dagger_s}A \ \text{ and } \ BB^{\dagger_s} = AB^{\dagger_s}. \tag{16.3}$$

From the definition of $\leq^{\dagger_s}$ relation and from the discussion proving Eq. 16.2, we get that $B \leq^{\dagger_s} A \implies B \leq^s A$. So, $\leq^{\dagger_s}$ relation is dominated by $\leq^s$ relation.

For convenience, we denote the class of matrices having s-g inverse by $\mathbb{R}^{\dagger_s}$ and the class of all s-cancellable matrices by $\mathbb{R}^s$. From Theorem 16.1, we have that $\mathbb{R}^{\dagger_s} = \mathbb{R}^s$. Further, Eq. 16.2 implies that $\leq^s$ and $\leq^{\dagger_s}$ coincide in this class of $\mathbb{R}^s (= \mathbb{R}^{\dagger_s})$. In other words,

$$B \leq^s A \Leftrightarrow B \leq^{\dagger_s} A, \ \text{for all } B \in \mathbb{R}^s = \mathbb{R}^{\dagger_s}. \tag{16.4}$$

For subspaces $\mathcal{M}$ and $\mathcal{N}$, we denote by $\mathcal{M} \perp_s \mathcal{N}$ to indicate that $\mathcal{M}$ is s-orthogonal to $\mathcal{N}$ (for every $x \in \mathcal{M}$ and for every $y \in \mathcal{N}$, $y^TVx = 0$) and by $\mathcal{M} \dot{+}_s \mathcal{N}$ to denote $\mathcal{M} \oplus \mathcal{N}$ with $\mathcal{M} \perp_s \mathcal{N}$. It is important to note that $\mathcal{M} \perp_s \mathcal{N}$ need not imply $\mathcal{M} \cap \mathcal{N} = \{0\}$. The following lemma provides different characterizations for $B \leq^{\dagger_s} A$, which is easily verified.

Lemma 16.2 *For $A = B + C$ and $B \in \mathbb{R}^{\dagger_s}$, the following statements are equivalent:*

(i) $C(B) \dot{+}_s C(A - B) = C(A)$ *and* $\mathcal{R}(B) \dot{+}_s \mathcal{R}(A - B) = \mathcal{R}(A)$
(ii) $B^s(A - B) = 0$ *and* $(A - B)B^s = 0$
(iii) $B^sB = B^sA$ *and* $BB^s = AB^s$
(iv) $B \leq^{\dagger_s} A$
(v) $B^{\dagger_s}B = B^{\dagger_s}A$ *and* $BB^{\dagger_s} = AB^{\dagger_s}$.

It may be noted from the following example that $\leq^s$ is not a partial order on the class of rectangular matrices.

Example 1

For

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

it may be noted that $A \leq^s B$ and $B \leq^s A$ as

$$AA^s = AB^s = A^sA = A^sB = B^sB = BB^s = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

But $A \neq B$, showing that $\leq^s$ is not antisymmetric in the class of rectangular matrices. In fact, $\leq^s$ is neither a preorder in the class of real rectangular matrices. For example, for the matrices

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \text{ and } C = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix},$$

it may be noted that $A \leq^s B$ and $B \leq^s C$, but $A \not\leq^s C$, and therefore $\leq^s$ is not transitive on the entire class of rectangular matrices.

Theorem 16.4 *The relation '$\leq^s$' ($\leq^{\dagger_s}$) is a partial order on the class of all rectangular matrices having s-g inverse (class of all s-cancellable matrices).*

Proof We continue to prove the theorem with the observation that $X \leq^s Y \Leftrightarrow X \leq^{\dagger_s} Y$ for all $X, Y \in \mathbb{R}^{\dagger_s}$. For any $A \in \mathbb{R}^{\dagger_s}$, it is clear from the definition that $A \leq^s A$, and therefore $\leq^s$ is reflexive. Now, let $A, B \in \mathbb{R}^{\dagger_s}$ such that $A \leq^s B$ and $B \leq^s A$. Note that $B \leq^s A \implies BB^{\dagger_s} = AB^{\dagger_s} \implies B = BB^{\dagger_s}B = AB^{\dagger_s}B$, which implies that $C(B) \subset C(A)$. Considering $A \leq^s B \implies A^sB = A^sA$ and $AA^{\dagger_s} = (AA^{\dagger_s})^s = A^{\dagger_s{}^s}A^s$, we get $C(B) \subset C(A)$ implies that

$$A = A^{\dagger_s^s}A^sA = A^{\dagger_s^s}A^sB = AA^{\dagger_s}B = B.$$

Hence the antisymmetry of $\leq^s$.

To prove the transitivity, consider the matrices $A, B, C \in \mathbb{R}^{\dagger_s}$ such that $A \leq^s B$ and $B \leq^s C$. Now, by the definition of $\leq^s$, we have that $AA^{\dagger_s} = BA^{\dagger_s}$ and therefore, $BA^{\dagger_s}A = AA^{\dagger_s}A = A$. This proves that $C(A) \subset C(B)$ or equivalently, $\mathcal{R}(A^s) \subset \mathcal{R}(B^s)$. Since $B^{\dagger_s} \underset{SP}{\simeq} B^s, \mathcal{R}(A^s) \subset \mathcal{R}(B^s) \implies \mathcal{R}(A^{\dagger_s}) \subset \mathcal{R}(B^{\dagger_s})$, and, therefore,

$$A^{\dagger_s}BB^{\dagger_s} = A^{\dagger_s}. \tag{16.5}$$

On similar lines, by taking $A^{\dagger_s}A = A^{\dagger_s}B$, we obtain $C(A^{\dagger_s}) \subseteq C(B^{\dagger_s})$ and

$$B^{\dagger_s} B A^{\dagger_s} = A^{\dagger_s}. \tag{16.6}$$

Hence, we get

$$\begin{aligned} A^{\dagger_s} A &= A^{\dagger_s} B && \because\ A \leq^s B \\ &= A^{\dagger_s} (B^{\dagger_s})^s B^s B) && \because\ B = BB^{\dagger_s} B = (BB^{\dagger_s})^s B \\ &= A^{\dagger_s} B^{\dagger_s})^s B^s C = (A^{\dagger_s} BB^{\dagger_s}) C && \because\ B \leq^s C \text{ and } B^{\dagger_s})^s B^s = BB^{\dagger_s} \\ &= A^{\dagger_s} C, && \text{using Eq. 26.4.} \end{aligned}$$

On similar lines, using Eq. 16.6 and $AA^{\dagger_s} = BA^{\dagger_s}$, we get $AA^{\dagger_s} = CA^{\dagger_s}$. Thus, $A \leq^s C$ proving that $\leq^s$ is transitive. Therefore, $\leq^s$ is a partial order on $\mathbb{R}^{\dagger_s}$ $(= \mathbb{R}^s)$.

In view of Theorem 16.4, all the matrices involved in the inequalities with reference to $\leq^{\dagger_s}$, in the future discussion, are from the class $\mathbb{R}^s = \mathbb{R}^{\dagger_s}$, unless indicated otherwise.

We shall now discuss some of the properties of $\leq^s$ and $\leq^{\dagger_s}$ in the following theorem.

Theorem 16.5 *For any rectangular matrix A and $B \in \mathbb{R}^{\dagger_s}$, the following statements hold:*

(i) $B \leq^s A$ *implies* $B \leq^- A$ *(i.e., $\leq^-$ dominates $\leq^s$)*
(ii) $B \leq^s A$ *if and only if* $B^s \leq^s A^s$
(iii) For $A \in \mathbb{R}^{\dagger_s}$, $B \leq^s A$ *if and only if* $B^{\dagger_s} \leq^s A^{\dagger_s}$
(iv) For $A \in \mathbb{R}^{\dagger_s}$, $B \leq^s A$ *if and only if* $A^{\dagger_s}$ *is a* $\{1, 3, 4\}$*-inverse of* B*, in which case* $B^{\dagger_s} = A^{\dagger_s} BA^{\dagger_s}$
(v) $B \leq^s A$ *if and only if* $B^{\dagger_s}$ *is a* $\{2, 3, 4\}$*-inverse of* A*, in which case* $B = AB^{\dagger_s} A$
(vi) For $A^{\dagger_s}$ *exists,* $B \leq^s A$ *if and only if* $C^{\dagger_s}$ *exists and* $C = A - B \leq^s A$*, in which case* $C^{\dagger_s} = A^{\dagger_s} CA^{\dagger_s}$ *and* $A^{\dagger_s} = B^{\dagger_s} + C^{\dagger_s}$.

Proof The statement (i) follows form definition of minus partial order and Eq. 16.2, by noting that $B^{\dagger_s}$ is a particular generalized inverse of B. Statement (ii) follows from the definition of $\leq^s$ as given in Definition 16.4.

Statement (iii) follows from the fact that $B = (B^{\dagger_s})^{\dagger_s}$ and the equivalent definition of $\leq^s$ as given in Eq. 16.2.

For $B \leq^s A$, we have from (i) that $B \leq^- A$. Now, by (iii) $\Longrightarrow$ (iv) of Theorem 16.2 we have that $A^{\dagger_s}$ is a generalized inverse of B. Further from (iii), we have $B \leq^s A \Leftrightarrow B^{\dagger_s} \leq^s A^{\dagger_s}$ and hence $A^{\dagger_s} B = B^{\dagger_s} B$ and $BA^{\dagger_s} = BB^{\dagger_s}$, which are s-symmetric. Thus $A^{\dagger_s}$ is a $\{1, 3, 4\}$-inverse of B. Conversely, if $A^{\dagger_s}$ is a $\{1, 3, 4\}$-inverse of B, then $BA^{\dagger_s}$ and $(A^{\dagger_s} B)^T$ are s-symmetric projectors onto $C(B)$ and $\mathcal{R}(B)$, respectively. By the uniqueness of the s-symmetric projector onto given space, we must have $BA^{\dagger_s} = BB^{\dagger_s}$ and $A^{\dagger_s} B = B^{\dagger_s} B$ and therefore, $B \leq^s A$. Now, the second part of statement (iv) follows from the fact that $A^{\dagger_s}$ is a $\{1, 3, 4\}$-inverse of B.

From definition, $B \leq^s A \implies B^{\dagger_s} B = B^{\dagger_s} A$ and $BB^{\dagger_s} = AB^{\dagger_s}$ and therefore the s-symmetry of $AB^{\dagger_s}$ and $B^{\dagger_s} A$ are trivial. Further, by $B^{\dagger_s} B = B^{\dagger_s} A$, we get

$B^{\dagger_s}AB^{\dagger_s} = B^{\dagger_s}BB^{\dagger_s} = B^{\dagger_s}$ and hence $B^{\dagger_s}$ is an outer inverse of A, thus proving only part of statement (v). Now to prove the converse part, let $B^{\dagger_s}$ be a $\{2, 3, 4\}$-inverse of A. Then, $B^{\dagger_s}A$ and $(AB^{\dagger_s})^T$ are s-symmetric projectors onto the spaces $C(B^s)$ and $\mathcal{R}(B^s)$, respectively. By the uniqueness of the s-symmetric projector onto a given space, we get $B^{\dagger_s}A = B^{\dagger_s}B$ and $AB^{\dagger_s} = BB^{\dagger_s}$ and therefore $B \leq^s A$, which proves the if part of statement (v).

To prove (vi), firstly let A and B be matrices such that $A^{\dagger_s}$ exists and $B \leq^s A$. Using (iii) $\Longrightarrow$ (iv), (v) and (vi) of Theorem 16.2, we get $BA^-C = CA^-B = 0$ and $CA^-C = C$ for all A^-, so for $A^{\dagger_s}$. In other words, $A^{\dagger_s}$ is a generalized inverse of C such that

$$BA^{\dagger_s}C = CA^{\dagger_s}B = 0. \tag{16.7}$$

Since $A^{\dagger_s}$ is a $\{1, 3, 4\}$-inverse of B as proved in statement (iv), we get that $A^{\dagger_s}CA^{\dagger_s}$ is the s-g inverse of C, satisfying $CC^{\dagger_s} = AC^{\dagger_s}$ and $C^{\dagger_s}C = C^{\dagger_s}A$ for $C^{\dagger_s} = A^{\dagger_s}CA^{\dagger_s} = A^{\dagger_s} - B^{\dagger_s}$. This proves that $C \leq^s A$. The converse part is symmetric.

Remark 16.1 As noted in Theorem 16.5 (i), the s-order is dominated by minus partial order. However, the s-order neither dominates the star order nor it is dominated by the star order. This can be seen from the following examples.

Example 2

For

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 2 \\ 2 & 1 & -2 \end{bmatrix},$$

we have $\text{rank}(AA^s) = \text{rank}(A^sA) = \text{rank}(A)$ and B is invertible. So, by Theorem 16.1 it follows that $A, B \in \mathbb{R}^{\dagger_s}$. Further, we observe that

$$AA^T = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = BA^T \text{ and } A^TA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} = A^TB,$$

and hence $A \leq^* B$. But we observe that

$$A^sA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \neq \begin{bmatrix} 2 & 1 & -2 \\ -1 & 1 & 3 \\ 2 & 1 & -2 \end{bmatrix} = A^sB,$$

which shows that $A \not\leq^s B$.

Example 3

Now, consider the example of

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 2 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix},$$

and observe that $A, B \in \mathbb{R}^{\dagger_s}$ as $\text{rank}(AA^s) = \text{rank}(A^sA) = \text{rank}(A) = 2$ and $\text{rank}(BB^s) = \text{rank}(B^sB) = \text{rank}(B) = 3$. Further, we have

$$AA^s = \begin{bmatrix} 0 & 0 & 2 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = BA^s \text{ and } A^sA = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} = A^sB.$$

Therefore, $A \leq^s B$. But in this case

$$A^TA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 & 2 \\ 1 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix} = A^TB,$$

which shows that $A \not\leq^* B$.

Remark 16.2 It is well known that if $A \leq^* B$, then $C = A - B \leq^* A$. Though the same is true in the case of $\leq^s$ for matrices in the class $\mathbb{R}^{\dagger_s}$, it is not the case in general. It may be noted that Theorem 16.5 (vi) may not remain on hold if we relax the condition that $A^{\dagger_s}$ exists. This can be seen in the following example.

Example 4

Consider the matrix

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix},$$

for which $\text{rank}(A) = 2 \neq 1 = \text{rank}(A^sA) = \text{rank}(AA^s)$ and hence $A^{\dagger_s}$ does not exist. Now, for $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ we have $B \in \mathbb{R}^{\dagger_s}$ and $B \leq^s A$, as

$$BB^s = AB^s = B^sB = B^sA = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Further, we have $C = A - B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. Note that $C^{\dagger_s}$ does not exist as $C^s C = 0 \implies \text{rank}(C^s C) = 0 \neq 1 = \text{rank}(C)$.

Remark 16.3 Given a complex matrix A, the conjugate transpose of A denoted by A^* is well known in the literature. Similarly, the involution of conjugate secondary-transpose on the class of complex rectangular matrices is defined by $A^\theta = VA^*V$, for any complex matrix A. The θ-order on the class of all complex rectangular matrices is defined on similar lines of s-order by replacing 's' with 'θ' in the definition. The properties of θ-order are similar to those of s-order discussed in this section.

16.5 Factors with Reference to the Order Relations

In Eagambaram et al. [6], it is noted that for a given rectangular matrix A and a proper subspace $\mathcal{M}$ of $C(A)$, there always exists a matrix B with $C(B) = \mathcal{M}$ such that $B \leq^- A$. Such choices of matrix B are infinitely many. Further, the authors of the same paper have characterized the subspace $\mathcal{M}$ of $C(A)$ for which there exists matrix B with $C(B) = \mathcal{M}$ and $B \leq^* A$. Further, note that such a matrix B, whenever it exists, is unique. Motivated by this, in this section, we probe the problem of characterizing the matrices $B \in \mathbb{R}^{\dagger_s}$, if it exists, such that $B \leq^s A$ with $C(B) = \mathcal{M}$, where A is any arbitrary rectangular matrix, not necessary that $A \in \mathbb{R}^{\dagger_s}$. In general, such a B does not exist, which can be seen from Example 5 below.

Example 5

Consider the matrix $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ (invertible) and $\mathcal{M} = span\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$. Any matrix B with $C(B) = \mathcal{M}$ is of the form $B = \begin{bmatrix} 2a & 2b \\ a & b \end{bmatrix}$, where a and b are real numbers, not both are zeros. If $B \leq^s A$, in which case, $B^s B = \begin{bmatrix} 4ab & 4b^2 \\ 4a^2 & 4ab \end{bmatrix} = \begin{bmatrix} 0 & 3b \\ 0 & 3a \end{bmatrix} = B^s A$, then $a = 0$ and $b = \frac{3}{4}$. In this case $BB^s = 0 \neq AB^s$, a contradiction.

In the following theorem, we prove that the matrix $B \in \mathbb{R}^{\dagger_s}$ such that $B \leq^s A$ with $C(B) = \mathcal{M}$, if it exists, is uniquely determined like in the case of star order.

Theorem 16.6 *If $B, C \in \mathbb{R}^{\dagger_s}$ such that $B, C \leq^s A$ and $C(B) = C(C)$, then $B = C$.*

Proof Since $C(B) = C(C)$, from Lemma 16.1 the s-symmetric projectors onto $C(B)$ and $C(C)$ given by $BB^{\dagger_s}$ and $CC^{\dagger_s}$, respectively, are the same. Therefore,

$$\begin{aligned} B &= BB^{\dagger_s}B \\ &= (BB^{\dagger_s})A \quad \because\ B \leq^s A \\ &= CC^{\dagger_s}A \quad \because\ BB^{\dagger_s} = CC^{\dagger_s} \\ &= C(C^{\dagger_s}C) \quad \because\ C \leq^s A \\ &= C, \end{aligned}$$

proving the theorem.

The following theorem provides a few necessary and sufficient conditions for the existence of a matrix $B \leq^s A$ such that $C(B) = \mathcal{M} \subseteq C(A)$.

Theorem 16.7 *For any rectangular matrix A and $\mathcal{M} \subseteq C(A)$, consider the following statements:*

(i) There exists a matrix $B \in \mathbb{R}^{\dagger_s}$ such that $B \leq^s A$ with $C(B) = \mathcal{M}$
(ii) There exists $P \in \mathcal{P}(\mathcal{M})$ such that $PAA^s(I - P) = 0$
(iii) There exists $P \in \mathcal{P}(\mathcal{M})$ such that P commutes with AA^s
(iv) There exists $P \in \mathcal{P}(\mathcal{M})$ such that PAA^s is s-symmetric.

Then (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iii) $\Longrightarrow$ (iv) *holds. Further,* (iv) $\Longrightarrow$ (i) *holds if $A^{\dagger_s}$ exists.*

Proof To prove (i) $\Longrightarrow$ (ii), consider $B \in \mathbb{R}^{\dagger_s}$ as given in (i). Now, set $P = BB^{\dagger_s}$, the s-projector onto the space $\mathcal{M}$. By substitution, we get $PA = B(B^{\dagger_s}A) = BB^{\dagger_s}B = B$, as $B \leq^s A$. Now, $B \leq^s A \implies BB^s = BA^s \implies PAA^sP = BB^s = BA^s = PAA^s$ proving (ii).

(ii) $\Longrightarrow$ (iii) is trivial, by noting that PAA^sP is s-symmetric.

Now let $P \in \mathcal{P}(\mathcal{M})$ as given in (iii). Then $PAA^s = AA^sP = (PAA^s)^s$, proving (iv). Hence (iii) $\Longrightarrow$ (iv).

Now, let P be the s-projector onto the space $\mathcal{M}$ satisfying (iv), i.e., PAA^s is s-symmetric. Set $B = PA$. Observe that

$$C(B) \subseteq C(A) \quad \text{and} \quad \mathcal{R}(B) \subseteq \mathcal{R}(A), \tag{16.8}$$

as $C(B) = \mathcal{M} = C(P) \subset C(A)$. Further,

$$BB^s = PAA^sP = (PAA^s)^sP = AA^sPP = AA^sP = AB^s \tag{16.9}$$

and

$$B^sB = A^sPPA = A^sPA = A^sB = B^sA, \tag{16.10}$$

which show that $B \leq^s A$. Now, to complete the proof of (iv) $\Longrightarrow$ (i), we need to prove that $B \in \mathbb{R}^{\dagger_s}$, whenever $A \in \mathbb{R}^{\dagger_s}$. From Eq. 16.8, we get $C(B^s) \subseteq C(A^s)$ and using Eq. 16.9 we get that $A^{\dagger_s}BB^s = A^{\dagger_s}AB^s = B^s$ (because $A^{\dagger_s} \underset{SP}{\simeq} A^s$), proving that $\text{rank}(BB^s) = \text{rank}(B^s) = \text{rank}(B)$. Similarly, from Eqs. 16.8 and 16.10, we get that $\text{rank}(BB^s) = \text{rank}(B)$. Now, referring to Theorem 16.1, we get that $B \in \mathbb{R}^{\dagger_s}$. Hence, (iv) $\Longrightarrow$ (i).

Remark 16.4 If we drop the condition that $A^{\dagger_s}$ exists in (iv) of Theorem 16.7, then (iv) may not imply (i), as seen in the following example.

Example 6

Consider the matrix $A = \begin{bmatrix} 0\,0\,0\,0 \\ 1\,1\,0\,0 \\ 1\,0\,0\,0 \end{bmatrix}$ and $\mathcal{M} = span\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \subseteq C(A)$. The s-symmetric projector P onto $\mathcal{M}$ is given by $P = \begin{bmatrix} 0\,0\,0 \\ 0\,1\,0 \\ 0\,0\,0 \end{bmatrix}$. Then P commutes with AA^s. Further, we have that $B = PA = \begin{bmatrix} 0\,0\,0\,0 \\ 1\,1\,0\,0 \\ 0\,0\,0\,0 \end{bmatrix}$, for which it is easily verified that $B \leq^s A$. However, note that $B \notin \mathbb{R}^{\dagger_s}$ as $\text{rank}(B^sB) = 1 \neq 0 = \text{rank}(BB^s)$. Therefore, (iv) does not imply (i) for the matrix A and the space $\mathcal{M}$.

The following theorem characterizes the subspaces $\mathcal{M}$ of $C(A)$ for which there exists matrix B with $C(B) = \mathcal{M}$ and $B \leq^{\dagger_s} A$.

Theorem 16.8 *For any rectangular matrix A and $\mathcal{M} \subseteq C(A)$, the following statements are equivalent:*

(i) There exists a matrix B such that $B \leq^{\dagger_s} A$ with $C(B) = \mathcal{M}$
(ii) There exists $P \in \mathcal{P}(\mathcal{M})$ such that $PAA^s(I - P) = 0$ and $\text{rank}(AA^sP) = \text{rank}(PAA^s) = \text{rank}(P)$
(iii) $\mathcal{M}$ is an invariant subspace of AA^s having s-symmetric projector (i.e., $AA^s(\mathcal{M}) = \mathcal{M}$ and $\mathcal{P}(\mathcal{M}) \neq \emptyset$).

Further, in the above case matrix B is given by

$$B = E(E^sE)^-E^sA \tag{16.11}$$

for any choice of $(E^sE)^-$.

Proof For the matrix B satisfying (i), the existence of $P \in \mathcal{P}(\mathcal{M})$ such that $PAA^s(I - P) = 0$ is proved as in the case of (i) $\Longrightarrow$ (ii) of Theorem 16.7, in which case $P = BB^{\dagger_s}$. The second part of (ii) of the theorem follows from the fact that B has s-g inverse, in which case $\text{rank}(BB^s) = \text{rank}(B) = \text{rank}(P)$. Hence (i) $\Longrightarrow$ (ii).

If P is s-symmetric projector satisfying (ii) of the theorem, then for the matrix $B = PA$ in which case $C(B) = C(P)$, $B \leq^s A$ is proved as in the case of (iv) $\Longrightarrow$ (i) of Theorem 16.7. Since P is an s-symmetric projector onto $C(B) = C(P)$, following from Lemma 16.1 (iii) we get that $\text{rank}(B^sB) = \text{rank}(B)$. Now, $\text{rank}(BB^s) = \text{rank}(B)$ follows from the fact that $PAA^sP = AA^sP$ and $\text{rank}(AA^sP) = \text{rank}(P)$,

as given in (ii). Therefore, $B^{\dagger_s}$ exists and hence it follows from Eq. 16.4 that $B \leq^{\dagger_s} A$. Hence (ii) $\Longrightarrow$ (i).

Note that

$$PAA^sP = AA^sP = PAA^s \Leftrightarrow C(AA^sP) = C(P).$$

Further, the equality on the right-hand side holds if and only if $\text{rank}(AA^sP) = \text{rank}(P)$, thus proving (iii)$\Leftrightarrow$(ii).

Further, Eq. 16.11 follows from the expression for s-symmetric projector as given in Lemma 16.1 (iii).

Remark 16.5 If we relax the condition of $\text{rank}(AA^sP) = \text{rank}(PAA^s) = \text{rank}(P)$ in (ii) of Theorem 16.8, then (ii) $\Longrightarrow$ (i) may not remain on hold. To observe this, consider the following example.

Example 7

For matrix $A = \begin{bmatrix} 0&0&0&1&0 \\ 0&0&1&0&0 \\ 1&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&1&0 \end{bmatrix}$ and $\mathcal{M} = span\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \subseteq C(A)$, we have that $P = \begin{bmatrix} \frac{1}{2}&0&0&0&\frac{1}{2} \\ 0&0&0&0&0 \\ 0&0&1&0&0 \\ 0&0&0&0&0 \\ \frac{1}{2}&0&0&0&\frac{1}{2} \end{bmatrix} \in \mathcal{P}(\mathcal{M})$. Observe that $\text{rank}(AA^sP) = \text{rank}(PAA^s) = 0 \neq 2 = \text{rank}(P)$. Now, for $B = PA = \begin{bmatrix} 0&0&0&1&0 \\ 0&0&0&0&0 \\ 1&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&1&0 \end{bmatrix}$, it is easily verified that $B \leq^s A$. However, note that $B \notin \mathbb{R}^{\dagger_s}$, as $\text{rank}(BB^S) = 0 \neq 2 = \text{rank}(B^SB)$. Therefore, for the matrix A and space $\mathcal{M}$, (ii) $\Longrightarrow$ (i) of Theorem 16.8 does not hold.

The following theorem characterizes the rank 1 factors under $\leq^{\dagger_s}$ of a given matrix and the proof follows as a corollary to (i)$\Leftrightarrow$(iii) of Theorem 16.8.

Theorem 16.9 *For any rectangular matrix A and a vector $0 \neq x \in C(A)$, there exists a matrix $B \in \mathbb{R}^{\dagger_s}$ such that $B \leq^{\dagger_s} A$ and $C(B) = span\{x\}$ if and only if x is an eigenvector of AA^s and $span\{x\}$ has s-symmetric projector.*

Remark 16.6 Observe that for a given eigenvector x of AA^s, the existence of a matrix $B \leq^{\dagger_s} A$ (or $B \leq^s A$) such that $C(B) = span\{x\}$ is not guaranteed. This can be seen in Example 8 below.

Example 8

Consider the matrix $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, for which $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is an eigenvector of $AA^s = I$. Any matrix B such that $B \leq^{\dagger_s} A$ with $C(B) = span\{x\}$ is of the form $B = \begin{bmatrix} a & b & c \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, not all a, b, c are zeros. Clearly, $B \leq^{\dagger_s} A$ is not possible, unless a, b, c are zeros, which is a contradiction. Similarly for $B \leq^s A$, in which case, $B^s B = B^s A$ and $BB^s = AB^s$ implies that $a, b, c = 0$, a contradiction to the fact that $C(B) = span\{x\}$.

Remark 16.7 For a given rectangular matrix A, Theorem 16.9 does not assure the existence of any eigenvector x of AA^s for which $span\{x\}$ has s-symmetric projector onto itself. To observe this, we consider the following example.

Example 9

Consider the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, for which $AA^s = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. The eigenvalues of AA^s are 0 with multiplicity one and 1 with multiplicity two. The eigenspace corresponding to the eigenvalue 1 has dimension 1 and is spanned by the vector $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ for which $span\{x\}$ does not have s-symmetric projector, and hence there is no matrix B of rank 1 such that $B \leq^{\dagger_s} A$. Further, there is no matrix B of rank 1 with $C(B) = span\{x\}$ such that $B \leq^s$, as seen in the earlier example.

16.6 A New Decomposition

The spectral decomposition of a symmetric matrix is very useful in computing Moore-Penrose inverse, but the analogous decomposition given in Theorem 2.1 of Varkady et al. [20] for an s-symmetric matrix is not so promising in computing s-g inverse, as seen in the following example.

Example 10

For matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, the decomposition $A = (1)\begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix} + (-1)\begin{bmatrix} 0 & -\frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$ as given in Theorem 2.1 of Varkady et al. [20] is in the form $\lambda_1 A_1 + \lambda_2 A_2$, where λ_1 and λ_2 are s-eigenvalues of A and A_1, A_2 are s-idempotent. Further, A_1 and A_2 have s-g inverses, as $\text{rank}(A_i^s A_i) = \text{rank}(A_i A_i^s) = \text{rank}(A_i) = 1$, for $i = 1, 2$. By using the expression for s-g inverse of rank 1 matrix as given in Theorem 16.1, we get $A_1^{\dagger S} = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ and $A_2^{\dagger S} = \begin{bmatrix} 0 & -2 & 2 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$, but A has no s-g inverse. In fact, the matrix $\lambda_1^{-1} A_1^{\dagger S} + \lambda_2^{-1} A_2^{\dagger S} = (1)A_1^{\dagger S} + (-1)A_2^{\dagger S}$ is not even a generalized inverse of A, as we observe that

$$A(A_1^{\dagger S} + (-1)A_2^{\dagger S})A = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq A.$$

This observation leads us to explore, in this section, a useful decomposition which could relate s-g inverse of a given matrix with s-g inverses of its components.

Theorem 16.10 *If $A_1, A_2, \ldots, A_k$ are any matrices such that*

$$A = \sum_{i=1}^{k} A_i, \quad \textit{and} \quad A_i {A_j}^s = 0, {A_i}^s A_j = 0 \ \textit{for } i \neq j,$$

then the following statements are equivalent:

(i) $A^{\dagger_s}$ *exists and* $\sum_{i=1}^{k} \text{rank}(A_i) = \text{rank}(A)$
(ii) ${A_i}^{\dagger_s}$ *exists for all* $i = 1, 2, \ldots, k$.

In the above case, we have

$$A^{\dagger_s} = \sum_{i=1}^{k} {A_i}^{\dagger_s} \quad \textit{and} \quad {A_i}^{\dagger_s} = A^{\dagger_s} A_i A^{\dagger_s}. \tag{16.12}$$

Proof Let $A = \sum_{i=1}^{k} A_i$ such that $A_i {A_j}^s = 0$ and ${A_i}^s A_j = 0$ for $i \neq j$ as given in the theorem and satisfy (i). Now to prove (i) $\Longrightarrow$ (ii), we shall prove that $\text{rank}(A_i^s A_i) = \text{rank}(A_i A_i^s) = \text{rank}(A_i)$, for each i.

Since $A^{\dagger_s}$ exists, we have that $\text{rank}(AA^s) = \text{rank}(A^s A) = \text{rank}(A)$. Therefore,

$$\begin{aligned}\operatorname{rank}(A) = \operatorname{rank}(AA^s) &= \operatorname{rank}\Big(\sum_{i=1}^{k} A_i A_i^s\Big) \quad (\because A_i A_j^s = 0)\\ &\le \sum_{i=1}^{k} \operatorname{rank}(A_i A_i^s)\\ &\le \sum_{i=1}^{k} \operatorname{rank}(A_i) = \operatorname{rank}(A), \quad \text{from rank additivity given in (i).}\end{aligned}$$

Therefore, we get

$$\sum_{i=1}^{k} \operatorname{rank}(A_i A_i^s) = \sum_{i=1}^{k} \operatorname{rank}(A_i), \tag{16.13}$$

which in turn implies that $\operatorname{rank}(A_i A_i^s) = \operatorname{rank}(A_i)$. Similarly, we prove that $\operatorname{rank}(A_i^s A_i) = \operatorname{rank}(A_i)$. Thus, $A_i^{\dagger_s}$ exists for each i and hence (i) $\Rightarrow$ (ii).

Now let ${A_i}^{\dagger_s}$ exist for each i, satisfying (ii). Equation 16.12 is easily verified by substitution and using the fact that $A_i^s \underset{SP}{\simeq} {A_i}^{\dagger_s}$ and given that $A_i A_j^s = 0$ and ${A_i}^s A_j = 0$ for $i \neq j$. Thus, $A^{\dagger_s}$ exists and it is a generalized inverse of A_i satisfying $A_i A^{\dagger_s} A_j = 0$, for each i and $j \neq i$.

Now, for $i = 1$ we have that $G = A^{\dagger_s} A_1 A^{\dagger_s} \in \{{A_1}^-\}$ such that $A_1 G = AG$ and $GA_1 = GA$. Referring to definition of $\le^-$, we have that $A_1 \le^- A$ and then

$$\operatorname{rank}(A) = \operatorname{rank}(A - A_1) + \operatorname{rank}(A_1)$$

follows from (ii)$\Leftrightarrow$(iii) of Theorem 16.2. Now, noting that $A - A_1 = A_2 + A_3 + \cdots + A_k$ such that $A_i^s A_j = 0$ and $A_i A_j^s = 0$ for $i, j \neq 1, i \neq j$, (ii) $\Longrightarrow$ (i) follows by induction.

Remark 16.8 It may be noted that (ii) of Theorem 16.10 does not hold, if we drop the condition of rank additivity in (i), which can be seen from the following example.

Example 11

Consider the matrix

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

which is an s-symmetric matrix and satisfies the condition $\operatorname{rank}(A^s A) = \operatorname{rank}(AA^s) = \operatorname{rank}(A^2) = 2 = \operatorname{rank}(A)$. Therefore, $A^{\dagger_s}$ exists. Now, consider a decomposition

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = A_1 + A_2(\text{ say}),$$

which satisfies the condition given in Theorem 16.10, but not satisfying the rank additivity property given in (i). It may be noted that (ii) does not hold as we observe that ${A_2}^{\dagger_s}$ does not exists.

The matrix decomposition $A = A_1 + A_2$ as given in the above example to demonstrate that $A_1 \leq^s A$ and $A^{\dagger_s}$ exists need not imply $A_1^{\dagger_s}$ exists.

The following example demonstrates Theorem 16.10.

Example 12

Consider

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

with a decomposition

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} = A_1 + A_2 \text{ (say)},$$

satisfying the condition $0 = {A_1}^s A_2 = A_1 {A_2}^s$, given in Theorem 16.10.

We also observe that $\text{rank}(A) = 3 = \text{rank}(A^s A) = \text{rank}(A A^s)$. Therefore, $A^{\dagger_s}$ exists. Further, we have

$$3 = \text{rank}(A) = \text{rank}(A_1) + \text{rank}(A_2) = (2 + 1),$$

and hence the decomposition satisfies (i) of the theorem. In fact, using the expression given in Eq. 3.5 of Varkady et al. [20, Theorem 3.5] for the s-g inverse, we get that

$$A^{\dagger_s} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

It is verified easily by substitutions that the matrices ${A_1}^s A_1$ and $A_1 {A_1}^s$ are of rank two, and the matrices ${A_2}^s A_2$ and $A_2 {A_2}^s$ are of rank one. Therefore, ${A_1}^{\dagger_s}$ and ${A_2}^{\dagger_s}$ exist. In fact, ${A_1}^{\dagger_s} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ and ${A_2}^{\dagger_s} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. Thus, (ii) of Theorem 16.10 is satisfied. Further, we observe that ${A}^{\dagger_s} = {A_1}^{\dagger_s} + {A_2}^{\dagger_s}$.

Acknowledgements Among the authors, Manjunatha Prasad Karantha acknowledges funding support by Science and Engineering Research Board (DST, Govt. of India) under MATRICS (MTR/2018/000156) and CRG (CRG/2019/000238) schemes, and Umashankara Kelathaya wishes to acknowledge funding support by Science and Engineering Research Board (DST, Govt. of India) under CRG (CRG/2019/000238).

References

1. Baksalary JK, Mitra SK (1991) Left-star and right-star partial orderings. Linear Algebra Appl 149:73–89
2. Baksalary OM, Trenkler G (2010) Core inverse of matrices. Linear Multilinear Algebra 58(6):681–697
3. Bapat RB (2012) Linear algebra and linear models. Springer Science & Business Media
4. Ben-Israel A, Greville TN (2003) Generalized inverses: theory and applications, vol 15. Springer Science & Business Media
5. Drazin MP (1978) Natural structures on semigroups with involution. Bull Am Math Soc 84(1):139–141
6. Eagambaram N, Manjunatha Prasad K, Mohana KS (2013) Column space decomposition and partial order on matrices. Electron J Linear Algebra 26:795–815
7. Hartwig RE (1980) How to partially order regular elements. Math Japon 25(1):1–13
8. Hartwig RE, Drazin MP (1982) Lattice properties of the star order for complex matrices. J Math Anal Appl 86(2):359–378
9. Krishnamoorthy S, Vijayakumar R (2009) Some characteristics on s-normal matrices. Int J Comput Appl Math 4(1):41–46
10. Krishnamoorthy S, Vijayakumar R (2009) Some equivalent conditions on s-normal matrices. Int J Contemp Math Sci 4(29):1449–1454
11. Lee A (1976) Secondary symmetric, skewsymmetric and orthogonal matrices. Period Math Hung 7(1):63–70
12. Manjunatha Prasad K (2012) An introduction to generalized inverse. In: Bapat RB, Kirkland S, Prasad KM, Puntanen S (eds) Lectures on matrix and graph methods. Manipal University Press, Manipal, India, chap 5, pp 43–60
13. Mitra SK (1986) The minus partial order and the shorted matrix. Linear Algebra Appl 83:1–27
14. Mitra SK (1987) On group inverses and the sharp order. Linear Algebra Appl 92:17–37
15. Mitra SK (1991) Matrix partial orders through generalized inverses: unified theory. Linear Algebra Appl 148:237–263
16. Mitra SK, Hartwig RE (1992) Partial orders based on outer inverses. Linear Algebra Appl 176:3–20
17. Mitra SK, Bhimasankaram P, Malik SB (2010) Matrix partial orders, shorted operators and applications, vol 10. World Scientific

18. Nambooripad KS (1980) The natural partial order on a regular semigroup. Proc Edinb Math Soc 23(3):249–260
19. Rao CR, Mitra SK (1971) Generalized inverse of matrices and its applications. Wiley, New York
20. Varkady S, Purushothama DS, Kelathaya U, Bapat RB (2022) Secondary transpose of matrix and generalized inverses. J Algebra Its Appl
21. Vijayakumar R (2010) Contributions to the study on s-normal matrices. PhD thesis, Bharathidasan University, Tiruchirappalli
22. Vijayakumar R (2016) s-g inverse of s-normal matrices. Int J Math Trends Technol 4:240–244
23. Wang H (2016) Core-EP decomposition and its applications. Linear Algebra Appl 508:289–300

Chapter 17
On Products of Graph Matrices



G. Sudhakara, Vinay Madhusudanan, and K. Arathi Bhat

Mathematics Subject Classification (2010) 05C50 · 05C76

17.1 Introduction

All graphs considered in this article are assumed to be simple and undirected. Let G be a graph with vertex set $V(G) = \{v_1, v_2, \ldots, v_n\}$ and edge set $E(G) = \{e_1, e_2, \ldots, e_m\}$. Then n is the order and m is the size of G. Two vertices v_i and v_j are adjacent in G if there is an edge between them, and this is denoted by $v_i \sim_G v_j$. If v_i and v_j are non-adjacent in G, we write $v_i \nsim_G v_j$. The complete graph on n vertices, denoted by K_n, is the graph in which every two vertices are adjacent, and its complement $\overline{K_n}$ is called the totally disconnected graph. If G and H are two graphs, then $G \cup H$ is the graph with $V(G \cup H) = V(G) \cup V(H)$ and $E(G \cup H) = E(G) \cup E(H)$. A cycle graph on n vertices, denoted by C_n, is a connected 2-regular graph. An isolated vertex is a vertex with degree zero, and a full degree vertex is a vertex that is adjacent to all other vertices. A tree on n vertices is a connected acyclic graph. An end vertex or a pendant vertex is a vertex of degree 1. A path graph is a tree with exactly two end vertices.

G. Sudhakara (✉)
Department of Mathematics, Poornaprajna College & Post Graduate Centre, Udupi, Karnataka 576101, India
e-mail: sudhakaragadiga@gmail.com

V. Madhusudanan · K. Arathi Bhat
Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India
e-mail: arathi.bhat@manipal.edu

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_17

Let G be a graph with $V(G) = \{v_1, v_2, \ldots, v_n\}$ and $E(G) = \{e_1, e_2, \ldots, e_m\}$. The adjacency matrix $A(G)$ of G is the $n \times n$ matrix whose rows and columns are indexed by the vertex set of G and whose (i, j)-entry, a_{ij}, is given by

$$a_{ij} = \begin{cases} 0 & \text{if } v_i \not\sim_G v_j \\ 1 & \text{if } v_i \sim_G v_j. \end{cases}$$

Thus, $A(G)$ is a symmetric $(0, 1)$-matrix whose diagonal entries are all equal to zero.

The incidence matrix of G, denoted by $B(G)$, is the $n \times m$ matrix whose rows are indexed by the vertex set and columns are indexed by the edge set of G. The (i, j)-element of $B(G)$ is given by

$$b_{ij} = \begin{cases} 0 & \text{if } e_j \text{ is not incident with the vertex } v_i \\ 1 & \text{if } e_j \text{ is incident with the vertex } v_i. \end{cases}$$

The $n \times m$ matrix in which every entry is 1 is denoted by $J_{n\times m}$. A permutation matrix of order n is an $n \times n$ matrix which contains exactly one 1 in each row and one 1 in each column.

17.1.1 Product of Graphs (Graph Product)

A graph product is a binary operation defined on the class of graphs. It is useful in constructing a new graph from given graphs.

Let G_1 and G_2 be two graphs with disjoint vertex sets $\{u_1, u_2, \ldots, u_m\}$ and $\{v_1, v_2, \ldots, v_n\}$, respectively. A graph product of G_1 and G_2 is a new graph whose vertex set is $V(G_1) \times V(G_2)$, the Cartesian product of $V(G_1)$ and $V(G_2)$. The adjacency of two distinct vertices (u_i, v_j) and (u_r, v_s) in the product graph is determined entirely by the adjacency/equality/non-adjacency of u_i and u_r in G_1 and that of v_j and v_s in G_2. Thus, one can define $256 = 2^8$ different types of graph products.

Section 17.1 introduces the basic definitions and concepts. In Sect. 17.2, results on matrix product of graphs are discussed. These results are taken from the article [6]. Section 17.3 deals with the realization of $A(G)A(H)$ where the multiplication is taken modulo 2. Results of this section are taken from the article [7]. Some equations involving matrices related to graphs are explored in Sect. 17.4. Results of this section have appeared in the article [4]. Discussion about the product of adjacency matrices of a graph and its generalized complement is given in Sect. 17.5, and results of this section have appeared in the article [3]. In Sect. 17.6, an algorithm which tells whether a given graph G has a companion or not is given, and the algorithm has appeared in the article [5].

Readers are referred to [9] for all the elementary notations and definitions not defined but used in this chapter.

17.2 Matrix Product of Graphs

A graph and a matrix representing the graph uniquely (up to isomorphism) can be treated as identical objects in the sense that the properties of one object can be studied using the properties of the other. It is so much so in the case of a graph and its adjacency matrix that the eigenvalues of a graph are nothing but the eigenvalues of its adjacency matrix. So, when graph products (graph operations) are defined to obtain a new graph, it is natural to ask what the adjacency matrix of the resulting graph is, in terms of the adjacency matrices of the factors. For a complete picture of the eigenvalues of the product graph in terms of the eigenvalues of its factors, one may refer to the beautiful survey article [2]. In this section, we address almost the reverse of this problem. Instead of asking for the adjacency matrix of the product, we ask for the graph (whenever one exists) representing the product of adjacency matrices of given graphs.

Since the adjacency matrix of a graph is a symmetric $(0, 1)$-matrix with zero diagonal, the following definition is natural.

Definition 17.1 (*Graphical Matrix,* [1]) A symmetric $(0, 1)$-matrix is said to be *graphical* if all its diagonal entries are equal to zero.

Note that a graphical matrix is the adjacency matrix of a graph. When B is graphical and is equal to $A(G)$, the adjacency matrix of some graph G, then we say that G is the realization of the graphical matrix B. We are interested in studying the cases where the product of adjacency matrices is graphical. In fact, we are interested in answering the following question: Given a graph G, when is it true that there exists a graph H such that $A(G)A(H)$ is graphical? We call such a graph H a *companion* of G.

Let G be a graph on n vertices and H be the graph $\overline{K_n}$. Since $A(\overline{K_n}) = 0_{n\times n}$, the zero matrix of order $n \times n$, $A(G)A(H) = 0_{n\times n}$ and the realizing graph is $\overline{K_n}$. Hence, $\overline{K_n}$ is a companion of every graph of order n. Also, an isolated vertex in either G or H results in an isolated vertex in the graph realizing $A(G)A(H)$, when $A(G)A(H)$ is graphical. So, for further discussion in this chapter, we consider graphs G and H which are neither totally disconnected nor contain any isolated vertex.

In characterizing the pairs of graphs for which the matrix product exists, we will make use of the concept of a G-H path, defined below.

Definition 17.2 (*G-HPath*) Given graphs G and H on the same set of vertices $\{v_1, v_2, \ldots, v_n\}$, two vertices v_i and v_j are said to have a *G-H path* between them if there exists a vertex v_k (necessarily different from v_i and v_j) such that $v_i \sim_G v_k$ and $v_k \sim_H v_j$ (see Fig. 17.1).

The name of a graph occurring beside an edge in the following as well in later figures does not represent the label of the edge, but instead represents the graph to which the edge belongs.

Note that a G-H path between two vertices v_i and v_j, whenever one exists, is a path $v_i v_k v_j$ of length two in $G \cup H$. The same G-H path also represents an H-G path $v_j v_k v_i$ between v_j and v_i through v_k.

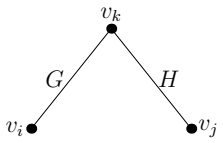


Fig. 17.1 G-H Path

Theorem 17.1 *Let G and H be graphs on the same set of vertices $\{v_1, v_2, \ldots, v_n\}$, with adjacency matrices $A(G)$ and $A(H)$, respectively. Then $C = A(G)A(H)$ is graphical if and only if all of the following hold:*

(i) G and H are edge disjoint.
(ii) For every two vertices v_i and v_j, there exists at most one G-H path from the vertex v_i to the vertex v_j.
(iii) For every two vertices v_i and v_j, if there exists a G-H path from the vertex v_i to the vertex v_j, then there exists an H-G path from v_i to v_j.

Proof Suppose that $C = A(G)A(H)$ is graphical. Since the diagonal entries of C are all zero, the ith row of $A(G)$ and the ith row of $A(H)$ are orthogonal for every i, $1 \leq i \leq n$. That is, G and H are edge disjoint. Note that the (i, j)-entry in C is nothing but the number of G-H paths from v_i to v_j. Since the (i, j)-entry of C is either 0 or 1, there exists at most one G-H path between v_i and v_j. Finally, since C is symmetric, whenever there is a G-H path between v_i and v_j, there is also an H-G path between v_i and v_j. The converse is verified easily using matrixbreak multiplication. □

The above result leads to the following definition.

Definition 17.3 (*Matrix Product of Graphs*) Given graphs G, H, and Γ on the same set of vertices, the graph Γ is said to be the *matrix product* of graphs G and H if $A(G)A(H)$ is graphical and $A(\Gamma) = A(G)A(H)$. Then we say that G and H are the graph factors of the matrix product Γ.

Remark 17.1 Let G and H be graphs on the same set of vertices and Γ be the matrix product of G and H. For any two vertices v_i and v_j, the G-H and H-G paths from v_i to v_j, when they exist, are not through the same vertex. Also note that the vertices v_i and v_j are adjacent in Γ if and only if $G \cup H$ has the following structure given in Fig. 17.2.

We shall now prove that the product graph Γ of G and H always has an even number of edges. To do this, we need the following definition.

Definition 17.4 Given two disjoint edges $e = (x, y)$ and $f = (u, v)$ in a matrix product Γ of the graphs G and H, f is said to be *parallel* to e if xuy is a G-H path (H-G path) and xvy is an H-G path (G-H path) between the vertices x and y, as shown in Fig. 17.3.

Theorem 17.2 *Given an edge $e = (x, y)$ in the matrix product Γ of the graphs G and H, the following statements are true:*

(i) There exists an edge $f = (u, v)$ in Γ, disjoint from e, such that f is parallel to e.
(ii) If f is parallel to e then e is parallel to f.
(iii) The edge f parallel to e is unique.

Proof (i) Let $e = (x, y)$ be an edge in Γ. Since Γ is a matrix product of G and H, there exist vertices u and v in Γ such that

$$x \sim_G u \sim_H y, \quad \text{and} \quad x \sim_H v \sim_G y. \tag{17.1}$$

But this implies that

$$u \sim_G x \sim_H v, \quad \text{and} \quad u \sim_H y \sim_G v. \tag{17.2}$$

In other words, $(u, v) = f$ is an edge in Γ and, from the definition of the parallel edge and from Eq. (17.1), f is parallel to e.

(ii) From the equivalence of Eqs. (17.1) and (17.2), we see that f is parallel to e if and only if e is parallel to f.

(iii) Suppose that $f = (u, v)$ and $f' = (u', v')$ are two edges parallel to e in Γ. Then $x \sim_G u \sim_H y$, $x \sim_H v \sim_G y$ and $x \sim_G u' \sim_H y$, $x \sim_H v' \sim_G y$ (see Fig. 17.4).

Hence, we get two G-H paths xuy and $xu'y$ between x and y and similarly, we get two H-G paths xvy and $xv'y$ between x and y. This contradicts (ii) of Theorem 17.1. This proves the uniqueness of the parallel edge f. □

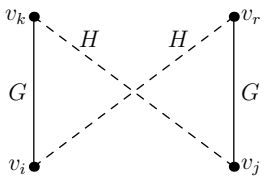


Fig. 17.2 Adjacency of v_i and v_j in Γ

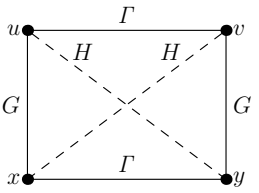


Fig. 17.3 Parallel edges in Γ

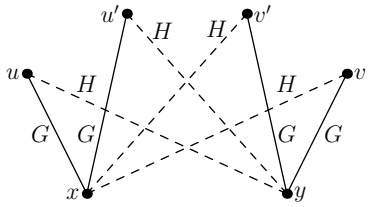


Fig. 17.4 Two G-H and H-G paths between x and y

Theorem 17.3 *If* Γ *is the matrix product of graphs* G *and* H*, then* Γ *has an even number of edges.*

Proof From Theorem 17.2, it is clear that every edge (x, y) in Γ has a unique parallel edge and the relation 'is parallel to' is symmetric. The relation is not an equivalence relation, but it partitions the set of edges of Γ and each part consists of exactly two edges that are parallel to each other. Hence, the number of edges in Γ is even. □

Let G, H, and Γ be graphs defined on the same set of vertices, and let Γ be the matrix product of G and H. Then it is natural to expect a formula for the degree of any vertex in Γ in terms of its degree in G and a degree in H. We address this in the next theorem.

First we prove the following lemma.

Lemma 17.1 *Given graphs* G*,* H*, and* Γ *on the same set of vertices such that* Γ *is a matrix product of* G *and* H*, for each vertex* v_i *of* Γ*,*

$$\deg_\Gamma v_i = \sum_{v_j \sim_H v_i} \deg_G v_j. \tag{17.3}$$

Proof Recall that every edge in the matrix product Γ corresponds to an H-G path. Thus, the number of edges of an incident with v_i is equal to the number of H-G paths of the form $v_i \sim_H v_j \sim_G v_k$, for some vertices v_j and v_k of Γ. The result follows. □

Note 17.1 Since the adjacency matrix of Γ is symmetric, $A(G)$ and $A(H)$ commute and therefore

$$\deg_\Gamma v_i = \sum_{v_j \sim_G v_i} \deg_H v_j. \tag{17.4}$$

Note 17.2 Let Γ be the matrix product of G and H. If there is a path between the vertices v_i and v_j in G, then $\deg_H v_i = \deg_H v_j$.

Theorem 17.4 *Let* G*,* H*, and* Γ *be graphs such that* Γ *is the matrix product of* G *and* H*. Then for each vertex* v_i *of* Γ*,*

$$\deg_\Gamma v_i = \deg_G v_i \cdot \deg_H v_i. \tag{17.5}$$

Proof From Lemma 17.1, we have $\deg_\Gamma v_i = \sum_{v_j \sim_H v_i} \deg_G v_j$. In the summation on the right-hand side, we consider all j such that $v_i \sim_H v_j$. Therefore, from Note 17.2, we get

$$\deg_G v_i = \deg_G v_j$$

for all j with $v_i \sim_H v_j$. Thus,

$$\begin{aligned}\deg_\Gamma v_i &= \sum_{v_i \sim_H v_j} \deg_G v_j \\ &= (\deg_G v_i)\left(\sum_{v_i \sim_H v_j} 1\right) \\ &= \deg_G v_i \cdot \deg_H v_i.\end{aligned}$$

□

Note 17.3 Let the graphs G and H be defined on the same set of vertices $\{v_1, v_2, \ldots, v_n\}$, and let Γ be the matrix product of G and H. If G and H are regular graphs with regularities r and s respectively, then Γ is regular with regularity rs.

The following is a corollary to Lemma 17.1.

Corollary 17.1 *Let Γ be the matrix product of the graphs G and H. If one of the graphs G and H is connected, then the other graph is regular.*

Proof First, we shall prove that $\deg_H v_1 = \deg_H v_2$ for every pair of vertices v_1 and v_2 adjacent in G, i.e., $v_1 \sim_G v_2$. For this purpose, we show that there is a one-to-one correspondence between H-adjacencies of v_1 and H-adjacencies of v_2. The case $\deg_H v_1 = \deg_H v_2 = 1$ is trivial.

To consider the non-trivial case, let $\deg_H v_2 \geq 2$. Consider any two vertices v_3 and v_4 that are adjacent to v_2 in H. Since $v_1 \sim_G v_2 \sim_H v_3$ and $v_1 \sim_G v_2 \sim_H v_4$ are G-H paths and Γ is the matrix product of G and H, we see that the vertices v_3 and v_4 are adjacent to v_1 in Γ.

Now $v_1 \sim_\Gamma v_3$ implies that, as in Remark 17.1, there exists an H-G path $v_1 \sim_H v_5 \sim_G v_3$ through a vertex $v_5 \neq v_2$. Similarly, $v_1 \sim_\Gamma v_4$ implies that there exists an H-G path $v_1 \sim_H v_6 \sim_G v_4$ through a vertex $v_6 \neq v_2$. Hence, the vertices v_5 and v_6 adjacent to v_1 in H respectively correspond to the vertices v_3 and v_4 adjacent to v_2 in H. Now, we shall prove that $v_5 \neq v_6$ in order to establish the one-to-one correspondence between adjacencies of v_1 and v_2 in H.

Suppose that $v_5 = v_6$. Then, v_5 is adjacent to v_3 and v_4 in G, and both v_3 and v_4 are adjacent to v_2 in H. Then there are two G-H paths $v_5 \sim_G v_3 \sim_H v_2$ and $v_5 \sim_G v_4 \sim_H v_2$ from v_5 to v_2 (see Fig. 17.5), which contradicts (ii) of Theorem 17.1.

This proves that $\deg_H v_1 = \deg_H v_2$, whenever v_1 and v_2 are adjacent in G. Thus, if G is connected, there exists a path between every pair of vertices and the degrees of all vertices of H are identical. In other words, H is regular. □

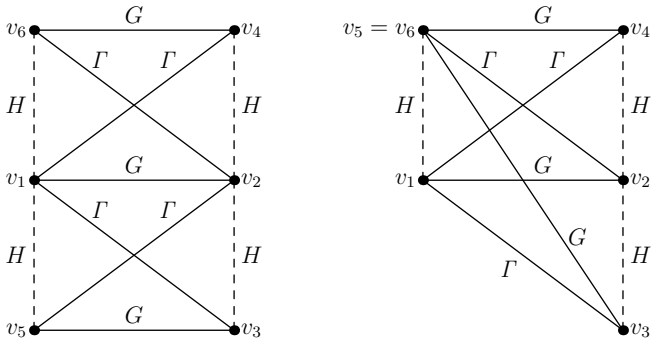


Fig. 17.5 G-H paths between v_5 and v_2

Note 17.4 From the proof of Corollary 17.1, the following is evident. If the matrix product of G and H exists, then vertices v_i and v_j are adjacent in one of the graphs G and H only if they are of the same degree in the other.

Note 17.5 Let Γ be the matrix product of G and H. Suppose that one of G and H, say G, has a full degree vertex v. Then G is connected, and by Corollary 17.1, H is regular. We also know that H is a subgraph of $\overline{G}$ and $\overline{G}$ has v as a vertex of degree zero. Hence, all the vertices of H are of degree zero, i.e., H is trivial, and so is Γ.

Now we define the concept of degree partition of a graph, which will be useful in establishing some further results regarding vertex degrees in the matrix product.

Definition 17.5 (*Degree Partition of a Graph*) For a graph G on the set of vertices $\{v_1, v_2, \ldots, v_n\}$, the *degree partition* of the graph G is a partition $\{V_0, V_1, \ldots, V_k\}$ of the vertex set $V(G)$ with $V_j = \{ v_i \mid \deg_G v_i = j \}$, $0 \le j \le k$ and $k = \Delta(G) = \max_{1 \le i \le n} \{\deg_G v_i\}$.

Note 17.6 Let Γ be the matrix product of G and H. From the definition of the degree partition of graph G and by Note 17.2, it is clear that every component of H lies entirely in one part of the degree partition of G.

Corollary 17.2 *Let G, H, and Γ be graphs such that Γ is the matrix product of G and H. Then the following hold:*

(i) If any two of G, H, and Γ are regular, then so is the third.
(ii) Γ is regular if and only if $\deg_G v_i \cdot \deg_H v_i$ is constant for all i, $1 \le i \le n$.
(iii) A connected graph Γ is Eulerian if and only if the degree of each vertex is even in at least one of the graphs G and H.

Corollary 17.3 *If Γ is the matrix product of G and H, where G or H is a regular graph, then Γ cannot be a tree.*

Proof Suppose that Γ is a tree. Then Γ has a pendant vertex, say v. By Theorem 17.4, $\deg_G v \cdot \deg_H v = \deg_\Gamma v = 1$, which implies that $\deg_G v = \deg_H v = 1$. Without loss of generality, let H be regular. Then H is 1-regular and hence contains an even number of vertices. Thus, Γ is a tree on an even number of vertices and hence contains an odd number of edges. This contradicts the fact that Γ has an even number of edges (see Theorem 17.3). □

Theorem 17.5 *A tree has no non-trivial companion.*

Proof Let G be a tree on n vertices. We prove the result by induction on n. The result is obvious when $n = 1$ or 2. Now, suppose that the result holds for all trees of order less than $n > 2$, and let G be a tree of order n.

Let P be the set of all pendant vertices of G, and note that $|P| \geq 2$. Let $G' = G \backslash P$ be the tree obtained by deleting all pendant vertices of G. Suppose, for the sake of contradiction, that G has a non-trivial companion H. As G is connected, H is regular, and hence has no isolated vertices. Therefore, if u is a pendant vertex of G, then $u \sim_H v$ for some vertex v, and by Note 17.4, v is also a pendant vertex of G. Now, let x and y be the vertices adjacent to u and v, respectively, in G. Then $x \sim_G u \sim_H v$ is a G-H path from x to v, and hence there exists a G-H path from v to x, which is necessarily $v \sim_G y \sim_H x$, since y is the unique vertex adjacent to v in G. Thus, $x \sim_H y$, which shows that H' is non-trivial (noting that x and y cannot be pendant vertices of G). The preceding argument also shows that if u is a pendant vertex of G, then for each edge (u, y) in the matrix product of G and H, the unique parallel edge of (u, y) also has a pendant vertex of G as one of its end vertices.

To see that H' is a companion of G', we verify that the three conditions given in Theorem 17.1 hold for G' and H'. Conditions (i) and (ii) obviously hold by virtue of the fact that G' and H' are subgraphs, respectively, of G and H, and H is a companion of G. For (iii), let s and t be any two vertices of G', and suppose that there exists a G'-H' path $s \sim_{G'} p \sim_{H'} t$ from s to t. We need to show that there exists a G'-H' path from t to s. Note that s, t, and p are non-pendant vertices of G, and hence $s \sim_G p \sim_H t$ is a G-H path from s to t. Since H is a companion of G, (iii) of Theorem 17.1 implies that there exists a G-H path $t \sim_G q \sim_H s$ from t to s. Then (s, q) and (t, p) are parallel edges in the matrix product of G and H. As observed earlier, any edge of the matrix product that has a pendant vertex of G as one of its end vertices must be parallel to another edge with the same property. Since s, t, and p are non-pendant in G, it follows that q is also not a pendant vertex of G, and hence is present in G'. Hence, $t \sim_{G'} q \sim_{H'} s$ is a G'-H' path, as required.

Thus, H' is a non-trivial companion of G', a tree of order less than n, which contradicts the induction hypothesis. □

Note 17.7 Let G, H, and Γ be graphs on the same set of vertices, and let Γ be the matrix product of G and H. Let m_1, m_2, and m_3 denote the number of edges in G, H, and Γ, respectively. Then, by Theorem 17.4,

$$\sum \deg_\Gamma v_i = \sum \deg_G v_i . \deg_H v_i .$$

Let G be a connected graph and r be the regularity of H. Then the above equation becomes $\sum \deg_\Gamma v_i = \sum r \deg_G v_i$. That is, $2m_3 = r(2m_1)$ or $m_3 = rm_1$.

But, by Theorem 17.3, Γ has an even number of edges, which implies that either r or m_1 is even.

Note 17.8 Let G, H, and Γ be graphs on the same set of vertices, and let Γ be the matrix product of G and H. Further, let G be connected and H be non-trivial. Let $\{V_1, V_2, \ldots, V_{\Delta(G)}\}$ be the degree partition of G. Then H is regular and the following hold:

(i) $|V_i| \neq 1$ for any i, $1 \leq i \leq \Delta(G)$.
(ii) If $|V_i| = 2$ for some i, $1 \leq i \leq \Delta(G)$, then H is a 1-factor graph and the number of vertices is even.
(iii) if $|V_i| = 3$, $1 \leq i \leq \Delta(G)$, then H is a union of cycles.
(iv) If $|V_i|$ is odd for some i, $1 \leq i \leq \Delta(G)$, then the regularity of H is even.

By Theorem 17.5, the path graph P_n has no non-trivial companion. Given a connected, non-complete graph G, the graph power G^k has more edges than G. We know that any companion of G is a spanning subgraph of the complement of G. Since $\overline{G^k}$ has fewer edges, it is natural to expect that G^k has less chance of having a non-trivial companion. We strengthen this by proving that P_n^m has no non-trivial companion. The formal definition of the mth power of a path graph P_n is given below.

Definition 17.6 The mth *power* of a path graph P_n, denoted by P_n^m, is a graph that has the same set of vertices as P_n, and in which two vertices are adjacent whenever the distance between them in P_n is at most m.

Since the diameter of P_n is $(n-1)$, for every $m \geq n-1$, $P_n^m = K_n$, the complete graph on n vertices. Since, either when $G = K_n$, or when $G = P_n$, there is no graph H such that $A(G)A(H)$ is realizable, we consider P_n^m, $2 \leq m < n-1$, in the following discussion.

Consider P_n with vertex set $\{v_1, v_2, \ldots, v_n\}$, where $v_i \sim v_{i+1}$ for $1 \leq i \leq n-1$. We note that when n is even and $n > 2m$, and when n is odd and $n > 2m+1$, there is no full degree vertex in P_n^m. If P_n^m has a full degree vertex, then P_n^m has no non-trivial companion. So, when n is even and $n > 2m$, and when n is odd and $n > 2m+1$, we observe the following.

In P_n^m, v_i and v_{n+1-i} are the only vertices with degree $m+i-1$, $1 \leq i \leq m$. All the other vertices are of degree $2m$. Also note that any $m+1$ consecutive vertices in P_n of degree at least m form a complete graph K_{m+1} in P_n^m. To prove that P_n^m, $2 \leq m < n-1$, and $n \geq 1$, has no companion, we make use of the following lemma.

Lemma 17.2 *Let G be a graph and H be a subgraph of $\overline{G}$ such that $A(G)A(H)$ is graphical. Then, for any two vertices u and v with $u \sim_H v$, the following property holds.*

For any k, corresponding to every vertex in $N_G(u)$ having degree k in G, there is a unique vertex in $N_G(v)$ with degree k in G and vice versa. In other words, the sequence of degrees in G of vertices in $N_G(u)$ is the same as the sequence of degrees in G of vertices in $N_G(v)$, when both of them are written in non-increasing order.

Proof Let $A(G)A(H) = A(\Gamma)$. Since $u \sim_H v$, by Note 17.4, $\deg_G(u) = \deg_G(v)$. Let $x \in N_G(u)$ be of degree k in G. Then, $x \sim_G u \sim_H v$ is a G-H path. Since $A(G)A(H) = A(\Gamma)$, there is a G-H path from v to x through a vertex, say y, i.e., $v \sim_G y \sim_H x$. As $y \sim_H x$, by Note 17.4, $\deg_G(y) = \deg_G(x) = k$.

Because of the uniqueness of the G-H path from v to x, it follows that y is the only such vertex in G with degree k in $N_G(v)$. Hence, for every vertex x in G with $\deg_G(x) = k$, there is a unique vertex y in G such that $\deg_G(y) = k$. The other side can be obtained similarly. □

Theorem 17.6 *For positive integers n and m, $2 \leq m < n-1$, there exists no non-trivial subgraph H of $\overline{P_n^m}$, such that $A(P_n^m)A(H)$ is graphical.*

Proof Let $G = P_n^m$. Suppose that there exists a non-trivial subgraph H of $\overline{G}$, such that $A(G)A(H)$ is graphical. Since G is connected, H is regular. Also, v_1 and v_n are the only vertices of degree m in G, and H is non-trivial, which together imply that $v_1 \sim_H v_n$ and H is 1-regular. Then $v_2 \sim_H v_{n-1}$, $v_3 \sim_H v_{n-2}, \ldots, v_m \sim_H v_{n-m+1}$, since each pair contains only two vertices of that particular degree in G. The remaining $n - 2m$ vertices in G are of degree $2m$. Since the vertices v_{m+1} and v_{n-m} are the only vertices of degree $2m$ which have the same neighboring degree sequence, by Lemma 17.2, v_{m+1} and v_{n-m} are adjacent in H. Similarly, the vertices v_{m+2} and v_{n-m-1} are adjacent in H. Continuing in this manner, we reach a stage where the vertices x and y of degree $2m$ in G are adjacent in H, where the distance between x and y in G is less than or equal to m. This contradicts the fact that when the distance between x and y in G is less than or equal to m, $x \sim_G y$ and H is a subgraph of $\overline{G}$. Hence, there is no non-trivial graph H such that $A(G)A(H)$ is graphical when $n \geq 1$ and $2 \leq m < n-1$. □

We know that a companion H of G is a subgraph of $\overline{G}$. It is therefore natural to ask what happens when $H = \overline{G}$. The next part of this section deals with this question. Note that for any G, either G or $\overline{G}$ is connected. Hence, if $A(G)A(\overline{G})$ is graphical then both G and $\overline{G}$ are regular. Therefore, it is enough to characterize regular graphs G for which $A(G)A(\overline{G})$ is graphical.

Theorem 17.7 *If G is a regular graph on the set of vertices $\{v_1, v_2, \ldots, v_n\}$, with regularity 1 or $n-2$, then the product $A(G)A(\overline{G})$ is graphical. In fact, if G is 1-regular, then*

$$A(G)A(\overline{G}) = A(\overline{G})A(G) = A(\overline{G})$$

and if G is $(n-2)$-regular, then

$$A(G)A(\overline{G}) = A(\overline{G})A(G) = A(G).$$

Proof We shall prove the result in the case where G is $(n-2)$-regular. The case of G being 1-regular follows by symmetry as $\overline{G}$ is then $(n-2)$-regular. Note that in these cases, n is necessarily greater than three.

Consider any two vertices v_1 and v_2 in G and two distinct cases based on whether or not v_1 and v_2 are adjacent in G.

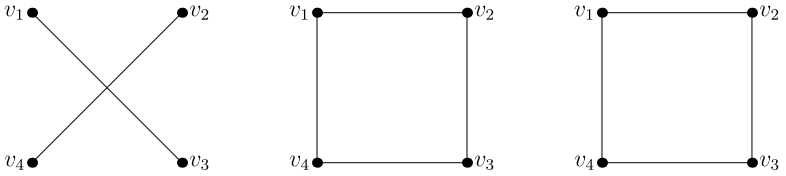


Fig. 17.6 $G = \overline{C_4}$ a 1-regular graph

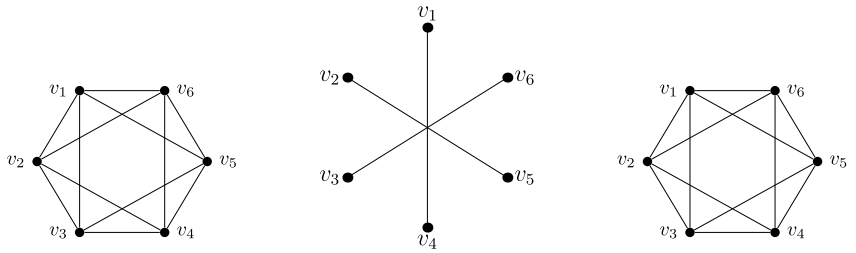


Fig. 17.7 G a cocktail party graph on six vertices, a (6-2)-regular graph

Case I. v_1 and v_2 are adjacent in G: Since G is $(n-2)$-regular, there exists a unique vertex u such that v_2 and u are adjacent in $\overline{G}$, but u is adjacent to v_1 in G. Hence, we get a G-$\overline{G}$ path $v_1 \sim_G u \sim_{\overline{G}} v_2$ between v_1 and v_2. Again by $(n-2)$-regularity of G, there is no vertex w different from u such that v_2 and w are adjacent in $\overline{G}$. Hence, the uniqueness of G-$\overline{G}$ path from v_1 to v_2. Similarly, we get a unique $\overline{G}$-G path from v_1 to v_2.

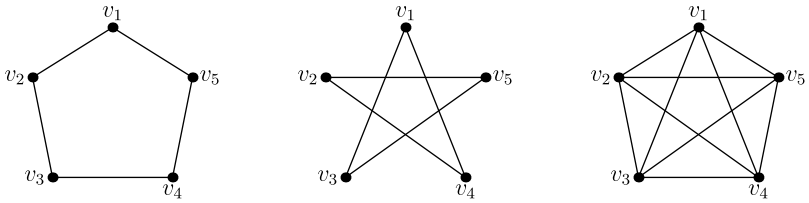
Case II. v_1 and v_2 are not adjacent in G: Since v_1 and v_2 are not adjacent in G, they are adjacent in $\overline{G}$ and not adjacent to any other vertex in $\overline{G}$. Therefore, there is no G-$\overline{G}$ or $\overline{G}$-G path from v_1 to v_2.

From Case I and Case II above, it is clear that for every pair of vertices there exists at most one pair of G-$\overline{G}$ and $\overline{G}$-G paths, and from Theorem 17.1, it follows that $A(G)A(\overline{G})$ is graphical. Again referring to the cases, we observe that a pair of vertices being adjacent (respectively not adjacent) in G implies that the vertices are adjacent (respectively not adjacent) in the matrix product Γ of G and $\overline{G}$. Therefore, $\Gamma = G$ and $A(G)A(\overline{G}) = A(G) = A(\overline{G})A(G)$. □

Figures 17.6 and 17.7 illustrate the cases $n = 4, 6$.

Remark 17.2 In Theorem 17.7, we observed that $A(G)A(\overline{G}) = A(G)$ whenever G is $(n-2)$-regular. In other words, $A(\overline{G})$, when $\overline{G}$ is a 1-factor, acts like a projector on the column space as well as the row space of $A(G)$. But it is not true that $A(\overline{G})$ is a projector.

The following theorem characterizes the regular graphs G for which $A(G)A(\overline{G})$ is graphical.



(a) G (b) H (c) Γ

Fig. 17.8 $G = C_5$, $H = \overline{C_5} \cong C_5$, $\Gamma = K_5$

Theorem 17.8 *Given a regular graph G, if the non-trivial product $A(G)A(\overline{G})$ is graphical then exactly one of the following holds:*

(i) G is 1-regular
(ii) G is $(n-2)$-regular (i.e., it is a cocktail party graph on n vertices)
(iii) $G = C_5$.

Proof Consider a regular graph G with regularity k such that $A(G)A(\overline{G})$ is graphical. Clearly, $\overline{G}$ is also regular with regularity $n-k-1$. From Note 17.3, the graph Γ realizing $A(G)A(\overline{G})$ is also regular with regularity $k(n-k-1)$. Therefore, $k(n-k-1) \leq n-1$ and we get

$$n \leq \frac{k}{k-1} + (k+1) \tag{17.6}$$

for $k \neq 1$.

In the case of $k = 1$, i.e., G is 1-regular and among the class of graphs described in (i), we know from Theorem 17.7 that $A(G)A(\overline{G})$ is graphical.

If $k \neq 1$, note that $1 \leq \frac{k}{k-1} \leq 2$ and by substituting the upper limit in (17.6) we get $n \leq k+3$ or $k \geq n-3$. That is,

$$k = n-1, \quad n-2, \quad \text{or} \quad n-3.$$

If $k = n-1$, G is the complete graph and $\overline{G}$ is the null graph, which is a trivial case.

If $k = n-2$, then the graph G is $(n-2)$-regular and among the class of graphs described in (ii) and hence, by Theorem 17.7, $A(G)A(\overline{G})$ is graphical.

If $k = n-3$ and $k \neq 1$, then the regularity of $\overline{G}$ is 2 and therefore $(n-3)2 \leq n-1$, which implies that $n = 5$. That is, $G = C_5$.

Note that for the graph $G = C_5$, $\overline{G} = \overline{C_5} \cong C_5$ and in $G \cup \overline{G}$, it is easy to observe that, between every two vertices in $\{v_1, v_2, v_3, v_4, v_5\}$ there exists a unique G-$\overline{G}$ path and a unique $\overline{G}$-G path. Thus, $A(G)A(\overline{G})$ is graphical and the matrix product Γ is the complete graph K_5 (see Fig. 17.8). This concludes the proof. □

Remark 17.3 Whenever H is a 1-factor graph and Γ is the matrix product of G and H, then it is also true that G is the matrix product of Γ and H. That is $A(\Gamma)A(H) = A(G)$. This follows from Theorem 17.7, and the fact that $A(H)$ is a self-inverse permutation matrix.

17.3 Realization of $A(G)A(H)$ (Modulo 2)

In this section, we discuss the realization of the matrix product under modulo 2 matrix multiplication.

Definition 17.7 (*Modulo* 2 *Matrix Product of Graphs*) The graph Γ is said to be the *modulo* 2 *matrix product* of graphs G and H if $A(G)A(H) \pmod 2$ is graphical and Γ is the realization of $A(G)A(H) \pmod 2$.

Note 17.9 As in the case of the usual matrix product of graphs, in the case of modulo 2 product also, we consider only graphs that are not totally disconnected and have no isolated vertices.

Note 17.10 Definition of G-H path between two vertices v_i and v_j $(i \neq j)$ remains the same as in the case of the general product.

Definition 17.8 (*Parity Regular Graph*) A graph is said to be *parity regular* if the degrees of its vertices are either all even or all odd.

Note 17.11 In the case of the usual product of the adjacency matrices of graphs G and H, a necessary and sufficient condition for realizability of $A(G)A(H)$ is that H must be a subgraph of $\overline{G}$ and for each ordered pair of distinct vertices, the number of G-H paths and the number of H-G paths are the same and are equal to 1 or 0. When the matrix product is taken modulo 2, this condition is no longer necessary. In fact, the class of graphs H for which $A(G)A(H) \pmod 2$ is graphical, is found to be a larger class. In Example 17.3, we have a graph H such that $A(G)A(H) \pmod 2$ is graphical but H is not a subgraph of $\overline{G}$.

Example 1

In the following graphs G and H (as shown in Fig. 17.9) on seven vertices, H is not a subgraph of $\overline{G}$. Note that the adjacency matrices of G and H are

$$A(G) = \begin{bmatrix} 0&0&1&0&0&1&0 \\ 0&0&0&1&0&0&1 \\ 1&0&0&0&1&0&0 \\ 0&1&0&0&0&1&0 \\ 0&0&1&0&0&0&1 \\ 1&0&0&1&0&0&0 \\ 0&1&0&0&1&0&0 \end{bmatrix} \text{ and } A(H) = \begin{bmatrix} 0&0&1&1&1&1&0 \\ 0&0&0&1&1&1&1 \\ 1&0&0&0&1&1&1 \\ 1&1&0&0&0&1&1 \\ 1&1&1&0&0&0&1 \\ 1&1&1&1&0&0&0 \\ 0&1&1&1&1&0&0 \end{bmatrix}. \text{ Then}$$

$$A(G)A(H) \pmod 2 = \begin{bmatrix} 0&1&1&1&1&1&1 \\ 1&0&1&1&1&1&1 \\ 1&1&0&1&1&1&1 \\ 1&1&1&0&1&1&1 \\ 1&1&1&1&0&1&1 \\ 1&1&1&1&1&0&1 \\ 1&1&1&1&1&1&0 \end{bmatrix}.$$

Clearly, $A(G)A(H)$ is graphical and the graph Γ realizing $A(G)A(H)$ is K_7.

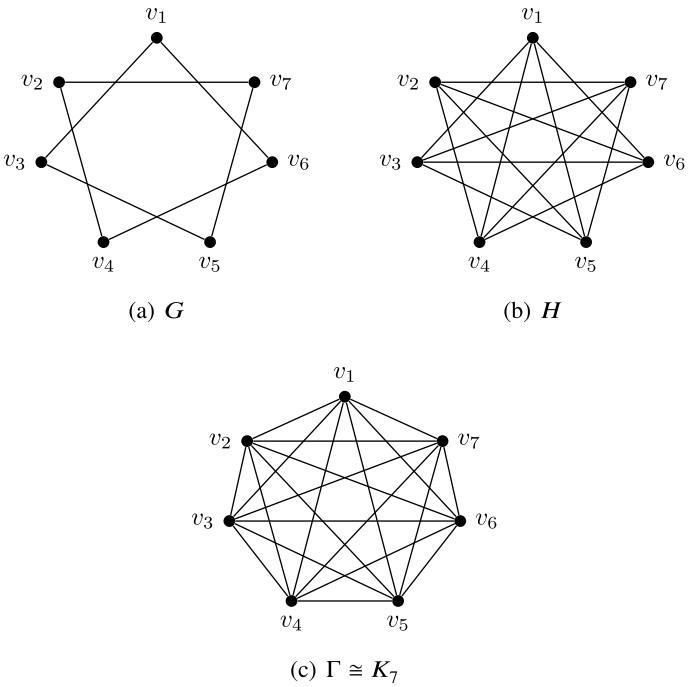


Fig. 17.9 Graphs G and H for which K_7 is the realization of $A(G)A(H)$

In the following results, we shall consider the graphs G and H on the same set of vertices $\{v_1, v_2, \ldots, v_n\}$.

Lemma 17.3 *The diagonal entries of the matrix product $A(G)A(H)$ are zero if and only if, for each vertex v_i, the number of G-H paths from v_i to itself is even.*

Proof Let $A = A(G)$, $B = A(H)$, and $C = B$. Then

$$c_{ii} = \sum_k a_{ik} b_{ki} \pmod 2. \tag{17.7}$$

As $a_{ik}b_{kj}$ is 1 if $v_i \sim_G v_k \sim_H v_j$, and is 0 otherwise, $\sum_k a_{ik}b_{kj}$ is the number of G-H paths from v_i to itself. The result follows. □

Lemma 17.4 *The (i, j)-entry $(i \neq j)$ of the matrix product $A(G)A(H)$ (mod 2) is either 0 or 1, depending on whether the number of G-H paths from v_i to v_j is even or odd, respectively.*

Proof Let $A(G) = [a_{ij}]$, $A(H) = [b_{ij}]$, and $A(G)A(H) = [c_{ij}]$. For $i \neq j$, note that $a_{ik}b_{jk} = 1$ if and only if there exists a G-H path $v_i \sim_G v_k \sim_H v_j$ from v_i to v_j. Now from (17.7), the result follows. □

Lemma 17.5 *The matrix product $A(G)A(H)$ is symmetric if and only if for each pair of distinct vertices v_i and v_j, the numbers of G-H paths and H-G paths from v_i to v_j have the same parity (i.e., either both are even or both are odd).* □

Proof Note that each of $A(G)$ and $A(H)$ is symmetric, and, therefore, their product $A(G)A(H)$ is symmetric if and only if $A(G)A(H) = A(H)A(G)$. Now the result follows from Lemma 17.4. □

The characterization theorem in the case of modulo 2 product takes the following form. The proof is omitted as it is similar to that of Theorem 17.1.

Theorem 17.9 *The product $A(G)A(H)$ is graphical if and only if the following statements are true.*

(i) For every i $(1 \leq i \leq n)$, there is an even number of vertices v_k such that $v_i \sim_G v_k$ and $v_k \sim_H v_i$.
(ii) For each pair of vertices v_i and v_j $(i \neq j)$, the numbers of G-H paths and H-G paths from v_i to v_j have the same parity. □

Note 17.12 The (i, j)-entry $(i \neq j)$ of the matrix product $A(G)A(H)$ is either 0 or 1 depending on whether the number of G-H paths from v_i to v_j is even or odd, respectively.

Recall that, under the usual multiplication, $A(G)A(\overline{G})$ is graphical only when G is regular with regularity 1 or $(n-2)$ regular or is C_5, the cycle on 5 vertices (Theorem 17.8). Example 17.3, given below, shows that even when G is none of the above three types of graphs, $A(G)A(\overline{G})$ can be graphical when the multiplication is taken modulo 2.

Example 2

Consider a 2-regular graph G on six vertices and its complement $\overline{G}$, as shown in Fig. 17.10. Note that

$$A(G)A(\overline{G}) = \begin{bmatrix} 0&1&1&0&1&1 \\ 1&0&1&1&0&1 \\ 1&1&0&1&1&0 \\ 0&1&1&0&1&1 \\ 1&0&1&1&0&1 \\ 1&1&0&1&1&0 \end{bmatrix},$$

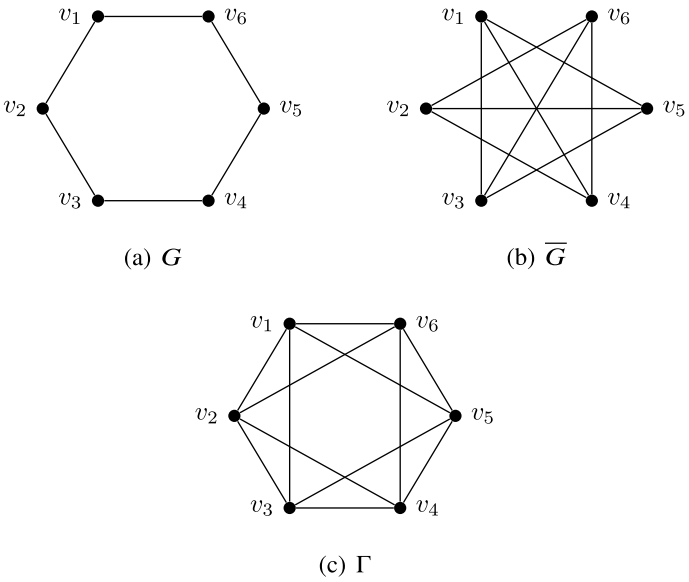


Fig. 17.10 A 2-regular graph G on 6 vertices, its complement $\overline{G}$, and their modulo 2 matrix product Γ

and the cocktail party graph shown in Fig. 17.10 is the graph realizing $A(G)A(\overline{G})$.

In the case of the modulo 2 matrix product, the following result characterizes the graphs G for which $A(G)A(\overline{G})$ is graphical.

Theorem 17.10 *For any graph G on the set of vertices $\{v_1, v_2, \ldots, v_n\}$, the following are equivalent:*

(i) The matrix product $A(G)A(\overline{G})$ is graphical.
(ii) For every i and j, $1 \le i, j \le n$, $\deg_G v_i \equiv \deg_G v_j \pmod 2$.
(iii) The graph G is parity regular.

Proof Note that (ii) $\iff$ (iii) follows from the definition of parity regular graphs. Now we shall prove (i) $\iff$ (ii). Let $A = A(G)$. From the definitions of the complement of a graph and the G-H path with $H = \overline{G}$,

$$\begin{aligned}\deg_G v_i = {} & \text{Number of} G - G \text{ paths from} v_i \text{ to} v_j + \\ & \text{Number of} G - \overline{G} \text{ paths from} v_i \text{ to} v_j + a_{ij},\end{aligned} \tag{17.8}$$

and similarly,

$$\deg_G v_j = \text{Number of} G - G \text{ paths from } v_j \text{ to} v_i + \text{Number of} G - \overline{G} \text{ paths from} v_j \text{ to} v_i + a_{ji} \tag{17.9}$$

for every pair of distinct vertices v_i and v_j. Note that a G-$\overline{G}$ path from v_j to v_i is a $\overline{G}$-G path from v_i to v_j. Then, from Theorem 17.9 and by comparing the right-hand sides of (17.8) and (17.9), we see that $A(G)A(\overline{G})$ is graphical if and only if $\deg_G v_i \equiv \deg_G v_j \pmod 2$. □

Remark 17.4 It is also possible for one to prove (i) $\iff$ (ii), by taking $A(\overline{G}) = J - A(G) - I$ in the matrix products $A(G)A(\overline{G})$ and $A(\overline{G})A(G)$, where J is the $n \times n$ all-ones matrix and I is the $n \times n$ identity matrix.

Example 3

For any two graphs G and H such that $A(G)A(H)$ is graphical under usual matrix multiplication, from Corollary 17.1, if any of G and H is connected, then the other is regular. When we consider matrix multiplication under modulo 2, we observe that the connected graphs G and $\overline{G}$ as shown in Fig. 17.11 do not satisfy this property, as both the graphs are merely parity regular but neither of them is regular. Note that

$$A(G)A(\overline{G}) = \begin{bmatrix} 0&0&1&0&1&0 \\ 0&0&0&1&0&1 \\ 1&0&0&0&1&0 \\ 0&1&0&0&0&1 \\ 1&0&1&0&0&0 \\ 0&1&0&1&0&0 \end{bmatrix}$$

realizes the graph Γ as shown in Fig. 17.11.

When the matrix multiplication is considered under modulo 2, the following theorem characterizes graphs G with $A(G)A(\overline{G}) = A(G)$.

Theorem 17.11 *Consider a graph G defined on the set of vertices $\{v_1, v_2, \ldots, v_n\}$. Then $A(G)A(\overline{G}) = A(G)$ if and only if $(A(G))^2$ is either a null matrix or the all-ones matrix J.*

Proof Let $A = A(G)$. From Lemma 17.5, taking $H = \overline{G}$, we see that $A(G)A(\overline{G}) = A(G)$ implies that the number of G-$\overline{G}$ paths from v_i to v_j, modulo 2, is a_{ij}. Now substituting in (17.8), we get, for each $j \neq i$,

$$\deg_G v_i \equiv (\text{number of} G - G \text{ paths from } v_i \text{ to} v_j) \pmod 2. \tag{17.10}$$

From Theorem 17.10, we know that G is parity regular and therefore $\deg_G v_i \equiv \deg_G v_j \pmod 2$. Thus, from (17.10), $(A(G))^2$ is either 0 or J.

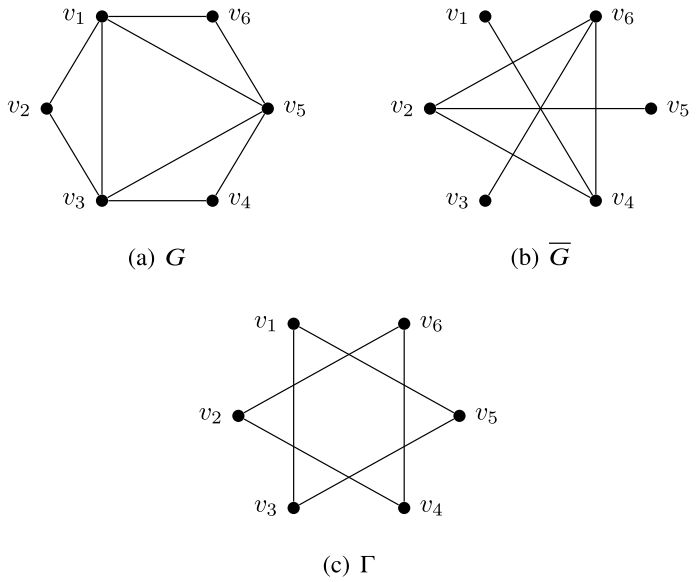


Fig. 17.11 Modulo 2 matrix product of a non-regular graph and its complement

Conversely, suppose that $(A(G))^2$ is either 0 or J. If $(A(G))^2 = 0$, then the degrees of all the vertices in G are even and if $(A(G))^2 = J$, then the degrees of all the vertices are odd. By taking $A(\overline{G}) = J - A(G) - I$ $(= J + A(G) + I$, since subtraction and addition are equivalent modulo 2), we get

$$\begin{aligned} A(G)A(\overline{G}) &= A(G)(J + A(G) + I) \\ &= A(G)J + (A(G))^2 + A(G). \end{aligned}$$

In either of the cases $(A(G))^2 = 0$ and $(A(G))^2 = J$, the right-hand side of the above reduces to $A(G)$ modulo 2. □

The corollary given below follows from Theorem 17.11, which also characterizes the graphs G with the property $A(G)A(\overline{G}) = A(G)$ in terms of properties of $\overline{G}$.

Corollary 17.4 *Consider a graph G such that $A(G) = A$ and $A(\overline{G}) = B$. Then the following statements are equivalent:*

(i) $AB = A$.
(ii) $B^2 = I$ *or* $J - I$.
(iii) $\overline{G}$ *is a graph with one of the following properties:*

- *(a) Each vertex has an odd degree and the number of paths of length* 2 *between each pair of vertices is even.*
- *(b) Each vertex has an even degree and the number of paths of length* 2 *between each pair of vertices is odd.*

Proof (i) $\Longrightarrow$ (ii) From Theorem 17.11, $AB = A$ implies that A^2 is either a null matrix or J. By taking $A = J - B - I$, we have $A^2 = J^2 + B^2 + I$. Further, note that J^2 is either a null matrix or J itself, depending on whether the number of vertices on which the graph defined is even or odd. In either case, $A^2 = 0$ or J implies $B^2 = I$ or $J - I$.

(ii) $\Longrightarrow$ (i) Noting that $J^2 = 0$ or J itself, by suitable substitution for B^2, we have $A^2 = J^2 + B^2 + I = 0$ or J. Hence, we obtain (i) from Theorem 17.11.

As (iii) is the graph theoretical interpretation of (ii), (ii) $\Longleftrightarrow$ (iii) is trivial. □

The following corollary characterizes the graphs G for which $A(G)A(\overline{G}) = A(\overline{G})$ and the proof is immediate from Corollary 17.4.

Corollary 17.5 *Consider a graph G such that $A(G) = A$ and $A(\overline{G}) = B$. Then the following statements are equivalent:*

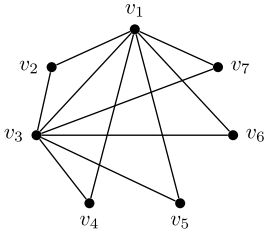
(i) $AB = B$.
(ii) $A^2 = I$ *or* $J - I$.

Note that the adjacency matrix $A(G)$ of a graph G is not idempotent under the usual multiplication, unless G is totally disconnected. The following theorem verifies when $(A(G))^2 = A(G)$ when the product is taken under modulo 2.

Theorem 17.12 *Let G be a graph with adjacency matrix $A(G)$. Then the following statements are equivalent:*

(i) $(A(G))^2 = A(G)$, *i.e., $A(G)$ is idempotent.*
(ii) $A(G)A(\overline{G}) = 0$ *and the degree of every vertex in G is even.*
(iii) The number of G-$\overline{G}$ paths of length 2 between every pair of vertices is even, and the degree of every vertex in G is even.

Proof (i) $\Longrightarrow$ ii) $(A(G))^2$ is graphical implies that the diagonal entries of $(A(G))^2$ are zeros and in turn, we see that degree of each vertex in G is zero modulo 2. In other words, degree of each vertex is even. So, we have $A(G)J = 0$ and



(a) G

Fig. 17.12 A graph G for which $A(G)A(G) = A(G)$

therefore $A(G)A(\overline{G}) = A(G)(J - A(G) - I) = -(A(G))^2 - A(G) = 0$ whenever $(A(G))^2 = A(G)$. (ii) $\Longrightarrow$ (iii) follows from Lemma 17.5 and (iii) $\Longrightarrow$ (i) follows from Theorem 17.10. □

Example 4

The graph G as shown in Fig. 17.12 is such that $(A(G))^2 = A(G)$ and the degree of every vertex of G is even. Further, $A(G)A(\overline{G}) = 0$.

17.4 Some Matrix Equations of Graphs

In the last section, we have seen the graph realizing the product of adjacency matrices of two graphs and the related results. In this section, we consider the product of other matrices associated with graphs and the equation involving this product. As a bi-product, we get the characterization of complete bipartite graphs and cycle graphs. In all the matrix equations we consider, the adjacency matrix $A(G)$ is a part on the left-hand side. An isolated vertex in a graph results in a zero row in its adjacency matrix, so the product on the left-hand side also has a zero row, which implies that the graph realizing the product has an isolated vertex. To avoid this situation, we consider graphs without isolated vertices.

For the sake of convenience, we say that an edge e is incident with a vertex v in a graph G if the vertex v is either incident with the edge e or else it is adjacent to any end vertex of e.

We mainly deal with the following matrix equations of graphs.

(i) Characterize the graphs G for which $A(G)B(G) = B(H)$, for some graph H.
(ii) Characterize the graphs G and the size of matrix J for which $A(G)B(G) = J$, where J is a rectangular matrix with all entries equal one.
(iii) Characterize the graphs G for which $A(G)B(G)$ is graphical, i.e., $A(G)B(G) = A(\Gamma)$ for some graph Γ.

Since the adjacency matrix of a cycle graph is circulant, we note the following properties of circulant matrices which are useful for later purpose.

Definition 17.9 An $n \times n$ matrix S is said to be a *circulant matrix* if

$$S_{i,j} = S_{1,k}, \quad \text{where} \quad 1 \leq k \leq n \text{ and } j - i + 1 \equiv k \pmod{n}. \tag{17.11}$$

Further, a circulant matrix is represented by its first row.

Remark 17.5 In the following, we remark on certain properties of circulant matrices which follow from the definition of circulant matrix.

(i) Sum and product of circulant matrices are circulant.

(ii) A (0, 1)-circulant matrix represented by $(a_1, a_2, \ldots, a_n)$ is graphical (is an adjacency matrix of some graph) if and only if

$$a_1 = 0 \quad \text{and} \quad a_i = a_k \quad \text{for all } k + i \equiv n + 2 \pmod{n}. \tag{17.12}$$

(iii) For the cycle graph $G = v_1 \sim v_4 \sim v_5 \sim v_2 \sim v_3 \sim v_1$, $A(G)$ is not circulant.

(iv) From Eq. (17.12) in (ii), it is clear that a circulant matrix $(a_1, a_2, \ldots, a_n)$ is $A(G)$ for some 2-regular graph, then $a_i = a_k = 1$ for exactly one pair of $i, k \geq 2$ such that $i + k \equiv n + 2 \pmod{n}$ and $a_j = 0$ for all other j.

(v) For $2 \leq i < k$, let $C_{(i,k)}$ represent a circulant matrix $(a_1, a_2, \ldots, a_n)$ with $a_i = a_k = 1$ and $a_j = 0$ for all other j. Then $C_{(2,n)}$ is the adjacency matrix of the cycle graph $v_1 \sim v_2 \sim \cdots \sim v_n \sim v_1$.

(vi) For $2 \leq i < k$, $C_{(i,k)}$ represents an adjacency matrix of a cycle graph G on n vertices if and only if $i + k \equiv n + 2 \mod n$ and, the integers $i - 1$ and n are relatively prime. Note that $i - 1$ and n are relatively prime $\iff$ the walk $v_1 \sim v_{1+(i-1)} \sim v_{1+2(i-1)} \sim \cdots \sim v_{1+p(i-1)} \sim \cdots$ has all the vertices of the graph $\iff$ the graph is connected and hence the two regular graph G is a cycle.

(vii) For $2 \leq i < n$, $C_{(i,i+1)}$ is an adjacency matrix of a graph G (so, $2i + 1 \equiv n + 2 \mod n$) if and only if G is a cycle on an odd number of vertices and $i = d + 1$, where d is the diameter of the graph.

(viii) For $1 \leq i \leq n$, let $C_{(i-1,i,i+1,i+2)}$ denote a circulant matrix $(a_1, a_2, \ldots, a_n)$ such that $a_{i-1} = a_i = a_{i+1} = a_{i+2} = 1$ and $a_j = 0$ for all other j. Note that for $n = 4$, the circulant matrix $C_{(i-1,i,i+1,i+2)}$ is trivially symmetric. For $n \geq 5$, we have $(i - 1) + (i + 2) = i + (i + 1) = n + 2$ for any n if and only if n is odd and $i = \frac{n+1}{2}$. Then $(a_1, a_2, \ldots, a_n)$ is graphical if and only if n is odd and $i = \frac{n+1}{2}$.

Let G be a graph with $V(G) = \{v_1, v_2, \ldots, v_n\}$ and $E(G) = \{e_1, e_2, \ldots, e_m\}$. Let $A(G)$ and $B(G)$ be the adjacency matrix and incidence matrix of G, respectively. Consider the equation $A(G)B(G) = C$. We try to understand the nature of C.

Theorem 17.13 *If $A(G)B(G) = C$, then the following assertions hold.*

(i) The entry c_{ip} is non-zero if and only if the vertex v_i is either incident or adjacent to the edge e_p.

(ii) The number of non-zero entries in the pth column of C represents the number of vertices adjacent to e_p.

(iii) Matrix C always dominates $B(G)$ entrywise, i.e., $(C)_{ij} \geq (B(G))_{ij}$ for all i, j.

(iv) C is a (0, 1)-matrix if and only if G is triangle free.

Proof Let $(A(G))_{ij} = a_{ij}$, $(B(G))_{kl} = b_{kl}$, and $(C)_{kl} = c_{kl}$ for all possible i, j, k, l. For any p $(1 \leq p \leq m)$, let the edge e_p be incident with the vertices v_k and v_l where $k < l$, in which case $b_{kp} = b_{lp} = 1$:

$$c_{ip} = \sum_j a_{ij} b_{jp}$$
$$= a_{ik} b_{kp} + a_{il} b_{lp} \tag{17.13}$$
$$= a_{ik} + a_{il}. \tag{17.14}$$

From (17.14), it is clear that c_{ip} represents the number of end vertices of e_p with which the vertex v_i is adjacent, i.e., the number of 1s among a_{ik} and a_{il}. So, (i) of the theorem holds. For the same reason, part (ii) holds.

For $b_{ip} = 1$, we have $i = k$ or l. If $i = k$, we have $a_{ik} = a_{ii} = 0$ and $a_{il} = 1$. So, from (17.14) we get $c_{ip} = 1$. Hence part (iii) of the theorem follows.

Again referring to (17.14), we see that $c_{ip} = 2$ if and only if both a_{ik} and a_{il} are equal to one, i.e., v_i, v_k, and v_l induce a triangle with $e_p = (v_k, v_l)$ as an edge in it. This proves part (iv) of the theorem. □

Note 17.13 The number of 2s in the matrix C equals three times the number of triangles in the graph G.

We notice that, when G has n vertices and m edges, $A(G)B(G)$ is a matrix of order $n \times m$. Now we consider the question, can $A(G)B(G)$ be $B(H)$ for some graph H on n vertices having m edges? The next theorem answers this question.

Theorem 17.14 *$A(G)B(G) = B(H)$ for some graph H if and only if $B(G) = B(H)$ and $G = H$ is a 1-factor graph.*

Proof From (17.14) in Theorem 17.13, it is clear that c_{ip} in the ith row of C, the product $A(G)B(G)$, is the number of end vertices of e_p with which the vertex v_i is adjacent, i.e., the number of 1s among a_{ik} and a_{il}. Now referring to (iii) of Theorem 17.13, we see that C dominates $B(G)$ entrywise. Since an incidence matrix has exactly two 1s in each of its columns, C is an incidence matrix of some graph H if and only if $C = B(G)$.

Again referring to (ii) of Theorem 17.13, $C = B(G)$ implies that vertices adjacent to any edge are only those vertices which are incident with it. So, no two edges in the graph are adjacent to each other. Since no vertex in the graph is an isolated vertex, degree of each vertex is exactly one and therefore the graph is a 1-factor graph. □

It is interesting to search for a graph G, if it exists, satisfying $A(G)B(G) = J$ where J is a $n \times m$ matrix with all entries equal to 1. We have the following theorem.

Theorem 17.15 *Given a graph G with n vertices and m edges, consider the matrix equation $A(G)B(G) = C$. Then the following statements are equivalent.*

(i) *$C = J_{n\times m}$, where $m = k(n-k)$ for some $1 \le k \le n$ and $J_{n\times m}$ is the matrix with all entries equal to one.*
(ii) *For every pair of vertex v_i and an edge e_j of the graph, the vertex v_i is adjacent to exactly one end vertex of e_j.*
(iii) *G is a complete bipartite graph with diameter 2.*

Proof (i) $\implies$ (ii). From the definition of J, the entries of every jth column are one, and from (ii) of Theorem 17.13, we know that every vertex v_i is adjacent to the edge e_j. Again referring to (iv) of Theorem 17.13, we see that the graph is triangle free and therefore v_i cannot be adjacent to both the end vertices of an edge e_j. So, (ii) holds.

(ii) $\implies$ (iii). If v_i and v_j are adjacent, then the vertices are at distance one. Otherwise, let v_i and v_j $(i \neq j)$ be any non-adjacent vertices in the graph. Since v_j is not an isolated vertex (from the choice of G), there exists an edge $v_j \sim v_k$ in the graph. Now from (ii) of the theorem, we see that v_i is adjacent to v_k and therefore v_i and v_j are at distance two from each other. Taking a vertex v_1 of the graph, partition the vertex set into V_1 and V_2 such that V_1 consists of all the vertices at distance one from v_1 and V_2 consists of v_1 and all the other vertices at distance two from v_1. From (ii), we see that G is triangle free and therefore no two vertices from V_1 are adjacent. Since the vertices from V_2 are not adjacent to v_1, from (ii) we know that they are adjacent to every vertex of V_1. Again the triangle free property of the graph proves the non-adjacency between any two distinct vertices of V_2. So, G is a complete bipartite graph.

(iii) $\implies$ (i). Let G be a complete bipartite graph with partition V_1 and V_2 having k and $n-k$ number of vertices, respectively. So, the graph has $k(n-k)$ number of edges, and the matrices $B(G)$ and C in the matrix product are of size $n \times m$, where $m = k(n-k)$. Since a complete bipartite graph is triangle free and every vertex is adjacent to every edge of the graph, we have $C = J$ with all the entries equal to one. □

Now we consider a matrix equation $A(G)B(G) = C$, where C is an adjacency matrix of some graph Γ. Note that G should be with n edges, so that the product is a square matrix. From the structures of $A(G)$ and $B(G)$, it is clear that $A(G)B(G) = C$ is graphical if and only if $A(H)B(H)$ is graphical, for every connected component H of G. So, we assume that G in the following discussions is connected, unless indicated otherwise.

If a connected graph G is with n vertices and n edges, then the graph is either a cycle graph or else it is a unicyclic graph. In the following theorem, we characterize a cycle graph on n vertices in terms of a matrix equation.

Theorem 17.16 *Given a graph G on n (≥ 4) vertices, the following statements are equivalent.*

(i) G is a cycle on four or more number of vertices.
(ii) $A(G)B(G)$ is an $n \times n$ matrix in which each column is with exactly four ones.
(iii) $A(G)B(G)$ is an $n \times n$ matrix in which each row is with exactly four ones.

Proof (i) $\implies$ (ii). Let G be a cycle on four or more vertices. Let $v_{i_1} \sim v_{i_2} \sim v_{i_3} \sim v_{i_4}$ and $e_p = v_{i_2} \sim v_{i_3}$ in which case from (17.13) we have

$$\left.\begin{aligned} c_{i_1 p} &= a_{i_1 j_1} b_{j_1 p} + a_{i_1 i_2} b_{i_2 p} &&= 1, \\ c_{i_2 p} &= a_{i_2 i_1} b_{i_1 p} + a_{i_2 i_3} b_{i_3 p} &&= 1, \\ c_{i_3 p} &= a_{i_3 i_2} b_{i_2 p} + a_{i_3 i_4} b_{i_4 p} &&= 1, \\ c_{i_4 p} &= a_{i_4 i_3} b_{i_3 p} + a_{i_4 j_2} b_{j_2 p} &&= 1, \end{aligned}\right\} \tag{17.15}$$

where v_{j_1} and v_{j_2} are the vertices, different from v_{i_2} and v_{i_3}, adjacent to v_{i_1} and v_{i_4}, respectively. Further, for every other $j \neq i_1, i_2, i_3, i_4$ we note that v_j is not adjacent to e_p and hence $c_{jp} = 0$. Therefore each column has exactly four 1s.

(ii) $\Longrightarrow$ (i). Let $A(G)B(G)$ be an $n \times n$ matrix in which each column is with exactly four 1s. From (iv) of Theorem 17.13, G is a triangle free graph and therefore the graph has four or more vertices. Suppose that G is not a cycle, then there exists at least one vertex not on the cycle subgraph of the unicyclic graph. Since the graph is with n vertices, n edges and triangle free, there exist distinct vertices w_1, w_2, w_3, w_4, and w_5 in the graph G such that $w_1 \sim w_2 \sim w_3 \sim w_4$ is on the cycle and $w_3 \sim w_5$ is not on the cycle. Now, the edge $w_2 \sim w_3$ has at least five distinct vertices adjacent to it. So the corresponding column has five 1s, a contradiction. Therefore G is a cycle.

(i) $\Longrightarrow$ (iii). Using the fact that every vertex is adjacent to exactly four edges in a cycle, the proof is in the similar lines of the proof of (i) $\Longrightarrow$ (ii).

(iii) $\Longrightarrow$ (i). If the graph is unicyclic, then take the vertices w_1, w_2, w_3, w_4, and w_5 as in the case of proof of (ii) $\Longrightarrow$ (i) and observe that the row corresponding to w_3 has five or more 1's. □

We adopt the lines in the proof of Theorem 17.16, suitably, to prove the following theorem.

Theorem 17.17 *Given a graph G on $n \geq 4$ vertices, the following statements are equivalent.*

(i) G has a cycle subgraph on the vertices $v_{i_1}, v_{i_2}, \ldots, v_{i_k}$ and with the edges $e_{p_1}, e_{p_2}, \ldots, e_{p_k}$ $(k \geq 4)$.

(ii) In the $k \times k$ sub-matrix of $A(G)B(G)$, with rows determined by $i_1, i_2, \ldots, i_k$ and columns determined by $p_1, p_2, \ldots, p_k$, each column has exactly four 1s.

(iii) In the $k \times k$ sub-matrix of $A(G)B(G)$, with rows determined by $i_1, i_2, \ldots, i_k$ and columns determined by $p_1, p_2, \ldots, p_k$, each row has exactly four 1s.

Note that if G is a cycle graph on three vertices, then $A(G)B(G)$ has three 2s, and if G is a cycle graph on four vertices then $A(G)B(G) = J_4$. In both cases, the product is not the adjacency matrix of any graph.

Lemma 17.6 *Let $v_{i-2} \sim v_{i-1} \sim v_i \sim v_{i+1} \sim v_{i+2} \sim v_{i+3}$ be a walk in a graph G and the vertices are distinct except possibly for the pair v_{i-2} and v_{i+3}. Let $C = A(G)B(G)$. Then*

$$e_p = v_i \sim v_{i+1} \quad \textit{if and only if} \quad c_{i-1p} = c_{ip} = c_{i+1p} = c_{i+2p} = 1. \tag{17.16}$$

Proof If $e_p = v_i \sim v_{i+1}$, from Eq. (17.15) we have

$$\left.\begin{aligned} c_{i-1,p} &= a_{i-1,i-2}b_{i-2,p} + a_{i-1i}b_{ip} &&= 1,\\ c_{ip} &= a_{ii-1}b_{i-1,p} + a_{i,i+1}b_{i+1,p} &&= 1,\\ c_{i+1,p} &= a_{i+1i}b_{ip} + a_{i+1,i+2}b_{i+2,p} &&= 1,\\ c_{i+2,p} &= a_{i+2,i+1}b_{i+1,p} + a_{i+2,i+3}b_{i+3,p} &&= 1. \end{aligned}\right\} \tag{17.17}$$

Conversely, if the equations in (26.7) hold, it is clear that the edge e_p in the cycle graph G is incident with two distinct vertices $v_{i-2}, v_{i-1}, v_i, v_{i+1}, v_{i+2}, v_{i+3}$. Since the vertices on the given walk are distinct except possibly for the pair v_{i-2} and v_{i+3}, then $e_p = v_{i-2} \sim v_{i-1}$ implies $b_{ip} = b_{i+2p} = 0 = c_{i+1p}$. In fact, for $e_p \neq v_i \sim v_{i+1}$, at least one equation in (26.7) does not hold. So, $e_p = v_i \sim v_{i+1}$. □

The following corollary follows immediately.

Corollary 17.6 *Let G be a graph free from any triangle and cycle of length four. If $v_{i_1} \sim v_{i_2} \sim v_{i_3} \sim v_{i_4}$ is a path, then $e_p = v_{i_2} \sim v_{i_3}$ if and only if $c_{i_1p} = c_{i_2p} = c_{i_3p} = c_{i_4p} = 1$.*

Theorem 17.18 *Let G be a cycle graph on $n \geq 5$ vertices with $A(G) = C_{(2,n)}$ and $d = \lfloor \frac{n}{2} \rfloor$, the integer part of $\frac{n}{2}$. Then the following statements are equivalent.*

(i) $A(G)B(G) = C$ *is graphical.*

(ii) C *is the circulant matrix* $C_{(k-1,k,k+1,k+2)}$, *where* $k = d + 1$ *and* $n = 2d + 1$. *In other words,* $C = A(\Gamma)$, *where* Γ *is a 4-regular graph in which* $v_i \sim_\Gamma v_{d+i-1}, v_{d+i}, v_{d+i+1}, v_{d+i+2}$.

(iii) G *is a graph on odd number of vertices and edge* e_i *is incident with the vertices* v_{d+i} *and* v_{d+i+1} *for all* i.

(iv) $B(G) = A(H)$ *for some 2-regular graph* H *in which* $v_i \sim_H v_{d+i}, v_{d+i+1}$ *for all* i. *In other words,* $A(H) = C_{(d+1,d+2)}$.

Proof (i) $\Longrightarrow$ (ii). Let $A(G)B(G) = C$ and C be graphical. From Theorem 17.16, G is a cycle graph on five or more vertices implying that the matrix $C = A(G)B(G)$ has exactly four 1s. Since $A(G)$ is $C_{(2,n)}$, from Eq. (26.8) in Lemma 17.6 we have

$$e_p = v_i \sim v_{i+1} \quad \text{if and only if} \quad c_{i-1p} = c_{ip} = c_{i+1p} = c_{i+2p} = 1. \tag{17.18}$$

Therefore, the ith row has leading non-zero entry only in the pth column and no two sequences of four 1s in different columns of C are in the same alignment. Now from the symmetry of C, a non-zero entry in any row is a part of a sequence of four 1s in that row and the leading 1 of a column is the terminating 1 of the corresponding row. So, in the $(p + 1)$th column we have

$$c_{ip+1} = c_{i+1p+1} = c_{i+2p+1} = c_{i+3p+1} = 1 \text{ and } \quad e_{p+1} = v_{i+1} \sim v_{i+2}. \tag{17.19}$$

So, the matrix $C = A(G)B(G)$ is a circulant matrix of the form $(a_1, a_2, \ldots, a_n)$ with $a_{k-1} = a_k = a_{k+1} = a_{k+2} = 1$ and $a_j = 0$ for all other j, whenever $e_1 = v_k \sim_G$

v_{k+1}. In other words, C is of the form $C_{(k-1,k,k+1,k+2)}$ as defined in (viii) of Remark 17.5. Therefore, $C = A(\Gamma)$ for some graph Γ (graphical) implies that n is odd, $k = d + 1$, and Γ is as described in (ii).

(ii) $\Longrightarrow$ (i) follows trivially.

(ii) $\Longrightarrow$ (iii). From (ii), it is clear that G is a cycle graph on odd number of vertices and $A(G)$ is $C_{(d,d+1,d+2,d+3)}$. Now referring to (26.8) in Lemma 17.6, we see that the edge e_i is incident with the vertices v_{d+i} and v_{d+i+1} for all i in the graph G.

(iii) $\Longrightarrow$ (iv). From (iii), n is odd and therefore $C_{(d+1,d+2)}$ is graphical (refer to (vii) of Remark 17.5). Again, from the description of $B(G)$ as in (iii) it is clear that $B(G) = C_{(d+1,d+2)}$ and equals $A(H)$, for the graph H as described in (iv).

(iv) $\Longrightarrow$ (ii). Let $C = A(G)B(G)$. From Eq. (26.8) of Lemma 17.6, $A(G) = C_{(2,n)}$ and $B(G) = C_{(d+1,d+2)}$ implies that C' is $C_{(d,d+1,d+2,d+3)}$ (C' denotes the transpose of C). From (iv), we see that $B(G) = C_{(d+1,d+2)}$ is graphical and therefore n is odd. Now referring to (viii) of Remark 17.5, we see that $C' = C_{(d,d+1,d+2,d+3)}$ is graphical, thus proving (ii). □

Remark 17.6 If G is a cycle graph on an odd number of vertices such that $A(G)$ is $C_{(2,n)}$ then, in Theorem 17.18, we have proved that the edges of G have a particular labeling such that $A(G)B(G)$ is graphical. In this case, $B(G) = A(H)$ for a particular cycle subgraph H of $\overline{G}$. In fact, every vertex is adjacent to two vertices in H which are at distance d (diagonally opposite vertices) in G. Similarly, the vertices in the graph Γ such that $A(G)B(G) = A(\Gamma)$ are adjacent to the four vertices that are at distance $d - 1$ or d in G. An example of G, H, and Γ which satisfy $A(G)B(G) = A(\Gamma)$ and $B(G) = A(H)$ are given below.

Note that

$$A(G) = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \end{array} \begin{array}{c} \begin{array}{ccccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{array},$$

$$B(G) = A(H) = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \end{array} \begin{array}{c} \begin{array}{ccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \end{array} \\ \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{pmatrix} \end{array}$$

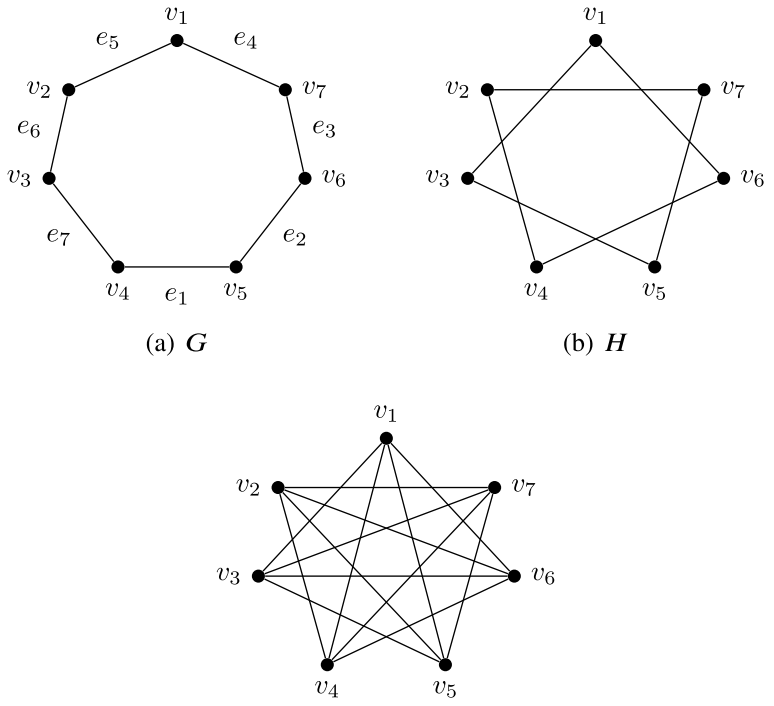


Fig. 17.13 Graphs G, H, and Γ satisfying $A(G)B(G) = A(\Gamma)$ and $B(G) = A(H)$

$$\text{and} A(\Gamma) = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \\ v_7 \end{array} \begin{array}{c} \begin{array}{ccccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 & v_7 \end{array} \\ \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{pmatrix} \end{array}$$

satisfying $A(G)B(G) = A(\Gamma)$.

In the following theorem, we provide a characterization of cycle graphs and cycle graphs on an odd number of vertices. The proof is immediate from the fact that given any cycle graph G with an adjacency matrix $A(G)$, there exists a permutation matrix, viz., P such that $PA(G)P' = C_{(2,n)}$, and if n is odd then there exists a permutation matrix Q such that $PB(G)Q = C_{(d+1,d+2)}$ (Fig. 17.13).

Theorem 17.19 *Let G be a connected graph on $n \geq 4$ vertices. For the adjacency matrix $A(G)$ and incidence matrix $B(G)$ with reference to the given labeling of vertices and edges, the following statements hold.*

(i) *G is a cycle if and only there exist permutation matrices P and Q such that $PA(G)B(G)Q = C_{(d,d+1,d+2,d+3)}$, where $d = \lfloor \frac{n}{2} \rfloor$. In this case, $PA(G)P' = C_{(2,n)}$ and $PB(G)Q = C_{(d+1,d+2)}$.*

(ii) *G is a cycle on an odd number of vertices if and only if $PA(G)B(G)Q = C_{(d,d+1,d+2,d+3)}$ in (i) is symmetric (graphical).*

The following theorem characterizes each component of the matrix equation $A(G)B(G) = A(\Gamma)$.

Theorem 17.20 *Let G be a connected graph on n vertices. Then $A(G)B(G) = A(\Gamma)$ if and only if the following holds.*

(i) *G is a cycle on an odd number of vertices.*

(ii) *There exists permutation matrix P such that $PA(G)P' = C_{(2,n)}$, $PB(G)P' = C_{(d+1,d+2)}$ and $PA(\Gamma)P' = C_{(d,d+1,d+2,d+3)}$, where $d = \lfloor \frac{n}{2} \rfloor$.*

Proof Let $A(G)B(G) = A(\Gamma)$, in which case G is triangle free and $n \geq 4$. First, we prove that the graph G is a cycle. Suppose that G is a unicyclic, then G has a unique proper cycle subgraph and $n > 4$. We claim that the vertices and edges on this cycle are labeled by the same set of indices. Otherwise, from Theorem 17.17 and the symmetry of $A(G)B(G)$ we will have two cycles in G, which is not possible in a connected unicyclic graph. Without loss of generality, assume that both the vertices and edges of the cycle subgraph are indexed by $1, 2, \ldots, k$ for some $4 \leq k < n$, i.e., $v_1, \ldots, v_k$ are the vertices on the cycle subgraph and $e_1, \ldots, e_k$ are the edges. Now $A(G)B(G)$ is of the form $\begin{pmatrix} C_1 & C_2 \\ C_3 & C_4 \end{pmatrix}$, where the block C_1 corresponds to the vertices and edges of the cycle subgraph. Now for any edge e_i not on the cycle (i.e., $k+1 \leq i \leq n$) but incident with some vertex of the cycle, e_i is adjacent to three vertices on the cycle. So, the corresponding column in C_2 has three 1s. But each vertex v_j not on the cycle (i.e., $k+1 \leq j \leq n$) is adjacent to at most two edges of the cycle. Therefore, each row of C_3 has at most two 1s. This contradicts the symmetry of the product $A(G)B(G)$. So, G must be a cycle. Further, note that given any cycle G and its adjacency matrix $A(G)$, there exists a permutation matrix P such that $PA(G)P' = C_{(2,n)}$. Also note that $PA(G)B(G)P' = PA(G)P'PB(G)P'$ and $PA(\Gamma)P'$ is also graphical. Now we refer to Theorem 17.18 for the rest of the proof. □

Let C_n be a cycle on n vertices $v_1, v_2, \ldots, v_n$, where n is even. We can observe that the 1-factor graph H with vertices $v_1, v_2, \ldots, v_n$ and $v_i \sim_H v_{d+i}$, where d is the diameter of the cycle C_n, is a companion of C_n with $v_i \sim_G v_{i+1}$ for $1 \leq i \leq n-1$, and $v_1 \sim_G v_n$. From Theorem 17.18 for a cycle graph C_n, $n \geq 5$ and n odd, there is a companion.

For the graph $\overline{K_m}$, every graph on m vertices is a companion. The generalized wheel $W_{m,n}$, which is the join of the above two graphs, has many more edges than the cycle graph. We show that when m and n are even and $n \geq 4$, the generalized wheel has a companion.

Let $v_1, v_2, \ldots, v_n$ be the peripheral vertices and $v'_1, v'_2, \ldots, v'_m$ be the central vertices of the generalized wheel $G = W_{m,n}$. The peripheral vertices $v_1, v_2, \ldots, v_n$ are such that $v_i \sim_G v_{i+1}$, $1 \leq i \leq n-1$ and $v_n \sim_G v_1$. So, the adjacency matrix of G can be viewed as follows:

$A(G) = \begin{pmatrix} C_{n\times n} & J_{n\times m} \\ J_{m\times n} & 0_{m\times m} \end{pmatrix}$, where $C_{n\times n}$ is the adjacency matrix of C_n, $J_{n\times m}$ and $J_{m\times n}$ are all-ones matrices of the respective orders, and $0_{m\times m}$ is a zero matrix of order m.

The following theorem determines positive integers m and n for which the graph $W_{m,n}$ has a companion.

Theorem 17.21 *Let $G = W_{m,n}$ be the generalized wheel graph with the central vertices $v'_1, v'_2 \ldots, v'_m$ and peripheral vertices $v_1, v_2, \ldots, v_n$. Then G has a companion H if and only if m and n are even and $n \geq 4$. The graph H is a 1-factor graph with $v_i \sim_H v_{d+i}$, $1 \leq i \leq \frac{n}{2}$, and $d = \frac{n}{2}$ is the diameter of the cycle C_n.*

Proof Assume that the graph $G = W_{m,n}$ has a companion H. Since G is connected, H is a regular subgraph of $\overline{G}$ by Corollary 17.1. Note that $\overline{G} = K_m \cup \overline{C_n}$ and the number of components of $\overline{G}$ is one more than that of $\overline{C_n}$. Thus, H is a subgraph of $K_m \cup \overline{C_n}$. The adjacency matrix of H can be viewed as

$A(H) = \begin{pmatrix} B_{n\times n} & 0_{n\times m} \\ 0_{m\times n} & D_{m\times m} \end{pmatrix}$, where $B_{n\times n}$ is the adjacency matrix of the subgraph of H induced by the vertices $v_1, v_2, \ldots, v_n$ and $D_{m\times m}$ is the adjacency matrix of the subgraph of H induced by the vertices $v'_1, v'_2, \ldots, v'_m$, and $0_{n\times m}$ and $0_{m\times n}$ are zero matrices of orders $n \times m$ and $m \times n$, respectively.

The product of $A(G)$ and $A(H)$ is given by

$$A(G)A(H) = \begin{pmatrix} C_{n\times n}B_{n\times n} & J_{n\times m}D_{m\times m} \\ J_{m\times n}B_{n\times n} & 0_{m\times m} \end{pmatrix}.$$

From the (1, 2) and (2, 1) blocks of $A(G)A(H)$, we can note that no row (column) of either $B_{n\times n}$ or $D_{m\times m}$ may have two or more non-zero entries, because otherwise there would be an entry greater than or equal to 2 in $A(G)A(H)$, a contradiction to the fact that $A(G)A(H)$ is graphical.

Hence, by regularity of H, it follows that $B_{n\times n}$ is a permutation matrix and $J_{m\times n}B_{n\times n} = J_{m\times n}$. By a similar argument, $D_{m\times m}$ is a permutation matrix. Since $B_{n\times n}$ and $D_{m\times m}$ are permutation matrices and are adjacency matrices of graphs, the graphs realizing them are 1-factor graphs, which implies that m and n are even.

Now we proceed to show that v_i is adjacent to $v_{\frac{n}{2}+i}$, $1 \leq i \leq \frac{n}{2}$, in the graph H.

Let $C_{n\times n}B_{n\times n} = \Gamma'_{n\times n}$. Since $B_{n\times n}$ is a permutation matrix, $\Gamma'_{i,j} \leq 1$ for $1 \leq i, j \leq n$.

Let $B_{i,j} = 1$ for some i and j. Then $\Gamma'_{i-1,j} = 1$ and $\Gamma'_{i+1,j} = 1$.

By the symmetry of Γ',

$$\begin{aligned} \Gamma'_{j,i-1} = (C_{n\times n}B_{n\times n})_{j,i-1} = 1, \\ \implies C_{j,j-1}B_{j-1,i-1} + C_{j,j+1}B_{j+1,i-1} = 1. \end{aligned}$$

The above implies that either $B_{j-1,i-1} = 0$ and $B_{j+1,i-1} = 1$, or $B_{j-1,i-1} = 1$ and $B_{j+1,i-1} = 0$.

Suppose that $B_{j-1,i-1} = 0$ and $B_{j+1,i-1} = 1$.

Because of symmetry, $B_{i-1,j+1} = 1$, which implies that $\Gamma'_{i-2,j+1} = \Gamma'_{i,j+1} = 1$.

By the symmetry of Γ', $\Gamma'_{j+1,i-2} = C_{j+1,j}B_{j,i-2} + C_{j+1,j+2}B_{j+2,i-2} = 1$.

The above implies that either $B_{j+2,i-2} = 1$ and $B_{j,i-2} = 0$, or $B_{j+2,i-2} = 0$ and $B_{j,i-2} = 1$. But, since $B_{j,i} = 1$ and B is a permutation matrix, $B_{j,i-2}$ must be zero and $B_{j+2,i-2} = 1$.

Continuing similarly, we get $B_{j+r,i-r} = 1$ for every r with $3 \leq r \leq k - j$. In general, $B_{k,i+j-k} = 1$.

When $i + j$ is an even integer and $k = \dfrac{i+j}{2}$, $B_{k,i+j-k} = 1$ becomes $B_{\frac{i+j}{2},\frac{i+j}{2}} = 1$, which is not possible. Also, when $i + j + 1$ is even, and $k = \dfrac{i+j+1}{2}$, $B_{k,i+j-k} = 1$ becomes $B_{\frac{i+j+1}{2},\frac{i+j+1}{2}-1} = 1$, which is not possible since the vertices $v_{\frac{i+j+1}{2}}$ and $v_{\frac{i+j+1}{2}-1}$ are adjacent in G, and H is a subgraph of $\overline{G}$.

Therefore $B_{j-1,i-1} = 1$ and $B_{j+1,i-1} = 0$.

Using the fact that B is a permutation matrix, one can show that $B_{i-1,j-1} = 1$, $B_{i-2,j-2} = 1, \ldots, B_{1,j-i+1} = 1$, and $B_{i+1,j+1} = 1, B_{i+2,j+2} = 1, \ldots, B_{i+n-j,n} = 1$, and $B_{i+n-j+1,1} = 1$.

We have $B_{1,j-i+1} = 1$ and $B_{i+n-j+1,1} = 1$. These two together imply that

$$\begin{aligned} j - i + 1 &= i + n - j + 1, \\ 2(j - i) &= n, \\ j &= \frac{n}{2} + i \text{ for } i = 1, 2, \ldots, \frac{n}{2}. \end{aligned}$$

Thus H is a 1-factor graph with $v_i \sim_H v_{\frac{n}{2}+i}$, $1 \leq i \leq \frac{n}{2}$ and a 1-factor induced by vertices $v'_1, v'_2, \ldots, v'_m$.

Conversely, let H be a subgraph of $\overline{G} = K_m \cup \overline{C_n}$ satisfying the conditions of the theorem. Then $A(H) = \begin{pmatrix} B'_{n\times n} & 0_{n\times m} \\ 0_{m\times n} & D'_{m\times m} \end{pmatrix}$, where $B'_{n\times n}$ is the adjacency matrix of a 1-factor graph on the vertices $v_1, v_2, \ldots, v_n$, in which the ith row has the non-zero entry in the $(\frac{n}{2} + i)$th column, $1 \leq i \leq \frac{n}{2}$. The matrix $D'_{m\times m}$ is the adjacency matrix of a 1-factor graph on the vertices $v'_1, v'_2, \ldots, v'_m$.

Then, it is easy to observe that $A(G)A(H)$ is a symmetric, (0, 1) matrix with zero diagonal. Hence $A(G)A(H)$ is graphical. □

17.5 Realization of $A(G)A(G_k^P)$

In this section, we introduce the notion of perfect matching property for a k-partition of the vertex set of a given graph. We consider non-trivial graphs G and G_k^P, the

k-complement of G with respect to a k-partition of $V(G)$, to prove that $A(G)A(G_k^P)$ is realizable as a graph if and only if P has the perfect matching property. Also, we discuss the characterization of complete k-partite graphs $K_{n_1,n_2,\ldots,n_k}$ that have a commuting decomposition into a perfect matching and its k-complement. The notion of k-complement of a graph G with respect to a partition P of $V(G)$ with k parts is a generalization of the concept of the complement of a graph G introduced by [8].

Definition 17.10 [8] Given a graph G and a k-partition $P = \{V_1, V_2, \ldots, V_k\}$ of $V(G)$, the *k-complement* of G with respect to the partition P is a graph H with $V(H) = V(G)$ in which the adjacency between $v \in V_i$ and $w \in V_j$ is defined by

$$v \sim_H w \text{ if } \begin{cases} v \nsim_G w, & \text{for } i \neq j \\ v \sim_G w, & \text{for } i = j. \end{cases}$$

Definition 17.11 A collection of subgraphs $H_1, H_2, \ldots, H_k$ of G is said to be a *decomposition* of G if $\bigcup_{i=1}^{k} H_i = G$ and $E(H_i) \cap E(H_j) = \Phi$ for $i \neq j$.

The above graph H is usually denoted by G_k^P and is also known as a generalized complement of graph G, where it reduces to the well-known complement of G when k is equal to n, the number of vertices in the graph G. From the definition, it is clear that the subgraph induced by V_i, $1 \leq i \leq k$, in G and G_k^P are the same.

Definition 17.12 A decomposition $H_1, H_2, \ldots, H_k$ of G is said to be a *commuting decomposition* of G if $H_1, H_2, \ldots, H_k$ are spanning subgraphs and $A(H_i)A(H_j) = A(H_j)A(H_i)$ for all i and j with $1 \leq i, j \leq k$.

Theorem 17.22 *Let G be a graph, and $P = \{V_1, V_2, \ldots, V_k\}$ be a k-partition of $V(G)$. The matrix product $A(G)A(G_k^P)$ is graphical if and only if the following statements are true.*

(i) The subgraph $\langle V_i \rangle$ is totally disconnected for every i, $1 \leq i \leq k$.
(ii) For every pair of vertices $u \in V_i$ and $v \in V_j$, there exists at most one vertex $w \in V_r$, $r \neq i$, j and $1 \leq i, j, r \leq k$, such that $u \sim_G w$ and $w \nsim_G v$.
(iii) For a pair of vertices u and v, if there exists w satisfying (ii), then there exists a unique $w' \in V_l$, $l \neq i, j$, such that $v \sim_G w'$ and $w' \nsim_G u$.

Proof Let $A(G)A(G_k^P)$ be graphical. If $u \sim_G v$ for any $u, v \in V_i$, then from the definition of G_k^P we see that $u \sim_{G_k^P} v$. This contradicts the fact that G_k^P is a subgraph of $\overline{G}$. This proves (i).

From (ii) of Theorem 17.1, we know that for every pair of vertices u and v, there exists a vertex w different from u and v such that $u \sim_G w$ and $w \sim_{G_k^P} v$. Since G_k^P is a subgraph of $\overline{G}$, it is trivial that $w \nsim_G v$. Further, if $w \in V_r$ for some r, then from (i), it follows that r is neither equal to i nor equal to j. Thus (ii) is proved.

Proof of (iii) follows immediately from (iii) of Theorem 17.1.

Conversely, let (i), (ii), and (iii) of the theorem be satisfied. Consider any two vertices $u \in V_i$ and $v \in V_j$. Since $\langle V_i \rangle$ and $\langle V_j \rangle$ are totally disconnected in G, from

the definition of G_k^P and from (ii), it follows that there is at most one G-G_k^P path between u and v. For the same reason, (iii) implies that there exists at most one G_k^P-G path between u and v whenever there exists one G-G_k^P path between u and v. Now, referring to Theorem 17.1, we see that $A(G)A(G_k^P)$ is graphical. □

Example 5

Consider the graph G given in Fig. 17.14, and its 4-complement G_4^P with respect to the partition $P = \{V_1, V_2, V_3, V_4\}$ as shown. It is clear that $\langle V_1\rangle$, $\langle V_2\rangle$, $\langle V_3\rangle$, and $\langle V_4\rangle$ are totally disconnected and also satisfy conditions (ii) and (iii) of Theorem 17.22. Further, by direct computation of $A(G)A(G_4^P)$, also given below, we observe that the product is graphical. The graphs G, G_4^P, and their matrix product Γ are given in Fig. 17.14.

$$A(G) = \begin{bmatrix} 0&0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0 \end{bmatrix}, A(G_4^{P_1}) = \begin{bmatrix} 0&0&0&0&1&1&1&1\\ 0&0&0&1&0&1&1&1\\ 0&0&0&1&1&0&1&1\\ 0&1&1&0&0&0&1&1\\ 1&0&1&0&0&0&1&1\\ 1&1&0&0&0&0&1&1\\ 1&1&1&1&1&1&0&0\\ 1&1&1&1&1&1&0&0 \end{bmatrix}, A(G)A(G_4^{P_1}) = \begin{bmatrix} 0&1&1&0&0&0&1&1\\ 1&0&1&0&0&0&1&1\\ 1&1&0&0&0&0&1&1\\ 0&0&0&0&1&1&1&1\\ 0&0&0&1&0&1&1&1\\ 0&0&0&1&1&0&1&1\\ 1&1&1&1&1&1&0&0\\ 1&1&1&1&1&1&0&0 \end{bmatrix}.$$

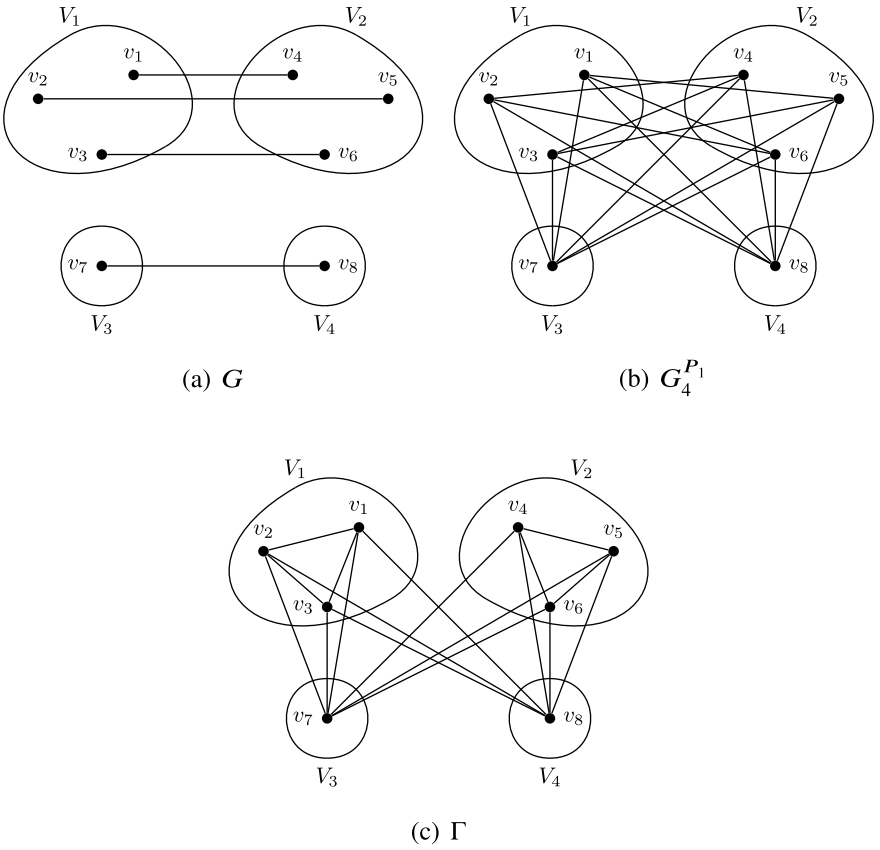
Let G be a graph on n vertices and $P = \{V_1, V_2, \ldots, V_n\}$ be a partition of $V(G)$ of size n. Then every part is a singleton set and the n-complement G_n^P is the same as the actual complement $\overline{G}$ of G. Then, by Theorem 17.8, $A(G)A(G_n^P) = A(G)A(\overline{G})$ is graphical if and only if G is either a 1-factor graph or an $(n-2)$-regular graph, or else C_5, the cycle on five vertices. So, while characterizing graph G and partition P of $V(G)$ satisfying the property that $A(G)A(G_k^P)$ is graphical, we consider $k \neq n$.

Definition 17.13 Given a graph G, a k-partition $P = \{V_1, V_2, \ldots, V_k\}$ of $V(G)$ is said to have *perfect matching property* in G if V_i is an independent set for every i, $1 \leq i \leq k$, and for every V_i there exists exactly one V_j, $1 \leq i, j \leq k$ and $i \neq j$, such that the graph induced by the set $V_i \cup V_j$ is a perfect matching. In such a case, V_i and V_j are said to be the *matching partite sets*.

Lemma 17.7 *Let G be a 1-factor graph and let $P = \{V_1, V_2, \ldots, V_k\}$ be a k-partition of $V(G)$ $(k < n)$. Then $A(G)A(G_k^P)$ is graphical if and only if the partition P has the perfect matching property.*

Proof Let G be a 1-factor graph, and suppose that $A(G)A(G_k^P)$ be graphical. Since $k \neq n$, there exists at least one V_i with two or more vertices. Consider any two vertices u and v from V_i, $1 \leq i \leq k$. Since $A(G)A(G_k^P)$ is graphical, from (i) of Theorem 17.22, we know that $\langle V_i\rangle$ is totally disconnected and therefore $u \nsim_G v$. Since G is a 1-factor graph, there exist distinct vertices w_1 and w_2 such that $u \sim_G w_1$ and $v \sim_G w_2$, and further, $w_1 \sim_G w_2$.

Now, to prove that P has perfect matching property, it is enough if we prove that both w_1 and w_2 are in the same partite set in P. Suppose, to the contrary, that w_1 and



(a) G (b) $G_4^{P_1}$

(c) Γ

Fig. 17.14 Graphs G, $G_4^{P_1}$, and Γ with respect to partition P of $V(G)$

w_2 are in different partite sets, say $w_1 \in V_r$ and $w_2 \in V_s$, $r \neq s$. From the definition of G_k^P, we observe that there exists a G-G_k^P path $u \sim_G w_1 \sim_{G_k^P} w_2$. But there is no G_k^P-G path between u and w_2, as v is the only vertex adjacent to w_2 in G and u and v remain non-adjacent in G_k^P. This implies that $A(G)A(G_k^P)$ is not graphical, which is a contradiction. Therefore, w_1 and w_2 must be in the same partite set, say V_r. Thus, $\langle V_i \cup V_r \rangle$ is a perfect matching and therefore P has perfect matching property.

Conversely, let the partition $P = \{V_1, V_2, \ldots, V_k\}$ of $V(G)$ have perfect matching property. Then, each partite set V_i is an independent set in the graph G and therefore, from the definition of G_k^P, we see that G_k^P is a subgraph of $\overline{G}$. Since G is a 1-factor graph, there exists at most one G-G_k^P path between any two vertices. Now, let there be a G-G_k^P path between the vertices u and v given by $u \sim_G x \sim_{G_k^P} v$. Let $u \in V_i$, $v \in V_j$, and $x \in V_r$. Since P has perfect matching property, we have $r \neq i$, and V_i and V_r are matching partite sets. Also, $r \neq j$ as V_j is an independent set in the graph G as well as in the graph G_k^P. Now for $y \sim_G v$, it is clear that y is not in V_i, as V_i and

V_j are not matching partite sets. So, y is not adjacent to u in G and therefore we obtain a G_k^P-G path between u and v, given by $u \sim_{G_k^P} y \sim_G v$. Hence, by Theorem 17.1, $A(G)A(G_k^P)$ is graphical. □

Theorem 17.23 *Let G be a graph on n vertices and $P = \{V_1, V_2, \ldots, V_k\}$ be a k-partition of $V(G)$ with $k \neq n$. Then $A(G)A(G_k^P)$ is graphical if and only if one of the following holds.*

(i) Either G or G_k^P is a null graph, in which case the other one is a complete k-partite graph.
(ii) The partition P has perfect matching property either in G or in G_k^P.

Proof Let $A(G)A(G_k^P)$ be graphical, in which case each of the parts is an independent set in G. Therefore, from the definition of k-complement of G, it is trivial that G is a null graph (respectively complete k-partite graph) if and only if G_k^P is a complete k-partite graph (respectively null graph). In this case (i) holds.

Suppose that G is neither the null graph nor a complete k-partite graph. Consider a pair of vertices u and v from one part of the partition, say V_1, which is possible as $k \neq n$. Since $A(G)A(G_k^P)$ is graphical, by Theorem 17.22, we have one of the following cases:

(a) For all $w \notin V_1$, $u \sim_G w$ and $w \sim_G v$ (equivalently, $u \sim_{G_k^P} w$ and $w \sim_{G_k^P} v$).
(b) There exists a unique $r \notin V_1$ such that $u \sim_G r$ and $r \sim_{G_k^P} v$, in which case G has a vertex s such that $u \sim_{G_k^P} s$ and $s \sim_G v$.

Case (a): For any $x \notin V_1$, it is not possible that both $u \sim_{G_k^P} x$ and $x \sim_{G_k^P} v$, for otherwise there would be two G-G_k^P paths between w and x. So, for every $x \notin V_1$, x is adjacent to both of u and v in G. This implies that $\deg_{G_k^P} u = 0$. By Note 17.2, $\deg_{G_k^P} u = \deg_{G_k^P} x = 0$ for all $x \notin V_1$. Further, from the definition of k-complement of G and the fact that V_1 is an independent set in G, it follows that $\deg_{G_k^P} v = 0$, since $\deg_{G_k^P} x = 0$ for all $x \notin V_1$. Hence, G_k^P is the null graph, which is not the case under discussion.

Case (b): Without loss of generality, let there exist a unique $r \notin V_1$ such that $u \sim_G r$ and $r \sim_{G_k^P} v$, in which case there exists a vertex s in G such that $u \sim_{G_k^P} s$ and $s \sim_G v$. For any $t \neq r, s$ and $t \notin V_1$, both u and v are simultaneously adjacent or simultaneously non-adjacent to t in G. Also, note that there is no pair of vertices p and q such that p is adjacent to both u and v in G, and q is non-adjacent to both u and v in G. Otherwise, there would be two G-G_k^P paths ($p \sim_G u \sim_{G_k^P} q$ and $p \sim_G v \sim_{G_k^P} q$) between p and q.

If both the vertices u and v are not adjacent to $t \notin V_1$ and $t \neq r, s$, then $\deg_G u = \deg_G v = 1$. As u and v are adjacent to t in G_k^P, from Note 17.2, we have $\deg_G t = \deg_G u = \deg_G v = 1$ for all $t \notin V_1$. Since r and v are adjacent to each other in G_k^P, $\deg_G r = 1$. For all $z \in V_1$ and $z \neq u$, we have $r \sim_{G_k^P} z$, and by Note 17.2, it implies that $\deg_G z = 1$.

Similarly, in case both the vertices u and v are adjacent to $t \notin V_1$ and $t \neq r, s$, we prove that G_k^P is 1-factor graph. So in the case (ii), either G or G_k^P is 1-factor

graph. Therefore, by Lemma 17.7, it follows that the partition P satisfies the perfect matching property.

The converse part is trivial if G and G_k^P satisfy the condition (i) given in the theorem. If G and G_k^P satisfy the condition (ii), the converse part follows from Lemma 17.7, as graph G with partition P satisfying perfect matching property is trivially a 1-factor graph. □

Next, we obtain a commuting decomposition of a complete k-partite graph G into a perfect matching and its k-complement with respect to the k-partition of G. First we note the following.

Note 17.14 From Theorem 17.1, one can conclude that the (i, j) entry of $A(G)A(H)$ represents number of G-H paths from v_i to v_j. Hence, G and H commute with each other if and only if for every two vertices v_i and v_j, $i \neq j$, $1 \leq i, j \leq n$, the number of G-H paths from v_i to v_j is same as the number of H-G paths from v_i to v_j.

Theorem 17.24 explains the commuting decomposition of $K_{n_1,n_2,\ldots,n_k}$ into a perfect matching and its k-complement. We consider the partition P to be the k-partition of the complete k-partite graph $K_{n_1,n_2,\ldots,n_k}$.

Theorem 17.24 *Let G be a complete k-partite graph $K_{n_1,n_2,\ldots,n_k}$ on n vertices with the k-partition of $V(G)$ given by $P = \{V_1, V_2, \ldots, V_k\}$ and let $|V_i| = n_i$, $1 \leq i \leq k$. Then G is decomposable into two commuting subgraphs, one of which is a perfect matching of G, say H_1, and the other one is the k-complement of H_1 with respect to the same partition $P = \{V_1, V_2, \ldots, V_k\}$, if and only if the number of partite sets in P with the same cardinality n_i is even for every i, $1 \leq i \leq k$.*

Proof Consider the complete k-partite graph $G = K_{n_1,n_2,\ldots,n_k}$. Suppose that G has a commuting decomposition into two subgraphs H_1 and H_2, one of which, say H_1, is a perfect matching of G and the other one is given by $H_2 = (H_1)_k^P$, the k-complement of H_1 with respect to the partition $P = \{V_1, V_2, \ldots, V_k\}$.

Since G has a perfect matching H_1, the number of vertices in G is even. If all the partite sets are singletons, then $k = n$ is also even and the assertion of the theorem is obviously true.

Suppose that there is a partite set, say V_i, $1 \leq i \leq k$, which has two or more vertices. Let $u \in V_i$ and $v \in V_j$ with $u \sim_{H_1} v$, $1 \leq i, j \leq k, i \neq j$. Let $w \in V_i$ and let $w \sim_{H_1} x$. If $x \in V_t, t \neq j$, then there is a H_1-H_2 path from w to v but no H_2-H_1 path from w to v which by Note 17.14 contradicts the assumption that H_1 and H_2 commute with each other. Hence, if $u \in V_i$ and $v \in V_j$, $1 \leq i, j \leq k, i \neq j$, if $u \sim_{H_1} v$, then for each vertex in V_i the only adjacent vertex of H_1 is also in V_j. The same is true for V_j which implies that $|V_i| = |V_j|$. Similarly, partite sets with the same cardinality are paired and since H_1 is a perfect matching, the number of partite sets with cardinality n_i is even for every i, $1 \leq i \leq k$.

Conversely, assume that the number of partite sets with the same cardinality is even. Then we can pair the partite sets having the same cardinality. Without loss of generality, assume that $|V_1| = |V_2| = n_1, |V_3| = |V_4| = n_2, \ldots, |V_{2r-1}| = |V_{2r}| =$

n_r, where $2r = k$. Observe that n_i may be equal to n_j for some i and j, $1 \leq i, j \leq r, i \neq j$. Since for every i, $1 \leq i \leq r$, $\langle V_{2i-1} \cup V_{2i} \rangle$ is a complete bipartite graph with $|V_{2i-1}| = |V_{2i}|$, there is a perfect matching in the induced graph. The union of all these perfect matching gives a perfect matching in G, denoted by H_1. In the graph $G \backslash H_1$, obtained by deleting the edges of H_1 from G, two vertices are adjacent if and only if they are in two different partite sets and they are not adjacent in H_1. Hence, $G \backslash H_1$ is same as $(H_1)_k^P$ which implies that $G = H_1 \cup (H_1)_k^P$ and $E(H_1) \cap E((H_1)_k^P) = \Phi$. Thus H_1 and $H_2 = (H_1)_k^P$ form a decomposition of G. To show that H_1 and $(H_1)_k^P$ form a commuting decomposition of G, we show that for any two vertices u and v in G, the number of H_1-H_2 path from u to v is same as the number of H_2-H_1 path from u to v.

Consider u, v in the same partite set V_i. Let $u \sim_{H_1} u'$ and $v \sim_{H_1} v'$. Then, $u \sim_{H_1} u' \sim_{H_2} v$ is the only H_1-H_2 path and $u \sim_{H_2} v' \sim_{H_1} v$ is the only H_2-H_1 path from u to v.

Consider $u \in V_i$ and let $u' \in V_j$ with $u \sim_{H_1} u'$. For any other vertex x in V_j, there neither exists a H_1-H_2 path nor exists a H_2-H_1 path from u to x. Consider a vertex $x \in V_l, l \neq i, j$, with $x \sim_{H_1} x'$. Then $u \sim_{H_1} u' \sim_{H_2} x$ is the only H_1-H_2 path and $u \sim_{H_2} x' \sim_{H_1} x$ is the only H_2-H_1 path from u to x. Hence the decomposition $\{H_1, H_2 = (H_1)_k^P\}$ of G is a commuting decomposition. □

Note 17.15 Let G be a complete k-partite graph $K_{n_1,n_2,\ldots,n_k}$ with respect to a partition $P = \{V_1, V_2, \ldots, V_k\}$ of $V(G)$. Then the graph G is decomposable into two commuting perfect matchings—i.e., both H_1 and $H_2 = (H_1)_k^P$ are perfect matchings only when $n = 4$ and $k = 2$. Equivalently, $K_{2,2}$ is the only complete k-partite graph, with $k = 2$ which is decomposable into two commuting perfect matchings as above.

Note 17.16 Let G be a complete k-partite graph $K_{n_1,n_2,\ldots,n_k}$ with respect to a partition $P = \{V_1, V_2, \ldots, V_k\}$ of $V(G)$. Then the graph G is decomposable into two commuting subgraphs, one of which is a perfect matching and the other one is a Hamiltonian cycle, i.e., H_1 is a perfect matching and $H_2 = (H_1)_k^P$ is a Hamiltonian cycle when either $n = 6$ and $k = 2$ or $n = 4$ and $k = 4$. Equivalently, $K_{3,3}$ and $K_{1,1,1,1}$ are the only complete k-partite graphs, with $k = 2$ and $k = 4$ respectively, which are decomposable into two commuting subgraphs, one of them being a perfect matching and the other one a Hamiltonian cycle.

17.6 Companion Search Algorithm

This section contains the companion search algorithm which, for a given graph G, constructs a non-trivial companion H of G whenever such a companion exists, and returns $H = \overline{K_n}$ otherwise. It also returns the matrix product Γ of G and H.

We know that whenever there is a G-H path from a vertex u to a vertex v, it results in an edge in Γ only when there also exists an H-G path from u to v. For the sake of convenience, we say that there is an *arc* (i.e., a directed edge) from u to v in Γ

whenever there is a G-H path from u to v. When there is an arc from u to v as well as an arc from v to u in Γ, there is a *symmetric pair of arcs* between u and v in Γ, i.e., there is an (undirected) edge between u and v in Γ. If there is a G-H path, but not an H-G path, from u to v, then we say that the arc from u to v in Γ is *asymmetric*.

The algorithm makes use of the results of Theorem 17.1. We assume that the input graph G contains no isolated vertex and that the vertex set of G is $V(G) = \{0, 1, \ldots, n-1\}$. As G and H are edge disjoint, the algorithm investigates the possibility of each edge in $\overline{G}$ to be an edge in H. So, it maintains a list of possible edges of H, $E(H)$, and a corresponding list of possible edges of Γ, $E(\Gamma)$.

The algorithm takes as input the adjacency list of the graph G. The graphs H and Γ are initialized to have empty adjacency lists, that is, the adjacency list of $\overline{K_n}$. The algorithm starts at a *root* vertex, say 0. It searches for the first edge $(0, j)$ that is not an edge of G, out of all possible edges $(0, 1), (0, 2), \ldots, (0, n-1)$ in lexicographic order. If no such edge exists, then no non-trivial companion exists for G, and the algorithm returns $H = \overline{K_n}$ and $\Gamma = \overline{K_n}$ as the output. If such an edge exists but is already included in the adjacency list of H, then the algorithm goes to the next possible selection of H edge.

Once any edge, say (i, j), is added to the adjacency list of H, then the algorithm will add edges to the adjacency list of Γ corresponding to all the G-H paths involving the edge (i, j) of H. That is, $E(H)$ is updated to $E(H) \cup \{(i, j)\}$ and $E(\Gamma)$ is updated to $E(\Gamma) \cup \{ (r, i) \mid r \in N_G(j) \} \cup \{ (k, j) \mid k \in N_G(i) \}$.

Next, the algorithm starts searching for the first asymmetric arc in $E(\Gamma)$. Suppose that there is an asymmetric arc from u to v in Γ. Now, to find an arc from v to u, the algorithm adds the first feasible edge (x, u), in lexicographic order, to $E(H)$, $x \in N_G(v)$. If no such feasible edge exists, then the algorithm goes to the next possible choice of edge for H, and otherwise it adds that edge to the adjacency list of H. Then it repeats the above steps until there are no more asymmetric arcs in $E(\Gamma)$. If there is no feasible edge for H, then it will come to the root vertex and select the next feasible edge. If there is no such asymmetric arc in Γ, then it returns the first non-null pair (H, Γ) as output or returns $(\overline{K_n}, \overline{K_n})$ as output.

Note that in the algorithm,

$$\overrightarrow{N}_{\Gamma}(u) = \{ v \in V(G) \mid \text{there exists } w \in V(G) \text{ with } u \sim_G w \sim_H v \}.$$

17.7 Conclusion and Open Problems

For the adjacency matrix $A(G)$, the nth power has many nice properties. But not even $((A(G))^2 = A(G)A(G)$ is graphical. We considered the problem of searching for a graph H with the property that $A(G)A(H)$ is graphical. In the continuation of our study, we obtained many results and some new characterizations of certain classes

Algorithm 3

```
1: function FINDHΓ(G)
2:     n := |V(G)|
3:     H_0 := K̄_n
4:     Γ_0 := K̄_n
5:     for j ∈ {1, ..., n − 1} do
6:         (H, Γ) := ADDEDGE(G, H_0, Γ_0, j, (0, j))
7:         if H ≠ K̄_n then                    ▷ Nontrivial companion obtained
8:             return (H, Γ)
9:         end if
10:    end for                                ▷ No non-trivial companion graph found
11:    return (H_0, Γ_0)
12: end function
```

of graphs in terms of the product of graph matrices. A few of these are mentioned below.

For a graph G on n (≥ 4) vertices, G is a cycle if and only if $A(G)B(G)$ is an $n \times n$ matrix in which every column and every row contains exactly four 1s.

For a graph on n (≥ 3) vertices and m edges, G is a complete bipartite graph if and only if $A(G)B(G)$ is an $n \times m$ matrix in which all entries are 1.

The graph $K_{2,2}$ is the only complete k-partite graph with $k = 2$ which is decomposable into two commuting perfect matchings. Also, $K_{3,3}$ and $K_{1,1,1,1}$ are the only complete k-partite graphs, with $k = 2$ and $k = 4$ respectively, which are decomposable into two commuting subgraphs, one of which is a perfect matching and the other one a Hamiltonian cycle.

The following problems could be a goal for future work.

(i) There is no closed-form formula for the number of edges in $A(G)A(H)$ in terms of the number of edges in G and the number of edges in H. One can attempt to find such a formula.
(ii) If $A(G)A(H)$ is graphical then H is a subgraph of $\overline{G}$. A few edges of $\overline{G}$ are to be removed to get H unless $H = \overline{G}$. One may try to characterize such edges.
(iii) We know that trees and P_n^m, $2 \leqslant m \leqslant n - 1$, have no non-trivial companions. One can find some more classes in which no graph has a non-trivial companion.
(iv) One may try to characterize the realizability of products of other graph matrices such as the distance matrix or Laplacian matrix, and explore the structural property of the realizing graph.
(v) One can study the relation between the distances in G, H, and the realizing graph Γ, and hence study the relation between some stress and associated properties of these graphs.
(vi) One may try to improve the companion search algorithm by making use of some properties of the graph factors.

```
1: function AddEdge(G, H, Γ, z, (i, j))
2:     n := |V(G)|
3:     if (i = 0 and j < z) or (j = 0 and i < z) then
4:         return (K̄_n, K̄_n)
5:     else if i = j then
6:         return (K̄_n, K̄_n)
7:     else if j ∈ N_G(i) then
8:         return (K̄_n, K̄_n)
9:     else
10:        E(H) := E(H) ∪ {(i, j)}
11:        for k ∈ N_G(i) do
12:            if (k, j) ∉ E(Γ) then
13:                E(Γ) := E(Γ) ∪ {(k, j)}
14:            else
15:                return (K̄_n, K̄_n)
16:            end if
17:        end for
18:        for k ∈ N_G(j) do
19:            if (k, i) ∉ E(Γ) then
20:                E(Γ) := E(Γ) ∪ {(k, i)}
21:            else
22:                return (K̄_n, K̄_n)
23:            end if
24:        end for
25:        for u ∈ V(Γ) do
26:            for v ∈ N⃗_Γ(u) do
27:                if u ∉ N⃗_Γ(v) then                ▷ Asymmetric arc
28:                    for x ∈ N_G(v) do
29:                        (H′, Γ′) := AddEdge(G, H, Γ, z, (x, u))
30:                        if H′ ≠ K̄_n then           ▷ Nontrivial companion
31:                            return (H′, Γ′)
32:                        else
33:                            continue
34:                        end if
35:                    end for
36:                    return (K̄_n, K̄_n)
37:                else
38:                    continue                       ▷ Check next v
39:                end if
40:            end for
41:        end for                                    ▷ No asymmetric arcs found
42:        return (H, Γ)
43:    end if
44: end function
```

References

1. Akbari S, Moazami F, Mohammadian A (2009) Commutativity of the adjacency matrices of graphs. Discret Math 309(3):595–600. https://www.sciencedirect.com/science/article/pii/S0012365X08005359, International workshop on design theory, graph theory, and computational methods
2. Barik S, Kalita D, Pati S, Sahoo G (2018) Spectra of graphs resulting from various graph operations and products: a survey. Spec Matrices 6(1):323–342
3. Bhat KA, Sudhakara G (2018) Commuting decomposition of $K_{n_1,n_2,\ldots,n_k}$ through realization of the product $A(G)A(G_k^P)$. Spec Matrices 6(1):343–356
4. Bhat KA, Prasad KM, Sudhakara G (2016) Some matrix equations of graphs. Adv Appl Discret Math 17(1):29–48
5. Bhat KA, Sudhakara G, Vinay M (2022) Algorithm to check the existence of H for a given G such that $A(G)A(H)$ is graphical. Discret Math Algorithms Appl 14(5)
6. Prasad KM, Sudhakara G, Sujatha H, Vinay M (2013) Matrix Product of Graphs. Springer India, India, pp 41–55
7. Prasad KM, Sudhakara G, Sujatha HS, Soumya KV (2014) Matrix product(modulo-2) of graphs. In: Indian Journal of Pure and Applied Mathematics, pp 851–860
8. Sampathkumar E, Pushpalatha L (1998) Complement of a graph: a generalization. Graphs Combinatorics 14:377–392
9. West DB (1996) Introduction to graph theory. Prentice Hall

Chapter 18
On Rao's Weighted Distributions for Modeling the Dynamics of Wildfires and Air Pollution



Saurav Guha, Alexander Petersen, Sumanta Ray, and Saumyadipta Pyne

Mathematics Subject Classification (2010) 00A71 · 86A08

18.1 Historical Background

Calyampudi Radhakrishna Rao, popularly known as "C. R. Rao" (henceforth, Rao), was born on September 10, 1920, in the Madras Presidency of British India. He is widely regarded as a "living legend" in the field of statistics, and known for Cramér-Rao Bound, Rao-Blackwell Theorem, Rao Score Test, Fisher-Rao Distance, Generalized Inverse, Quadratic Entropy, and Orthogonal Arrays, to name only a few among his numerous path-breaking contributions to the subject. In comparison, the topic of weighted distributions, as formulated by Rao in 1963, is probably less well known. Yet, it has led to great advances in the subsequent decades both in theoretical as well as applied areas of statistics; and curiously, in the arc of Rao's remarkably long

S. Guha
ICAR-Indian Agricultural Statistics Research Institute, New Delhi, India
e-mail: saurav.iasri@gmail.com; sguha@healthanalytics.net

S. Guha · S. Ray · S. Pyne
Health Analytics Network, Pittsburgh, PA, USA
e-mail: sumanta.ray@aliah.ac.in; sray@healthanalytics.net

A. Petersen
Department of Statistics, Brigham Young University, Provo, UT, USA
e-mail: petersen@stat.byu.edu

S. Ray
Department of Computer Science and Engineering, Aliah University, Kolkata, WB, India

S. Pyne (✉)
Department of Statistics and Applied Probability, University of California Santa Barbara, Santa Barbara, CA, USA
e-mail: spyne@ucsb.edu; spyne@healthanalytics.net

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_18

career, as the present study suggests, this particular topic perhaps made a uniquely historic contribution.

In 1911–1914, Henry Wellcome, the British pharmaceutical entrepreneur, led, over four seasons, an archeological expedition at the Jebel Moya site in the southern Gezira plain of Sudan [4]. It is the site of the largest pastoralist cemetery in Northeast Africa, currently dated to 5000-500 BCE [9]. The excavated skeletal remains were shipped to London. Here, these were warehoused poorly and, in fact, even suffered from flooding. After the death of Wellcome in 1936, and by the end of World Wars I and II, the remains and the excavation records were transferred to the Duckworth Laboratory in the University Museum of Archaeology and Ethnology at Cambridge University under the curatorship of J. C. Trevor.

In March 1946, P. C. Mahalanobis at the Indian Statistical Institute (ISI) in Kolkata received a telegram from Trevor asking him to send someone to help with the anthropomorphic analysis of the collection. Incidentally, Mahalanobis had assigned to Rao a project on the analysis of an anthropometric survey, which used the D^2 distance for grouping of Indian populations [23]. Thus, Rao, along with Ramkrishna Mukherjee, an anthropologist, went to England in August of 1946 to work as visiting scholars at the Cambridge University Museum of Archaeology and Anthropology. The aim of their study was "to undertake laboratory examination of identifiable and usable adult specimens, to analyze the measurements and observations of the field physical anthropologists, and to determine the relationship between the Jebel Moya inhabitants and other African peoples" [9].

The long overdue report of this historic project was finally published owing to Mukherjee, Rao, and Trevor in 1955 [27]. Not surprisingly, the study posed certain technical challenges. In particular, the 40-year hiatus between the excavation and the anthropometric data analysis had proved to be catastrophic for the remains, which, according to Trevor, had "disintegrated beyond hope of repair" [9]. Out of more than 3,000 skeletal parts that were originally excavated, only 98 crania, 139 mandibles, and a few post-cranial elements had survived intact.

Rao's task was to estimate the unknown mean cranial capacity and other features of the *original* Jebel Moya population from the damaged remains, many of which had measurements missing due to damage. Toward maximum likelihood estimation, one could write the likelihood function using a multivariate (normal) distribution based on the samples with complete measurements, and the derived marginal distribution for those with incomplete sets of measurements. However, such estimation is based on the assumption that each skull can be considered to belong to a *random* sample from the original population of skulls. The key question would then be: was such an assumption valid?

While looking at the samples that survived, Rao (Fig. 18.6) made a curious observation, "are only small skulls preserved?" [38]. If $w(c)$ is the probability that a skull of capacity c is unbroken, then in archeological recovery, i.e., during the data gathering process, it is known that $w(c)$ is a decreasing function of c. The larger a skull, the greater is its chance of being damaged upon burial and recovery, as Rao noted in his study of yet another collection [41]. This will lead to a larger representation of small skulls among the unbroken cranial remains, and, therefore, the mean of the available

measurements of the corresponding random variable C will be an underestimate of the mean cranial capacity of the original population.

Alongside this full-time job, Rao, also a student at King's College, Cambridge, was "suggested" by his Ph.D. advisor R. A. Fisher to work simultaneously at the latter's genetics laboratory. Fisher had long been interested in the concept of "ascertainment"—a mode of sampling that depended on the outcome which one wanted to analyze as a dependent variable. In a classical paper, Fisher studied how the methods of ascertainment can influence the form of the distribution of recorded observations [12]. One can assume a model that has been adjusted for ascertainment will estimate parameters in the *general* population from which the sample was drawn. In usual statistical practice, it is generally assumed that a random sample from a population to be studied *can be* observed in data. Not surprisingly, therefore, the key specification of "what population does a sample represent?" might be taken for granted.

However, in many situations, obtaining a random sample may be practically too difficult, or too costly, or indeed, even less preferable to the non-random data that are actually available, e.g., from field observations or non-experimental data or a survey lacking a suitable sampling frame, as in many "big data" scenarios. Sometimes, the events may be observed only in modified form, e.g., in damage models. Indeed, certain events may be rendered unobservable by the adoption of the very method that is used for making observations, and are, therefore, missed unwittingly. Or they might be observable but only with certain probabilities or weights that may depend on sample-specific characteristics, such as their inherent conspicuousness, as well as other unknown parameters.

Fisher's problem of ascertainment was formally addressed by Rao when, during August 15–20, 1963, G. P. Patil organized the "First International Symposium on Classical and Contagious Distributions" at McGill University in Montreal. Patil recollected in the symposium proceedings that Rao had advised him to study discrete distributions when he visited ISI a decade earlier. At a second-day session chaired by Jerzy Neyman, Rao presented the first paper that formulated and unified weighted (discrete) distributions in general terms, which appeared in the proceedings edited by Patil [34], and was later reprinted in *Sankhyā* [35].

18.2 Weighted Distributions

Let X be a random variable with probability density function (pdf) $g(x;\theta)$ with parameters θ. The traditional statistical analysis assumes that an identically (as well as independently) distributed random sample $X_1, \ldots, X_n$ can always be observed. Weighted distributions arise when $X = x$ enters the sample with a non-zero weight $w(x, \alpha)$ that depends on the observed value x and possibly also on some unknown parameter α. Then x is not an observation on X but on the resulting random variable X^w which has the weighted pdf:

$$f(x;\theta,\alpha) = \frac{w(x,\alpha)\cdot g(x;\theta)}{\omega}.$$

The denominator ω is the normalizing factor chosen to be $E[w(X;\alpha)]$ so that $f(x;\theta,\alpha)$ integrates to 1. The weight $w(x,\alpha)$ could be any non-negative function for which $E[w(X;\alpha)]$ exists. When x is univariate and non-negative, then the weighted distribution

$$f(x;\theta) = \frac{x\cdot g(x;\theta)}{E(X)},$$

for $w(x,\alpha) = x$ is called *size-biased*; and *length-biased* for $w(x,\alpha) = |x|$ where $|x|$ is some measure of "length" of x. Many length-biased distributions could be shown to belong to the same family as their unweighted versions [34]. Size-biased sampling and related form-invariant weighted distributions were studied by Patil and Ord [28]. Patil and Rao [30] studied size-biased sampling with various applications to wildlife populations and human families.

Weighted distributions are useful for modulating the probabilities of the events as they are observed by means of collecting data possibly under less-than-perfect conditions. In this regard, prominent application areas of weighted distributions include truncation and censoring, damage models, size-biased sampling, nonresponse in surveys, "file-drawer" problem in meta-analysis, etc. Even mixtures of distributions can be shown to belong to this general formulation [22]. Asymmetric data bias is common in many scenarios that may have serious practical implications such as for unemployment statistics which fail to account for the discouraged workers who have stopped looking for jobs during a recession [46]. Interestingly, sometimes biased samples available from observational studies may contain more (Fisher) information than their randomly drawn counterparts [7]. Rao's studies, e.g., Rao and Rubin [40], Patil and Rao [29], and Rao [37] further advanced the area, which he surveyed during his post-ISI years as the University Professor at the University of Pittsburgh [38, 39]. For a more recent review of weighted distributions, see Saghir et al. [42].

As an example of popular special cases of weighted distributions, we consider the well-known approach of Azzalini [6] that allows us to adjust the shape of and introduce skewness to a symmetric distribution, say, with pdf g and cumulative distribution function (cdf) G. Thus, a random variable $X \in R$ is said to have a skew distribution if its pdf is given by

$$f_X(x) = 2g(x)G(\alpha x),$$

based on a shape parameter, α, which helps in adjusting g to account for the asymmetry observed in the data, which are, thereby, better represented by f. In general, a p-variate Generalized Skew Elliptical (GSE) distribution of random variable $X \in R^p$ has pdf of the form

$$2g(x;\xi;\Omega)\pi(x-\xi),$$

where g is an elliptically contoured pdf with location parameter ξ, scale matrix Ω, and skewing function π. An example of a GSE pdf due to Branco and Dey [8] is

$$\frac{2}{\sqrt{\pi}} g(\Omega^{-\frac{1}{2}}(x-\xi)) \cdot \pi(\Omega^{-\frac{1}{2}}(x-\xi)),$$

which could be formulated as a weighted distribution as follows:

$$f(x;\theta) = \frac{g(\Omega^{-\frac{1}{2}}x)}{|\Omega|^{-\frac{1}{2}}},$$

$$w(x) = \pi(\Omega^{-\frac{1}{2}}x),$$

$$E[w(X)] = \frac{1}{2} \text{ (assuming } \xi = 0).$$

Here, the weight function w distorts the elliptical (symmetric) contours of f through the presence of asymmetric "outliers" in the observed sample due to GSE [15].

Azzalini's approach could be also extended to distributions that are asymmetric. Gupta and Kundu [16] introduced the new class of weighted exponential (WE) distribution by applying Azzalini's method to the exponential distribution. Let U and V be two non-negative continuous independent random variables with pdfs f and g and cdfs F and G, respectively. Then for a shape parameter $\alpha > 0$, the pdf of random variable $X = U$ if $V < \alpha U$ is given by

$$f_X(x) = \frac{f(x)G(\alpha x)}{\omega}, x > 0,$$

where $\omega = \int_0^\infty f(x)G(\alpha x)\,dx$ and $0 < \omega < \infty$. When U and V follow an exponential distribution with mean $1/\lambda$, a random variable X is said to have a WE distribution, denoted by $WE(\alpha, \lambda)$, if its pdf is given by

$$f_X(x) = (\frac{\alpha+1}{\alpha})\lambda e^{-\lambda x}(1 - e^{-\lambda\alpha x}), x > 0, \alpha > 0, \lambda > 0.$$

Here, α and λ are the shape and scale parameters respectively [24].

As shown by Gupta and Kundu [16], WE distribution possesses useful likelihood ratio properties and all its moments can be calculated explicitly. The shape parameter governs the shape of the pdf of the WE distribution, which is similar to those of the Weibull, gamma, or generalized exponential distributions. This makes WE suitable for analyzing positively skewed data. While the exponential distribution is among the most popular models used in reliability engineering and life data analysis, when wearing out or aging effects are present, the WE distribution could be very effective. In fact, it can be viewed as a hidden truncation model, as was also observed earlier

in the case of the skew-normal distribution [5]. Further advances based on the *WE* distribution can be found in recent reviews, e.g., Tomy et al. [44].

18.3 Weighted Systems for Data Fusion

Recent studies have extended the concept of weighted distributions to a weighted system of distributions. Using this general approach, we developed a computational framework for modeling the combined dynamics of environmental variables as observed by multiple sources, e.g., spatially distributed multivariate data streams, automated sensors, and surveillance networks [48, 49]. Often, environmental monitoring sites are not located randomly, the system may contain built-in redundancies, and could be difficult to maintain without failures. Therefore, by systematic fusion of a moderate number of nearby data sources, we can increase the system's predictive capability with a combined model.

We assume a reference event distribution g_0, and its possible distortions or "tilt"-ed forms $g_1, \ldots, g_m$ due to m different sources. This gives us a weighted system of distributions:

$$\begin{aligned} g_1(x) &= w_1(x) \cdot g_0(x) \\ &\vdots \\ g_m(x) &= w_m(x) \cdot g_0(x). \end{aligned}$$

Assume that we have data from each of $g_0, g_1, \ldots, g_m$. Then, the relationship between a baseline distribution and its distortions or tilts $w_k, k = 1, \ldots m$ allows us to do inference on the system's overall behavior based on the fusion of data observed from multiple, and possibly dependent, sources. In a weighted system, this is realized by a model for multiple distributions g_k to be simultaneously "regressed" against a common reference distribution g_0, thus letting our data fusion approach estimate the model parameters, including the densities, by using all data sources and not just the reference sample.

It was noted by Vardi [45] that data from one distribution could form a biased sample from another, and this could be modeled by the ratio of their densities. In recent years, the original idea of the density ratio model (DRM) has been developed into a powerful tool, e.g., Qin [31], Fokianos [13]. In particular, it has been found ideally suited for pooling information from multiple populations by Chen and Liu [11], Zhang and Chen [47], and Zhang et al. [49]. Other applications include estimating tail probabilities of exceeding a pre-specified event threshold, especially with few observations of extreme events, e.g., de Carvalho and Davison [10], Zhang et al. [48, 50], Kedem et al. [20]. In general terms, if an environmental variable X of interest has only moderate amounts of data from any given source, then, based on

such data, what is the probability of any observation of X exceeding a high threshold T?

To address this problem, we developed a data fusion approach that pools information across multiple sources using a multi-sample DRM given by

$$\frac{g_k(x)}{g_0(x)} = exp(\alpha_k + \beta_k^T \boldsymbol{h}_k(x)); k = 1, \ldots, m,$$

where g_0 represents the density of X (say, the level of a pollutant measured at a monitoring site) and $g_1, \ldots, g_m$ represent those at its m nearby sites. Instead of making parametric assumptions on each of these densities, DRM captures the parametric structure of their ratios with a common model. To prevent bias, large standard errors, and loss of power in inference, it is important to appropriately select the multi-dimensional distortion or tilt functions $\boldsymbol{h}_k$'s. The choice of such tilt could allow flexibility in parameterization or even be data-driven as shown by Kedem et al. [19]. Recently, a functional principal component analysis approach was used to identify an effective approximation for a tilt function in low-dimensional functional subspace [47].

Let $\boldsymbol{X}_0, \ldots, \boldsymbol{X}_m$ be the samples from the area of interest and its m nearby sites with sample sizes $n_0, \ldots, n_m$, respectively. The sample $\boldsymbol{X}_0$ is referred to as the reference sample; let G denote the corresponding reference cumulative distribution function (CDF). The fused sample is defined as $t = (\boldsymbol{X}_0^T, \ldots \boldsymbol{X}_m^T)^T$, with size $n = \sum_{k=0}^m n_k$. Zhang, Pyne, and Kedem [49] described inference based on the following empirical likelihood due to t:

$$L(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{G}) = \prod_{i=1}^n p_i \prod_{k=1}^m \prod_{j=1}^{n_k} exp(\alpha_k + \beta_k^T \boldsymbol{h}_k(X_{kj})),$$

where $p_i = d(G(t_i))$, and the estimates $\hat{\alpha}$, $\hat{\beta}$, and hence $\hat{p}_i$'s are obtained by maximizing the above likelihood with constraints:

$$\sum_{i=1}^n p_i = 1; \quad \sum_{i=1}^n p_i \, exp(\alpha_k + \beta_k^T \boldsymbol{h}_k(t_i)) = 1, k = 1, \ldots, m.$$

This gives the estimated reference CDF $\hat{G}(t) = \sum_{i=1}^n \hat{p}_i I[t_i \leq t]$ and the asymptotic result

$$\sqrt{n}(\hat{G}(t) - G(t)) \xrightarrow{d} N(0, \sigma(t)),$$

as $n \to \infty$. The expression of $\sigma(t)$ and other details regarding estimation and asymptotic result can be found in our recent paper [48]. Based on the above result, a 95% confidence interval of the tail probability $1 - G(T)$ for a given threshold T is given by

$$(1 - \hat{G}(t) - z_{0.025}\sqrt{\frac{\hat{\sigma}(t)}{n}}, 1 - \hat{G}(t) + z_{0.025}\sqrt{\frac{\hat{\sigma}(t)}{n}}).$$

Finally, an optimal choice of tilt function may be made based on a criterion that ensures better specification of the density ratio structure. For instance, such selection can be made using Akaike Information Criterion (AIC) given by $-2logL(\hat{\alpha}, \hat{\beta}, \hat{G}) + 2q$ where q is the number of free parameters in the model [49]. Applications of the weighted systems framework in modeling augmented reality and estimation of exceedance probabilities can be found in our recent papers such as Kedem and Pyne [18], Kedem et al. [20], Zhang et al. [50].

18.4 Application: Wildfires and Air Pollution Data

As described above, the capacity of introducing weighting mechanisms to adjust the shape or suitably "distort" the standard distributions to be able to model real-world data is a powerful tool. To monitor a dynamic phenomenon, modeling the change between the distributions of observations made before and after its onset is useful in many important applications such as inference of dynamic patterns as they emerge in data over space and/or time, e.g., environmental monitoring, statistical ecology, public health surveillance, and economics [21]. For describing the changes over time in the distribution of an environmentally important variable X, a *propagation function* (PF) models how the frequency of sampling units with the value $X = x$ at one point in time must change in order to produce the distribution that occurs at a later point in time. Thus, PF provides a useful tool in long-term monitoring studies since all changes in a distribution can be examined together rather than just changes in single parameters such as the mean.

Let X and Y be the values of an environmental variable on a population unit at two consecutive time-points, with the marginal densities $f_X(\cdot)$ and $g_Y(\cdot)$ [17]. The PF is defined as

$$w(x) = \frac{g_Y(x)}{f_X(x)}$$

.
This gives a weighted distribution:

$$g_Y(x) = w(x) \cdot f_X(x)$$

where $E_f[w] \equiv E[w(X)] = 1$. A similar concept of *resource selection function* (RSF) in wildlife habitat modeling was defined as a logistic discriminant function in terms of a ratio of pdf-s for used (f_u) and available (f_a) resources in k habitats:

$$\frac{f_u(x)}{f_a(x)} = w(x) = exp(\sum\nolimits_{i=1}^{k} \beta_i x_i)$$

where β_i is the model coefficient for the habitat i's covariate x_i [25]. We extended this approach of connecting the log of the density ratio of a monitored variable with

a linear model (that includes site-specific covariates) via outcomes that we represent as flexibly shaped curves or functional data.

In the era of climate change, the problem of wildfires, their usually complex dynamics, and their impact on human and environmental health via air pollution are of growing concern worldwide. For this study, a dataset on wildfires that took place over the past decade in the U.S. state of California (CA) was obtained from the website of the California Department of Forestry and Fire Protection [1]. From the dataset, we filtered minor events by selecting those with a duration of at least 14 days. Thus, we obtained a final list of $n = 73$ wildfires. For each wildfire, we used its spatial coordinates to assign a geographical Zone as per the California State Plane Coordinate System [2]. Here, a Zone is a label that represents a collection of contiguous counties in CA and takes values from 1 (northernmost) through 6 (southernmost). As per NAD83 specification, Los Angeles county (previously Zone 7) was assigned to Zone 5.

We also used the location of each wildfire to identify its nearest surface air pollution monitoring site from a nationwide network of stations of the U.S. Environmental Protection Agency (EPA) [3]. Data in the form of hourly measurements of PM2.5 level (i.e., fine particulate matter of width up to 2.5 μm) at the selected site, for 7 days right before ("week0") and after ("week1") the starting date of the wildfire, was obtained from EPA. If the number of missing values exceeded 25% of the PM2.5 data, then the next nearest monitoring site was considered. If none of the 3 nearest sites have sufficient data for the duration of a wildfire, then the event is discarded from the analysis. Otherwise, the distance (D_i) between a wildfire i's location and its selected monitoring site was calculated. We used this covariate to study the effects of "visibility" of any change in air pollution levels in terms of the distortion in the distributions of PM2.5 data collected before and after the onset of wildfires in California.

In the present study, we (i) applied density ratio to model the change in an environmental variable measured as the PM2.5 level in the air at the onset of a given wildfire, and (ii) gave a functional representation of the ratio of the density of this variable for week1 to that of week0, which serves as the baseline density. Thus, for each wildfire i in our dataset, we created a functional outcome variable that describes the log density ratio $Y_i(t)$ (Fig. 1) based on a variety of flexible shapes that could be used for this representation [43]. We defined t as equally spaced points on a common functional domain, e.g., the interval [0, 1]. Thus, we modeled the log density ratio not by selecting any explicit functional form but with nonparametric Gaussian kernel-based estimates computed with the R package "densratio", which is run with default parameter settings.

Upon deriving the wildfire-specific curves $Y_i(t), i = 1 \ldots n$, we conducted functional data analysis to identify outliers and test for potential association of this outcome with the "visibility" of the wildfire i to its nearest monitoring station with sufficient measurements. For testing association, the covariate x_i used in the regression model is the distance D_i (as defined above) for wildfire i. The model is given by

$$Y_i(t) = \beta_0(t) + \beta_1(t)x_i + \epsilon_i(t). \tag{18.1}$$

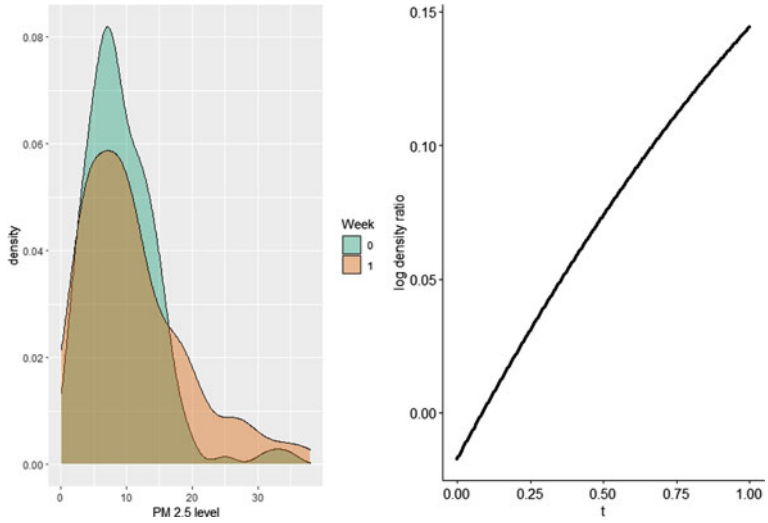


Fig. 18.1 **a** Densities of PM2.5 for week0 (shown in green) and week1 (shown in orange), and **b** Gaussian kernel estimated log density ratio curve of a selected fire

We assume i.i.d. mean zero errors with finite variance. We examined two different fits for model (18.1) [26] (i) *Point-wise Ordinary Least Squares*: For a fixed t, model (18.1) is a simple linear regression model, so we fit the model separately for each t and combined the estimates. (ii) *PACE Method*: We began with functional principal component analysis on the outcome variables (using the "FPCA" function in the R package "fdapace") which yielded a sequence of K basis functions $\phi_1, \ldots, \phi_K$. Then, with $\overline{Y}(t) = n^{-1} \sum_{i=1}^{n} Y_i(t)$, we have (Fig. 18.1)

$$Y_i(t) \approx \overline{Y}(t) + \sum_{k=1}^{K} \xi_{ik}\phi_k(t), \;\; \xi_{ik} = \int_0^1 (Y_i(t) - \overline{Y}(t))\phi_k(t)dt, \tag{18.2}$$

$$\epsilon_i(t) \approx \sum_{k=1}^{K} \varepsilon_{ik}\phi_k(t), \;\; \varepsilon_{ik} = \int_0^1 \epsilon_i(t)\phi_k(t)dt, \tag{18.3}$$

$$\beta_0(t) \approx \sum_{k=1}^{K} b_{0k}\phi_k(t), \tag{18.4}$$

$$\beta_1(t) \approx \sum_{k=1}^{K} b_{1k}\phi_k(t). \tag{18.5}$$

The basis approximation provides a form of regularization since some of the noise in the original curves is filtered out. Sixteen outlier curves were removed. The non-outlier curves are shown in Fig. 18.2. Furthermore, these basis approximations

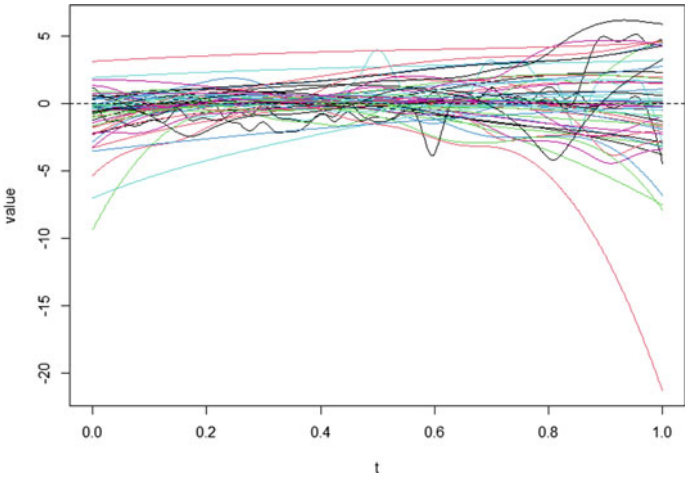


Fig. 18.2 Functional representation as non-outlier curves based on PM2.5 levels due to 57 fires

yielded a collection of K simple linear regression models

$$\xi_{ik} = b_{0k} + b_{1k}x_i + \varepsilon_{ik},\ k = 1 \ldots K \tag{18.6}$$

which were fit using ordinary least squares. Here, based on the scree plot in Fig. 18.3, we selected the first $K = 3$ functional principal components that together explain more than 90% of the variation among the curves, and their corresponding scores were used in the above model. The results, in Fig. 18.4, show the estimates of the slope parameter $\beta_1(t)$ using both OLS and PACE methods. We note that larger values of D_i, the distance between the wildfire i and its monitoring site, appears to have a negative effect on the functional outcome as values of t exceed 0.5 and reach the higher end of the PM2.5 range. The significance of this effect is assessed by the point-wise confidence bands for these estimates, assuming homoscedastic (across i, but not necessarily across t), normally distributed errors.

Finally, for the purpose of exploratory data analysis, we plotted the scores of the first two functional principal components of all wildfire-specific curves as 2D points in Fig. 18.5. We note significant heterogeneity among these curves, which is particularly interesting since the wildfires could thus be grouped into at least 4 clusters according to the variations in their PM2.5 density ratio "signatures". In fact, some of these unsupervised clusters appear to be associated with the geographical Zones of the wildfires that they contain. Given the role of the depicted components in capturing the variation among PM2.5 outcomes, this could lead to a more systematic analysis of wildfire dynamics in future work.

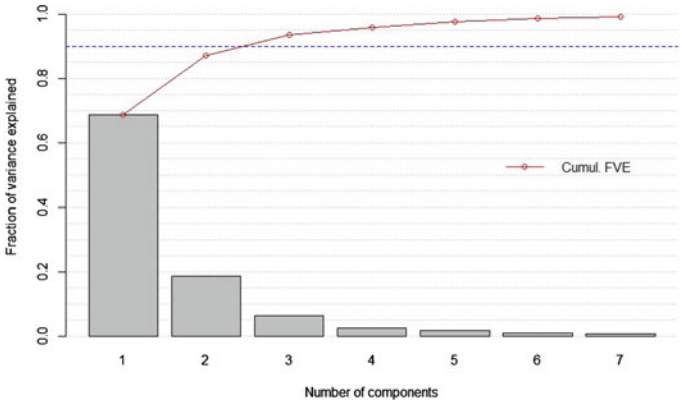


Fig. 18.3 Scree plot based on functional principal component analysis. The fraction of variance (FVA) explained (y-axis) by each principal component (x-axis) is shown. The cumulative FVA is shown (red curve), which for the first 3 components exceeds 90% (blue dashed line)

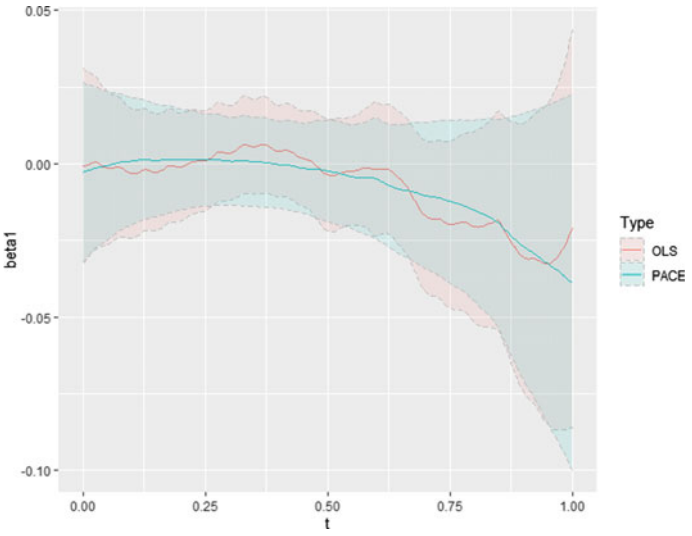


Fig. 18.4 The estimates for the slope parameter obtained from OLS (red curve) and PACE (blue curve) methods of model fitting. The corresponding confidence bands are shown

18.5 Discussion

In statistical inference, Rao noted that "wrong specification may lead to wrong inference, which is sometimes called the third kind of error in statistical parlance. The problem of specification is not a simple one" [39]. The aim of this study, therefore, is to draw the attention of students and researchers to the fascinating area of weighted distributions, which should gain relevance in the current age of data sci-

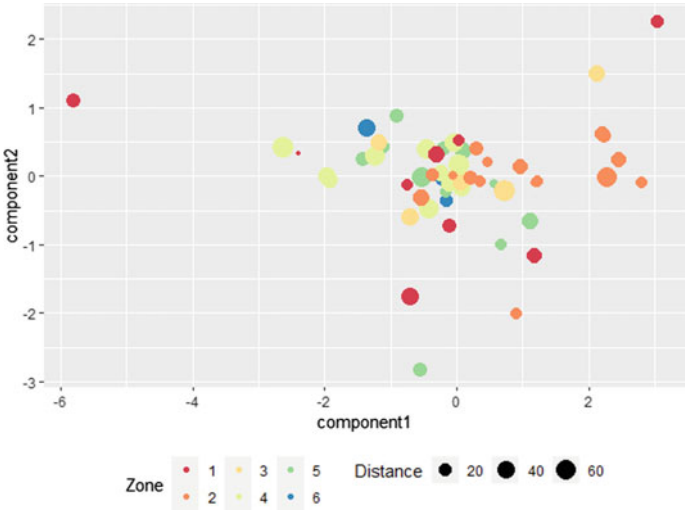


Fig. 18.5 Scatterplot of the first two functional PCA scores (x- and y-axes respectively) of the curves due to the wildfires. The Zone and the Distance to the monitoring site (in km) of each wildfire are depicted respectively by the color and the size of the corresponding point

ence. Yet, as observed by Patil and Rao, "although the situations that involve weighted distributions seem to occur frequently in various fields, the underlying concept of weighted distributions as a major stochastic concept does not seem to have been widely recognized" [29]. As more datasets of value that seemingly lack rigorous design are being collected, and possibly shared through unconventional and occasionally biased sources, the need for working with such less-than-ideal yet practically useful observations will have to be addressed timely and systematically by statistical pedagogy.

In this paper, we explored some aspects of the fascinating topic of weighted distributions with the purpose of introducing the students and researchers to the rich historical context, and the profound impact, of one of Rao's key contributions. Interestingly, our use of functional data analysis (FDA) presents a further opportunity to note that Rao's pioneering work on growth curves had, in fact, provided an early theoretical foundation for FDA. Motivated by a practical problem of analyzing the patterns of growth in a cohort of babies in Kolkata, Rao introduced a novel inferential framework [33] for analyzing growth curves for their multivariate structure. Further, he addressed many related issues in this area such as estimation, variable selection, prediction, and discriminant analysis. In his classic textbook "LSI" [36], Rao treated the latter problem in a more general form known as discrimination between composite hypotheses. It is, however, beyond the scope of the present study to elaborate on this yet another profound and path-breaking contribution of Rao in applied statistics [14].

Scientists addressing complex challenges such as monitoring climate change often face various data and metadata issues including visibility of rare phenomena, inspection paradox, citizen science bias, hidden truncation, and selective reporting. As this paper shows, weighted distributions offer a historical perspective for the visionary,

inter-disciplinary nature of Rao's early research, some of which appeared in his 1952 title, "*Advanced Statistical Methods in Biometric Research*" [32]. The cover of this text (C. A. B. Smith: "one of the best textbooks in statistics available anywhere") contained a remarkably futuristic subtitle for its time: "A unified treatment of mathematical distributions, statistical inference and diverse computational techniques". Befittingly, decades later, Rao was honored by the establishment of an eponymous Advanced Institute of Mathematics, Statistics and Computer Science (where one of the present authors, S. P., was fortunate to have him as a colleague and collaborator). We take this opportunity to wish Prof. Rao a longer, healthy, and happy life.

Acknowledgements The authors thank Ryan Stauffer for providing data for the study. Some parts of the paper were adapted from a lecture delivered by S. P. at the C. R. Rao Birth Centenary Session of the 22nd Annual Conference of the Society of Statistics, Computer and Applications, Pune, 2020.

Appendix

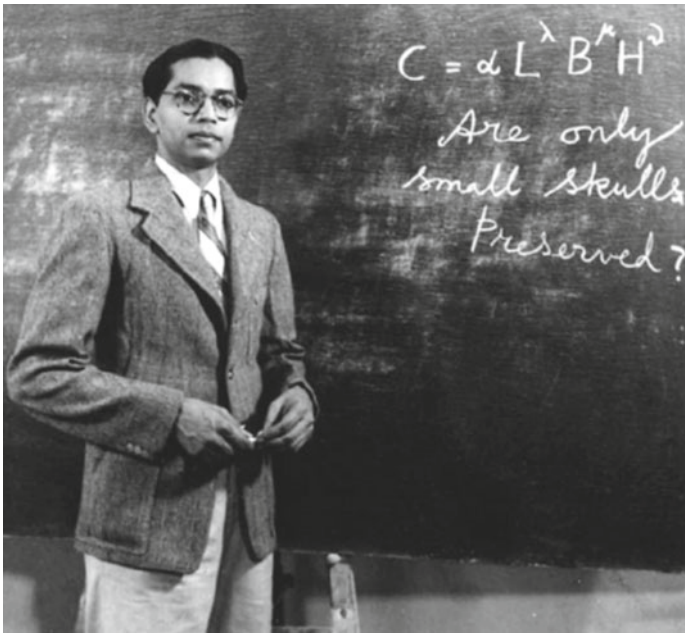


Fig. 18.6 C. R. Rao describing a formula on cranial capacity. For details, see Rao and Shaw [41], Rao [36]

References

1. California Department of Forestry and Fire Protection. https://www.fire.ca.gov/
2. California State Plane Coordinate System. https://www.conservation.ca.gov/cgs/rgm/state-plane-coordinate-system
3. United States Environmental Protection Agency. https://www.epa.gov/
4. Addison F (1949) The Wellcome excavations in the Sudan, vol I, Jebel Moya, 1910–1914. Oxford University Press, Oxford
5. Arnold B, Beaver R (2000) Hidden truncation models. Sankhyā, Ser A 62:23–35
6. Azzalini A (1985) A class of distributions which includes the normal ones. Scand J Stat 12:171–178
7. Bayarri M, DeGroot M (1992) A "BAD" view of weighted distributions and selection models. In: Bernardo J, Berger J, Dawid A, Smith A (eds) Bayesian statistics 4. Oxford University Press, Oxford, pp 17–33
8. Branco M, Dey D (2001) A general class of multivariate skew-elliptical distributions. J Multivar Anal 79:99–113
9. Brass M (2016) Reinterpreting chronology and society at the mortuary complex of Jebel Moya (Sudan). Archaeopress Publishing Ltd., Oxford
10. de Carvalho M, Davison A (2014) Spectral density ratio models for multivariate extremes. J Am Stat Assoc 109:764–776
11. Chen J, Liu Y (2013) Quantile and quantile-function estimations under density ratio model. Ann Stat 41:1669–1692
12. Fisher R (1934) The effect of methods of ascertainment upon the estimation of frequencies. Ann Eugen 6:13–25
13. Fokianos K (2007) Density ratio model selection. J Stat Comput Simul 77:805–819
14. Fujikoshi Y (2021) Contributions to multivariate analysis due to C.R. Rao and associated developments. In: Arnold B, Balakrishnan N, Coelho C (eds) Methodology and applications of statistics. Contributions to statistics. Springer, Cham, pp 239–257
15. Genton M (2005) Skew-elliptical distributions and their applications: a journey beyond normality. Chapman and Hall/CRC, Boca Raton
16. Gupta R, Kundu D (2009) A new class of weighted exponential distributions. Statistics 43:621–634
17. Kaur A, Di Consiglio L, Patil G, Taillie C (1995) Propagation functions for monitoring distributional changes. Environ Ecol Stat 2:239–269
18. Kedem B, Pyne S (2021) Estimation of tail probabilities by repeated augmented reality. J Stat Theory Pract 15:1–16
19. Kedem B, De Oliveira V, Sverchkov M (2017) Statistical data fusion. World Scientific Publishing Co., Singapore
20. Kedem B, Stauffer R, Zhang X, Pyne S (2021) On the probabilities of environmental extremes. Int J Stat Med Res 10:72–84
21. Kneip A, Utikal K (2001) Inference for density families using functional principal component analysis. J Am Stat Assoc 96:519–542
22. Larose D, Dey D (1996) Weighted distributions viewed in the context of model selection: a Bayesian perspective. TEST 5:227–246
23. Mahalanobis P, Majumdar D, Rao C (1949) Anthropometric survey of the united provinces: a statistical study. Sankhyā 9:89–324
24. Mahdavi A (2015) Two weighted distributions generated by exponential distribution. J Math Ext 9:1–12
25. McDonald L, Gonzalez L, Manly B (1995) Using selection functions to describe changes in environmental variables. Environ Ecol Stat 2:225–237
26. Morris JS (2015) Functional regression. Annu Rev Stat Its Appl 2:321–359
27. Mukherjee R, Rao C, Trevor J (1955) The ancient inhabitants of Jebel Moya (Sudan). Cambridge University Press, Cambridge

28. Patil G, Ord J (1975) On size-biased sampling and related form-invariant weighted distributions. Sankhyā, Ser B 38:48–61
29. Patil G, Rao C (1977) The weighted distributions: a survey of their applications. In: Krishnaiah P (ed) Applications of statistics. North Holland, Amsterdam, pp 383–405
30. Patil G, Rao C (1978) Weighted distributions and size biased sampling with applications to wildlife populations and human families. Biometrics 34:179–189
31. Qin J (1998) Inferences for case-control and semiparametric two-sample density ratio models. Biometrika 85:619–630
32. Rao C (1952) Advanced statistical methods in biometric research. Wiley, New York
33. Rao C (1958) Some statistical methods for comparison of growth curves. Biometrics 14:1–17
34. Rao C (1965) On discrete distributions arising out of methods of ascertainment. In: Patil G (ed) Classical and contagious discrete distributions. Statistical Publishing Society, Calcutta, pp 320–333
35. Rao C (1965) On discrete distributions arising out of methods of ascertainment. Sankhyā Ser A 27:311–324
36. Rao C (1973) Linear statistical inference and its applications. Wiley, New York
37. Rao C (1977) A natural example of a weighted binomial distribution. Am Stat 31:24–26
38. Rao C (1985) Weighted distributions arising out of methods of ascertainment: what population does a sample represent? In: AC A, E FS (eds) A Celebration of statistics: the ISI centenary volume. Springer, New York, pp 320–333
39. Rao C (1988) Weighted and clouded distributions. Technical report, No. 88-01, Center for multivariate analysis. University of Pittsburgh, Pittsburgh, PA 15260
40. Rao C, Rubin H (1964) On a characterization of the Poisson distribution. Sankhyā Ser A 25:295–298
41. Rao C, Shaw D (1948) On a formula for the prediction of cranial capacity. Biometrics 4:247–253
42. Saghir A, Hamedani G, Tazeem S, Khadim A (2017) Weighted distributions: a brief review, perspective and characterizations. Int J Stat Probab 6:109–131
43. Sugiyama M, Suzuki T, Kanamori T (2012) Density ratio estimation in machine learning. Cambridge University Press, Cambridge
44. Tomy L, Jose M, Veena G (2020) A review on recent generalizations of exponential distribution. Biom Biostat Int J 9:152–156
45. Vardi Y (1982) Nonparametric estimation in the presence of length bias. Ann Stat 10:616–620
46. Vinod H (1991) Rao's weighted distribution in econometrics: an application to unemployment statistics and Okun's law. J Quant Econ 7:247–254
47. Zhang A, Chen J (2022) Density ratio model with data-adaptive basis function. J Multivar Anal 191:105043
48. Zhang X, Pyne S, Kedem B (2020) Estimation of residential radon concentration in Pennsylvania counties by data fusion. Appl Stoch Model Bus Ind 36:1094–1110
49. Zhang X, Pyne S, Kedem B (2020) Model selection in radon data fusion. Stat Transit New Ser 21:159–165
50. Zhang X, Pyne S, Kedem B (2021) Multivariate tail probabilities: predicting regional pertussis cases in Washington state. Entropy 23:675

Chapter 19
Characterization of Q-Matrices Using Bordered Matrix Algorithm

N. Eagambaram

Mathematics Subject Classification (2010) 90C33

19.1 Introduction

Given a matrix $M \in \mathscr{R}^{n\times n}$ and a vector $q \in \mathscr{R}^n$, the linear complementarity problem $LCP(q, M)$ is to find a solution (w, z) that satisfies

$$w - Mz = q, w, z \geq 0 \tag{19.1}$$

and

$$w^T z = 0. \tag{19.2}$$

Let

$$M = \begin{bmatrix} A & v \\ u^T & \alpha \end{bmatrix} \tag{19.3}$$

where $A \in \mathscr{R}^{(n-1)\times(n-1)}$, $u, v \in \mathscr{R}^{n-1}$ and $\alpha \in \mathscr{R}$.

In the LCP(q, M), if we take $w = \begin{bmatrix} \overline{w} \\ w_n \end{bmatrix}$,$z = \begin{bmatrix} \overline{z} \\ \theta \end{bmatrix}$ and $q = \begin{bmatrix} \overline{q} \\ q_n \end{bmatrix}$, where $\overline{w}$, $\overline{z}$ and $\overline{q} \in \mathscr{R}^{n-1}$, we have

$$\overline{w} - A\overline{z} = \overline{q} + \theta v \tag{19.4}$$

N. Eagambaram (✉)
Former Deputy Director General, Indian Statistical Service; Guest faculty, Indian Institute of Technology, Tirupati, India
e-mail: ekambaramn@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*, Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_19

and

$$w_n - u^T \overline{z} = q_n + \theta\alpha. \tag{19.5}$$

By applying Lemke's algorithm to LCP($\overline{q} + \theta v$, A), we get solutions to (19.4) for various values of θ obtained from the algorithm. Solution to LCP(q, M) is obtained when (19.5) is satisfied for some solution ($\overline{w}$, $\overline{z}$, θ) of (19.4). An algorithm based on the above procedure is presented in this article. The algorithm can be called Bordered Matrix Algorithm (BMA). Though BMA is essentially a variant of Lemke's algorithm, an example is given to illustrate as to how it differs from Lemke's algorithm.

In the process of ascertaining the existence of solution to the LCP and developing a method for obtaining a solution, if any, various classes of matrices have been identified. One such a class is the class of Q-matrices. If the matrix M is such that LCP(q, M) has a solution for every q, then M is a Q-matrix. A basic problem in LCP is to characterize Q-matrices. In this paper, an algorithmic approach for characterization of Q-matrices is proposed by using BMA. The algorithm gives a better understanding of the problems involved in the Q-characterization by comparing M with its major principal sub-matrix A.

BMA is based on the continuity property of solution to LCP when Lemke's algorithm is applied to the LCP. The notations used in the paper, Lemke's algorithm, and the continuity property of solution are explained in the Preliminaries section. BMA and its applications in Q-characterization are presented in the Main section. The utility of BMA and its prospects in Q-characterization are pointed out in the Conclusion.

19.2 Preliminaries

In a solution (w, z) to an LCP, w is a function of z and as such z itself is called a solution to the LCP. We denote the jth column of $B \in \mathscr{R}^{k\times k}$ as $B_{.j}$. A matrix $C \in \mathscr{R}^{k\times k}$ is called a complementary matrix if the $C_{.j} \in \{I_{.j}, B_{.j}\}$ where I is the identity matrix of order k. Let $\overline{k} = \{1, 2, \ldots, k\}$ and for $J \subseteq \overline{k}$, let $J' = \overline{k} - J$. Each C is uniquely defined by a $J \subseteq \overline{k}$ where $C_{.l} = B_{.l}$ if $l \in J$ and $C_{.l} = I_{.l}$ otherwise. The vectors $I_{.j}$ and $B_{.j}$ are complementary vectors. The notation $C(J)$ is used to denote the complementary matrix indexed by J. B_{JK} stands for the sub-matrix of B, whose rows and columns are obtained by deleting the rows and columns of B indexed by J' and K' respectively. B_{JJ} is denoted in short as B_J. A complementary cone generated by a complementary matrix C denoted as $pos(C)$ is the set of all non-negative linear combinations of the columns of C. The union of all complementary cones is denoted by $K(B)$. $pos(C)$ is a full cone if rank(C) is k. Two cones $pos(C(J_1))$ and $pos(C(J_2))$ are adjacent if $J_1 \Delta J_2 = \{i\}$ for some $i \in J_1 \cup J_2$. The cone generated by r complementary vectors is called an r-facet. A complementary cone $pos(C)$ is degenerate if C is singular. When C is singular and if there exists a non-zero non-negative x such that $Cx = 0$, $pos(C)$ is called a strongly degenerate cone. $pos(C)$ is weakly degenerate if C is singular and $pos(C)$ is not strongly degenerate.

19.2.1 Lemke's Algorithm and Continuity of Solution z

(2.1.1) The properties of Lemke's algorithm as narrated here are based on the material that can be found in Cottle et al. [1]. Let $B \in \mathscr{R}^{k\times k}$. If $p \geq 0$, the LCP(p, B) has a solution: $w = p, z = 0$. If at least one component of p is negative, the Lemke's algorithm applied to $LCP(p, B)$ with artificial vector $v > 0$ begins with a solution to LCP$(p + \theta_0 v, B)$ where $p + \theta_0 v \geq 0$ lies on a $(k-1)$-facet of I for some $\theta_0 \geq 0$. In the subsequent iterations, a sequence of $\theta_i, i = 1, 2, \ldots$ is obtained where the algorithm fetches solution to LCP$(p + \theta_i v, B)$ lying on a $(k-1)$-facet common to the adjacent cones $pos(C_i)$ and $pos(C_{i+1})$. By adopting some standard procedure to avoid cycling, the algorithm terminates in a finite number, say s of iterations after passing through a unique sequence of $\{C_i\}, i = 0, 1, 2, \ldots, s$ where $C_0 = I$. Unless otherwise specified, hereafter, we shall take C_s as the terminal cone of the algorithm, and C_0 as the initial cone. The algorithm can start from a full cone $pos(C_0)$ instead of $pos(I)$ if there exists a $\theta_0 \geq 0$ such that $p + \theta v$ belongs to the interior of $pos(C_0)$ for $\theta > \theta_0$.

Proposition 19.1 *Lemke's algorithm applied to LCP(p, B) using artificial vector $v > 0$ has the following properties:*

The algorithm starts with a solution (w, z) to LCP$(p + \theta_0 v, B)$ where $\theta_0 = min\{\theta : p + \theta v \geq 0\}$. $(w = p + \theta v, z = 0)$, $\theta \geq \theta_0$ gives a ray of solutions, called primary ray of solutions to LCP$(p + \theta v, B)$.
The algorithm terminates in one of the following three ways.

(T_1) *$pos(C_s)$ contains the vector p:*
If $C_s = C(J)$, the algorithm provides a solution to the LCP, viz.,$z_J = (C_s^{-1})_J$ with $z_{J'} = 0$.

(T_2) *(w_{s-1}, z_{s-1}) is a solution to LCP$(p + \theta_{s-1} v, B)$ at the $(s-1)$th iteration and $C_s = C(J)$ contains v:*
The algorithm fails to find a solution, but gets a solution to LCP(v, B), viz, $z_J = (C_s^{-1} v)_J$ with $z_{J'} = 0$. The algorithm provides solution to LCP$(p + \theta v, B)$ for all $p + \theta v$, $\theta \geq \theta_{s-1}$. The set of such solutions $(z_J = (C_s^{-1} p)_J + \theta(C_s^{-1} v)_J, z_{J'} = 0)$, $\theta \geq \theta_{s-1}$ is called a secondary ray of solutions to the LCP.

(T_3) *(w_{s-1}, z_{s-1}) is a solution to LCP$(p + \theta_{s-1} v, B)$ at the $(s-1)$th iteration and $C_s(J)$ is a strongly degenerate cone:*
The algorithm reaches a point $p + \theta_s v$ lying on a facet of $pos(C_s)$. There exists a non-negative vector x such that $C_s(J)x = 0$. $\tilde{z}_J = x_J$ with $\tilde{z}_{J'} = 0$ is a solution to LCP$(0, B)$. $z_{s-1} + \gamma \tilde{z}$, $\gamma \geq 0$ is a secondary ray of solutions given by the algorithm.

In Lemke's algorithm, solutions are obtained to the points $p + \theta v$ by moving through a sequence of adjacent cones $C_i, i = 0, 1, 2, \ldots, s$ in an interval of the line $\{p + \theta v : \theta \in [\theta_{min}, \infty)\}$ where $\theta_{min} \geq 0$ is the smallest possible θ for which LCP$(p + \theta v, B)$ has a solution in the implementation of the algorithm. In general,

the algorithm can start from any full cone $pos(C_0)$ other than $pos(I)$, containing the artificial vector v in its interior. We see that the algorithm terminates with a solution at pos(C_s) or terminates at $pos(C_s)$ which is either a strongly degenerate cone or a cone containing v.

Time Parameter

(2.1.2) Let the first iteration take place at $t = t_0 = 0$ and the ith iteration at time t_{i-1}. Let $\theta_i = \theta(t_i)$. For $t_i < t < t_{i+1}$, let us define $\theta(t) = \theta_i + [(\theta_{i+1} - \theta_i)/(t_{i+1} - t_i)](t - t_i)$, $i = 0, 1, 2, \ldots, s-1$. If the algorithm finds a solution to LCP(p, B), then there exists a t, say, $t^* \geq t_{s-1}$ such that $\theta(t^*) = 0$. If C_s is a strongly degenerate cone, then $\theta(t) = \theta(t_{s-1})$ for all $t \geq t_{s-1}$. If C_s contains v, define $\theta(t) = \theta(t_{s-1}) + t - t_{s-1}$ for $t \geq t_{s-1}$. For $t < 0$, $\theta(t) = -t$. The range of t is either $(-\infty, t^*]$ or $(-\infty, \infty)$. $\theta(t)$ is a continuous function of t. The range of θ is $[\theta_{min}, \infty)$ for some $\theta_{min} \geq 0$.

Let $z(t)$ be the solution to LCP$(p + \theta(t)v, B)$. If C_s is a strongly degenerate cone, $z(t) = z(t_{s-1}) + (t - t_{s-1})\tilde{z}_s$ where $\tilde{z}_s$ is a solution to LCP$(0, B)$ given by C_s. If C_s contains v, $z(t) = z(t_{s-1}) + (t - t_{s-1})\hat{z}_s$ where $\hat{z}_s$ is a solution to LCP(v, B) given by C_s. For $t < 0$, $z(t) = z(t_0) - t\hat{z}$ where $\hat{z}$ is the solution to LCP(v, B) given by C_0. $z(t)$ is a one-one continuous function of t, which has been pointed out by Eagambaram and Mohan [2].

Let v_J and u_J denote the vectors of v and u constituted by taking elements from J.

Definition 19.1 For $J \subset \overline{n-1}$, $\overline{M}_J = \begin{bmatrix} A_J & v_J \\ u_J^T & \alpha \end{bmatrix}$. For A_J non-singular,

$$S(\overline{M}_J/A_J) = \alpha - u_J^T A_J^{-1} v_J$$

is the Schur complement of $\overline{M}_J$ with respect to A_J.

19.2.2 Some Q and Q_0-Matrices

A matrix B is called a Q_0-matrix if $K(B)$ is convex. If $K(B) = \mathscr{R}^k$, then B is called a Q-matrix. For every strictly positive vector p, if LCP(p, B) has a unique solution, then B is called a semi-monotone matrix or an E_0-matrix. If all the principal minors of B are positive, then B is called a P-matrix. The class of matrices having a particular property is denoted by the same symbol used to denote the property.

A principal pivot transform (PPT) of B with respect to an index set $J \subset \overline{k}$ is the matrix $-C(J)^{-1}C(J')$ where $C(J)$ and $C(J')$ are complementary matrices related to B, and $C(J)$ is non-singular. If B is a Q_0-matrix, a PPT of B is also a Q_0-matrix.

19.3 Main Results

19.3.1 Bordered Matrix Algorithm (BMA)

Algorithm 1: Let

$$M = \begin{bmatrix} A & v \\ u^T & \alpha \end{bmatrix}$$

where $v > 0$. Let $q = [\overline{q}, q_n]^T$. Apply the parametric Lemke's algorithm to LCP($\overline{q} + \theta v$, A) with time parameter t. Let $(\overline{w}(t), \overline{z}(t))$ be a solution to the LCP($\overline{q} + \theta(t)v$, A).

Let

$$q_n(t) = w_n(t) - u^T \overline{z}(t) - \alpha\theta(t). \tag{19.6}$$

We note that if $\theta(t) > 0$, $w_n(t) = 0$. Since $(\overline{w}(t), \overline{z}(t))$ is a continuous function of t, $q_n(t)$ is also a continuous function of t. The bordered matrix algorithm (BMA) aims to solve LCP(q, M) by using the solutions to LCP($\overline{q} + \theta(t)v$, A).

Step 1: $\overline{q} \geq 0$.
(1.1) If $q_n \geq 0$, the solution is $w = q$, and $z = 0$.
(1.2) If $q_n < 0$ and $\alpha > 0$, solution is $\overline{w} = \overline{q} + (-(q_n/\alpha)v)$, $w_n = 0$, $\overline{z} = 0$, $z_n = -q_n/\alpha$.
Otherwise go to Step 2.
Step 2: At least one component of $\overline{q}$ is negative.
Carry out Lemke's Algorithm on LCP($\overline{q}$, A) with artificial vector v.
(2.1) In the first iteration $\theta = \theta_0$.
(2.1.1) If $\alpha > 0$ and $q_n \leq -\theta_0\alpha$, find $\theta \geq \theta_0$ such that $q_n = -\theta\alpha$. The solution is $\overline{w} = \overline{q} + \theta v$, $w_n = 0$, $\overline{z} = 0$ and $z_n = \theta$.
(2.1.2) If $\alpha < 0$ and $q_n > -\alpha\theta_0$, let $\theta = -q_n/\alpha$. The solution is $\overline{w} = \overline{q} + \theta v$, $w_n = 0$, $\overline{z} = 0$ and $z_n = \theta$.
Otherwise, proceed with the parametric LCP($\overline{q} + \theta(t)v$) and go to Step 3.
Step 3: According to the time parametric version, let $z(t)$ be the solution at time t and $q_n(t) = w_n(t) - u^T z(t) - \alpha\theta(t)$
(3.1) At the ith iteration $q_n(t_i) = -u^T \overline{z}(t_i) - \alpha\theta(t_i)$.
(3.1.1) If q_n lies between $q_n(t_i)$ and $q_n(t_{i+1})$, obtain t lying between t_i and t_{i+1} such that $q_n(t) = q_n$. The solution is $\overline{z} = \overline{z}(t)$ and $z_n = \theta(t)$.
(3.1.2) If $t = t^*$ so that $\theta(t^*) = 0$ at any stage of iteration, check $q_n(t^*)$. If q_n lies between $q_n(t_i)$ and $q_n(t^*)$ choose t such that $q_n(t) = q_n$. The solution is $\overline{z} = \overline{z}(t)$ and $z_n = \theta(t)$.
If $q_n(t^*) \leq q_n$, the solution is $(\overline{w}, \overline{z})$ is the solution to LCP($\overline{q}$, A). $w_n = q_n - q_n(t^*)$, and $z_n = 0$.
Otherwise, go to Step 4.

Step 4:

(4.1) The algorithm terminates at a strongly degenerate cone C_s or the cone C_s contains v.

(4.1.1) C_s is strongly degenerate. $\tilde{z}$ is solution to LCP(0, A).

(4.1.1(a)) $\beta = -u^T\tilde{z} > 0$ and $q_n \geq q_n(t_{s-1})$. Solution is $\overline{z}(t_{s-1}) + [[q_n - q_n(t_{s-1})]/\beta]\tilde{z}$ and $z_n = \theta(t_{s-1})$.

4.1.1(b)) $\beta = -u^T\tilde{z} < 0$ and $q_n \leq q_n(t_{s-1})$. Solution is $\overline{z}(t_{s-1}) - [[q_n(t_{s-1}) - q_n]/\beta]\tilde{z}$ and $z_n = \theta(t_{s-1})$.

(4.1.2) C_s contains v. $\hat{z}$ is solution to LCP(v, A) corresponding to C_s.

(4.1.2(a)) $\beta = -u^T\hat{z} - \alpha > 0$ and $q_n \geq q_n(t_{s-1})$. Solution is $\overline{z}(t_{s-1}) + [[q_n - q_n(t_{s-1})]/\beta]\hat{z}$ and $z_n = \theta(t_{s-1}) + [q_n - q_n(t_{s-1})]/\beta$.

(4.1.2(b)) $\beta = -u^T\hat{z} - \alpha < 0$ and $q_n \leq q_n(t_{s-1})$. Solution is $\overline{z}(t_{s-1}) - [[q_n(t_{s-1}) - q_n]/\beta]\hat{z}$ and $z_n = \theta(t_{s-1}) - [q_n(t_{s-1}) - q_n]/\beta$.

Otherwise, the algorithm terminates without finding a solution.

BMA is essentially based on Lemke's algorithm and the continuity property of $q_n(t)$. Though the description of the algorithm is lengthy, it is easy to visualize and implement by following the continuity property of $q_n(t)$. In BMA, solutions to q are obtained by fixing the first $n-1$ components of q. The set of q's for which solutions obtained are perpendicular to R^{n-1}. The direction vector of the algorithm is the n-dimensional unit vector with the nth component unity. BMA is used in this article only as a tool to study the Q-properties of matrices. However, we give below an example of a matrix M and a vector q for which BMA solves LCP(q, M), but Lemke's algorithm fails.

Example 1

Let

$$M = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 2 & -1 & -1 \end{bmatrix}$$

LCP(q, M) where $q = [-2, -1, -2]^T$ is not solvable by Lemke's algorithm with a positive artificial vector. But, BMA solves it and gets the solution $z = [3, 2.5, 1.5]^T$.

Let us see how BMA works with regard to the above example. The initial table with artificial vector $v = [1, 1]^T$ is

Initial table with v as artificial vector						
x1	x2	x3	x4	x5	x6= −v	q
1	0	0	−1	1	−1	−2
0	1	0	1	−1	−1	−1
0	0	1	−2	1	1	−2

Lemke's algorithm is applied to LCP($\overline{q}$, A) where $\overline{q} = [-2, -1]^T$. $v = [1, 1]^T$. $u^T = [2, -1]$ and $\alpha = -1$. The most negative component of $\overline{q}$ is -2. The pivot

element is the first component of $x6$. After pivoting, the basic variables are w_2 and θ. The entering variable is $z_1 = x4$.

Iteration-0								
Basic variable	x1	x2	x3	x4	x5	x6	q	Ratio
θ	-1	0	0	1	-1	1	2	$2/1 = 2$
w_2	-1	1	0	2	-2	0	1	$1/2$
$-u^T \rightarrow$	0	0	1	-2	1	1	$q_n(0) = 2$	

The solution at the first iteration is $z_1 = 0$, $z_2 = 0$ and $z_3 = \theta = 2$ and $q_n(0) = z_1 * (-u_1) + z_2 * (-u_2) + \theta * (-\alpha) = 0 * (-2) + 0 * (1) + 2 * (1) = 2$. $x1$ is driven out and $x4 = z_1$ enters. The ratios $q_i/(x4)_i$ for the rows 1 and 2 are computed and the minimum ratio is $1/2$. The pivot element is the second component of x4 which is 2. In the next iteration, we get the following table. $q_n(1) = (-2) *$

Iteration-1								
Basic variable	x1	x2	x3	x4	x5	x6	q	Ratio
θ	$-1/2$	$-1/2$	0	0	0	1	3/2	0
z_1	$-1/2$	1/2	1	1	-1	0	1/2	
$-u^T \rightarrow$	0	0	1	-2	1	1	$q_n(1) = 1/2$	

$(1/2) + 1 * 0 + 1 * (3/2) = 1/2$. $x2 = w_2$ is driven out and $x5 = z_2$ enters. The first and second components of x5 are non-positive. Lemke's algorithm terminates for LCP($\overline{q}$, A) at the strongly degenerate cone $C(\{1, 2\})$. $(1, 1)$ is the non-trivial solution to LCP(0, A). $\beta = -u^T \tilde{z} = (-2) * 1 + 1 * (1) = -1 < 0$. $q_n(1) + 2.5 * \beta = 1/2 + 2.5 * (-1) = -2$. Therefore the solution is $(1/2, 0, 3/2) + 2.5(1, 1, 0) = (3, 2.5, 1.5)$.

Let us continue to use M as a bordered matrix with A as the leading $n - 1$ principal sub-matrix. BMA can be applied even though the first $n - 1$ components of the last column of M are not positive, as seen from the following remarks.

Remark 19.1 If one of the columns of M is positive except the diagonal element, then there exists a permutation matrix P such that PMP^T has its last column positive except the last diagonal element. BMA can be applied to PMP^T to get solutions to the original LCP with M.

Remark 19.2 If v is in the interior of a complementary cone, the corresponding Principal Pivot Transform yields a bordered matrix with v positive. Then BMA can be applied to process the LCP.

19.3.2 Characterization of Q-Matrices Using BMA

The following theorem gives a necessary condition for the matrix M to be a Q-matrix.

Theorem 19.1 *Let*

$$M = \begin{bmatrix} A & v \\ u^T & \alpha \end{bmatrix}.$$

Then the following are the two equivalent necessary conditions for M to be a Q-matrix.

(i) *For $\overline{q} \in \mathscr{R}^{n-1}$ there exists a θ_0 and a $C_0 \in \mathscr{C}(A)$ such that $\overline{q} + \theta v \in pos(C_0)$ for all $\theta \geq \theta_0$.*
(ii) *v is in the interior of $K(A)$.*

Proof (i) Let $(\begin{bmatrix} \overline{w} \\ w_n \end{bmatrix}, \begin{bmatrix} \overline{z} \\ z_n \end{bmatrix})$ be a solution to LCP(M, q) where $q = \begin{bmatrix} \overline{q} + \beta v \\ 0 \end{bmatrix}$, $\beta > 0$. Then $\overline{w} - A\overline{z} = \overline{q} + (z_n + \beta)v$. As $\beta \to \infty, \overline{q} + (z_n + \beta)v$ is in a particular full cone pos(C_0), $C_0 \in \mathscr{C}(A)$, for all $\beta \geq \beta_0$ for some large positive β_0, because the complementary cones are finite in number.

$(i) \Rightarrow (ii)$ If v does not lie in the interior of $K(A)$, then there exists a $\overline{q}$ and $\epsilon_0 > 0$ such that $v + \epsilon\overline{q}$ is not in $K(A)$ for all $\epsilon < \epsilon_0$. This leads to the contradiction that $\overline{q} + (1/\epsilon)v$ is not in $K(A)$ for all $\epsilon < \epsilon_0$.

$(ii) \Rightarrow (i)$ If v is in the interior of $K(A)$, then for a given $\overline{q}$, there exists ϵ_0 such that $v + \epsilon\overline{q}$ is in the interior of $K(A)$ for $0 < \epsilon < \epsilon_0$. This implies that $\overline{q} + \beta v$ is in the interior of $K(A)$ for $\beta > 1/\epsilon_0$. As there are only a finite number of complementary cones, we see that (i) holds. □

It follows from Theorem 19.1 that

Corollary 19.1 *If M is a Q-matrix, then there exists a PPT $\overline{M}$ of M such that the last column of $\overline{M}$ is non-negative except the last element which can be of any sign.*

Example 2

Let $M = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ a & b & c \end{bmatrix}$. Here $v = [1, -1]^T$ is not in the interior of $K(A)$ and so M is not a Q matrix for any a, b, c.

Now we obtain some additional necessary conditions for M to be a Q-matrix as follows.

Theorem 19.2 *M is a Q-matrix only if for a given $\overline{q} \in \mathscr{R}^{n-1}$,*

(i) *there exists a full cone $C_0 = C(J) \in \mathscr{C}(A)$ and $\theta_0 \geq 0$ such that $\overline{q} + \theta v$ is in pos$(C(J))$ for all $\theta \geq \theta_0$*
and
(ii) *$-u^T \hat{z}_0 - \alpha = -S(\overline{M}_J / A_J) < 0$ where $\hat{z}_0$ is the solution to LCP(v, A) corresponding to $C(J)$.*

Proof Necessity of (i) follows from Theorem 19.1. Additional necessity of (ii) is as follows:

Let $q = [\overline{q}^T, \beta]^T$. If M is a Q-matrix, then a fixed $\overline{q}$, LCP(q, M) has a solution for any β, especially for a large negative β.

We see that there exists a large $\theta_0 > 0$, such that for $\theta \geq \theta_0$, $\overline{q} + \theta v$ does not belong to any complementary cone that does not contain v. For such a θ_0, $\overline{q} + \theta v$ with $\theta \geq \theta_0$ belongs to each non-singular complementary cone containing v.

For $\theta \geq \theta_0$, consider solution $\overline{z}(\theta)$ to LCP$(\overline{q} + \theta v, A)$ given by the complementary cone C_J that contains v.

$$\overline{z}_J(\theta) = (C_0^{-1}\overline{q})_J + \theta(C_0^{-1}v)_J,$$

and $\overline{z}_{J'}(\theta) = 0$. Corresponding to the above solution, we get a solution $[z(\theta)^T, \theta]^T$ to LCP(q, M) where

$$\beta = -u^T\overline{z}(\theta) - \theta\alpha = -u_J^T(C_0^{-1})_J + \theta(\alpha - u_J^T\overline{z}_J(\theta)).$$

When $\theta = 0$, LCP(q, M) has solution for all$q$ with $\beta = w_n - u^T\overline{z}(0)$, $w_n \geq 0$. This ensures that β can take all values greater than $-u^T z(0)$. If β has to take all large negative values, then

$$\alpha - u_J^T\overline{z}_J(0) = \alpha - u_J^T(C_0^{-1}v)_J = S(\overline{M}_J/A_J) > 0.$$

Therefore, one of the non-singular complementary cones C_0 containing $\overline{q} + \theta v$ for $\theta \geq \theta_0$ should satisfy Condition(ii). Hence the proof. □

Example 3

Let $M = \begin{bmatrix} A & v \\ u^T & \alpha \end{bmatrix}$ where A is an E_0-matrix, $v \geq 0$, and $\alpha < 0$. Then M is not a Q-matrix.

Proof pos(I)= pos(C_ϕ) is the only cone containing v. $S(\overline{M}_\phi/A_\phi) = \alpha < 0$. The necessary condition (ii) of Theorem 19.2 for M to be Q is not satisfied. □

Example 4

Let $M = \begin{bmatrix} -1 & 1 & 1 \\ 3 & -2 & 1 \\ 2 & -1 & -3 \end{bmatrix}$. Here, v is in more than one cone of $K(A)$. But none of such cones satisfies Condition (ii) of Theorem 19.2. Hence M is not a Q-matrix. For example, LCP(q, M) with $q = [1, 1, -10]^T$ has no solution. Here, as another example, we note that $-M$ is not a Q-matrix if M is a P-matrix, because $S(\overline{-M}_J/(-A)_J) < 0$ for all $J \subset \overline{n-1}$ when M is a P-matrix.

The following theorem appearing in Sivakumar et al. [7] can be proved alternatively by using Theorem 19.2.

Theorem 19.3 *The matrix M of rank unity is a Q matrix if and only if each element of M is positive*

Proof A complementary cone $C(J)$ related to A is non-singular implies that $|J| \leq 1$. If $|J| = 1$ and $C(J)$ is non-singular, then $S(\overline{M}_J/A_J) = 0$. Therefore, the $C(J)$ that satisfies Conditions (i) and (ii) of Theorem 19.2 is $C_\phi = I$ and the corresponding Schur complement is α which must be positive. Again, by Theorem 19.2, v is in pos(I) and so v is non-negative. Since any principal rearrangement is a Q-matrix, we see that all the diagonal entries are positive, and all other elements are non-negative. This is possible for a rank one matrix only if each element of M is positive. On the other hand, if each element of M is positive, it is known that M is a $P_0 \cap Q$-matrix Katta [4]. Hence the result. □

The necessary conditions in Theorems 19.1 and 19.2 help us to rule out that M is a Q-matrix. The next result includes one more condition to bring out a set of conditions sufficient for M to be a Q-matrix.

Theorem 19.4 *M is a Q-matrix if for a given $\overline{q} \in \mathscr{R}^{n-1}$,*

(i) *there exists a full cone pos$(C(J)) \in K(A)$ and $\theta_0 \geq 0$ such that $\overline{q} + \theta v$ is in pos$(C(J))$ for all $\theta \geq \theta_0$,*
(ii) *$-u^T \hat{z}_0 - \alpha < 0$ where $\hat{z}_0$ is the solution to LCP(v, A) corresponding to $C(J)$ and*
(iii) *if C_s is the terminal cone in Lemke's Algorithm to solve LCP$(\overline{q}, A)$ with artificial vector v, then*

   (a) *C_s contains $\overline{q}$ and Lemke's algorithm terminates with a solution*
   or
   (b) *C_s is a strongly degenerate cone, $\tilde{z}$ is the non-trivial solution to LCP$(0, A)$ obtained from C_s and $-u^T \tilde{z} > 0$*
   or
   (c) *C_s contains v, $\hat{z}$ is the solution to LCP(v, A) corresponding to C_s and $-u^T \hat{z} - \alpha > 0$.*

Proof Consider

$$q_n(t) = w_n(t) - u^T \overline{z}(t) - \alpha\theta(t)$$

defined in the BMA applied to solve LCP(q, M). Conditions (i) and (ii) ensure that $q_n(-\infty) = -\infty$. $q_n(t)$ is a continuous function of t and it attains ∞ under the conditions $(iii)(b)$ or $(iii)(c)$. Under $(iii)(a)$ let the solution to LCP($\overline{q}$) be obtained at $t = t^*$. Then $w = [0, \gamma]^T$, $z = [\overline{z}(t^*), 0]^T$ is a solution to $[\overline{q}, q_n(t^*) + \gamma]^T$, for $\gamma \geq 0$, which shows that the nth component of q takes all real values. Since $\overline{q}$ is arbitrary, we see that LCP(q, M) has solution for all q. □

The example below is an application of Theorem 19.4 to obtain a range of α for which M is a Q-matrix.

Example 6

$M = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 2 & \alpha \end{bmatrix}$ is a Q-matrix when $\alpha \in (-1, 0) \cup (-3, -2)$.

Proof Here, $n = 3$, $A = -I$, $v = [1, 1]^T$, $\overline{q} = [q_1, q_2]^T$, $q = [q_1, q_2, q_3]^T$ and M is singular for $\alpha = -3$

Case 1: $\alpha \in (-1, 0)$ and $q_1 \leq q_2$
Start BMA from $C_0 = C(\{1\})$ which contains $\overline{q} + \theta v$ for all large θ. $C(\{1\})$ is non-singular, which gives solution to LCP(v, A) as $\hat{z}(v) = [1, 0]^T$. $-u^T \hat{z}(v) - \alpha = -1 - \alpha < 0$. So, Conditions (i) and (ii) of Theorem 19.4 are satisfied. Condition(iii)(a) of Theorem 19.4 is satisfied if $\overline{q} > 0$. Otherwise, the next adjacent cone of Lemke's algorithm is $C_s = I$. The solution to LCP(v, A) corresponding to $C_s = I$ is $\hat{z}(v) = 0$ and so $-u^T \hat{z}(v) - \alpha > 0$, which satisfies Condition (iii)(c) of Theorem 19.4. Therefore, for the given $\overline{q}$, LCP(q, M) has solution for all q_3.

Case 2: $\alpha \in (-1, 0)$ and $q_1 > q_2$
The proof is similar to Case 1. Here we have $C_0 = C(\{2\})$ and $C_s = I$. $-u^T \hat{z}(v) - \alpha < 0$ with respect to C_0 and is positive with respect to C_s.

Case 3: $\alpha \in (-3, -2)$ and $q_1 \leq q_2$
We take $C_0 = C(J)$, where $J = \{1, 2\}$. The Schur complement $S(\overline{M}(J)/\overline{A}(J)) > 0$. If $\overline{q}$ is in $pos(C(J))$ Condition (iii) is invoked and if $\overline{q}$ lies outside $pos(C(J))$, Condition (iii)(c) is invoked, to show that q_3 can take all values for a given $\overline{q}$.

Case 4: $\alpha \in (-3, -2)$ and $q_1 > q_2$
This case is similar to Case 3. Here, again $C_0 = C(\{1, 2\})$ and $C_s = C(\{1\})$ when $\overline{q}$ lies outside $pos(C(J))$. □

Example 7

$M = \begin{bmatrix} A & 0 \\ u^T & \alpha \end{bmatrix}$ is a Q-matrix if A is a Q-matrix and $\alpha > 0$.
Here, 0 is in the interior of $K(A)$. If z is a solution to LCP$(A, \overline{q})$, q_n can take values exceeding $-u^T z$ by increasing w_n; and can take values less than $-u^T z$ by increasing θ. Therefore, M is a Q-matrix.

Example 8

Let $M = \begin{bmatrix} A & v \\ -u^T & \alpha \end{bmatrix}$, where A is an E_0-matrix, $\alpha > 0$ and $u, v > 0$. It follows from Theorem 19.4 that M is a Q matrix. It can also be seen that M is an E_0 matrix as well. A particular case is that $A = 0$.

Example 9

It follows by induction, from the previous example that a matrix M where $M_{ij} > 0$ for $j \geq i$ and $M_{ij} < 0$ for $j < i$. Then M is an $E_0 \cap Q$-matrix.

Example 10

Let $M = \begin{bmatrix} A & v \\ u^T & \alpha \end{bmatrix}$ where A is a Q matrix processable by Lemke's algorithm with artificial vector $v > 0$ and $\alpha > 0$. Applying BMA from pos(I), we see that M is a Q-matrix.

Whether A is a Q-matrix or not is relevant in examining the Q-property of M. We give below a necessary condition for M to be a Q-matrix when A is not a Q-matrix.

Theorem 19.5 *Let A be not a Q-matrix. Then M is a Q-matrix only if for $\overline{q}$ not in $K(A)$, there exists a $\theta > 0$ such that $\overline{q} + \theta v$ is in a cone pos(C).*
Either

(i) *pos(C) is a strongly degenerate cone, the solution $\tilde{z}$ to LCP$(0, A)$ given by C satisfies $-u^T \tilde{z} > 0$*
or
(ii) *pos(C) is a full cone containing v and the solution $\hat{z}$ to LCP(v, A) given by C satisfies $-u^T \hat{z} - \alpha > 0$.*

Proof If M is a Q-matrix, then for a given $\overline{q}$ not in $K(A)$, LCP(M, q) where $q = [\overline{q}, \delta]^T$ has solution for all real δ. Since LCP$(A, \overline{q})$ has no solution, for every $\delta > 0$

there is a $\theta > 0$ such that LCP$(A, \overline{q} + \theta v)$ has a solution $\overline{z}$ and $\delta(\theta) = -u^T \overline{z}(\theta) - \alpha\theta = \delta$. In order that there exists a $\delta(\theta)$ as large positive as desired, $\overline{q} + \theta v$ lies in a cone pos(C), for some $\theta > 0$ such that either (i) pos(C) is a strongly degenerate with $-u^T \tilde{z} > 0$ where $\tilde{z}$ is a non-trivial solution to LCP$(A, 0)$; or (ii) pos(C) contains v and $-u^T \hat{z} - \alpha > 0$, where $\hat{z}$ is a solution to LCP(A, v) corresponding to pos(C). Hence the proof. □

Theorem 19.5 gives a necessary condition which implies that if A is not a Q-matrix, there must be a strongly degenerate cone or a cone containing v satisfying conditions of the theorem in order that M is a Q-matrix. So, as a corollary, we have the result given by Jeter and Pye [3]:

Corollary 19.2 *If A is not a Q-matrix and M is a Q-matrix, then there exists a solution z to LCP(q, M) with $q = [0, 0, \ldots, 1]^T$, such that $[u^T, \alpha]z < 0$.*

19.4 Conclusion

BMA is a variant of Lemke's algorithm and is based on the continuity property of solution to the parametric LCP. The algorithm can be carried out when one of the columns of M is positive except the diagonal element. As seen from Example 1, there are matrices processable by BMA but not by Lemke's algorithm. The classes of matrices processable by BMA are yet to be explored.

The motivation for this article is to characterize Q-matrices by studying the relationships between an $n \times n$ matrix M and its $(n-1) \times (n-1)$ leading principal sub-matrix A in terms of LCP properties. Cottle et al. [1] presents an augmented matrix approach to solve LCP$(\overline{q}, A)$ by means of solution to LCP(q, M). Jeter and Pye [3] had done such a study and obtained some necessary conditions on Q-property involving M and A. Projesh et al. [6] characterized M as a Q-matrix where A is of a particular type. Here, BMA gives an algorithmic approach for such studies. Some necessary conditions for a matrix to be a Q-matrix are obtained using BMA. The utility of BMA in identifying Q-matrices is illustrated by examples.

Corollary 19.2 in the case of Q_0-matrices is proved by Murthy and Parthasarathy [5]. One may try to obtain further results on necessary conditions for Q_0-properties of matrices on the lines of the theorems for Q-matrices given in this paper, which would help in the characterization of Q_0-matrices in general. The study of completely-Q_0 and fully Q_0- matrices is another area where the algorithmic approach given in this paper can be useful. The invariance of Q-property under principal rearrangement and PPT is an added advantage in the application of BMA.

Acknowledgements The author is thankful to Professor T. Parthasarathy and Professor K. C. Sivakumar for discussions during the preparation of this article.

References

1. Cottle RW, Jong-Shi P, Richard ES (2009) The linear complementarity problem. Society for Industrial and Applied Mathematics, Philadelphia
2. Eagambaram N, Mohan SR (1990) On some classes of linear complementarity problems of order n and rank $n - 1$. Math Oper Res 15(2)
3. Jeter MW, Pye WC (1984) Some properties of Q matrices. Linear Algebra Appl 57:169–180
4. Katta GM (1972) On the number of solutions of the linear complementarity problem and spanning properties of complementary cones. Linear Algebra Appl 5:65–108
5. Murthy GSR, Parthasarathy T (1995) Some properties of fully semi-monotone, Q_0-matrices. SIAM J Matrix Anal Appl 16(4):1268–1286
6. Projesh NC, Eagambaram N, Sivakumar KC, Sushmitha P (2021) On the matrix class Q_0 and inverse monotonicity properties of bordered matrices. Linear Algebra Appl 612(1):206–222
7. Sivakumar KC, Sushmitha P, Tsatsomeros M (2021) $Q_{\#}$-matrices and $Q_{\dagger}$-matrices: two extensions of the Q-matrix concept. Linear Multilinear Algebra

Chapter 20
Descending Endomorphisms of Some Families of Groups



Vinay Madhusudanan, Arjit Seth, and G. Sudhakara

Mathematics Subject Classification (2010) 20E15 · 20B30 · 20D45 · 20E36

20.1 Introduction

Descending endomorphisms of groups were defined in [4] and used for developing an alternative proof of a result earlier proved by Miller in [5], characterising the normal subgroups of direct products. A descending endomorphism is a group endomorphism that descends to every quotient group. Equivalently, it is an endomorphism that leaves every normal subgroup invariant, which immediately implies that all power endomorphisms [2] and inner automorphisms are descending endomorphisms. This also implies that in any Dedekind group (in particular, any Abelian group), the descending endomorphisms are exactly the power endomorphisms. We are interested in the question: *how can we determine the descending endomorphisms of non-Abelian groups?* In [4], it is shown that every descending endomorphism of a direct product of finitely many groups is a pointwise product of direct products of the direct factors, and a condition is provided for determining exactly which of the pointwise products define descending endomorphisms of the product group. In this paper, we prove a general result showing that all endomorphisms of a certain class of groups, which includes all symmetric groups, are descending. We then apply this together with the result on

V. Madhusudanan (✉) · G. Sudhakara
Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India
e-mail: vinay.m2000@gmail.com

G. Sudhakara
e-mail: sudhakara.g@manipal.edu

A. Seth
Department of Mechanical and Aerospace Engineering, Hong Kong University of Science and Technology, Hong Kong, China
e-mail: ajseth@ust.hk

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_20

descending endomorphisms of direct products in order to determine all descending endomorphisms of direct products of finitely many symmetric groups. Next, we give an explicit description and enumeration of all descending endomorphisms of dihedral groups and direct products of finitely many dihedral groups. In the final two sections, we determine the descending endomorphisms of Hamiltonian groups and dicyclic groups.

We begin by reviewing the definition and some of the basic results that will be used in proving the main results of this paper.

Definition 20.1 [4, Definition 2.1] An endomorphism δ of a group G is a *descending endomorphism* if, for every epimorphism $\varphi\colon G \twoheadrightarrow Q$ of G onto a group Q, there exists an endomorphism $\overline{\delta}$ of Q such that the diagram in Eq. (20.1) commutes:

$$\begin{array}{ccc} G & \xrightarrow{\delta} & G \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle\varphi} \\ Q & \xrightarrow[\overline{\delta}]{} & Q \end{array} \tag{20.1}$$

Then $\overline{\delta}$ is unique and is said to be the *descended endomorphism* or *descent* of δ to Q along φ, or the endomorphism *induced by* δ.

By the first isomorphism theorem, epimorphic images of a group G are isomorphic to quotients of G. This leads us to the following characterisation of descending endomorphisms in terms of normal subgroups. We write $H \leq G$ (respectively, $H \trianglelefteq G$) to mean that H is a subgroup (respectively, normal subgroup) of G. We call a subgroup H of a group G *invariant* under an endomorphism ε of G if $\varepsilon(H) \leq H$.

Lemma 20.1 [4, Lemma 2.4] *An endomorphism δ of a group G is a descending endomorphism of G if and only if for every normal subgroup N of G, $\delta(N) \leq N$. Then the descent of δ to any quotient G/N along the canonical projection is defined by $\overline{\delta}(xN) = \delta(x)N$ for all $xN \in G/N$.*

Proof Consider a group Q onto which there is an epimorphism from G. Then by the first isomorphism theorem, we can identify Q with a quotient G/N of G where $N \trianglelefteq G$, with the epimorphism being replaced by the canonical projection $\varphi\colon G \twoheadrightarrow G/N$. Now, let δ be an endomorphism of G. Then by (20.1), δ is a descending endomorphism of G if and only if G/N has an endomorphism $\overline{\delta}$ such that $\overline{\delta} \circ \varphi = \varphi \circ \delta$. But $\varphi(x) = xN$ for all $x \in G$. Thus, δ is descending if and only if G/N has an endomorphism $\overline{\delta}$ defined by $\overline{\delta}(xN) = \delta(x)N$ for all $x \in G$. Observe that $\overline{\delta}$ is well-defined if and only if $xN = yN$ implies $\delta(x)N = \delta(y)N$ for all $x, y \in N$. However this is equivalent to the condition that $x \in N$ implies $\delta(x) \in N$ for all $x \in N$, i.e., $\delta(N) \leq N$. □

Recall that a *power endomorphism*, defined as an endomorphism that leaves all subgroups invariant, can be characterised as an endomorphism that maps every element to some power of itself. From Lemma 20.1, we obtain an analogous characterisation of descending endomorphisms. For an element x of a group G, let $\langle x^G \rangle$ denote the normal closure of x in G, i.e., the intersection of all normal subgroups of G that contain x.

Corollary 20.1 *An endomorphism δ of a group G is descending if and only if for every element $x \in G$, $\delta(x) \in \langle x^G \rangle$.*

Proof If δ is a descending endomorphism of G, then $\langle x^G \rangle \trianglelefteq G$ is left invariant by δ, and hence $\delta(x) \in \langle x^G \rangle$. Conversely, suppose δ is an endomorphism of G such that $\delta(x) \in \langle x^G \rangle$ for all $x \in G$, and let N be any normal subgroup of G. Then for all $x \in N$, $\langle x^G \rangle \leq N$ and hence $\delta(x) \in N$, which implies that $\delta(N) \leq N$. Thus, δ is descending. □

Corollary 20.1 provides a general answer to the question posed in Sect. 20.1 as to how to determine the descending endomorphisms of non-Abelian groups. It proves to be especially useful in obtaining the automorphisms of groups defined by presentations, as will be seen in Sects. 20.3 and 20.5.

Using the characterisation of descending endomorphisms given in Lemma 20.1, we can show that the property of being a descending endomorphism is inherited by the descent as well. The proof given here is different from that in [4] where this result originally appears.

Corollary 20.2 [4, Theorem 2.2] *Let G be a group, and let $\varphi\colon G \twoheadrightarrow Q$ be an epimorphism. If δ is a descending endomorphism of G, then its descent $\overline{\delta}$ to Q is a descending endomorphism of Q.*

Proof As in the proof of Lemma 20.1, we identify Q with a quotient G/N for some $N \trianglelefteq G$, and φ with the canonical projection from G to G/N. Now, we show that $\overline{\delta}$ leaves all normal subgroups of G/N invariant, which would imply, by Lemma 20.1, that $\overline{\delta}$ is a descending endomorphism of G/N. By the correspondence theorem, every normal subgroup of G/N is of the form M/N for some normal subgroup M of G such that $N \leq M$. To see that $\overline{\delta}(M/N) \leq M/N$, consider any $xN \in M/N$ with $x \in M$ (without loss of generality) and observe that $\overline{\delta}(xN) = \delta(x)N$. Since $M \trianglelefteq G$ is left invariant by δ, $\delta(x) \in M$, which implies that $\delta(x)N \in M/N$. □

The set of all descending endomorphisms of G, denoted by $\operatorname{Des} G$, is a submonoid of the endomorphism monoid $\operatorname{End} G$. The identity automorphism and the trivial endomorphism that map every element to the identity element are descending endomorphisms of any group. Lemma 20.1 shows that all inner automorphisms and power endomorphisms are descending endomorphisms. Power endomorphisms are studied in [2, 3, 6]. If A is an Abelian group then all its subgroups are normal, and hence $\operatorname{Des} A$ is the set of all power endomorphisms of A. A power endomorphism ε of A is *universal* if there exists an integer k such that for all $x \in A$, $\varepsilon(x) = x^k$. The following result proved in [4] will be used here.

Theorem 20.1 [4, Theorem 2.5] *Suppose that A is an Abelian group satisfying any one of the following properties:*

(i) *A is a direct product of cyclic groups.*
(ii) *A has finite exponent.*
(iii) *A has an element of infinite order.*

Then the descending endomorphisms of A are exactly its universal power endomorphisms.

A non-Abelian group may have descending endomorphisms that are neither power endomorphisms nor inner automorphisms, as shown in the following example.

Example

Consider the symmetric group S_3 generated by the permutations $r = (123)$ and $s = (12)$. Let δ be the endomorphism of S_3 that maps r to 1 and fixes s. Then δ is neither an inner automorphism nor a power endomorphism. The only proper, non-trivial normal subgroup of S_3 is $\langle r\rangle$, and $\delta(\langle r\rangle) = 1 \leq \langle r\rangle$. Hence, $\delta \in \operatorname{Des} S_3$.

Now we define some notation that will be followed throughout the rest of the paper. Let $G = \prod_{i=1}^{m} G_i$ be the direct product of m groups $G_1, \ldots, G_m$. Let π_i be the canonical projection G onto G_i. For $\delta \in \operatorname{Des} G$, the restriction of δ to G_i, denoted by $\delta|_{G_i}$, is a descending endomorphism of G_i.

Definition 20.2 If $\delta_i \in \operatorname{Des} G_i, i = 1, \ldots, m$, then their *pointwise product*, denoted by $\delta = \prod_{i=1}^{m} \delta_i$ is the endomorphism of $G = \prod_{i=1}^{m} G_i$ defined by $\delta(x_1, \ldots, x_m) = (\delta_1(x_1), \ldots, \delta_m(x_m))$, for all $(x_1, \ldots, x_m) \in G$.

Let N be a fixed normal subgroup of G, and let $N_i = \pi_i(N)$ be its projection into G_i. Then $\prod_{i=1}^{m}[G_i, N_i] \leq N \leq \prod_{i=1}^{m} N_i$, with all three subgroups normal in G.

If $H \leq G$ is such that $\prod_{i=1}^{m}[G_i, N_i] \leq H$, then with respect to N, $\overline{H}$ denotes the unique subgroup of $G/(\prod_{i=1}^{m}[G_i, N_i])$ corresponding to H under the bijection defined by the correspondence theorem, i.e., $\overline{H} = H/(\prod_{i=1}^{m}[G_i, N_i])$. In particular, $\overline{G} = G/(\prod_{i=1}^{m}[G_i, N_i])$.

Note that $H \trianglelefteq G$ if and only if $\overline{H} \trianglelefteq \overline{G}$. Since $\overline{N_1 \times N_2}$ itself is central in $\overline{G}$, every subgroup H of G lying between $\prod_{i=1}^{m}[G_i, N_i]$ and $\prod_{i=1}^{m} N_i$ is normal in G. We identify $\overline{G}$ with the product $\prod_{i=1}^{m} \overline{G_i}$ via their canonical isomorphism, where $\overline{G_i} = G_i/[G_i, N_i]$. Let δ_i be a fixed descending endomorphism of G_i. Then the pointwise product $\prod_{i=1}^{m} \delta_i$ is an endomorphism of G, but is not necessarily descending. However, δ_i descends to $\overline{\delta_i} \in \operatorname{Des} \overline{G_i}$ along the canonical epimorphism from G_i to $\overline{G_i}$, and $\prod_{i=1}^{m} \overline{\delta_i} \in \operatorname{End} \overline{G}$. Thus, $\prod_{i=1}^{m} \overline{\delta_i}$ maps each element $(x_1, \ldots, x_n)(\prod_{i=1}^{m}[G_i, N_i]) \in \overline{G}$ to $(\delta_1(x_1), \ldots, \delta_m(x_m))(\prod_{i=1}[G_i, N_i])$. The following results appearing in [4] provide a complete description of the descending endomorphisms of direct products.

Theorem 20.2 [4, Theorem 3.9] *Let $G = \prod_{i=1}^{k} G_i$ be the direct product of k groups $G_1, \ldots, G_k$. Let $\delta_i \in \operatorname{Des} G_i$, $i = 1, \ldots, k$, and let $\delta = \prod_{i=1}^{k} \delta_i$ denote their pointwise product. Then δ is a descending endomorphism of G if and only if for all $N_i \trianglelefteq G_i$, $i = 1, \ldots, k$, the restriction $\overline{\delta}|_{\overline{N}}$ is a descending endomorphism of $\overline{N}$, where $\overline{N} = \prod_{i=1}^{k} (N_i/[G_i, N_i])$ and $\overline{\delta} = \prod_{i=1}^{k} \overline{\delta}_i$.*

Theorem 20.3 [4, Theorem 3.10] *Let $A = \prod_{i=1}^{k} A_i$ be the direct product of k torsion Abelian groups $A_1, \ldots, A_k$. If $\delta_i \in \operatorname{Des} A_i$, $i = 1, \ldots, k$, then*

(i) *for each $a_i \in A_i$ there exists $m_i \in \mathbb{Z}$ such that $\delta_i(a_i) = a_i^{m_i}$, $i = 1, \ldots, k$, and*
(ii) *the pointwise product $\delta_1 \times \cdots \times \delta_k$ is a descending endomorphism of A if and only if for every pair $i, j \in \{1, \ldots, k\}$, and for all $a_i \in A_i$ and $a_j \in A_j$, $m_i \equiv m_j \mod \gcd\left(|a_i|, |a_j|\right)$.*

Remark 20.1 In Theorem 20.3, whenever A_i has finite exponent n_i, we may replace $|a_i|$ in (ii) by n_i, thus making the condition independent of individual elements $a_i \in A_i$.

20.2 Descending Endomorphisms of Symmetric Groups

In this section, we will first show that all endomorphisms of S_n are descending, and then determine the descending endomorphisms of direct products of symmetric groups. We begin by recalling some elementary facts about symmetric groups. Let S_n be the symmetric group, considered as the group of all permutations of the set $X = \{1, 2, \ldots, n\}$, where $n \geq 5$. Let A_n be the alternating group, which is the subgroup of S_n consisting of all even permutations in S_n. Then A_n is the unique proper, non-trivial normal subgroup of S_n, has index 2 in S_n, is non-Abelian, and $[S_n, A_n] = [S_n, S_n] = A_n$.

S_n is centreless. For $n \neq 6$, all automorphisms of S_n are inner automorphisms, and $\operatorname{Aut} S_n \cong S_n$. For $n = 6$, the inner automorphism group has index 2 in $\operatorname{Aut} S_6$, which is isomorphic to a semidirect product of S_6 and C_2. Here, C_2 denotes the cyclic group $\{1, -1\}$ under multiplication.

Consider any endomorphism δ of S_n that is neither trivial nor an automorphism. Then $\ker \delta = A_n$. Let $s \in S_n$ be the permutation (12) that interchanges 1 and 2 and fixes all other elements of X. Then δ is uniquely determined by its action on s, as $|S_n : A_n| = 2$. Moreover, any mapping of s to an element of order 2 defines such an endomorphism δ, since we may regard δ as the composition of a homomorphism from S_n to C_2 sending s to $-1 \in C_2$, with an embedding of C_2 into S_n sending -1 to the desired element of order 2. Thus, the non-trivial, non-bijective endomorphisms of S_n are in one-to-one correspondence with the elements of S_n that have order 2.

First, we establish a result from which it will follow that all endomorphisms of S_n are descending.

Theorem 20.4 *If the normal subgroups of a finite group G form a chain under inclusion, then all endomorphisms of G are descending.*

Proof Since G is finite, its chain of normal subgroups has a finite length, say m. The result is obvious for $m = 1, 2$. Assume for the sake of induction that the result holds for a group whenever its normal subgroups form a chain of length less than m.

Let δ be any endomorphism of G. If δ is injective, then it is an automorphism of G, and hence maps each normal subgroup of G to a normal subgroup of the same order. However, since the normal subgroups form a chain, no two distinct normal subgroups can have the same order, and, therefore, δ maps each to itself.

If δ is not injective, let $K = \ker \delta$, and consider G/K. The canonical projection from G to G/K defines a bijection between the normal subgroups of G containing K and the normal subgroups of G/K, and, therefore, the normal subgroups of G/K form a chain of length less than m (since $K > 1$).

Define $\overline{\delta}\colon G/K \to G/K$ given by $\overline{\delta}(gK) = \delta(g)K$ for all $g \in G$. This map is a well-defined endomorphism of G/K, since $K = \ker \delta$.

Now let N be any normal subgroup of G. Then either $N \leq K$ or $K < N$. If $N \leq K$, then $\delta(N) = 1 \leq N$. Otherwise, N/K is a normal subgroup of G/K and by the induction hypothesis, $\overline{\delta}(N/K) \leq N/K$. Thus, for all $x \in N$, $\overline{\delta}(xK) = \delta(x)K = yK$ for some $y \in N$, which implies that $y^{-1}\delta(x) \in K \leq N$, and therefore, $\delta(x) \in N$. Thus, $\delta(N) \leq N$ in this case as well.

Since $\delta(N) \leq N$ for all $N \trianglelefteq G$, it follows from Lemma 20.1 that δ is a descending endomorphism of G. □

As observed earlier, S_n has a unique proper, non-trivial normal subgroup, A_n, and therefore the normal subgroups of S_n form the chain $1 < A_n < S_n$. Hence, Theorem 20.4 applies to S_n, and $\operatorname{Des} S_n = \operatorname{End} S_n$. Now, any descending endomorphism of a direct product of symmetric groups is a pointwise product of endomorphisms of the direct factors. The following result characterises all such pointwise products.

Theorem 20.5 *Let $G = \prod_{i=1}^{k} G_i$ where $G_i \cong S_{n_i}$ and $n_i \geq 5$, $i = 1, \ldots, k$. Let $s_i \in G_i$ be the permutation* (12)*, and let $M_i \cong A_{n_i}$ denote the alternating subgroup of G_i. If $\delta_i \in \operatorname{End} G_i = \operatorname{Des} G_i$, $i = 1, \ldots, k$, then the pointwise product $\delta = \prod_{i=1}^{k} \delta_i$ is a descending endomorphism of G if and only if either $\delta_i(s_i) \in M_i$ for all $i \in \{1, \ldots, k\}$, or $\delta_i(s_i) \notin M_i$ for all $i \in \{1, \ldots, k\}$.*

Proof Let $\delta_i \in \operatorname{End} G_i$, $i = 1, \ldots, k$, and let $\delta = \prod_{i=1}^{k} \delta_i$. Since $G_i \cong S_{n_i}$, $\delta_i \in \operatorname{Des} G_i$, by Theorem 20.4. We shall use Theorem 20.2 to establish the result. For this purpose, we consider all possible normal subgroups N_i of G_i, $i = 1, \ldots, k$, and apply the condition in Theorem 20.2 to them. Let $\overline{N} = \prod_{i=1}^{k} (N_i/[G_i, N_i])$.

Recall that each $N_i \trianglelefteq G_i$ must be 1, $M_i \cong A_{n_i}$, or $G_i \cong S_{n_i}$. Let $\overline{N}_i = N_i/[G_i, N_i]$, and let $\overline{\delta}_i$ be the descent of δ_i to $\overline{N}_i$, which is a finite Abelian group. By Theorem 20.1, for each $i = 1, \ldots, k$, there exists an integer m_i such that for all $a_i \in \overline{N}_i$, $\delta_i(a_i) = a_i^{m_i}$.

If $N_i = 1$ or M_i, then $[G_i, N_i] = N_i$, so $\overline{N}_i = 1$. In this case, the integer m_i mentioned in the preceding paragraph can be taken to be any integer, and therefore we may ignore this case, as it is independent of the condition in Theorem 20.3, (ii).

If $N_i = G_i$, then $[G_i, N_i] = M_i$, so $\overline{N}_i \cong S_{n_i}/A_{n_i}$ has order 2. Now, if $\delta_i(s_i) \in M_i$, then $\overline{\delta}_i$ is the trivial endomorphism, and therefore $m_i \equiv 0 \mod 2$. If $\delta_i(s_i) \notin M_i$,

then $\overline{\delta}_i$ is the identity automorphism of $\overline{N}_i$, and therefore $m_i \equiv 1 \mod 2$. Now, from Theorem 20.3 and Remark 20.1, δ is a descending endomorphism of G if and only if $m_i \equiv m_j \mod 2$ for all $i, j = 1, \ldots, k$, and according to our preceding observation, this is true if and only if δ_i maps s_i to M_i either for all i or for no i. This concludes the proof. □

20.3 Descending Endomorphisms of Dihedral Groups

Consider the dihedral group D_n of order $2n$ as defined by the following presentation:

$$D_n = \langle r, s \mid r^n = s^2 = 1,\ srs = r^{-1} \rangle.$$

Then D_n is a semidirect product of its subgroups $\langle r \rangle \trianglelefteq D_n$ and $\langle s \rangle$, of orders n and 2, respectively. Thus, every element of D_n can be written uniquely in the form $r^i s^j$ for some $i \in \{0, \ldots, n-1\}$ and $j \in \{0, 1\}$.

Remark 20.2 Any endomorphism ε of D_n can be defined by specifying its action on the generators r and s, and extending the action to all elements $r^u s^v$ as $\varepsilon(r^u s^v) = (\varepsilon(r))^u (\varepsilon(s))^v$. Such a map is well-defined if it is consistent with the relations in the presentation of D_n, i.e., if $(\varepsilon(r))^n = (\varepsilon(s))^2 = 1$ and $\varepsilon(s)\varepsilon(r)\varepsilon(s) = (\varepsilon(r))^{-1}$. Then it is an endomorphism by definition.

We will show that the descending endomorphisms of D_n can be classified into two types of endomorphisms, each defined by two parameters. The possible values of the parameters for which these endomorphisms are descending depend on whether n is odd, singly even, or doubly even.

Definition 20.3 For every pair $i, j \in \{0, \ldots, n-1\}$, define maps $\delta^{(1;ij)}$ and $\delta^{(2;ij)}$ on D_n by their actions on the generators r and s, as follows:

(i) $\delta^{(1;ij)} : r \mapsto r^i, s \mapsto r^{2j}s$.
(ii) $\delta^{(2;ij)} : r \mapsto r^i, s \mapsto r^{2j}$.

When $\delta^{(1;ij)}$ and $\delta^{(2;ij)}$ are well-defined endomorphisms of D_n, we shall call them endomorphisms of the *first kind* and *second kind*, respectively.

We shall first determine the values of the parameters i and j for which these are well-defined maps on D_n, and hence endomorphisms.

Lemma 20.2 *Let $G = D_n$, and let $\delta^{(1;ij)}$ and $\delta^{(2;ij)}$ be the maps on G as given in Definition 20.3 for $i, j \in \{0, \ldots, n-1\}$. Then the following hold:*

(i) *For each n, $\delta^{(1;ij)}$ is well-defined for all values of i and j.*
(ii) *If n is odd, $\delta^{(2;ij)}$ is well-defined if and only if $i = j = 0$.*
(iii) *If $n \equiv 2 \mod 4$, then $\delta^{(2;ij)}$ is well-defined if and only if $i \in \left\{0, \frac{n}{2}\right\}$ and $j \in \left\{0, \frac{n}{2}\right\}$.*
(iv) *If $n \equiv 0 \mod 4$, then $\delta^{(2;ij)}$ is well-defined if and only if $i \in \left\{0, \frac{n}{2}\right\}$ and $j \in \left\{0, \frac{n}{4}, \frac{n}{2}, \frac{3n}{4}\right\}$.*

Proof Note that δ, whether it is of the first kind or the second kind, maps r to r^i, which implies that $(\delta(r))^n = 1$. Thus, it is sufficient to check the other two conditions given in Remark 20.2.

Firstly, if $\delta = \delta^{(1;ij)}$, then $(\delta(s))^2 = (r^{2j}s)^2 = 1$ and $\delta(s)\delta(r)\delta(s) = (r^{2j}s)r^i(r^{2j}s) = r^{-i} = (\delta(r))^{-1}$, for all i and j. Hence, $\delta^{(1;ij)}$ is always well-defined.

Now let $\delta = \delta^{(2;ij)}$. Then $(\delta(s))^2 = r^{4j} = 1$ if and only if $4j \equiv 0 \mod n$, and $\delta(s)\delta(r)\delta(s) = r^i$, which will be equal to $\delta(r)^{-1} = r^{-i}$ if and only if $2i \equiv 0 \mod n$. When n is odd, these two congruences hold only when $i = j = 0$. When n is even, these imply that $i \in \{0, \frac{n}{2}\}$, and $j \in \left\{0, \frac{n}{2}\right\}$ or $j \in \left\{0, \frac{n}{4}, \frac{n}{2}, \frac{3n}{4}\right\}$ depending on whether $n \equiv 2 \mod 4$ or $n \equiv 0 \mod 4$, respectively. □

Now, to determine the values of i and j for which $\delta^{(1;ij)}$ and $\delta^{(2;ij)}$ are descending endomorphisms of D_n, we use the fact that an endomorphism is descending if and only if it maps every element to some element in its normal closure (Corollary 20.1). Observe that in D_n, the normal closure of an arbitrary element of the form r^k is the cyclic subgroup $\langle r^k \rangle$, and the normal closure of an arbitrary element of the form $r^k s$ is $\langle r^2, r^k s \rangle$, which can be written in the simpler form $\langle r^2, r^t s \rangle$, where $t \in \{0, 1\}$ and $t \equiv k \mod 2$. In particular, the normal closures of r and s are $\langle r \rangle$ and $\langle r^2, s \rangle$, respectively. Note that when n is odd, then for all k, $\langle r^2, r^k s \rangle$ is D_n itself.

Theorem 20.6 *The descending endomorphisms of D_n are completely described and enumerated by the following:*

(i) *If $n \equiv 1 \mod 2$, then*

$$\operatorname{Des} D_n = \left\{ \delta^{(1;ij)} \;\middle|\; 0 \le i \le n-1,\ 0 \le j \le n-1 \right\} \cup \left\{\delta^{(2;00)}\right\}$$

and $|\operatorname{Des} D_n| = n^2 + 1$.

(ii) *If $n \equiv 2 \mod 4$, then*

$$\operatorname{Des} D_n = \left\{ \delta^{(1;ij)} \;\middle|\; i \in \{1, 3, \ldots, n-1\},\ 0 \le j \le \frac{n}{2} - 1 \right\} \cup \left\{ \delta^{(2;00)} \right\}$$

and $|\operatorname{Des} D_n| = \dfrac{n^2}{4} + 1$.

(iii) *If* $n \equiv 0 \mod 4$, *then*

$$\operatorname{Des} D_n = \left\{ \delta^{(1;ij)} \;\middle|\; i \in \{1, 3, \ldots, n-1\},\ 0 \le j \le \frac{n}{2} - 1 \right\} \cup \left\{ \delta^{(2;ij)} \;\middle|\; i \in \left\{0, \frac{n}{2}\right\},\ j \in \left\{0, \frac{n}{4}\right\} \right\}$$

and $|\operatorname{Des} D_n| = \dfrac{n^2}{4} + 4.$

Proof We begin by noting that any descending endomorphism of D_n must necessarily map r to r^i for some $i \in \{0, \ldots, n-1\}$, and s to either $r^{2j}s$ or r^{2j} for some $j \in \{0, \ldots, n-1\}$. This is evident from the structure of the normal closures of r and s. It is also clear that when n is odd, r^{2j} has n distinct values for $j = 0, \ldots, n-1$, whereas when n is even, r^{2j} has exactly $\frac{n}{2}$ distinct values, which are obtained by taking $j = 0, \ldots, \frac{n}{2} - 1$. Therefore, to determine $\operatorname{Des} D_n$, it is sufficient to consider only maps of the form $\delta^{(1;ij)}$ and $\delta^{(2;ij)}$ for the values of i and j in the ranges specified.

Now, by considering the normal closure of an arbitrary element of D_n, we show that the values of i and j for which $\delta^{(1;ij)}$ and $\delta^{(2;ij)}$ are descending, in each of the cases given in (i), (ii), and (iii), are indeed as claimed. Both kinds of endomorphisms define power maps on the subgroup $\langle r \rangle$, and hence map its elements to their normal closures. Thus, we need to consider only an arbitrary element of the form $r^k s$. Recall that the normal closure of $r^k s$ can be written as $\langle r^2, r^t s \rangle$, where $t \in \{0, 1\}$ and $t \equiv k \mod 2$.

First, consider $\delta = \delta^{(1;ij)}$, applied to the element rs, which gives $\delta(r)\delta(s) = r^i r^{2j} s = r^{i+2j} s$. If δ is descending, the values of i and j should be such that $r^{i+2j} s \in \langle r^2, rs \rangle$, the normal closure of rs. If n is odd, then $\langle r^2, rs \rangle = D_n$, and therefore i and j can take any value. If n is even, then $r^{i+2j} s \in \langle r^2, rs \rangle$ only if $i + 2j$ is odd, i.e., i is odd. To see that this is sufficient to make δ descending, observe that $\delta(r^k s) = r^{ik+2j} s \in \langle r^2, r^t s \rangle$, where $t \in \{0, 1\}$ and $t \equiv k \mod 2$, if and only if i is odd. This proves that the values of i and j must be as given, in each case, for δ of the first kind.

Next, consider $\delta = \delta^{(2;ij)}$, which when applied to rs gives $\delta(r)\delta(s) = r^i r^{2j} = r^{i+2j}$. From Lemma 20.2, we know that δ is trivial if n is odd. This completes the proof of (i). Now let n be even. Recall that $i \in \{0, \frac{n}{2}\}$, and $j \in \left\{0, \frac{n}{2}\right\}$ or $j \in \left\{0, \frac{n}{4}, \frac{n}{2}, \frac{3n}{4}\right\}$ accordingly as $n \equiv 2 \mod 4$ or $n \equiv 0 \mod 4$. Now r^{i+2j} belongs to the normal closure of rs, namely $\langle r^2, rs \rangle$, only if $i + 2j$ is even, and hence i is even. In this case, $\delta(r^k s) = r^{ik+2j}$, which belongs to $\langle r^2, r^t s \rangle$, where $t \in \{0, 1\}$ and $t \equiv k \mod 2$. Thus, δ is descending if and only if i is even. If $n \equiv 2 \mod 4$, then $\frac{n}{2}$ is odd, and this forces i to be 0, which proves (ii). If $n \equiv 0 \mod 4$, then both $\frac{n}{2}$ and 0 are even, and hence i can assume both these values.

As noted, the respective ranges given for i and j define distinct descending endomorphisms of D_n, which immediately lead to the enumerations of $|\operatorname{Des} D_n|$ given in (i), (ii), and (iii). □

We can now use results on direct products to compute the descending endomorphisms of all finite direct products of dihedral groups. In order to apply Theorem 20.2, we need to determine, for every $N \trianglelefteq D_n$, the quotient $\overline{N} = N/[D_n, N]$ as well as, for every $\delta \in \mathrm{Des}\, D_n$, the restriction $\overline{\delta}|_{\overline{N}}$ of the descent $\overline{\delta} \in \mathrm{Des}(D_n/[D_n, N])$. Then, in the direct product $\prod_{k=1}^{m} D_{n_k}$, for every normal subgroup N_k of each factor D_{n_k} and every $\delta_k \in \mathrm{Des}\, D_{n_k}$, we may use the results of this general computation in conjunction with Theorem 20.2, to identify all the pointwise products of δ_k that define descending endomorphisms of the direct product.

Let 1 denote the trivial group, C_2 the cyclic group of order 2, and $C_2^2 = C_2 \times C_2$ the Klein 4-group.

Lemma 20.3 *If $G = D_n$ and $N \trianglelefteq G$, then $N/[G, N] \cong 1$, C_2, or C_2^2.*
If $N/[G, N] \cong C_2$, then one of the following holds:

(i) $N \leq \langle r \rangle$, *n is even, and* $[G, N] = \langle x^2 \mid x \in N \rangle$.
(ii) $N = \langle r^2, s \rangle$ *or* $\langle r^2, rs \rangle$, *n is even, and* $[G, N] = \langle r^2 \rangle$.
(iii) $N = G$ *and n is odd, and* $[G, N] = \langle r \rangle$.

If $N/[G, N] \cong C_2^2$, then n is even, $N = G$, and $[G, N] = \langle r^2 \rangle$.

Proof Observe that the commutator of any two elements $r^i s^t$ and $r^j s^u$, i.e., $[r^i s^t, r^j s^u]$, is an element of the form $r^{2\alpha i + 2\beta j}$, where α and β may be 0 or 1 depending on the values of t and u. Hence, for any $N \trianglelefteq G$, $[G, N] \leq \langle r^2 \rangle$. Now we consider all non-trivial normal subgroups N.

If $N \leq \langle r \rangle$, then the general form of the commutator shows that $[G, N] = \langle x^2 \mid x \in N \rangle$. Clearly, in this case, $|N : [G, N]|$ is 2 if n is even and 1 if n is odd.

If $N = \langle r^2, s \rangle$ or $\langle r^2, rs \rangle$, then $[G, N] = \langle r^2 \rangle$, since $[r, s] = [r, rs] = r^2 \in [G, N] \leq \langle r^2 \rangle$. In this case, if n is even, then $\langle r^2 \rangle < N < G$ and $|N : [G, N]| = 2$, whereas if n is odd, then $N = G$ and $\langle r^2 \rangle = \langle r \rangle$, which again implies that $|N : [G, N]| = 2$.

The only remaining case is $N = G$ with n even. In this case, $[G, N] = \langle r^2 \rangle < \langle r \rangle$, by the same argument as in the previous case, and then $N/[G, N]$ is the quotient group isomorphic to C_2^2 generated by $s[G, N] = s\langle r^2 \rangle$ and $rs[G, N] = rs\langle r^2 \rangle$. □

Let $G = D_n$, $N \trianglelefteq G$, and $\delta \in \mathrm{Des}\, D_n$. Following the notation of Theorem 20.2, $\overline{\delta}$ denotes the descent of δ to $\overline{G} = G/[G, N]$, and $\overline{\delta}|_{\overline{N}}$ denotes the restriction of $\overline{\delta}$ to the subgroup $\overline{N} = N/[G, N]$. $\overline{N}$ is central in $\overline{G}$ and hence $\overline{\delta}|_{\overline{N}} \in \mathrm{Des}\, \overline{N}$. From Lemma 20.3, we know that $\overline{N}$ is 1, C_2, or C_2^2. Thus, $\overline{\delta}|_{\overline{N}} \in \mathrm{Des}\, \overline{N}$ must be either the trivial or the identity endomorphism. Observing that $\overline{\delta}|_{\overline{N}}$ is trivial if and only if $\delta(N) \leq [G, N]$, we obtain the following corollary to Lemma 20.3.

Corollary 20.3 *Let $G = D_n$, $\delta = \delta^{(t;ij)} \in \mathrm{Des}\, G$, for some i and j, and $t \in \{1, 2\}$. Let $N \trianglelefteq G$.*

(i) *If $N \leq \langle r \rangle$, then $\overline{\delta}|_{\overline{N}}$ is trivial if and only if i is even.*
(ii) *If $N = \langle r, s \rangle$, $\langle r^2, rs \rangle$, or G, then $\overline{\delta}|_{\overline{N}}$ is trivial if and only if $t = 2$.*

Using the cases listed in Theorem 20.6, we can now show that $\overline{\delta}|_{\overline{N}}$ is the identity or trivial endomorphism accordingly as δ is of the first or second kind.

First, suppose that δ is of the first kind. If N is not contained in $\langle r \rangle$, then by Corollary 20.3, $\overline{\delta}|_{\overline{N}}$ is the identity endomorphism of $\overline{N}$. If $N \leq \langle r \rangle$, then we have two cases depending on the value of n. If n is odd, then $\overline{N} = 1$, and hence $\delta_{\overline{N}}$ is the identity endomorphism of the trivial group. If n is even, then by Theorem 20.6, $\delta = \delta^{(1;ij)}$ where i is odd, and hence by Corollary 20.3, $\overline{\delta}|_{\overline{N}}$ is again the identity endomorphism.

Suppose on the other hand that δ is of the second kind. Then by reasoning along the same lines, we see that $\overline{\delta}_{\overline{N}}$ must be trivial in all cases. Note that when n is odd and $N \leq \langle r \rangle$, $\overline{N} = 1$ as in the previous case, but the identity endomorphism of the trivial group is the same as its trivial endomorphism. Thus, we obtain the following second corollary to Theorem 20.6 and Lemma 20.3.

Corollary 20.4 *Let $G = D_n$ and $\delta \in \operatorname{Des} G$. Then for any $N \trianglelefteq G$, $\overline{\delta}_{\overline{N}}$ is the identity endomorphism of $\overline{N}$ if δ is of the first kind, and is the trivial endomorphism of $\overline{N}$ if δ is of the second kind.*

Applying Corollary 20.4, we prove that the descending endomorphisms of any finite direct product of dihedral groups are the pointwise products of descending endomorphisms of the same kind.

Theorem 20.7 *Let $G = \prod_{k=1}^{m} G_k$, where $G_k = D_{n_k}$, $i = 1, \ldots, m$. Then the descending endomorphisms of G are all the pointwise products of the form $\prod_{k=1}^{m} \delta_k$ where each $\delta_k \in \operatorname{Des} G_k$ and the δ_k are all of the same kind.*

Proof We know, from Theorem 20.2, that every descending endomorphism δ of G is a pointwise product $\delta = \prod_{k=1}^{m} \delta_k$ of descending endomorphisms δ_k of G_k, $k = 1, \ldots, m$, and that any such pointwise product is a descending endomorphism precisely when, for all $N_k \trianglelefteq G_k, k = 1, \ldots, m$, it holds that $\prod_{k=1}^{m} \overline{\delta}_k|_{\overline{N}_k}$ is a descending endomorphism of $\prod_{k=1}^{m} \overline{N}_k$. Let $\overline{N}$ denote $\prod_{k=1}^{m} \overline{N}_k$.

From Lemma 20.3, we know that for each k, $\overline{N}_k$ is 1, C_2, or C_2^2. Thus, $\overline{N} \cong C_2^l$, for some non-negative integer l. In particular, when $N_k = G_k$ for at least one $k \in \{1, \ldots, m\}$, $\overline{N} \cong C_2^l$ for some $l > 0$, since $\overline{N}_k$ is C_2 or C_2^2. Thus, $\prod_{k=1}^{m} \overline{\delta}_k|_{\overline{N}_k}$ is descending if and only if it is either the trivial endomorphism or the identity endomorphism of $\overline{N}$, or equivalently, $\overline{\delta}_k|_{\overline{N}_k}$ is trivial either for all k, or for no k. From Corollary 20.4, this is equivalent to the requirement that all δ_k be of the same kind. □

20.4 Descending Endomorphisms of Hamiltonian Groups

A *Hamiltonian* group is a non-Abelian group in which every subgroup is normal. The smallest Hamiltonian group is the quaternion group Q_8 of order 8. Reinhold

Baer, in [1], showed that a group G is Hamiltonian if and only if it is a direct product of the quaternion group, an elementary Abelian 2-group, and a torsion Abelian group in which every element has odd order (for proof of this result, see [8, Theorem 9.7.4]). Applying Theorems 20.2 and 20.3, we shall now determine all descending endomorphisms of Hamiltonian groups. Observe that since all subgroups of a Hamiltonian group are normal, every descending endomorphism of such a group is a power endomorphism, and vice versa.

The quaternion group is a non-Abelian group of order 8, defined by $Q_8 = \{1, -1, i, -i, j, -j, k, -k\}$ where $i^2 = j^2 = k^2 = ijk = -1$ and 1 is the identity element. The set $\{i, j\}$ is a generating set of Q_8. Therefore, we may define all endomorphisms of Q_8 by specifying their actions on i and j. If $\tau \in \operatorname{Des} Q_8$, then it must be a power endomorphism, as noted earlier, and hence is completely defined by the powers of i and j to which it maps them.

Definition 20.4 For $u, v \in \{0, 1, 2, 3\}$, define $\tau^{(uv)}$ to be the unique endomorphism of Q_8 such that $\tau^{(uv)}(i) = i^u$ and $\tau^{(uv)}(j) = j^v$, provided it is well-defined.

We can determine $\operatorname{Des} Q_8$ by a simple and direct computation. We present the result below, omitting the details (also see Theorem 20.9 and Remark 20.3).

Proposition 20.1 *The quaternion group Q_8 has eight descending endomorphisms, given by*

$$\operatorname{Des} Q_8 = \{\, \tau^{(uv)} \mid u, v \in \{0, 1, 2, 3\}, u \equiv v \mod 2 \,\}.$$

Note that every descending endomorphism of Q_8 is a power endomorphism that either maps all elements to odd powers or maps all elements to even powers. Also note that for each non-trivial normal subgroup N of Q_8, $[Q_8, N] = \langle -1 \rangle$ and $\overline{N} = N/[Q_8, N] \cong 1$ (for $N = \langle -1 \rangle$), C_2 (for $N = \langle i \rangle$, $\langle j \rangle$, or $\langle k \rangle$), or C_2^2 (for $N = Q_8$). Thus, for all $\tau \in \operatorname{Des} Q_8$, $\overline{\tau}|_{\overline{N}}$ (the descent of τ to $Q_8/[Q_8, N]$, restricted to $\overline{N}$) is either the identity automorphism or the trivial endomorphism of $\overline{N}$. In fact, we observe the following.

Lemma 20.4 *Let $\tau \in \operatorname{Des} Q_8$. Then for any non-trivial $N \trianglelefteq Q_8$, $\overline{\tau}|_{\overline{N}}$ is the identity automorphism of $\overline{N}$ if τ is an odd-power endomorphism of Q_8, and is the trivial endomorphism of $\overline{N}$ if τ is an even-power endomorphism of Q_8.*

The following result shows that the condition in Theorem 20.3 can be simplified in the case of a direct product where one factor is Abelian.

Lemma 20.5 *Let $G = G_1 \times G_2$, where G_2 is Abelian, and let $\delta_i \in \operatorname{Des} G_i$, $i = 1, 2$. Then $\delta_1 \times \delta_2 \in \operatorname{Des} G$ if and only if, for all $N_1 \trianglelefteq G_1$, $\overline{\delta}_1|_{\overline{N}_1} \times \delta_2 \in \operatorname{Des} \overline{N}_1 \times G_2$.*

Proof The implication in one direction is obvious. For the converse, suppose that for all $N_1 \trianglelefteq G_1$, $\overline{\delta}_1|_{\overline{N}_1} \times \delta_2 \in \operatorname{Des} \overline{N}_1 \times G_2$. To prove that $\delta_1 \times \delta_2 \in \operatorname{Des} G_1 \times G_2$, we need to show that for all $N_1 \trianglelefteq G_1$ and $N_2 \trianglelefteq G_2$, $(\overline{\delta}_1 \times \overline{\delta}_2)|_{\overline{N}_1 \times \overline{N}_2} \in \operatorname{Des} \overline{N}_1 \times \overline{N}_2$ (Theorem 20.2). But as G_2 is Abelian, for each $N_2 \trianglelefteq G_2$, $\overline{N}_2 = N_2$, and hence

$(\overline{\delta}_1 \times \overline{\delta}_2)|_{\overline{N}_1 \times \overline{N}_2}$ is the restriction of $\overline{\delta}_1|_{\overline{N}_1} \times \delta_2$ to $\overline{N}_1 \times N_2 = \overline{N}_1 \times \overline{N}_2$. Again, since $\overline{N}_1 \times G_2$ is Abelian the restriction of $\overline{\delta}|_{\overline{N}_1} \times \delta_2 \in \operatorname{Des} \overline{N}_1 \times G_2$ to any subgroup is a descending endomorphism of the subgroup, completing the proof. □

Lemma 20.5 can be applied to Hamiltonian groups, since they are direct products of the non-Abelian group Q_8 with an Abelian group, the latter itself being the direct product of two Abelian groups.

Theorem 20.8 *Let G be a Hamiltonian group with direct product decomposition $G = G_1 \times G_2 \times G_3$, where $G_1 \cong Q_8$, G_2 is elementary 2-Abelian, and G_3 is torsion Abelian with all elements having odd order. Let id_{G_2} and 0_{G_2} denote the identity automorphism and trivial endomorphism, respectively, of G_2. Then*

$$\begin{aligned}\operatorname{Des} G = &\{\, \tau^{(uv)} \times 0_{G_2} \times \delta_3 \mid u, v \in \{0, 2\}, \delta_3 \in \operatorname{Des} G_3 \,\} \cup \\ &\{\, \tau^{(uv)} \times \mathrm{id}_{G_2} \times \delta_3 \mid u, v \in \{1, 3\}, \delta_3 \in \operatorname{Des} G_3 \,\}\end{aligned}$$

where $\operatorname{Des} G_3$ *is the set of all power endomorphisms of G_3. If G_3 has finite exponent n, then* $|\operatorname{Des} G| = 8n$.

Proof We know, from Proposition 20.1, that the descending endomorphisms of $G_1 \cong Q_8$ are of the form $\tau^{(uv)}$ where $0 \le u, v \le 3$ and $u \equiv v \mod 2$, and that these are power endomorphisms that map all elements of G_1 to odd or even powers according to whether u and v are both odd or both even, respectively.

Now consider the torsion Abelian group $A = G_2 \times G_3$, all of whose descending endomorphisms are power endomorphisms. By Lemma 20.5, the descending endomorphisms of $G = G_1 \times A$ are all the endomorphisms of the form $\delta_1 \times \theta$, with $\delta_1 \in \operatorname{Des} G_1$ and $\theta \in \operatorname{Des} A$, such that for all $N \trianglelefteq G_1$, $\overline{\delta}_1|_{\overline{N}_1} \times \theta \in \operatorname{Des} \overline{N}_1 \times A$. We know that for all non-trivial $N_1 \trianglelefteq G_1 \cong Q_8$, $\overline{\delta}_1|_{\overline{N}_1}$ is the identity automorphism if u and v are odd and otherwise it is the trivial endomorphism. Hence $\overline{\delta}_1|_{\overline{N}_1} \times \theta \in \operatorname{Des} \overline{N}_1 \times A$ if and only if either $\delta_1 = \tau^{(uv)}$ with u and v odd and θ is a power endomorphism of A mapping all elements to odd powers, or $\delta_1 = \tau^{(uv)}$ with u and v even and θ is a power endomorphism of A mapping all elements to even powers (by Theorem 20.3). Since $A = G_2 \times G_3$, every power endomorphism of A, i.e., every descending endomorphism of A, is of the form $\delta_2 \times \delta_3$ where δ_2 is either id_{G_2} or 0_{G_2} (the only descending endomorphisms of an elementary Abelian 2-group) and δ_3 is a power endomorphism of G_3. As every element of G_3 has odd order, any power endomorphism δ_3 of G_3 can be expressed as an odd-power endomorphism as well as an even-power endomorphism. Hence, the condition in Theorem 20.3 always holds for any $\delta_2 \times \delta_3$, and $\theta = \delta_2 \times \delta_3$ is a power endomorphism of $A = G_2 \times G_3$ that maps all elements to odd powers if $\delta_2 = \mathrm{id}_{G_2}$ and even powers if $\delta_2 = 0_{G_2}$. Then, the result follows on the application of Theorem 20.3 to $\delta_1 \times \theta$. If G_3 has finite exponent n, then $|\operatorname{Des} G_3| = n$, and hence $|\operatorname{Des} G| = 8n$. □

20.5 Descending Endomorphisms of Dicyclic Groups

In this section, we determine the descending endomorphisms of dicyclic groups. These groups are similar in structure to dihedral groups, and this similarity is reflected in their descending endomorphisms as well. The dicyclic group of order $4n$ is defined by the presentation

$$\mathrm{Dic}_n = \langle x, y \mid x^{2n} = 1, x^n = y^2, yxy^{-1} = x^{-1} \rangle.$$

The element $x^n = y^2$ is the unique involution of Dic_n, the centre of Dic_n is $\langle x^n \rangle$, and $\mathrm{Dic}_n / \langle x^n \rangle \cong D_n$ [7]. In the lemma given below, we make some observations about Dic_n that can be easily verified from the given presentation.

Lemma 20.6 *Consider the dicyclic group* Dic_n.

(i) *Every element of* Dic_n *can be written uniquely in the form* $x^u y^v$ *where* $0 \leq u \leq 2n-1$ *and* $0 \leq v \leq 1$.
(ii) *The normal closure of any element of* Dic_n *of the form* x^u *is the cyclic subgroup* $\langle x^u \rangle$.
(iii) *The normal closure of any element of* Dic_n *of the form* $x^u y$ *is* $\langle x^2, x^u y \rangle$. *If* n *is odd, then this is equal to* Dic_n *itself, and if* n *is even, then it is equal to* $\langle x^2, y \rangle$ *or* $\langle x^2, xy \rangle$ *depending on whether* u *is even or odd, respectively.*

As in the case of dihedral groups, the descending endomorphisms of dicyclic groups can also be classified into two types in terms of their action on the two generators. Let $\eta^{(1;ij)}$ and $\eta^{(2;ij)}$ be maps on Dic_n defined by the prescription

$$\begin{aligned} \eta^{(1;ij)}\colon &\quad x \mapsto x^i, \quad y \mapsto x^j y \\ \eta^{(2;ij)}\colon &\quad x \mapsto x^i, \quad y \mapsto x^j, \end{aligned}$$

$0 \leq i, j \leq 2n-1$.

Lemma 20.7 *The map* $\eta^{(1;ij)}$ *defines an endomorphism of* Dic_n *if and only if* i *is odd, and the map* $\eta^{(2;ij)}$ *defines an endomorphism of* Dic_n *if and only if* $i \equiv 0 \mod n$ *and* $2j \equiv in \mod 2n$.

Proof If $\delta = \eta^{(1;ij)}$ or $\eta^{(2;ij)}$, then δ defines an endomorphism of Dic_n if and only if the images of x and y under δ satisfy the relations given in the presentation of Dic_n, i.e., $\delta(x)^{2n} = 1$, $\delta(x)^n = \delta(y)^2$, and $\delta(y)\delta(x)\delta(y)^{-1} = \delta(x)^{-1}$. First, observe that for $\delta = \eta^{(1;ij)}$ as well as $\delta = \eta^{(2;ij)}$, $\delta(x)^{2n} = 1$.

Now, consider $\delta = \eta^{(1;ij)}$. Then $\delta(y)^2 = \delta(x)^n$ reduces to $x^n = x^{in}$, which holds if and only if $i \equiv 1 \mod 2$. Clearly, $\delta(y)\delta(x)\delta(y)^{-1} = \delta(x)^{-1}$ holds for all values of i and j.

Next, consider $\delta = \eta^{(2;ij)}$. Then $\delta(y)^2 = \delta(x)^n$ reduces to $x^{2j} = x^{in}$, which is equivalent to $2j \equiv in \mod 2n$, and $\delta(y)\delta(x)\delta(y)^{-1} = \delta(x)^{-1}$ reduces to $x^i = x^{-i}$, which holds if and only if $i \equiv 0 \mod n$. □

Theorem 20.9 *The descending endomorphisms of* Dic_n *are described and enumerated by the following:*

(i) *If* n *is odd, then*

$$\mathrm{Des}\,\mathrm{Dic}_n = \{\,\eta^{(1;ij)} \mid i \in \{1, 3, \ldots, 2n-1\}, 0 \le j \le 2n-1\,\} \cup \{\eta^{(2;00)}, \eta^{(2;0n)}\}$$

and $|\,\mathrm{Des}\,\mathrm{Dic}_n| = 2n^2 + 2$.

(ii) *If* n *is even, then*

$$\mathrm{Des}\,\mathrm{Dic}_n = \{\,\eta^{(1;ij)} \mid i \in \{1, 3, \ldots, 2n-1\},\ j \in \{0, 2, \ldots, 2n-2\}\,\} \cup \{\,\eta^{(2;ij)} \mid i, j \in \{0, n\}\,\}$$

and $|\,\mathrm{Des}\,\mathrm{Dic}_n| = n^2 + 4$.

Proof First, observe that $\eta^{(1;ij)}$ and $\eta^{(2;ij)}$ leave all subgroups of $\langle x\rangle$ invariant. To check if they are descending endomorphisms, therefore, we only need to verify that they map any element of the form $x^t y$ to an element in its normal closure.

Suppose that n is odd. Then by Lemma 20.6, the normal closure of any element of the form $x^t y$ is Dic_n itself, and hence every well-defined endomorphism of the form $\eta^{(1;ij)}$ or $\eta^{(2;ij)}$ is a descending endomorphism of Dic_n. From Lemma 20.7, we know that $\eta^{(1;ij)}$ defines an endomorphism of Dic_n if and only if i is odd, and $\eta^{(2;ij)}$ defines an endomorphism if and only if $i \equiv 0 \mod n$ and $2j \equiv in \mod 2n$. Since n is odd, these two congruences hold simultaneously exactly when $i = 0$ and $j \in \{0, n\}$. This proves (i).

Now, suppose that n is even. Recall that the normal closure of $x^t y$ is $\langle x^2, x^t y\rangle$. Consider $\delta = \eta^{(1;ij)}$. From Lemma 20.7, i is necessarily odd. We know that $\overline{\mathrm{Dic}_n} = \mathrm{Dic}_n/\langle x^n\rangle$ is the dihedral group of order n generated by the rotation $\overline{x}$ and the reflection $\overline{y}$ (the cosets of $\langle x^n\rangle$ with respect to x and y, respectively). If $\delta = \eta^{(1;ij)} \in \mathrm{Des}\,\mathrm{Dic}_n$, then it descends to $\overline{\delta} \in \mathrm{Des}\,\overline{\mathrm{Dic}_n}$, which maps $\overline{x}$ to $\overline{x}^i$ and $\overline{y}$ to $\overline{x}^j\overline{y}$. Then Theorem 20.6 implies that j is even. To see that this condition is sufficient for δ to be a descending endomorphism of Dic_n, observe that $\delta(x^t y) = x^{it+j} y$, and since i is odd and j is even, $it + j \equiv t \mod 2$, which implies that $x^{it+j} y \in \langle x^2, x^t y\rangle$, the normal closure of $x^t y$. Thus, δ maps every element of Dic_n to an element in its normal closure.

Next, consider $\delta = \eta^{(2;ij)}$. From Lemma 20.7, δ defines an endomorphism of Dic_n if and only if $i \in \{0, n\}$ and $2j \equiv in \mod 2n$. Since n is even, so is i (from the first condition), and hence the second condition reduces to $j \equiv 0 \mod n$. Thus, $j \in \{0, n\}$. To see that $\delta = \eta^{(2;ij)}$ is a descending endomorphism of Dic_n for all $i, j \in \{0, n\}$, we verify that $\delta(x^t y)$ is in the normal closure of $x^t y$. Indeed, $\delta(x^t y) = x^{it+j}$, and since both i and j are even, $x^{it+j} \in \langle x^2, x^t y\rangle$, the normal closure of $x^t y$. This proves (ii). □

Remark 20.3 We note that when $n = 2$, Dic_n becomes $\mathrm{Dic}_2 \cong Q_8$, the quaternion group. In the case corresponding to even n in the above proof, the quotient $\overline{\mathrm{Dic}_n}$ will

be isomorphic to the Klein 4-group. However, the Klein 4-group is defined by the presentation of the dihedral group D_n for $n = 2$, and is sometimes considered to be D_2. As such, Theorem 20.6 applies to this case and hence the above proof is still valid for $n = 2$.

References

1. Baer R (1933) Situation der Untergruppen und Struktur der Gruppe. Sitz Ber Heidelberg Akad 2:12–17
2. Cooper CD (1968) Power automorphisms of a group. Math Z 107(5):335–356
3. Levi FW (1946) The ring of endomorphisms for which every subgroup of an abelian group is invariant. J Indian Math Soc 10:29–31
4. Madhusudanan V, Seth A, Sudhakara G (2023) Descending endomorphisms of groups. Palest J Math 12(1): 318–325
5. Miller MD (1975) On the lattice of normal subgroups of a direct product. Pac J Math 60(2):153–158
6. Newell ML (1976) Normal and power endomorphisms of a group. Math Z 151(2):139–142
7. Roman S (2011) Fundamentals of group theory: an advanced approach. SpringerLink, Bücher, Birkhäuser Boston
8. Scott WR (1987) Group theory. Dover books on mathematics. Dover Publications

Chapter 21
On Circulant Partial Hadamard Matrices



Pankaj Kumar Manjhi and Mahendra Kumar Rana

Mathematics Subject Classification (2010) 05B15 · 05B20 · 05B30

21.1 Introduction

Ryser's conjecture [12] is one of the interesting open problems since more than 50 years. It states that there is no circulant Hadamard matrix of the order $n > 4$. It plays a crucial role in the study of r-row regular circulant partial Hadamard matrix $r - H(k \times n)$ of combinatorial design theory. In this continuation, Craigen et al. [2] have given many results. With the help of particle swarm optimization, Lin et al. [9] linked circulant partial Hadamard matrices with general difference sets and generated new $r - H(k \times n)$ for $r = 0$ and $r = 2$. It provides one more find to analyse circulant partial Hadamard matrices. These matrices have direct application in the construction of functional magnetic resonance imaging (fMRI) designs. The fMRI is used for monitoring brain activity in response to mental stimuli given to the test subject [7]. Manjhi and Rana [10] worked on examples of circulant partial Hadamard matrices using sign patterns and elaborated on the term inequivalent on such matrices.

In this research article, we worked on circulant partial Hadamard matrices. All the results in this paper are proved analytically with the hope of advancement in the theoretical part of the circulant partial Hadamard matrices.

P. K. Manjhi (✉) · M. K. Rana
University Department of Mathematics, Vinoba Bhave University, Hazaribag, India
e-mail: 19pankaj81@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_21

21.2 Preliminaries

21.2.1 Hadamard Matrix

Hadamard matrix is a ± 1 square matrices H of order n such that $HH^T = nI_n$. The first example of the Hadamard matrix was forwarded by Sylvester in 1867, and he observed that *H is a Hadamard matrix then so is* $\begin{pmatrix} H & H \\ H & -H \end{pmatrix}$.

In 1893, J. Hadamard showed that the maximum determinant of a square matrix $H = (h_{ij})$ of order n with $\|h_{ij}\| \leq 1$ is $n^{\frac{n}{2}}$. This maximum determinant value was obtained under the following conditions:

(i) $HH^T = nI_n$, where $n = 1, 2$ or some multiple of 4.
(ii) The entries in H are ± 1.

J. Hadamard also gave examples of Hadamard matrices of order 12 and 20. These results of Hadamard revealed a famous and challenging conjecture on Hadamard matrices that is named Hadamard matrix conjecture [6, 13].

Conjecture 21.1 (Hadamard matrix conjecture) *If r is a positive integer, then there exists a Hadamard matrix of order $4r$.*

The Hadamard conjecture remains one of the most famous unsolved problems in mathematics, despite the fact that there are numerous methods for constructing Hadamard matrices. Currently, unknown orders of the Hadamard matrix below 1000 are 668, 716, 892 and 956, the last new Hadamard matrix of order 448 was constructed by Kharaghani et al. [8] in 2005.

21.2.2 Partial Hadamard Matrices

A partial Hadamard matrix $H(r \times n)$ is a row orthogonal matrix, satisfying $HH^T = nI_r$ where I_r is the identity matrix of order r. Following are some examples of partial Hadamard matrices: $\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & -1 & -1 & 1 \\ 1 & 1 & -1 & -1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}$.

Partial Hadamard matrices are preserved by the permutations of rows and columns. They are also preserved by the multiplication of rows and columns with -1. Two partial Hadamard matrices are called equivalent if and only if one is obtained from the other by a sequence of operations of permutation of rows or columns and multiplication of rows or columns by -1. Each partial Hadamard matrix is equivalent to a matrix in which all the entries in its first row as well as first column are 1's; such form is called the normal form of partial Hadamard matrix [2].

By eliminating rows of Hadamard matrices, we can obtain a number of partial Hadamard matrices but the reverse process is not an easy task. Completion

of some special form of partial Hadamard matrices into a Hadamard matrix is quite interesting. For example, we can easily extend a partial Hadamard matrix $H((n-1)\times n)=(x_{ij})$ to a Hadamard matrix $H(n\times n)$ by inserting a new last row $(x_{n1}, x_{n2}, x_{n3}, \ldots, x_{nn})$ such that $x_{nk}=\Pi_{i=1}^{n-1}x_{ik}$ for all $k=1,2,3,\ldots,n$. [Since $(\Pi_{i=1}^{n-1})x_{ik}x_{nk}=1$ for all $k=1,2,\ldots,n$.].

In 1977, Hall Jr. [5] had shown that a partial Hadamard matrix $H((n-4)\times n)$ can be extended to a Hadamard matrix of order n. In 1978, Verheiden [15] gave some results in this regard. He had shown that every $H((n-7)\times n)$ partial Hadamard matrix can be extended to a Hadamard matrix of order n. In 2009, Bulutoglu and Kaziska [1] had shown that the extension of partial Hadamard matrix $H((n-8)\times n)$ to a Hadamard matrix of order n is not possible in general.

21.2.3 Row Regular Matrices

Matrices with constant row sum are called row regular matrices. Following are some examples of row regular matrices: $\begin{pmatrix}1&2&3\\3&1&2\\2&3&1\end{pmatrix}$, $\begin{pmatrix}1&-1&1&1\\1&1&1&-1\end{pmatrix}$ and $\begin{pmatrix}a&b\\b&a\end{pmatrix}$. A matrix with constant row sum r is called a r-row regular matrix.

21.2.4 Circulant Partial Hadamard Matrices

If we add one more restriction to the partial Hadamard matrix, such as it must be a row regular circulant matrix, then such partial Hadamard matrices are known as Circulant Partial Hadamard matrices. A r-row regular circulant partial Hadamard matrix with k rows and n columns is denoted by $r-H(k\times n)$. If $r-H(k\times n)$ is a r-row regular circulant partial Hadamard matrix with k rows and n columns, then following are some elementary relations among r, k, and n:

(i) $k>1\Rightarrow n$ is even.
(ii) $r\leq\frac{n}{2}$.
(iii) $k>2\Rightarrow n$ is divisible by 4.

Ryser [12] conjectured that there is no circulant partial Hadamard matrix of order greater than 4. It is one of the famous conjectures on circulant partial Hadamard matrices. In 2021, Gallardo [4] has shown that there is no circulant partial Hadamard matrix of order $n(n>4)$ if $\sqrt{\frac{n}{4}}$ is not a square-free integer. This gives partial proof of Ryser's conjecture. However, the Ryser conjecture is still an open challenge for Mathematicians. In 2022, Manjhi and Rana [11] provided some results associated with $r-H(k\times 2k)$ and gave a construction method of such matrices.

In 2013, Craigen et al. [2] forwarded some new results on circulant partial Hadamard matrices. These results strengthen the relationship among r, n, and k

in a circulant partial Hadamard matrix $r - H(k \times n)$. They introduce a relation $r\sqrt{k} \leq n$ and with the help of this inequality, they have shown the following results on $r - H(k \times n)$:

(i) $r \leq k$.
(ii) $\frac{n}{2} \leq r \leq n \Rightarrow k = 1$.
(iii) $\frac{n}{\sqrt{5}} \leq r \leq \frac{n}{2} \Rightarrow k = 4$.

Craigen et al. [2] also gave a table relating the parameters r, k and n in a circulant partial Hadamard matrix $r - H(k \times n)$. In this table, they have obtained different values of k up to $r = 32$ and $n = 64$. Some interesting conjectures given by Craigen et al. [2] are given below.

Conjecture 21.2 *Let c be a positive rational number. Among the even values of r such that $rc \equiv 0(mod 4)$*

(i) *The largest value of k is such that $r - H(k \times rc)$ exist which is a strictly increasing function of r.*
(ii) *For the largest value of r for which $r - H(k \times rc)$ exist, we must have $c^2 - 1 \leq k \leq c^2$.*

Conjecture 21.3 *A $2 - H(k \times 2k)$ exist if and only if $k - 1$ is an odd prime power.*

As a consequence, we take the opportunity to learn more about partial Hadamard matrices and circulant partial Hadamard matrices. In the next section (Theorem 10), we have given the proof of one part of conjecture 2.

21.3 Main Work

The table given by Craigen et al. [2] in 2013 relating the parameters r, k and n in a $r - H(k \times n)$ by computer search and some results reflected by the table have already been done. In order to give a better insight into the table given in [2], we now prove the following results on $r - H(k \times n)$.

21.3.1 New Bounds of r for Circulant Partial Hadamard Matrix $r - H(k \times n)$ in Terms of n Which Implies $k = 4$

Craigen et al. (Lemma 5 of [2]) gave bounds of r in $r - H(k \times n)$ that implies $k = 4$. Here in terms of the following theorem, we have given new bounds of r in $r - H(k \times n)$ that implies $k = 4$.

Theorem 21.1 *Let $r - H(k \times n)$ be a circulant partial Hadamard matrix with $k > 1$ and r divide n, then*

(i) $\frac{n}{3} < r \leq \frac{n}{2} \Rightarrow k = 4$.
(ii) *If* $r = \frac{n}{2} - c, c \geq 0$ *and* $n > 6c \Rightarrow k = 4$.

Proof (i) $k \neq 1 \Rightarrow r \leq \frac{n}{2}$. Since n is a multiple of r, $\frac{n}{r}$ is an integer. Now $\frac{n}{3} < r \leq \frac{n}{2} \Rightarrow 2 \leq \frac{n}{r} < 3 \Rightarrow \frac{n}{r} = 2 \Rightarrow k = 4$.
(ii) We have $r = \frac{n}{2} - c$ and $c < \frac{n}{6}$. Since $\frac{n}{2} - \frac{n}{6} = \frac{n}{3}$, $\frac{n}{2} - c = r$ and $c < \frac{n}{6} \Leftrightarrow \frac{n}{3} < r \leq \frac{n}{2} \Rightarrow k = 4$.
Clearly (i) and (ii) are equivalent statements. □

21.3.2 Some Results on Circulant Partial Hadamard Matrix $r - H(k \times n)$ Obtained with the Help of Column Sum Restrictions and $k = 3$

The main purpose of this section is to picturize the column sum pattern in a circulant partial Hadamard matrix $r - H(k \times n)$ for $k = 3$. The pattern is shown in terms of the following theorem:

Theorem 21.2 *Let $r - H(k \times n)$ be a circulant partial Hadamard matrix, and let C_i be the ith column sum of H for all $i = 1, 2, \ldots, n$. Then the following results hold:*

(i) *If $C_i > 0$ for all $i = 1, 2, \ldots, n$, then $r > 0$.*
(ii) *If $C_i \in \{1, -1\}$, for all $i = 1, 2, \ldots, n$, then $k = 1$.*
(iii) *If $k = 3$, then at least one C_i must be equal to ± 3.*
(iv) *If $k = 3$, then at least one C_i must be equal to ± 1.*
(v) *If $k = 3$, then the difference between number of C_i's with $+1$ and -1 is a multiple of 3.*
(vi) *If $k = 3$, then $n \geq 8$.*

Proof (i) For given circulant Partial Hadamard matrix $r - H(k \times n)$, we have

$$C_1 + C_2 + C_3 + \cdots + C_n = rk. \tag{21.1}$$

If possible, let $r = 0$, then $C_1 + C_2 + C_3 + \cdots + C_n = 0$ (from (21.1) and $k \geq 1$) which contradicts that $C_i > 0$ for all $i = 1, 2, \ldots, n$. So, $r > 0$.
(ii) $C_i \in \{1, -1\}$ for all $i = 1, 2, \ldots, n \Rightarrow \Sigma_{i=1}^{n} C_i^2 = n \Rightarrow nk = n \Rightarrow k = 1$.

Note 21.1 From the above results, we can conclude that $k > 1$ implies that all column sums cannot be ± 1. □

(iii) If $k = 3$, then possible columns are given by

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \text{ or } \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}.$$

It is to notice that in all the above cases $C_i = \pm 3$ or $\pm 1 \Rightarrow C_i = \pm 3$ for at least one $i \in \{1, 2, \ldots, n\}$, otherwise k will become 1.

(iv) We prove this result by contradiction. For $C_i \in \{3, -3\}$, we must have $\sum_{i=1}^{n} C_i^2 = 9n \Rightarrow 3 = 9$ which is a contradiction.

(v) Let x, y, z and w be the number of C_i's with $1, -1, 3$ and -3, respectively. Then $x - y + 3(z - w) = 3r$. This implies that $(x - y)$ is a multiple of 3.

(vi) Suppose if possible $n = 4$, then $C_1^2 + C_2^2 + C_3^2 + C_4^2 = 12 \Rightarrow C_1 + C_2 + C_3 + C_4 \in \{0, \pm 2, \pm 4, \pm 6\}$. Again since $C_1 + C_2 + C_3 + C_4 = 3r$, therefore, $(C_1 + C_2 + C_3 + C_4)$ is divisible by 3. This implies that $(C_1 + C_2 + C_3 + C_4) \in \{0, \pm 6\}$. Now, $C_1 + C_2 + C_3 + C_4 = 6 \Rightarrow rk = 6 \Rightarrow r3 = 6 \Rightarrow r = 2 \Rightarrow r = \frac{n}{2}$ (since $n = 4$).
Thus, $k = 4$ which is a contradiction, therefore $(C_1 + C_2 + C_3 + C_4) \neq \pm 6$. Next, if $C_1 + C_2 + C_3 + C_4 = 0 \Rightarrow r = 0$, then *one of the column sums is* 3 *and others are* $-1, -1, -1$ or *one column sum is* -3 *and the others are* 1, 1, 1. Column sum 3 implies that one of the columns is $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = V_i$ (say). Since matrix is a circulant matrix, therefore $V_{i+1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ or $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$; this implies that $C_{i+1} = 3$ or 1. This arrived in contradiction. Hence, $n \neq 4 \Rightarrow n > 4$. Since we have a $0 - H(3 \times 8)$ matrix given by

$$\begin{pmatrix} -1 & 1 & 1 & 1 & -1 & -1 & -1 & 1 \\ 1 & -1 & 1 & 1 & 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & 1 & 1 & 1 & -1 & -1 \end{pmatrix},$$

therefore, for $k = 3$, $n \geq 8$. □

21.3.3 Results on Circulant Partial Hadamard Matrices Which Reflect Types of Column Sums

In this section, we have given a picture of different column sums in $r - H(k \times n)$ under some relations among r, n and k in terms of the following theorems:

Theorem 21.3 *If $n = rk$ and $k > 1$, then all column sums of circulant partial Hadamard matrix $r - H(k \times n)$ cannot be equal.*

Proof Suppose if possible $C_i = C$ for all $i = 1, 2, \ldots, n$ in $r - H(k \times n)$. Then we have $nC = rk = n \Rightarrow C = 1$ and $\Sigma_{i=1}^{n} C_i^2 = nC^2 = nk$. This implies that k equals 1, which is a contradiction. □

Theorem 21.4 *Let $n = rk$ in a circulant partial Hadamard matrix $r - H(k \times n)$. If x and y are the column sums in H each of multiplicity $\frac{n}{2}$, then $k \leq 2$.*

Proof We must have $\frac{n}{2}(x+y) = n = rk$ and $\frac{n}{2}(x^2+y^2) = nk$. This implies that $x+y = 2$ and $\frac{n}{2}((x+y)^2 - 2xy) = nk$. It follows that $2-k = xy$. Since $xy \geq 0$, therefore, $k \leq 2$. □

Theorem 21.5 *Let $r - H(k \times n)$ be a circulant partial Hadamard matrix with constant column sum C and $r\sqrt{k} = n$, then $C = \sqrt{k}$.*

Proof Constant column sum in H implies that $nC^2 = nk$. It follows that $C = \sqrt{k}$. □

21.3.4 Some Results on Augmentation of Circulant Partial Hadamard Matrices

In this section, we have given techniques to increase the number of columns in a circulant partial Hadamard matrix $r - H(k \times n)$ with the help of proper augmentations. The results have been shown in terms of the following theorems.

Theorem 21.6 *Existence of an r-row regular circulant partial Hadamard matrix of order $(k \times n)$ implies the existence of a circulant partial Hadamard matrix $rm - H(k \times mn)$ for all positive integer m.*

Proof $rm - H(k \times mn)$ is given by m copies of r-row regular circulant partial Hadamard matrix of order $A(k \times n)$.
That is, $rm - H(k \times mn) = (A\ A \cdots A) = m$ copies of A. □

Corollary 21.1 *Existence of a circulant Hadamard matrix of order n implies existence of a circulant partial Hadamard matrix $m\sqrt{n} - H(n \times mn)$.*

Theorem 21.7 *Let $circ_{(k\times n)}(x_1, x_2, x_3, \ldots, x_n)$ and $circ_{(k\times n)}(y_1, y_2, y_3, \ldots, y_m)$ be circulant partial Hadamard matrices with row sum r_1 and r_2, respectively.*
Then $(circ_{(k\times n)}(x_1, x_2, x_3, \ldots, x_n)|circ_{(k\times n)}(y_1, y_2, y_3, \ldots, y_m))$ is a circulant partial Hadamard matrix if $x_{(n-i)} = y_{(m-i)}$ for all $i = 0, 1, 2, \ldots, (k-1)$ and $m, n \geq k$.

Proof of this theorem is obvious, and a generalized version of this theorem can be stated as follows.

Theorem 21.8 *Let $A_i(k \times n_i) = circ(x_1^i, x_2^i, x_3^i, \ldots, x_{n_i}^i)$ be an r_i-row regular circulant partial Hadamard matrix with $k \leq n_i$ for all $i = 1, 2, 3, \ldots, t$. Then the augmented matrix $\left(A_1\ A_2\ A_3 \cdots A_t\right)$ is a circulant partial Hadamard matrix with row sum $\sum_{i=1}^{t} r_i$ if $x_{n_i-u}^v = x_{n_i-u}^{v+1}$ for all $u = 0, 1, 2, \ldots, (k-1)$ and for all $v = 1, 2, \ldots, (t-1)$.*

Proof Since $x_{n_i-u}^v = x_{n_i-u}^{v+1}$ for all $u = 0, 1, 2, \ldots, (k-1)$ and for all $v = 1, 2, \ldots, (t-1)$, and $k \leq n_i$ for all $i = 1, 2, 3, \ldots, t$. Therefore, the augmented matrix $\left(A_1\ A_2\ A_3 \cdots A_t\right)$ is a circulant partial Hadamard matrix with k rows. Hence, $\left(A_1\ A_2\ A_3 \cdots A_t\right)$ is a circulant partial Hadamard matrix with row sum $\sum_{i=1}^{t} r_i$. □

21.3.5 Non-existence of Circulant Skew Hadamard Matrix of Order $n > 1$

Theorem 21.9 *There does not exist any circulant skew Hadamard matrix of order* $n > 1$.

Proof Let H be a circulant skew Hadamard matrix of order $n > 1$. Let $C_1, C_2, \ldots, C_n$ be column sums of H, then $\sum_{i=1}^{n} C_i = n\sqrt{n}$. This implies that the sum of all entries of $H + H^T$ is $2n\sqrt{n}$. Since H is a skew Hadamard matrix, $H + H^T = 2I_n$. It follows that $2n\sqrt{n} = 2n$. This implies that $n = 1$. We arrived at a contradiction. □

21.3.6 One Part of Conjecture Given by Craigen et al. [2] for the Existence of 2-$H(k \times 2k)$

Theorem 21.10 *A* $2 - H(k \times 2k)$ *exist if* $k - 1$ *is an odd prime power.*

Proof The result is implied by the following two theorems:

(i) For every odd prime power p^r, there exist a negacyclic $W(p^r + 1, p^r)$. (Theorem 1.6 of [3, 14])
(ii) There exist a $2 - H(k \times 2k)$ if and only if there exist a negacyclic $W(k, k - 1)$. (Theorem 8 of [2]) □

21.4 Thoughts and Conclusion

For finding complete circulant partial Hadamard matrices $H(n)$ for $n > 4$, the statement of Theorem 21.1 strengthens Ryser's conjecture for large n where k is always 4 in the described range. The results that have been obtained in this paper are mainly derived analytically from the basic properties of circulant partial Hadamard matrices. In spite of that, this type of analytical approach towards the development of theory may find some new paths for the invention of new techniques. Part (i) of Theorem 21.2 gives new evidence that if $C_i > 0$ for all i then $r \neq 0$ and therefore $H(n)$ never exist for $r = 0$; existence of least order of n for $k = 3$ is given in Part (vi) of Theorem 21.2, and remaining results are obtained on some restrictions of column sums C_i's in circulant partial Hadamard matrices. Theorems 21.3, 21.4 and 21.5 have a remarkable approach as they reflect the pictures of different column sums in $r - H(k \times n)$ under some relation between r, n and k. Theorems 21.6, 21.7 and 21.8 are based on the augmentation of partial Hadamard matrices. Theorem 21.9 gives non-existing evidence of circulant skew Hadamard matrices of order greater than 1. Theorem 21.10 gives proof of one part of the conjectures given by Craigen et al. [2] on the existence of $2 - H(k \times 2k)$. The column sum approach may give a

new style of thinking for the solutions to many problems in this field, which has been used in this paper. All the theorems stated in this research paper are strengthening the theory part of circulant partial Hadamard matrices. Overall, the finding of this research may be useful for scholars.

Acknowledgements The presented work is sponsored by Council of Scientific and Industrial Research, India (Project Number: 25(0305)/19/*EMR-II*, dated: 16.05.2019). The authors express their gratitude to the editor and reviewers for their valuable suggestions.

References

1. Bulutoglu DA, Kaziska DM (2009) A counter example to Beder's conjecture about Hadamard matrices. J Stat Plan Interf 139:3381–3383
2. Craigen R, Faucher G, Low R, Wares T (2013) Circulant partial Hadamard matrices. Linear Algebr Appl 439:3307–3317
3. Delsarte P, Geothals JM, Seidel JJ (1971) Orthogonal matrices with zero diagonal ii. Canad J Math 23:816–832
4. Gallardo LH (2021) Under a mild condition Ryser conjecture holds for every $n = 4h^2$ with $h>1$ odd and non-square free. Math Commun 26:1–8
5. Hall MJ (1977) Integral matrices for which $aa^t = mi$. Number theory and Algebra. Academic Press, New York, pp 119–134
6. Horadam KJ (2007) Hadamard matrices and their application. Princeton University Press, Princeton
7. Kao MH (2015) Universally optimal FMRI designs for comparing hemodynmic response function. Stat Sin 25:499–506
8. Kharaghani H, Tayfeh-Rezaie B (2005) A Hadamard matrix of order 428. J Comb Des 13:435–440
9. Lin YL, Phoa FKH, Kao MH (2017) Circulant partial Hadamard matrices: construction via general difference sets and its application to FMRI experiments. Stat Sin 27:1715–1724
10. Manjhi PK, Rana MK (2022) Some new examples of circulant partial Hadamard matrices of type 4-(k×n). Adv Appl Math Sci 21:129–136
11. Manjhi PK, Rana MK (2022) Some results on r-row regular circulant partial Hadamard matrices of order (k×2k). Commun Math Appl 13:129–136
12. Ryser HJ (1963) Combinatorial mathematics. The carus mathematical monographs. Wiley
13. Saberry JJ, Yamada MM (2020) Hadamard matrices: constructions using number theory and linear algebra. Wiley
14. Sabery JJ (2016) Remarks on negacyclic matrices. J Theor Comput Math 2:71–79
15. Verheiden E (1978) Integral and rational completion of combinatorial matrices. J Comb Theory Ser A 25:267–276

Chapter 22
On Weak Hypervector Spaces Over a Hyperfield



Pallavi Panjarike, Syam Prasad Kuncham, Madeleine Al-Tahan, Vadiraja Bhatta, and Harikrishnan Panackal

Mathematics Subject Classification (2010) 20N20 · 97H60

22.1 Introduction

In an algebraic structure, the composition of two elements is again an element in the underlying set. Hyperoperation is a generalization of binary operation in a classical algebraic system, wherein the composition of two elements is again a non-empty set. For a non-empty set $\mathbb{S}$, let $\mathcal{P}^*(\mathbb{S})$ be its power set excluding empty set. A hypergroupoid on $\mathbb{S}$ is a mapping $\circ : \mathbb{S} \times \mathbb{S} \to \mathcal{P}^*(\mathbb{S})$. A semihypergroup $(\mathbb{S}, \circ)$ is a hypergroupoid which satisfies

$$(x \circ y) \circ w = x \circ (y \circ w) \text{ i.e. } \bigcup_{v \in x \circ y} v \circ w = \bigcup_{u \in y \circ w} x \circ u$$

Dedicated to Prof. Calyampudi Radhakrishna Rao and Prof. Arbind Kumar Lal

P. Panjarike · S. P. Kuncham · V. Bhatta · H. Panackal (✉)
Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka, India
e-mail: pk.harikrishnan@manipal.edu

P. Panjarike
e-mail: pallavipanjrk@gmail.com

S. P. Kuncham
e-mail: syamprasad.k@manipal.edu

V. Bhatta
e-mail: vadiraja.bhatta@manipal.edu

M. Al-Tahan
Department of Mathematics, Abu Dhabi University, Abu Dhabi, UAE
e-mail: altahan.madeleine@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_22

for all x, y, w in $\mathbb{S}$. Using the notion of hyperoperations and H_v-operations, the classical vector spaces are generalized in several ways. Scafati-Tallini [12] introduced the idea of hypervector spaces over fields, in which the external operation was considered as a hyperoperation. Vougiouklis [15] introduced the notion of H_v-vector spaces and discussed the properties such as linear independence and spanning set. Ameri and Dehghan [2] studied further the concept of hypervector spaces over fields. They extended some of the fundamental results of vector spaces to hypervector spaces over fields, and explored the concept of dimension. Taghavi and Hosseinzadeh [13] studied weak hypervector spaces over a field using the idea of essential points. Roy and Samanta [11] explored hypervector spaces over a hyperfield by considering the action of a canonical hypergroup over a hyperfield with the external operation as a hyperoperation. Consequently, they have studied the quotient structure of these hypervector spaces. Furthermore, when the external operation is considered as a classical operation, Al-Tahan and Davvaz [1] investigated the properties of linear transformations and established the Rank Nullity theorem in hypervector spaces. The remaining part of the paper is organized as follows. In Sect. 22.2, we provide necessary definitions and examples given by Al-Tahan and Davvaz [1], Viro [14], and Davvaz and Leoreanu-Fotea [3] that are used throughout the paper. In Sect. 22.3, we illustrate some remarkable examples of a weak hypervector space over a hyperfield. In Sect. 22.4, we discuss the properties involving the notions such as direct sums and complements of subhyperspaces. In Sect. 22.5, we start with the notion of linear transformation between two hypervector spaces and prove isomorphism theorems. Unlike in classical vector spaces, an n-dimensional hypervector space need not be isomorphic to F^n (as shown in Example 22), where F is a hyperfield. In Sect. 22.6, we obtain the annihilator properties of linear functionals in hypervector spaces. Throughout this paper, F denotes a Krasner hyperfield as studied by Al-Tahan and Davvaz [1], and $\mathbb{V}$ denotes a hypervector space unless otherwise specified. For the basic concepts and related results in classical vector spaces, we refer to Hoffman and Kunze [6].

22.2 Preliminaries

In this section, we present necessary concepts that are used throughout the paper.

Definition 22.1 [3] A semihypergroup $(H, \circ)$ is a hypergroup if for all $x \in H$; we have

$$x \circ H = H \circ x = H.$$

Definition 22.2 [3] A hypergroup $(H, +)$ is called a quasicanonical hypergroup if

(i) there is $0 \in H$ such that $u + 0 = 0 + u = u$, for all $u \in H$,
(ii) for every $u \in H$, there exists one and only one $u' \in H$ such that $0 \in (u + u') \cap (u' + u)$, u' is denoted by $-u$.
(iii) $w \in u + v$ implies $v \in -u + w$ and $u \in w - v$, for all $u, v, w \in H$.

A canonical hypergroup is a hypergroup $(H, +)$ which is quasicanonical and commutative.

Definition 22.3 [3] A weak hyperring is an algebraic hyperstructure $(R, +, \cdot)$ which satisfies the following axioms:

(i) $(R, +)$ is a canonical hypergroup.
(ii) $(R, \cdot)$ is a semigroup with zero as a bilaterally absorbing element, i.e., $u \cdot 0 = 0 \cdot u = 0$,
(iii) $u(v + w) \subseteq uv + uw$ and $(u + v)w \subseteq uw + vw$,

for all $u, v, w \in R$.

A weak hyperring is called a Krasner hyperring if Condition (iii) of Definition 22.3 holds with equality. A Krasner hyperring $(R, +, \cdot)$ is called commutative (with unit element) if $(R, \cdot)$ is a commutative semigroup (with unit element).

Definition 22.4 [3] A Krasner hyperring is called a hyperfield, if $(R \setminus \{0\}, \cdot)$ is a group.

Example 1 (Viro [14])

Let V be the set of all non-negative real numbers. We define the following hyperoperation on V:

$$r_1 \triangle r_2 = \{r \in V : |r_1 - r_2| \leq r \leq r_1 + r_2\},$$
$$r_1 \triangledown r_2 = \begin{cases} \max\{r_1, r_2\}, & \text{if } r_1 \neq r_2; \\ [0, r_1] & \text{otherwise} \end{cases}$$

and

$$r_1 \odot r_2 = r_1 r_2$$

for all $r_1, r_2 \in V$. Then $(V, \triangle, \odot)$ and $(V, \triangledown, \odot)$ are hyperfields (these are called triangle and ultra triangle hyperfields, respectively).

Example 2 (Jun [8])

Let $F_2 = \{0, 1\}$. Define the hyperoperation $+$ and the operation $\cdot$ on F_2 by following tables:

$+$	0	1
0	0	1
1	1	F_2

$\cdot$	0	1
0	0	0
1	0	1

Then $(F_2, \oplus, \times)$ is a hyperfield.

Example 3 (Viro [14])

Let $\mathbb{C}$ be the set of all complex numbers with the following hyperoperation:

$$x \diamond y = \begin{cases} \{x\}, & \text{if } |x| > |y|; \\ \{y\}, & \text{if } |y| > |x|; \\ \{c \in \mathbb{C} : c \in S_{xy}\}, & \text{if } x + y \neq 0 \text{ and } |x| = |y|; \\ \{c \in \mathbb{C} : |c| \leq |x|\}, & \text{if } x + y = 0 \end{cases}$$

and

$$x \odot y = xy.$$

Here, S_{xy} is the shortest arc connecting x to y on the circle with $|x|$ as radius. Then $(\mathbb{C}, \diamond, \odot)$ is a hyperfield (called tropical hyperfield, it is denoted by $\mathcal{T}_{\mathbb{C}}$).

Definition 22.5 [1] Let F be a hyperfield. A canonical hypergroup $(\mathbb{V}, +)$ together with a map $\cdot : F \times \mathbb{V} \to \mathbb{V}$ is called a hypervector space over F if the following conditions hold:

(i) $\alpha \cdot (u_1 + u_2) = \alpha \cdot u_1 + \alpha \cdot u_2$;
(ii) $(\alpha + \beta) \cdot u_1 = \alpha \cdot u_1 + \beta \cdot u_1$;
(iii) $\alpha \cdot (\beta \cdot u_1) = (\alpha\beta) \cdot u_1$;
(iv) $\alpha \cdot (-u_1) = (-\alpha) \cdot u_1 = -(\alpha \cdot u_1)$;
(v) $1 \cdot u_1 = u_1$,

for all $\alpha, \beta \in F, u_1, u_2 \in \mathbb{V}$ and $1 \in F$.

Example 4 (Al-Tahan and Davvaz [1])

Let F be a hyperfield. Then F is trivially a hypervector space over itself.

Example 5 (Viro [14])

Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{C}$. We define $\smile : V \times V \to V$ by

$$u \smile v = \begin{cases} \{v\}, & \text{if } \|v\| > \|u\|; \\ \{u\}, & \text{if } \|u\| > \|v\|; \\ \left\{ \dfrac{\|v\|}{\|av + bu\|}(av + bu) : a, b \in \mathbb{R}^+ \right\} \cup \{v, u\}, & \text{if } v + u \neq 0 \text{ and } \|v\| = \|u\|; \\ \{w \in V : \|u\| \leq \|v\|\}, & \text{if } v + u = 0. \end{cases}$$

Then $\mathbb{V} = (V, \smile)$ is a canonical hypergroup. Furthermore, $\mathbb{V}$ is a hypervector space over $\mathcal{T}_{\mathbb{C}}$ with respect to the external operation: $(a, v) \mapsto av$ for all $a \in \mathcal{T}_{\mathbb{C}}$ and $v \in \mathbb{V}$.

Definition 22.6 [1] Let F be a hyperfield. A canonical hypergroup $(\mathbb{V}, +)$ together with a map $\cdot : F \times \mathbb{V} \to \mathbb{V}$ is called a weak hypervector space over F if the following conditions hold:

(i) $\alpha \cdot (u_1 + u_2) \subseteq \alpha \cdot u_1 + \alpha \cdot u_2$;
(ii) $(\alpha + \beta) \cdot u_1 \subseteq \alpha \cdot u_1 + \beta \cdot u_1$;
(iii) $\alpha \cdot (\beta \cdot u_1) = (\alpha\beta) \cdot u_1$;
(iv) $\alpha \cdot (-u_1) = (-\alpha) \cdot u_1 = -(\alpha \cdot u_1)$;
(v) $1 \cdot u_1 = u_1$,

for all $\alpha, \beta \in F, u_1, u_2 \in \mathbb{V}$ and $1 \in F$.

Definition 22.7 [1] Let $\mathbb{V}$ be a (weak) hypervector space over F. Then a non-empty set $U \subseteq \mathbb{V}$ is called a subhyperspace of $\mathbb{V}$ if

(i) $u_1 - u_2 \subseteq U$;
(ii) $\alpha \cdot u_1 \in U$,

for all $u_1, u_2 \in U$ and $\alpha \in F$.

Proposition 22.1 ([1]) *Let $\mathbb{V}$ be a (weak) hypervector space over F. Then $\emptyset \neq W \subseteq \mathbb{V}$ is a subhyperspace of $\mathbb{V}$ if and only if $\alpha \cdot u_1 + \beta \cdot u_2 \subseteq W$ for all $u_1, u_2 \in W$ and $\alpha, \beta \in F$.*

Definition 22.8 [1] Let $\mathbb{V}$ be a (weak) hypervector space. Then $u_1, u_2, \cdots, u_p \in \mathbb{V}$ are said to be linearly independent if $0 \in \sum_{i=1}^{p} \alpha_i u_i$ implies $\alpha_i = 0$ for all $1 \leq i \leq p$.

Definition 22.9 [1] Let $\mathbb{V}$ be a (weak) hypervector space and $S = \{u_1, u_2, \cdots, u_p\} \subseteq \mathbb{V}$. Then S is said to span $\mathbb{V}$ if for each $v \in \mathbb{V}$, there exist $\alpha_1, \alpha_2 \cdots, \alpha_p$ in F such that $v \in \sum_{i=1}^{p} \alpha_i u_i$.

Definition 22.10 [1] Let W be a subhyperspace of $\mathbb{V}$. Define $\frac{\mathbb{V}}{W} = \{v + W : v \in \mathbb{V}\}$ with the hyperoperation $v + W \oplus u + W = (v + u) + W$ and $a \star (u + W) = au + W$, for all $\alpha \in F$ and $u, v \in \mathbb{V}$.

Remark 22.1 [1] For a subhyperspace W of $\mathbb{V}$, $\left(\frac{\mathbb{V}}{W}, \oplus, \star\right)$ is a hypervector space over F.

Proposition 22.2 ([1]) *Let U_1, U_2 be subhyperspaces of $\mathbb{V}$. Then $U_1 \cap U_2$ and $U_1 + U_2 = \{u + v : u \in U_1, v \in U_2\}$ are subhyperspaces of $\mathbb{V}$.*

22.3 Some Examples of Weak Hypervector Spaces

Throughout this paper, let $\mathbb{V}, \mathbb{W}$ denote hypervector spaces unless otherwise specified. In this section, we provide some examples, and we study a few properties of weak hypervector spaces that are different from classical vector spaces.

Example 6

Let $\mathcal{M} = \left\{ \begin{bmatrix} k_1 & k_2 \\ k_3 & k_4 \end{bmatrix} : k_1, k_2, k_3, k_4 \in F_2 \right\}$ be the collection of all 2×2 matrices over F_2. The hyperoperation addition $+_{\mathcal{M}}$ on $\mathcal{M}$ is defined as follows. For any $A = \begin{bmatrix} k_1 & k_2 \\ k_3 & k_4 \end{bmatrix}$, $B = \begin{bmatrix} k_1' & k_2' \\ k_3' & k_4' \end{bmatrix} \in \mathcal{M}$,

$$A +_{\mathcal{M}} B = \left\{ \begin{bmatrix} t_1 & t_2 \\ t_3 & t_4 \end{bmatrix} : t_i \in k_i \oplus k_i', i = 1, 2, 3, 4 \right\}.$$

For instance,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} +_{\mathcal{M}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \left\{ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\},$$
$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} +_{\mathcal{M}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \left\{ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\}.$$

We claim that $(\mathcal{M}, +_{\mathcal{M}})$ is a canonical hypergroup. Clearly $+_{\mathcal{M}}$ is well defined. Let $A, B, C \in \mathcal{M}$. Now

$$\begin{aligned}
(A +_{\mathcal{M}} B) +_{\mathcal{M}} C &= \left\{ \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} : t_{ij} \in a_{ij} \oplus b_{ij}, i, j = 1, 2 \right\} +_{\mathcal{M}} \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \\
&= \left\{ \begin{bmatrix} t_{11}' & t_{12}' \\ t_{21}' & t_{22}' \end{bmatrix} : t_{ij}' \in t_{ij} \oplus c_{ij}, t_{ij} \in a_{ij} \oplus b_{ij}, i, j = 1, 2 \right\} \\
&= \left\{ \begin{bmatrix} t_{11}' & t_{12}' \\ t_{21}' & t_{22}' \end{bmatrix} : t_{ij}' \in (a_{ij} \oplus b_{ij}) \oplus c_{ij} = a_{ij} \oplus (b_{ij} \oplus c_{ij}), i, j = 1, 2 \right\} \\
&= A +_{\mathcal{M}} (B +_{\mathcal{M}} C).
\end{aligned}$$

Clearly $A +_{\mathcal{M}} \mathcal{M} \subseteq \mathcal{M}$. Let $B = [b_{ij}] \in \mathcal{M}$ where $b_{ij} \in F_2$. As F_2 is a hypergroup, we get $b_{ij} = a_{ij} \oplus F_2$. So, there exists $t_{ij} \in F_2$ such that $b_{ij} \in a_{ij} \oplus t_{ij}$. Take $T = \begin{bmatrix} t_{11}' & t_{12}' \\ t_{21}' & t_{22}' \end{bmatrix}$. Then $B \in A +_{\mathcal{M}} T \subseteq A +_{\mathcal{M}} \mathcal{M} = \mathcal{M} +_{\mathcal{M}} A$. Therefore $(\mathcal{M}, +_{\mathcal{M}})$ is a hypergroup. Further we can verify that $(\mathcal{M}, +_{\mathcal{M}})$ is a canonical hypergroup. Define an external operation $\cdot : F_2 \times \mathcal{M} \to \mathcal{M}$ by

$$\alpha \cdot A = \alpha \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} \alpha a_{11} & \alpha a_{12} \\ \alpha a_{21} & \alpha a_{22} \end{bmatrix}.$$

Then $(\mathcal{M}, +_{\mathcal{M}})$ forms a weak hypervector space over F_2. In fact,

$$\begin{aligned} \alpha_1 \cdot (A +_{\mathcal{M}} B) &= \left\{ \alpha_1 \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} : t_{ij} \in a_{ij} \oplus b_{ij}, i, j = 1, 2 \right\} \\ &= \left\{ \begin{bmatrix} t'_{11} & t'_{12} \\ t'_{21} & t'_{22} \end{bmatrix} : t'_{ij} \in \alpha_1(a_{ij} \oplus b_{ij}) = \alpha_1 a_{ij} \oplus \alpha_1 b_{ij}, i, j = 1, 2 \right\} \\ &= \alpha_1 \cdot A +_{\mathcal{M}} \alpha_1 \cdot B, \end{aligned}$$

for all $\alpha_1 \in F_2$, $A, B \in \mathcal{M}$.

$$\begin{aligned} (\alpha_1 \oplus \alpha_2) \cdot A &= \left\{ t \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} : t \in \alpha_1 \oplus \alpha_2 \right\} \\ &\subseteq \left\{ \begin{bmatrix} t'_{11} & t'_{12} \\ t'_{21} & t'_{22} \end{bmatrix} : t'_{ij} \in \alpha_1 a_{ij} \oplus \alpha_2 a_{ij}, i, j = 1, 2 \right\} \\ &= \alpha_1 \cdot A +_{\mathcal{M}} \alpha_2 \cdot A \end{aligned}$$

for all $\alpha_1, \alpha_2 \in F_2$, $A \in \mathcal{M}$. In a similar way, we can verify the other axioms of weak hypervector spaces. So $(\mathcal{M}, +_{\mathcal{M}}, \cdot)$ is a weak hypervector space over F_2.
Evidently, $U = \left\{ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$ is a subhyperspace of $\mathcal{M}$.

More generally, let F be any hyperfield and let $\mathcal{M}_{m\times n}$ denote the set of all $m \times n$ matrices with entries from the hyperfield F. Define $+_{\mathcal{M}_{m\times n}}$ on $\mathcal{M}_{m\times n}$

$$A +_{\mathcal{M}_{m\times n}} B = \{(t_{ij}) : t_{ij} \in a_{ij} \oplus b_{ij}\}$$

for all $A, B \in \mathcal{M}_{m\times n}$. We define scalar multiplication $\cdot : F \times \mathcal{M}_{m\times n} \to \mathcal{M}_{m\times n}$ by $\alpha \cdot A = \alpha \cdot (a_{ij}) = (\alpha a_{ij})$ for all $\alpha \in F$ and $A \in \mathcal{M}$. Then $(\mathcal{M}_{m\times n}, +_{\mathcal{M}_{m\times n}}, \cdot)$ is a weak hypervector space over F.

Example 7

Let $X \neq \emptyset$ and F^X be the set of all functions from X to F. Then F^X is a weak hypervector space over F, where $\oplus : F^X \times F^X \to \mathcal{P}^*(F^X)$ and $\star : F \times F^X \to F^X$ are defined as $(f \oplus g)(u) = f(u) + g(u)$ and $(a \star f)(u) = a \cdot f(u)$, for all $f, g \in F^X$, $a \in F$, $u \in X$. We give the following illustration to show that Condition ii of Definition 22.5 fails to hold. Take $E = \{a, b\}$. Then F_2^E is a weak hypervector space over F_2. Here $F_2^E = \{f_0, f_1, f_2, f_3\}$ where $f_0(a) = f_0(b) = 0$, $f_1(a) = 0$, $f_1(b) = 1$, $f_2(a) = 1$, $f_2(b) = 0$, and $f_3(a) = f_3(b) = 1$. Now we can see that $(1 + 1) f_3 = \{0, f_3\} \subseteq 1 \star f_3 \oplus 1 \star f_3 = F_2^E$.

Example 8

From Example 1, $\mathbb{V} = (V, \triangle)$ is a canonical hypergroup and $K = (V, \triangledown, \odot)$ is a hyperfield. With the external operation $(a, v) \mapsto av$, $\mathbb{V}$ is a weak hypervector space over K. Here Condition ii of Definition 22.5 fails to satisfy. For instance, $2, 3 \in V$, $5 \in \mathbb{V}$, $(2 \triangledown 3)5 = 15$, and $2 \cdot 5 \triangle 3 \cdot 5 = [10, 15]$.

Proposition 22.3 *Let $\mathbb{V}$ be a weak hypervector space over K, and $0 \neq a \in F$. Then $a \cdot 0 = 0$.*

Proof For any $t \in a \cdot 0 - a \cdot 0$, we have $a \cdot 0 \in t + a \cdot 0$. Then $0 \in a^{-1} \cdot t + 0 = \left\{a^{-1} \cdot t\right\}$. This implies $0 = a^{-1} \cdot t$ and so $a \cdot 0 = t$. That is $a \cdot 0 - a \cdot 0 = \{a \cdot 0\}$. Note that $a \cdot 0 = a \cdot (-0) = -a \cdot 0$, and so $0 \in a \cdot 0 - a \cdot 0 = \{a \cdot 0\}$ which gives $0 = a \cdot 0$. □

From the axioms of a classical vector space, we can conclude that $0 \cdot x = 0$. This property is preserved in hypervector spaces as well. However, in a weak hypervector space, this property is no longer true as we show in the following example.

Example 9

Consider F_2 as discussed in Example 2. We define an external operation $\star : F_2 \times F_2 \to F_2$ by $a \star x = x$ for all $a, x \in F_2$. Then we can see that F_2 is a weak hypervector space over F_2 (in fact, $(0 + 1) \star 1 = 1 \in 1 + 1 = (0 \star 1) + (1 \star 1)$, whereas $0 \cdot 1 = 1 \neq 0$).

This observation leads to the following generalization.

Proposition 22.4 *Let F be a hyperfield and $(\mathbb{V}, +)$ be a non-trivial canonical hypergroup such that $x = -x$ and $x \in x + x$ for all $x \in \mathbb{V}$. Then $(\mathbb{V}, +)$ is a weak hypervector space with respect to the external operation $\cdot : F \times \mathbb{V} \to \mathbb{V}$ defined by $\alpha \cdot x = x$ for all $\alpha \in F$ and $x \in \mathbb{V}$.*

Proof Let $\alpha, \beta \in F$ and $v, w \in \mathbb{V}$. Then $\alpha(v + w) = v + w = \alpha \cdot v + \alpha \cdot w$. Also, we have $(\alpha + \beta) \cdot v = v \in v + v = \alpha \cdot v + \beta \cdot v$. In a similar manner, we can prove the rest of the conditions of a weak hypervector space. Hence, $\mathbb{V}$ is a weak hypervector space over F. But by definition, for $0 \neq x \in \mathbb{V}$, $0 \cdot x = x \neq 0$. □

Definition 22.11 A weak hypervector space $\mathbb{V}$ over K is said to be torsion-free if $0 \cdot v = 0$ for all $v \in \mathbb{V}$.

Remark 22.2 (i) Every hypervector space is torsion-free.
(ii) The weak hypervector spaces discussed in Proposition 22.4 are not torsion-free.

Example 10

$\mathcal{T}_{\mathbb{C}}$ is a torsion-free weak hypervector space over the phase hyperfield ϕ defined in Example 12.

Theorem 22.1 ([14]) *Any subset A of $\mathcal{T}_{\mathbb{C}}$ containing 0, invariant under the involution $x \mapsto -x$ and closed with respect to multiplication, inherits the structure of weak hyperring from $\mathcal{T}_{\mathbb{C}}$. Furthermore, if $A \setminus \{0\}$ is invariant under the involution $x \mapsto \frac{1}{x}$, then A with the inherited structure is a hyperfield.*

Example 11 (Viro [14])

$\mathbb{R}$ inherits the structure of hyperfield from $\mathcal{T}_{\mathbb{C}}$ with

$$r_1 \diamond_{\mathbb{R}} r_2 = r_1 \diamond r_2 \cap \mathbb{R} = \begin{cases} \{r_1\}, & \text{if } |r_1| > |r_2|; \\ \{r_2\}, & \text{if } |r_2| > |r_1|; \\ \{r_1\}, & \text{if } r_1 + r_2 \neq 0 \text{ and } |r_1| = |r_2|; \\ [-|r_1|, |r_1|], & \text{if } r_1 + r_2 = 0. \end{cases}$$

Example 12 (Viro [14])

The set $\phi = \{z \in \mathbb{C} : \|z\| = 1\} \cup \{0\}$ also inherits the hyperfield structure from $\mathcal{T}_{\mathbb{C}}$. The hyperoperation addition is as follows:

$$z_1 \diamond_{\phi} z_2 = z_1 \diamond z_2 \cap \phi = \begin{cases} \{c \in \mathbb{C} : c \in S_{z_1 z_2}\}, & \text{if } z_1 + z_2 \neq 0; \\ \phi & \text{otherwise.} \end{cases}$$

The hyperfield $(\phi, \diamond_{\phi})$ discussed in Example 12 is called the phase hyperfield.

Example 13

Consider the tropical hyperfield $\mathcal{T}_{\mathbb{C}}$ in Example 3. The set $I = \{1, i, -1, -i\} \cup \{0\}$ also inherits the hyperfield structure from $\mathcal{T}_{\mathbb{C}}$. The hyperoperation addition on I is as follows:

$$x_1 \diamond_I x_2 = x_1 \diamond x_2 \cap I = \begin{cases} \{x_1, x_2\}, & \text{if } x_1 + x_2 \neq 0; \\ I & \text{otherwise.} \end{cases}$$

Example 14 (Viro [14])

Take $\mathbb{S} = \{-1, 0, 1\}$. Then $\mathbb{S}$ inherits the hyperfield structure from $\mathcal{T}_{\mathbb{C}}$, with $a \diamond_{\mathbb{S}} b = a \diamond b \cap \mathbb{S}$. This hyperfield is also called hyperfield of signs.

Theorem 22.2 *Let $(A, \diamond_A)$ be a hyperfield inherited from $(\mathcal{T}_{\mathbb{C}}, \diamond)$. $\mathcal{T}_{\mathbb{C}}$ is a weak hypervector space over A with respect to the external operation $\cdot : A \times \mathcal{T}_{\mathbb{C}} \to \mathcal{T}_{\mathbb{C}}$ defined by $a \cdot v = av$ for all $a \in A$ and $v \in \mathcal{T}_{\mathbb{C}}$.*

Proof We have $a \cdot (v \diamond w) = a(v \diamond w) = av \diamond aw$ for all $a \in A$ and $v, w \in \mathcal{T}_{\mathbb{C}}$. Also, $(a \diamond_A b) \cdot v = \big((a \diamond b) \cap A\big)v \subseteq (a \diamond b)v = av \diamond bv$, for all $a, b \in A$ and $v \in V$. Rest of the axioms follow directly from the properties of $\mathcal{T}_{\mathbb{C}}$. □

Example 15

From Example 13, we have I is an inherited hyperfield. By Theorem 22.2, it is interesting to note that $\mathcal{T}_{\mathbb{C}}$ is a hypervector space over the finite hyperfield I.

Definition 22.12 Let $\mathbb{V}$ be a weak hypervector space and $B = \{v_1, v_2, \cdots, v_n\} \subseteq \mathbb{V}$. B is called a weak basis if it is a linear independent spanning subset. A weak basis B is called a basis if every element of $\mathbb{V}$ is in a unique linear combination of elements of B.

Remark 22.3 If $\mathbb{V}$ is a hypervector space, then the notion of weak basis and basis coincide.

Remark 22.4 In a classical finite dimensional vector space, every element can be expressed as a unique linear combination of elements in its basis. This property plays a key role in proving many structural results of a vector space. Analogously, in a hypervector space, every element belongs to a unique linear combination of elements in its basis. However, this property fails to hold in a weak hypervector space. Indeed, in a weak hypervector space every linearly independent and spanning set need not inherit this property.

Consider the following examples.

Example 16

Let F_2 be the hyperfield of Example 2 and $F_2^2 = \{\underline{0}, u_1, u_2, u_3\}$ where $\underline{0} = (0, 0)$, $u_1 = (0, 1), u_2 = (1, 0), u_3 = (1, 1)$. Define $\oplus$ on F_2^2 as follows.

$\oplus$	$\underline{0}$	u_1	u_2	u_3
$\underline{0}$	$\{\underline{0}\}$	$\{u_1\}$	$\{u_2\}$	$\{u_3\}$
u_1	$\{u_1\}$	$\{\underline{0}, u_1\}$	$\{u_3\}$	$\{u_2, u_3\}$
u_2	$\{u_2\}$	$\{u_3\}$	$\{\underline{0}, u_2\}$	$\{u_1, u_3\}$
u_3	$\{u_3\}$	$\{u_2, u_3\}$	$\{u_1, u_3\}$	F_2^2

Then $(F_2^2, \oplus)$ is a canonical hypergroup.

We define $\cdot : F_2 \times F_2^2 \to F_2^2$ by $a \cdot (x, y) = (a \cdot x, a \cdot y)$. Then $(F_2^2, \oplus)$ is a weak hypervector space over F_2. Here $\{u_1, u_2\}$ is a weak basis for F_2^2. We can observe that every element can be expressed as a unique linear combination of u_1 and u_2. But this uniqueness need not be achieved with respect to every basis of F_2^2. For example, we consider the basis $\{u_1, u_3\}$ where u_3 belongs to two distinct linear combinations, that is, $u_3 \in 1 \cdot u_1 \oplus 1 \cdot u_3$ and $u_3 \in 0 \cdot u_1 \oplus 1 \cdot u_3$.

Example 17

In Example 13, one can see that $v = 1 + i$ and $w = 2$ are linearly independent elements and $2i \in span\{v, w\}$. But we can observe that $2i$ is in two distinct linear combinations of v and w. That is, $2i \in v \diamond (-w)$ and $2i \in iv \diamond w$.

If a weak hypervector space $\mathbb{V}$ has a finite weak basis B, then we say that V is finite dimensional and we write $\dim = |B|$, otherwise $\mathbb{V}$ is infinite dimensional.

Example 18

$\mathcal{T}_{\mathbb{C}}$ is a weak hypervector space over $(\mathbb{R}, \diamond_{\mathbb{R}})$ of dimension 2 with $\{1, i\}$ as its weak basis.

Example 19

$\mathcal{T}_{\mathbb{C}}$ is weak hypervector space over the phase hyperfield $(\phi, \diamond_\phi)$ as discussed in Example 10. It is an infinite dimensional hypervector space with $\mathbb{R}^+$ as its weak basis.

Example 20

$\mathcal{T}_{\mathbb{C}}$ is an infinite dimensional weak hypervector space over $(\mathbb{S}, \diamond_{\mathbb{S}})$. Here, $B = \{x, yi : x, y \in \mathbb{R}^+\}$ is a basis for $\mathcal{T}_{\mathbb{C}}$.

22.4 Properties of Subhyperspaces

let $\mathbb{V}, \mathbb{W}$ denote hypervector spaces unless otherwise specified. In this section, we define the direct sum of subhyperspaces, its complement, and finally we prove $\dim(U_1) + \dim(U_2) = \dim(U_1 \cap U_2) + \dim(U_1 + U_2)$.

Definition 22.13 (i) A subhyperspace W of $\mathbb{V}$ is said to be a direct sum of subhyperspaces S and S' if to each $w \in W$, there exist unique $u \in S$ and $v \in S'$ such that $w \in u + v$; we denote this by $W = S \bigoplus S'$.
(ii) If $\mathbb{V} = S \bigoplus S'$ then we say S' is a complement of S.

Theorem 22.3 *$\mathbb{V}$ is a direct sum of the subhyperspaces U_1 and U_2 if and only if $\mathbb{V} = U_1 + U_2$ and $U_1 \cap U_2 = \{0\}$.*

Proof Suppose that $\mathbb{V} = U_1 + U_2$ and $U_1 \cap U_2 = \{0\}$.

Let $v \in \mathbb{V}$ such that $v \in w_1 + w_2$ and $v \in w_1' + w_2'$, where $w_i, w_i' \in U_i$, $i = 1, 2$. Then $0 \in v - v \subseteq w_1 - w_1' + w_2 - w_2'$. The latter implies that there exist $t_1 \in w_1 - w_1'$ and $t_2 \in w_2 - w_2'$ such that $0 \in t_1 + t_2$. Now, $t_1 = -t_2 \in w_1 - w_1' \subseteq U_1$ implies $t_2 \in U_1$. Therefore, $t_2 \in U_1 \cap U_2 = \{0\}$, and so $t_2 = 0$ and consequently $t_1 = 0$. This shows that $0 \in w_1 - w_1'$ and $0 \in w_2 - w_2'$ and hence, $w_1 = w_1'$ and $w_2 = w_2'$.
Conversely, let $x \in U_1 \cap U_2$. Then, $x = x + 0$ where $x \in U_1$ and $0 \in U_2$, and $x = 0 + x$ where $0 \in U_1$ and $x \in U_2$. By uniqueness of representation, we get $x = 0$. Hence $U_1 \cap U_2 = \{0\}$. □

Lemma 22.1 ([1]) *Every linearly independent subset of $\mathbb{V}$ set is contained in its basis.*

Proposition 22.5 ([1]) *Let $B = \{v_1, v_2, \cdots, v_p\}$ be a basis of $\mathbb{V}$. Then each $v \in \mathbb{V}$ belongs to a unique linear combination of $v_1, v_2, \cdots, v_p$.*

The following theorem shows the existence of complementary subhyperspace in hypervector spaces.

Theorem 22.4 *Every subhyperspace of a hypervector space $\mathbb{V}$ has a complement.*

Proof Let S be a subhyperspace of $\mathbb{V}$ and $B = \{u_1, u_2, ..., u_k\}$ be a basis of S. Then by Lemma 22.1, B is contained in a basis B' of $\mathbb{V}$, say $B' = \{u_1, u_2, ..., u_k, w_1, w_2, ..., w_m\}$. Take $S' = span\{w_1, w_2, ..., w_m\}$. We prove that $\mathbb{V} = S \bigoplus S'$. Let $v \in \mathbb{V}$. By Proposition 22.5, there exists a unique linear combination $\sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{m} b_j w_j$ such that $v \in \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{m} b_j w_j$. Then there exist $w \in \sum_{i=1}^{k} a_i u_i$ and $w' \in \sum_{j=1}^{m} b_j w_j$ such that $v \in w + w' \subseteq S + S'$. The other inclusion is obvious. Let $u \in S \cap S'$. Then $u \in \sum_{i=1}^{k} a_i' u_i$ and $u \in \sum_{j=1}^{m} b_j' w_j$ for some $a_i', b_j' \in F$, $1 \le i \le k$, $1 \le j \le m$. This implies $0 \in \sum_{i=1}^{k} a_i' u_i - \left(\sum_{j=1}^{m} b_j w_j \right)$. As B' is a basis, we get $a_i' = 0$ for all $1 \le i \le k$ and $b_j' = 0$ for all $1 \le j \le m$. Hence, $u = 0$.

Theorem 22.5 *Let U_1 and U_2 be two subhyperspaces of $\mathbb{V}$. Then*

$$\dim(U_1) + \dim(U_2) = \dim(U_1 \cap U_2) + \dim(U_1 + U_2).$$

Proof Let $\dim(U_1) = m$, $\dim(U_2) = n$ and $\dim(U_1 \cap U_2) = r$.

Let $B = \{u_1, u_2, ..., u_r\}$ be a basis of $U_1 \cap U_2$. Then by Lemma 22.1, B is contained in a basis B' of U_1 and a basis B'' of U_2 where

$$B' = \{u_1, u_2, ..., u_r, w_1, w_2, ..., w_{m-r}\}$$

and

$$B'' = \{u_1, u_2, ..., u_r, w_1', w_2', ..., w_{n-r}'\}$$

Consider $L = \{u_1, u_2, ..., u_r, w_1, w_2, ..., w_{m-r}, w_1', w_2', ..., w_{n-r}'\}$. We prove that L is a basis for $U_1 + U_2$. Suppose $0 \in \sum_{i=1}^{r} a_i u_i + \sum_{j=1}^{m-r} b_j w_j + \sum_{k=1}^{n-r} c_k w_k'$. Then there exist $t \in \sum_{k=1}^{n-r} c_k w_k'$ and $t' \in \sum_{i=1}^{r} a_i u_i + \sum_{j=1}^{m-r} b_j w_j$ such that $0 \in t - t'$ implies that $-t = t'$, and shows that $t \in U_1$, and $t \in U_2$. Therefore, $t \in U_1 \cap U_2$. This gives $-t \in \sum_{i=1}^{r} d_i u_i$ for some d_i where $1 \leq i \leq r$. Now, $0 \in t - t \subseteq \sum_{i=1}^{r} d_i u_i + \sum_{k=1}^{n-r} c_k w_k'$. Since B'' is linearly independent, we get $d_i = 0$ for all $1 \leq i \leq r$ and $c_k = 0$ where $1 \leq k \leq n - r$. So, $0 \in \sum_{j=1}^{m-r} b_j w_j$. As B' is linearly independent, we get $b_j = 0$ for all $1 \leq j \leq m - r$. Hence, L is linearly independent. It remains to show that $span(L) = U_1 + U_2$. Let $v \in U_1 + U_2$. Then $v \in w_1 + w_2$ for some $w_1 \in U_1$ and $w_2 \in U_2$. Since B' is a basis for U_1, there exist unique $a_i'(1 \leq i \leq r)$ and $b_j'(1 \leq j \leq m - r)$ such that

$$w_1 \in \sum_{i=1}^{r} a_i' u_i + \sum_{j=1}^{m-r} b_j' w_j.$$

Since B'' is a basis for U_2, there exists a unique $d_i'(1 \leq i \leq r)$ and $c_k'(1 \leq k \leq m - r)$ such that

$$w_2 \in \sum_{i=1}^{r} d_i' u_i + \sum_{k=1}^{m-r} c_k' w_k'.$$

Therefore,

$$v \in w_1 + w_2 \subseteq \sum_{i=1}^{r} a_i' u_i + \sum_{j=1}^{m-r} b_j' w_j + \sum_{i=1}^{r} d_i' u_i + \sum_{k=1}^{m-r} c_k' w_k'$$

$$= \sum_{i=1}^{r} \left(a_i^{'} + d_i^{'}\right) u_i + \sum_{j=1}^{m-r} b_j^{'} w_j + \sum_{k=1}^{m-r} c_k^{'} w_k^{'}.$$

This means that $v \in span(L)$.

Hence, L is a basis for $U_1 + U_2$, and $\dim(U_1 + U_2) = m + n - r$. □

Corollary 22.1 *If $\mathbb{V}$ is a hypervector space, U_1 and U_2 are subhyperspaces of $\mathbb{V}$ with $\mathbb{V} = U_1 \oplus U_2$, then* $\dim(\mathbb{V}) = \dim(U_1) + \dim(U_2)$.

22.5 Properties of Linear Transformations

In this section, we study linear transformations on (weak) hypervector spaces, and prove that the set of all weak linear transformations form a weak hypervector space.

Definition 22.14 [1] Let $\mathbb{V}$ and $\mathbb{U}$ be two hypervector spaces over a hyperfield F. A map $\mu : \mathbb{V} \to \mathbb{U}$ is called a linear transformation if

(i) $\mu(v + v') = \mu(v) + \mu(v')$;
(ii) $\mu(\alpha v) = \alpha\mu(v)$,

for all $v, v' \in \mathbb{V}$ and $\alpha \in F$.
A bijective linear transformation is an isomorphism.

Definition 22.15 Let $\mathbb{V}$ and $\mathbb{U}$ be two weak hypervector spaces over a hyperfield F. A map $\mu : \mathbb{V} \to \mathbb{U}$ is called a weak linear transformation if

(i) $\mu(v + v') \subseteq \mu(v) + \mu(v')$;
(ii) $\mu(\alpha v) = \alpha\mu(v)$,

for all $v, v' \in \mathbb{V}$ and $\alpha \in F$.

Remark 22.5 Every linear transformation between two hypervector spaces is a weak linear transformation.

Example 21

As discussed in Example 16, F_2^2 is a weak hypervector space over F_2^2. Define $\mu_1 : F_2^2 \to F_2$ by $\mu_1(\underline{0}) = \mu_1(u_1) = 0$, $\mu_1(u_2) = \mu_1(u_3) = 1$. Then μ_1 is a linear transformation. The map $\mu_2 : F_2^2 \to F_2$ by $\mu_2(\underline{0}) = 0$, $\mu_2(u_1) = 1$, $\mu_2(u_2) = 1$, $\mu_2(u_3) = 1$ is a weak linear transformation, since $\mu_2(u_1 \oplus u_3) = \mu_2(\{u_2, u_3\}) = 1$ and $\mu_2(u_1) + \mu_2(u_3) = 1 + 1 = \{0, 1\}$, and hence $\mu_2(u_1 \oplus u_3) \subseteq \mu_2(u_1) + \mu_2(u_3)$.

Remark 22.6 In classical sense, an n-dimensional vector space over a field $\mathbb{K}$ is isomorphic to $\mathbb{K}^n$. This is not true in general for weak hypervector spaces. In Example 1, $\mathbb{V}$ is a weak hypervector space over K with $\dim \mathbb{V} = 1$. But $\mathbb{V}$ is not isomorphic to K because K is a hypervector space over K.

Example 22

From Example 7, we have $F_2^E = \{f_0, f_1, f_2, f_3\}$. Now we define a map $\eta : F_2^E \to F_2^2$ by $\eta(f_0) = \underline{0}$, $\eta(f_1) = u_1$, $\eta(f_3) = u_2$, and $\eta(f_3) = u_3$. Then η is an isomorphism.

Proposition 22.6 ([1]) *Let $\mathbb{V}$ and $\mathbb{U}$ be two hypervector spaces over a hyperfield F and $\mu : \mathbb{V} \to \mathbb{U}$ be a (weak) linear transformation. Then $\mu(0_{\mathbb{V}}) = 0_{\mathbb{V}}$ and $\mu(-x) = -\mu(x)$ for all $x \in \mathbb{V}$.*

Definition 22.16 [1] Let $\mathbb{V}$ and $\mathbb{U}$ be two hypervector spaces over a hyperfield F and $\mu : \mathbb{V} \to \mathbb{U}$ be a linear transformation. Then we define the kernel of μ as

$$\ker \mu = \{v \in \mathbb{V} : \mu(v) = 0\},$$

and the image of μ as

$$Im(\mu) = \{\mu(v) : v \in \mathbb{V}\}.$$

Proposition 22.7 ([1]) *Let $\mu : \mathbb{V} \to \mathbb{U}$ be a linear transformation. Then* $\ker \mu$ *is a subhyperspace of $\mathbb{V}$ and $Im(\mu)$ is a subhyperspace of $\mathbb{U}$.*

Proposition 22.8 ([1]) *Let $T : \mathbb{U} \to \mathbb{V}$ be a linear transformation. Then T is one-one if and only if* $\ker T = \{0\}$.

This property fails if T is a weak linear transformation (see Example 23).

Example 23

Consider the map μ_2 as defined in Example 21. Here, $\ker \mu = \{\underline{0}\}$ but μ is not one-one.

The following Proposition shows that a weak linear transformation between weak hypervector spaces carries a spanning set to a spanning set.

Proposition 22.9 *Let $\mathbb{U}$ and $\mathbb{V}$ be two weak hypervector spaces, $S = \{u_1, u_2, \cdots, u_p\}$ be a spanning set of $\mathbb{U}$, and $T : \mathbb{U} \to \mathbb{V}$ be an onto weak linear transformation. Then $T(S)$ is a spanning set of $\mathbb{V}$.*

Proof Let $v \in \mathbb{V}$. Since T is onto, there exists $u \in \mathbb{U}$ such that $T(u) = v$. We can find $a_i \in F$, $1 \leq i \leq p$ such that $u \in \sum_{i=1}^{p} a_i u_i$.

Then $v = T(u) \in T(\sum_{i=1}^{p} a_i u_i) \subseteq \sum_{i=1}^{p} a_i T(u_i)$, as desired. □

Corollary 22.2 *Let $T : \mathbb{U} \to \mathbb{V}$ be an isomorphism and B a basis of $\mathbb{U}$. Then $T(B)$ is a basis of $\mathbb{V}$.*

Theorem 22.6 ([1]) *Let $\mathbb{U}$ and $\mathbb{V}$ be finite dimensional hypervector spaces, and $T : \mathbb{U} \to \mathbb{V}$ be a linear transformation. Then* $\dim(\ker T) + \dim(ImT) = \dim \mathbb{U}$.

We summarize Proposition 22.8, Proposition 22.9, and Theorem 22.6 as follows.

Theorem 22.7 *Let $\mathbb{U}$ and $\mathbb{V}$ be two hypervector spaces with* $\dim \mathbb{U} = \dim \mathbb{V} = n$ *and $T : \mathbb{U} \to \mathbb{V}$ be a linear transformation. Then the following are equivalent:*

(i) *T is bijective.*
(ii) *T is one-one.*
(iii) *T is onto.*

Theorem 22.8 *Let $\mu : \mathbb{V} \to \mathbb{W}$ be an onto linear transformation of hypervector spaces $\mathbb{V}$ and $\mathbb{W}$. Then $\dfrac{\mathbb{V}}{\ker \mu}$ is isomorphic to $\mathbb{W}$.*

Proof Define $\eta : \dfrac{\mathbb{V}}{\ker \mu} \to \mathbb{W}$ by $\eta(v + \ker \mu) = \mu(v)$ for any $v + \ker \mu \in \dfrac{\mathbb{V}}{\ker \mu}$. To see η is well-defined, let $v + \ker \mu = v' + \ker \mu$ for any $v, v' \in \mathbb{V}$. Then $v \in v + 0 \in v + \ker \mu = v' + \ker \mu$ implies that $v \in v' + t$ for some $t \in \ker \mu$. This means, $\mu(v) \in \mu(v' + t) = \mu(v') + \mu(t)$. That is, $\mu(v) \in \mu(v') + 0 = \mu(v')$, and so $\eta(v) = \eta(v')$. Suppose that $\eta(v_1 + \ker \mu) = \eta(v_2 + \ker \mu)$. Now $\mu(v_1) = \mu(v_2)$ implies that $0 \in \mu(v_1) - \mu(v_2) = \mu(v_1 - v_2)$. This means there exists $x \in v_1 - v_2$ such that $\mu(x) = 0$. From this, $v_1 \in x + v_2$ and $v_2 \in v_1 - x$. So, $v_1 \in v_2 + \ker \mu$ and $v_2 \in v_1 + \ker \mu$. Therefore, $v_1 + \ker \mu \subseteq v_2 + \ker \mu$ and $v_2 + \ker \mu \subseteq v_1 + \ker \mu$. The latter implies that $v_1 + \ker \mu = v_2 + \ker \mu$ and hence, η is one-one. Let $w \in W$. Since μ is onto, there exists $v \in \mathbb{V}$ such that $\mu(v) = w$. Now, $\eta(v + \ker \mu) = \mu(v) = w$. Therefore, η is onto.
Let $v_1 + \ker \mu, v_2 + \ker \mu \in \dfrac{\mathbb{V}}{\ker \mu}$ and $\alpha, \beta \in F$. Then

$$\begin{aligned}\eta(\alpha v_1 + \ker \mu + \beta v_2 + \ker \mu) &= \eta(\alpha v_1 + \beta v_2 + \ker \mu)\\ &= \mu(\alpha v_1 + \beta v_2)\\ &= \alpha\mu(v_1) + \beta\mu(v_2)\\ &= \alpha\eta(v_1 + \ker \mu) + \beta\eta(v_2 + \ker \mu).\end{aligned}$$

Therefore, η is a linear transformation. Hence $\dfrac{\mathbb{V}}{\ker \mu}$ is isomorphic to $\mathbb{W}$. □

Proposition 22.10 *If U_1 and U_2 are subhyperspaces of $\mathbb{V}$, then $\dfrac{U_2}{U_1 \cap U_2}$ is isomorphic to $\dfrac{U_1 + U_2}{U_1}$.*

Proof Define a map $\mu : U_2 \to \dfrac{U_1 + U_2}{U_1}$, by $\mu(v) = v + U_1$ for every $v \in U_2$. We prove that μ is a bijective linear transformation. Let $u, v \in U_2$ and $a, b \in F$. Then

$$\mu(au+bv) = (au+bv)+U_1 = (au+U_1)+(bv+U_1)$$
$$= a(u+U_1)+b(v+U_1) = a\mu(u)+b\mu(v).$$

Therefore, μ is linear.

Let $v+U_1 \in \dfrac{U_1+U_2}{U_1}$. Then, $v \in U_1+U_2$. So, $v \in w_1+w_2$ for some $w_1 \in U_1$ and $w_2 \in U_2$. Now $v+U_1 \in (w_1+w_2)+U_1 = (w_1+U_1)+(w_2+U_1)$. This means that $v+U_1 \in \{w_2+U_1\}$ implies that $v+U_1 = w_2+U_1 = \mu(w_2)$, and so $\mu(w_2) = v+U_1$. Therefore, μ is onto.
Suppose that $\mu(v) = 0+U_1$ where $v \in U_2$. Then, $v \in v+0 \in v+U_1 = 0+U_1$ implies $v \in U_1$ so, $v \in U_1 \cap U_2$. On the other hand, $v' \in U_1 \cap U_2$ implies $\mu(v') = v'+U_1 = U_1$ and hence, $\ker \mu = U_1 \cap U_2$. The rest follows from Theorem 22.8. □

Corollary 22.3 *If U_1 and U_2 are complementary subhyperspaces of a hypervector space $\mathbb{V}$ then U_2 is isomorphic to $\dfrac{\mathbb{V}}{U_1}$.*

Proof The proof follows from Proposition 22.10. □

Theorem 22.9 *Let $(\mathbb{V}, +, \cdot)$ be a (weak) hypervector space over F and let $\mathbb{U}$ be a non-empty-set such that there is a bijection $f : \mathbb{V} \to \mathbb{U}$. Then $\mathbb{U}$ is a (weak) hypervector space over F under the hyperoperation $\oplus$ and the external operation $\circ$ induced by f, defined as follows:*

$$u \oplus v = f\left(f^{-1}(u)+f^{-1}(v)\right) \text{ and } \alpha \circ u = f\left(\alpha \cdot f^{-1}(u)\right)$$

for all $u, v \in \mathbb{U}$ and $\alpha \in F$.
Furthermore, f is an isomorphism between $\mathbb{V}$ and $\mathbb{U}$.

Proof First, we show that f gives rise to a canonical hypergroup structure for the set $\mathbb{U}$. Let $u_1, u_2, u_3 \in \mathbb{U}$. Then

$$\begin{aligned} u_1 \oplus (u_2 \oplus u_3) &= f\Big[f^{-1}(u_1)+f^{-1}\Big(f\big(f^{-1}(u_2)+f^{-1}(u_3)\big)\Big)\Big] \\ &= f\big[f^{-1}(u_1)+f^{-1}(u_2)+f^{-1}(u_3)\big] \\ &= f\Big[f^{-1}\Big(f\big(f^{-1}(u_1)+f^{-1}(u_3)\big)\Big)+f^{-1}(u_3)\Big] \\ &= (u_1 \oplus u_2) \oplus u_3. \end{aligned}$$

Let $u_1, u_2 \in \mathbb{U}$. Then $f^{-1}(u_2) \in f^{-1}(u_1)+\mathbb{V}$, and so $f^{-1}(u_2) \in f^{-1}(u_1) \oplus v$ for some $v \in \mathbb{V}$. This gives $u_2 \in f\left(f^{-1}(u_1)+v\right) = u_1 \oplus f(v) \subseteq u_1 \oplus \mathbb{U}$. Similarly, we can show that $\mathbb{U} \subseteq \mathbb{U} \oplus u_1$. Hence, $(\mathbb{U}, \oplus)$ is a hypergroup. It is easy to verify that $(\mathbb{U}, \oplus)$ is a canonical hypergroup. Now it is clear that f is an isomorphism between the hypergroups $(\mathbb{V}, +)$ and $(\mathbb{U}, \oplus)$. Further $\mathbb{U}$ becomes a (weak) hypervector space, following from the fact f is a homomorphism and that $\mathbb{V}$ is a (weak) hypervector space. In this case, f is a (weak) linear transformation and hence an isomorphism between the (weak) hypervector spaces $\mathbb{V}$ and $\mathbb{U}$. □

Now onwards in this section, a weak hypervector space $\mathbb{V}$ stands for a torsion-free weak hypervector space which has a basis.

Proposition 22.11 *Let $\mathbb{V}$ and $\mathbb{W}$ be two weak hypervector spaces over F and $L(\mathbb{V}, \mathbb{W})$ denote the collection of all weak linear transformations from $\mathbb{V}$ to $\mathbb{W}$. For any $\mu_1, \mu_2 \in L(\mathbb{V}, \mathbb{W})$, we define the hyperoperation $\oplus$ by*

$$\mu_1 \oplus \mu_2 = \{\mu \in L(\mathbb{V}, \mathbb{W}) : \mu(v) \in \mu_1(v) + \mu_2(v) \text{ for all } v \in \mathbb{V}\}$$
$$\text{where } (\mu_1 \oplus \mu_2)(v) = \mu_1(v) + \mu_2(v),$$

and the scalar multiplication $\cdot : F \times L(\mathbb{V}, \mathbb{W}) \to L(\mathbb{V}, \mathbb{W})$ by

$$(\alpha \cdot \mu)(v) = \alpha\mu(v) \quad \text{for all } \mu \in L(\mathbb{V}, \mathbb{W}), \alpha \in F, \text{ and } v \in \mathbb{V}.$$

Then $L(\mathbb{V}, \mathbb{W})$ is a weak hypervector space over F with respect to the hyperoperation addition and the scalar multiplication defined above.

Proof For $\mu_1, \mu_2, \mu_3 \in L(\mathbb{V}, \mathbb{W})$,

$$\begin{aligned}\Big((\mu_1 \oplus \mu_2) \oplus \mu_3\Big)(v) &= \Big(\mu_1 \oplus \mu_2\Big)(v) + \mu_3(v)\\ &= \Big(\mu_1(v) + \mu_2(v)\Big) + \mu_3(v)\\ &= \mu_1(v) + \Big(\mu_2(v) + \mu_3(v)\Big)\\ &= \mu_1(v) + \Big(\mu_2 \oplus \mu_3\Big)(v)\\ &= \Big(\mu_1 \oplus (\mu_2 \oplus \mu_3)\Big)(v)\end{aligned}$$

for all $v \in \mathbb{V}$.

This implies $(\mu_1 \oplus \mu_2) \oplus \mu_3 = \mu_1 \oplus (\mu_2 \oplus \mu_3)$. So, $(L(\mathbb{V}, \mathbb{W}), \oplus)$ is a semihypergroup. Let $\mu_1, \mu_2 \in L(\mathbb{V}, \mathbb{W})$. Then

$$(\mu_1 \oplus \mu_2)(v) = \mu_1(v) + \mu_2(v) = \mu_2(v) + \mu_1(v) = (\mu_2 \oplus \mu_1)(v)$$

for all $v \in \mathbb{V}$.

Let $\mu_1 \in L(\mathbb{V}, \mathbb{W})$ and take $\mathbf{0}(v) = 0$ for all $v \in \mathbb{V}$. Then $\mathbf{0} \in L(\mathbb{V}, \mathbb{W})$. Now for any $v \in \mathbb{V}$, $(\mu_1 \oplus \mathbf{0})(v) = (\mathbf{0} \oplus \mu_1)(v) = \mu_1(v) + \mathbf{0}(v) = \mu_1(v)$. Clearly, $\mathbf{0} \in \mu_1 - \mu_1$ as $0 \in \mu_1(v) - \mu_1(v)$ for all $v \in \mathbb{V}$. Now, suppose $h \in \mu_1 \oplus \mu_2$. Then $h(v) \in \mu_1(v) \oplus \mu_2(v)$ for all $v \in \mathbb{V}$. Then $\mu_1(v) \in h(v) - \mu_2(v)$ and $\mu_2(v) \in \mu_1(v) + h(v)$ for all $v \in \mathbb{V}$. Therefore $(L(\mathbb{V}, \mathbb{W}), \oplus)$ is a canonical hypergroup.
Let $\alpha, \beta \in F$ and $\mu_1, \mu_2 \in L(\mathbb{V}, \mathbb{W})$. Then

$$\begin{aligned}\alpha \cdot (\mu_1 \oplus \mu_2) &= \{\mu \in L(\mathbb{V}, \mathbb{W}) : \mu(v) \in \alpha(\mu_1(v) + \mu_2(v)) \text{ for all } v \in \mathbb{V}\} \\ &\subseteq \{\mu \in L(\mathbb{V}, \mathbb{W}) : \mu(v) \in \alpha\mu_1(v) + \alpha\mu_2(v) \text{ for all } v \in \mathbb{V}\} \\ &= \alpha \cdot \mu_1 \oplus \alpha \cdot \mu_2.\end{aligned}$$

On the other hand,

$$\begin{aligned}(\alpha + \beta) \cdot \mu_1 &= \{\mu \in L(\mathbb{V}, \mathbb{W}) : \mu(v) \in (\alpha + \beta)\mu_1(v) \text{ for all } v \in \mathbb{V}\} \\ &\subseteq \{\mu \in L(\mathbb{V}, \mathbb{W}) : \mu(v) \in \alpha\mu_1(v) + \beta\mu_1(v), \text{ for all } v \in \mathbb{V}\} \\ &= \alpha \cdot \mu_1 \oplus \beta \cdot \mu_1.\end{aligned}$$

In a similar way, we can verify the remaining axioms for hypervector spaces. Thus $(L(\mathbb{V}, \mathbb{W}), \oplus)$ is a weak hypervector space over F. □

Example 24

Consider the hypervector space defined in Example 21. Define $\mu_0 : F_2^2 \to F_2$ by $\mu_0(\underline{0}) = \mu_0(u_1) = \mu_0(u_2) = \mu_0(u_3) = \underline{0}$ and $\mu_3 : F_2^2 \to F_2$ by $\mu_3(\underline{0}) = \underline{0}$, $\mu_3(u_2) = \underline{0}$, $\mu_3(u_1) = \mu_3(u_3) = 1$. Then $L(F_2^2, F_2) = \{\mu_0, \mu_1, \mu_2, \mu_3\}$ is a weak hypervector space.

Definition 22.17 In a finite dimensional (weak) hypervector space $\mathbb{V}$, an ordered basis is a finite sequence of vectors which forms a basis for $\mathbb{V}$.

Theorem 22.10 *Let $\{v_1, v_2, \cdots, v_n\}$ be an ordered basis for $\mathbb{V}$. If $\{w_1, w_2, \cdots, w_n\}$ is an ordered set of elements in $\mathbb{W}$ with at most one non-zero w_i, then there exists a unique linear transformation φ from $\mathbb{V}$ to $\mathbb{W}$ such that $\varphi(v_i) = w_i$ for every $1 \leq i \leq n$.*

Proof Let $v \in \mathbb{V}$. Then there exist $a_1, a_2, ..., a_n$ such that $v \in \sum_{i=1}^{n} a_i v_i$.

Let $\{w_1, w_2, \cdots, w_n\}$ be elements in $\mathbb{W}$ with $w_1 = 0, w_2 = 0, \cdots, w_{i-1} = 0$, $w_{i+1} = 0, \cdots, w_n = 0$ and $w_i \neq 0$. Now define $\varphi : \mathbb{V} \to \mathbb{W}$ by $\varphi(v) = a_i w_i$. It is obvious that φ is well-defined. Now, $\varphi(v_j) = \begin{cases} 0, & \text{if } i \neq j \\ w_i, & \text{if } i = j \end{cases}$. To prove φ is linear, let $u, v \in \mathbb{V}$. Then $u \in \sum_{i=1}^{n} a_i v_i$ and $v \in \sum_{i=1}^{n} b_i v_i$ for some a_i, b_i $(1 \leq i, j \leq n)$ implies $u + v \subseteq \sum_{i=1}^{n} a_i v_i + \sum_{i=1}^{n} b_i v_i$. This gives $\varphi(u + v) \subseteq \varphi(u) + \varphi(v)$. Similarly, $\varphi(\delta u) = \delta\varphi(u)$ for all $\delta \in F$.

Let $\varphi' : \mathbb{V} \to \mathbb{W}$ be a weak linear transformation such that $\varphi'(v_i) = w_i$. Let $v \in \mathbb{V}$ with $v \in \sum_{i=1}^{n} a_i v_i$. Then

$$\varphi'(v) \in \varphi'\left(\sum_{i=1}^{n} a_i v_i\right) \subseteq a_i \left(\sum_{i=1}^{n} \varphi'(v_i)\right) \quad (\text{since } \phi' \text{ is weak linear})$$
$$= a_i w_i = \varphi(v).$$

Therefore, $\varphi' = \varphi$. □

Definition 22.18 Let $\mathbb{V}$ be a weak hypervector space over a hyperfield F. A (weak) linear functional is a (weak) linear transformation from $\mathbb{V}$ to F.

Corollary 22.4 *Let $\{v_1, v_2, \cdots, v_n\}$ be an ordered basis for $\mathbb{V}$. If $\{\alpha_1, \alpha_2, \cdots, \alpha_n\}$ is an ordered set of scalars with at most one $\alpha_i \neq 0$ then there exists a unique linear functional φ on $\mathbb{V}$ such that $\varphi(v_i) = \alpha_i$ for every $1 \leq i \leq n$.*

Proof It follows from Theorem 22.10. □

For a hypervector space $\mathbb{V}$, we denote $L(\mathbb{V}, F)$ by $\mathbb{V}^*$.

Theorem 22.11 *Let $\{v_1, v_2, \cdots, v_n\}$ be a basis of a hypervector space $\mathbb{V}$. Then there exists a unique basis $B^* = \{\varphi_1, \varphi_2, \cdots, \varphi_n\}$ of $\mathbb{V}^*$ such that* $\dim(V^*) = \dim(V)$. *Furthermore, we have*

$$f \in \sum_{i=1}^{n} f(v_i)\varphi_j \quad \textit{where } f \textit{ is a linear functional}$$

and for each $x \in \mathbb{V}$, we have

$$x \in \sum_{i=1}^{n} \varphi_i(x) v_i.$$

Proof Define $\varphi_i(v_j) = \delta_{ij} = \begin{cases} 0, \text{ if } i \neq j \\ 1, \text{ if } i = j \end{cases}$. Then by Corollary 22.4, φ_i is a weak linear functional. We prove that $B = \{\varphi_1, \varphi_2, ..., \varphi_n\}$ is linearly independent. Suppose that $f = 0$ with $f \in \sum_{i=1}^{n} a_i \varphi_i$ for some a_i $(i \leq i \leq n)$. Then $0 = f\left(v_j\right) \in \sum_{i=1}^{n} a_i \varphi_i\left(v_j\right)$ for each j. This gives, $0 = f\left(v_j\right) \in \sum_{i=1}^{n} a_i \delta_{ij} = a_j$ for each j, implies $a_j = 0$ for $(1 \leq j \leq n)$. Therefore, $\{\varphi_1, \varphi_2, ..., \varphi_n\}$ is linearly independent.
Now we show that $\{\varphi_1, \varphi_2, ..., \varphi_n\}$ spans $\mathbb{V}^*$. Let $f \in \mathbb{V}^*$. Put $k_i = f(v_i)$. Then for any $x \in \mathbb{V}$, $x \in \sum_{i=1}^{n} a_i v_i$. Then,

$$\varphi_j(x) \in \varphi_j\left(\sum_{i=1}^{n} a_i v_i\right) \subseteq \sum_{i=1}^{n} a_i \varphi_j\left(v_i\right) = \sum_{i=1}^{n} a_i \delta_{ij} = a_j.$$

So, $\varphi_j(x) = a_j$. Therefore, $x \in \sum_{i=1}^{n} \varphi_i(x) v_i$.

We have

$$f(x) \in f\left(\sum_{i=1}^{n} a_i v_i\right) = \sum_{i=1}^{n} a_i f(v_i) \quad \text{(since } f \text{ is a linear functional)}$$
$$= \sum_{i=1}^{n} \varphi_i(x) f(v_i).$$

So,

$$f(x) \in \sum_{i=1}^{n} \varphi_i(x) f(v_i) = \sum_{i=1}^{n} k_i \varphi_i(x) \text{ for every } x.$$

Therefore, $f \in \sum_{i=1}^{n} k_i \varphi_j = \sum_{i=1}^{n} f(v_i)\varphi_j$. This shows that $\{\varphi_1, \varphi_2, ..., \varphi_n\}$ spans $\mathbb{V}^*$. Furthermore, the scalars are unique and hence B is a basis. Thus, $\dim(\mathbb{V}^*) = \dim(\mathbb{V})$. □

Example 25

From Example 24, $(F_2^2)^* = \{\mu_0, \mu_1, \mu_2, \mu_3\}$. Now we define a map $\Gamma : (F_2^2)^* \to F_2^2$ by $\Gamma(\mu_0) = \underline{0}$, $\Gamma(\mu_1) = u_1$, $\Gamma(\mu_2) = u_2$, and $\Gamma(\mu_3) = u_3$. Then Γ is an isomorphism.

Theorem 22.12 *Let* $\mathbb{V}$ *and* $\mathbb{W}$ *be two hypervector spaces with* $\dim(\mathbb{V}) = m$ *and* $\dim(\mathbb{W}) = n$. *Then* $\dim(L(\mathbb{V}, \mathbb{W})) = mn$.

Proof Let $\{v_1, v_2, \cdots, v_m\}$ and $\{w_1, w_2, \cdots, w_n\}$ be the basis of $\mathbb{V}$ and $\mathbb{W}$, respectively. For $1 \le i \le n$, $1 \le j, k \le m$, define

$$T_{ij}(v_k) = \begin{cases} 0, & \text{if } j \neq k \\ w_i, & \text{if } j = k \end{cases} = \delta_{jk} w_i,$$

where δ_{jk} is as defined in the proof of Theorem 22.11. Then by Theorem 22.10, T_{ij} is a weak linear transformation. We prove $B = \{T_{ij}; 1 \le i \le n, 1 \le j \le m\}$ is a basis for $L(\mathbb{V}, \mathbb{W})$. Suppose $f = 0 \in \sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} T_{ij}$, where $a_{ij} \in F$. Now for $1 \le k \le m$,

$$f(v_k) = 0 \in \sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} T_{ij}(v_k)$$
$$= \sum_{i=1}^{n} \sum_{j=1}^{m} a_{ij} \delta_{jk} w_i$$

$$= \sum_{i=1}^{n} a_{ik} w_i.$$

That is, $0 \in \sum_{i=1}^{n} a_{ik} w_i$. Since $\{w_1, w_2, \cdots, w_n\}$ is linearly independent, we have $a_{ik} = 0\ (1 \leq i \leq n)$ for every k. Therefore, B is linearly independent.

Next, let $T \in L(\mathbb{V}, \mathbb{W})$. Now for $1 \leq k \leq m$, $T(v_k) \in \mathbb{W}$, and so, $T(v_k) \in \sum_{i=1}^{n} a_{ik} w_i$.

For any $x \in \mathbb{V}$, we have $x \in \sum_{j=1}^{m} d_j v_j,\ d_j \in F,\ (1 \leq j \leq m)$. Then,

$$\begin{aligned} T(x) \in T\left(\sum_{j=1}^{m} d_j v_j\right) &\subseteq \sum_{j=1}^{m} d_j T(v_j) \\ &\subseteq \sum_{j=1}^{m} d_j \sum_{i=1}^{n} a_{ij} w_i \\ &= \sum_{j=1}^{m} \sum_{i=1}^{n} a_{ij} d_j w_i. \end{aligned}$$

Also, for $1 \leq i \leq n$, $1 \leq j \leq m$,

$$T_{ij}(x) \in T_{ij}\left(\sum_{k=1}^{m} d_k v_k\right) \subseteq \sum_{j=1}^{m} d_k T_{ij}(v_k) = \sum_{j=1}^{m} d_k \delta_{jk} w_i = d_j w_i.$$

Therefore, $T(x) \in \sum_{j=1}^{m} \sum_{i=1}^{n} a_{ij} T_{ij}(x)$, and so $T \in \sum_{j=1}^{m} \sum_{i=1}^{n} a_{ij} T_{ij}$. Further, we can observe that a_{ij}'s are unique and so B is a basis. Therefore, $\dim(\mathbb{V}, \mathbb{W}) = mn$. □

Example 26

Consider Example 24. The basis for F_2^2 is $\{u_1, u_2\}$ and basis for $L(F_2^2, F_2)$ is $\{\mu_1, \mu_3\}$. Then $\dim L(F_2^2, F_2) = 2 = \dim(F_2^2)\ \dim(F_2)$.

22.6 Annihilators in Hypervector Spaces

In this section, we discuss some properties of annihilators of linear functionals. We introduce a notion projection on a weak hypervector space and prove related results.

Definition 22.19 Let $\mathbb{V}$ be a hypervector space and $\emptyset \neq U \subseteq \mathbb{V}$. The annihilator of U is the set of all linear functionals on $\mathbb{V}$ such that $f(v) = 0$ for every $v \in U$, and it is denoted by U^o.

Example 27

Consider Example 24. Let $\mathbb{V} = F_2^2$ and $U = \{0, u_2\}$. Then $U^o = \{\mu_0, \mu_3\}$.

Proposition 22.12 *U^o is a subhyperspace of $\mathbb{V}^*$.*

Proof $U^o \neq \emptyset$, since $f = 0 \in U^o$. For any $f, g \in U^o$, $(f - g)(x) = f(x) - g(x) = 0$ for every $x \in U$, and we get $f - g \subseteq U^o$. For $a \in F$, we have $(af)(x) = af(x) = 0$ for every $x \in U$. Hence, $af \in U^o$. □

Theorem 22.13 *Let $\mathbb{V}$ be a hypervector space over the hyperfield F and U a subhyperspace of $\mathbb{V}$. Then* $\dim(U) + \dim(U^o) = \dim(\mathbb{V})$.

Proof Let $B = \{s_1, s_2, \cdots, s_m\}$ be a basis of U. Then by Lemma 22.1, B is contained in a basis $B' = \{s_1, s_2, \cdots, s_m, s_{m+1}, ..., s_n\}$ of $\mathbb{V}$. Let $\{\varphi_1, \varphi_2, \cdots, \varphi_n\}$ be the dual basis of $\mathbb{V}^*$. We prove that $\{\varphi_{m+1}, \varphi_{m+2}, \cdots, \varphi_n\}$ is a basis of U^o. Clearly, $\varphi_k(s_i) = 0$ for every $1 \leq i \leq m$ and $m + 1 \leq k \leq n$. For any $x \in U$, we have $x \in \sum_{i=1}^{m} a_i s_i$. So, $\varphi_k(x) \in \sum_{i=1}^{m} a_i \varphi(s_i) = 0$ implies, $\varphi_k(x) = 0$ for every $m + 1 \leq k \leq n$, which shows that $\varphi_k \in U^o$. Trivially, $\{\varphi_{m+1}, \varphi_{m+2}, \cdots, \varphi_n\}$ is linearly independent. Let $f \in U^o \subseteq \mathbb{V}^*$. By Theorem 22.11, we get

$$f \in \sum_{i=1}^{n} f(s_i)\varphi_i = \sum_{i=1}^{m} f(s_i)\varphi_i + \sum_{i=m+1}^{n} f(s_i)\varphi_i = \sum_{i=m+1}^{n} f(s_i)\varphi_i.$$

Therefore, $f \in span\{\varphi_{m+1}, \varphi_{m+2}, \cdots, \varphi_n\}$. Hence, $\{\varphi_{m+1}, \varphi_{m+2}, \cdots, \varphi_n\}$ is a basis for U^o. Furthermore, $\dim(U^o) = n - m = \dim(\mathbb{V}) - \dim(U)$. □

Theorem 22.14 *Let U_1 and U_2 be two subhyperspaces of a hypervector space $\mathbb{V}$ over the hyperfield F. Then $U_1 = U_2$ if and only if $U_1^o = U_2^o$.*

Proof If $U_1 = U_2$, then $U_1^o = U_2^o$. Conversely, suppose that $U_1^o = U_2^o$. On a contrary, assume that $U_1 \neq U_2$. Then there exists $w_1 \in U_1 \setminus U_2$. Now we can find a linear functional φ such that $\varphi(w_1) \neq 0$ and $\varphi(v) = 0$ for every $v \in U_2$. This gives $\varphi \in U_1^o$ but $\varphi \notin U_2^o$, a contradiction. □

The following lemma is easy.

Lemma 22.2 *Let $U_1 \subset U_2$ be two subhyperspaces of a hypervector space $\mathbb{V}$. Then $U_2^o \subset U_1^o$.*

Theorem 22.15 *Let U_1 and U_2 be two subhyperspaces of a hypervector space $\mathbb{V}$ over F. Then, $(U_1 + U_2)^o = U_1^o \cap U_2^o$.*

Proof Since $U_1, U_2 \subseteq U_1 + U_2$, by Lemma 22.2, we have $(U_1 + U_2)^o \subseteq U_1^o, U_2^o$, which implies that $(U_1 + U_2)^o \subseteq U_1^o \cap U_2^o$. Let $\varphi \in U_1^o \cap U_2^o$. Then $\varphi(v) = 0$ for every $v \in U_1, U_2$. Now for any $x \in U_1 + U_2$, we have $v \in w_1 + w_2$, and so $\varphi(x) \in \varphi(w_1) + \varphi(w_2) = 0$ implies $\varphi(x) = 0$ for every $x \in U_1 + U_2$. Therefore, $\varphi \in (U_1 + U_2)^o$. □

Definition 22.20 Let $\mathbb{V}$ be a hypervector space. A (weak) linear transformation $\mathcal{P} : \mathbb{V} \to \mathbb{V}$ with $\mathcal{P}^2 = \mathcal{P} \circ \mathcal{P} = \mathcal{P}$ is called a (weak) projection on $\mathbb{V}$.

Theorem 22.16 *Let $\mathbb{V}$ be a hypervector space and $\mathcal{P}$ be a projection on $\mathbb{V}$ with $Im(\mathcal{P}) = M$ and $\ker \mathcal{P} = N$. Then $\mathbb{V} = M \bigoplus N$.*

Proof It is obvious that M, N are subhyperspaces of $\mathbb{V}$ and $M \bigoplus N \subseteq \mathbb{V}$. Let $x \in \mathbb{V}$. Then $\mathcal{P}(v - \mathcal{P}(v)) = \mathcal{P}(v) - \mathcal{P}(\mathcal{P}(v)) = \mathcal{P}(v) - \mathcal{P}^2(v) = \mathcal{P}(v) - \mathcal{P}(v)$. That is, $0 \in \mathcal{P}(v) - \mathcal{P}(v) = \mathcal{P}(v - \mathcal{P}(v))$. Therefore, there exists $t \in v - \mathcal{P}(v)$ such that $\mathcal{P}(t) = 0$. Now $t \in N$ and $\mathcal{P}(t) \in M$ and $v \in t + \mathcal{P}(v) \subseteq M + N$. It follows that $\mathbb{V} \subseteq M + N$. Hence, $\mathbb{V} = M + N$. Let $u \in M \cap N$. Then $u = \mathcal{P}(x)$ for some $x \in \mathbb{V}$ and $\mathcal{P}(u) = 0$. Now $0 = \mathcal{P}(u) = \mathcal{P}(\mathcal{P}(x)) = \mathcal{P}^2(x) = \mathcal{P}(x) = u$. Therefore, $u = 0$. □

Theorem 22.17 *Let $\mathbb{V}$ be a hypervector space and M, N be subhyperspaces of $\mathbb{V}$ such that $\mathbb{V} = M \bigoplus N$. Then there exists a weak projection $\mathcal{P}$ on $\mathbb{V}$ such that $Im(\mathcal{P}) = M$ and $\ker \mathcal{P} = N$.*

Proof Since $\mathbb{V} = M \bigoplus N$, for each $u \in \mathbb{V}$ we have $u \in w_1 + w_2$ for unique $w_1 \in M$ and $w_2 \in N$. Define $\mathcal{P} : \mathbb{V} \to \mathbb{V}$ by $\mathcal{P}(u) = w_1$. Clearly $\mathcal{P}$ is well defined. For any $w_1 \in M$, we have $w_1 \in w_1 + 0$ and so, $\mathcal{P}(w_1) = w_1$. Similarly, for any $w_2 \in N$, $w_2 \in 0 + w_2$ so, $\mathcal{P}(w_2) = 0$. Also, for $t \in w_1 + w_2$, where $w_1 \in M$ and $w_2 \in N$, we have $\mathcal{P}(t) = w_1$ and $\mathcal{P}(\mathcal{P}(t)) = \mathcal{P}(w_1) = w_1 = \mathcal{P}(t)$. Therefore, $\mathcal{P}^2 = \mathcal{P}$.
Let $\alpha \in F$ and $u, u' \in \mathbb{V}$. Let $u \in w_1 + w_2$ and $u' \in w_1' + w_2'$ for unique $w_1, w_1' \in M$ and $w_2, w_2' \in N$. Now, $u + u' \in (w_1 + w_1') + (w_2 + w_2')$. So,

$$\begin{aligned}\mathcal{P}(u + u') &\subseteq \mathcal{P}\left((w_1 + w_1') + (w_2 + w_2')\right)\\ &= w_1 + w_1'\\ &= \mathcal{P}(w_1) + \mathcal{P}(w_1')\\ &= \mathcal{P}(u) + \mathcal{P}(u').\end{aligned}$$

In fact for any $t \in (w_1 + w_1') + (w_2 + w_2')$, $t \in w + w'$ such that $w \in w_1 + w_1' \subseteq M$ and $w' \in w_2 + w_2' \subseteq N$, we get $\mathcal{P}(t) = w$ and so $\mathcal{P}(w_1 + w_1') + (w_2 + w_2') = w_1 + w_1'$.
For $\alpha \in F$, $u \in \mathbb{V}$, we have $u \in w_1 + w_2$ and so, $\mathcal{P}(\alpha u) = \alpha w_1 = \alpha \mathcal{P}(w_1)$. Therefore, $\mathcal{P}$ is a weak linear transformation and $\mathcal{P}^2 = \mathcal{P}$. □

22.7 Conclusion

We have proved the properties of subhyperspaces involving the notions such as direct sums and complements, and have given some examples and counter-examples. These results were the generalizations of respective aspects in classical vector spaces. We have established isomorphism theorems and annihilator properties of linear functionals in hypervector spaces. As future scope, one can study the hypernorms in hypervector spaces, and in similar structures. The authors Guillen and Harikrishnan [4], Harikrishnan et al. [5] have introduced various norms in different normed linear spaces, which can be generalized to hypervector spaces. Also, Joy and Thomas [7] have introduced the concept of lattice vector space; hence, we wish to extend these notions to the algebraic hyperstructures like hyperlattices as discussed by Pallavi et al. [9, 10].

Acknowledgements The authors express their deep gratitude to the referee(s)/editor(s) for their careful reading of the manuscript, and valuable suggestions that have definitely improved the paper. The first author acknowledges the Manipal Academy of Higher Education, Manipal, for providing the scholarship under Dr T M A Pai Fellowship. All the authors thank the Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, for their kind encouragement. The author Madeleine Al-Tahan acknowledges Abu Dhabi University, UAE. The first author thanks the organizers of ICLAA-2021 for providing an opportunity to present the talk.

References

1. Al-Tahan M, Davvaz B (2020) Hyper vector spaces over krasner hyperfields. J Algebr Hyperstructures Log Algebr 1(32020):61–72
2. Ameri R, Dehghan O (2008) On dimension of hypervector spaces. Eur J Pure Appl Math 1(2):32–50
3. Davvaz B, Leoreanu-Fotea V (2007) Hyperring theory and applications, vol 347. International Academic Press, USA
4. Guillen BL, Harikrishnan P (2014) Probabilistic normed spaces. World Scientific
5. Harikrishnan P, Guillen BL, Ravindran K (2011) Accretive operators and banach alaoglu theorem in linear 2-normed spaces. Proyecciones (Antofagasta, On line) 30(3):319–327
6. Hoffman K, Kunze R (1971) Linear algebra. Prentice-Hall
7. Joy G, Thomas KV (2019) Lattice vector spaces and linear transformations. Asian Eur J Math 12(02):1950031. https://doi.org/10.1142/S1793557119500311
8. Jun J (2017) Geometry of hyperfields. arXiv:1707.09348
9. Pallavi P, Kuncham S, Vadiraja G, Harikrishnan P (2022) Computation of prime hyperideals in meet hyperlattices. Bull Comput Appl Math 10(01):33–58
10. Pallavi P, Kuncham S, Vadiraja G, Harikrishnan P (2023) Prime hyperideals and twin zeros in join hyperlattices. Riv Mat Univ Parma 14(2)
11. Roy S, Samanta TK (2010) A note on hypervector spaces. arXiv:1002.3816
12. Scafati-Tallini M (2002) Characterization of remarkable hypervector space. In: Proceeding of the 8th congress on "Algebraic hyperstructures and applications". Samothraki, Greece, pp 231–237
13. Taghavi A, Hosseinzadeh R (2012) Operators on weak hypervector spaces. Rat Math 22(1):37–43

14. Viro O (2010) Hyperfields for tropical geometry I. hyperfields and dequantization. arXiv:1006.3034
15. Vougiouklis T (1994) Hyperstructures and their representations. Hadronic Press

Chapter 23
Generalized Lie Triple Derivations of Trivial Extension Algebras



Mohammad Ashraf, Mohammad Afajal Ansari, and Md Shamim Akhter

Mathematics Subject Classification (2010) 16W25 · 15A78 · 47B47

23.1 Introduction

Let $\mathcal{A}$ be an algebra (over a fixed commutative unital ring $\mathcal{R}$) with center $\mathcal{Z}(\mathcal{A})$. A linear mapping $d : \mathcal{A} \to \mathcal{A}$ is said to be a *derivation* if $d(uv) = d(u)v + ud(v)$ holds for all $u, v \in \mathcal{A}$; is said to be a *Lie derivation* if $d([u, v]) = [d(u), v] + [u, d(v)]$ holds for all $u, v \in \mathcal{A}$; is said to be a *Lie triple derivation* if $d([[u, v], w]) = [[d(u), v], w] + [[u, d(v)], w] + [[u, v], d(w)]$ holds for all $u, v, w \in \mathcal{A}$, where $[u, v] = uv - vu$ is the usual Lie product. Lie (triple) derivations on various algebras have been studied by many authors over the past decades (see [3, 4, 11, 12, 15–20, 22, 23]).

The notion of derivation can be further extended to generalized derivation. A linear mapping $G_d : \mathcal{A} \to \mathcal{A}$ is called a *generalized derivation* if there exists a derivation $d : \mathcal{A} \to \mathcal{A}$ such that $G_d(uv) = G_d(u)v + ud(v)$ for all $u, v \in \mathcal{A}$. Similarly, a linear mapping $G_d : \mathcal{A} \to \mathcal{A}$ is called a *generalized Lie triple derivation* if there exists a Lie triple derivation $d : \mathcal{A} \to \mathcal{A}$ such that $G_d([[u, v], w]) = [[G_d(u), v], w] + [[u, d(v)], w] + [[u, v], d(w)]$ for all $u, v, w \in \mathcal{A}$. A generalized Lie triple derivation is called *proper* if it can be written as the sum of a generalized derivation and a linear mapping from the algebra into its center that vanishes on all second commutators.

In the year 1991, Brešar [10] introduced the notion of a generalized derivation. Many results obtained earlier for derivations have been extended to generalized derivations (see [1, 2, 5–7, 9, 13, 14, 21] and references therein). In the year 2007, Hvala [14] introduced the notion of a generalized Lie derivation and proved that each generalized Lie derivation of a prime ring is the sum of a generalized derivation and

M. Ashraf (✉) · M. A. Ansari · M. S. Akhter
Department of Mathematics, Aligarh Muslim University, Aligarh 202002, India
e-mail: mashraf80@hotmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_23

a central map which vanishes on all commutators. Motivated by the above result, Benkovič [7] investigated the problem of characterizing generalized Lie derivations of triangular algebras. Mokhtari et al. in [20] provided some conditions under which every Lie derivation of a trivial extension algebra is proper, that is, it can be expressed as the sum of a derivation and a central map which vanishes on all commutators. Bennis et al. [9] introduced the concept of Lie generalized derivation and extended the above result in the setting of Lie generalized derivations of trivial extension algebras. The first author together with Jabeen [5] introduced and studied generalized Lie triple derivations of triangular algebras. Recently, the authors [1] gave a description of generalized Lie triple derivations of generalized matrix algebras. Of course, the class of generalized Lie triple derivations covers both the classes of Lie triple derivations and Lie generalized derivations. Therefore, studying generalized Lie triple derivations enables us to treat both important classes of Lie triple derivations and Lie generalized derivations simultaneously.

The objective of this chapter is to investigate the problem of describing the form of generalized Lie triple derivations of trivial extension algebras. More precisely, we prove that under certain appropriate conditions every generalized Lie triple derivation of a trivial extension algebra is proper. Finally, as a consequence of our main results (Theorem 23.1 and Theorem 23.2), we describe the form of generalized Lie triple derivations of triangular algebras.

23.2 Preliminaries

From now on, we assume that $\mathcal{A}$ is a unital algebra and $\mathcal{T}$ an $\mathcal{A}$-bimodule. Recall that a trivial extension algebra is a unital algebra $\mathcal{A} \times \mathcal{T}$ together with the pairwise addition, scalar multiplication, and the multiplication defined by $(u, t_1)(v, t_2) = (uv, ut_2 + t_1v)$ for all $u, v \in \mathcal{A}$, $t_1, t_2 \in \mathcal{T}$. It is denoted by $\mathcal{A} \ltimes \mathcal{T}$. The center of $\mathcal{A} \ltimes \mathcal{T}$ is given by

$$\mathcal{Z}(\mathcal{A} \ltimes \mathcal{T}) = \{(u, t) \mid u \in \mathcal{Z}(\mathcal{A}), \ \ [v, t] = 0 = [u, t'] \ \text{ for all } \ v \in \mathcal{A}, t' \in \mathcal{T}\}.$$

In addition, if $\mathcal{A}$ has a nontrivial idempotent e_1 such that $e_1te_2 = t$ for all $t \in \mathcal{T}$, where $e_2 = 1 - e_1$, then we see that

$$\mathcal{Z}(\mathcal{A} \ltimes \mathcal{T}) = \{(u, 0) \mid u \in \mathcal{Z}(\mathcal{A}), \ \ [u, t] = 0 \ \text{ for all } \ t \in \mathcal{T}\}. \tag{23.1}$$

We now state some known results, which shall be used in proving the main results of this chapter.

Lemma 23.1 ([8, Proposition 2.5]) *Let $\mathcal{A}$ be an algebra with a nontrivial idempotent e_1 and set $e_2 = 1 - e_1$. For an $\mathcal{A}$-bimodule $\mathcal{T}$, the following statements are equivalent:*

(i) *For every* $t \in \mathcal{T}$, $e_1te_2 = t$.
(ii) *For every* $t \in \mathcal{T}$, $e_2t = 0 = te_1$.
(iii) *For every* $t \in \mathcal{T}$, $e_1t = t = te_2$.
(iv) *For every* $t \in \mathcal{T}$ *and* $u \in \mathcal{A}$, $ut = e_1ue_1t$ *and* $tu = te_2ue_2$.

Note that if $\mathcal{A}$ contains a nontrivial idempotent e_1, then it can be expressed as $\mathcal{A} = e_1\mathcal{A}e_1 + e_1\mathcal{A}e_2 + e_2\mathcal{A}e_1 + e_2\mathcal{A}e_2$, where $e_1\mathcal{A}e_1$ and $e_2\mathcal{A}e_2$ are subalgebras of $\mathcal{A}$ with identity elements e_1 and e_2, respectively, $e_1\mathcal{A}e_2$ is an $(e_1\mathcal{A}e_1, e_2\mathcal{A}e_2)$-bimodule and $e_2\mathcal{A}e_1$ is an $(e_2\mathcal{A}e_2, e_1\mathcal{A}e_1)$-bimodule. Hence each element $u \in \mathcal{A}$ can be written as $u = e_1ue_1 + e_1ue_2 + e_2ue_1 + e_2ue_2$. We define two projection maps $\pi_{e_1\mathcal{A}e_1} : \mathcal{A} \ltimes \mathcal{T} \to \mathcal{A}$ and $\pi_{e_2\mathcal{A}e_2} : \mathcal{A} \ltimes \mathcal{T} \to \mathcal{A}$ by $\pi_{e_1\mathcal{A}e_1}(u, t) = e_1ue_1$ and $\pi_{e_2\mathcal{A}e_2}(u, t) = e_2ue_2$.

Lemma 23.2 ([9, Lemma 3.11]) *Let* $\mathcal{A}$ *be an algebra with a nontrivial idempotent* e_1 *such that* $e_1te_2 = t$ *for all* $t \in \mathcal{T}$. *Then* $[\mathcal{Z}(\mathcal{A}), t] = 0$ *for all* $t \in \mathcal{T}$, *if one of the following statements holds:*

(i) $\mathcal{Z}(e_1\mathcal{A}e_1) = \pi_{e_1\mathcal{A}e_1}(\mathcal{Z}(\mathcal{A} \ltimes \mathcal{T}))$ *and* $e_1\mathcal{A}e_2$ *is a faithful right* $e_2\mathcal{A}e_2$*-module.*
(ii) $\mathcal{Z}(e_2\mathcal{A}e_2) = \pi_{e_2\mathcal{A}e_2}(\mathcal{Z}(\mathcal{A} \ltimes \mathcal{T}))$ *and* $e_1\mathcal{A}e_2$ *is a faithful left* $e_1\mathcal{A}e_1$*-module.*

The following lemma can be seen in [3] and [20].

Lemma 23.3 *Let* $\delta : \mathcal{A} \ltimes \mathcal{T} \to \mathcal{A} \ltimes \mathcal{T}$ *be a linear mapping. Then* δ *is of the form*

$$\delta(u, t) = (\mathrm{f}_{\mathcal{A}}(u) + h_{\mathcal{A}}(t), \mathrm{f}_{\mathcal{T}}(u) + h_{\mathcal{T}}(t)), \tag{23.2}$$

where $\mathrm{f}_{\mathcal{A}} : \mathcal{A} \to \mathcal{A}$; $h_{\mathcal{A}} : \mathcal{T} \to \mathcal{A}$; $\mathrm{f}_{\mathcal{T}} : \mathcal{A} \to \mathcal{T}$; *and* $h_{\mathcal{T}} : \mathcal{T} \to \mathcal{T}$ *are linear mappings. Moreover,*

- δ *is a Lie triple derivation if and only if*

(i) $\mathrm{f}_{\mathcal{A}}$ *and* $\mathrm{f}_{\mathcal{T}}$ *are Lie triple derivations;*
(ii) $h_{\mathcal{A}}([u_1, u_2], t]) = [[u_1, u_2], h_{\mathcal{A}}(t)]$ *and* $h_{\mathcal{A}}([u_1, t], u_2]) = [[u_1, h_{\mathcal{A}}(t)], u_2]$ *for all* $u_1, u_2 \in \mathcal{A}$, $t \in \mathcal{T}$;
(iii) $h_{\mathcal{T}}([u_1, u_2], t]) = [[\mathrm{f}_{\mathcal{A}}(u_1), u_2], t] + [[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], t] + [[u_1, u_2], h_{\mathcal{T}}(t)]$ *and* $h_{\mathcal{T}}([u_1, t], u_2]) = [[\mathrm{f}_{\mathcal{A}}(u_1), t], u_2] + [[u_1, h_{\mathcal{T}}(t)], u_2] + [[u_1, t], \mathrm{f}_{\mathcal{A}}(u_2)]$ *for all* $u_1, u_2 \in \mathcal{A}$, $m \in \mathcal{T}$;
(iv) $[[u, h_{\mathcal{A}}(t_1)], t_2] + [[u, t_1], h_{\mathcal{A}}(t_2)] = 0$ *and* $[[h_{\mathcal{A}}(t_1), t_2] + [t_1, h_{\mathcal{A}}(t_2)], u] = 0$ *for all* $u \in \mathcal{A}$ *and* $t_1, t_2 \in \mathcal{T}$.

- δ *is a derivation if and only if*

(a) $\mathrm{f}_{\mathcal{A}}$ *and* $\mathrm{f}_{\mathcal{T}}$ *are derivations;*
(b) $h_{\mathcal{A}}(ut) = uh_{\mathcal{A}}(t)$ *and* $h_{\mathcal{A}}(tu) = h_{\mathcal{A}}(t)u$ *for all* $u \in \mathcal{A}$ *and* $t \in \mathcal{T}$;
(c) $h_{\mathcal{T}}(ut) = \mathrm{f}_{\mathcal{A}}(u)t + uh_{\mathcal{T}}(t)$ *and* $h_{\mathcal{T}}(tu) = h_{\mathcal{T}}(t)u + t\mathrm{f}_{\mathcal{A}}(u)$ *for all* $u \in \mathcal{A}$ *and* $t \in \mathcal{T}$;
(d) $h_{\mathcal{A}}(t_1)t_2 + t_1h_{\mathcal{A}}(t_2) = 0$ *for all* $t_1, t_2 \in \mathcal{T}$.

Lemma 23.4 *Suppose that there exists a nontrivial idempotent e_1 in $\mathcal{A}$ such that $e_1te_2 = t$ for all $t \in \mathcal{T}$. Then the following statements hold:*

(i) If $\Delta : \mathcal{A} \to \mathcal{A}$ is a generalized derivation, then $[\Delta(e_1ue_2) + \Delta(e_2ue_1), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.

(ii) If $F_{\mathcal{A}} : \mathcal{A} \to \mathcal{A}$ is a generalized Lie triple derivation, then $[F_{\mathcal{A}}(e_1ue_2), t] = 0$ and $[F_{\mathcal{A}}(e_2ue_1), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.

Proof (i) Assume that Δ is a generalized derivation with associated derivation d. Then $td(e_1) = 0 = d(e_2)t$ for all $t \in \mathcal{T}$. Indeed, using Lemma 23.1, $td(e_1) = t(d(e_1)e_1 + e_1d(e_1)) = 0$. Similarly, we can prove that $d(e_2)t = 0$. Since d is a derivation, using Lemma 23.1, we have

$$\begin{aligned}[d(e_1ue_2) + d(e_2ue_1), t] &= [d(e_1)e_1ue_2 + e_1d(e_1ue_2) + d(e_2)e_2ue_1 + e_2d(e_2ue_1), t] \\ &= ud(e_2)t - tud(e_1) = 0.\end{aligned}$$

Similarly, $[\Delta(e_1ue_2) + \Delta(e_2ue_1), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.
(ii) Assume that $F_{\mathcal{A}}$ is a generalized Lie triple derivation with associated Lie triple derivation $\mathrm{f}_{\mathcal{A}}$. Then for all $a \in \mathcal{A}$, we see that

$$\begin{aligned}F_{\mathcal{A}}(e_1ue_2) &= F_{\mathcal{A}}([[e_1, e_1ue_2], e_2]) \\ &= [[F_{\mathcal{A}}(e_1), e_1ue_2], e_2] + [[e_1, \mathrm{f}_{\mathcal{A}}(e_1ue_2)], e_2] + [[e_1, e_1ue_2], \mathrm{f}_{\mathcal{A}}(e_2)] \\ &= [\mathrm{f}_{\mathcal{A}}(e_1)e_1ue_2 - e_1ue_2\mathrm{f}_{\mathcal{A}}(e_1), e_2] + [e_1\mathrm{f}_{\mathcal{A}}(e_1ue_2) - \mathrm{f}_{\mathcal{A}}(e_1ue_2)e_1, e_2] \\ &\quad + [e_1ue_2, \mathrm{f}_{\mathcal{A}}(e_2)] \\ &= \mathrm{f}_{\mathcal{A}}(e_1)e_1ue_2 - e_1ue_2\mathrm{f}_{\mathcal{A}}(e_1)e_2 - e_2\mathrm{f}_{\mathcal{A}}(e_1)e_1ue_2 + e_1\mathrm{f}_{\mathcal{A}}(e_1ue_2)e_2 \\ &\quad + e_2\mathrm{f}_{\mathcal{A}}(e_1ue_2)e_1 + e_1ue_2\mathrm{f}_{\mathcal{A}}(e_2) - \mathrm{f}_{\mathcal{A}}(e_2)e_1ue_2.\end{aligned}$$

Hence, using Lemma 23.1, we have $[F_{\mathcal{A}}(e_1ue_2), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$. Similarly, $[F_{\mathcal{A}}(e_2ue_1), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$. □

23.3 Generalized Lie Triple Derivations

Inspired by Brešar's idea, Bennis et al. [9] introduced the concept of a Lie generalized derivation as follows: A linear mapping $G_d : \mathcal{A} \to \mathcal{A}$ is called a *Lie generalized derivation* if there exists a Lie derivation $d : \mathcal{A} \to \mathcal{A}$ such that $G_d([u, v]) = [G_d(u), v] + [u, d(v)])$ for all $u, v \in \mathcal{A}$. Further, they showed that under certain restrictions every Lie generalized derivation of a trivial extension algebra is the sum of a generalized derivation and a central map which vanishes on all commutators. It is not difficult to verify that a Lie generalized derivation is a generalized Lie triple derivation but the converse is not true in general. For example, if $\delta : \mathcal{A} \to \mathcal{A}$ is a Lie triple derivation but not necessarily a Lie derivation (note that such map exists, see [22, Counterexample 3.5]), then the map $G_\delta(x) = \lambda x + \delta(x)$ for all $x \in \mathcal{A}$, where

$\lambda \in \mathcal{Z}(\mathcal{A})$, is a generalized Lie triple derivation but not necessarily a Lie generalized derivation.

In this section, we extend the main results of [9] to the setting of generalized Lie triple derivations of trivial extension algebras. Let us begin our investigation with the following lemma which gives the structures of generalized derivations and generalized Lie triple derivations of trivial extension algebras.

Lemma 23.5 *Let $G_\delta : \mathcal{A} \ltimes \mathcal{T} \to \mathcal{A} \ltimes \mathcal{T}$ be a linear mapping. Then G_δ is of the form*

$$G_\delta(u, t) = (F_{\mathcal{A}}(u) + H_{\mathcal{A}}(t), F_{\mathcal{T}}(u) + H_{\mathcal{T}}(t)), \tag{23.3}$$

where $F_{\mathcal{A}} : \mathcal{A} \to \mathcal{A}$; $H_{\mathcal{A}} : \mathcal{T} \to \mathcal{A}$; $F_{\mathcal{T}} : \mathcal{A} \to \mathcal{T}$ and $H_{\mathcal{T}} : \mathcal{T} \to \mathcal{T}$ are linear mappings. Moreover,

- *G_δ is a generalized Lie triple derivation with associated Lie triple derivation δ (as given in Lemma 23.3) if and only if*

  (i) *$F_{\mathcal{A}}$, $F_{\mathcal{T}}$ are generalized Lie triple derivations with associated Lie triple derivations $\mathrm{f}_{\mathcal{A}}$, $\mathrm{f}_{\mathcal{T}}$, respectively;*

  (ii) $H_{\mathcal{A}}([u_1, u_2], t_3]) = [[u_1, u_2], h_{\mathcal{A}}(t_3)]$, $H_{\mathcal{A}}([u_1, t_2], u_3]) = [[u_1, h_{\mathcal{A}}(t_2)], u_3]$ *and* $H_{\mathcal{A}}([t_1, u_2], u_3]) = [[H_{\mathcal{A}}(t_1), u_2], u_3]$ *for all* $u_1, u_2, u_3 \in \mathcal{A}$ *and* $t_1, t_2, t_3 \in \mathcal{T}$;

  (iii) $H_{\mathcal{T}}([u_1, u_2], t_3]) = [[F_{\mathcal{A}}(u_1), u_2], t_3] + [[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], t_3] + [[u_1, u_2], h_{\mathcal{T}}(t_3)]$,
  $H_{\mathcal{T}}([u_1, t_2], u_3]) = [[F_{\mathcal{A}}(u_1), t_2], u_3] + [[u_1, h_{\mathcal{T}}(t_2)], u_3] + [[u_1, t_2], \mathrm{f}_{\mathcal{A}}(u_3)]$ *and*
  $\mathcal{T}([t_1, u_2], u_3]) = [[H_{\mathcal{T}}(t_1), u_2], u_3] + [[t_1, \mathrm{f}_{\mathcal{A}}(u_2)], u_3] + [[t_1, u_2], \mathrm{f}_{\mathcal{A}}(u_3)]$ *for all* $u_1, u_2, u_3 \in \mathcal{A}$ *and* $t_1, t_2, t_3 \in \mathcal{T}$;

  (iv) $[[u_1, h_{\mathcal{A}}(t_2)], t_3] + [[u_1, t_2], h_{\mathcal{A}}(t_3)] = 0$, $[[H_{\mathcal{A}}(t_1), u_2], t_3] + [[t_1, u_2], h_{\mathcal{A}}(t_3)]$
  $= 0$ *and* $[[H_{\mathcal{A}}(t_1), t_2], u_3] + [[t_1, h_{\mathcal{A}}(t_2)], u_3] = 0$ *for all* $u_1, u_2, u_3 \in \mathcal{A}$ *and* $t_1, t_2, t_3 \in \mathcal{T}$.

- *G_δ is a generalized derivation with associated derivation δ (as given in Lemma 23.3) if and only if*

  (a) *$F_{\mathcal{A}}$, $F_{\mathcal{T}}$ are generalized derivations with associated derivations $\mathrm{f}_{\mathcal{A}}$, $\mathrm{f}_{\mathcal{T}}$, respectively;*

  (b) $H_{\mathcal{A}}(ut) = uh_{\mathcal{A}}(t)$ *and* $H_{\mathcal{A}}(tu) = H_{\mathcal{A}}(t)u$ *for all* $u \in \mathcal{A}$ *and* $t \in \mathcal{T}$;

  (c) $H_{\mathcal{T}}(ut) = F_{\mathcal{A}}(u)t + uh_{\mathcal{T}}(t)$ *and* $H_{\mathcal{T}}(tu) = H_{\mathcal{T}}(t)u + t\mathrm{f}_{\mathcal{A}}(u)$ *for all* $u \in \mathcal{A}$ *and* $t \in \mathcal{T}$;

  (d) $H_{\mathcal{A}}(t_1)t_2 + t_1h_{\mathcal{A}}(t_2) = 0$ *for all* $t_1, t_2 \in \mathcal{T}$.

Proof We only prove the assertion for generalized Lie triple derivation. The second assertion can be proved in a similar way. Let G_δ be a generalized Lie triple derivation with associated Lie triple derivation δ (as given in Lemma 23.3). Then

$$G_\delta([[T_1, T_2], T_3]) = [[G_\delta(T_1), T_2], T_3] + [[T_1, \delta(T_2)], T_3] + [[T_1, T_2], \delta(T_3)] \tag{23.4}$$

for all $T_1, T_2, T_3 \in \mathcal{A} \ltimes \mathcal{T}$.

Taking $T_1 = (u_1, 0)$, $T_2 = (u_2, 0)$ and $T_3 = (u_3, 0)$, we have

$$G_\delta([[T_1, T_2], T_3]) = (F_{\mathcal{A}}([u_1, u_2], u_3]), F_{\mathcal{T}}([u_1, u_2], u_3]))$$

and

$$\begin{aligned}
&[[G_\delta(T_1), T_2], T_3] + [[T_1, \delta(T_2)], T_3] + [[T_1, T_2], \delta(T_3)] \\
&= ([[F_{\mathcal{A}}(u_1), u_2], u_3], [[F_{\mathcal{T}}(u_1), u_2], u_3]) + ([[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], u_3], [[u_1, \mathrm{f}_{\mathcal{T}}(u_2)], u_3]) \\
&\quad +([[u_1, u_2], f_{\mathcal{A}}(u_3)], [[u_1, u_2], \mathrm{f}_{\mathcal{T}}(u_3)]) \\
&= ([[F_{\mathcal{A}}(u_1), u_2], u_3] + [[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], u_3] + [[u_1, u_2], \mathrm{f}_{\mathcal{A}}(u_3)], \\
&\quad [[F_{\mathcal{T}}(u_1), u_2], u_3]) + [[u_1, \mathrm{f}_{\mathcal{T}}(u_2)], u_3] + [[u_1, u_2], \mathrm{f}_{\mathcal{T}}(u_3)]).
\end{aligned}$$

Hence, it follows from (23.4) that $F_{\mathcal{A}}$, $F_{\mathcal{T}}$ are generalized Lie triple derivations with associated Lie triple derivations $\mathrm{f}_{\mathcal{A}}$, $\mathrm{f}_{\mathcal{T}}$, respectively.

Choosing $T_1 = (u_1, 0)$, $T_2 = (u_2, 0)$ and $T_3 = (0, t_3)$, we get

$$G_\delta([[T_1, T_2], T_3]) = (H_{\mathcal{A}}([u_1, u_2], t_3]), H_{\mathcal{T}}([u_1, u_2], t_3]))$$

and

$$\begin{aligned}
&[[G_\delta(T_1), T_2], T_3] + [[T_1, \delta(T_2)], T_3] + [[T_1, T_2], \delta(T_3)] \\
&= ([[u_1, u_2], h_{\mathcal{A}}(t_3)], [[F_{\mathcal{A}}(u_1), u_2], t_3] + [[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], t_3] + [[u_1, u_2], h_{\mathcal{T}}(t_3)]).
\end{aligned}$$

Thus, using (23.4), we get

$$H_{\mathcal{A}}([u_1, u_2], t_3]) = [[u_1, u_2], h_{\mathcal{A}}(t_3)];$$

$$H_{\mathcal{T}}([u_1, u_2], t_3]) = [[F_{\mathcal{A}}(u_1), u_2], t_3] + [[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], t_3] + [[u_1, u_2], h_{\mathcal{T}}(t_3)].$$

Similarly, we can obtain

$$H_{\mathcal{A}}([u_1, t_2], u_3]) = [[u_1, h_{\mathcal{A}}(t_2)], u_3];$$

$$H_{\mathcal{T}}([u_1, t_2], u_3]) = [[F_{\mathcal{A}}(u_1), t_2], u_3] + [[u_1, h_{\mathcal{T}}(t_2)], u_3] + [[u_1, t_2], \mathrm{f}_{\mathcal{A}}(u_3)];$$

$$H_{\mathcal{A}}([t_1, u_2], u_3]) = [[H_{\mathcal{A}}(t_1), u_2], u_3];$$

$$H_{\mathcal{T}}([t_1, u_2], u_3]) = [[H_{\mathcal{T}}(t_1), u_2], u_3] + [[u_1, \mathrm{f}_{\mathcal{A}}(u_2)], u_3] + [[t_1, u_2], \mathrm{f}_{\mathcal{A}}(u_3)].$$

Considering $T_1 = (u_1, 0)$, $T_2 = (0, t_2)$ and $T_3 = (0, t_3)$ in (23.4), we obtain

$$(0, 0) = (0, [[u_1, h_{\mathcal{A}}(t_2)], t_3] + [[u_1, t_2], h_{\mathcal{A}}(t_3)]).$$

Therefore, we have

$$[[u_1, h_{\mathcal{A}}(t_2)], t_3] + [[u_1, t_2], h_{\mathcal{A}}(t_3)] = 0.$$

Similarly, one can obtain

$$[[H_{\mathcal{A}}(t_1), (u_2)], t_3] + [[t_1, u_2], h_{\mathcal{A}}(t_3)] = 0;$$

$$[[H_{\mathcal{A}}(t_1), t_2], u_3] + [[t_1, h_{\mathcal{A}}(t_2)], u_3] = 0.$$

Conversely, suppose that G_δ is a linear mapping of the form (23.3) satisfying conditions $(i) - (iv)$. Then it is easy to see that G_δ is a generalized Lie triple derivation of $\mathcal{A} \ltimes \mathcal{T}$ with associated Lie triple derivation δ. □

The first main result of the chapter states as follows:

Theorem 23.1 *Let $\mathcal{A} \ltimes \mathcal{T}$ be a trivial extension algebra. Suppose there exists a nontrivial idempotent e_1 in $\mathcal{A}$ such that $e_1 t e_2 = t$ for all $t \in \mathcal{T}$. Then every generalized Lie triple derivation G_δ of the form (23.3) is proper if and only if the following statements hold:*

1. *There exists a linear mapping $\ell_{\mathcal{A}} : \mathcal{A} \to \mathcal{Z}(\mathcal{A})$ such that*
   - *(i) $F_{\mathcal{A}} - \ell_{\mathcal{A}}$ is a generalized derivation,*
   - *(ii) $[\ell_{\mathcal{A}}(e_1 u e_1), t] = 0 = [\ell_{\mathcal{A}}(e_2 u e_2), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.*
2. *$F_{\mathcal{T}}(e_2 u e_1) = 0$ and $e_2 H_{\mathcal{A}}(t) e_1 = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.*

Proof Suppose that G_δ is a generalized Lie triple derivation of the form (23.3) and there exists a linear mapping $\ell_{\mathcal{A}} : \mathcal{A} \to \mathcal{Z}(\mathcal{A})$ such that the statements (1) and (2) hold.

Let us define two mappings $\Delta, \ell : \mathcal{A} \ltimes \mathcal{T} \to \mathcal{A} \ltimes \mathcal{T}$ by

$$\Delta(u, t) = ((F_{\mathcal{A}} - \ell_{\mathcal{A}})(u) + H_{\mathcal{A}}(t), F_{\mathcal{T}}(u) + H_{\mathcal{T}}(t)) \text{ and } \ell(u, t) = (\ell_{\mathcal{A}}(u), 0).$$

Then, Δ and ℓ are linear mappings and $G_\delta = \Delta + \ell$. Moreover, it follows from (23.1) and $\ell_{\mathcal{A}}(\mathcal{A}) \subseteq \mathcal{Z}(\mathcal{A})$ that $\ell(\mathcal{A} \ltimes \mathcal{T}) \subseteq \mathcal{Z}(\mathcal{A} \ltimes \mathcal{T})$. Now, we prove Δ is a generalized derivation through the following claims.

Claim 1. $H_{\mathcal{A}}(ut) = u h_{\mathcal{A}}(t)$, $H_{\mathcal{A}}(tu) = H_{\mathcal{A}}(t)u$ and $H_{\mathcal{A}}(t_1)t_2 + t_1 h_{\mathcal{A}}(t_2) = 0$.

Using the assumption (2), we get $H_{\mathcal{A}}(t) = H_{\mathcal{A}}([[t, e_2], e_2]) = [[H_{\mathcal{A}}(t), e_2], e_2] = e_1 H_{\mathcal{A}}(t) e_2$ for all $t \in \mathcal{T}$. Then $e_1 H_{\mathcal{A}}(t) e_1 = 0 = e_2 H_{\mathcal{A}}(t) e_2$. Moreover, $H_{\mathcal{A}}(t) = H_{\mathcal{A}}([[e_1, t], e_2]) = [[e_1, h_{\mathcal{A}}(t)], e_2] = e_1 h_{\mathcal{A}}(t) e_2 + e_2 h_{\mathcal{A}}(t) e_1$. Again, using the fact that $e_2 H_{\mathcal{A}}(t) e_1 = 0$, we get $e_2 h_{\mathcal{A}}(t) e_1 = 0$ for all $t \in \mathcal{T}$. Thus, $H_{\mathcal{A}}(t) = e_1 h_{\mathcal{A}}(t) e_2$ for all $t \in \mathcal{T}$. Now, we have $H_{\mathcal{A}}(ut) = e_1 h_{\mathcal{A}}(ut) e_2 = u h_{\mathcal{A}}(t)$. Similarly, we can show that $H_{\mathcal{A}}(tu) = H_{\mathcal{A}}(t)u$. Using Lemma 23.1, we get $H_{\mathcal{A}}(t_1)t_2$

$+ t_1 h_{\mathcal{A}}(t_2) = e_1 H_{\mathcal{A}}(t_1) e_1 t_2 + t e_2 h_{\mathcal{A}}(t_2) e_2 = 0.$

Claim 2. $F_{\mathcal{T}}$ is a generalized derivation.

Let us define a linear mapping $\mathcal{T}_{\mathcal{T}}$ as follows: $\mathcal{T}_{\mathcal{T}}(u) = F_{\mathcal{T}}(e_1 u e_1) + F_{\mathcal{T}}(e_2 u e_2)$ for all $u \in \mathcal{A}$. We assert that $\mathcal{T}_{\mathcal{T}}$ is an inner derivation. Since $F_{\mathcal{T}}$ is a generalized Lie triple derivation, we get $0 = F_{\mathcal{T}}([[e_1 u e_1, e_2 u e_2], e_2]) = F_{\mathcal{T}}(e_1 u e_1)u + u\mathrm{f}_{\mathcal{T}}(e_2 u e_2) = -u F_{\mathcal{T}}(e_2 u e_2) - \mathrm{f}_{\mathcal{T}}(e_1 u e_1)u$. Taking $u = e_2 u e_2 + e_1$ in the above equation, we get $F_{\mathcal{T}}(e_2 u e_2) = \mathrm{f}_{\mathcal{T}}(e_2)u$. Similarly if we take $u = e_1 u e_1 + e_2$, we get $F_{\mathcal{T}}(e_1 u e_1) = -u\mathrm{f}_{\mathcal{T}}(e_2)$. Hence $\mathcal{T}_{\mathcal{T}}(u) = -u\mathrm{f}_{\mathcal{T}}(e_2) + \mathrm{f}_{\mathcal{T}}(e_2)u$. We now show that $\mathrm{g}_{\mathcal{T}}$ defined by $a \mapsto F_{\mathcal{T}}(e_1 u e_2)$ is a generalized derivation. We have

$$\begin{aligned}
\mathrm{g}_{\mathcal{T}}(uv) &= F_{\mathcal{T}}(e_1 uv e_2) \\
&= F_{\mathcal{T}}(e_1 u e_1 v e_2) + F_{\mathcal{T}}(e_1 u e_2 v e_2) \\
&= F_{\mathcal{T}}([[e_1 u, e_1 v e_2], e_2]) + F_{\mathcal{T}}([[e_1 u, e_2 v e_2], e_2]) \\
&= [[F_{\mathcal{T}}(e_1 u), e_1 v e_2], e_2] + [[e_1 u, \mathrm{f}_{\mathcal{T}}(e_1 v e_2)], e_2] + [[e_1 u, e_1 v e_2], \mathrm{f}_{\mathcal{T}}(e_2)] \\
&\quad + [[F_{\mathcal{T}}(e_1 u), e_2 v e_2], e_2] + [[e_1 u, \mathrm{f}_{\mathcal{T}}(e_2 v e_2)], e_2] + [[e_1 u, e_2 v e_2], \mathrm{f}_{\mathcal{T}}(e_2)] \\
&= ea\mathrm{f}_{\mathcal{T}}(e_1 v e_2) + F_{\mathcal{T}}(e_1 u) e_2 v e_2 + e_1 u \mathrm{f}_{\mathcal{T}}(e_2 v e_2) \\
&= u\mathrm{f}_{\mathcal{T}}(e_1 v e_2) + (F_{\mathcal{T}}(e_1 u e_1) + F_{\mathcal{T}}(e_1 u e_2))v + u\mathrm{f}_{\mathcal{T}}(e_2 v e_2) \\
&= F_{\mathcal{T}}(e_1 u e_2)v + u(\mathrm{f}_{\mathcal{T}}(e_1 v e_2) - \mathrm{f}_{\mathcal{T}}(e_2)v + \mathrm{f}_{\mathcal{T}}(e_2 v e_2)) \\
&= \mathrm{g}_{\mathcal{T}}(u)v + u(\mathrm{f}_{\mathcal{T}}(v e_2) - \mathrm{f}_{\mathcal{T}}(e_2)v).
\end{aligned}$$

Since $F_{\mathcal{T}}(u) = \mathcal{T}_{\mathcal{T}}(u) + \mathrm{g}_{\mathcal{T}}(u)$, $F_{\mathcal{T}}$ is a generalized derivation.

Claim 3. $H_{\mathcal{T}}(ut) = (\mathrm{F}_{\mathcal{A}} - \ell_{\mathcal{A}})(u)t + uh_{\mathcal{T}}(t)$ and $H_{\mathcal{T}}(tu) = H_{\mathcal{T}}(t)u + t(\mathrm{f}_{\mathcal{A}} - \ell_{\mathcal{A}})(u)$.

Using Lemma 23.1, Lemma 23.5 and the fact that $F_{\mathcal{A}} - \ell_{\mathcal{A}}$ is a generalized derivation with associated derivation $\mathrm{f}_{\mathcal{A}} - \ell_{\mathcal{A}}$, we see that

$$\begin{aligned}
H_{\mathcal{T}}(ut) &= H_{\mathcal{T}}([[e_1 u e_1, t], e_2]) \\
&= [[F_{\mathcal{A}}(e_1 u e_1), t], e_2] + [[e_1 u e_1, h_{\mathcal{T}}(t)], e_2] + [[e_1 u e_1, t], \mathrm{f}_{\mathcal{A}}(e_2)] \\
&= [F_{\mathcal{A}}(e_1 u e_1), t] + [e_1 u e_1, h_{\mathcal{T}}(t)] + [ut, \mathrm{f}_{\mathcal{A}}(e_2)] \\
&= [(F_{\mathcal{A}} - \ell_{\mathcal{A}})(e_1 u e_1), t] + [\ell_{\mathcal{A}}(e_1 u e_1), t] + [e_1 u e_1, h_{\mathcal{T}}(t)] \\
&= (F_{\mathcal{A}} - \ell_{\mathcal{A}})(u)t + [\ell_{\mathcal{A}}(e_1 u e_1), t] + uh_{\mathcal{T}}(t). \qquad (23.5)
\end{aligned}$$

Using the hypothesis (2), we get $H_{\mathcal{T}}(ut) = (F_{\mathcal{A}} - \ell_{\mathcal{A}})(u)t + uh_{\mathcal{T}}(t)$. Similarly, one can prove that $H_{\mathcal{T}}(tu) = H_{\mathcal{T}}(t)u + t(\mathrm{f}_{\mathcal{A}} - \ell_{\mathcal{A}})(u)$. Thus, using Lemma 23.5 and Claims 1–3, we conclude that Δ is a generalized derivation.

Conversely, suppose that G_δ is proper, that is, $G_\delta = \Delta + \ell$, where Δ is a generalized derivation and ℓ is a linear mapping from $\mathcal{A} \ltimes \mathcal{T}$ into its center. Then, it follows from (23.1) that $\ell(u, t) = (\ell_{\mathcal{A}}(u), 0)$ $((u, t) \in \mathcal{A} \ltimes \mathcal{T})$ for some linear map-

ping $\ell_{\mathcal{A}} : \mathcal{A} \to \mathcal{Z}(\mathcal{A})$ with $[\ell_{\mathcal{A}}(u), t] = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$. On the other hand, $G_\delta - \ell = \Delta$ is a generalized derivation. Hence, by Lemma 23.5, $F_{\mathcal{A}} - \ell_{\mathcal{A}}$, $F_{\mathcal{T}}$ are generalized derivations and $H_{\mathcal{A}}$ satisfies $H_{\mathcal{A}}(tu) = H_{\mathcal{A}}(t)u$ for all $u \in \mathcal{A}$ and $m \in \mathcal{T}$. Therefore, using Lemma 23.1, one can easily prove that $F_{\mathcal{T}}(e_2 u e_1) = 0$ and $e_2 H_{\mathcal{A}}(t) e_1 = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$. □

Applying Theorem 23.1, we now obtain the second main result of the chapter. Let $\mathcal{W}_{\mathcal{A}}$ denote the smallest subalgebra of $\mathcal{A}$ containing all idempotents and second commutators.

Theorem 23.2 *Assume that $\mathcal{A} \ltimes \mathcal{T}$ is a trivial extension algebra and there exists a nontrivial idempotent e_1 in $\mathcal{A}$ such that $e_1 t e_2 = t$ for all $t \in \mathcal{T}$. Then every generalized Lie triple derivation G_δ of the form (23.3) is proper if the following statements hold:*

1. *Every generalized Lie triple derivation of $\mathcal{A}$ is proper.*
2. *$F_{\mathcal{T}}(e_2 u e_1) = 0$ and $e_2 H_{\mathcal{A}}(t) e_1 = 0$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.*
3. *One of the following three conditions holds:*

   (i) $\mathcal{W}_{e_1\mathcal{A}e_1} = e_1\mathcal{A}e_1$ *and* $\mathcal{W}_{e_2\mathcal{A}e_2} = e_2\mathcal{A}e_2$.
   (ii) $\mathcal{Z}(e_1\mathcal{A}e_1) = \pi_{e_1\mathcal{A}e_1}(\mathcal{Z}(\mathcal{A} \ltimes \mathcal{T}))$ *and $e_1\mathcal{A}e_2$ is a faithful right $e_2\mathcal{A}e_2$-module.*
   (iii) $\mathcal{Z}(e_2\mathcal{A}e_2) = \pi_{e_2\mathcal{A}e_2}(\mathcal{Z}(\mathcal{A} \ltimes \mathcal{T}))$ *and $e_1\mathcal{A}e_2$ is a faithful left $e_1\mathcal{A}e_1$-module.*

Proof Let G_δ be a generalized Lie triple derivation of the form (23.3). Since $F_{\mathcal{A}}$ is a generalized Lie triple derivation on $\mathcal{A}$, by hypothesis (1), there exists a linear mapping $\ell_{\mathcal{A}} : \mathcal{A} \to \mathcal{Z}(\mathcal{A})$ vanishing on all second commutators of $\mathcal{A}$ such that $F_{\mathcal{A}} - \ell_{\mathcal{A}}$ is generalized derivation. It is sufficient to show that $\ell_{\mathcal{A}}$ satisfies the condition $(1)(ii)$ of Theorem 23.1, that is, $[\ell_{\mathcal{A}}(e_1 u e_1), t] = 0 = [\ell_{\mathcal{A}}(e_2 u e_2), t]$ for all $u \in \mathcal{A}$ and $t \in \mathcal{T}$.

Set $\mathcal{A}' = \{e_1 u e_1 \in e_1\mathcal{A}e_1 \mid [\ell_{\mathcal{A}}(e_1 u e_1), t] = 0 \text{ for all } t \in \mathcal{T}\}$. First, we prove that $\mathcal{A}'$ is subalgebra of $e_1\mathcal{A}e_1$. Note that $\mathcal{A}'$ is $\mathcal{R}$-submodule of $e_1\mathcal{A}e_1$, since $\ell_{\mathcal{A}}$ is an $\mathcal{R}$-linear mapping. To prove $\mathcal{A}'$ is closed under multiplication, we establish the following identity:

$$[\ell_{\mathcal{A}}(e_1 u e_1 v e_1), t] = [\ell_{\mathcal{A}}(e_1 u e_1), vt] + [\ell_{\mathcal{A}}(e_1 v e_1), ut] \tag{23.6}$$

for all $u, v \in \mathcal{A}, t \in \mathcal{T}$. From the expression (23.5), we have

$$H_{\mathcal{T}}(ut) = (F_{\mathcal{A}} - \ell_{\mathcal{A}})(u)t + [\ell_{\mathcal{A}}(e_1 u e_1), t] + u h_{\mathcal{T}}(t). \tag{23.7}$$

Replacing u by uv in (23.7), we get

$$H_{\mathcal{T}}(uvt) = (F_{\mathcal{A}} - \ell_{\mathcal{A}})(uv)t + [\ell_{\mathcal{A}}(e_1 uve_1), t] + uvh_{\mathcal{T}}(t). \tag{23.8}$$

Using the fact that $F_{\mathcal{A}} - \ell_{\mathcal{A}}$ is a generalized derivation with associated derivation $\mathrm{f}_{\mathcal{A}} - \ell_{\mathcal{A}}$ and Eq. (23.7), we have

$$\begin{aligned} H_{\mathcal{T}}(uvt) &= (F_{\mathcal{A}} - \ell_{\mathcal{A}})(u)vt + [\ell_{\mathcal{A}}(e_1ue_1), vt] + uh_{\mathcal{T}}(vt) \\ &= (F_{\mathcal{A}} - \ell_{\mathcal{A}})(u)vt + [\ell_{\mathcal{A}}(e_1ue_1), vt] + u(\mathrm{f}_{\mathcal{A}} - \ell_{\mathcal{A}})(v)t + [\ell_{\mathcal{A}}(e_1ve_1), ut] \\ &\quad + uvh_{\mathcal{T}}(t) \\ &= (F_{\mathcal{A}} - \ell_{\mathcal{A}})(uv)t + [\ell_{\mathcal{A}}(e_1ue_1), vt] + [\ell_{\mathcal{A}}(e_1ve_1), ut] + uvh_{\mathcal{T}}(t). \end{aligned} \tag{23.9}$$

Comparing (23.8) and (23.9), we obtain

$$[\ell_{\mathcal{A}}(e_1uve_1), t] = [\ell_{\mathcal{A}}(e_1ue_1), vt] + [\ell_{\mathcal{A}}(e_1ve_1), ut] \quad \text{for all} \quad u, v \in \mathcal{A}, t \in \mathcal{T},$$

which yields the desired expression (23.6). Hence $\mathcal{A}'$ is subalgebra of $e_1\mathcal{A}e_1$. Taking $u = v$ in (23.6), we get

$$[\ell_{\mathcal{A}}((e_1ue_1)^2), t] = [\ell_{\mathcal{A}}(e_1ue_1), 2ut] \ \ (u \in \mathcal{A}, t \in \mathcal{T}), \tag{23.10}$$

which implies that

$$[\ell_{\mathcal{A}}((e_1ue_1)^3), t] = [\ell_{\mathcal{A}}((e_1ue_1)^2(e_1ue_1)), ut] = [\ell_{\mathcal{A}}(e_1ue_1), 3u^2t]. \tag{23.11}$$

Suppose that $e_1ue_1 \in e_1\mathcal{A}e_1$ is an idempotent; that is, $(e_1ue_1)^2 = e_1ue_1$. By (23.10) and (23.11), we arrive at $[\ell_{\mathcal{A}}(e_1ue_1), t] = [\ell_{\mathcal{A}}(3(e_1ue_1)^2 - 2(e_1ue_1)^3), t] = 0$, which implies that $e_1ue_1 \in \mathcal{A}'$.

Moreover, $\mathcal{A}'$ contains all second commutators, since $\ell_{\mathcal{A}}$ vanishes on all second commutators. Now, if $\mathcal{W}_{e_1\mathcal{A}e_1} = e_1\mathcal{A}e_1$, then $\mathcal{A}' = e_1\mathcal{A}e_1$, that is, $[\ell_{\mathcal{A}}(e_1ue_1), t] = 0$ for every $u \in \mathcal{A}, t \in \mathcal{T}$. In a similar fashion, we can show that if $\mathcal{W}_{e_2\mathcal{A}e_2} = e_2\mathcal{A}e_2$, then $[\ell_{\mathcal{A}}(e_2ue_2), t] = 0$ for all $u \in \mathcal{A}, t \in \mathcal{T}$. Furthermore, in view of Lemma 23.2, if either $(3)(ii)$ or $(3)(iii)$ holds, then $[\mathcal{Z}(\mathcal{A}), t] = 0$ for all $t \in \mathcal{T}$, as required. □

23.4 Application to Triangular Algebras

Let $\mathcal{U}, \mathcal{V}$ be unital algebras and $\mathcal{T}$ a nonzero $(\mathcal{U}, \mathcal{V})$-bimodule. An algebra

$$\mathcal{A} = \mathrm{Tri}(\mathcal{U}, \mathcal{T}, \mathcal{V}) = \left\{ \begin{pmatrix} u & t \\ 0 & v \end{pmatrix} \mid u \in \mathcal{U}, t \in \mathcal{T}, v \in \mathcal{V} \right\}$$

under the usual matrix-like operations is called a *triangular algebra*. Note that every triangular algebra $\mathrm{Tri}(\mathcal{U}, \mathcal{T}, \mathcal{V})$ can be identified with the trivial extension algebra $(\mathcal{U} \oplus \mathcal{V}) \ltimes \mathcal{T}$ (see [3] for details).

Furthermore, one can easily see that every generalized Lie triple derivation of $(\mathcal{U} \oplus \mathcal{V}) \ltimes \mathcal{T}$ has the form

$$G_\delta((u, v), t) = (F_{\mathcal{U}\oplus\mathcal{V}}(u, v), F_{\mathcal{T}}(u, v) + H_{\mathcal{T}}(t)).$$

That is, $H_{\mathcal{U}\oplus\mathcal{V}}(t) = 0$ for all $t \in \mathcal{T}$. Therefore, as consequences of Theorem 23.1 and Theorem 23.2, we have the following results:

Corollary 23.1 *Let G_δ be a generalized Lie triple derivation of a triangular algebra* $\mathrm{Tri}(\mathcal{U}, \mathcal{T}, \mathcal{V})$ *of the form $G_\delta((u, v), t) = (F_{\mathcal{U}\oplus\mathcal{V}}(u, v), F_{\mathcal{T}}(u, v) + H_{\mathcal{T}}(t))$. Then G_δ is proper if and only if there exists a linear mapping $\ell_{\mathcal{U}\oplus\mathcal{V}} : \mathcal{U} \oplus \mathcal{V} \to \mathcal{Z}(\mathcal{U} \oplus \mathcal{V})$ such that the following statements hold:*

(i) $F_{\mathcal{U}\oplus\mathcal{V}} - \ell_{\mathcal{U}\oplus\mathcal{V}}$ *is a generalized derivation,*
(ii) $[\ell_{\mathcal{U}\oplus\mathcal{V}}(u, v), t] = 0$ *for all for all* $(u, v) \in \mathcal{U} \oplus \mathcal{V}$ *and* $t \in \mathcal{T}$.

Corollary 23.2 *Every generalized Lie triple derivation of a triangular algebra* $\mathrm{Tri}(\mathcal{U}, \mathcal{T}, \mathcal{V})$ *is proper if the following two statements are satisfied:*

(i) Every generalized Lie triple derivation on $\mathcal{U}$ (resp. on $\mathcal{V}$) is proper,
(ii) $\mathcal{W}_{\mathcal{U}} = \mathcal{U}$ *and* $\mathcal{W}_{\mathcal{V}} = \mathcal{V}$.

Acknowledgements The authors would like to thank the referee for his/her valuable comments and suggestions. This research is partially supported by a research grant from NBHM (No. 02011/5/2020 NBHM(R.P.) R & D II/6243) and DST (No. DST/INSPIRE/03/2017/IF170834).

References

1. Akhtar MS, Ashraf M, Ansari MA (2022) Generalized Lie triple derivations on generalized matrix algebras. Comm Algebra 50(5):2279–2289
2. Ali S, Dhara B, Fahid B, Raza MA (2019) On semi(prime) rings and algebras with automorphisms and generalized derivations. Bull Iranian Math Soc 45:1805–1819
3. Ansari MA, Ashraf M, Akhtar MS (2022) Lie triple derivations on trivial extension algebras. Bull Iranian Math Soc 48:1763–1774
4. Ashraf M, Akhtar MS (2020) Characterizations of Lie triple derivations on generalized matrix algebras. Comm Algebra 48(9):3651–3660
5. Ashraf M, Jabeen A (2017) Nonlinear generalized Lie triple derivation on triangular algebras. Comm Algebra 45(10):4380–4395
6. Ashraf M, Akhtar MS, Ansari MA (2021) Generalized Lie (Jordan) triple derivations on arbitrary triangular algebras. Bull Malaysian Math Sci Soc 44:3767–3776
7. Bencovič D (2011) Generalized Lie derivations on triangular algebras. Linear Algebra Appl 434:1532–1544
8. Bennis D, Fahid B (2017) Derivations and the first cohomology group of trivial extension algebras. Mediterr J Math 14(150). https://doi.org/10.1007/s00009-017-0949-z
9. Bennis D, Ebrahimi Vishki HR, Fahid B, Khadem-Maboudi AA, Mokhtari AH (2019) Lie generalized derivations on trivial extension algebras. Boll Unione Mat Ital 12:441–452
10. Brešar M (1991) On the distance of the composition of two derivations to the generalized derivations. Glasgow Math J 33:89–93
11. Cheung WS (2003) Lie derivation of triangular algebras. Linear Multilinear Algebr 51(3):299–310
12. Du YQ, Wang Y (2012) Lie derivations of generalized matrix algebras. Linear Algebr Appl 437:2719–2726
13. Hvala B (1998) Generalized derivations in rings. Commun Algebr 26(4):1147–1166
14. Hvala B (2001) Generalized Lie derivations in prime rings. Taiwan J Math 11:1425–1430

15. Li J, Shen Q (2012) Characterization of Lie higher and Lie triple derivation on triangular algebra. J Korean Math Soc 49:419–433
16. Liu L (2015) Lie triple derivations on factor von neumann algebras. Bull Korean Math Soc 52:581–591
17. Martindale WS III (1964) Lie derivations of primitive rings. Mich Math J 11:183–187
18. Miers CR (1973) Lie derivations of von Neumann algebras. Duke Math J 40:403–409
19. Miers CR (1978) Lie triple derivations of von Neumann algebras. Proc Am Math Soc 71:57–61
20. Mokhtari AH, Moafian F, Ebrahimi Vishki HR (2017) Lie derivations on trivial extension algebras. Ann Math Sil 31:141–153
21. Qi XF, Hou JC (2009) Additive Lie (ξ-Lie) derivations and generalized Lie (ξ-Lie) derivations on nest algebras. Linear Algebr Appl 431:843–854
22. Xiao ZK, Wei F (2012) Lie triple derivations of triangular algebras. Linear Algebr Appl 437:1234–1249
23. Zhang J, Wu BW, Cao HX (2006) Lie triple derivations of nest algebras. Linear Algebr Appl 416(2–3):559–567

Chapter 24
On Ideals of Compatible ΘΓ N-Group



Hamsa Nayak, Syam Prasad Kuncham, and Babushri Srinivas Kedukodi

Mathematics Subject Classification (2010) 16Y30

24.1 Introduction

A nearring is a generalized ring with relaxed conditions that only one of the distributive laws need to hold and the addition operation is not expected to be abelian. If $(G, +)$ is a non-abelian group then the set of all mappings from G to itself forms a nearring (which is not a ring) under the usual operations of addition and composition of mappings. In 1984, Bhavanari [20] introduced an extension of nearring called a gamma nearring, wherein a set Γ of binary operations handles the multiplication between every pair of elements. This structure was further studied and developed by several authors including Booth [5], Booth and Groenewald [6], Kuncham [24]. Booth and Groenewald [6] introduced concepts such as equiprime gamma nearrings and contributed to the radical theory of gamma nearrings. Bhavanari and Kuncham [4] studied fuzzy cosets of gamma nearring. Booth, Groenewald and Veldsman [7], Veldsman [25] studied different types of prime ideals of nearrings such as completely prime, 3-prime, and equiprime ideals and formulated corresponding radicals. Jagadeesha, Kedukodi, Kuncham [9] further related these concepts to lattice theory by defining interval valued L-fuzzy ideals of nearrings based on t-norms and

Dedicated to Prof. Calyampudi Radhakrishna Rao and Prof. Arbind Kumar Lal

H. Nayak · S. P. Kuncham · B. S. Kedukodi (✉)
Department of Mathematics, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India
e-mail: babushrisrinivas.k@manipal.edu

H. Nayak
e-mail: nayakhamsa@gmail.com

S. P. Kuncham
e-mail: syamprasad.k@manipal.edu

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_24

t-conorms. Jagadeesha, Kedukodi, Kuncham [13] studied homomorphic images of interval valued L-fuzzy ideals and proved isomorphism theorems. Bhavanari, Kuncham, and Kedukodi [10] linked prime ideals to graph theoretic aspects of nearrings. Deepa and Anita Kumari [22] characterized rings for which the co-maximal graph is a line graph of some graph G and determined rings for which is Eulerian or Hamiltonian. Amritha and Selvakumar [2] analyzed certain basic properties of k-maximal hypergraph of a commutative ring. N-Groups are modules over a nearring (Pilz [18]). Nayak, Kuncham, and Kedukodi [15] introduced a generalization of gamma nearring and N-Group called a $\Theta\Gamma$ N-Group. Nayak, Kuncham and Kedukodi [8, 16] and Harikrishnan, Nayak, Vadiraja Bhatta, Kedukodi and Kuncham [17] further developed and discussed $\Theta\Gamma$ N-Group. It is worth noting that a $\Theta\Gamma$ N-Group even handles non-associative binary operations. This paper derives interesting properties of $\Theta\Gamma$ N-Group G when N is an additive subgroup of G.

24.2 Preliminaries

We refer to Pilz [18] and Koppula, Kedukodi and Kuncham [11, 12] for basic definitions. For recent developments in nearrings, we refer to Kuncham, Kedukodi, Panackal and Bhavanari [19], Anshel and Clay [3], Rao and Veldsman [23] and Maxson and Meyer [14]. Computations in nearrings can be done using SONATA (Aichinger et al. [1]).

Definition 24.1 Let $(G, +)$ be a group with additive identity 0. G is said to be an N-Group if there exists a nearring $(N, +, .)$, and mapping $N \times \theta \times G \to G$ (the image $(n, g) \in N \times \theta \times G$ is denoted by $n\theta g$ where θ is an operation), satisfying the following conditions:

(i) $(n + m)\theta g = n\theta g + m\theta g$ and
(ii) $(nm)\theta g = n\theta(m\theta g)$

for all $g \in G$ and $n, m \in N$.

We denote N-Group by ${}_N G$

Definition 24.2 (*Bhavanari and Kuncham* [21]) Let $(M, +)$ be a group (not necessarily abelian). Let Γ be a non-empty set. Then M is said to be a Γ-nearring if $\exists$ a mapping $M\Gamma M \to M$ (denote the image of (m_1, ρ_1, m_2) by $m_1\rho_1 m_2$ for $m_1, m_2 \in M$ and $\rho_1 \in \Gamma$) with the following conditions:

(i) $(m_1 + m_2)\rho_1 m_3 = m_1\rho_1 m_3 + m_2\rho_1 m_3$ and
(ii) $(m_1\rho_1 m_2)\rho_2 m_3 = m_1\rho_1(m_2\rho_2 m_3)$

for all $m_1, m_2, m_3 \in M$ and for all $\rho_1, \rho_2 \in \Gamma$.

Definition 24.3 (Nayak, Kuncham and Kedukodi [15]) Let $(G, +_G)$ be a group. G is called a $\Theta\Gamma$ N-Group if there exists a nearring $(N, +, \cdot)$ and there exist maps $\Theta(N \times \Theta \times G \to G)$, $\Gamma(N \times \Gamma \times N \to N)$ containing nearring multiplication $\cdot$, $\Delta_\Gamma(N \times \Delta_\Gamma \times G \to G)$ satisfying the following conditions:

(i) $(n + m)\theta g = n\theta g +_G m\theta g$ for all $n, m \in N, g \in G, \theta \in \Theta$ (θ is *right distributive*),
(ii) for every $n, m \in N, \gamma \in \Gamma$, there exists $\delta_\gamma \in \Delta_\Gamma$ such that $(n\gamma m)\theta g = n\delta_\gamma (m\theta g)$ for all $g \in G, \theta \in \Theta$ (θ is *quasi-associative*).

Definition 24.4 (*Nayak, Kuncham and Kedukodi* [15]) Let G be a $\Theta\Gamma$ N-Group. $H \subseteq G$ is said to be $\Theta\Gamma$ N-subgroup of G if $N\Theta H \subseteq H$.

24.3 Compatible ΘΓ N-Group

Definition 24.5 Let $(G, +_G)$ be a group. G is called a compatible $\Theta\Gamma$ N-Group if there exists a nearring $(N, +, .)$ and there exist maps $\Theta(N \times \Theta \times G \to G)$, $\Gamma(N \times \Gamma \times N \to N)$ containing nearring multiplication $\cdot$, $\Delta_\Gamma(N \times \Delta_\Gamma \times N \to G)$ satisfying the following conditions:

(i) $N \subseteq G$ and $+$ and $+_G$ coincide in G (N is an *additive subgroup* of G),
(ii) $(n + m)\theta g = n\theta g + m\theta g$ for all $n, m \in N, g \in G, \theta \in \Theta$ (θ is *right distributive*),
(iii) for every $n, m \in N, \gamma \in \Gamma$, there exists $\delta_\gamma \in \Delta_\Gamma$ such that $(n\gamma m)\theta g = n\delta_\gamma (m\theta g)$ for all $g \in G, \theta \in \Theta$ (θ is *quasi-associative*).

Definition 24.6 Let G be a compatible $\Theta\Gamma$ N-Group. A subset H of G is said to be a subgroup of G if $N\Theta H \subseteq H$.

Definition 24.7 Let H be a normal subgroup of a compatible $\Theta\Gamma$ N-Group $(G, +)$. Then H is called

(i) a *left ideal* of G if $n\theta(x + a) - n\theta x \in H$ for all $n \in N$, $x \in G$, $a \in H$ and $\theta \in \Theta$.
(ii) a *right ideal* of G if $a\theta x \in H$ for all $x \in G$ and $a \in H$.
(iii) an *ideal* of G if H is both a left ideal and a right ideal.

Note 24.1 A left ideal of compatible $\Theta\Gamma$ N-Group is a Θ N-ideal of $\Theta\Gamma$ N-Group.

First, we provide some examples of compatible $\Theta\Gamma$ N-Groups.

Example 1

Let $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $N = \{0, 2, 4, 6\}$. $(N, +)$ is defined as follows:

$+$	0	2	4	6
0	0	2	4	6
2	2	4	6	0
4	4	6	0	2
6	6	0	2	4

Define $\Gamma = \{\gamma_1, \gamma_2\}$, $\Theta = \{\theta_1, \theta_2\}$ and $\Delta_\Gamma = \{\delta_{\gamma_1}, \delta_{\gamma_2}\}$ as follows:

γ_1	0	2	4	4
0	0	0	0	0
2	0	4	0	4
4	0	0	0	0
6	0	4	0	4

γ_2	0	2	4	6
0	0	0	0	0
2	0	6	4	2
4	0	4	0	4
6	0	2	4	6

θ_1	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
2	0	2	4	6	0	2	4	6
4	0	4	0	4	0	4	0	4
6	0	6	4	2	0	6	4	2

θ_2	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
2	0	2	0	6	0	2	0	6
4	0	4	0	4	0	4	0	4
6	0	6	0	2	0	6	0	2

δ_{γ_1}	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
2	0	0	4	0	0	0	4	0
4	0	0	0	0	0	0	0	0
6	0	0	4	0	0	0	4	0

δ_{γ_2}	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
2	0	0	6	0	4	0	2	0
4	0	0	4	0	0	0	4	0
6	0	0	2	0	4	0	6	0

Then G is a compatible $\Theta\Gamma$ N-Group. $I = \{0, 2, 4, 6\}$ is an ideal of G.

Example 2

Consider $Z_8 = \{0, 1, 2, 3, ..., 7\}$, the group of integers modulo 8, and a set $X = \{a, b\}$.
Let $G = \{g \mid g : X \to Z_8 \text{ and } g(a) = 0\}$ and $N = \{g \mid g : X \to Z_4 \text{ and } g(a) = 0\}$.
Then $G = \{g_0, g_1, g_2, ..., g_7\}$, where g_i is defined by $g_i(a) = 0$ and $g_i(b) = i$ for $0 \leq i \leq 7$ and $N = \{g_0, g_1, g_2, g_3\}$, where g_i is defined by $g_i(a) = 0$ and $g_i(b) = i$ for $0 \leq i \leq 3$.
Define mappings $f_0, f_1 : Z_4 \to X$ by setting $f_0(i) = a$ for all $i \in Z_4$ and $f_1(i) = a$ if $i \neq 3$, $f_1(i) = b$ if $i = 3$. Write $\Gamma = \{f_0, f_1\}$.
Define $\Theta = \{f_2, f_3\}$ where $f_2, f_3 : Z_8 \to X$ defined as $f_2(i) = a$ for all $i \in Z_8$ and $f_3(i) = a$ if $i \notin \{3, 7\}$, $f_3(i) = b$ if $i \in \{3, 7\}$.
Now, let $\Delta_\Gamma = \{\delta_{f_0}, \delta_{f_1}\}$ where $\delta_{f_0}, \delta_{f_1} : Z_8 \to X$ by setting $\delta_{f_0}(i) = a$ for all $i \in Z_8$ and $\delta_{f_1}(i) = a$ if $i \notin \{3, 7\}$, $\delta_{f_1}(i) = b$ if $i \in \{3, 7\}$.
Then G is a compatible $\Theta\Gamma$ N-Group. $I = \{g_0, g_2, g_4, g_6\}$ is an ideal of G.

Example 3

Let $(\mathbb{R}, +)$ be the group. Take $N = (\mathbb{R}, +, \cdot)$ and $a, b, c \in \mathbb{R}$.
Define $a \; div \; b = \begin{cases} 0 \; if \; b = 0 \\ \frac{a}{b} \; if \; b \neq 0 \end{cases}$
Corresponding to the operation div, define div_c by $a \; div_c b = a \; div \; (bc^2)$.
Let $\Theta = \{\cdot, div\}$, $\Gamma = \{\cdot, div\}$ and $\Delta_\Gamma = \{\cdot, div_c\}$. Clearly, the operations in Θ are right distributive. We have

$$a \; div_c(b \; div \; c) = a \; div \; ((b \; div \; c)c^2) = \frac{a}{\frac{b}{c} \cdot c^2} = (a \; div \; b) \; div \; c.$$

This implies div is quasi-associative. Hence the operations in Θ are quasi-associative. Therefore, $\mathbb{R}$ is a compatible $\Theta\Gamma$ $\mathbb{R}$ group. $\{0\}$ and $\mathbb{R}$ are the only ideals of $\mathbb{R}$.

Note 24.2 Every gamma nearring is a compatible $\Theta\Gamma$ N-Group (Take $N = G, \Theta = \Gamma$ and $\Delta_\Gamma = \Gamma$).

Proposition 24.1 *Let G be a compatible $\Theta\Gamma$ N group. If $I \trianglelefteq G$ then $G/I = \{g + I | g \in G\}$ is an compatible $\Theta\Gamma$ N-Group.*

Proof To show G/I is a compatible $\Theta\Gamma$ N-Group, we first define operations $+, \theta$ on G/I as follows:

$$(g_1 + I) + (g_2 + I) = (g_1 + g_2) + I,$$

$$(n + I)\theta(g_1 + I) = n\theta g_1 + I.$$

It is easy to show that $+$ is well-defined. We will prove that θ is well-defined. Let $n\theta g_1 + I = n\theta g_1^{'} + I$ and $x \in n\theta g_1 + I$. Then $x = n\theta g_1 + i = (n + i_1)\theta(g_1 + i_2) = (n + i_1)\theta(g_1^{'} + i_2^{'}) = n\theta g_1^{'} + i^{'}$. Hence $n\theta g_1 + I \subseteq n\theta g_1^{'} + I$. Similarly, $n\theta g_1^{'} + I \subseteq n\theta g_1 + I$. Hence $n\theta g_1 + I = n\theta g_1^{'} + I$. Now, we see that for $n, m \in N, \gamma \in \Gamma$, there exists $\delta_\gamma \in \Delta_\Gamma$ such that $(n\gamma m)\theta g + I = n\delta_\gamma(m\theta(g + I))$.

Clearly θ is right distributive and quasi-associative. Hence G/I is a compatible $\Theta\Gamma$ N-Group. □

Definition 24.8 Let G be a compatible $\Theta\Gamma$ N-Group. If $I \trianglelefteq G$, then $G/I = \{g + I | g \in G\}$ is called the factor group.

Definition 24.9 Let N be a nearring and $G, G^{'}$ be compatible $\Theta\Gamma$ N-Groups. Then $h : G \to G^{'}$ is called a homomorphism if it satisfies

(i) $h(x + y) = h(x) + h(y)$ and
(ii) $h(n\theta g) = h(n)\theta h(g)$ for all $n \in N, x, y \in G$ and $\theta \in \Theta$.

$Ker\ h = \{x \in G | h(x) = 0^{'}\}$.

Proposition 24.2 *Let G and $G^{'}$ be compatible $\Theta\Gamma$ N-Groups and $f : G \to G^{'}$ be a homomorphism. Then $ker\ f$ is an ideal of G. Conversely, every ideal is a kernel of a homomorphism.*

Proof We have $f(0) = 0^{'}$. Hence $0 \in ker\ f$. Let $g \in G, n \in N, a \in ker\ f$. Then $f(a) = 0^{'}$. Now,

$$f(g + a - g) = f(g) + f(a) - f(g) = 0^{'} \Rightarrow g + a - g \in ker\ f,$$

$$f(n\theta(x + a) - n\theta x) = f(n\theta(x + a)) - f(n\theta x) = f(n)\theta f(x + a) - f(n)\theta f(x)$$

$$= f(n)\theta(f(x) + f(a)) - f(n)\theta f(x) = 0^{'} \Rightarrow n\theta(x + a) - n\theta x \in ker\ f.$$

$$f(a\theta g) = f(a)\theta f(g) = 0\theta f(g) = 0^{'} \Rightarrow a\theta g \in ker\ f.$$

Hence $ker\ f$ is an ideal of G. To prove the converse, define $\phi : G \to G/I$ by $\phi(g) = g + I$. We prove that ϕ is well-defined and one-one. We have $g_1 = g_2 \Leftrightarrow g_1 + I = g_2 + I \Leftrightarrow \phi(g_1) = \phi(g_2)$. Let $g + I \in G/I$. Then $\phi(g) = g + I$. Hence ϕ is onto. ϕ is a homomorphism because

$$\phi(g_1 + g_2) = (g_1 + g_2) + I = g_1 + I + g_2 + I = \phi(g_1) + \phi(g_2),$$

$$\phi(n\theta g) = n\theta g + I = (n + I)\theta(g + I) = \phi(n)\theta\phi(g).$$

Now, $Ker\ \phi = \{x \in G | \phi(x) = 0 + I\} = \{x \in G | g + I = 0 + I\} = I$. Hence ideal is kernel of a homomorphism. □

Theorem 24.1 *Let G and $G^{'}$ be compatible ΘΓ N-Groups, $f : G \to G^{'}$ a surjection and $K = kerf$. Then K is an ideal of G and $G/K \cong G^{'}$.*

Proof Define $\phi : G/K \to G^{'}$ by $\phi(a + K) = f(a)$. We will show that ϕ is well-defined and one-one. Let $a, b \in G$. We have $a + K = b + K \Leftrightarrow a - b \in K \Leftrightarrow f(a - b) = 0 \Leftrightarrow f(a) - f(b) = 0 \Leftrightarrow f(a) = f(b) \Leftrightarrow \phi(a + K) = \phi(b + K)$. Now, we prove that ϕ is onto. Let $y \in G^{'}$. As f is onto, $y = f(a)$ for some $a \in G$. Now $a + K \in G/K$ and $\phi(a + K) = f(a) = y$. Now we prove that ϕ is homomorphism. We have

$$\phi((a + K) + (b + K)) = \phi((a + b) + K) = f(a + b) = f(a) + f(b) = \phi(a + K) + \phi(b + K),$$

and for any $n \in N, \theta \in \Theta$, we have

$$\phi((n + K)\theta(a + K)) = \phi(n\theta a + K) = f(n\theta a) = f(n)\theta f(a) = \phi(n + K)\theta\phi(a + K).$$

Thus $G/K \cong G^{'}$. □

Theorem 24.2 *(1) Let G and $G^{'}$ be compatible ΘΓ N-Groups, $f : G \to G^{'}$ be homomorphism and $H = ker\ f$. If $K^{'}$ is subgroup (resp. ideal) of G and $K = \{x \in G : f(x) \in K^{'}\} = f^{-1}(K^{'})$, then K is a subgroup(resp. ideal) of G by $H \subseteq K$ and*

$$G/K \cong G^{'}/K^{'}.$$

(2) Let H and K be ideals of compatible ΘΓ N-Group G by $H \subseteq K$. Then

$$G/K \cong (G/H)/(K/H).$$

Proof (1) Define $\phi : G \to G^{'}/K^{'}$ by $\phi(x) = f(x) + K^{'}$. First, we show that ϕ is a homomorphism. Let $a, b \in G$. We have $\phi(a + b) = f(a + b) + K^{'} = (f(a) + f(b)) + K^{'}$. We claim $(f(a) + f(b)) + K^{'} = (f(a) + K^{'}) + (f(b) + K^{'})$. Note that $x \in (f(a) + f(b)) + K^{'} \Rightarrow x = f(a) + f(b) + k_1^{'} = f(a) + 0 + f(b) + k_1^{'}$ $\Rightarrow x \in f(a) + K^{'} + f(b) + K^{'} \Rightarrow (f(a) + f(b)) + K^{'} \subseteq (f(a) + K^{'}) + (f(b) + K^{'})$. Let $y \in (f(a) + K^{'}) + (f(b) + K^{'})$. Hence there exist $k_1^{'}, k_2^{'} \in K^{'}$ such that

$y = f(a) + k_1^{'} + f(b) + k_2^{'}$. Then $y = f(a) + f(b) - f(b) + k_1^{'} + f(b) + k_2^{'} \in f(a) + f(b) + K^{'} \Rightarrow (f(a) + K^{'}) + (f(b) + K^{'}) \subseteq (f(a) + f(b)) + K^{'}$. Hence $\phi(a + b) = (f(a) + f(b)) + K^{'} = (f(a) + K^{'}) + (f(b) + K^{'}) = \phi(a) + \phi(b)$. Now, $\phi(n\theta a) = f(n\theta a) + K^{'} = (f(n) + 0^{'})\theta(f(a) + K^{'}) = (f(n) + k^{'})\theta(f(a) + K^{'}) = \phi(n)\theta\phi(a)$. Now we prove that ϕ is onto. Let $a^{'} + K^{'} \in G^{'}/K^{'}$, $a^{'} \in G^{'}$. Since f is onto, there exists $a \in G$ such that $f(a) = a^{'}$. Hence $\phi(a) = f(a) + K^{'} = a^{'} + K^{'}$. By Theorem 24.1, we get $G/ker\ g \cong G^{'}/K^{'}$. Now, we have

$$ker\ \phi = \{x|\phi(x) = e + K^{'}\} = \{x|f(x) + K^{'} = e + K^{'}\} = \{x|f(x) \in K^{'}\} = K.$$

Let $x \in G, k \in K, n \in N$ and $\theta \in \Theta$. As $\phi(k) = 0$, we get

$$\phi(x + k - x) = \phi(x) + \phi(k) - \phi(x) = 0 \Rightarrow x + k - x \in K;$$
$$\phi(k\theta g) = \phi(k)\theta\phi(g) = 0\theta\phi(k) = 0 \Rightarrow k\theta g \in K;\ \phi(n\theta(x + k) - n\theta x) = \phi(n\theta(x + k)) - \phi(n\theta x) = \phi(n)\theta(\phi(x) + \phi(k)) - \phi(n)\theta\phi(x) = 0$$
$$\Rightarrow n\theta(x + k) - n\theta x \in K.$$

Hence K is subgroup (resp. ideal) of G and $G/K \cong G^{'}/K^{'}$. Let $x \in H$. Then we have $f(x) = e^{'} \in K^{'}$. This implies $\phi(x) = f(x) + K^{'} = e^{'} + K^{'} = K^{'}$. We get $x \in ker\ \phi = K$. Hence $H \subseteq K$.

(2) H is a normal subgroup of G and K is normal subgroup of G containing H. Hence K/H and G/H are factor $\Theta\Gamma\ N$-Groups. First, we prove that K/H is a normal subgroup of G/H. Define $f : G/H \to G/K$ by $f(x + H) = x + K$. We prove f is well-defined. We have $x + H = y + H \Rightarrow x - y \in H \Rightarrow x - y \in K \Rightarrow x + K = y + K \Rightarrow f(x + H) = f(y + H)$. We now show that f is an onto homomorphism with $ker\ f = K/H$. We have

$$\begin{aligned} f[(x + H) + (y + H)] &= f[(x + y) + H] \\ &= (x + y) + K \\ &= (x + K) + (y + K) \\ &= f(x + H) + f(y + H), \quad \text{and} \\ f(n\theta(x + H)) &= f(n\theta x + H) \\ &= n\theta x + K \\ &= (n + 0)\theta(x + K) \\ &= f(n + 0)\theta f(x + H) \\ &= f(n)\theta f(x + H). \end{aligned}$$

Let $x + K \in G/K$. Then there exists $x + H \in G/H$ such that $f(x + H) = x + K$. We have

$$ker\ f = \{g + H|f(g + H) = e + K\} = \{g + H|g \in K\} = K/H.$$

Hence K/H is a normal subgroup of G/H. By Theorem 24.1, $(G/H)/ker\ f \cong G/K$. Thus, $(G/H)/(K/H) \cong G/K$. □

Acknowledgements The authors thank the reviewers for their suggestions. All authors acknowledge Manipal Institute of Technology, Manipal Academy of Higher Education for their encouragement. The first author acknowledges Manipal Academy of Higher Education for the financial support under MU-PhD Scholarship.

References

1. Aichinger E, Binder F, Ecker J, Mayr P, Nöbauer C (2015) SONATA—system of near-rings and their applications, GAP package, Version 2.8. http://www.algebra.uni-linz.ac.at/Sonata/
2. Amritha V, Selvakumar K (2022) Ideal based hypergraph of a commutative ring. AKCE Int J Graphs Comb 19(1):18–23
3. Anshel M, Clay JR (1968) Planar algebraic systems: some geometric interpretations. J Algebr 10(2):166–173
4. Bhavanari S, Kuncham SP (2005) On fuzzy cosets of gamma nearrings. Turk J Math 29(1):11–22
5. Booth G (1988) A note on r-near rings. Stud Sci Math Hung 23:471–475
6. Booth G, Groenewald N (1990) On radicals of gamma-near-rings. Math soc Japan 35:417–425
7. Booth G, Groenewald N, Veldsman S (1990) A kurosh-amitsur prime radical for near-rings. Commun Algebr 18(9):3111–3122
8. Hamsa N, Srinivas KB, Prasad KS (2019) On central boolean rings and boolean type fuzzy ideals. Kuwait J Sci 46(4):23–32
9. Jagadeesha B, Srinivas KB, Prasad KS (2015) Interval valued l-fuzzy ideals based on t-norms and t-conorms. J Intell & Fuzzy Syst 28(6):2631–2641
10. Kedukodi BS, Jagadeesha B, Kuncham SP (2017) Different prime graphs of a nearring with respect to an ideal
11. Koppula K, Srinivas KB, Prasad KS (2020) On prime strong ideals of a seminearring. Mat Vesnik 72:243–256
12. Koppula K, Kedukodi BS, Kuncham SP (2021) On perfect ideals of seminearrings. Beiträge zur Algebra und Geometrie/Contributions to Algebra and Geometry 62(4):823–842
13. Kuncham SP, Jagadeesha B, Kedukodi BS (2016) Interval valued l-fuzzy cosets of nearrings and isomorphism theorems. Afrika Matematika 27(3):393–408
14. Maxson C, Meyer J (2021) Rings of congruence preserving functions, II. Commun Algebr 49(2):579–589
15. Nayak H, Kuncham SP, Kedukodi BS (2018) $\theta\gamma$ n-group. Mat Vesnik 70(1):64–78
16. Nayak H, Kuncham SP, Kedukodi BS (2019) Extensions of boolean rings and nearrings. J Sib Fed Univ Math Phys 12(1):58–67
17. Panackal H, Nayak H, Vadiraja Bhatta G, Kedukodi B, Kuncham S (2018) A note on algebraic structures. Manipal J Sci Tech 3(2):12–19
18. Pilz G (2011) Near-rings: the theory and its applications. Elsevier
19. Prasad KS, Srinivas KB, Harikrishnan P, Satyanarayana B, Neuerburg K, Booth G, Davvaz B, Farag M, Juglal S, Badawi A (2016) Nearrings, nearfields and related topics. World Scientific
20. Satyanarayana B (1984) Contributions to near-rings theory doctoral thesis. Nagarjuna University
21. Satyanarayana B, Prasad KS (2013) Near rings, fuzzy ideals, and graph theory. CRC Press, New York, NY, USA
22. Sinha D, Rao AK (2018) A note on co-maximal graphs of commutative rings. AKCE Int J Graphs Comb 15(1):112–114

23. Srinivasa Rao R, Veldsman S (2019) Right representations of right near-ring radicals. Afrika Matematika 30(1):37–52
24. Syam Prasad K (2000) Contributions to near-ring theory II. PhD thesis, Doctoral thesis, Nagarjuna University
25. Veldsman S (1992) On equiprime near-rings. Commun Algebr 20(9):2569–2587

Chapter 25
The Range Column Sufficiency and the Pseudo-SSM Property of Linear Transformation on Euclidean Jordan Algebra

Punit Kumar Yadav and K. Palpandi

Mathematics Subject Classification (2010) 90C33 · 17C55 · 15A09

25.1 Introduction

Consider a matrix $B = [b_{ij}]_{n\times n},\ b_{ij} \in \mathbb{R}$ and a vector q in $\mathbb{R}^n$; the standard linear complementarity problem (LCP), denoted by LCP (B,q), is to find a vector $z \in \mathbb{R}^n$ such that

$$z \in \mathbb{R}^n_+,\ Bz + q \in \mathbb{R}^n_+ \quad \text{and} \quad z^T(Bz + q) = 0.$$

The LCP has many applications in many branches of optimization, economics, game theory, and marketing. The monograph Cottle et al. [2] is a typical reference for the standard LCP. In recent years, various authors have introduced different classes of matrices for studying the structure of solution sets to LCP; see, for example, Jeyaraman and Sivakumar [11], Pang and Venkateswaran [13], Cottle et al. [2], Tao [17]. One of the most important classes of matrices is strictly semimonotone matrix. We say a matrix $B \in \mathbb{R}^{n\times n}$ is strictly semimonotone if

$$z \geq 0,\ z * Bz \leq 0 \implies z = 0,$$

where the notation $*$ is for the Hadamard product and $z \geq 0$ denotes that each component of z is in $\mathbb{R}_+$. Such a class of matrix gives a unique solution to LCP on $\mathbb{R}^n_+$ (for example, Cottle et al. [2]). Motivated by this, a new class of matrices called the

P. K. Yadav (✉) · K. Palpandi
Malaviya National Instiute of Technology, Jaipur 302017, India
e-mail: punitjrf@gmail.com

K. Palpandi
e-mail: kpalpandi.maths@mnit.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_25

strictly range semimonotone was introduced and studied in Jeyaraman and Sivakumar [11] by I. Jeyaraman and K. C. Sivakumar in the context of LCP. A matrix B is strictly range semimonotone if

$$0 \leq z \in R(B),\ z * Bz \leq 0 \implies z = 0,$$

where $R(B)$ is the range of matrix B. Jeyaraman and Sivakumar [11] proved that the strictly range semimonotone matrix B gives the unique solution of LCP(B, q) on $R(B)$ when $q \in \mathbb{R}^n_+$. Due to the importance of the strictly range semimonotone matrix in LCP, it is generalized and studied for linear transformation on $\mathcal{S}^n$ (the spaces of real symmetric matrices) and $\mathcal{H}^n$ (the space of the Hermitian matrices) in Jeyaraman et al. [12].

Another important class of matrices, so-called column sufficient matrix, was introduced in Pang and Venkateswaran [13]. The matrix $B \in \mathbb{R}^{n\times n}$ is called column sufficient if

$$z * Bz = 0 \text{ whenever } z * Bz \leq 0.$$

The column sufficient matrix gives equivalence relation with the convexity property of the collection of solutions of the corresponding LCP. Recently, Jeyaraman and Sivakumar [11] has extended the column sufficient matrix to the range column sufficient matrix. The matrix $B \in \mathbb{R}^{n\times n}$ is the range column sufficient matrix if

$$z \in R(B),\ z * Bz \leq 0 \implies z * Bz = 0.$$

Further, they have proved the convexity of the intersection of the solution set of LCP and $R(B)$, if B is the range column sufficient. For the range column sufficient matrix B, they showed that the intersection of the solution set of LCP(B, q) and $R(B)$ is a zero set for all $q \in \mathbb{R}^n_{++}$. They proved that the strictly range semimonotone matrix and the range column sufficient matrix are equivalent on $\mathbb{R}^n_+$ for a range monotone Z-matrix.

Motivated by this, we study the pSSM and the rCS property over Euclidean Jordan algebra. We study the solution set to LCP when the underlying linear transformation has the pSSM property (rCS property), and we study the interconnections of the pSSM and the range cone column sufficiency (rCCS) property for some special transformations.

Here is an outline of the paper: In the next section, we give some preliminaries and definitions useful for our discussion. In Sect. 25.3, the range column sufficiency (rCS) and the pSSM properties are introduced and studied in the context of Euclidean Jordan algebra. In Sect. 25.4, we show an equivalence relation between the SSM property and the rCCS property for some special transformations. Section 25.5 characterizes the rCS and the pSSM property for the relaxation transformation.

25.2 Preliminaries

Throughout the paper, we use the following notations:

(i) The $\mathbb{R}^n$ denotes the Euclidean n-space and $\mathbb{R}^n_+$ ($\mathbb{R}^n_{++}$) denotes the nonnegative orthant (positive orthant).
(ii) The set of all complex (real) matrices of order n is denoted by $\mathbb{C}^{n\times n}$ ($\mathbb{R}^{n\times n}$).
(iii) For $B \in \mathbb{R}^{n\times n}$, B^* denotes the conjugate transpose of B. If B is positive semidefinite (positive definite), then we write $B \succeq 0$ ($\succ 0$).
(iv) The spectrum of matrix M and the spectral radius of M are denoted by $\sigma(M)$ and $\rho(M)$, respectively.
(v) We use $[n]$ to denote the set $\{1, 2, ..., n\}$.

25.2.1 Euclidean Jordan Algebra

We state some basic concepts, structure, and results of Euclidean Jordan algebra (EJA) from Faraut and Korainy [3] and Gowda et al. [9].

(i) Let $(\mathcal{V}, \circ, \langle\cdot,\cdot\rangle)$ be a Euclidean Jordan algebra. Here $'\circ'$ is a given Jordan product. We say an element $e \in \mathcal{V}$ is a unit element if $x \circ e = x$ for all $x \in \mathcal{V}$. We define $\mathcal{K} = \{x \circ x : x \in \mathcal{V}\}$ as the symmetric cone in $\mathcal{V}$, and $\mathcal{K}$ is a self-dual closed convex cone. For $x \in \mathcal{V}$, we say that

$$x \in \mathcal{K}\ (\mathcal{K}^\circ) \text{if and only if } x \geq 0 (> 0).$$

(ii) A $Jordan\ frame$ $\{e_i\}_{i=1}^m$ is a finite set of primitive idempotent in $\mathcal{V}$ which satisfies

$$e_i \circ e_j = 0 \text{ if } i \neq j \text{ and } \sum_{i=1}^{m} e_i = e.$$

(iii) Given any $x \in \mathcal{V}$, there exists a Jordan frame $\{e_i\}_{i=1}^r$ such that

$$x = \sum_{i=1}^{r} \alpha_i e_i,$$

where $\alpha_i \in \mathbb{R}$. This expression is called the spectral decomposition of x and the real numbers α_i are called eigenvalues of $x \in \mathcal{V}$. This can be rewritten as

$$x = x^+ - x^- \text{ with } \langle x^+, x^-\rangle = 0,$$

$$\text{where } x^+ = \sum_{i=1}^{r} \alpha_i^+ e_i, x^- = \sum_{i=1}^{r} \alpha_i^- e_i, \alpha_i^+ := \max\{0, \alpha_i\} \text{ and } \alpha_i^- := \alpha_i^+ - \alpha_i.$$

(iv) Fix $a \in \mathcal{V}$; we define a linear transformation L_a in $\mathcal{V}$ as

$$L_a(v) = a \circ v,$$

for all $v \in \mathcal{V}$. Any two elements $u, v \in \mathcal{V}$ operator commute if $L_u L_v = L_v L_u$. From the Lemma X.2.2 in Faraut and Korainy [3], we say that x and y operators commute if and only if x and y have their spectral decomposition with respect to a common Jordan frame. By a proposition in Gowda et al. [9], we observe that $\langle x, y\rangle = 0$ if and only if $x \circ y = 0$ for $x,\ y \in \mathcal{K}$. In each case x and y operators commute.

(v) For a fixed Jordan frame $\{e_i\}_{i=1}^r$ in $\mathcal{V}$ and $1 \le i, j \le r$, we write eigenspaces of $\mathcal{V}$ as

$$\mathcal{V}_{ii} := \{x \in \mathcal{V} : x \circ e_i = x\} = \mathbb{R}e_i,$$

when $i \neq j$,

$$\mathcal{V}_{ij} := \{x \in \mathcal{V} : x \circ e_i = \frac{1}{2}x = x \circ e_j\}.$$

By theorem IV.2.1 in Faraut and Korainy [3], for given any Jordan frame $\{e_i\}_{i=1}^r$, we write for any $x \in \mathcal{V}$ as

$$x = \sum_{i=1}^{r} x_i e_i + \sum_{i<j} x_{ij},$$

where x_i is real number and $x_{ij} \in \mathcal{V}_{ij}$. This expression is called the Peirce decomposition.

(vi) An *algebra automorphism* Λ on $\mathcal{V}$ is a invertible linear transformation which satisfies

$$\Lambda(x \circ y) = \Lambda(x) \circ \Lambda(y) \text{ for all } x, y \in \mathcal{V}.$$

We denote Aut($\mathcal{V}$) as the set of all algebra automorphism of $\mathcal{V}$. We say Λ is an orthogonal automorphism if $\Lambda\Lambda^T(x) = \Lambda^T\Lambda(x) = x$ for all $x \in \mathcal{V}$.

25.2.2 The Linear Complementarity Concepts

Given a transformation $\mathcal{L}$ on $\mathcal{V}$ and a vector $q \in \mathcal{V}$, the LCP($\mathcal{L}, q$) is to find $x \in \mathcal{V}$ such that

$$x \in \mathcal{K},\ \ y := \mathcal{L}(x) + q \in \mathcal{K} \ \text{ and } \ \langle x, y\rangle = 0.$$

If such x exists then x is called a solution to LCP($\mathcal{L}, q$), and we denote $SOL(\mathcal{L}, q)$ as the set of all solutions to LCP($\mathcal{L}, q$). For $\mathcal{V} = \mathbb{R}^n$ and $\mathcal{K} = \mathbb{R}^n_+$, the LCP becomes the standard LCP. If $\mathcal{V} = \mathcal{S}^n$ and $\mathcal{K} = \mathcal{S}^n_+$, then LCP is called the semidefinite LCP

(SDLCP); see (Gowda and Parthasarathy [4], Gowda and Song [6], Tao [17]). Here are some properties of LCP.

Definition 25.1 Let $\mathcal{L}$ be linear on $\mathcal{V}$. Then we say the $\mathcal{L}$ has the

(i) monotone property if $\langle x, \mathcal{L}(x)\rangle \geq 0$, for all $u \in \mathcal{V}$.
(ii) P property if

$$\{x \text{ operator commutes with } \mathcal{L}(x) \text{ and } x \circ \mathcal{L}(x) \leq 0\} \Rightarrow x = 0.$$

(iii) strict semimonotone (SSM) property if

$$\{x \in \mathcal{K},\ x \text{ operator commutes with } \mathcal{L}(x) \text{ and } x \circ \mathcal{L}(x) \leq 0\} \Rightarrow x = 0.$$

(iv) column sufficiency property if

$$\{x \text{ operator commutes with } \mathcal{L}(x) \text{ and} x \circ \mathcal{L}(x) \leq 0\} \Rightarrow x \circ \mathcal{L}(x) = 0.$$

25.2.3 Z Property

We now recall the Z property of linear transformation from Gowda and Tao [8]. A linear transformation $\mathcal{L} : \mathcal{V} \to \mathcal{V}$ has the Z property on $\mathcal{K}$, if

$$x, y \in \mathcal{K} \text{ and } \langle x, y\rangle = 0 \implies \langle \mathcal{L}(x), y\rangle \leq 0.$$

In short, we write it as $\mathcal{L} \in Z$. Here are some examples of Z-transformation.

Example 1

All the matrices with non-positive off diagonal entries are Z-matrices concerning $\mathbb{R}^n_+$.

Example 2

Let $\mathcal{V} = \mathcal{H}^n$ and $\mathcal{K} = \mathcal{H}^n_+$ (the cone of positive semidefinite Hermitian matrices). Then for $A \in \mathbb{C}^{n\times n}$, the Lyapunov transformation $L_A(X) = AX + XA^*$ and the Stein transformation $S_A(X) = X - AXA^*$ are Z-transformations.

We recall a well-known result from Schneider and Vidyasagar [16], which will be used later.

Theorem 25.1 *Let $\mathcal{L} \in Z$. Then $\{Re(\mu) : \mu \in \sigma(\mathcal{L})\} \in \sigma(\mathcal{L})$ with a corresponding eigenvector in $\mathcal{K}$. Here* $\mathrm{Re}(\mu)$ *is the real part of* μ.

25.2.4 Group Inverse

We say a matrix $A \in \mathbb{C}^{n\times n}$ has a group inverse if there exists a unique matrix $Y \in \mathbb{C}^{n\times n}$ which satisfies $A = AYA,\ Y = YAY,\ AY = YA$. It is denoted by $A^{\#}$. The group inverse of A is the same as the inverse of matrix A when A is invertible. For more details, refer to the book of Ben-Israel and Greville [1]. The group inverse for a linear transformation has been discussed by Robert [15]. The group inverse of a linear transformation $\mathcal{L}$ on $\mathcal{V}$ is a linear transformation T on $\mathcal{V}$ that satisfies $T\mathcal{L}T = T,\ \mathcal{L}T\mathcal{L} = \mathcal{L},\ \mathcal{L}T = T\mathcal{L}$. It is denoted by $\mathcal{L}^{\#}$. We recall the following characterizations and definitions from Robert [15], which will be used later.

(T1) The $\mathcal{L}^{\#}$ exists if and only if $\mathcal{V} = R(\mathcal{L}) \oplus N(\mathcal{L})$, i.e., $\mathcal{V}$ is the direct sum of range space and null space of $\mathcal{L}$. Here, $R(\mathcal{L})$ and $N(\mathcal{L})$ denote the range space and null space of $\mathcal{L}$, respectively.
(T2) The $R(\mathcal{L})$, $R(\mathcal{L}^{\#})$, and range of $\mathcal{L}\mathcal{L}^{\#}$ are the same.
(T3) If $\mathcal{L}^{\#}$ exists and $x \in R(\mathcal{L})$, then $\mathcal{L}\mathcal{L}^{\#}(x) = x$.

We end this section with an easy observation that the group inverse exists for any self-adjoint transformation.

25.3 The Range Column Sufficiency Property and the pSSM Property

In this section, we extended the range column sufficient and the pSSM property on Euclidean Jordan algebra.

Definition 25.2 Let $\mathcal{L}$ be linear on $\mathcal{V}$. Then we say $\mathcal{L}$ has the

(i) range column sufficiency (rCS) property if

$$\{x \in R(\mathcal{L}),\ x \text{ operator commutes with } \mathcal{L}(x) \text{ and } x \circ \mathcal{L}(x) \leq 0\} \Rightarrow x \circ \mathcal{L}(x) = 0.$$

(ii) range cone column sufficiency (rCCS) property if

$$\{0 \leq x \in R(\mathcal{L}),\ x \text{ operator commutes with } \mathcal{L}(x) \text{ and } x \circ \mathcal{L}(x) \leq 0\} \Rightarrow x \circ \mathcal{L}(x) = 0.$$

Proposition 25.1 *Let $\mathcal{L}$ be linear on $\mathcal{V}$. Then,*

(i) If the group inverse of $\mathcal{L}$ exists, then $\mathcal{L}$ has the rCS property if and only if the group inverse of $\mathcal{L}$ has the rCS property.
(ii) $\mathcal{L}$ has the rCS property if and only if $\hat{\mathcal{L}} := \Lambda^T \mathcal{L}\Lambda$ has the rCS property for all the orthogonal automorphism Λ.

Proof (i). Assume $\mathcal{L}$ has the rCS property. Let $x \in R(\mathcal{L}^{\#})$ such that x operator commutes with $\mathcal{L}^{\#}(x)$ and $x \circ \mathcal{L}^{\#}(x) \leq 0$. Let $z := \mathcal{L}^{\#}(x)$. As $x \in R(\mathcal{L}^{\#})$, then from (T2) and (T3) we have $\mathcal{L}(z) = x$. Since x operator commutes with $\mathcal{L}^{\#}(x)$ and $x \circ \mathcal{L}^{\#}(x) \leq 0$, z and $\mathcal{L}(z)$ operator commute with each other $z \circ \mathcal{L}(z) \leq 0$. From the rCS property of $\mathcal{L}$, we get $z \circ \mathcal{L}(z) = 0$ which implies that $x \circ \mathcal{L}^{\#}(x) = 0$. Hence $\mathcal{L}^{\#}$ has the rCS property. By using a similar argument, the reverse implication can be proved.

We now prove (ii). Let $x \in R(\hat{\mathcal{L}})$ such that x operator commutes with $\hat{\mathcal{L}}(x)$ and $x \circ \hat{\mathcal{L}}(x) \leq 0$. Then there exists a common Jordan frame $\{e_i\}_{i=1}^r$ such that

$$x = \sum_{i=1}^{r} \lambda_i e_i,\ \hat{\mathcal{L}}(x) = \sum_{i=i}^{r} \mu_i e_i \text{ and } \lambda_i \mu_i \leq 0\ \forall\ i \in [r].$$

Our aim is to prove that $\lambda_i \mu_i = 0$ for all $i \in [r]$. As $x \in R(\hat{\mathcal{L}})$, then we can find $z \in V$: $x = \Lambda^T \mathcal{L}\Lambda(z)$ and also Λ is an orthogonal automorphism which gives that $\Lambda x = \mathcal{L}(\Lambda z)$ and hence $\Lambda(x) \in R(\mathcal{L})$. By Proposition 4.2 in Gowda and Sznajder [7], $\{\Lambda(e_i)\}_{i=1}^r$ is a Jordan frame in $\mathcal{V}$ and $(\Lambda^T)^{-1}(e_i) = \Lambda(e_i)$ for all $i \in [r]$. So we have $\Lambda(x) = \sum_{i=1}^{r} \lambda_i \Lambda(e_i) \in R(\mathcal{L})$ and $\mathcal{L}(\Lambda(x)) = \sum_{i=1}^{r} \mu_i \Lambda(e_i)$. Here, $\Lambda(x)$ and $\mathcal{L}(\Lambda(x))$ have their spectral decomposition with respect to a common Jordan frame. Hence, $\Lambda(x)$ operator commutes with $\mathcal{L}(\Lambda(x))$ and $\Lambda(x) \circ \mathcal{L}(\Lambda(x)) = \sum_{i=1}^{r} \lambda_i \mu_i \Lambda(e_i)$. As $\lambda_i \mu_i \leq 0$ for all $i \in [r]$, $\Lambda(x) \circ \mathcal{L}(\Lambda(x)) \leq 0$. Since $\mathcal{L}$ has the rCS property, we have $\lambda_i \mu_i = 0$ for all $i \in [r]$. Hence $\hat{\mathcal{L}}$ has the rCS property. The reverse implication follows immediately by choosing Λ as the identity transformation. □

Definition 25.3 Let $\mathcal{L}$ be linear on $\mathcal{V}$ and $\mathcal{S}$ be a subset of $\mathcal{V}$. Then we say $\mathcal{L}$ has the range cross commutative property on $\mathcal{S}$ if for any $q \in \mathcal{S}$, x_1 operator commutes with y_2 and x_2 operator commutes with y_1, where $x_1, x_2 \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ and $y_i := \mathcal{L}(x_i) + q,\ i = 1, 2$.

Next, we give a sufficient condition for the convexity of the $SOL(\mathcal{L}, q) \cap R(\mathcal{L})$.

Theorem 25.2 *Let $\mathcal{L}$ be linear on $\mathcal{V}$. If $\mathcal{L}$ has the rCS property and range cross commutative property on $\mathcal{V}$, then the following are true.*

(i) $SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ is a convex set for all $q \in \mathcal{V}$.
(ii) $SOL(\mathcal{L}, q) \cap R(\mathcal{L}) = \{0\}$ for all $q > 0$.

Proof (i): Let $x_1, x_2 \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$. As $R(\mathcal{L})$ is a subspace, we have $tx_1 + (1-t)x_2 \in R(\mathcal{L})$ for all $0 \leq t \leq 1$. Note that for all $0 \leq t \leq 1$,

$$tx_1 + (1-t)x_2 \geq 0 \text{ and } \mathcal{L}(tx_1 + (1-t)x_2) + q \geq 0.$$

Let $y_i := \mathcal{L}(x_i) + q$. As $x_1, y_2 \geq 0$, by the range cross commutative property on $\mathcal{V}$, x_1 operator commutes with y_2 and $x_1 \circ y_2 \geq 0$. Similarly, $x_2 \circ y_1 \geq 0$. Therefore, $(x_1 - x_2)$ operator commutes with $\mathcal{L}(x_1 - x_2)$ and $(x_1 - x_2) \circ (\mathcal{L}(x_1 - x_2)) = (-x_1 \circ y_2 - x_2 \circ y_1) \leq 0$. Since $x_1, x_2 \in R(\mathcal{L})$ then $x_1 - x_2 \in R(\mathcal{L})$ and from the rCS property of $\mathcal{L}$, we have $(x_1 - x_2) \circ \mathcal{L}(x_1 - x_2) = 0$ which implies that

$$x_1 \circ y_2 = 0 \text{ and } x_2 \circ y_1 = 0. \tag{25.1}$$

From Eq. (25.1), we have

$$(tx_1 + (1-t)x_2) \circ (\mathcal{L}(tx_1 + (1-t)x_2) + q) = t(1-t)[x_1 \circ y_2 + x_2 \circ y_1] = 0.$$

Therefore, $tx_1 + (1-t)x_2 \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ for all $0 \leq t \leq 1$. This proves item (i).

We now prove (ii): Let $q > 0$ with $0 \neq x \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$. As $q > 0$, $0 \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ and by item (i), $tx \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ for all $0 \leq t \leq 1$. So we have $x \circ (\mathcal{L}(tx) + q) = 0$ for all $0 \leq t \leq 1$. As $t \to 0$, $x \circ q = 0$ which implies $x = 0$ as $q > 0$. Thus we get a contradiction. Hence we have our claim. □

The following example illustrates that we cannot drop the range cross commutative property in the above theorem to show that $SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ is equal to 0 whenever $q > 0$.

Example 3

Let $A = \begin{bmatrix} -1 & 3 \\ -3 & 3 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 & x_2 \\ x_2 & x_3 \end{bmatrix} \in \mathcal{S}^2$. Then

$$L_A(X) = \begin{bmatrix} -2x_1 + 6x_2 & 2x_2 + 3x_3 - 3x_1 \\ 2x_2 + 3x_3 - 3x_1 & -6x_2 + 6x_3 \end{bmatrix}.$$

Here we can see easily that A is a positive stable matrix. Hence from a result from Gowda and Song [6], L_A has the P property. L_A has the rCS property.

Assume $Q = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \in \mathcal{S}^2_{++}$. Then $Z = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in SOL(L_A, Q) \cap R(L_A)$. Now let us take $Y = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in \mathcal{S}^2_+$ and for $U = \begin{bmatrix} 5/8 & 3/8 \\ 3/8 & 3/8 \end{bmatrix} \in \mathcal{S}^2$, we get $Y = L_A(U)$ which implies $Y \in R(L_A)$. By an easy computation we get $L_A(Y) + Q = \begin{bmatrix} 0 & 0 \\ 0 & 5 \end{bmatrix} \in \mathcal{S}^2_+$ and $Y(L_A(Y) + Q) = 0$. Therefore, we can see that $Z,\ Y \in SOL(L_A, Q) \cap R(L_A)$ and

$Y(L_A(Z) + Q) \neq Z(L_A(Y) + Q)$. So L_A does not have the range cross commutative property and $SOL(L_A, Q) \cap R(L_A) \neq \{0\}$ for $Q \succ 0$.

Next, we extend the pSSM property on Euclidean Jordan algebra.

Definition 25.4 Let $\mathcal{L}$ be linear on $\mathcal{V}$ then $\mathcal{L}$ has the pseudo strict semimonotone (pSSM) property if

$$\{x \text{ operator commutes with } \mathcal{L}(x),\ 0 \leq x \in R(\mathcal{L}) \text{ and } x \circ \mathcal{L}(x) \leq 0\} \Rightarrow x = 0.$$

In the next theorem, we give an equivalence relation for the intersection of the solution set of the LCP and the range of linear transformation with the pSSM property.

Theorem 25.3 *Let $\mathcal{L}$ be linear on $\mathcal{V}$. Then the following are equivalent:*

(i) $\mathcal{L}$ has the pSSM and the range cross commutative property on $\mathcal{K}$.
(ii) $SOL(\mathcal{L}, q) \cap R(\mathcal{L}) = \{0\}$ for all $q \in \mathcal{K}$.

Proof (i) $\Longrightarrow$ (ii): Let $q \in \mathcal{K}$ and $x \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$. Then x operator commutes with $\mathcal{L}(x) + q$ and $x \circ (\mathcal{L}(x) + q) = 0$ so $x \circ \mathcal{L}(x) = -x \circ q$. As $0 \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$ and by the range cross commutative property of $\mathcal{L}$ on $\mathcal{K}$, x operator commutes with q and hence $x \circ q \geq 0$ which implies $x \circ \mathcal{L}(x) \leq 0$. Since $\mathcal{L}$ has the pSSM property, we have $x = 0$. Hence $SOL(\mathcal{L}, q) \cap R(\mathcal{L}) = \{0\}$.
(ii) $\Longrightarrow$ (i): Suppose $SOL(\mathcal{L}, q) \cap R(\mathcal{L}) = \{0\}$ for all $q \in \mathcal{K}$. We can easily check that $\mathcal{L}$ has the range cross commutative property on $\mathcal{K}$. Suppose there is a vector $x \in R(\mathcal{L}) \cap \mathcal{K}$ such that x and $\mathcal{L}(x)$ operator commute and $x \circ \mathcal{L}(x) \leq 0$. This gives that x and $\mathcal{L}(x)^+$ operator commute and $x \circ \mathcal{L}(x)^+ = 0$. Now set $q := -\mathcal{L}(x) + \mathcal{L}(x)^+$, then $q = \mathcal{L}(x)^-$ and hence $q \geq 0$. As $x \circ \mathcal{L}(x)^+ = 0$ and $\mathcal{L}(x) + q = \mathcal{L}(x) + \mathcal{L}(x)^- = \mathcal{L}(x)^+ \geq 0$, we get $x \circ (\mathcal{L}(x) + q) = x \circ \mathcal{L}(x)^+ = 0$. So we have $x \in SOL(\mathcal{L}, q) \cap R(\mathcal{L})$. Then $x = 0$. Hence $\mathcal{L}$ has the pSSM property. □

We say a matrix is semipositive stable (positive stable) if real parts of all the eigenvalues of the matrix are nonnegative (positive). Now we discuss a spectral property of the range cone column sufficient Z-transformation.

Theorem 25.4 *Let $\mathcal{L} \in Z$. If $\mathcal{L}$ has the rCCS property, then $\mathcal{L}$ is semipositive stable.*

Proof Assume $\mathcal{L}$ has the rCCS property. As $\mathcal{L}$ is a Z-transformation then from Theorem 25.1, there will be a non-zero vector $x \in \mathcal{K}$ such that $\mathcal{L}(x) = \mu_{\min} x$, where $\mu_{\min} = \min\{Re(\mu) : \mu \in \sigma(L)\}$. It is enough to show that $\mu_{\min}$ is nonnegative. Assume the contrary, i.e., $\mu_{\min} < 0$. Then

$$x \circ \mathcal{L}(x) = \mu_{\min} x^2 \leq 0.$$

As $x \in R(\mathcal{L})$ and from the hypothesis, $x = 0$, which contradicts the fact that $x \neq 0$. So $\mu_{\min} \geq 0$. Hence proved. □

Remark 25.1 In this example, we can see that for Z-transformation, the converse of the above theorem is not true. Let $\mathcal{L} = \begin{bmatrix} 0 & -1 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} : \mathbb{R}^3 \to \mathbb{R}^3$ and $x = (1, 1, 0)^T \in \mathbb{R}^3$. For $y = (0, 1, -1)^T$, we get $\mathcal{L}(y) = x$ so $x \in R(\mathcal{L})$. Here, all the eigenvalues of $\mathcal{L}$ are zero so $\mathcal{L}$ is semipositive stable and off diagonal entries of $\mathcal{L}$ are non-positive then $\mathcal{L}$ is a Z-transformation on $\mathbb{R}^3_+$ and

$$0 \leq x \in R(\mathcal{L}) \text{ and } x * \mathcal{L}(x) = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \leq 0.$$

But $x * \mathcal{L}(x) \neq 0$. So $\mathcal{L}$ is not range cone column sufficient.

25.4 Equivalence of rCCS and pSSM Properties for Some Special Transformations

In matrix theory, a matrix $A \in \mathbb{R}^{n \times n}$ is a range monotone matrix if $Ax \geq 0$, $x \in R(A)$ implies $x \geq 0$. Various characterizations have been given for range monotone matrices; see Peris and Subiza [14]. For a range monotone Z-matrix, the strictly range semimonotone is equivalent to the range column sufficient on $\mathbb{R}^n_+$; see in Jeyaraman and Sivakumar [11]. A natural question arises whether this relation will hold for linear transformation in general or not. This will be answered in the present section. To proceed further, first, we define the *range monotone* property for linear transformation.

Definition 25.5 Let $\mathcal{L}$ be linear on $\mathcal{V}$ then $\mathcal{L}$ has range monotone property if

$$\mathcal{L}(x) \in \mathcal{K},\ x \in R(\mathcal{L}) \implies x \in \mathcal{K}.$$

We say here $\mathcal{L}$ is range monotone transformation.

Theorem 25.5 *Let $\mathcal{L}$ be a range monotone Z-transformation on $\mathcal{V}$. Then the following are equivalent.*

1. *$\mathcal{L}$ has the pSSM property.*
2. *$\mathcal{L}$ has the rCCS property.*

Proof (i) $\implies$ (ii) is obvious.

(ii) $\implies$ (i): Let $0 \leq x \in R(\mathcal{L})$ and x operator commute with $\mathcal{L}(x)$ such that $x \circ \mathcal{L}(x) \leq 0$. Then there exists a Jordan frame $\{e_i\}_{i=1}^r$ such that

$$x = \sum_{i=1}^{r} \alpha_i e_i,\ \mathcal{L}(x) = \sum_{i=1}^{r} \beta_i e_i \text{ and } \alpha_i \beta_i \leq 0,\ i \in [r].$$

From the rCCS-property of $\mathcal{L}$, $x \circ \mathcal{L}(x) = 0$ and therefore we have $\alpha_i \beta_i = 0$, $i \in [r]$. If $x > 0$, then $\mathcal{L}(x) = 0$ and hence $\mathcal{L}(-x) = 0$. As $\mathcal{L}(-x) \in \mathcal{K}$ and $-x \in R(\mathcal{L})$, then from the range monotone property of $\mathcal{L}$, $-x \in \mathcal{K}$ which implies $x = 0$. Thus it gives us a contradiction. Therefore, x must have a zero eigenvalue. We are done if all the eigenvalues of x are zero. If not, that is, x contains a non-zero eigenvalue. Without loss of generality, we suppose $\alpha_i > 0$ for all $1 \leq i \leq k$ and $\alpha_i = 0$ for all $k + i \in [r]$. Note that $\beta_i = 0$ for all $1 \leq i \leq k$. As $\alpha_i = 0$ for all $k + i \in [r]$, we have $\langle x, e_i \rangle = 0$ for all $k + i \in [r]$, which implies $\langle (x), e_i \rangle \leq 0$ for all $k + 1 \leq i \leq r$, as $\mathcal{L}$ is Z-transformation. So $\beta_i \leq 0$ for all $k + 1 \leq i \leq r$. Hence we get $\mathcal{L}(x) \leq 0$. As $\mathcal{L}(-x) \in \mathcal{K}$ and $-x \in R(\mathcal{L})$, therefore from range monotone property of $\mathcal{L}$, we have $-x \in \mathcal{K}$ which implies $x = 0$. It gives a contradiction. Hence $\mathcal{L}$ has the pSSM property. □

Next, we show the equivalence of the pSSM property, the range column sufficiency, and the semipositive stability for the Lyapunov transformation L_A.

Theorem 25.6 *Let $A \in \mathbb{C}^{n \times n}$. If $L_A^{\#}$ exists, the following are equivalent:*

(i) A is semipositive stable.
(ii) L_A has the pSSM property.
(iii) L_A has the rCCS property.
(iv) L_A is semipositive stable.

Proof (i) $\Longrightarrow$ (ii): If A is semipositive stable then from Theorem 3.11 in Jeyaraman et al. [12], L_A has the pSSM property.
(ii) $\Longrightarrow$ (iii): By definition, the pSSM property implies the rCCS property.
(iii) $\Longrightarrow$ (iv): It follows from Theorem 25.4.
(iv) $\Longrightarrow$ (i): Let $Av = \lambda v$. Then we have

$$L_A(vv^*) = Avv^* + vv^*A^* = (\lambda + \bar{\lambda})vv^* = 2Re(\lambda)vv^*.$$

As L_A is semipositive stable so $Re(\lambda) \geq 0$. Hence A is semipositive stable. □

In the following theorem, we show equivalence between the pSSM property and rCCS property for S_A.

Theorem 25.7 *Let $A \in \mathbb{C}^{n \times n}$ such that $S_A^{\#}$ exists. Then the following are equivalent:*

1. *$\rho(A) \leq 1$, i.e., all the eigenvalues of A lie in the closed unit disk.*
2. *S_A has the pSSM property.*
3. *S_A has the rCCS property.*
4. *S_A is semipositive stable.*

Proof (i) $\Longrightarrow$ (ii): From Theorem 3.16 in Jeyaraman et al. [12], S_A has the pSSM property.
(ii) $\Longrightarrow$ (iii): Follows from definition.
(iii) $\Longrightarrow$ (iv): By Theorem 25.4, S_A is semipositive stable.

(iv) $\Longrightarrow$ (i): Suppose there exist $\lambda \in \sigma(A)$ and non-zero vector u such that $Au = \lambda u$, and $|\lambda| > 1$. Now

$$S_A(uu^*) = uu^* - Auu^*A^* = (1 - |\lambda|^2)uu^*.$$

Then $|\lambda| > 1$ implies $(1 - |\lambda|^2) < 0$. So we have a contradiction of our assumption. Hence $\rho(A) \leq 1$. This completes the proof. □

Let $B,\ C_1,\ C_2, ...,\ C_k$ be in $\mathbb{C}^{n\times n}$. We define a linear transformation T on $\mathcal{H}^n$ by

$$T(X) = BXB^* - \sum_{i=1}^{k} C_i X C_i^*.$$

This class of operators has been studied in the context of positive operators in Hans [10]. When $B = A + I, k = 2, C_1 = A, C_2 = I$, the operator T reduces to the Lyapunov transformation and for $B = I, k = 1, C_1 = A$, we get the Stein transformation. From Theorems 25.6 and 25.7, the pSSM property and the rCCS property are equivalent for the Lyapunov and the Stein transformation. Motivated by this, one can ask whether the same result holds for the operator T. We now study a particular case of T. By considering $B = A + I, k = 3, C_1 = I, C_2 = A, C_3 = A$ in T, we get

$$L(X) = AX + XA^* - AXA^*.$$

We prove an equivalence relation between the pSSM property and the rCCS property, for $L(X) = AX + XA^* - AXA^*$.

Theorem 25.8 *Let $L(X) = AX + XA^* - AXA^*$. If $L^{\#}$ exists, then the following are equivalent:*

(i) L has the pSSM property.
(ii) L has the rCCS property.

Proof (i) $\Longrightarrow$ (ii): It is obvious.
(ii) $\Longrightarrow$ (i): Suppose $0 \neq X \in R(L)$ such that $X \succeq 0$ and $XL(X) = L(X)X \preceq 0$. Since $XL(X) = L(X)X$, there will be a unitary matrix U, i.e., $U^*U = UU^* = I$, such that

$$D = U^*XU \ \text{ and } \ E = U^*L(X)U,$$

where D is a diagonal matrix with nonnegative entries and E is a diagonal matrix with $ED \preceq 0$. As we have $X \succeq 0$ and $XL(X) = L(X)X \preceq 0$, so by the rCCS property of L, $XL(X) = 0$ which implies $ED = 0$.
Case I: Suppose X is invertible and $X \in R(L)$. Then $L(X) = 0$ which gives us $X = 0$ as $L^{\#}L(X) = X$. Thus we get a contradiction.
Case II: Suppose X is not invertible. Then

$$D = \begin{bmatrix} D_1 & 0 \\ 0 & 0 \end{bmatrix} \ \text{ and } \ E = \begin{bmatrix} 0 & 0 \\ 0 & E_1 \end{bmatrix}$$

where $D_1 \succ 0$. Let $U^*AU = \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$, where B_1 compatible with D_1. As $E = U^*L(X)U$, we get

$$\begin{bmatrix} B_1D_1 + D_1B_1^* - B_1D_1B_1^* & D_1B_3^* - B_1D_1B_3^* \\ B_3D_1 - B_3D_1B_1^* & -B_3D_1B_3^* \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & E_1 \end{bmatrix}.$$

This gives that

$$B_1D_1 + D_1B_1^* - B_1D_1B_1^* = 0,\ B_3D_1B_3^* = -E_1 \text{ and } B_3D_1 - B_3D_1B_1^* = 0.$$

We will prove here that $I - B_1^*$ is invertible. On the contrary, let us assume that v is a non-zero vector such that $(I - B_1^*)v = 0$. This implies $B_1^*v = v$, so

$$\langle (B_1D_1 + D_1B_1^* - B_1D_1B_1^*)v, v\rangle = 0 \implies \langle D_1v, v\rangle = 0.$$

It contradicts the fact that $D_1 \succ 0$. Hence $I - B_1^*$ is invertible. As $B_3D_1 - B_3D_1B_1^* = B_3D_1(I - B_1^*) = 0$, this implies $B_3 = 0$. So $E_1 = 0$ which gives that $E = 0$. Therefore, we get $L(X) = 0$ which implies that $X = 0$ from (T3). Thus we get a contradiction. Hence L has the pSSM property. □

In Theorem 25.6, we proved the equivalence for the rCCS and the pSSM property for the Lyapunov transformation. Motivated by this, we prove the same relation for the Lyapunov-like transformation. A linear transformation $\mathcal{L}$ on $\mathcal{V}$ is called Lyapunov-like if

$$x, y \in \mathcal{K} \text{ and } \langle x, y\rangle = 0 \implies \langle \mathcal{L}(x), y\rangle = 0.$$

Theorem 25.9 *Let $\mathcal{L}$ be a Lyapunov-like transformation. If $\mathcal{L}^\#$ exists, then the following are equivalent:*

(i) $\mathcal{L}$ has the pSSM property.
(ii) $\mathcal{L}$ has the rCCS property.

Proof (i) $\implies$ (ii): This is obvious.
(ii) $\implies$ (i): Let $0 \le x \in R(\mathcal{L})$ and x operator commute with $\mathcal{L}(x)$ such that $x \circ \mathcal{L}(x) \le 0$. As $\mathcal{L}$ has the rCCS property, $x \circ \mathcal{L}(x) = 0$ and from this we get $\langle x, \mathcal{L}(x)\rangle = 0$. Now $x \ge 0$ and $x \circ \mathcal{L}(x) = x \circ (\mathcal{L}(x)^+ - \mathcal{L}(x)^-) = 0$ which implies $x \circ \mathcal{L}(x)^+ = x \circ \mathcal{L}(x)^- = 0$. From this we get $\langle x, \mathcal{L}(x)^+\rangle = \langle x, \mathcal{L}(x)^-\rangle = 0$. Here $x \ge 0$, $\mathcal{L}(x)^+ \ge 0$ and $\langle x, \mathcal{L}(x)^+\rangle = 0$. As $\mathcal{L}$ is a Lyapunov-like transformation, then we have

$$\langle \mathcal{L}(x), \mathcal{L}(x)^+\rangle = 0.$$

Similarly $\langle \mathcal{L}(x), \mathcal{L}(x)^-\rangle = 0$. Therefore, $\langle \mathcal{L}(x), \mathcal{L}(x)\rangle = 0$ which implies $\|\mathcal{L}(x)\|^2 = 0$. Hence $\mathcal{L}(x) = 0$. As $x \in R(\mathcal{L})$ and $\mathcal{L}^\#$ exists, $x = 0$, as $\mathcal{L}^\#\mathcal{L}(x) = x$. Thus we have our claim. □

Proposition 25.2 Let $\mathcal{L}$ be linear on $\mathcal{V}$ with $\langle x, \mathcal{L}(x)\rangle \geq 0$ for all $x \in R(\mathcal{L})$. Then $\mathcal{L}$ has the rCS property.

Proof Let $x \in R(\mathcal{L})$. Assume x operator commutes with $\mathcal{L}(x)$ such that $x \circ \mathcal{L}(x) \leq 0$. This implies $\langle x, \mathcal{L}(x)\rangle \leq 0$. Therefore, from our hypothesis $\langle x, \mathcal{L}(x)\rangle = 0$. By the operator commutativity of x and (x), we get $x \circ \mathcal{L}(x) = 0$. Thus $\mathcal{L}$ has the rCS property. □

Theorem 25.10 *Let $\mathcal{L}$ be linear on $\mathcal{V}$ such that $\mathcal{L}^{\#}$ exists. If $\|\mathcal{L}\| \leq 1$, then $\mathcal{L}^{\#}\mathcal{L} - \mathcal{L}$ has the rCS property.*

Proof Let u be in $R(\mathcal{L})$. Then $\mathcal{L}\mathcal{L}^{\#}(u) = u$, which implies that

$$\langle u, (\mathcal{L}\mathcal{L}^{\#} - \mathcal{L})u\rangle = \langle u, \mathcal{L}\mathcal{L}^{\#}(u)\rangle - \langle u, \mathcal{L}(u)\rangle = \langle u, u\rangle - \langle u, \mathcal{L}(u)\rangle \geq (1 - \|\mathcal{L}\|)\|u\|^2 \geq 0.$$

By Proposition 25.2, $\mathcal{L}$ has the rCS property. □

Theorem 25.11 *Let $\mathcal{L}$ be a self-adjoint Z-transformation. Then the following are equivalent:*

(i) *$\mathcal{L}$ is monotone.*
(ii) *$\mathcal{L}$ has the column sufficiency property.*
(iii) *$\mathcal{L}$ has the rCS property.*
(iv) *$\mathcal{L}$ has the rCCS property.*
(v) *$\mathcal{L}$ has the pSSM property.*

Proof (i) $\Longrightarrow$ (ii): From Theorem 6.1 in Tao and I. Jeyaraman [18], it holds.
(ii) $\Longrightarrow$ (iii) and (iii) $\Longrightarrow$ (iv) follow from the definition.
(iv) $\Longrightarrow$ (v): Let $x \in R(\mathcal{L}) \cap \mathcal{K}$ such that x operator commutes with $\mathcal{L}(x)$ and $x \circ \mathcal{L}(x) \leq 0$. From the rCCS property of $\mathcal{L}$, $x \circ \mathcal{L}(x) = 0$ and this implies $\langle x, \mathcal{L}(x)\rangle = 0$. Since $\mathcal{L}$ is a self-adjoint Z-transformation, by Theorem 25.4, all the eigenvalues of $\mathcal{L}$ are real and nonnegative, giving that $\mathcal{L}(x) = 0$. As $\mathcal{L}$ is self-adjoint, $\mathcal{L}^{\#}$ exists. Since $x \in R(\mathcal{L})$, then $\mathcal{L}^{\#}\mathcal{L}(x) = x$. This gives that $x = 0$.
(v) $\Longrightarrow$ (i): Let $\mathcal{L}$ has the pSSM property and $\mathcal{L}$ be a self-adjoint Z-transformation. So by Lemma 3.5 in Jeyaraman et al. [12], all the eigenvalues of $\mathcal{L}$ are semipositive stable. As $\mathcal{L}$ is self-adjoint, all the eigenvalues of $\mathcal{L}$ are real and nonnegative. Hence $\mathcal{L}$ is monotone. □

25.5 The Relaxation Transformation

Given a fixed Jordan frame in $\mathcal{V}$, we define the relaxation transformation as follows: For $A \in \mathbb{R}^{r\times r}$ and for any $x \in \mathcal{V}$, the Peirce decomposition

$$x = \sum_{i=1}^{r} x_i e_i + \sum_{i<j} x_{ij}$$

then we write the relaxation transformation as

$$R_A(x) = \sum_{i=1}^{r} y_i e_i + \sum_{i<j} x_{ij},$$

where $[y_1, y_2, ..., y_r]^T = A([x_1, x_2, ..., x_r]^T)$. This transformation is the generalization of a concept studied in Gowda and Song [5], for $\mathcal{V} = \mathcal{S}^n$. Various authors have studied this transformation; see Tao and I. Jeyaraman [18], Tao [17]. Here, we characterize the rCS and pSSM properties of R_A.

Theorem 25.12 *Let $A \in \mathbb{R}^{r\times r}$. Then for the R_A, the following are equivalent.*

(i) A is a range column sufficient matrix.
(ii) R_A has the rCS property.

Proof (i) $\Longrightarrow$ (ii): Let $\{e_i\}_{i=1}^r$ be a given Jordan frame. Suppose $x \in \mathcal{V}$ such that x is in range of R_A, x operator commutes with $R_A(x)$, and $x \circ R_A(x) \leq 0$. Then there will be some $v \in \mathcal{V}$ such that

$$v = \sum_{i=1}^{r} v_i e_i + \sum_{i<j} v_{ij} \text{ and } R_A(v) = x,$$

where $[x_1, x_2, ..., x_r]^T = A[v_1, v_2, ..., v_r]^T$. Let $[y_1, y_2, ..., y_r]^T = A[x_1, x_2, ..., x_r]^T$, then we can write

$$x = \sum_{i=1}^{r} x_i e_i + \sum_{i<j} v_{ij} \text{ and } R_A(x) = \sum_{i=1}^{r} y_i e_i + \sum_{i<j} v_{ij}.$$

By an easy calculation, we get $x \circ e_l = x_l e_l + \frac{1}{2}\sum_{1\leq i<l} v_{il} + \frac{1}{2}\sum_{l<j\leq r} v_{lj}$ for $1 \leq l \leq r$,

which implies

$$0 \geq \langle x \circ R_A(x), e_l\rangle = \langle R_A(x), x \circ e_l\rangle = x_l y_l \|e_l\|^2 + \frac{1}{2}\sum_{1\leq i<l} \|v_{il}\|^2 + \frac{1}{2}\sum_{l<j\leq r} \|v_{lj}\|^2. \tag{25.2}$$

This gives that $x_l y_l \leq 0$. Since $[x_1, x_2, ..., x_r]^T$ is in range of A, and by our assumption A is range column sufficient, then $x_l y_l = 0$. From Eq. (25.2), we have $v_{il} = v_{lj} = 0$ for all $1 \leq i < l$ and $l < j \leq r$. Then $x \circ R_A(x) = 0$. Thus R_A has the rCS property.
(ii) $\Longrightarrow$ (i): Let $x = [x_1, x_2, ..., x_r]^T$ be in $R(A)$ and $x * Ax \leq 0$. Then there will be $y \in \mathbb{R}^r$ such that $A(y) = x$. Now set $z := \sum_{i=1}^{r} x_i e_i$, then $R_A(z) = \sum_{i=1}^{r} (Ax)_i e_i$. Here $z = \sum_{i=1}^{r} (Ay)_i e_i$, as $A(y) = x$ in $\mathbb{R}^r$ which gives z is in range of R_A and $z \circ R_A(z) =$

$\sum_{i=1}^{r} x_i(Ax)_i e_i \leq 0$. Since R_A has the rCS property, $z \circ R_A(z) = 0$ which gives that $x * Ax = 0$. Thus A is range column sufficient. □

Theorem 25.13 *Let $A \in \mathbb{R}^{r\times r}$. Then A is strictly range semimonotone if and only if R_A has the pSSM property.*

Proof By using a similar argument as in the above theorem, this can be proved easily. □

25.6 Conclusion

In this paper, we have introduced and studied the rCS property and the pSSM property on EJA. We have concluded that the intersection of the solution set of LCP with the range of linear transformation is convex if it has the rCS property. We showed that the linear transformation $\mathcal{L}$ has the pSSM property with the range cross commutative property on $\mathcal{K}$ if and only if the intersection of the solution set of LCP with the range of linear transformation is a zero set on the symmetric cone. In particular, we have shown the equivalence of the pSSM and the rCCS property for some special transformations. Finally, we have characterized the rCS and the pSSM property for the relaxation transformation.

Declaration of Competing Interest The authors have no competing interests.

Acknowledgements The first author is a CSIR-SRF fellow, and he wants to thank the Council of Scientific & Industrial Research (CSIR) for the financial support.

References

1. Ben-Israel A, Greville TNE (2002) Generalized inverses: theory and applications. Springer-Verlag, New york
2. Cottle RW, Pang JS, Stone RE (1992) The linear complementarity problem. Academic Press
3. Faraut J, Korainy A (1994) Analysis on symmetric cones. Oxford University Press
4. Gowda MS, Parthasarathy T (2000) Complementarity forms of theorems of Lyapunov and Stein, and related results. Linear Algebra Appl 320:131–144
5. Gowda MS, Song Y (2002) Semidefinite relaxations of linear complementarity problems. Technical Report, Department of Mathematics and Statistics, University of Maryland Baltimore County, Baltimore MD 21250
6. Gowda MS, Song Y (2010) On semidefinite linear complementarity problems. Math Progam Ser A 88:575–587
7. Gowda MS, Sznajder R (2006) Automorphism invariance of p-and gus-properties of linear transformations on euclidean jordan algebras. Math Oper Res 30:109–123
8. Gowda MS, Tao J (2009) Z-transformations on proper and symmetric cones. Math Program Ser B117:195–221

9. Gowda MS, Sznajder R, Tao J (2004) Some P-properties for linear transformations on euclidean jordan algebras. Linear Algebra Appl 393:20–23
10. Hans S (1965) Positive operators and an inertia theorem. Numerische Mathematik 7:11–17
11. Jeyaraman I, Sivakumar KC (2016) Complementarity properties of singular M-matrices. Linear Algebr Appl 510:42–63
12. Jeyaraman I, Bisht K, Sivakumar KC (2017) Extensions of P-property, R_0 property and semidefinite linear complementarity problems. Yugosl J Oper Res 27:135–151
13. Pang JS, Venkateswaran (1989) Sufficient matrices and the linear complementarity problem. Linear Algebra App 114(115):231–249
14. Peris JE, Subiza B (1992) A characterization of weak-monotone matrices. Linear Algebra Appl 166:167–184
15. Robert P (1998) On the group inverse of a linear transformation. J Math Anal Appl 20(3):658–669
16. Schneider H, Vidyasagar M (1970) Cross-positive matrices. SIAM J Numer Anal 7:508–519
17. Tao J (2010) Strict semimonotonicity property of linear transformations on euclidean jordan algebras. J Optim Theory Appl 144:575–596
18. Tao J, Jeyaraman I, Ravindran G (2016) More results on column sufficiency property in Euclidean Jordan algebras. Ann Oper Res 243:229–243

Chapter 26
On the Proofs of Formulae by Mahāvīra and Brahmagupta

Narayana Acharya

Abstract In this article, independent proofs for the formulae enunciated by Indian mathematician Mahāvīra for the sum of squares and the sum of cubes of n terms of an arithmetic sequence are given. Also, a detailed study of cyclic quadrilaterals is made in the light of the works done by another Indian mathematician Brahmagupta. The formulae given by Brahmagupta for the area of a cyclic quadrilateral and lengths of the diagonals of a cyclic quadrilateral are proved. Further, a formula for the area of a general quadrilateral, which is not cyclic, is obtained.

Keywords Arithmetic sequence · Quadrilaterals · Cyclic quadrilaterals

Mathematics Subject Classification (2010) 01A32 · 40-03 · 11B25 · 51A05 · 97G60

26.1 On Mahāvīra Formulae for Sum of the Squares and Sum of the Cubes of n Terms of an Arithmetic Sequence

Mahāvīrācārya (850 A. D) in his Gaṇita-sāra-saṅgraha (see, [2]), Chap. 6, shloka 298 says as follows:

द्विगुणैकोनपदोत्तर कृतिहती षष्टांश मुखचयाहतयुतिः

व्योकपदघ्ना मुखाकृतिसहिता पदताडितेष्ट कृतिविथिका

This shloka enunciates a formula to find the sum of squares of the first n terms of an arithmetic sequence whose first term is a and common difference d. If we denote the sum of the squares of n terms as S_2 it says

N. Acharya (✉)
Retired Professor, Bhandarkars' Arts & Science College Kundapura, Udupi, KA, India
e-mail: n.acharya1938@gmail.com

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2023

R. B. Bapat et al. (eds.), *Applied Linear Algebra, Probability and Statistics*,
Indian Statistical Institute Series,
https://doi.org/10.1007/978-981-99-2310-6_26

$$S_2 = \left[\left\{\frac{2n-1}{6}d^2 + ad\right\}(n-1) + a^2\right]n,$$

where n stands for the number of terms of the arithmetic series. Also, in another shloka No. 304 of the same chapter, he says

चित्यादि हति मुखचय शोषघ्ना प्रचयनिघ्न चितिवर्गे

आदौप्रचयादूने वियुतामुक्ताधिके तु घन चितिका

which means sum S_3 of the cubes of n terms of the same arithmetic sequence is equal to $S_1^2 d + S_1 a(a-d)$, where S_1 denotes the sum of n terms of the arithmetic sequence.

This formula enunciated by Mahāvīrācārya is not proved so far anywhere. We present the proof for these here below.

First, we recall that a sequence of numbers $t_1, t_2, \cdots, t_n, \cdots$ is an arithmetic sequence if $t_1 = a$ and $t_{i+1} - t_i = d$ (a constant) for all $i \geq 1$. Term t_1 is called the first term and t_i denotes the ith term of the sequence and d is called the common difference of the arithmetic sequence.

We make the following notation:

$$t_1 + t_2 + \cdots + t_n = \sum_{i=1}^{n} t_i = S_1$$

$$t_1^2 + t_2^2 + \cdots + t_n^2 = \sum_{i=1}^{n} t_i^2 = S_2$$

$$t_1^3 + t_2^3 + \cdots + t_n^3 = \sum_{i=1}^{n} t_i^3 = S_3.$$

First, we get the formula for t_n and formula for S_1.

From the definition of the arithmetic sequence

$$t_n - t_1 = t_n - a = \sum_{i=1}^{n-1}(t_{i+1} - t_i) = (t_n - t_{n-1}) + (t_{n-1} - t_{n-2}) + \cdots + (t_2 - t_1).$$

Since $t_{i+1} - t_i = d$ for all i, $t_n - a = (n-1)d$. Therefore

$$t_n = a + (n-1)d. \tag{26.1}$$

Now consider

$$\begin{aligned} t_{n+1}^2 - t_1^2 &= t_{n+1}^2 - a^2 \\ &= \sum_{i=1}^{n} (t_{i+1}^2 - t_i^2) \\ &= (t_{n+1}^2 - t_n^2) + (t_n^2 - t_{n-1}^2) + \cdots + (t_2^2 - t_1^2). \end{aligned}$$

Use the algebraic identity $a^2 - b^2 = (a - b)(a + b)$ on both the sides of the above equation. We get

$$\begin{aligned} (a + nd)^2 - a^2 &= (a + nd + a)(a + nd - a) \\ &= \sum_{i=1}^{n} (t_{i+1} - t_i)(t_{i+1} + t_i) \\ &= \sum_{i=1}^{n} d(t_i + d + t_i) \\ &= d \sum_{i=1}^{n} (2t_i + d) \\ &= d(2S_1 + nd) \\ \text{i.e.,} \quad nd(2a + nd) &= d(2S_1 + nd) \\ \Longrightarrow \quad n(2a + nd) &= 2S_1 + nd \end{aligned}$$

or

$$n(2a + nd - d) = 2S_1$$

or

$$S_1 = \frac{n}{2}(2a + (n - 1)d). \tag{26.2}$$

It can also be observed that

$$S_1 = n\left(\frac{t_1 + t_n}{2}\right). \tag{26.3}$$

Now we proceed to get the formula for S_2.
Consider

$$\begin{aligned} t_{n+1}^3 - t_1^3 &= t_{n+1}^3 - a^3 \\ &= \sum_{i=1}^{n} (t_{i+1}^3 - t_i^3) \\ &= (t_{n+1}^3 - t_n^3) + (t_n^3 - t_{n-1}^3) + (t_{n-1}^3 - t_{n-2}^3) + \cdots + (t_2^3 - t_1^3). \end{aligned}$$

Now use the algebraic identity $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ on either side of the above equation.

$$(t_{n+1}-a)(t_{n+1}^2+at_{n+1}+a^2)=\sum_{i=1}^{n}(t_{i+1}-t_i)(t_{i+1}^2+t_it_{i+1}+t_i^2)$$

$$\implies (a+nd-a)(\overline{a+nd}^2+a(\overline{a+nd}+a^2))=\sum_{i=1}^{n}d\Big[(t_i+d)^2+t_i(t_i+d)+t_i^2\Big]$$

$$\implies nd(3a^2+3and+n^2d^2)=d\sum_{i=1}^{n}(3t_i^2+3dt_i+d^2)$$

$$\implies n(3a^2+3and+n^2d^2)=3\sum_{i=1}^{n}t_i^2+3d\sum_{i=1}^{n}t_i+d^2n=3S_2+3dS_1+d^2n$$

$$\implies S_2+dS_1+\frac{d^2n}{3}=n\Big[a^2+and+\frac{n^2d^2}{3}\Big]$$

Therefore

$$\begin{aligned}
S_2&=n\left\{a^2+and+\frac{n^2d^2}{3}\right\}-d\frac{n(2a+\overline{n-1}d)}{2}-\frac{d^2n}{3}\\
&=n\left\{a^2+\frac{n^2d^2}{3}-\frac{(n-1)d^2}{2}-\frac{d^2}{3}+ad(n-1)\right\}\\
&=n\left\{a^2+(n-1)\Big((n+1)\frac{d^2}{3}-\frac{d^2}{2}\Big)+ad(n-1)\right\}\\
&=n\left\{a^2+(n-1)\Big[ad+d^2\Big(\frac{2n-1}{6}\Big)\Big]\right\}\\
&=n\left\{a^2+(n-1)\Big(ad+\frac{2n-1}{6}d^2\Big)\right\}
\end{aligned}\tag{26.4}$$

This is the formula enunciated by Mahāvīrācārya mentioned in the first shloka above.

Now we proceed further to simplify this formula to a more elegant form.

$$\begin{aligned}
S_2&=n\left\{a^2+(n-1)\Big(ad+\frac{2n-1}{6}d^2\Big)\right\}\\
&=\frac{n}{6}\Big[6a^2+6(n-1)ad+(2n-1)d^2(n-1)\Big]\\
&=\frac{n}{6}\left\{6a^2+6(n-1)ad+\Big[2(n-1)+1\Big]d^2(n-1)\right\}\\
&=\frac{n}{6}\left\{2a^2+4a^2+6(n-1)ad+(n-1)d^2+2(n-1)^2d^2\right\}\\
&=\frac{n}{6}\left\{(2a^2+\overline{n-1}d^2)+2(2a+\overline{n-1}d)\Big[a+\overline{n-1}d\Big]\right\}\\
&=\frac{1}{3}\left\{\frac{n(2a^2+\overline{n-1}d^2)}{2}+2\Big[\frac{n}{2}(2a+\overline{n-1}d)\Big](a+\overline{n-1}d)\right\}.
\end{aligned}$$

Now recall that $\frac{n}{2}(2a + \overline{n-1}d) = S_1$, the sum of first n terms of the given arithmetic sequence whose first term is a and common difference d and $\frac{n}{2}(2a^2 + \overline{n-1}d^2)$ is the sum denoted by $\overline{S}_1$ of the first n terms of the associated arithmetic sequence whose first term is a^2 and common difference d^2 and $a + \overline{n-1}d = t_n$ the nth term of the given arithmetic sequence. Then it follows that

$$S_2 = \frac{1}{3}\Big\{\overline{S}_1 + 2t_n S_1\Big\}.$$

This is the most elegant form for the sum S_2 identified by the present author.

Now we proceed further to get the Mahāvīra formula for S_3, the sum of the cubes of n terms of the given arithmetic sequence whose first term is a and common difference d.

Consider

$$\begin{aligned}\sum_{i=1}^{n}(t_{i+1}^4 - t_i^4) &= (t_{n+1}^4 - \cancel{t_n^4}) + (\cancel{t_n^4} - \cancel{t_{n-1}^4}) + \cdots + (\cancel{t_2^4} - t_1^4)\\ &= t_{n+1}^4 - t_1^4\\ &= (a + nd)^4 - a^4.\end{aligned}$$

Observe that $A^4 - B^4 = (A^2 + B^2)(A^2 - B^2) = (A^2 + B^2)(A + B)(A - B)$. We use this identity on either side of the above equation. Then we get

$$\begin{aligned}\text{R. H. S} = (a + nd)^4 - a^4 &= \Big[(a + nd)^2 + a^2\Big]\Big[a + nd + a\Big][nd]\\ &= nd(2a + nd)(2a^2 + 2and + n^2d^2)\\ &= dn(2a + \overline{n-1}d + d)\big[2a^2 + nd(2a + \overline{n-1}d + d)\big]\\ &= d(2S_1 + nd)\big[2a^2 + d(2S_1 + nd)\big].\end{aligned}$$

$$\begin{aligned}\text{L. H. S} &= \sum_{i=1}^{n}(t_{i+1}^2 - t_i^2)(t_{i+1}^2 + t_i^2)\\ &= \sum_{i=1}^{n}(t_{i+1} - t_i)(2t_i + d)\big[(t_i + d)^2 + t_i^2\big]\\ &= d\sum_{i=1}^{n}(2t_i + d)(2t_i^2 + 2dt_i + d^2)\\ &= d\sum_{i=1}^{n}(4t_i^3 + 6dt_i^2 + 4d^2t_i + d^3)\\ &= d(4S_3 + 6dS_2 + 4d^2S_1 + d^3n).\end{aligned}$$

Equating the simplified results on L. H. S and R. H. S, we get

$$d(4S_3 + 6dS_2 + 4d^2S_1 + d^3n) = d(2S_1 + nd)\big[2a^2 + d(2S_1 + nd)\big]$$
$$\therefore\ 4S_3 + 6dS_2 + 4d^2S_1 + d^3n = (2S_1 + nd)(2a^2) + d(2S_1 + nd)^2$$

Substituting for S_2 from the previous Mahāvīra formula as modified, we get

$$\begin{aligned}
&4S_3 + d(2\overline{S}_1 + 4(a + \overline{n-1}d))S_1 + 4d^2S_1 + d^3n \\
&= 4a^2S_1 + 2a^2nd + d(4S_1^2 + 4S_1nd + n^2d^2) \\
&= 4a^2S_1 + nd(2a^2 + (\overline{n-1} + 1)d^2) + 4S_1^2d + 4S_1nd^2 \\
&= 4a^2S_1 + 4S_1^2d + 2\overline{S}_1d + nd^3 + 4S_1nd^2
\end{aligned}$$

$$\begin{aligned}
&\therefore\ 4S_3 = 4a^2S_1 + 4S_1^2d + \cancel{2d\overline{S}_1} + \cancel{4S_1nd^2} - \cancel{2d\overline{S}_1} - 4adS_1 - \cancel{4(n-1)d^2S_1} \\
&\implies\ 4S_3 = 4a^2S_1 - 4S_1ad + 4S_1^2d \\
&\implies\ S_3 = (a^2 - ad)S_1 + dS_1^2 \\
&\implies\ S_3 = S_1^2d + a(a-d)S_1.
\end{aligned}$$

This is Mahāvīra's formula enunciated in the second shloka above to find the sum of the cubes of n terms of the given arithmetic sequence.

If we put $a = d = 1$ in these formulae

$$S_2 = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$
$$S_3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = 1 \cdot S_1^2 = \left(\frac{n(n+1)}{2}\right)^2$$

because then

$$S_1 = \frac{n(n+1)}{2}$$
$$\overline{S}_1 = \frac{n(n+1)}{2}.$$

These formulae for this special case has been mentioned by Āryabhaṭa in his Āryabhaṭīyam ([4]) as Krticitika Sūtra and Ghanaciti Sūtra.

26.2 A Study of Cyclic Quadrilaterals

While discussing cyclic quadrilaterals, Brahmagupta in his book Brāhma-sphuṭa-siddhānta (see [3]) gives two formulae one for finding the area of the cyclic quadrilateral and the other for finding out the measures of the diagonals of the cyclic quadrilateral.

In Chap. 12, Shloka 21, Brahmagupta says:

स्थूलफलम् त्रिचतुर्भुज बाहु प्रतिबाहु योगदलाघात:

भुजयोगार्धचतुष्टय भुजोनघातात् पदम् सूक्ष्मम्

Second line of the shloka says if s is the semi perimeter of the quadrilateral (cyclic) and a, b, c, d are the sides, the exact area is $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.

Also, in Shloka 28 of the same chapter, he says

कर्णाश्रित भुजघातैक्यमुभयान्योन्य भाजितम् गुनायेत्

योगेन भुजप्रतिभूज नधायों: कर्णे पदे विषमे

This says that the lengths of the diagonals of the cyclic quadrilaterals are

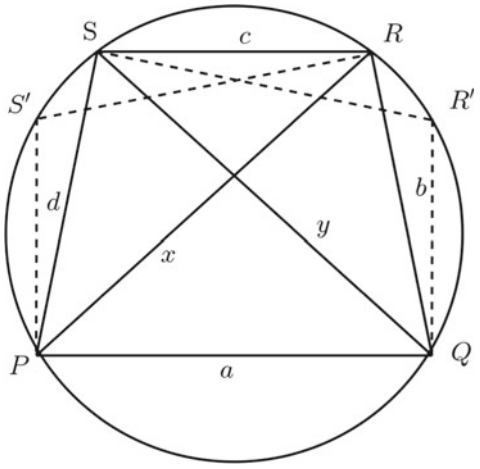
$$\sqrt{\frac{ab+cd}{ad+bc}(ac+bd)} \quad \text{and} \quad \sqrt{\frac{ad+bc}{ab+cd}(ac+bd)}.$$

These formulae have been only enunciated and not proved. In this chapter, we make a study in detail of the cyclic quadrilateral where we present the proofs for the above results and also get some more relevant formulae that can be used beneficially.

First of all, we fix some notations and give certain definitions that we are going to use in our study.

Let $\mathcal{S}$ be a circle of a certain chosen radius r. Let P, Q, R, and S be four points (distinct) on this circle taken for convenience in the anticlockwise order. Then the quadrilateral obtained by joining PQ, QR, RS, and SP is called a cyclic quadrilateral. In other words, a quadrilateral $PQRS$ whose vertices all lie on one and the same circle is called a cyclic quadrilateral. The circle $\mathcal{S}$ which passes through the vertices P, Q, R, S is called the circumcircle of the quadrilateral. Let $PQ = a,\ QR = b,\ RS = c$, and $SP = d$ be the measures of the sides of this quadrilateral. P, Q, R, and S are vertices of the quadrilateral. PQ, QR, RS, and SP are the sides of the quadrilateral. The pairs (P, R) and (Q, S) are pairs of opposite vertices. The line segments joining PR and QS are called the diagonals of the quadrilateral. (PQ, RS), (PS, QR) are pairs of opposite sides. Let $PR = x$ and $QS = y$ be the measures of the diagonals. Largest side of the quadrilateral (without loss of generality can be chosen to be $PQ = a$) will be called the base of the quadrilateral. Its opposite side $RS = c$ is called the face of the quadrilateral. Remaining two sides

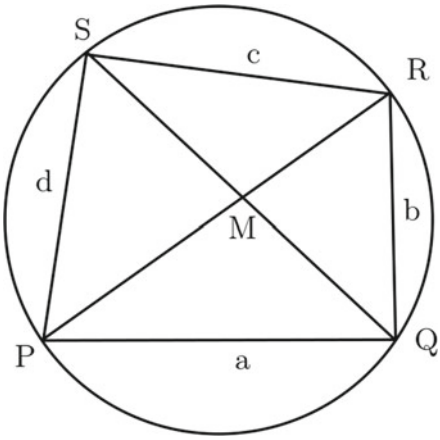
b and d which meet the base at Q and P, respectively, are called adjacent sides of the quadrilateral. The quadrilateral $PQRS$ will be denoted by $\mathcal{Q}_c$, the letter $\mathcal{Q}$ represents the quadrilateral and the subscript c attached to $\mathcal{Q}$ denotes the face of the quadrilateral. Now, mark two points R' and S' on the arc of the circle determined by the base PQ which contains R and S, such that $QR' = PS' = RS = c$. Then, arc $\widehat{QR'} = \widehat{PS'} = \widehat{SR}$. Now, arc SR' = arc QS − arc QR' = arc QS − arc RS = arc QR, and therefore, arc SR' = arc QR. Hence, chord $R'S$ = chord $QR = b$. Similarly, chord RS' = chord $PS = d$. Thus, we get two more cyclic quadrilaterals $PQR'S$ and $PQRS'$ both of which have same base. The face of $PQR'S$ is $R'S = b$ and face of $PQRS'$ is $RS' = d$. Thus, we have got these two quadrilaterals both of which are again cyclic and whose sides have same measure as those of $PQRS$ but they have faces b and d, respectively, and we denote them by $\mathcal{Q}_b$ and $\mathcal{Q}_d$. The quadrilaterals $\mathcal{Q}_c$, $\mathcal{Q}_b$, $\mathcal{Q}_d$ have the following properties: The diagonals of $\mathcal{Q}_c$ are PR and QS, diagonals of $\mathcal{Q}_b$ are PR' and QS, and diagonals of $\mathcal{Q}_d$ are QS' and PR. Thus, $\mathcal{Q}_b$ and $\mathcal{Q}_d$ have one diagonal common with that of $\mathcal{Q}_c$ the uncommon diagonals PR' and QS' are got in the process of changing the face c with b and c with d, respectively, are such that arc PR' = arc PQR' = arc PQ + arc QR' = arc PQ + arc PS' ($\because$ $\widehat{PS'} = \widehat{QR'} = \widehat{QR}$) = arc QS'. Therefore, chord QS' = chord PR'. Hence, these uncommon diagonals are congruent to each other and their measure will be denoted by z which will be called para-diagonal or transcendental diagonal of $\mathcal{Q}_c$. Thus, $\mathcal{Q}_c$ has two actual diagonals x, y and para-diagonal z. Similarly, $\mathcal{Q}_b$ has actual diagonals y, z and para-diagonal x and $\mathcal{Q}_d$ has actual diagonals x, z and para-diagonal y.



Thus, every cyclic quadrilateral is associated with three diagonals two of them are the actual diagonals and the third is called para-diagonal which can be got by changing the face of that quadrilateral with one of its adjacent sides.

Now we proceed to associate with each cyclic quadrilateral three pairs of supplimental angles and find the connection between these three pairs of supplimental angles with the three diagonals of the quadrilateral.

We know that any chord of a circle subtends angle at the center and also at a point on the circumcircles and if θ_a is the measure subtended by a at the center, the angle subtended by a at a point on the circumference is $\frac{\theta_a}{2}$. Therefore



$$\angle PSQ = \angle PRQ = \frac{\theta_a}{2}.$$

Similarly

$$\angle QSR = \angle QPR = \frac{\theta_b}{2}$$
$$\angle RPS = \angle RQS = \frac{\theta_c}{2}$$

and

$$\angle PQS = \angle PRS = \frac{\theta_d}{2}.$$

We see that

$$\theta_a + \theta_b + \theta_c + \theta_d = 360^\circ$$
$$\therefore \ \frac{\theta_a + \theta_b + \theta_c + \theta_d}{2} = 180^\circ.$$

Thus,

$$\left(\frac{\theta_a+\theta_b}{2},\frac{\theta_c+\theta_d}{2}\right),\left(\frac{\theta_a+\theta_c}{2},\frac{\theta_b+\theta_d}{2}\right),\text{ and }\left(\frac{\theta_b+\theta_c}{2},and\frac{\theta_a+\theta_d}{2}\right)$$

are three pairs of supplementary angles.

Observe that

$$\frac{\theta_a+\theta_b}{2}=\angle PSR \text{ and } \frac{\theta_c+\theta_d}{2}=\angle PQR.$$

These are pair of vertically opposite angle at Q and S of the quadrilateral which is angle made by diagonal PR at the remaining two vertices Q and S facing it.

Similarly,

$$\frac{\theta_a+\theta_d}{2}=\angle QRS \text{ and } \frac{\theta_a+\theta_c}{2}=\angle QPS$$

and this second pair of opposite angle at the vertices P and R made by diagonal QS at the pair of vertices P and Q facing it and the third pair

$$\frac{\theta_a+\theta_c}{2}=\angle RMS=\angle PMQ,\ \frac{\theta_b+\theta_d}{2}=\angle PMS=\angle QMR,$$

where M is the point of intersection of PR and QS.

This pair of supplementary angles are linear pair of angles between the actual diagonals of the quadrilateral $PQRS$. We can also see that this pair of angles is actually the pair of opposite angles of the quadrilateral $\mathcal{Q}_b$ facing the transcendental diagonal z at the vertices P and S.

Now we proceed to prove the Brahmagupta formula for diagonals of the quadrilateral.

To prove this, we need the cosine formula of trigonometry. If $\triangle ABC$ is any triangle with sides $BC=a,\ CA=b,\ andAB=c$, then $c^2=a^2+b^2-2ab\ \cos C$. This we assume without proof. The proof of this can be given by using Bodhayana Sutra (the so called Pythagoras theorem) which we will not give here.

Let $PQRS$ be the cyclic quadrilateral with side $PQ=a$, $QR=b$, $RS=c$, and $SP=d$. Let the diagonal $PQ=x$ and $QS=y$. Using cosine rule for $\triangle PQR$ and $\triangle PSR$ separately,

$$\begin{aligned} PR^2=x^2&=PQ^2+QR^2-2PQ\cdot QR\cdot\cos Q\\ &=a^2+b^2-2ab\cos Q \qquad (26.5)\\ x^2&=PS^2+SR^2-2PS\cdot SR\cos S\\ &=c^2+d^2-2cd\cos S \qquad (26.6)\end{aligned}$$

Multiply equation (26.5) by cd and (26.6) by ad and add the two. We get

$$(ab+cd)x^2=cd(a^2+b^2)+ab(c^2+d^2)-4abcd(\cos Q+\cos S).$$

Since Q and S are supplementary angles, $S = 180 - Q$. Therefore, $\cos S = -\cos Q$ and hence $\cos Q + \cos S = 0$.

Therefore

$$\begin{aligned}(ab+cd)x^2 &= cd(a^2+b^2)+ab(c^2+d^2)\\ &= ac(ad+bc)+bd(ad+bc)\\ &= (ad+bc)(ac+bd)\\ \therefore\ x^2 &= \left(\frac{ad+bc}{ab+cd}\right)(ac+bd)\\ \text{or}\quad x &= \sqrt{\left(\frac{ad+bc}{ab+cd}\right)(ac+bd)}. \end{aligned} \tag{26.7}$$

Now, we apply cosine formula for $\triangle PSQ$ and $\triangle RSQ$ separately. We get

$$y^2 = a^2+d^2+-2ad\cos P \tag{26.8}$$

$$= b^2+c^2-2bc\cos R \tag{26.9}$$

$$= b^2+c^2-2bc\cos P,$$

as $\cos R = -\cos P$, P and Q are supplementary pairs. Therefore

$$\begin{aligned}(bc+ad)y^2 &= bc(a^2+d^2)+ad(b^2+c^2)\\ &= (ab+cd)(ac+bd)\\ \therefore\ y^2 &= \frac{ab+cd}{ad+bc}(ac+bd)\\ \text{or}\quad y &= \sqrt{\frac{ab+cd}{ad+bc}(ac+bd)} \end{aligned} \tag{26.10}$$

The value of x and y in (26.7) and (26.10) are actually the Brahmagupta formulae enunciated by the relevant shloka given above.

If we use these results for the quadrilateral $PQR'S$ or $PQRS'$, we will get

$$z = \sqrt{\frac{(ab+cd)(ad+bc)}{ac+bd}} \tag{26.11}$$

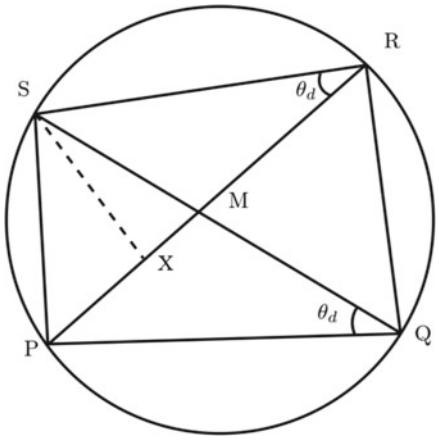
Thus, we have got all the three diagonals of the quadrilateral.

Now, if we multiply the results in (26.7) and (26.10) above, we get $xy = ac+bd$ which is well known as Ptolemy's theorem (see, [5, pp. 50]).

Theorem 26.1 *The product of the diagonals of the cyclic quadrilateral is equal to the sum of the products of the pairs of opposite sides of the quadrilateral.*

Now we prove Theorem 26.1 independently (without using trigonometry) using the properties of similar triangles.

Pert. enunciation $PQRS$ is a cyclic quadrilateral with sides $PQ = a$, $QR = b$, $RS = c$, and $SP = d$. PR and QS are diagonals x and y, respectively, intersecting at M.



Construction Draw SX such that $\angle PSQ = \angle RSX$ to meet PQ at X.

Proof Consider $\triangle PSQ$ and $\triangle XSR$, where $\angle PSQ = \angle XSR$, by construction, $\angle SRX = \angle SRP = \angle SQP = \theta_d$. Therefore, the third angle of these triangles, namely, $\angle SPQ = \angle SXR$. Hence, these two triangles are similar. Thus

$$\frac{PQ}{XR} = \frac{SQ}{SR} \implies \frac{a}{XR} = \frac{y}{c}$$
$$\text{or} \quad y \cdot XR = ac. \tag{26.12}$$

Similarly, $\triangle PSX$ and $\triangle QSR$ have the corresponding angles $\angle PSX = \angle QSR$ and $\angle SPR = \angle SQR = \theta_c$.

Therefore, thr third angle of these triangles are equal. Hence, they are similar. Therefore

$$\frac{XP}{QR} = \frac{PS}{QS} \implies \frac{XQ}{b} = \frac{d}{y}$$
$$\text{or} \quad y \cdot XP = bd. \tag{26.13}$$

Adding (26.12) and (26.13)

$$ac + bd = y \cdot XP + y \cdot XR = y(XP + XR) = y \cdot x.$$

This proves the Theorem 26.1. □

Having proved Theorem 26.1 independently, we can use this theorem separately to $\mathcal{Q}_c$, $\mathcal{Q}_b$, and $\mathcal{Q}_d$. From $\mathcal{Q}_c$, we have $xy = ac + bd$, from $\mathcal{Q}_b$, we have $yz = ab + cd$, and from $\mathcal{Q}_c$, we have $xz = ad + bc$.

We can use these results to prove Brahmagupta's formulae for the measure of the diagonals as follows:

$$x^2 = \frac{xy \cdot xz}{yz} = \frac{(ac + bd)(ad + bc)}{ab + cd}$$
$$\implies x = \sqrt{\frac{(ac + bd)(ad + bc)}{ab + cd}}. \tag{26.14}$$

Similarly

$$y^2 = \frac{xy \cdot yz}{xz} = \frac{(ac + bd)(ab + cd)}{ad + bc}$$
$$\implies x = \sqrt{\frac{(ac + bd)(ab + cd)}{ad + bc}}. \tag{26.15}$$

Also

$$z^2 = \frac{xz \cdot yz}{xy} = \frac{(ab + cd)(ad + bc)}{ac + bd}$$
$$\implies x = \sqrt{\frac{(ab + cd)(ad + bc)}{ac + bd}}. \tag{26.16}$$

This proof of the Brahmagupta formula is independent of the application of the trigonometric formula and is simple and elegant and purely geometrical.

Now we proceed to prove the Brahmagupta formula for the area enveloped by the quadrilateral $PQRS$.

Using the cosine formula for the triangle $\triangle PQR$ and $\triangle PRS$ separately, we have seen that

$$x^2 = a^2 + b^2 - 2ab\cos Q = c^2 + d^2 - 2cd\cos S$$
$$= (a + b)^2 - 2ab(1 + \cos Q) = (a + b)^2 - 4ab\cos^2\frac{Q}{2} \tag{26.17}$$
$$= (a - b)^2 + 2ab(1 - \cos Q) = (a - b)^2 + 4ab\sin^2\frac{Q}{2}. \tag{26.18}$$

Similarly

$$x^2 = (c + d)^2 - 4cd\cos^2\frac{S}{2} \tag{26.19}$$
$$= (c - d)^2 + 4cd\sin^2\frac{S}{2}. \tag{26.20}$$

From (26.17) and (26.20)

$$\begin{aligned}(a+b)^2 - 4ab\cos^2\frac{Q}{2} &= (c-d)^2 + 4cd\sin^2\frac{S}{2}\\ \implies (a+b)^2 - (c-d)^2 &= 4(ab\cos^2\frac{Q}{2} + cd\sin^2\frac{S}{2})\\ &= 4(ab+cd)\cos^2\frac{Q}{2} \qquad \because \sin\frac{S}{2} = \cos\frac{Q}{2}.\end{aligned}$$

From (26.18) and (26.19)

$$\begin{aligned}(c+d)^2 - 4cd\cos^2\frac{S}{2} &= (a-b)^2 + 4ab\sin^2\frac{Q}{2}\\ \implies (c+d)^2 - (a-b)^2 &= 4(ab+cd)\sin^2\frac{Q}{2} \qquad (26.21)\end{aligned}$$

$\because \sin\frac{Q}{2} = \cos\frac{S}{2}$ as S and Q are supplementary angles.

$$\text{Or} \quad (a+b+c-d)(a+b-c+d) = 4(ab+cd)\cos^2\frac{Q}{2} \qquad (26.22)$$

$$\text{and} \quad (c+d+a-b)(c+d-a+b) = 4(ab+cd)\sin^2\frac{Q}{2}. \qquad (26.23)$$

If we denote

$$\frac{a+b+c+d}{2} = s,$$

we have

$$a+b+c-d = a+b+c+d-2d = 2(s-d).$$

Similarly

$$\begin{aligned}a+b+d-c &= 2(s-c)\\ a+c+d-b &= 2(s-b)\\ b+c+d-a &= 2(s-a).\end{aligned}$$

Substituting in (26.22) and (26.23) and simplifying, we get

$$\begin{aligned}2(s-d)\cdot 2(s-c) &= 4(ab+cd)\cos^2\frac{Q}{2}\\ \implies (s-c)(s-d) &= (ab+cd)\cos^2\frac{Q}{2} \qquad (26.24)\\ 2(s-a)2(s-b) &= 4(ab+cd)\sin^2\frac{Q}{2}\\ \implies (s-a)(s-b) &= (ab+cd)\sin^2\frac{Q}{2}. \qquad (26.25)\end{aligned}$$

Multiplying (26.24) and (26.25), we get

$$\begin{aligned}(s-a)(s-b)(s-c)(s-d) &= (ab+cd)^2 \cdot \sin^2\frac{Q}{2}\cos^2\frac{Q}{2}\\ &= \left[\frac{1}{2}(ab+cd)\sin Q\right]^2\\ &= \left[\frac{1}{2}ab\sin Q+\frac{1}{2}cd\sin Q\right]^2\\ &= \left[\frac{1}{2}ab\sin Q+\frac{1}{2}cd\sin S\right]^2 \quad \because \sin Q = \sin S\\ &= (\text{Area of } \triangle PQR + \text{Area of } \triangle PRS)^2\\ &= (\text{Area of the quadrilateral } PQRS)^2.\end{aligned}$$

Therefore, area of the cyclic quadrilateral $PQRS = \sqrt{(s-a)(s-b)(s-c)(s-d)}$.

Thus, we have obtained the Brahmagupta formula for the area mentioned in the above-mentioned shloka at the beginning of this section.

Remark 26.1 If we allow the point S to move along the circle toward P and ultimately coincide with P, we will have $d \to 0$ and the quadrilateral $PQRS$ collapses to triangle PQR and the Brahmagupta's formula for the area of cyclic quadrilateral $PQRS$ reduces to the formula for the area of $\triangle ABC$ by putting $d = 0$.

Therefore, area of the triangle whose sides are a, b, c, $s = \frac{a+b+c}{2}$ is

$$\text{Area of } \triangle PQR = \sqrt{(s-a)(s-b)(s-c)s}.$$

This is Heron's formula which is also Bhaskara's formula for the area of the triangle. This occurs as a particular case of the Brahmagupta's formula for area of the cyclic quadrilateral by putting $d = 0$.

We now observe that $\mathcal{Q}_c$, $\mathcal{Q}_b$, and $\mathcal{Q}_d$ have same area.

Area of $\mathcal{Q}_b$ = Area of $\triangle PQS$ + Area of $\triangle PR'S$. By construction of $PQR'S$, we know $\triangle PR'S$ is congruent to $\triangle QSR$. Therefore

$$\begin{aligned}\text{Area of } \triangle PR'S &= \text{Area of } \triangle QSR\\ \therefore\ \text{Area of } \mathcal{Q}_b &= \text{Area of } PQS + \triangle QSR\\ &= \text{Area of } PQRS\\ &= \text{Area of } \mathcal{Q}_c.\end{aligned}$$

Similarly,

$$\text{Area of } \mathcal{Q}_d = \text{Area of } \mathcal{Q}_c.$$

Therefore, area enclosed by $\mathcal{Q}_b$, $\mathcal{Q}_c$ and $\mathcal{Q}_d$ are equal.

Now we use the results derived above to find the formula for measures of vertex angles of the cyclic quadrilateral.

Now, from (26.24) and (26.25) above

$$\frac{(s-a)(s-b)}{(s-c)(s-d)} = \tan^2\frac{Q}{2} = \cot^2\frac{S}{2} \quad \because Q \text{ and } S \text{ are supplementary angles.}$$

$$\tan\frac{Q}{2} = \sqrt{\frac{(s-a)(s-b)}{(s-c)(s-d)}}, \quad \text{which gives a formula for } Q.$$

$$Q = 2\arctan\sqrt{\frac{(s-a)(s-b)}{(s-c)(s-d)}}$$

$$\tan\frac{S}{2} = \sqrt{\frac{(s-c)(s-d)}{(s-a)(s-b)}}$$

$$\text{or} \quad S = 2\arctan\sqrt{\frac{(s-c)(s-d)}{(s-a)(s-b)}}.$$

If we use cosine formula for triangles $\triangle PQS$, $\triangle RQS$ and proceed as before, we can get

$$\tan\frac{P}{2} = \sqrt{\frac{(s-a)(s-d)}{(s-b)(s-c)}}$$

and

$$\tan\frac{R}{2} = \sqrt{\frac{(s-b)(s-c)}{(s-a)(s-d)}}$$

so that

$$P = 2\arctan\sqrt{\frac{(s-a)(s-d)}{(s-b)(s-c)}}$$

and

$$S = 2\arctan\sqrt{\frac{(s-b)(s-c)}{(s-a)(s-d)}}.$$

If we do a similar exercise with $\mathcal{Q}_6 = PQR'S$, we can get vertex (para) angles of this quadrilateral facing the transcendental diagonals of $PQRS$ which happens to be the linear pair of angles between the diagonals as

$$2\arctan\sqrt{\frac{(s-c)(s-a)}{(s-b)(s-d)}}, \quad 2\arctan\sqrt{\frac{(s-b)(s-d)}{(s-a)(s-c)}}.$$

Thus, when the sides of a cyclic quadrilateral are given, we can find 3 quadrilaterals of the same area and we can find the vertex angles of the quadrilateral too, and also the lengths of the diagonals of these quadrilaterals.

Now we use the sine formula of trigonometry to find the relation between the sides, area, and the circumradius of the cyclic quadrilateral.

If we apply the sine formula to the triangles $\triangle PQR$, $\triangle PQS$, $\triangle PRS$, $\triangle QRS$, and $\triangle PR'S$

$$\frac{x}{\sin Q} = \frac{x}{\sin S} = \frac{y}{\sin P} = \frac{y}{\sin R} = \frac{z}{\sin \phi} = 2r,$$

where ϕ is the angle between actual diagonals. We also know that

$$\begin{aligned}
&\text{Area of the cyclic quadrilateral} = \frac{1}{2}(xy \sin \phi) \\
&\therefore \sqrt{(s-a)(s-b)(s-c)(s-d)} = \frac{1}{2} xy \frac{z}{2r} \quad \because z = 2r \sin \phi \\
\Longrightarrow\ & \frac{xyz}{4\sqrt{(s-a)(s-b)(s-c)(s-d)}} = r \\
&= \sqrt{\frac{x^2y^2z^2}{2(s-a) \cdot 2(s-b) \cdot 2(s-c) \cdot 2(s-d)}} \\
&= \sqrt{\frac{xy \cdot yz \cdot zx}{(a+b+c-d)(a+c+d-e)(b+c+d-a)(a+b+d-c)}} \\
&= \left(\frac{(ab+cd)(ad+bc)(ac+bd)}{(a+b+c-d)(a+c+d-b)(b+c+d-a)(a+b+d-c)} \right)^{\frac{1}{2}}.
\end{aligned}$$

This is the formula enunciated by Parameśvara in his commentary on Līlāvatī (reader may refer to [1]).

Finally, we make the following observation.

In any quadrilateral $PQRS$, the sum of any three sides is greater than the fourth, i.e.

$$\begin{aligned}
a + b + c &> d \\
b + c + d &> a \\
c + d + a &> b \\
d + a + b &> c.
\end{aligned}$$

Now if a, b, c, d are any four numbers satisfying these properties, i.e., sum of any three is greater than the fourth, we ask the following converse question: Is there a cyclic quadrilateral with a, b, c, d as lengths of its sides?

We answer this question below.

Call

$$s = \frac{a+b+c+d}{2}.$$

Then $\sqrt{(s-a)(s-b)(s-c)(s-d)}$ is meaningful because $(s-a)$, $(s-b)$, $(s-c)$, $(s-d)$ are all positive since the sum of any three of a, b, c, d is greater than the fourth.

Call

$$\frac{(ab+cd)(ac+bd)(ad+bc)}{4\sqrt{(s-a)(s-b)(s-c)(s-d)}} = r.$$

This r is a positive number if we draw a circle with this r as radius and on this circle. Choose arbitrarily a point P. Mark Q, R, and S such that $PQ = a$, $QR = b$, and $RS = c$ then by monotonicity of r as a function of $PS = d$ and we get the cyclic quadrilateral $PQRS = \mathcal{Q}_c$ by changing face with b or d we get two more quadrilaterals $\mathcal{Q}_b$ and $\mathcal{Q}_d$.

Thus, there will be up to congruence three quadrilaterals which are cyclic with these sides a, b, c, d.

Further, if we demand that the base is the largest and the face is the smallest, there is a unique cyclic quadrilateral with a as base, c as face, and b and d as adjacent sides $a > b, d > c$.

Such a quadrilateral has been considered by Brahmagupta where the base is the largest and the face is the shortest and is called the Brahmagupta quadrilateral.

Brahmagupta in his Brāhma-sphuṭa-siddhānta ([3]) explains a technique of finding a rational cyclic quadrilateral (Brahmagupta quadrilateral).

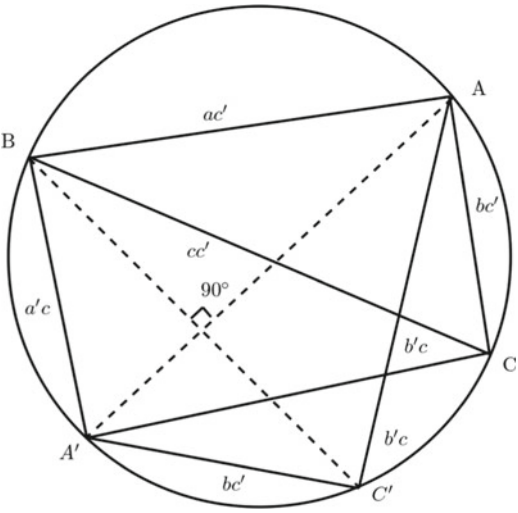
According to him, if (a, b, c) and (a', b', c') are two rational triples such that $a^2 + b^2 = c^2$ and $a'^2 + b'^2 = c'^2$, the numbers $ac', bc', a'c$, and $b'c$ are sides of a cyclic quadrilateral with largest among them as base and shortest as face.

This statement of Brahmagupta can be proved as follows: Since

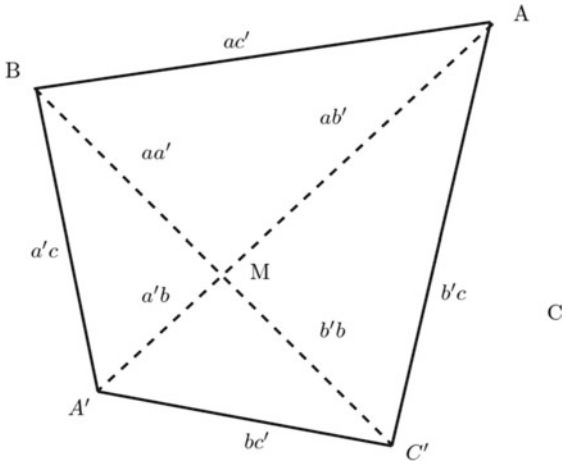
$$a^2 + b^2 = c^2 \quad \text{and} \quad a'^2 + b'^2 = c'^2$$
$$a^2c'^2 + c^2c'^2 = c^2c'^2 \quad \text{and} \quad a'^2c^2 + b'^2c^2 = c^2c'^2.$$

Therefore, (ac', bc') and $(a'c, b'c)$ are arms of two right angled triangles whose hypotenuse has measure cc'.

Now draw two right angled triangles ABC and $A'BC$ with $BC = cc'$. AB and AC are of measures ac' and bc', respectively. $A'B$ and $A'C$ are of measures $a'c$ and $b'c$. See that A and A' are chosen to be on opposite sides of BC as shown in the figure.



Now clearly $\angle A = 90° = \angle A'$. Therefore, circle on BC as diameter passes through A and A'. Therefore, $ABA'C$ is a cyclic quadrilateral. In this quadrilateral, let the largest side be ac' (without loss of generality we can assume this), then bc' will be the shortest and the face will be $A'C$. If we interchange the face $A'C$ with the adjacent side bc', we get the Brahmagupta quadrilateral $ABA'C$.



According to our discussion, the diagonals of this new quadrilateral (got by exchanging the face with the shortest side of the above quadrilateral) will have their diagonals AA' and BC' at right angles, BC will be the transcendental (para) diagonal of this new quadrilateral and its sides will be ac' base, bc' face, and $a'c$ and

$b'c$ will be adjacent sides. If the diagonals intersect at M, then $AM = ab',\ A'M = a'b,\ MB = aa',\ MC' = bb'$ and all sides and diagonals of the quadrilateral will be rational which is known as Brahmagupta quadrilateral. With this, we end the study of cyclic quadrilaterals.

We close this section on the study of cyclic quadrilaterals. We pass on to an appendix where we discuss the formula for the area of a general quadrilateral (not necessarily cyclic) and the formulae for diagonals when its sides are known and observe the formula we arrive at for the area of a general quadrilateral is just an extension of the Brahmagupta formula and we get Brahmagupta formula as a particular case of the generalized one.

Appendix

In this appendix, we make an attempt to get a formula for the area of a general quadrilateral which is not cyclic. Let $ABCD$ be a general non cyclic quadrilateral.

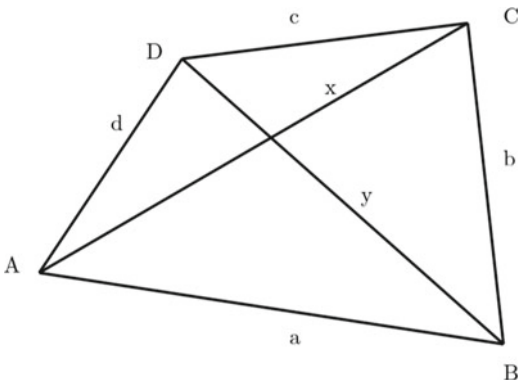
Since quadrilateral $ABCD$ is cyclic if and only if sum of the measures of pair of opposite angles is $180\,^\circ$, i.e., $A + C = B + D = 180\,^\circ$; for a noncyclic quadrilateral (general quadrilateral) $A + C \neq B + D$. Therefore, either $A + C > B + D$ or $B + D > A + C$. Without loss of generality for the sake of definiteness assume $A + C > B + D$. Then $\overline{A + C} - \overline{B + D} = 2\theta$ (say) is positive or $\frac{\overline{A+C}-\overline{B+D}}{2} = \theta$ is positive. Quadrilateral $ABCD$ will be cyclic if and only if $\theta = 0$. Therefore, $\theta = \frac{\overline{A+C}-\overline{B+D}}{2}$ gives a measure of deviation of the quadrilateral from cyclicity. Therefore, we call θ as acyclic deviation or measure of deviation from the cyclicity of the quadrilateral. Then

$$\begin{aligned} \overline{A + C} + \overline{B + D} &= 360^\circ \quad \text{(as already known)} \\ \overline{A + C} - \overline{B + D} &= 2\theta \\ \therefore\ A + C &= \frac{360^\circ + 2\theta}{2} = 180^\circ + \theta \end{aligned}$$

and

$$B + D = \frac{360^\circ - 2\theta}{2} = 180^\circ - \theta.$$

Now, let $AC = x$, $BD = y$, $AB = a$, $BC = b$, $CD = c$ and $DA = d$.



So, the area of the quadrilateral $ABCD$ = Area $\triangle ABC$ + Area $\triangle ADC$

$$= \frac{1}{2}ab\sin B + \frac{1}{2}cd\sin D.$$

From $\triangle ABC$ using cosine formula

$$\begin{aligned} x^2 &= a^2 + b^2 - 2ab\cos B \\ &= (a-b)^2 + 2ab(1-\cos B) = (a-b)^2 + 4ab\sin^2\frac{B}{2} \qquad (26.26) \\ &= (a+b)^2 - 2ab(1+\cos B) = (a+b)^2 - 4ab\cos^2\frac{B}{2}. \qquad (26.27) \end{aligned}$$

From $\triangle ADC$ using cosine rule

$$\begin{aligned} x^2 &= c^2 + d^2 - 2cd\cos D \\ &= (c-d)^2 + 2cd\sin^2\frac{D}{2} \qquad (26.28) \\ &= (c+d)^2 - 2cd\cos^2\frac{D}{2}. \qquad (26.29) \end{aligned}$$

Using (26.26) and (26.29)

$$\begin{aligned} &(c+d)^2 - 2cd\cos^2\frac{D}{2} = (a-b)^2 + 2ab\sin^2\frac{B}{2} \\ \Longrightarrow\ &(c+d)^2 - (a-b)^2 = 4ab\sin^2\frac{B}{2} + 4cd\cos^2\frac{D}{2} \\ \Longrightarrow\ &(c+d+a-b)(c+d-a+b) = 4ab\sin^2\frac{B}{2} + 4cd\cos^2\frac{D}{2}. \qquad (26.30) \end{aligned}$$

Using (26.27) and (26.28)

$$(a+b)^2-(c-d)^2=4ab\cos^2\frac{B}{2}+4cd\sin^2\frac{D}{2}$$
$$(a+b+c-d)(a+b-c+d)=4ab\cos^2\frac{B}{2}+4cd\sin^2\frac{D}{2}. \tag{26.31}$$

Now if we denote $a+b+c+d=2s$

$$a+b+c-d=a+b+c+d-2d=2(s-d)$$
$$a+b+d-c=a+b+c+d-2c=2(s-c)$$
$$a-b+c+d=a+b+c+d-2b=2(s-b)$$
$$b+c+d-a=a+b+c+d-2a=2(s-a).$$

Substituting these in (26.30) and (26.31), we get

$$\not{4}(s-c)(s-d)=\not{4}(ab\cos^2\frac{B}{2}+cd\sin^2\frac{D}{2}) \tag{26.32}$$

$$\not{4}(s-a)(s-b)=\not{4}(ab\sin^2\frac{B}{2}+cd\cos^2\frac{D}{2}). \tag{26.33}$$

From (26.32) and (26.33), we get

$$(s-a)(s-b)(s-c)(s-d)=(ab\cos^2\frac{B}{2}+cd\sin^2\frac{D}{2})(ab\sin^2\frac{B}{2}+cd\cos^2\frac{D}{2})$$
$$=a^2b^2\sin^2\frac{B}{2}\cos^2\frac{B}{2}+c^2d^2\sin^2\frac{D}{2}\cos^2\frac{D}{2}+abcd\cos^2\frac{B}{2}\cos^2\frac{D}{2}+abcd\sin^2\frac{B}{2}\sin^2\frac{D}{2}$$
$$=\Big(ab\sin\frac{B}{2}\cos\frac{B}{2}+cd\sin\frac{D}{2}\cos\frac{D}{2}\Big)^2+abcd\Big(\cos\frac{B}{2}\cos\frac{D}{2}-\sin\frac{B}{2}\sin\frac{D}{2}\Big)^2$$
$$=\Big(\frac{ab\sin B}{2}+\frac{cd\sin D}{2}\Big)^2+abcd\cos^2\frac{B+D}{2}.$$

We already know

$$\frac{1}{2}ab\sin B=\text{Area }\triangle ABC$$
$$\frac{1}{2}cd\sin D=\text{Area }\triangle ACD$$

and

$$\cos\frac{B+D}{2}=\cos\frac{180-\theta}{2}=\sin\frac{\theta}{2},$$

where θ is acyclic deviation.

Therefore,

$$(s-a)(s-b)(s-c)(s-d)-abcd\sin^2\frac{\theta}{2}=(\text{Area of quadrilateral }ABCD)^2,$$

which implies

$$\text{Area of the quadrilateral } ABCD = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd\sin^2\frac{\theta}{2}}.$$

Therefore, we see that to know the area of the quadrilateral $ABCD$, we need the measure of acyclic deviation of the quadrilateral in addition to the measure of four sides.

We get the Brahmagupta formula for the area of the cyclic quadrilateral as a particular case when $\theta = 0$, then we know essentially quadrilateral becomes cyclic.

Therefore, the Brahmagupta formula for area $\sqrt{(s-a)(s-b)(s-c)(s-d)}$ is true only for cyclic quadrilaterals.

We can use cosine formula for $\triangle ABD$ and $\triangle CBD$ and get

$$y^2 = a^2 + d^2 - 2ad\cos A \tag{26.34}$$
$$= b^2 + c^2 - 2bc\cos C. \tag{26.35}$$

Then from (26.26) and (26.27)

$$cd(a^2+b^2-2ab\cos B) + ab(c^2+d^2-2cd\cos D) = (cd+ab)x^2$$
$$= (cd)(a^2+b^2) + ab(c^2+d^2) - 2abcd(\cos B + \cos D)$$

or

$$x^2 = \frac{(cd)(a^2+b^2)+ab(c^2+d^2)-2abcd(\cos B+\cos D)}{(cd+ab)}$$
$$= \frac{(ad+bc)(ac+bd)}{ab+cd} - \frac{4abcd}{ab+cd}\left(\cos\frac{B+D}{2}\cdot\cos\frac{B-D}{2}\right)$$
$$= \frac{(ad+bc)}{ab+cd}(ac+bd) - \frac{4abcd}{ab+cd}\sin\frac{\theta}{2}\cdot\cos\frac{B-D}{2}. \tag{26.36}$$

Similarly, from (26.34) and (26.35), we get

$$(ad+bc)y^2 = (ab+cd)(ac+bd) - 2abcd(\cos A + \cos C)$$
$$\text{or } y^2 = \frac{(ab+cd)(ac+bd)}{ad+bc} + \frac{4abcd}{ad+bc}\sin\frac{\theta}{2}\cdot\cos\frac{A-C}{2}. \tag{26.37}$$

Expressions (26.36) and (26.37) will give the formula for the diagonals x and y from where Brahmagupta formula for the diagonals of a cyclic quadrilateral can be obtained as a particular case by putting $\theta = 0$.

But these formulae can be used only if one angle of the quadrilateral in addition to four sides are known.

For if B is known, then from (26.26) x can be found. Knowing x, we can find D from (26.27). Therefore, knowing B, $\frac{B+D}{2}$ and $\frac{B-D}{2}$ are known. Therefore, θ is also

known. Now to know y, we need the value of the $\frac{A-C}{2}$, which can be found using the following method:

$$\frac{A-C}{2} = \frac{\overline{A_1+A_2} - \overline{C_1+c_2}}{2},$$

where $A_1 = \angle CAB,\ A_2 = \angle CAD,\ C_1 = \angle ACB,\ C_2 = \angle ACD.$

Napier's rule applied to $\triangle ABC$ gives

$$\tan\frac{A_1-C_1}{2} = \frac{b-a}{b+a}\cot\frac{B}{2}.$$

Napier's rule applied to $\triangle ADC$ gives

$$\tan\frac{A_2-C_2}{2} = \frac{c-d}{c+d}\cot\frac{D}{2}.$$

Therefore

$$\begin{aligned}\sec^2\frac{A_1-C_1}{2} &= 1+\left(\frac{b-a}{b+a}\right)^2\cdot\cot^2\frac{B}{2}\\ &= \frac{(b+a)^2\sin^2\frac{B}{2}+(b-a)^2\cos^2\frac{B}{2}}{(b+a)^2\sin^2\frac{B}{2}}\\ \therefore\ \frac{1}{\cos^2\frac{A_1-C_1}{2}} &= \frac{(b+a)^2\sin^2\frac{B}{2}}{(b+a)^2\sin^2\frac{B}{2}+(b-a)^2\cos^2\frac{B}{2}}.\end{aligned}$$

Note that $(b+a)^2\sin^2\frac{B}{2}+(b-a)^2\cos^2\frac{B}{2} = x^2$ so that

$$\begin{aligned}\cos\frac{A_1-C_1}{2} &= \frac{(b+a)\sin\frac{B}{2}}{x}\\ \sin^2\frac{A_1-C_1}{2} &= 1-\cos^2\frac{A_1-C_1}{2} = \frac{(a-b)^2\cos^2\frac{B}{2}}{x^2}\\ \Longrightarrow\ \sin\frac{A_1-C_1}{2} &= \frac{(a-b)\cos\frac{B}{2}}{x}.\end{aligned}$$

Similarly, we get from Napier's formula applied to $\triangle ADC$

$$\begin{aligned}\cos\frac{A_2-C_2}{2} &= \frac{(c+d)\sin\frac{D}{2}}{x}\\ \sin\frac{A_2-C_2}{2} &= \frac{(c-d)\cos\frac{D}{2}}{x}.\end{aligned}$$

From these two results, we get

$$\begin{aligned}
\cos\frac{A-C}{2} &= \cos\frac{A_1-C_1}{2}\cdot\cos\frac{A_2-C_2}{2} - \sin\frac{A_1-C_1}{2}\cdot\sin\frac{A_2-C_2}{2}\\
&= \frac{(a+d)\sin\frac{B}{2}(c+d)\sin\frac{D}{2}}{x^2} - \frac{(a-d)\cos\frac{B}{2}(c-d)\cos\frac{D}{2}}{x^2}\\
&= \left\{\frac{\left[(ac+bd)+(ad+bc)\right]\sin\frac{B}{2}\sin\frac{D}{2}}{x^2}\right\} - \left\{\frac{\left[(ac+bd)-(ad+bc)\right]\cos\frac{B}{2}\cos\frac{D}{2}}{x^2}\right\}\\
&= \frac{(ac+bd)(\sin\frac{B}{2}\sin\frac{D}{2}-\cos\frac{B}{2}\cos\frac{D}{2})+(ad+bc)(\cos\frac{B}{2}\cos\frac{D}{2}+\sin\frac{B}{2}\sin\frac{D}{2})}{x^2}\\
&= \frac{(ad+bc)\cos\frac{B+D}{2}-(ac+bd)\cos\frac{B-D}{2}}{x^2}\\
&= \frac{(ad+bc)\sin\theta-(ac+bd)\cos\frac{B-D}{2}}{x^2}.
\end{aligned}$$

Since B and D are already known, $\frac{B-D}{2}$ is known. Therefore, y^2 can be found. Thus, y is also known.

Hence, if a, b, c, d, B or a, b, c, d, x given, we have the above formulae to find all remaining elements, namely, A, C, B, D and θ.

References

1. Gupta RC (1977) Parameśvara's rule for the circumradius of a cyclic quadrilateral. Hist Math 4:67–74
2. Rangacarya M (1912) The Ganita-sāra-saṅgraha of Mahāvīrācārya, edited with English Translation and Notes. Govt. of Madras, Madras (now Chennai)
3. Sharma RS et al (1966) Śrī Brahmagupta viracita Brāhma-sphuṭa-siddhānta. Indian Institute of Astronomical and Sanskrit Research, New Delhi
4. Shukla KS, Sarma KV (1976) Āryabhaṭiya of Āryabhaṭa. The Indian National Science Academy, New Delhi
5. Toomer GJ (1984) Ptolemy's almagest. Gerald Duckworth & Co., Ltd